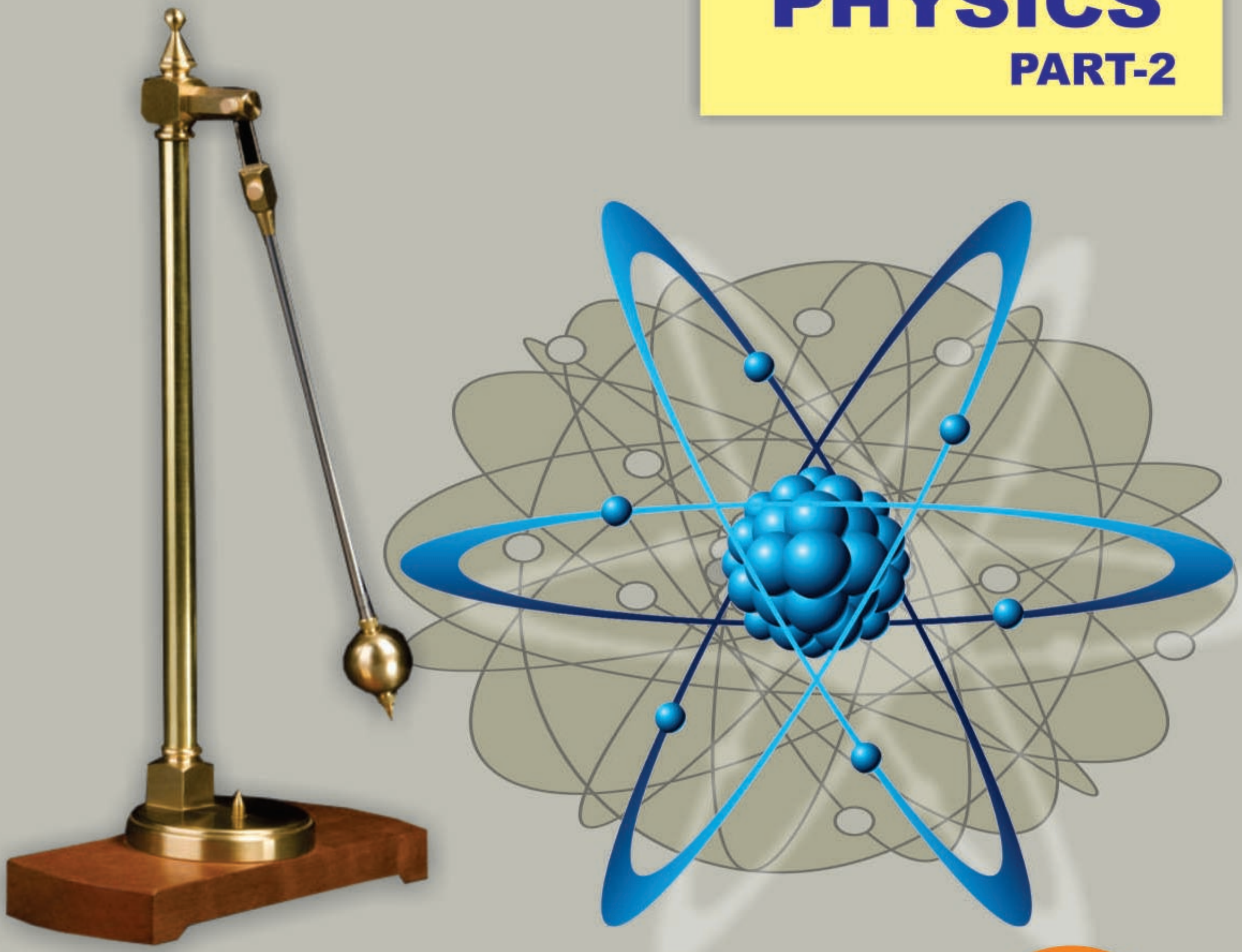
Gujarat Secondary and Higher Secondary Education Board, Gandhinagar

PHYSICS

PART-2



Guiding Question Bank (MCQ) for
Preparation of JEE/NEET Examination

Price
₹95/-

Secretary
Gujarat Secondary and Higher Secondary Education Board
Sector 10-B, Near Old Sachivalaya, Gandhinagar-382 010.

Copyright of this book is reserved by Gujarat Secondary and Higher Secondary Education Board, Gandhinagar. No reproduction of this book in whole or in part, or in any form is permitted without written permission of the Secretary, Gujarat Secondary and Higher Secondary Education Board, Gandhinagar.

Gujarat Secondary and Higher Secondary Education Board, Gandhinagar

QUESTION BANK
PHYSICS

Price : ₹ 95.00

Published by :
Secretary
Gujarat Secondary and Higher Secondary Education Board,
Gandhinagar

Contribution

1	Dr. Hasmukh Adhiya (IAS)	Principal Secretary , Education Department Gandhinagar
2	Shri R. R. Varsani (IAS)	Chairman , G.S&H.S.E. Bord, Gandhinagar
3	Shri H. K. Patel (G.A.S)	Dy. Chairman, G.S&H.S.E. Bord, Gandhinagar
4	Shri M. I. Joshi (G.E.S)	Secretary , G.S&H.S.E. Bord, Gandhinagar

Coordination

1	Shri B. K. Patel	O.S.D., G.S&H.S.E. Bord, Gandhinagar
2	Shri D. A.Vankar	Assistant Secretary (Retd.), G.S&H.S.E. Bord, Gandhinagar
5	Shri G. M. Rupareliya	Assistant Secretary, G.S&H.S.E. Bord, Gandhinagar

Expert Teachers

1.	Shri J. M. Patel	Shree J. M. Chaudhary Sarvajanik Vidhyalaya, Mehsana
2.	Shri K. D. Patel	J. N. Balika Vidhyalaya, Saraspur
3.	Shri Mayur M. Raval	P. J. Vakharia High School, Kalol
4.	Shri S. G. Patel	Sarkari Schook, Sector-12, Gandhinagar
5.	Shri J. P. Joshi	Diwan Ballubhai High School, Ahmedabad
6.	Shri Vasudev B. Raval	Vidhya Mandir High School, Palanpur
7.	Shri Surendrabhai M. Rajkutir	Convent of Jesus And Merry
8.	Shri Sureshchandra H. Patel	Alambic Vidhyalaya, Vadodara
9.	Shri C. D. Patel	Lalbahadur Shastri Vidhyalaya, Vadodara
10.	Shri Mukesh N. Gandhi	New English School, Nadiad
11.	Shri Dineshbhai V. Suthar	Retired Teacher
12.	Shri S. S. Patel	J. M. Chaudhary Sarvajanik Vidhyalaya, Mehsana
13.	Shri Jayesh M. Purohit	Ankur Vidhyalaya, Ahmedabad
14.	Smt. Asha M. Patel	Shree M.B. Vamdot Sarvajanik High School, Bardoli
15.	Shri Maheshbhai Dhandhla	Bhavnagar
16.	Shri Mukesh M. Bhatt	Bhavnagar
17.	Shri Anilkumar Trivedi	Anand
18.	Shri Anand Thakkar	Navchetan High School, Ahmedbad
19.	Shri Sudhirkumar G. Patel	Nutan High School, Visnagar
20.	Smt. Anita Pillai	Surat

PREFACE

Uptil now , the Students had to appear in various entrance examinations for engineering and medical courses after std-12. The burden of examinations on the side of the students was increasing day-by-day. For alleviating this difficulty faced by the students, from the current year, the Ministry of Human Resource Development , Government of India, has Introduced a system of examination covering whole country. For entrance to engineering colleges, JEE(Main) and JEE(Advanced) examinations will be held by the CBSE. The Government of Gujarat has except the new system and has decided to follow the examinations to be held by the CBSE.

Necessary information pertaining to the proposed JEE (Main) and JEE(Advanced) examination is available on CBSE website www.cbse.nic.in and it is requested that the parents and students may visit this website and obtain latest information – guidance and prepare for the proposed examination accordingly. The detailed information about the syllabus of the proposed examination, method of entrances in the examination /centers/ places/cities of the examinations etc. is available on the said website. You are requested to go through the same carefully. The information booklet in Gujarati for JEE(Main) examination booklet has been brought out by the Board for Students and the beneficieries and a copy of this has been already sent to all the schools of the state. You are requested to take full advantage of the same also However, it is very essential to visit the above CBSE website from time to time for the latest information – guidance . An humble effort has been made by the Gujarat secondary and Higher Secondary Education Boards, Gandhinagar for JEE and NEET examinations considering the demands of the students and parents , a question bank has been prepared by the expert teachers of the science stream in the state. The MCQ type Objective questions in this Question Bank will provide best guidance to the students and we hope that it will be helpful for the JEE and NEET examinations.

It may please be noted that this "Question Bank" is only for the guidance of the Students and it is not a necessary to believe that questions given in it will be asked in the examinations. This Question Bank is only for the guidance and practice of the Students. We hope that this Question Bank will be useful and guiding for the Students appearing in JEE and NEET entrance examinations. We have taken all the care to make this Question Bank error free, however, if any error or omission is found, you are requested to refer to the text – books.

M.I. Joshi — **Secretary**

R.R. Varsani (IAS) — **Chairman**

Date: 02/01/2013

INDEX

Unit No.	Unit Name	GSEB	NEET	JEE	Page No.
PART - II					
11	Electrostatics	Y	Y	Y	1
12	Current Electricity	Y	Y	Y	53
13	Magnetic Effects of Current and Magnetism	Y	Y	Y	107
14	Electromagnetic Induction and Alternating Currents	Y	Y	Y	173
15	Electromagnetic Waves	Y	Y	Y	197
16	Optics	Y	Y	Y	211
17	Dual Nature of Matter and Radiation	Y	Y	Y	235
18	Atoms and Nuclii	Y	Y	Y	285
19	Electronic Devices	Y	Y	Y	321
20	Communication Systems	Y	Y	Y	347
21	Experimental Skills (1 to 21)	Y	Y	Y	359

UNIT - 11
ELECTROSTATICS

SUMMARY

1. **Electric Charge :** Just as masses of two particles are responsible for the gravitational force, charges are responsible for the electric force. Electric charge is an intrinsic property of a particle.

   Charges are of two types : (1) Positive charges (2) Nagative charges.

   The force acting between two like charges is repulsive and two unlike charges it is attractive between .

2. **Quantization of Electric Charge :** The magnitude of all charges found in nature are an integral multiple of a fundamental charge. $Q = ne$, where e is the fundamental unit of charge.

3. **Conservation of Electric Charge :** Irrespective of any process taking place, the algebraic sum of electric charges in an electrically isolated system always remains constant.

4. **Coulomb's Law :** The electric force between two stationary point charges is directly proportional to the product of their charges and inversely proportional to the square of the distance between them.

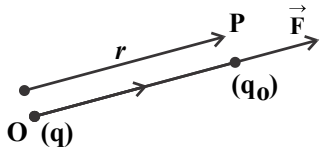
$$F = k\frac{q_1 q_2}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

If $q_1 q_2 > 0$ then there is a repulsion between the two charges and for $q_2 q_2 < 0$, there is a attraction between the charges.

5. **Equation for Force using Columb's Law, when two charges are placed in a medium having dielectric constant k.**

(1) The electric force $\left(\vec{F}\right)$ experienced by a test charge (q_0) due to a source charge (q) when both are placed in a medium having dielectric constant k and separated by a distance r, is given by :

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{kr^2}\hat{r}$$



Here $\hat{r}$ is the unit vector directed from q to q_0.

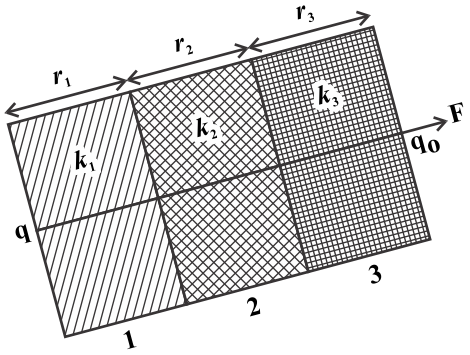
(2) The equation of coulomb's force may be written as follows : $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{\left(\sqrt{k}\, r^2\right)}\hat{r}$

(3) If the source charge and test charge are separated by a number of medium of thickness d_1 , d_2 , d_3..... having dielectric constants k_1, k_2, k_3.....respectively, then the electric force on charge q_0 due to a charge q is given by

$$\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{\left(\sqrt{k_1}d_1^2 + \sqrt{k_2}d_2^2 + \sqrt{k_3}d_3^2\right)}\hat{r} \quad \textbf{OR} \quad \vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{\left[\sum \sqrt{k_i}\ d_i^2\right]}\hat{r}$$

In this equation k_i is dielectric constant of medium which spreads through the distance d_i along the line joining q and q_0.

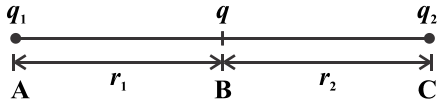
For example, see the figure below :



Here, the space between the charges q and q_0 is filled with medium (1, 2, 3). The thickness of medium 1 is d_1 and its dielectric consant is k_1 Similarly the thickness of medium 2 and 3 is d_2 and d_3 of medium 3 and their dielectric constants are k_2 and k_3 respectively.

6. Conditions for Equilibrium in Various Cases :

Suppose three charges q_1, q_2 and q are situated on a straight line as shown below :



If q_1 and q_2 are like charges and q is of unlike charge then,

(1) Force on $q_1 = F_1 = \dfrac{q_1}{4\pi\varepsilon_0}\left[\dfrac{q_1}{(r_1 + r_2)^2} - \dfrac{q}{r_1^2}\right]$

(2) Force on $q_2 = F_2 = \dfrac{q_2}{4\pi\varepsilon_0}\left[\dfrac{q_1}{(r_1 + r_2)^2} - \dfrac{q}{r_2^2}\right]$

(3) Force on $q = F = \dfrac{1}{4\pi\varepsilon_0}\left[\dfrac{q_1}{r_1^2} - \dfrac{q_2}{r_2^2}\right]$

Now, from above equations, it is clear that various equilibrium conditions can be as follows :

(a) Condition for F_1 to be zero is, $\dfrac{q}{r_1^2} = \dfrac{q_2}{(r_1 + r_2)} \Rightarrow \dfrac{q}{q_2} = \dfrac{r_1^2}{(r_1 + r_2)^2}$

(b) Condition for F_2 to be zero is, $\dfrac{q}{r_2^2} = \dfrac{q_1}{(r_1 + r_2)^2} \Rightarrow \dfrac{q}{q_1} = \dfrac{r_2^2}{(r_1 + r_2)^2}$

(c) Condition for F to be zero is, $\dfrac{q_1}{r_1^2} = \dfrac{q_2}{r_2^2} \Rightarrow \dfrac{q_1}{r_2} = \dfrac{r_1^2}{r_2^2}$

If q_1, q_2 and q are of same type charges in nature, then,

(1) Charge q will be in equilibrium, if

$$F = \frac{q}{4\pi\varepsilon_0}\left[\frac{q_1}{r_1^2} - \frac{q_2}{r_2^2}\right] = 0 \Rightarrow \frac{q_1}{r_1^2} = \frac{q_2}{r_2^2} \Rightarrow \boxed{\frac{q_1}{q_2} = \frac{r_1^2}{r_2^2}}$$

(2) Charges q_1 and q_2 will not be in equilibrium.

7. **Electric Field Intensity :** The electric force acting on a unit positive charges at a given point in an electric field of a system of charges is called the electric field or the intensity of electric field $\left(\vec{E}\right)$ at that point.

$$\vec{E} = \frac{\vec{F}}{q}$$

The SI unit of $\vec{E}$ is $\frac{N}{C}$ or Vm^{-1}.

If $\vec{r_1}, \vec{r_2}, \ldots\ldots\ldots\vec{r_n}$ are the position vectors of the charges $q_1, q_2, \ldots\ldots\ldots q_n$ respectively, then the resultant electric field at a point of position vector $\vec{r}$ is,

$$\vec{E} = k \sum_{j=1}^{n} \frac{q_j}{\left|\vec{r} - \vec{r_j}\right|^3} \left(\vec{r} - \vec{r_j}\right)$$

8. **Electric Dipole :** A system of two equal and opposite charge, separated by a finite distance is called **electric dipole**.

Electric dipole moment $\vec{p} = q\left(2\vec{a}\right)$

The direction of $\vec{p}$ is from the negative electric charge to the positive electric charge.

9. Electric field of a dipole on the axis of the dipole at point $z = z$

$$\vec{E}\ (z) = \frac{2kp}{z^3}\ \hat{p}\ (\text{for } z >> a)$$

Electric field of a dipole on the equator of the dipole at point $y = y$

$$\vec{E}\ (y) = \frac{kp}{y^3}\ \hat{p}\ (\text{for } y >> a)$$

10. The torque acting on the dipole place in an uniform the electric field at an angle θ,

$\vec{\tau} = \vec{p} \times \vec{E}$, $|\vec{\tau}| = p\, E \sin\theta$

11. **Electric Flux :** Electric flux associated with surface of area $\vec{A}$, placed in the uniform electric field.

$\phi = \vec{E} \cdot \vec{A} = EA\cos\theta$ where, θ is the angle between $\vec{E}$ and $\vec{A}$,

Its SI unit is $\frac{Nm^2}{C}$ or V.m.

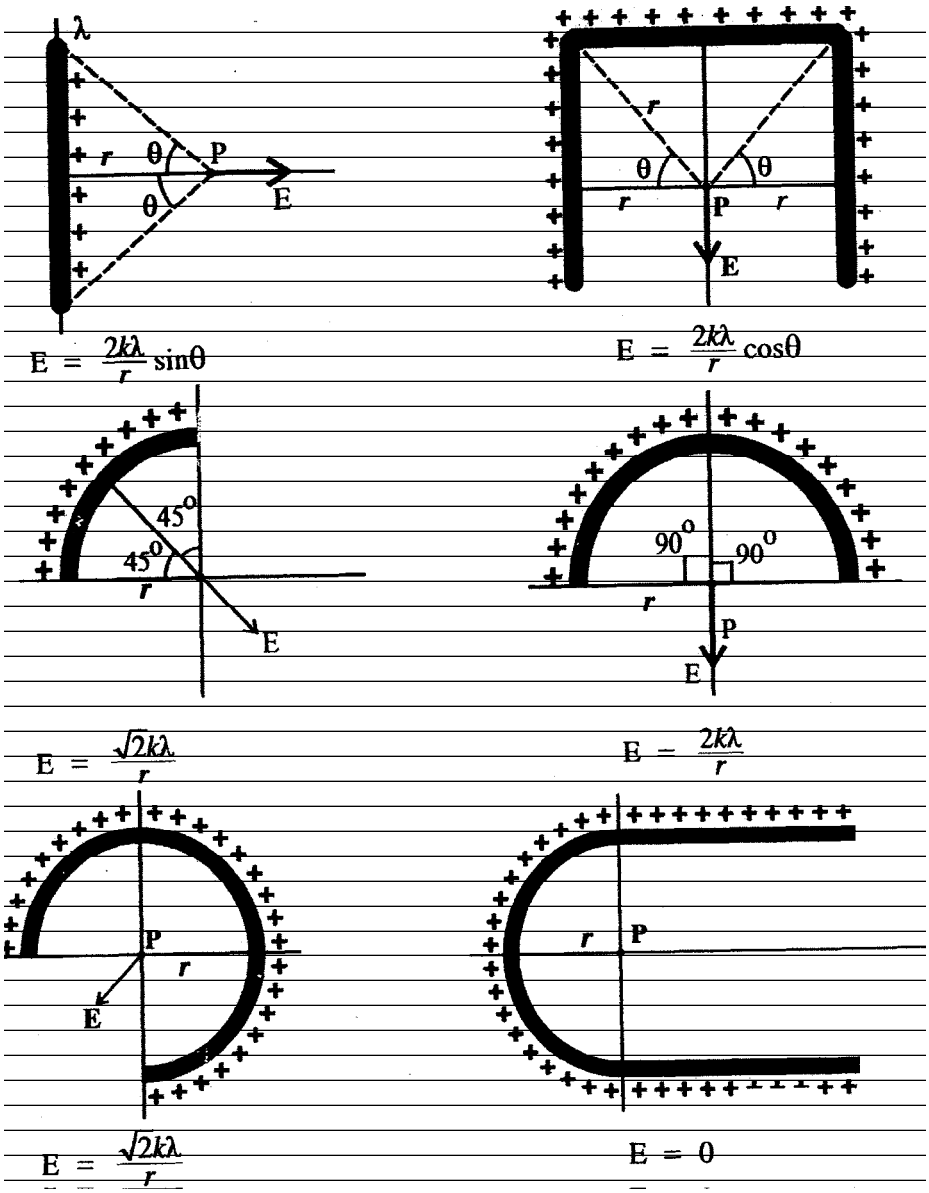
12. **Gauss's Law :** The total electric flux associated with the closed surface,

$$\phi = \int_S \vec{E} \cdot d\vec{a} = \frac{\Sigma q}{\varepsilon_0}$$ where, Σq is the net charge enclosed by the surface.

13. Electric field due to an infinitely long straight charged wire,

$\vec{E} = \frac{\lambda}{2\pi\varepsilon_0}\frac{1}{r}\hat{r}$, where, r is the perpendicular distance from the charged wire.

14. Electric field due to bending of charged rod,



15. Electric field due to uniformly charged thin spherical shell,

(1) Electric field inside the shell $\vec{E} = 0$

(2) Electric field at a distance r from the centre outside the shell,

$E = k\frac{q}{r^2} = \frac{\sigma}{\varepsilon_0}\frac{R^2}{r^2}$ where, R = radius of spherical shell.

16. Electric field due to a uniformly charged density sphere of radius R,

(1) Electric field inside the region of the sphere, $E = \frac{Q}{4\pi\varepsilon_0}\frac{r}{R^3}\frac{\rho r}{3\varepsilon_0}$

(2) Electric field outside the sphere, $E(r) = \frac{Q}{4\pi\varepsilon_0}\frac{r}{r^2} = \frac{R^3\rho}{3r^2\varepsilon_0}$

where, Q is the total charge inside the sphere.

17. The information about the work done to take an electric charge from one point to the other in a given electric field, obtained from the quantities called electric potential and electric potential energy.

18. $\int_A^B \vec{E}\cdot\vec{dr}$ is the line-integral of electric field between point A and B and it shows the work done by the electric field in taking a unit positive charge from A and B. Moreover, it does not depend on the path and $\oint \vec{E}\cdot\vec{dr} = 0$.

19. "The work required to be done against the electric field to bring a unit positive charge from infinite distance to the given point in the electric field, is called the electric potential (V) at that point".

Electric potential at point P is $V_p = -\int_\infty^P \vec{E}\cdot\vec{dr}$

It unit is $\frac{\text{Joule}}{\text{coulomb}} = \text{volt.}$ Symbolically $V = \frac{J}{C}$

Its dimensional formula is $M^1\ L^2\ T^{-3}\ A^{-1}$

Absolute value of electric potential has no importance but only the change in it is important.

20. "The work required to be done against the electric field to bring a given change (q) from infinite distance to the given point in the electric field is called the electric potential energy of that electric charge at that point."

$U_p = -q\int_\infty^P \vec{E}\cdot\vec{dr} = qV_p$

The absolute value of electric potential energy has no importance, only the change in it is important.

21. Electric potential at point P, lying at a distance r from a point charge q is $V_p = \frac{kq}{r}$

22. The electric potential at a point at distance r from an electric dipole is

$v(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}$, (For $r >> 2a$)

Potential on its axis is $V = \pm\frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}$, Potential on its equator is $V = 0$

23. Electric potential at a point $\vec{r}$ due to a system of point charge $q_1, q_2, \ldots\ldots q_n$ situated at position at position $r_1, r_2, \ldots\ldots r_n$ is $V = \sum_{i=1}^{n}\frac{kq_i}{\left|\vec{r} - \vec{r_i}\right|}$

The electric potential at point $\vec{r}$, due to a continuous charge distribution is

$$V\left(\vec{r}\right) = \frac{1}{4\pi\varepsilon_0} \int_{\text{volume}} \frac{\sigma\left(\vec{r}\right) d\tau'}{\left|\vec{r} - \vec{r}_i\right|}$$

The electric potential due to a spherical shell is

$$V = \frac{1}{4\pi\varepsilon_0} \frac{q}{r} \left(\text{For } r \geq R\right) \text{ and } V = \frac{1}{4\pi\varepsilon_0} \frac{q}{R} \left(\text{For } r \leq R\right)$$

24. A surface on which electric potential is equal at all points is called an equipotential surface. The direction of electric field is normal to the equipotential surface.

25. $E = \frac{-dV}{dl}$ gives the magnitude of electric field in the direction of $\vec{dl}$.

To find E from V, in general, we can use the equation

$$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{-\partial V}{\partial y}\hat{j} + \frac{-\partial V}{\partial z}\hat{k}\right)$$

The direction of electric field is that in which the rate of decrease of electric potential with distance $\frac{-dV}{dl}$ is maximum, and this direction is always normal to the equipotential surface.

26. The electrostatic potential energy of a system of point charges $q_1, q_2, \ldots\ldots q_n$ situated at positions $r_1, r_2, \ldots\ldots r_n$ is

$$U = \sum_{\substack{i=1 \\ i<j}}^{n} \frac{kq_i\ q_j}{r_{ij}} \text{ where } r_{ij} = r_j - r_i$$

27. The electrostatic potential energy of an electric dipole in an external electric field E, is

$$U = -\vec{E} \cdot \vec{p} = -Ep\cos\theta$$

28. When a metallic conductor is placed in an external electric field,

(i) A stationary charge distribution is induced on the surface of the conductor.

(ii) The resultant electric field inside the conductor is zero.

(ii) The net electric charge inside the conductor is zero.

(iv) The electric field at every point on the outer surface of conductor is locally normal to the surface.

(v) The electric potential inside the region of conductor is the same every where.

(vi) If there is a cavity in the conductor then, even when the conductor is placed in an external electric field, the resultant electric field inside the conductor and also inside the cavity is always zero.

This fact is called the electrostatic shielding.

When electric charge is placed on the metallic conductor :

(i) The electric field inside the conductor is zero everywhere.

(ii) The charge is distributed only on the outer surface of the conductor.

(iii) The electric field on the surface is locally normal, and is equal to $\vec{E} = \left(\frac{\sigma}{\varepsilon_0}\right)\hat{n}$.

(iv) If a charge is placed inside the cavity in the conductor, the electric field in the conductor remains zero.

29. "A device formed by two conductors seprated from each other is called a capacitor." Its capacitance is $C = \frac{Q}{V}$ = constant. The unit of C is $\frac{\text{coulomb}}{\text{volt}}$ which is also called farad.

$1\ \mu F = 10^{-6}$ F. $pF = 10^{-12}$ F

30. The effective capacitance in series connection is C then,

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \text{.....}$$

The effective capacitance in parallel connection is C then,

$$C = C_1 + C_2 + C_3 + \text{.....}$$

31. The capacitance of the parallel plate capacitor is $C = \frac{\varepsilon_0 A}{d}$.

32. The energy stored in the capacitor is $U = \frac{Q^2}{2C} = \frac{CV^2}{2} = \frac{VQ}{2}$ and the energy density = energy stored per unit volume $= \frac{1}{2}\varepsilon_0 E^2$, where E = electric field.

33. When a dielectric is placed in an external electric field E_0, polarisation of dielectric occurs due to electrical induction. The electric field produced by these induced charges is in the opposite direction to the direction of external electric field. Hence the resultant electric field E, inside the dielectric is less than the external electric field E_0.

The dipole moment produced per uint volume is called the intensity of polarisation or in short polarisation

$\vec{P}\hat{n} = \sigma_b$.

Since $P \alpha E$, $P = \varepsilon_0 x_e E \bullet x_e$ is called the electric susceptibility of the dielectric medium. $\varepsilon_0 (1 + x_e)$ is called the permittivitty ε of the dielectric medium. $\frac{\varepsilon}{\varepsilon_0}$ is called the relative permittivity of that medium and it is also called the dielectric constant K.

i.e. $\frac{\varepsilon}{\varepsilon_0} = \varepsilon_r = \mathrm{K}$

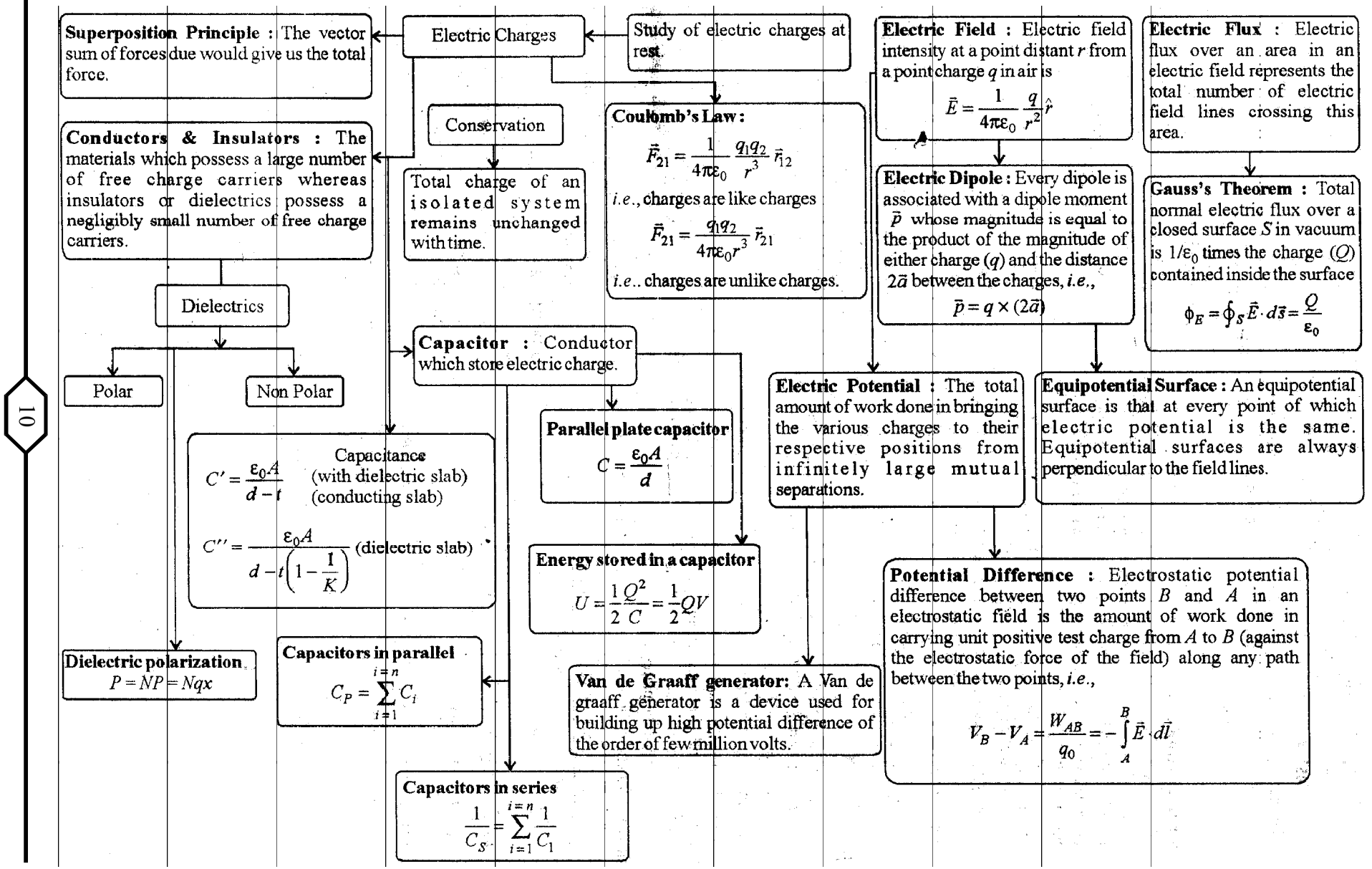
$\mathrm{K} = 1 + x_e$, $\mathrm{E} = \frac{\mathrm{E}_0}{\mathrm{K}}$. Thus in the dielectric the electric field reduces to the Kth part.

$\vec{\mathrm{D}} = \varepsilon_0 \vec{\mathrm{E}} + \vec{\mathrm{P}}$ is called the electric displacement. Gauss Law in the presence of dielectric is written as $\oint \vec{\mathrm{D}} \cdot \vec{\mathrm{ds}} = \mathrm{q}$, where q is only the net free charge.

34. when there is air (or vacuum) between the plates of a parallel plate capacitor, the capacitance is $\mathrm{C} = \frac{\varepsilon_0 \mathrm{A}}{\mathrm{d}}$. On placing a medium of dielectric constant K, the capacitance is $\mathrm{C}' = \mathrm{CK}$. Thus the capacitance becomes K times, due to the presence of the dielectric.

35. With the help of Van-De-Graf generator a potential differance of a few nillion volt can be established.

CONCEPT MAP


Electric Charges
Study of electric charges at rest.
Superposition Principle : The vector sum of forces due would give us the total force.
Conservation
Total charge of an isolated system remains unchanged with time.
Coulomb's Law :
$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^3}\vec{r}_{12}$
i.e., charges are like charges
$\vec{F}_{21} = \frac{q_1 q_2}{4\pi\varepsilon_0 r^3}\vec{r}_{21}$
i.e.. charges are unlike charges.
Electric Field : Electric field intensity at a point distant r from a point charge q in air is
$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{r}$
Electric Flux : Electric flux over an area in an electric field represents the total number of electric field lines crossing this area.
Gauss's Theorem : Total normal electric flux over a closed surface S in vacuum is $1/\varepsilon_0$ times the charge (Q) contained inside the surface
$\phi_E = \oint_S \vec{E}\cdot d\vec{s} = \frac{Q}{\varepsilon_0}$
Conductors & Insulators : The materials which possess a large number of free charge carriers whereas insulators or dielectrics possess a negligibly small number of free charge carriers.
Dielectrics
Polar
Non Polar
Electric Dipole : Every dipole is associated with a dipole moment $\vec{p}$ whose magnitude is equal to the product of the magnitude of either charge (q) and the distance $2\vec{a}$ between the charges, i.e.,
$\vec{p} = q \times (2\vec{a})$
Capacitor : Conductor which store electric charge.
Parallel plate capacitor
$C = \frac{\varepsilon_0 A}{d}$
Capacitance
$C' = \frac{\varepsilon_0 A}{d-t}$ (with dielectric slab)
(conducting slab)
$C'' = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{K}\right)}$ (dielectric slab)
Electric Potential : The total amount of work done in bringing the various charges to their respective positions from infinitely large mutual separations.
Equipotential Surface : An equipotential surface is that at every point of which electric potential is the same. Equipotential surfaces are always perpendicular to the field lines.
Energy stored in a capacitor
$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}QV$
Potential Difference : Electrostatic potential difference between two points B and A in an electrostatic field is the amount of work done in carrying unit positive test charge from A to B (against the electrostatic force of the field) along any path between the two points, i.e.,
$V_B - V_A = \frac{W_{AB}}{q_0} = -\int_A^B \vec{E}\cdot d\vec{l}$
Dielectric polarization
$P = NP = Nqx$
Capacitors in parallel
$C_P = \sum_{i=1}^{i=n} C_i$
Van de Graaff generator: A Van de graaff generator is a device used for building up high potential difference of the order of few million volts.
Capacitors in series
$\frac{1}{C_S} = \sum_{i=1}^{i=n}\frac{1}{C_1}$

MCQ

For the answer of the following questions choose the correct alternative from among the given ones.

1. When a Piece of Polythene is rubbed with wool, a charge of -2×10^{-7} is developed on polythene. The mass transferred to polythene is ..... kg.

   (A) 11.38×10^{-19} (B) 5.69×10^{-19} (C) 2.25×10^{-19} (D) 9.63×10^{-19}

2. The protonic charge in 100 gm of water is c

   (A) 4.8×10^{5} (B) 5.4×10^{6} (C) 3.6×10^{4} (D) 4.9×10^{6}

3. A copper sphere of mass 2 gm contains about 2×10^{22} atoms. The charge on the nucleus of each atom is 29e. The fraction of electrons removed.

   (A) 2×10^{-10} (B) 1.19×10^{-12} (C) 1.25×10^{-11} (D) 2.16×10^{-11}

4. The rate of alpha partical falls on neutral spheare is 10^{12} per second. The time in which sphere gets charged by 2 c is ..... sec.

   (A) 2.25 (B) 3.15 (C) 6.25 (D) 1.66

5. A charge Q is divided into two parts and then they are placed at a fixed distance. The force between the two charges is always maximum when the charges are .....

   (A) $\frac{Q}{3}, \frac{Q}{3}$ (B) $\frac{Q}{2}, \frac{Q}{2}$ (C) $\frac{Q}{4}, \frac{3Q}{4}$ (D) $\frac{Q}{5}, \frac{4Q}{5}$

6. Two point charges repel each other with a force of 100 N. One of the charges is increased by 10% and other is reduced by 10%. The new force of repulsion at the same distance would be ..... N.

   (A) 121 (B) 100 (C) 99 (D) 89

7. Given that $q_1 + q_2 = q$ if the between q_1 and q_2 is maximum, $\frac{q_1}{q}$ .....

   (A) 1 (B) 0.75 (C) 0.25 (D) 0.5

8. Two small conducting sphere of equal radius have charges $+1$ c and -2 c respectively and placed at a distance d from each other experience force F_1. If they are brought in contact and separated to the same distance, they experience force F_2. The ratio of F_1 to F_2 is .....

   (A) –8 : 1 (B) 1 : 2 (C) 1 : 8 (D) –2 : 1

9. Three charges, each of value Q, are placed at the vertex of an equilateral triangle. A fourth charge q is placed at the centre of the triangle. If the charges remains stationery then, q = ...

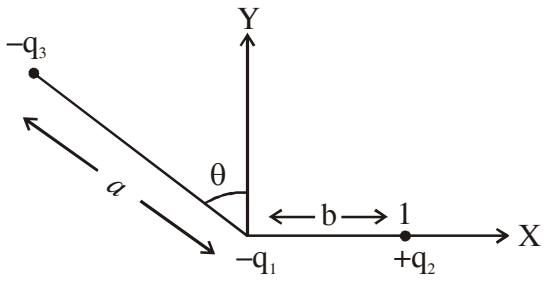
   (A) $\frac{Q}{\sqrt{2}}$ (B) $-\frac{Q}{\sqrt{3}}$ (C) $-\frac{Q}{\sqrt{2}}$ (D) $\frac{Q}{\sqrt{3}}$

10. Two small charged spheres repal each other with a force 2×10^{-3}N . The charge on one sphere is twice that of the other. When these two spheares displaced 10 cm further apart the force is 5×10^{-4}N , then the charges on both the spheres are .....

(A) 1.6×10^{-19}C, 3.2×10^{-19}C (B) 3.4×10^{-19}C, 11.56×10^{-19}C

(C) 33.33×10^{-19}C, 66.66×10^{-19}C (D) 2.1×10^{-19}C, 4.41×10^{-19}C

11. Three charges $-q_1$, $+q_2$ and $-q_3$ are placed as shown in figure. The *x* component of the force on $-q_1$ is proportional to .....



(A) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\sin\theta$ (B) $\frac{q_2}{b^2}-\frac{q_3}{a^2}\cos\theta$

(C) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\sin\theta$ (D) $\frac{q_2}{b^2}+\frac{q_3}{a^2}\cos\theta$

12. Two equal negative charges –q are fixed at points (o, a) and (o, –a). A positive charge Q is released from rest at the point (2a, o) on the X - axis. The charge Q will .....

(A) move to the origin and remain at rest there

(B) execute simple harmonic motion about the origin

(C) move to infinity

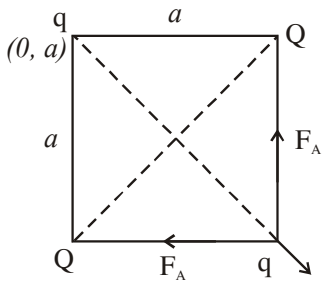
(D) execute oscillations but not simple harmonic motion

13. Four charges, each equal to –Q, are placed at the corners of a square and a charge +q is placed at its centre. If the system is in equilibrium, the value of q is .....

(A) $\frac{Q}{4}\left(1+2\sqrt{2}\right)$ (B) $-\frac{Q}{4}\left(1+2\sqrt{2}\right)$

(C) $-\frac{Q}{2}\left(1+2\sqrt{2}\right)$ (D) $\frac{Q}{2}\left(1+2\sqrt{2}\right)$

14. For the system shown in figure, if the resultant force on q is zero, then Q = ...



(A) $-2\sqrt{2}Q$ (B) $2\sqrt{2}Q$

(C) $2\sqrt{3}Q$ (D) $-3\sqrt{2}Q$

15. Two point positive charges q each are placed at (–a, o) and (a, o). A third positive charge q_o is placed at (o, y). For which value of y the force at q_o is maximum .....

(A) a (B) 2a (C) $\frac{a}{\sqrt{2}}$ (D) $\frac{a}{\sqrt{3}}$

16. Two identical charged spheres suspended from a common point by two massless strings of length *l* are initially a distance d (d << *l*) apart because of their mutual repulsion. The charge begins to leak from both the spheres at a constant rate. As a result the spheres approach each other with a velocity υ. Then function of distance *x* between them becomes .....

(A) $v \alpha x$ (B) $v \alpha x^{\frac{-1}{2}}$ (C) $v \alpha x^{-1}$ (D) $v \alpha x^{\frac{1}{2}}$

17. Three identical spheres each having a charge q and radius R, are kept in such a way that each touches the other two spheares. The magnitude of the electric force on any sphere due to other two is .....

(A) $\frac{R}{2}\frac{1}{4\pi \in_0}\frac{\sqrt{5}}{4}\left(\frac{q}{R}\right)^2$ (B) $\frac{1}{8\pi \in_0}\sqrt{\frac{2}{3}}\left(\frac{q}{R}\right)^2$ (C) $\frac{1}{4\pi \in_0}\frac{\sqrt{3}}{4}\left(\frac{q}{R}\right)^2$ (D) $-\frac{1}{8\pi \in_0}\sqrt{\frac{3}{2}}\left(\frac{q}{R}\right)^2$

18. Two equal negative charges –q are fixed at points (o, a) and (o, –a) on the Y axis. A positive charge q is released from rest at the point *x* (*x* < < a) on the X-axis, then the frequency of motion is .....

(A) $\sqrt{\frac{q^2}{\pi \in_0 ma^3}}$ (B) $\sqrt{\frac{2q^2}{4\pi \in_0 ma^3}}$ (C) $\sqrt{\frac{4q^2}{2\pi \in_0 ma^3}}$ (D) $\sqrt{\frac{q^2}{2\pi \in_0 ma^3}}$

19. Two identical balls having like charges and placed at a certain distance apart repel each other with a certain force. They are brought in contact and then moved apart to a distance equal to half their initial separation. The force of repulsion between them increases 4.5 times in comparison with the initial value. The ratio of the initial charges of the balls is .....

(A) 4 : 1 (B) 6 : 1 (C) 3 : 1 (D) 2 : 1

20. A point charge q is situated at a distance r from one end of a thin conducting rod of length L having a charge Q (uniformly distributed along its length). The magnitude of electric force between the two, is .....

(A) $\frac{2kqQ}{r(r+L)}$ (B) $\frac{kqQ}{r(r+L)}$ (C) $\frac{kqQ}{r(r-L)}$ (D) $\frac{kQ}{r(r+L)}$

21. Two point charges of +16 c and –9 c are placed 8 cm apart in air. ..... distance of a point from –9 c charge at which the resultant electric field is zero.

(A) 24 cm (B) 9 cm (C) 16 cm (D) 35 cm

22. Point charges 4 c and 2 c are placed at the vertices P and Q of a right angle triangle PQR respectively. Q is the right angle, $PR = 2\times10^{-2}m$ and $QR = 10^{-2}m$. The magnitude and direction of the resultant electric field at c is .....

(A) $4.28\times10^{9}\ NC^{-1}, 45^{0}$ (B) $2.38\times10^{8}\ NC^{-1}, 40.9^{0}$

(C) $1.73\times10^{4}\ NC^{-1}, 34.7^{0}$ (D) $4.9\times10^{10}\ NC^{-1}, 34.7^{0}$

23. An inclined plane making an angle of 30° with the horizontal is placed in an uniform electric field E = 100 Vm^{-1}. A particle of mass 1 kg and charge 0.01 c is allowed to slide down from rest from a height of 1m. If the coefficient of friction is 0.2 the time taken by the particle to reach the bottom is ..... sec.

(A) 2.337 (B) 4.337 (C) 5 (D) 1.337

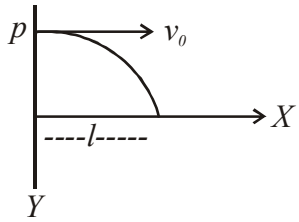
24. A small sphere whose mass is 0.1 gm carries a charge of $3\times10^{-10}C$ and is tieup to one end of a silk fibre 5 cm long. The other end of the fibre is attached to a large vertical conducting plate which has a surface charge of $25\times10^{-6}Cm^{-2}$, on each side. When system is freely hanging the angle fibre makes with vertical is .....

(A) 41.8^{0} (B) 45^{0} (C) 40.8^{0} (D) 45.8^{0}

25. A Semicircular rod is charged uniformly with a total charge Q coulomb. The electric field intensity at the centre of curvature is .....

(A) $\frac{2KQ}{\pi r^2}$ (B) $\frac{3KQ}{\pi r^2}$ (C) $\frac{KQ}{\pi r^2}$ (D) $\frac{4KQ}{\pi r^2}$

26. The electron is projected from a distance d and with initial velocity υ_0 parallel to a uniformly charged flat conducting plate as shown in figure. It strikes the plate after travelling a distance *l* along the direction. The surface charge density of conducting plate is equal to



(A) $\frac{2d\in_0 m\upsilon_0}{el}$ (B) $\frac{d\in_0 m\upsilon^2_0}{el}$ (C) $\frac{d\in_0 m\upsilon_0}{el}$ (D) $\frac{2d\in_0 m\upsilon^2_0}{el^2}$

27. Two point masses m each carrying charge –q and +q are attached to the ends of a massless rigid non-conducting rod of length *l*. The arrangement is placed in a uniform electric field E such that the rod makes a small angle 5^{0} with the field direction. The minimum time needed by the rod to align itself along the field is .....

(A) $t = \pi\sqrt{\frac{2ml}{3qE}}$ (B) $t = \frac{\pi}{2}\sqrt{\frac{ml}{2qE}}$ (C) $t = \sqrt{\frac{ml}{qE}}$ (D) $t = 2\pi\sqrt{\frac{ml}{E}}$

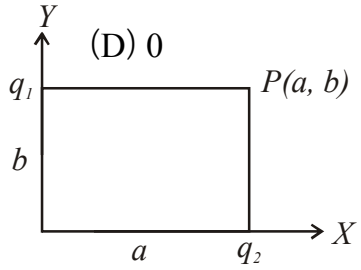
28. Two uniformaly charged spherical conductors A and B having radius 1mm and 2mm are separated by a distance of 5 cm. If the spheres are connected by a conducting wire then in equilibrium condition, the ratio of the magnitude of the electric fields at the surfaces of spheres A and B is .....

(A) 4 : 1 (B) 1 : 2 (C) 2 : 1 (D) 1 : 4

29. Let $P(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point 'P' inside the sphere at distance r_1 from the centre of the sphere the magnitude of electric field is

(A) $\frac{Q}{4\pi \in_0 r_1^2}$ (B) $\frac{Qr_1^2}{4\pi \in_0 R^4}$ (C) $\frac{Qr_1^2}{3\pi \in_0 R^4}$ (D) 0

30. Two point charges $q_1 = 2$ c and $q_2 = 1$ c are placed at distance b = 1cm and a = 2 cm from the origin on the y and x axes as shown in figure. The electric field vector at point P (a, b) will subtend an angle θ with the X - axis given by,



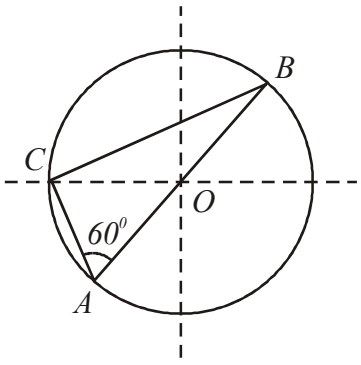
(A) $\tan\theta = 4$ (B) $\tan\theta = 1$

(C) $\tan\theta = 3$ (D) $\tan\theta = 2$

31. A simple pendulum consists of a small sphere of mass m suspended by a thread of length *l*. The sphere carries a positive charge q. The pendulum is placed in a uniform electric field of strength E directed Vertically upwards. If the electrostatic force acting on the sphere is less than gravitational force the period of pendulum is

(A) $T = 2\pi\left[\frac{l}{g - \frac{qE}{m}}\right]^{\frac{1}{2}}$ (B) $T = 2\pi\left(\frac{l}{g}\right)^{\frac{1}{2}}$ (C) $T = 2\pi\left[\frac{l}{g + \frac{qE}{m}}\right]^{\frac{1}{2}}$ (D) $T = 2\pi\left[\frac{ml}{qE}\right]^{\frac{1}{2}}$

32. Consider a system of three charges q/3, q/3 and –2a/3 placed at points A, B and C respectively as shown in the figure. It the radius of the circle is R and $\angle CAB = 60^0$ then the electric field at centre 0 is .....



(A) $\frac{q}{8\pi \in_0 R^2}$ (B) $\frac{q^2}{54\pi \in_0 R^2}$

(C) $\frac{q}{6\pi \in_0 R^2}$ (D) 0

33. In Millikan's oil drop experiment an oil drop carrying a charge Q is held stationary by a p.d. 2400 v between the plates. To keep a drop of half the radius stationary the potential difference had to be made 600 v. What is the charge on the second drop ?

(A) $\frac{3Q}{2}$ (B) $\frac{Q}{4}$ (C) Q (D) $\frac{Q}{2}$

34. Equal charges q are placed at the vertices A and B of an equilateral triangle ABC of side a. The magnitude of electric field at the point c is .....

(A) $\frac{Kq}{a^2}$ (B) $\frac{\sqrt{3}Kq}{a^2}$ (C) $\frac{\sqrt{2}Kq}{a^2}$ (D) $\frac{q}{2\pi t\varepsilon_0 a^2}$

35. An electric dipole is placed along the x-axis at the origin o. A point P is at a distance of 20 cm from this origin such that OP makes an angle $\frac{\pi}{3}$ with the x-axis. If the electric field at P makes an angle θ with the x-axis, the value of θ would be .....

(A) $\frac{\pi}{3} + \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (B) $\frac{\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

36. A particle having a charge of 1.6×10^{-19}C enters between the plates of a parallel plate capaciter. The initial velocity of the particle is parallel to the plates. A potential difference of 300v is applied to the capacitor plates. If the length of the capacitor plates is 10cm and they are separated by 2cm, Calculate the greatest initial velocity for which the particle will not be able to come out of the plates. The mass of the particle is 12×10^{-24}kg .

(A) $10^4\frac{m}{s}$ (B) $10^2\frac{m}{s}$ (C) $10^{-1}\frac{m}{s}$ (D) $10^3\frac{m}{s}$

37. A charged particle of mass 1 kg and charge 2 c is thrown from a horizontal ground at an angle $\theta = 45^o$ with speed 20m/s. In space a horizontal electric field $E = 2\times10^7$V/m exist. The range on horizontal ground of the projectile thrown is ..... .

(A) 100 m (B) 50 m (C) 200 m (D) 0 m

38. If electron in ground state of H-atom is assumed in rest then dipole moment of electron proton system of H-atom is ..... .

Orbit radius of H atom in ground state is $0.56\overset{0}{A}$.

(A) 0.253×10^{-29} cm (B) 0.848×10^{-29} cm (C) 0.305×10^{-29} cm (D) 1.205×10^{-28} cm

39. At what angle θ a point P must be located from dipole axis so that the electric field intensity at the point is perpendicular to the dipole axis ?

(A) 53^0 to 54^0 (B) 50^0 to 51^0 (C) 45^0 to 46^0 (D) 52^0 to 53^0

40. An electric dipole is placed at an angle of 60° with an electric field of intensity 10^{5} NC^{-1}. It experiences a torque equal to $8\sqrt{3}Nm$. If the dipole length is 2cm then the charge on the dipole is ..... c.

(A) -8×10^{3} (B) 8.54×10^{-4} (C) 8×10^{-3} (D) 0.85×10^{-6}

41. An electric dipole coincides on z axis and its mid point is on origin of the cartesian co-ordinate system. The electric field at an axial point at a distance z from origin is $\vec{E}(z)$ and electric field at an equatorial point at a distance y from origin is $\vec{E}_{(y)}$

$\left|\frac{\vec{E}_{(z)}}{\vec{E}_{(y)}}\right|$ $(y = z >> a)$ = ...

(A) 1 (B) 2 (C) 4 (D) 3

42. An oil drop of 12 excess electrons is held stationary under a constant electric field of 2.55×10^{4} Vm^{-1}. If the density of the oil is 1.26 gm/cm^3 then the radius of the drop is ..... m.

(A) 9.81×10^{-7} (B) 9.29×10^{-7} (C) 9.38×10^{-8} (D) 9.34×10^{-8}

43. A Charge q is placed at the centre of the open end of cylindrical vessel. The flux of the electric field through the surface of the vessel is .....

(A) $\frac{q}{\epsilon_0}$ (B) $\frac{q}{2\epsilon_0}$ (C) $\frac{2q}{\epsilon_0}$ (D) Zero

44. The inward and outward electric flux for a closed surface in units of Nm^2/C are respectively 8×10^{3} and 4×10^{3}. Then the total charge inside the surface is ..... c.

(A) $\frac{-4\times10^{3}}{\epsilon_0}$ (B) -4×10^{3} (C) 4×10^{3} (D) $-4\times10^{3}\,\epsilon_0$

45. A sphere of radius R has a uniform distribution of electric charge in its volume. At a distance x from its centre, (for x < R), the electric field is directly proportional to .....

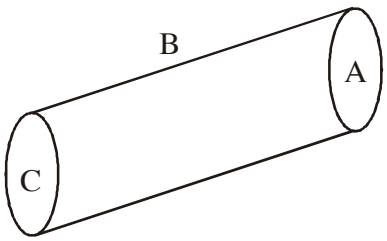
(A) x (B) x^{-1} (C) x^{-2} (D) x^{2}

46. The electric flux for gaussian surface A that enclose the chrged particles in free space is .....

(given $q_1 = -14nc$, $q_2 = 78.85$ nc, $q_3 = -56nc$)

(A) 10^{4} Nm^2/C (B) 10^{3} Nm^2/C (C) 6.2×10^{3} Nm^2/C (D) 6.3×10^{4} Nm^2/C

47. A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in units of volt-meter associated with the curved surface B, the flux linked with the plane surface A in units of volt-meter will be .....



(A) $\frac{1}{2}\left(\frac{q}{\epsilon_0} - \phi\right)$ (B) $\frac{q}{2\epsilon_0}$

(C) $\frac{\phi}{3}$ (D) $\frac{q}{\epsilon_0} - \phi$

48. An infinitly long thin straight wire has uniform linear charge density of $\frac{1}{3}$ c/m. Then, the magnitude of the electric intiensity at a point 18 cm away is ..... NC^{-1}.

(A) 0.66×10^{11} (B) 1.32×10^{11} (C) 0.33×10^{11} (D) 3×10^{11}

49. Two points are at distances a and b (a < b) from a long string of charge per unit length λ. The potential difference between the points in proportional to .....

(A) $ln\left(\frac{b}{a}\right)$ (B) $\frac{\lambda}{\pi \epsilon_0} ln\left(\frac{b^2}{a^2}\right)$ (C) $\frac{\lambda}{2\pi \epsilon_0} ln\sqrt{\frac{b}{a}}$ (D) $\frac{\lambda}{2\pi \epsilon_0} ln\left(\frac{b}{a}\right)$

50. A long string with a charge of λ per unit length passes through an imaginary cube of edge *l*. The maximum possible flux of the electric field through the cube will be .....

(A) $\sqrt{3}\frac{\lambda l}{\epsilon_0}$ (B) $\frac{\lambda l}{\epsilon_0}$ (C) $\sqrt{2}\frac{\lambda l}{\epsilon_0}$ (D) $\frac{6\lambda l^2}{\epsilon_0}$

51. Two Points P and Q are maintained at the Potentials of 10 v and –4 v, respectively. The work done in moving 100 electrons from P to Q is .....

(A) 2.24×10^{-16} J (B) -9.60×10^{-17} J (C) -2.24×10^{-16} J (D) 9.60×10^{-17} J

52. The electric Potential V at any Point o (x, y, z all in metres) in space is given by $V = 4x^2$ volt. The electric field at the point (1m, 0.2m) in volt/metre is .....

(A) 8, along negative *x* - axis (B) 8, along positives *x* - axis

(C) 16, along negative *x* - axis (D) 16, along positives *x* - axis

53. Charges of $+\frac{10}{3} \times 10^{-9}$ C are placed at each of the four corners of a square of side 8cm. The potential at the intersection of the diagonals is ....

(A) $150\sqrt{2}$ Volt (B) $900\sqrt{2}$ Volt (C) $1500\sqrt{2}$ Volt (D) $900\sqrt{2} \cdot \sqrt{2}$ Volt

54. Three charges 2q, –q, –q are located at the vertices of an equilateral triangle. At the centre of the triangle.

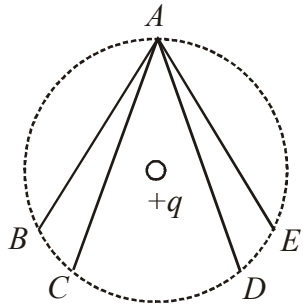
(A) The Field is Zero but Potential is non - zero

(B) The Field is non - Zero but Potential is zero

(C) Both field and Potential are Zero

(D) Both field and Potential are non- Zero

55. In the electric field of a point charge q, a certain charge is carried from point A to B, C, D and E. Then the work done ....



(A) Is least along the Path AB

(B) Is least along the Path AD

(C) Is Zero along all the Path AB, AC, and

(D) Is least along AE

56. Three concentric spherical shells have radii a, b and c (a < b < c) and have surface charge densities $\sigma, -\sigma$ and σ respectively. If V_A, V_B and V_C denote the Potentials of the three shells, then for c = a + b, we have

(A) $V_C = V_B = V_A$ (B) $V_c = V_B \neq V_A$ (C) $V_c \neq V_B \neq V_A$ (D) $V_c = V_A \neq V_B$

57. The electric Potential at a point P (x, y, z) is given by $V = -x^2y - xz^3 + 4$ The electric field $\vec{E}$ at that point is .....

(A) $\hat{i}\,(2xy + z^3) + \hat{j}\,x^2 + \hat{k}3xz^2$ (B) $\hat{i}\,2xy + \hat{j}\,(x^2 + y^2) + \hat{k}\,(3xy - y^2)$

(C) $\hat{i}\,z^3 + \hat{j}\,xyz + \hat{k}z^2$ (D) $\hat{i}\,(2xy - z^3) + \hat{j}\,xy^2 + \hat{k}3z^2x$

58. Three particles, each having a charge of 10 c are placed at the corners of an equilateral triangle of side 10 cm. The electrostatic potential energy of the system is

$\left(\text{Given } \frac{1}{4\pi \in_o} = 9 \times 10^9 \text{ N.m}^2/\text{c}^2\right)$.

(A) 100 J (B) 27 J (C) Zero (D) Infinite

59. Four equal charges Q are placed at the four corners of a square of each side is 'a'. Work done in removing a charge - Q from its centre to infinity is ...

(A) 0 (B) $\frac{\sqrt{2}Q^2}{\pi \in_0 a}$ (C) $\frac{\sqrt{2}Q^2}{4\pi \in_0 a}$ (D) $\frac{Q^2}{2\pi \in_0 a}$

60. Two charged spheres of radii R_1 and R_2 having equal surface charge density. The ratio of their potential is ...

(A) $\frac{R_2}{R_1}$ (B) $\left(\frac{R_2}{R_1}\right)^2$ (C) $\left(\frac{R_1}{R_2}\right)^2$ (D) $\frac{R_1}{R_2}$

61. Two equal charges q are placed at a distance of 2a and a third charge -2q is placed at the midpoint. The potential energy of the system is ....

(A) $\frac{9q^2}{8\pi \in_0 a}$ (B) $\frac{q^2}{8\pi \in_0 a}$ (C) $-\frac{7q^2}{8\pi \in_0 a}$ (D) $\frac{6q^2}{8\pi \in_0 a}$

62. Two point charges 100 c and 5 c are placed at points A and B respectively with AB = 40 cm. The work done by external force in displacing the charge 5 c from B to C where BC = 30 cm, angle ABC = $\frac{\pi}{2}$ and $\frac{1}{4\pi \in_0} = 9 \times 10^9$ Nm^2/c^2.

(A) 9 J (B) $\frac{9}{25}$ J (C) $\frac{81}{20}$ J (D) $-\frac{9}{4}$ J

63. The electric potential V is given as a function of distance x (metre) by $V = (5x^2 + 10x - 9)$ volt. Value of electric field at $x = 1$ is .....

(A) $-20\frac{v}{m}$ (B) $6\frac{v}{m}$ (C) $11\frac{v}{m}$ (D) $-23\frac{v}{m}$

64. A sphere of radius 1cm has potential of 8000 v, then energy density near its surface will be ...

(A) $64 \times 10^5 \frac{J}{m^3}$ (B) $2.83\frac{J}{m^3}$ (C) $8 \times 10^3 \frac{J}{m^3}$ (D) $32\frac{J}{m^3}$

65. If a charged spherical conductor of radius 10cm has potential v at a point distant 5 cm from its centre, then the potential at a point distant 15cm from the centre will be .....

(A) $\frac{1}{3}V$ (B) $\frac{3}{2}V$ (C) 3V (D) $\frac{2}{3}V$

66. Electric charges of +10 c, 5 c, –3 c and 8 c are placed at the corners of a square of side $\sqrt{2}$m the potential at the centre of the square is .....

(A) 1.8 V (B) 1.8×10^5V (C) 1.8×10^6V (D) 1.8×10^4V

67. Two positive point charges of 12 c and 8 c are 10 cm apart each other. The work done in bringing them 4cm closer is .....

(A) 5.8 J (B) 13eV (C) 5.8eV (D) 13 J

68. The displacement of a charge Q in the electric field $\vec{E} = e_1\hat{i} + e_2\hat{j} + e_3\hat{k}$ is $\vec{r} = a\hat{i} + b\hat{j}$. The work done is .....

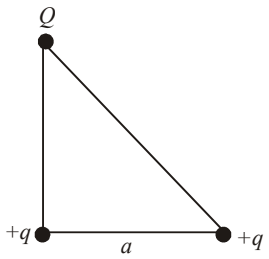
(A) $Q(e_1 + e_2)\sqrt{a^2 + b^2}$ (B) $Q\left(\sqrt{e_1^2 + e_2^2}\right)(a + b)$

(C) $Q(ae_1 + be_2)$ (D) $Q\sqrt{(ae_1)^2 + (be_2)^2}$

69. If an electron moves from rest from a point at which potential is 50 volt to another point at which potential is 70 volt, then its kinetic energy in the final state will be .....

(A) 1 N (B) 3.2×10^{-18} J (C) 3.2×10^{-10} J (D) 1 dyne

70. Three charges Q, + q and + q are placed at the verticles of a right-angled triangle as shown. The net electrostatic energy of the configuration is zero if Q is equal to .....



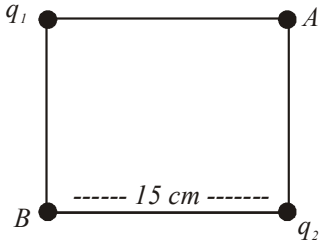
(A) –2q (B) $\frac{-q}{1+\sqrt{2}}$

(C) + q (D) $\frac{-2q}{2+\sqrt{2}}$

71. Two electric charges 12 c and –6 c are placed 20cm apart in air. There will be a point P on the line joining these charges and outside the region between them, at which the electric potential is zero. The distance of P from –6 c charge is .....

(A) 0.20 m (B) 0.10 m (C) 0.25 m (D) 0.15 m

72. In the rectangle, shown below, the two corners have charges $q_1 = -5$ c and $q_2 = +2.0$ c. The work done in moving a charge 3 c from B to A is $\left(\text{take } \frac{1}{4\pi \in_0} = 10^{10}\ \text{Nm}^2/\text{c}^2\right)$.



(A) 5.5 J (B) 2.8 J (C) 3.5 J (D) 4.5 J

73. 4 Points charges each +q is placed on the circumference of a circle of diameter 2d in such a way that they form a square. The potential at the centre is .....

(A) 0 (B) $\frac{4d}{q}$ (C) $\frac{q}{4d}$ (D) $\frac{4q}{d}$

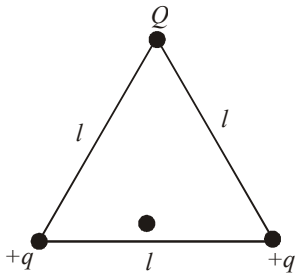
74. Three identical charges each of 2 c are placed at the vertices of a triangle ABC as shown in the figure. If AB + AC = 12 cm and AB . AC = 32cm^2, the potential energy of the charge at A is .....

(A) 1.53 J (B) 5.31 J (C) 1.35 J (D) 3.15 J

75. A ball of mass 1 gm and charge 10^{-8} c moves from a point A, where the potential is 600 volt to the point B where the potential is zero. Velocity of the ball of the point B is 20cm/s. The velocity of the ball at the point A will be .....

(A) $16.8\frac{m}{s}$ (B) $22.8\frac{cm}{s}$ (C) $228\frac{cm}{s}$ (D) $168\frac{m}{s}$

76. Three charges Q, +q and +q are placed at the vertices of an equilateral triangle of side *l* as shown in the figure. It the net electrostatic energy of the system is zero, then Q is equal



(A) –q (B) +q

(C) Zero (D) $-\frac{q}{2}$

77. Electric potential at any point is $V = -5x + 3y + \sqrt{15}z$, then the magnitude of the electric field is ..... N/C.

(A) $3\sqrt{2}$ (B) $4\sqrt{2}$ (C) ৩ (D) $5\sqrt{2}$

78. A small conducting sphere of radius r is lying concentrically inside a bigger hollow conducting sphere of radius R. The bigger and smaller sphere are charged with Q and q (Q > q) and are insulated from each other. The potential difference between the sphers will be .....

(A) $\frac{1}{4\pi\in_0}\left(\frac{q}{r}-\frac{Q}{R}\right)$ (B) $\frac{1}{4\pi\in_0}\left(\frac{q}{r}-\frac{q}{R}\right)$ (C) $\frac{1}{4\pi\in_0}\left(\frac{Q}{R}+\frac{q}{r}\right)$ (D) $\frac{1}{4\pi\in_0}\left(\frac{q}{R}-\frac{Q}{r}\right)$

79. If 3 charges are placed at the vertices of equilateral triangle of charge 'q' each. What is the net potential energy, if the side of equilateral triangle is *l* cm.

(A) $\frac{1}{4\pi\in_0}\frac{3q^2}{l}$ (B) $\frac{1}{4\pi\in_0}\frac{2q^2}{l}$ (C) $\frac{1}{4\pi\in_0}\frac{q^2}{l}$ (D) $\frac{1}{4\pi\in_0}\frac{4q^2}{l}$

80. If identical charges (–q) are placed at each corner of a cube of side b, then electric potential energy of charge (+q) which is placed at centre of the cube will be .....

(A) $-\frac{4q^2}{\sqrt{3}\pi\in_0 b}$ (B) $\frac{8\sqrt{2}q^2}{4\pi\in_0 b}$ (C) $\frac{-8\sqrt{2}q^2}{\pi\in_0 b}$ (D) $\frac{-4\sqrt{2}q^2}{4\pi\in_0 b}$

81. A simple pendulum of period T has a metal bob which is negatively charged. If it is allowed to ascillate above a positively charged metal plate, its period will .....

(A) Remains equal to T (B) Less than T

(C) Infinite (D) Greater than T

82. A charged particale of mass m and charge q is released from rest in a uniform electric field E. Neglecting the effect of gravity, the kinetic energy of the charged particale after 't' second is .....

(A) $\frac{Eq^2m}{2t^2}$ (B) $\frac{E^2q^2t^2}{2m}$ (C) $\frac{2E^2t^2}{qm}$ (D) $\frac{Eqm}{t}$

83. A thin spherical conducting shell of radius R has a charge q. Another charge Q is placed at the centre of the shell. The electrostatic potential at a point p a distance $\frac{R}{2}$ from the centre of the shell is .....

(A) $\frac{(q+Q)}{4\pi\in_0}\frac{2}{R}$ (B) $\frac{2Q}{4\pi\in_0 R}-\frac{2q}{4\pi\in_0 R}$ (C) $\frac{2Q}{4\pi\in_0 R}+\frac{q}{4\pi\in_0 R}$ (D) $\frac{2Q}{4\pi\in_0 R}$

84. Two point charges –q and +q are located at points (o, o, –a) and (o, o, a) respectively. The potential at a point (o, o, z) where z > a is .....

(A) $\frac{2qa}{4\pi\in_0\left(z^2+a^2\right)}$ (B) $\frac{q}{4\pi\in_0 a}$ (C) $\frac{qa}{4\pi\in_0 z^2}$ (D) $\frac{2qa}{4\pi\in_0\left(z^2-a^2\right)}$

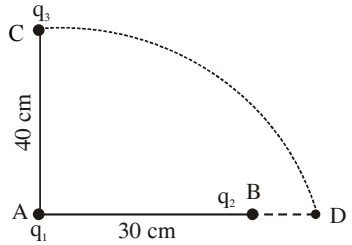
85. Point charges $q_1 = 2$ c and $q_2 = -1$ c are kept at points x = 0 and x = 6 respectively. Electrical potential will be zero at points .....

(A) x = –2, x = 2 (B) x = 1, x = 5 (C) x = 4, x = 12 (D) x = 2, x = 9

86. Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are +q and –q. The potential difference between the centres of the two rings is ....

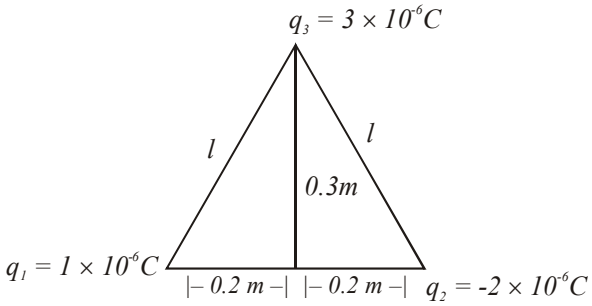
(A) 0 (B) $\frac{Q}{2\pi\in_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+d^2}}\right]$ (C) $\frac{Q}{4\pi\in_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+d^2}}\right]$ (D) $\frac{QR}{4\pi\in_0 d^2}$

87. Two charges q_1 and q_2 are placed 30cm apart, as shown in the figure. A third charge q_3 is moved along the arc of a circle of radius 40 cm from C to D. The change in the potential energy of the system is $\frac{q_3}{4\pi\in_0}K$, where k is .....



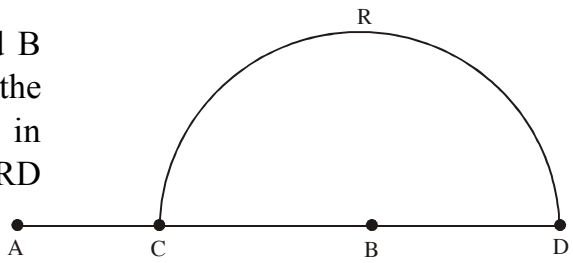
(A) $8q_2$ (B) $8q_1$ (C) $6q_2$ (D) $6q_1$

88. Figure shows a triangular array of three point charges. The electric potencial v of these source charges at the midpoint P of the base of the triangle is $\left[\frac{1}{4\pi \in_0} = 9\times10^9\ \text{Nm}^2/\text{c}^2\right]$



(A) 55 KV (B) 63 KV
(C) 49 KV (D) 45 KV.

89. Charges +q and –q are placed at point A and B respectively which are a distance 2L apart, C is the midpoint between A and B. The work done in moving a charge +Q along the semicircle CRD is .....



(A) $\frac{qQ}{2\pi \in_0 L}$ (B) $\frac{-qQ}{6\pi \in_0 L}$ (C) $\frac{qQ}{6\pi \in_0 L}$ (D) $\frac{qQ}{4\pi \in_0 L}$

90. N identical drops of mercury are charged simultaneously to 10 volt. when combined to form one large drop, the potential is found to be 40 volt, the value of N is .....

(A)4 (B)6 (C) 8 (D) 10

91. Two parallel plate air capacitors have their plate areas 100 and 500 cm^2 respectively. If they have the same charge and potential and the distance between the plates of the first capacitor is 9.5 mm, what is the distance between the plates of the second capacitor ?

(A) 0.25 cm (B) 0.50 cm (C) 0.75 cm (D) 1 cm

92. The effective capacitances of two capacitors are $3\ \text{F}$ and $16\ \text{F}$, when they are connected in series and parallel respectively. The capacitance of each capacitor is

(A) $2\ \text{F}, 14\ \text{F}$ (B) $4\ \text{F}, 12\ \text{F}$ (C) $6\ \text{F}, 8\ \text{F}$ (D) $10\ \text{F}, 6\ \text{F}$

93. An electrical technician requires a capacitance of $2\ \text{F}$ in a circuit across a potential difference of 1KV. A large number of $1\ \text{F}$ capacitors are available to him, each of which can withstand a potential difference of not than 400 V. suggest a possible arrangement that requires a minimum number of capacitors.

(A) 2 rows with 2 capacitors (B) 4 rows with 2 capacitors
(C) 3 rows with 4 capacitors (D) 6 rows with 3 capacitors

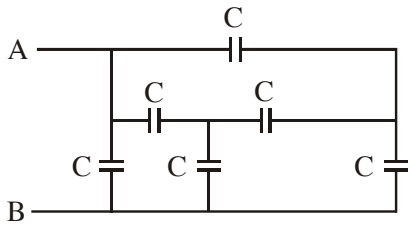
94. Two spherical conductors of radii r_1 and r_2 are at potentials V_1 and V_2 respectively, then what will be the common potential when the conductors are brought in contant ?

(A) $\frac{r_1v_1 + r_2v_2}{r_1 + r_2}$ (B) $\frac{r_1v_1 + r_2v_2}{r_1 - r_2}$ (C) $\frac{r_1v_1 - r_2v_2}{r_1 + r_2}$ (D) None of these

95. A $5\ \mu F$ capacitor is charged by a 220 V supply. It is then disconnected from the supply and is connected to another uncharged $2.5\ \mu F$ capacitor. How much electrostatic energy of the first capacitor is lost in the form of heat and electromagnetic radiation ?

(A) 0.02 J (B) 0.121 J (C) 0.04 J (D) 0.081 J

96. Find the equivalent capacitance of the system across the terminals A and B. All the capacitors have equal capacitances.



(A) 2C (B) 4C
(C) 3C (D) 5C

97. Capacitance of a parallel plate capacitor becomes $\frac{4}{3}$ times its original value if a dielectric slab of thickness t = d/2 is inserted between the plates (d is the separation between the plates). The dielectric constant of the slab is

(A) 8 (B) 4 (C) 6 (D) 2

98. The plates of a parallel capacitor are charged up to 100 V. If 2 mm thick plate is inserted between the plates, then to maintain the same potential difference, the distance between the capacitor plates is increased by 1.6mm the dielectric constant of the plate is

(A) 5 (B) 4 (C) 1.25 (D) 2.5

99. A parallel plate air capacitor has a capacitance $18\ \mu F$. If the distance between the plates is tripled and a dielectric medium is introduced, the capacitance becomes $72\ \mu F$. The dielectric constant of the medium is

(A) 4 (B) 12 (C) 9 (D) 2

100. Taking earth to be a metallic spheres, its capacity will approximately be

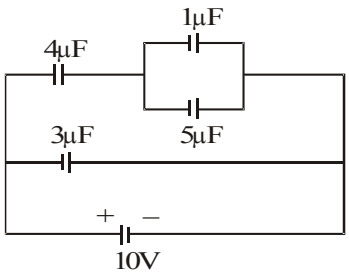
(A) $6.4\times10^{6}F$ (B) 700 F (C) 711 F (D) 700 pF

101. A parallel plate capacitor has the space between its plates filled by two slabs of thickness $\frac{d}{2}$ each and dielectric constant K_1 and K_2. If d is the plate separation of the capacitor, then capacity of the capacitor is .....

(A) $\frac{2d\in_0}{A}\left(\frac{K_1+K_2}{K_1K_2}\right)$ (B) $\frac{2A\in_0}{d}\left(\frac{K_1K_2}{K_1+K_2}\right)$

(C) $\frac{2A\in_0}{d}\left(\frac{K_1+K_2}{K_1K_2}\right)$ (D) $\frac{2A\in_0}{d}(K_1+K_2)$

102. For the circuit shown in figure the charge on 4 F capacitor is



(A) 20 c (B) 40 c

(C) 30 c (D) 54 c

103. The capacitors of capacitance 4 F, 6 F and 12 F are connected first in series and then in parallel. What is the ratio of equivalent capacitance in the two cases ?

(A) 2 : 3 (B) 11 : 1 (C) 1 : 11 (D) 1 : 3

104. Large number of capactors of rating 10 F/200V V are available. The minimum number of capacitors required to design a 10 F/700V capacitor is

(A) 16 (B) 8 (C) 4 (D)7

105. A variable condenser is permanently connected to a 100 V battery. If capacitor is changed from 2 F to 10 F. then energy changes is equal to

(A) 2×10^{-2}J (B) 2.5×10^{-2}J (C) 6.5×10^{-2}J (D) 4×10^{-2}J

106. Two positive point charges of 12 c and 8 c are placed 10 cm apart in air. The work done to bring them 4 cm closer is

(A) Zero (B) 4.8 J (C) 3.5 J (D) 5.8 J

107. 1000 similar electrified rain drops merge together into one drop so that their total charge remains unchanged. How is the electric energy affected ?

(A) 100 times (B) 200 times (C) 400 times (D) 102 times

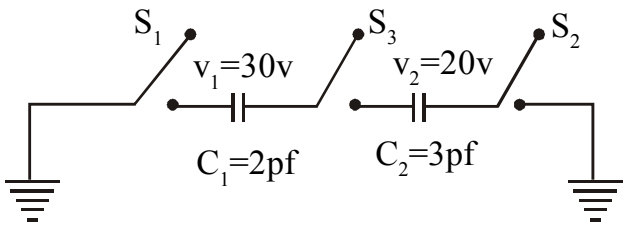
108. There are 10 condensers each of capacity 5 F. The ratio between maximum and minimum capacities obtained from these condensers will be

(A) 40 : 1 (B) 25 : 5 (C) 60 : 3 (D) 100 : 1

109. A parallel plate capacitor is made by stocking n equally spaced plates connected alternately. If the capacitance between any two plates is x, then the total capacitance is,

(A) nx (B) nx^2 (C) $\frac{n}{x}$ (D) (n – 1)x

110. For the circuit shown figure, which of the following statements is true ?



(A) With S_1closed $V_1 = 15V$, $V_2 = 20V$

(B) With S_3 closed$V_1 = V_2 = 20V$

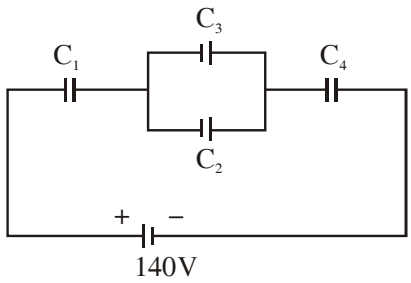
(C) With S_1 and S_3 closed $V_1 = V_2 = 0$

(D) With $S_1$and S_3closed $V_1 = 30V$, $V_2 = 20V$

111. Two identical metal plates are given positive charges Q_1 and Q_2 ($< Q_1$) respectively. If they are now brought close to gether to form a parallel plate capacitor with capacitance C, the potential difference between them is

(A) $\frac{(Q_1 + Q_2)}{2c}$ (B) $\frac{(Q_1 + Q_2)}{c}$ (C) $\frac{(Q_1 - Q_2)}{2c}$ (D) $\frac{(Q_1 - Q_2)}{c}$

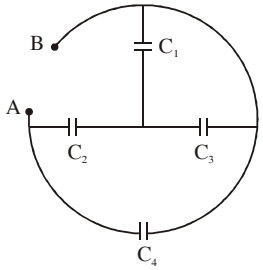
112. In the circuit arrangement shown in figure, the value of $C_1 = C_2 = C_3 = 30$ pF and $C_4 = 120$ pF. If the combination of capacitors is charged with 140V DC supply, the potencial differences across the four capacitors will be respectively



(A) 80, 40, 40 and 20V (B) 20, 40, 40 and 80V

(C) 35, 35, 35 and 35V (D) 80, 20, 20 and 20V

113. In the arrangement of capacitors shown in figure, each capacitor is of 9 F, then the equivalent capacitance between the points A and B is



(A) 18 F (B) 9 F

(C) 15 F (D) 4.5 F

114. The electric potencial V at any point *x*, *y*, *z* (all in metre) in space is given by $V = 4x^2$ volt. The electric field at the point (1m, 0, 2m) in Vm^{-1} is

(A) $+8\hat{i}$ (B) $-8\hat{i}$ (C) $-16\hat{i}$ (D) $+16\hat{i}$

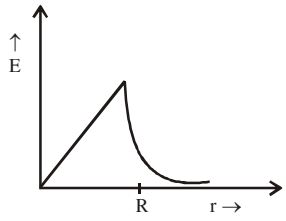
115. Two air capacitors A = 1 F, B = 4 F are connected in series with 35 V source. When a medium of dielectric constant K = 3 is introduced between the plates of A, change on the capacitor changes by

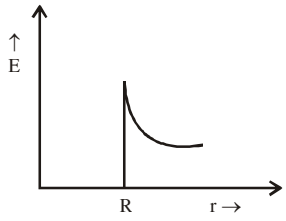
(A) 16 c (B) 32 c (C) 28 c (D) 60 c

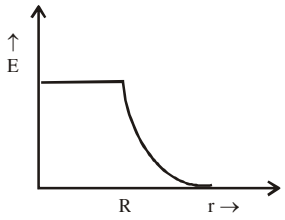
116. A parallel plate capacitor with air between the plates has a capacitance of 9 pF. The separation between its plates is d. The space between the plates is now filled with two dielectrics. One of the dielectric constant $K_1 = 3$ and thickness d/3 while the other one has dielectric constant $K_2 = 6$ and thickness 2d/3. Capacitance of the capacitor is now

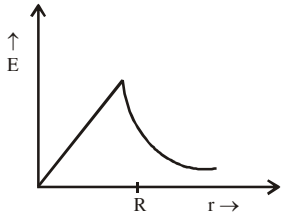
(A) 1.8 pF (B) 20.25 pF (C) 40.5 pF (D) 45 pF

117. A thin spherical shell of radius R has change Q spread uniformly over its surface. Which of the following graphs, figure most closely represents the electric field E (r) produced by the shell in the range $0 \le r < \infty$, where r is the distance from the centre of the shell.

(A) 

(B) 

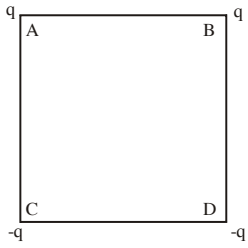
(C) 

(D) 

118. A parallel plate condenser with dielectric of constant K between the plates has a capacity C and is charged to potential V volt. The dielectric slab is slowly removed from between the plates and reinserted. The network done by the system in this process is

(A) Zero (B) $\frac{1}{2}(K-1)cv^2$ (C) $(K-1)cv^2$ (D) $cv^2\frac{(K-1)}{K}$

119. Charges are placed on the vertices of a square as shown. Let $\vec{E}$ be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C respectively then



(A) $\vec{E}$ Change V remains unchanged

(B) $\vec{E}$ remains unchanged , V changes

(C) Both $\vec{E}$ and V change

(D) $\vec{E}$ and $\vec{V}$ remain unchanged

120. The potencial at a point *x* (measured in m) due to some charges situated on the *x*-axis is given by $V(x) = \frac{20}{x^2-4}$ Volt. The electric field at $x = 5$ m is given by

(A) $\frac{5}{3}$ Vμm^{-1} and in positive *x* - direction

(B) $\frac{10}{9}$ Vμm^{-1} and in positive *x* - direction

(C) $\frac{10}{9}$ Vμm^{-1} and in positive *x* - direction

(D) $\frac{5}{3}$ Vμm^{-1} and in positive *x* - direction

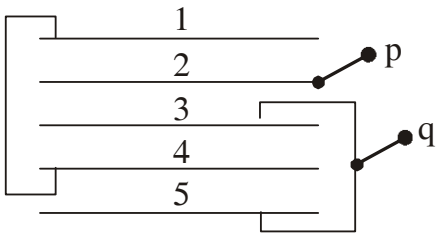
121. A battery is used to charge a parallel plate capacitor till the potential difference between the plates becomes equal to the electromotive force of the battery. The ratio of the energy stored in the capacitor and work done by the battery will be

(A) $\frac{1}{2}$ (B) $\frac{2}{1}$ (C) 1 (D) $\frac{1}{4}$

122. Two spherical conductors A and B of radii 1mm and 2mm are separated by a distance of 5mm and are uniformly charged. If the spheres are connected by a conducting wire then in equilibrium condition, the ratio of the magnitude of the electric fields at the surfaces of sphere of A and B is

(A) 1 : 2 (B) 2 : 1 (C) 4 : 1 (D) 1 : 4

123. The following arrangement consists of five identical metal plates marked 1, 2, 3, 4 and 5 parallel to each other. Area of each plate is A and separation between the successive plates is d. The capacitance between P and Q is

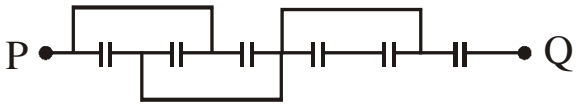


(A) $5\frac{A\in_0}{d}$ (B) $\frac{7}{3}\frac{A\in_0}{d}$ (C) $\frac{5}{3}\frac{A\in_0}{d}$ (D) $\frac{4}{3}\frac{A\in_0}{d}$

124. A parrallel plate capacitor of capacitance 5 F and plate separation 6 cm is connected to a 1 V battery and charged. A dielectric of dielectric constant 4 and thickness 4 cm is introduced between the plates of the capacitor. The additional charge that flows into the capacitor from the battery is

(A) 2 c (B) 5 c (C) 3 c (D) 10 c

125. For circuit the equivalent capacitance between points P and Q is



(A) 6 C (B) 4 C

(C) $\frac{3}{2}$C (D) $\frac{6}{11}$C

126. Four identical capacitors are connected in series with a 10 V battery as shown in the figure. Potentials at A and B are

(A) 10 V, 0 V (B) 5 V, –5V (C) 7.5 V, –2.5 V (D) 7.5 V, 2.5 V

127. 64 identical drops of mercury are charged simultaneously to the same potential of 10 volt. Assuming the drops to be spherical, if all the charged drops are made to combine to form one large drop, then its potential will be

(A) 100 V (B) 320 V (C) 640 V (D) 160 V

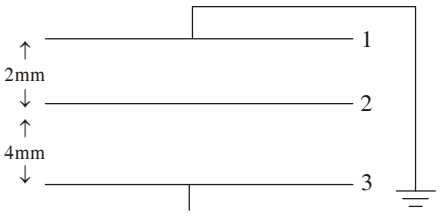
128. Two metal plate form a parallel plate capacitor. The distance between the plates is d. A metal sheet of thickness d/2 and of the same area is introduced between the plates. What is the ratio of the capacitance in the two cases ?

(A) 4 : 1 (B) 3 : 1 (C) 2 : 1 (D) 5 : 1

129. The circular plates A and B of a parrallel plate air capacitor have a diameter of 0.1 m and are 2×10^{-3} m apart. The plates C and D of a similar capacitor have a diameter of 0.12 m and are 3×10^{-3} m apart. Plate A is earthed. Plates B and D are connected together. Plate C is connected to the positive pole of a 120 V battery whose negative is earthed, The energy stored in the system is

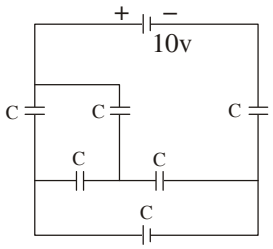
(A) 0.1224 J (B) 0.2224 J (C) 0.4224 J (D) 0.3224 J

130. Two parallel conducting plates of area $A = 2.5m^2$ each are placed 6 mm apart and are both earthed. A third plate, identical with the first two, is placed at a distance of 2 mm from one of the earthed plates and is given a charged of 1 C. The potencial of the central plate is



(A) 6×10^7 V (B) 3×10^7 V (C) 2×10^7 V (D) 4×10^7 V

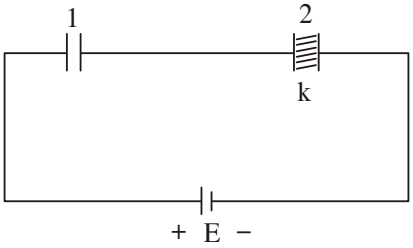
131. What is the energy stored in the capacitor between terminals A and B of the network shown in the figure ? (Capacitance of each capacitor $C = 1\mu F$)



(A)Zero (B) 50 J

(C) 12.5 J (D) 25 J

132. Two identical capacitors 1 and 2 are connected in series to a battery as shown in figure. Capacitor 2 contains a dielectric slab of constant K. Q_1 and Q_2 are the charges stored in 1 and 2. Now, the dielectric slab is removed and the corresponding charges are Q'_1 and Q'_2. Then



(A) $\frac{Q_1^1}{Q_1} = \frac{K+1}{K}$ (B) $\frac{Q_1^1}{Q_1^1} = \frac{K}{2}$

(C) $\frac{Q_2^1}{Q_2} = \frac{K+1}{2K}$ (D) $\frac{Q_2^1}{Q_2} = \frac{K+1}{K}$

133. A parallel plate capacitor has plate of area A and separation d. It is charged to a potential difference V_o. The charging battery is disconnected and the plates are pulled apart to three times the initial separation. The work required to separate the plates is

(A) $\frac{A \in_0 V_0^2}{d}$ (B) $\frac{A \in_0 V_0^2}{2d}$ (C) $\frac{A \in_0 V_0^2}{3d}$ (D) $\frac{A \in_0 V_0^2}{4d}$

134. Two identical capacitors have the same capacitance C. one of them is charged to a potential V_1 and the other to V_2. The negative ends of the capacitors are connected together. When the positive ends are also connected, the decrease in energy of the combined system is

(A) $\frac{1}{4}c\left(v_1^2 - v_2^2\right)$ (B) $\frac{1}{4}c\left(v_1^2 + v_2^2\right)$ (C) $\frac{1}{4}c\left(v_1 - v_2\right)^2$ (D) $\frac{1}{4}c\left(v_1 + v_2\right)^2$

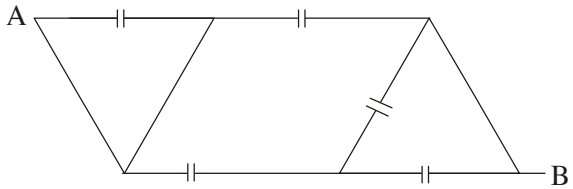
135. A parallel plate air capacitor has a capacitance C. When it is half filled with a dielectric of dielectric constant 5, the percentage increase in the capacitance will be

(A) 200 % (B) 33.3 % (C) 400 % (D) 66.6 %

136. A network of six identical capacitors, each of value C is made as shown in figure. Equivalent capacitance between points A and B is

(A) $\frac{C}{4}$ (B) $\frac{3C}{4}$

(C) $\frac{4C}{3}$ (D) 3 C



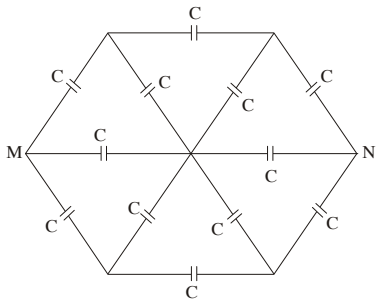
137. The capacities of three capacitors are in the ratio 1 : 2 : 3. Their equivalent capacity when connected in parallel is $\frac{60}{11}$ F more then that when they are connected in series. The individual capacitors are of capacities in μF

(A) 4, 6, 7 (B) 1, 2, 3 (C) 1, 3, 6 (D) 2, 3, 4

138. In the given arrangement of capacitors equivalent capacitance between points M and N is

(A) $\frac{5}{4}$c (B) $\frac{4}{5}$c

(C) $\frac{4}{3}$c (D) $\frac{3}{4}$c



139. An electric circuit requires a total capacitance of 2 F across a potencial of 1000 V. Large number of 1 F capacitances are available each of which would breakdown if the potential is more then 350 V. How many capacitances are required to make the circuit ?

(A) 24 (B) 12 (C) 20 (D) 18

140. Read the assertion and reason carefully to mark the correct option out of the options given below :

(a) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(c) If assertion is true but reason is false.

(d) If the assertion and reason both are false.

(e) If assertion is false but reason is true.

1. Assertion : The coulomb force is the dominating force in the universe.
   Reason : The coulomb force is weaker than the gravitational force.
2. Assertion : If three capacitors of capacitance $C_1 < C_2 < C_3$ are connected in parallel then their equivalent capacitance $C_p > C_3$
   Reason : $\frac{1}{C_p} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$
3. Assertion : A metalic shield in form of a hollow shell may be built to block an electric field.
   Reason : In a hollow shherical shield, the electric field inside it is zero at every point.
4. Assertion : Electrons move away from a low potential to high potential region.
   Reason : Because electrons have negative charge
5. Assertion : If the distance between parallel plates of a capacitor is halved and dielectric constant is made three times, then the capacitance becomes 6 times.
   Reason : Capacity of the capacitor does not depend upon the nature of the material.
6. Assertion : A parallel plate capacitor is connected across battery through a key. A di-electric slab of constant *K* is introduced between the plates. The energy which is stored becomes *K* times.
   Reason : The surface density of charge on the plate remains constant or unchanged.
7. Assertion : Electric lines of force cross each other.
   Reason : Electric field at a point superimpose to give one resultant electric field.
8. Assertion : If a proton and an electron are placed in the same uniform electric field. They experience different acceleration.
   Reason : Electric force on a test charge is independent of its mass.
9. Assertion : Dielectric breakdown occrus under the influence of an intense light beam.
   Reason : Electromagnetic radiations exert pressure.
10. Assertion : When charges are shared between any two bodies, no charge is really lost, but some loss of energy does occur.
    Reason : Some energy disappeares in the form of heat, sparking etc.

KEY NOTES

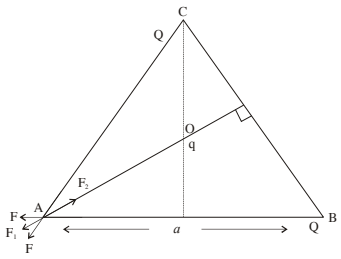
1(A)	17(C)	33(D)	49(D)	65(D)	81(B)	97(D)	113(C)	129(A)
2(B)	18(A)	34(C)	50(A)	66(B)	82(B)	98(A)	114(B)	130(A)
3(D)	19(D)	35(A)	51(A)	67(D)	83(C)	99(B)	115(C)	131(C)
4(C)	20(B)	36(A)	52(A)	68(C)	84(D)	100(B)	116(C)	132(C)
5(B)	21(A)	37(C)	53(C)	69(B)	85(C)	101(B)	117(B)	133(A)
6(C)	22(B)	38(B)	54(B)	70(D)	86(B)	102(B)	118(A)	134(C)
7(D)	23(D)	39(D)	55(C)	71(A)	87(A)	103(C)	119(A)	135(D)
8(A)	24(C)	40(C)	56(D)	72(B)	88(D)	104(A)	120(B)	136(C)
9(B)	25(A)	41(B)	57(A)	73(D)	89(B)	105(D)	121(A)	137(B)
10(C)	26(D)	42(A)	58(B)	74(C)	90(C)	106(D)	122(B)	138(A)
11(C)	27(B)	43(D)	59(B)	75(B)	91(A)	107(A)	123(C)	139(D)
12(D)	28(C)	44(D)	60(D)	76(D)	92(C)	108(D)	124(B)	
13(B)	29(B)	45(A)	61(C)	77(C)	93(D)	109(D)	125(D)	
14(A)	30(D)	46(B)	62(D)	78(B)	94(A)	110(D)	126(C)	
15(C)	31(A)	47(A)	63(A)	79(A)	95(C)	111(D)	127(D)	
16(B)	32(C)	48(C)	64(B)	80(A)	96(A)	112(A)	128(C)	

140 :	1(D)	2(C)	3(A)	4(A)	5(B)
	6(C)	7(E)	8(B)	9(B)	10(B)

(9) Distance OA = $\frac{2}{3}\left(a\sin 60^\circ\right)$

$= \frac{2}{3}.a.\frac{\sqrt{3}}{2}$

$= a/\sqrt{3}$



The force on the Charge at A due to those at B and C have magnitude F = $k\frac{Q^2}{a^2}$

The Resultant of theses force is

$F_1 = 2F\cos 30^\circ$

$F_1 = 2\frac{kQ^2}{a^2}\cdot\frac{\sqrt{3}}{2}$

$\therefore\ F_2 = \frac{\sqrt{3}KQ^2}{q^2}$

The force on the charge Q at A due to Charge q at o is, $F_2 = \frac{3KQq}{a^2}$

Now for equilibrium of the Charges, $F_1 = F_2$ Now Calculate q.

(10) Let q and 2q be the Charges and r the distance between them.

Then $2\times10^{-3} = \frac{2kq^2}{r^2}$ and $5\times10^{-4} = \frac{2kq^2}{(r+0.1)^2}$

So, $\frac{20}{5} = \frac{(r+0.1)^2}{r^2}$

$\therefore\ 2 = \frac{r+0.1}{r^2} \Rightarrow r = 0.1\,m.$

Now, Substitating the Value of r So obtained

$2\times10^{-3} = 9\times10^9\ \frac{2q^2}{(0.1)^2}$

$q^2 = \frac{10^{-14}}{9} \Rightarrow q = \frac{10^{-7}}{3}$

Now find out q and 2q.

(11) Force on $-q_1$ due to q_2 is $F_{12} = \frac{kq_1q_2}{b^2}$ along $\times$ axis

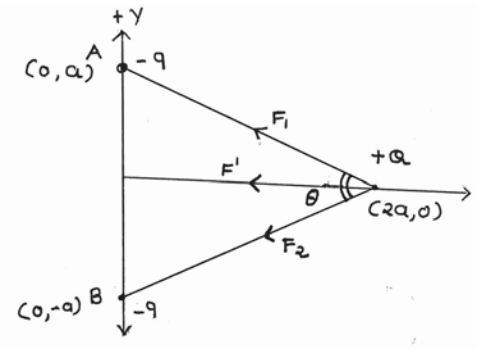
Force on $-q_1$ due to $-q_3$ is $F_{13} = \frac{kq_1q_3}{a^2}$ at $\angle\theta$ negative direction of yaxis.

$\therefore$ x component of force on -q, is $F_x = F_{12} + F_{13}\sin\theta$

$= Fq_1\left[\frac{q_2}{b^2} + \frac{q_3}{a^2}\sin\theta\right]$

i.e $= F_x\ \alpha\ \frac{q_2}{b^2} + \frac{q_3}{a^2}\sin\theta$

(12) Force exerted by charges -q at A and B on Charge Q are F_1 amd F_2 which are equal and have a magnitude $F = \frac{KQq}{r^2}$



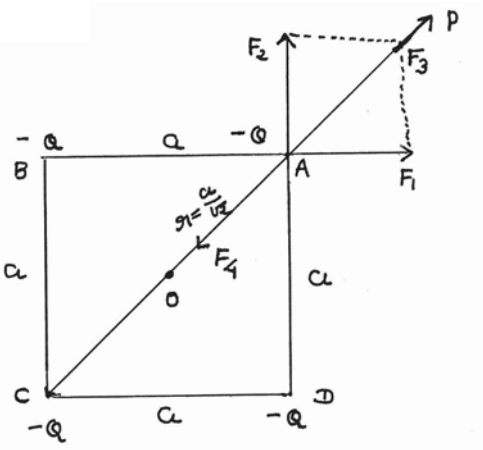
The resultant of these equal forces equally inclined with the X-axis is along the -X direction to wards the origin.

so, $F'^2 = F^2 + F^2 + 2F^2\cos\theta = 2F^2(1+\cos\theta)$

since $F \propto \frac{1}{r^2}, F^1$ is also Proportional to $\frac{1}{r^2}$,

Hence charge Q will move towards the origin and because of its inertia it will overshoot the origin o. Thus charge Q will oscillate about o but its motion is not simple harmonic.

(13) $OA = OC = r = \frac{a}{\sqrt{2}}$



$F_1 = F_2 = \frac{KQ^2}{a^2} \quad F_3 = \frac{KQ^2}{2a^2} \quad F_4 = \frac{-2KQq}{a^2}$

The resultant of $F_1 \& F_2$ is $F = \sqrt{2}F_1 = \sqrt{3}\frac{KQ^2}{a^2}$

F and F_3 act along AP So, $F^1 = F + F_3$ find.

Now for equilibrium $F^1 = F_4$.

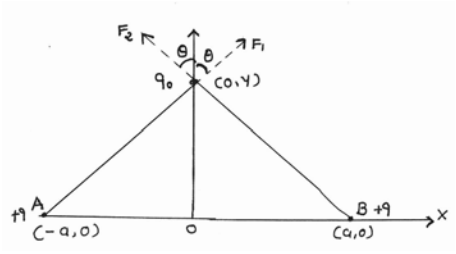
From this find out q.

(14) Net attraction force on q due to Q = repulsion force due to q

$\sqrt{2}F_A = F^1$ from find out q.

(15) $F_1 = \frac{kq_0q}{\left(a^2+y^2\right)} F_2 = \frac{kq_0q}{\left(a^2+y^2\right)}$

$\therefore F_1 = F_2$



By symmetry, the x components of force will cancel each other while along y axis will add up

$\therefore$ The resultant force on $+q_\circ$ is

$F = 2F_1\cos\theta = \frac{2kqq_0}{a^2+y^2}\cdot\frac{y}{\sqrt{a^2+y^2}}$

Now, Force on charge q_0 will be maximum, when $\frac{dF}{dy} = 0$

$\therefore\ 2kq_0q\left[\frac{1}{(a^2+y^2)^{\frac{3}{2}}} - \frac{\left(\frac{3}{2}y\right)(2y)}{(a^2+y^2)^{\frac{5}{2}}}\right] = 0$

$\therefore \frac{1}{\left(a^2+y^2\right)^{3/2}} - \frac{3y^2}{\left(a^2+y^2\right)^{5/2}} = 0$ Now find out y.

(16) $T\cos\theta = mg$ and $T\sin\theta = F$

$$\therefore \tan\theta = \frac{F}{mg}$$

=

$$\therefore \frac{x}{2\ell} \propto \frac{q^2}{x^2}$$

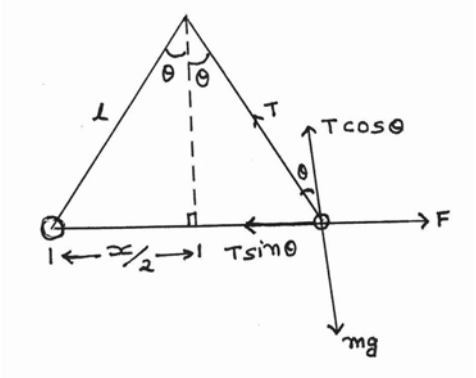
$$\therefore q^2 \propto x^3$$

$$\therefore q^2 \propto x^{3/2}$$

$$\frac{dq}{dt} \propto \frac{3}{2} x^{1/2} \frac{dx}{dt}$$

$$\therefore V \alpha x^{-\frac{1}{2}}$$

$$\left[\therefore \frac{dq}{dt} = \text{Constant}\right]$$



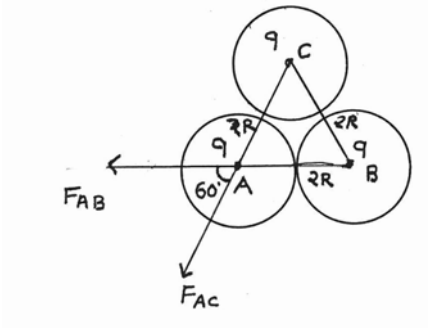
(17) $F_{AB} = \frac{1}{4\pi\varepsilon_o}\frac{9^2}{4R^2}$ (along $\overrightarrow{BA}$)

$$F_{AC} = \frac{1}{4\pi\varepsilon_o}\frac{9^2}{4R^2} \text{ (along } \overrightarrow{CA})$$

$$F_{AB} = F_{AC} = F$$

$$\therefore F^1 = \sqrt{F^2 + F^2 + 2F^2\cos 60°}$$

Now find out F^1.

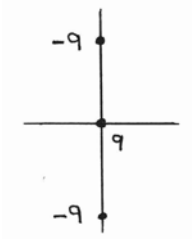


(18) Force on q is $F = \frac{kq^2}{(a+x)^2} - \frac{kq^2}{(a-x)^2}$

$$\text{ma} = F = kq^2\left[\frac{1}{(a+x)^2} - \frac{1}{(a-x)^2}\right]$$

$$a = \frac{-q^2 x}{\pi\varepsilon_0 m a^3}$$

Now compare with a = $-\omega^2 x$ and find out frequenly.



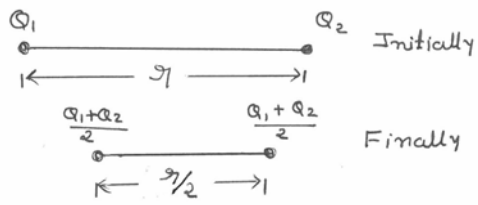
(19) Suppose the balls having charges Q_1 and Q_2 respectively.

Initially,..... $F = \frac{kQ_1Q_2}{r^2}$

It is given that $F^1 = 4.5F$

$$\frac{K(Q_1+Q_2)^2}{r^2} = 4.5\ \frac{Q_1\ Q_2}{r^2}$$

$(Q_1+Q_2)^2 = 4.5\ Q_1Q_2$ Now solve it, and find Q_1/Q_2.



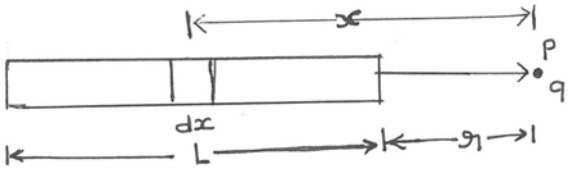
(20) Consider a small element of the rod of length dx, at a distance x from the point charge q.

The force between q and charge element will be,

$dF = \frac{kqdQ}{x^2}$ But $dQ = \frac{Q}{L}dx$,

$\therefore dF = \frac{kqQ}{L}\frac{dx}{x^2}$

$F = \int_r^{r+L} dF = \frac{kqQ}{L}\int_r^{r+L} \frac{1}{x^2}dx$



Now solve it.

(21) $\frac{k.q_1}{(x+0.08)^2} = \frac{kq_2}{x^2}$ then find x

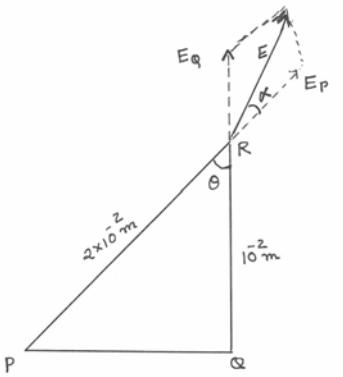
(22) $E_p = \frac{kq_p}{(PQ)^2} = 18\times10^7 \text{ NC}^{-1}$

$E_Q = \frac{kq_Q}{(QR)^2} = 18\times10^7 \text{ NC}^{-1}$

in ΔPQR, $\cos\theta = \frac{QR}{PR} = \frac{1}{2} = \theta = 60°$

$\therefore E = \sqrt{E_p^{\,2} + E_Q^{\,2} + 2E_pE_Q \cos 60^0}$ and

$\tan\alpha = \frac{E_Q \sin\theta}{E_p + E_Q\cos\theta}$



find out E and α .

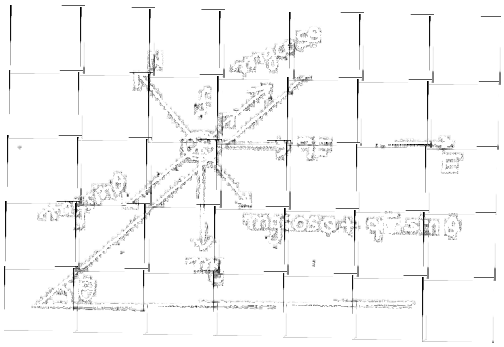
(23) From the fig, Net force F acting along the inclined plance ...(i)

$F = mg\sin\theta - qE\cos\theta - f$

$\therefore ma = mg\sin\theta - qE\cos\theta - \mu N$

So, $ma = mg\sin\theta - qE\cos\theta - \mu(mg\cos\theta + qE\sin\theta)$

from this, find out a.

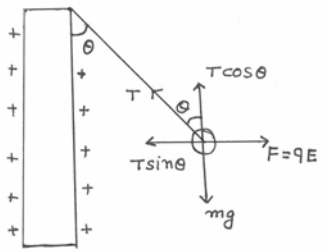


Now, $d = v_ot + \frac{1}{2}at^2$ use this equation and calculate t.

(24) From the fig $T_{\sin\theta} = qE\frac{E_\sigma}{\epsilon_o}$

$\therefore T_{\cos\theta} = mg \quad \therefore T_{\cos\theta} = \frac{q_\sigma}{mg\varepsilon_o}$

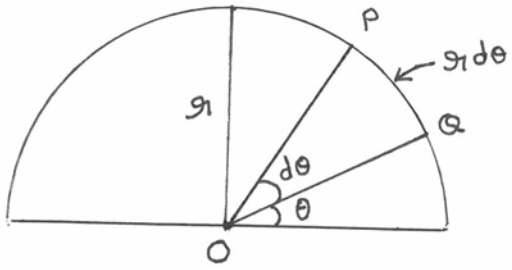
Now calculate θ.



(25) Let λ be the charge Per unit length so, charge on Portion PQ ...

= Intensity at o is $dE = \dfrac{k\lambda r d\theta}{r^2}$

$$dE = \frac{k\lambda d\theta}{r}$$



$\therefore$ Total intensity E = $\int_0^{\pi} dE$ and

total cahrge on the rod, Q = $\lambda \pi r$

Now find out E.

(26) Here E = $\dfrac{\sigma}{\varepsilon_o}$ and $t = l / v_o$

Along y axis v_o = 0, a = $\dfrac{F}{M} = eE / M$

So, $d = v_o t + \dfrac{1}{2} a t^2$

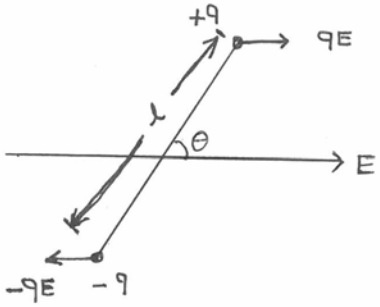
Now Put the values

(27) $\tau = qE\left(l \sin\theta\right)$

$\tau = qEl\theta$

The moment of inertia of the rod is

$$I = m\left(\frac{l}{2}\right)^2 + m\left(\frac{l}{2}\right)^2 = \frac{ml^2}{2}$$



Now, $\tau = I\alpha \Rightarrow \alpha = \dfrac{\tau}{I}$

Calculate $\propto$ and compale with $\omega^2\theta$ then $\omega^2 = \dfrac{2qE}{ml}$

So, T = $\dfrac{2\pi}{\omega}$ and rod will become Paralled to E in a time t = $\dfrac{T}{4}$ Now calculate.

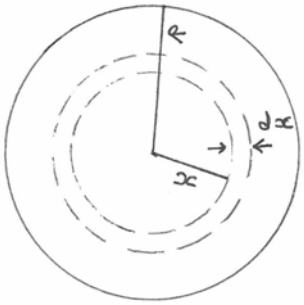
(28) After Connection $V_1 = V_2$, $\dfrac{KQ_1}{r_1} = \dfrac{KQ_2}{r_2} \Rightarrow \dfrac{Q_1}{Q_2} = \dfrac{r_2}{r_1}$

The ratio of electric fields $\dfrac{E_1}{E_2} = \dfrac{\frac{KQ_1}{r_1^2}}{\frac{KQ_2}{r_2^2}} = \dfrac{Q_1}{r_1^2} \times \dfrac{r_2^2}{Q_2}$

Now Calculate

(29) Consider a Spherical Shell of thickness dx and radius x.

The area of this Spherical shell = $4\pi x^2 dx$



$$\left[\frac{Qx}{\pi R^4}\right]4\pi x^2 dx = \frac{4Q}{R^4}x^3 dx$$

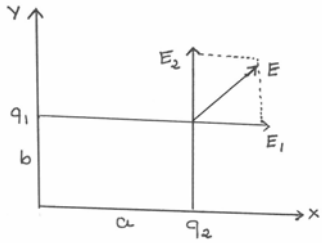
The Charge enclosed in a Sphere of

radius r_1 is $= \frac{4Q}{R^4}\int_0^r x^3 dx = \frac{4Q}{R^4}\left[\frac{x^4}{4}\right]_0^{r_1} = \frac{Q}{R^4}r_1^4$

$$= \frac{1}{4\pi\varepsilon_o}\cdot\left[\frac{\frac{Q}{R^4}r_1^4}{r_1^2}\right]$$

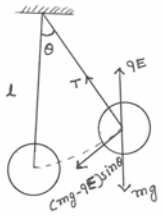
Now find out E.

(30) $E_1 = \frac{Kq_1}{a^2}$ and $E_2 = \frac{Kq_2}{b^2}$



Now, $\tan\theta = \frac{E_2}{E_1} = \frac{q_2}{q_1}\cdot\frac{a^2}{b^2}$ Now, Put the values.

(31) Net acceleration $g^1 = g - \frac{gE}{m}$



So, $T = 2\pi\sqrt{\frac{l}{g'}}$ Now Put the Value of g'.

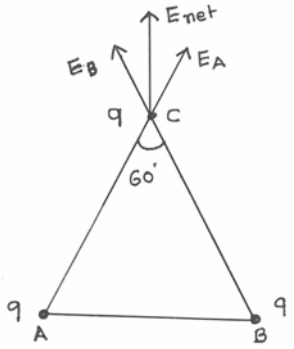
(32) Net electric field due to both charges $\frac{q}{3}$ will get cancelled.

So. electric field of O is $E = \frac{4\left(\frac{2q}{3}\right)}{R^2} = \frac{q^2}{6\pi\varepsilon_o R^2}$

(33) In balance condition

$$QE = mg \Rightarrow Q\frac{V}{d} = \left(\frac{4}{3}\pi r^3\rho\right)g$$

$Q \propto \frac{r^3}{V}$ Now find out Q_2

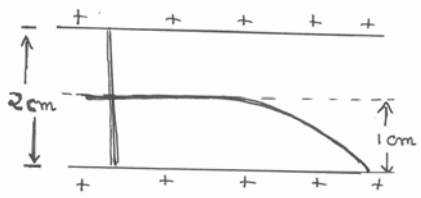


(34) Use equation, $E = \sqrt{E^2 + E_B{}^2 + 2E_A\cos 60°}$

(35) $\theta = \frac{\pi}{3} + \propto, \tan\propto = \frac{1}{2}\tan\pi/3$ $\propto = \tan^{-1}\sqrt{3}/2$

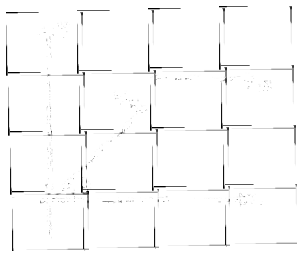
So, $\theta = \frac{\pi}{3} + \tan^{-1}\sqrt{3}/2$

(36) $d = \frac{1}{2}at^2 = \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{x}{v_o}\right)^2$



$\therefore\ V_0^2 = \frac{qE_x^{\ 2}}{2md}$ Now find out v_o.

(37) Time of flight is $t_F = \frac{2v_{oy}}{g}$ Now range $R = V_{ox}t_F + \frac{1}{2}a_x\, t^2{}_F$

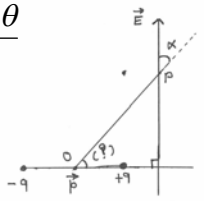


Where $a_x = \frac{qE}{m}$

Now, Calculate range.

(38) $P = e \times ro$ Calculate P.

(39) As $\tan \propto = \frac{\tan\theta}{2}$, Shown in fig $\propto = 90° - \theta$ $\tan(90° - \theta) = \frac{\tan\theta}{2}$



$\cot\theta = \frac{\tan\theta}{2}$ $\frac{1}{\tan\theta} = \frac{\tan\theta}{2}$ $\tan^2\theta = 2$

$\theta = \tan^{-1}(2)\theta = 52°\ to\ 53°$

(40) $\tau = PE\sin\theta = q(2q)E\ sin\theta$

$q = \frac{\tau}{(2a)E\sin\theta}$ Now find out q.

(41) Theory related quesition.

(42) As the drop is stationary, weight of drop = froce due to electric field.

$\frac{4}{3}\pi r^3 \rho g = neE$ So, $r^3 = \frac{3neE}{4\pi\rho g}$ now find out r.

(43) Theory related question.

(44) $\phi = \frac{\text{Q enclosed}}{\in_0} = \text{Q enclosed} = \text{f}\in_0 = (-8\times10^{-3} + 4\times10^3)\in_0$

Now find Q.

(45) Applysing gauss law

$E.4\pi x^2 = \frac{q}{\in_0} = \frac{\rho v}{\in_0} = \frac{\rho}{\in_0}\frac{4}{3}\pi x^3 = E \propto x$

(46) Applying gauss law $\phi = \frac{\Sigma q}{\in_0}$

(47) ϕ total $= \phi_A + \phi_B + \phi_C = \frac{q}{\in o}$ Assume $\phi_b = \phi$ and $\phi_A = \phi_C = \phi^1$

So, $2\phi^1 + \phi = \frac{\Sigma}{\in o}$ $\phi^1 = \frac{1}{2}\left(\frac{q}{\in_0} - \varnothing\right)$

(48) $\int \vec{E} \cdot \overrightarrow{da} = \frac{\sum q}{\in o}$ $\qquad E.2\pi rl = \frac{q}{\in_0}$

$E = \frac{q}{2\pi \in_o rl}$ $\qquad E = \frac{2\lambda}{4\pi \in_0 r}$ Now find out E.

(49) $E = -\frac{dv}{dr}$ $\qquad dv = \frac{-\lambda}{2\pi \in_0 r}$ $\qquad \frac{\lambda}{2\pi \in_o r} = -\frac{dv}{dr}$

$\int_{va}^{vb} dv = -\frac{\lambda}{2\pi \in_o} \int_a^b \frac{1}{r} dr$

Now calculate.

(50) maximum length of string = $\sqrt{3}l$

maximan enclosed charge = $\sqrt{3}\, l\, \lambda$ $\quad \therefore \ \phi = \frac{\sqrt{3}\, l\, \lambda}{\varepsilon_0}$

(51) (A) W = 100e (-4-10) = -1400ev
= -1400(-1.6 $\times$ 10^{-19}) J = 2.24 $\times$ 10^{-16} J

(52) (A) The electric Potential V $(x,y,z) = 4x^2$ Volt

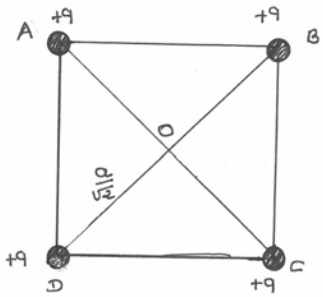
Now $\vec{E} = \left(\hat{i}\frac{dv}{dx} + \hat{j}\frac{dv}{dy} + \hat{k}\frac{dv}{dz} \right)$

Now $\frac{dv}{dx} = 8x$, $\frac{dv}{dy} = 0$ and $\frac{dv}{dz} = 0$

Hence $\vec{E} = -8x\hat{i}$, So at Point (1m,0,2m)

$\vec{E} = -8\hat{i}$ Volt/metre or 8 along negative *x*-axis

(53) (c) Potential at the centre o, $V = 4 \times {}^{1}\!/_{4\pi\varepsilon o} \times {}^{q}\!/_{\sqrt{2}}$

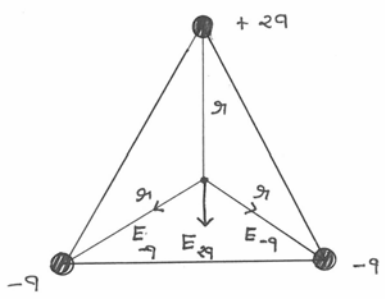


Where Q = ${}^{10}\!/_{3} \times 10^{-9}$ C and a = 8cm = $8 \times 10^{-2} m$

So v = $4 \times 9 \times 10^9 \frac{10}{3} \times 10^{-9}$

$\frac{8 \times 10^{-2}}{\sqrt{2}} = 1500\sqrt{2}\ Volt$

(54) Obviously, from charge configuration, at the centre electric field is non - zero. Potential at the centre due to 2q charge $V_{2q} = \frac{2q}{r}$ and



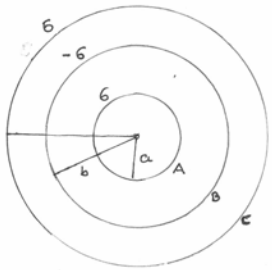
Potential due to –q Charge $V - q = {}^{-q}\!/_{r}$

(r = distance of centre Point)

$\therefore$ *Total Potential* $V = V_{2q} + V_{-q} + V_{-q} = 0$

(55) ABCDE is an equipotential surface, on equipotential surface no work is done in shifing a chrge from one place to another.

(56) $V_A = \dfrac{\sigma}{\varepsilon o}(a-b+c)$



$V_B = \dfrac{\sigma}{\varepsilon o}\left(\dfrac{a^2}{b}-b+c\right)$

$V_C = \sigma/\varepsilon o\left(a^2/c - b^2/c + C\right)$

on Putting $C = a + b \Rightarrow V_A = V_C \neq V_B$

(57) $E_x = \dfrac{-dv}{dx} = -\left[-2xy - z^3\right]$

$Ey = \dfrac{-dv}{dy} = -\left[-x^2\right]$

$Ez = \dfrac{-dv}{dz} = -\left[-3xz^2\right]$

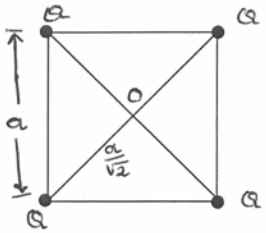
$\therefore\ \vec{E} = E_x\hat{i} + E_y\hat{j} + C_z\hat{k} = \left(2xy + z^3\right)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$

(58) (B) For Pair of Charge

$$U = \frac{1}{4}\pi\varepsilon_\circ\left[\frac{10\times10^{-6}\times10\times10^{-6}}{10/100}\right] + \left[\frac{10\times10^{-6}\times10\times10^{-6}}{10/100}\right] + \left[\frac{10\times10^{-6}\times10\times10^{-6}}{10/100}\right]$$

$$= \frac{3\times9\times10^9\times100\times10^{-2}\times100}{10} = 27J$$

(59) (B) Potential at Centre o of the square Vo = $4\left(\dfrac{Q}{4\pi\varepsilon_\circ\left(\frac{a}{\sqrt{2}}\right)}\right)$



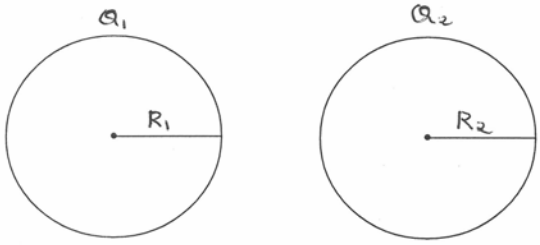
Work done in shifting (-Q) Charge from Centre to infinity

$W = -Q\left(V_\infty - V_0\right) QV_0$

$= \dfrac{4\sqrt{2}Q^2}{4\pi\varepsilon_0 a} = \dfrac{\sqrt{2}Q^2}{\pi\varepsilon_0 a}$

(60) (D) Let Q_1, and Q_2 are the Charges on Sphere of radii R_1 and R_2 respectively

Surface Charge density $\sigma = \dfrac{ch\,\text{arg}\,e}{Area}$



According to given Problem, $\sigma_1 = \sigma_2$

$= \dfrac{Q_1}{4\pi R_2^{\,2}} = \dfrac{Q_2}{4\pi R_2^{\,2}}$

$\therefore\ \dfrac{Q_1}{Q_2} = \dfrac{R_1^{\,2}}{R_2^{\,2}}$ .....(1)

In Case of a charged sphere, vs = $\frac{1}{4\pi\varepsilon_\circ}\ \frac{Q}{R}$

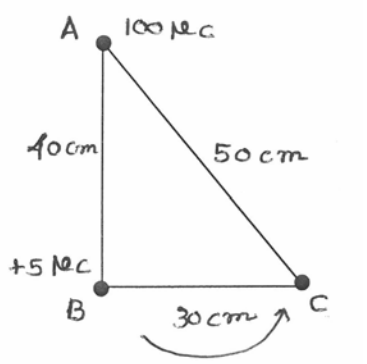
$\therefore\ V_1 = \frac{1}{4\pi\varepsilon_\circ}\ \frac{Q_1}{R_1}$, $V_2 = \frac{1}{4\pi\varepsilon_\circ}\frac{Q_2}{R_2}\ \Rightarrow \frac{V_1}{V_2} = \frac{Q_1}{R_1}\times\frac{R_2}{Q_2}\ = \frac{Q_1}{Q_2}\times\frac{Rz}{R_1}$

$= \left(\frac{R_1}{R_2}\right)^2 \times \frac{R_2}{R_1} = \frac{R_1}{R_2}$ .....(using (i))

(61) Usystem = $\frac{1}{4\pi\varepsilon_\circ}\ \frac{(q)(-2q)}{q} + \frac{1}{4\pi\varepsilon_\circ}\frac{(-2q)(q)}{a} + \frac{1}{4\pi\varepsilon_\circ}\frac{(q)(q)}{2a} = \frac{-7q^2}{8\pi\varepsilon_o a}$

(62) Work done indisplacing charge of 5lic from B to C is

$W = 5\times10^9(V_c - V_b)$ where



$V_B = \frac{9\times10^9\times100\times10^{-6}}{0.4} = \frac{9}{4}\times10^6 v$

$V_C = \frac{9\times10^9\times100\times10^{-6}}{0.5} = \frac{9}{5}\times10^6 v$

So $W = 5\times10^{-6}\times\left(\frac{9}{5}\times10^{-6}\frac{9}{4}\times10^6\right) = \frac{-9}{4}J$

(63) $E = \frac{-dv}{dx} = \frac{-d}{dx}\left(5x^2 + 10x - 9\right) = -10x - 10$

$\therefore\ \mathrm{E}_{(x=1)} = -10\times1 - 10 = -20\frac{v}{m}$

(64) Energy density ue = $\frac{1}{2}\varepsilon_o E^2 = \frac{1}{2}8.86\times10^{-12}\times\left(\frac{v}{r}\right)^2 = 2.83\ J/m^3$

(65) Potential inside the Sphere will be same as that on its Surface i.e. v = $V_{surface}$

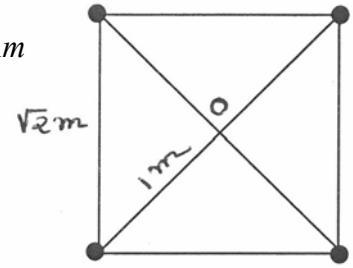
$= \frac{q}{10}$ volt

$Vout = \frac{q}{15}\ volt$

$\therefore \frac{vout}{v} = \frac{2}{3}\ \Rightarrow Vout = \frac{2}{3}V$

(66) Length of each side of square is $\sqrt{2}m$

so distance of it's centre from each cornet is $\frac{\sqrt{2}}{\sqrt{2}} = 1m$



Potential at the centre

$V = 9\times10^9\left[\frac{10\times10^{-6}}{1} + \frac{5\times10^{-6}}{1} - \frac{3\times10^{-6}}{1} + \frac{8\times10^{-6}}{1}\right]$

$\mathrm{V} = 1.8\times10^5$ V

(67) $W = Uf - U_1 = 9 \times 10^9 \times Q_1 Q_2 \left[\frac{1}{r_2} - \frac{1}{r_1}\right]$

$= W = 9 \times 10^9 \times 12 \times 10^{-6} \times 8 \times 10^{-6} \left[\frac{1}{4 \times 10^{-2}} - \frac{1}{10 \times 10^{-2}}\right]$

$= 12.96 \times 10° J = 13J$

(68) By using W = $Q\left(\vec{E} \,.\, \overrightarrow{\Delta r}\right)$

$W = Q\left[\left(e_1\hat{i} + e_2\hat{j} + e_3\hat{k}\right) \,.\, \left(a\hat{i} + b\hat{j}\right)\right] = Q\left(e_1 a + e_2 b\right)$

(69) K.E. = $q_0(V_A - V_B) = 1.6 \times 10^{-6}(70 - 50) = 3.2 \times 10^{-18} J$

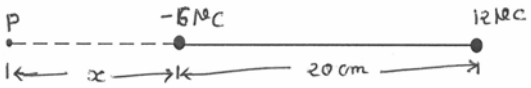
(70) Net electrostatic energy $U = \frac{KQq}{a} + \frac{Kq^2}{a} + \frac{KQq}{a\sqrt{2}} = 0$

$\frac{Kq}{a}\left(Q + q + \frac{Q}{\sqrt{2}}\right) = 0$

Now find out Q.

(71) Point P will lie near the Charge which is smaller in magnitude i.e. -6lic Hence Potential at P.

$V = \frac{1}{4\pi\varepsilon_0}\frac{(-6 \times 10^{-6})}{x} + \frac{1}{4\pi\varepsilon_0}\frac{(12 \times 10^{-6})}{(0.2 + x)} = 0$



$\therefore \; x = 0.2m$

(72) Calculate as MCQ 67.

(73) Calculate as MCQ 66.

(74) (C) AB+AC = 12 cm .....(i)

$= AB \,.\, AC = 32\ cm^2$

$= AB - AC = \sqrt{(AB - AC)^2 - 4\,AB \,.\, AC}$

$= AB - AC = 4$

From equation (i) and (ii)

AB = 8 cm; AC = 4cm

Potential energy at Point A

$V_A = \frac{1}{4\pi\varepsilon_0}\, q_1\, q_2 \left[\frac{1}{AB} + \frac{1}{AC}\right]$

$V_A = 1.35J$

(75) Use the equation $\frac{1}{2}$m $\left(V_1^2 - V_2^2\right)$ = QV

(76) Potential energy of the system

$$U = \frac{kQq}{l} + \frac{kq^2}{l} + \frac{kqQ}{l} = 0$$

$$\therefore \quad \frac{KQ}{l}(Q + q + Q) = 0 \qquad \therefore \; Q = \frac{-q}{2}$$

(77) $E_x = \frac{-dv}{dx} = -(-5) = 5$; also find E_y an E_z

$$E_{net} = \sqrt{E_x^{\,2} + E_y^{\,2} + E_z^{\,2}} = \sqrt{(5)^2 + (-3)^2 + \left(-\sqrt{15}\right)^2} = 7$$

(78) Van - d - glaph generater principal

(79) $$U = \frac{1}{4\pi\varepsilon_o}\left[\frac{Q_1Q_2}{r_1} + \frac{Q_2Q_3}{r_2} + \frac{Q_1Q_3}{r_3}\right]$$

$$U_{net} = 3 \times \frac{1}{4\pi\varepsilon_0} \cdot \frac{q^2}{l}$$

(80) Length of the diagonal of a cube having each side b is $\sqrt{3}$b. So distance of centre of cube from each vertex is $\frac{\sqrt{3}b}{2}$.

Hence Potential energy of the given system of charge is

$$U = 8 \times \left[\frac{1}{4\pi\varepsilon_0}\frac{(-q)(q)}{\sqrt{3}\,b/2}\right] = \frac{-4q^2}{\sqrt{3}\pi\varepsilon_0 b}$$

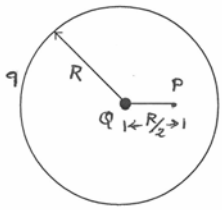
(81) When a negatively charged Pendylum oscillates overa Positively charged Plate then effective Value of 9 increases so, according to $T = 2\pi\sqrt{\frac{2}{g}}$ decreases

(82) When charge q is released in uniform electric field E then its acceleration $a = \frac{qE}{m}$ (is constant) so it's motion will be uniformly accelerated motion and it's Velocity after time is given by

$$V = at = \frac{qE}{m}t = K = \frac{1}{2}mv^2 = \frac{1}{2}\left(\frac{qq}{m}t\right)^2 = \frac{q^2E^2t^2}{2m}$$

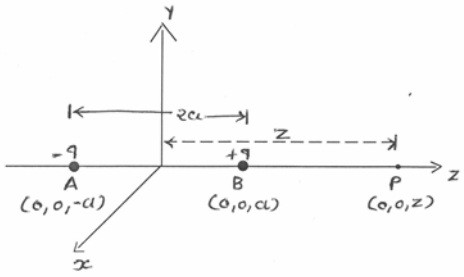
(83) Electric Potential at P

$$V = \frac{KQ}{\frac{R}{2}} + \frac{Kq}{R} = \frac{2Q}{4\pi\varepsilon_o R} + \frac{q}{4\pi\varepsilon_0 R}$$



(84) Potential at P due to (+q) charge

$$V_1 = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{(z-a)}$$



Potential at P due to (-q) Charge

$$= V_2 = \frac{1}{4\pi\varepsilon_o} \cdot \frac{-q}{(z+a)}$$

Total Potential at P due (AB) electric dipole $V = V_1 + V_2$

Now calculate

(85) (C) Potential will be zero at two Points .....

$$\text{At internal Point(M)} = \frac{1}{4\pi\varepsilon_o}\left[\frac{2\times10^{-6}}{(6-l)} + \frac{\left(-1\times10^{-6}\right)}{l}\right] = 0$$

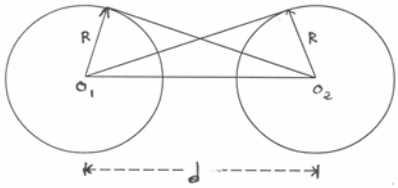
$\Rightarrow l = 2$

So distance of M from origin; $x = 6 - 2 = 4$ At exteriot Point (N)

$$\frac{1}{4\pi\varepsilon_o}\left[\frac{2\times10^{-6}}{(6+l)} + \frac{(-1\times10^{-6})}{l'}\right] = 0 \Rightarrow l' = 6$$

So distance of N from origin $x = 6 + 6 = 12$

(86) Potential at the centre of rings are



$$V_{O_1} = \frac{k.q}{R} + \frac{k(-q)}{\sqrt{R^2+d^2}}, \qquad V_{O_2} = \frac{k(-q)}{R} + \frac{kq}{\sqrt{R^2+d^2}}$$

$$\Rightarrow V_{O_1} - V_{O_2} = 2kq\left[\frac{1}{R} - \frac{1}{\sqrt{R^2+d^2}}\right] = \frac{q}{2\pi\varepsilon_0}\left[\frac{1}{R} - \frac{1}{\sqrt{R^2+d^2}}\right]$$

(87) (A) Change in Potential energy (Δu) = u_f - u_d

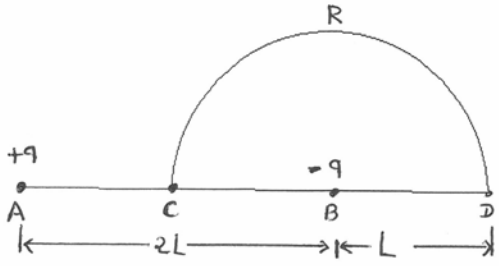
Now calculate

(88) Theory base Question

(89) Potential at C, $V_C = 0$

$\therefore$ Potential difference VD -VC = $-\frac{2}{3}\frac{kq}{L}$

$\Rightarrow W = Q(V_D - V_C)$ put the values



(90) $V = N^{2/3}\ V \Rightarrow 40 = N^{2/3} \times 10$

$N^{2/3} = 4 \Rightarrow N^2 = 64 \Rightarrow N = 8$

(91) $C_1 = C_2$

$\frac{A_1\varepsilon_o}{d_1} = \frac{A_2\varepsilon_o}{d_2}$ $\therefore\ d_2 = \frac{A_2}{A_1}d_1$ Now find d_1.

(92) Here $C_s = 3l_1F$ *and* $C_p = 16\ l_1F$ Now in Parallel connection $C_p = C_1 + C_2$

So, $C_1 + C_2 = 16 + 11$

in series connection $C_5 = \frac{C_1C_2}{C_1 + C_2}$ So, $= \frac{C_1C_2}{C_1 + C_2}$ .....(2)

or C_1C_2 Now, $(C_1 - C_2)^2 = (C_1 + C_2)^2 - 4C, C_2$ or

$C_1 - C_2 = 8$.....(3)

Adding the question (1) and (3) we have

$(C_1 + C_2) + (C_1 - C_2)\ 16 + 8$

$\therefore\ a = 12\mu F\ \&\ C_2 = 16 - C_1 = 4\mu F$

(93) Suppose that the techinician makes a combination use of N Capacitors and connects them in m rows, each row having m Capacitors Then N=mn

Capacitance of each capacitor = $1l_1F$ required Capacitance of the combination C = $2l_1F$

Voltage rating of each < apalitor = 400V

required Voltage rating of the combination = 10^3V

When capacitars are connected in series P.d. a cross their Plates get added. for n capacitors

Voltage ...to $400 \times nv$

$\therefore\ 400n = 10^3$

$\therefore\ n = \frac{10^3}{400} = 2.5$ *or* $n = 3$

The total Capucitae $\frac{1}{C_1} = \frac{1}{1} + \frac{1}{1} + \frac{1}{1} = 3$ $C' = \frac{1}{3}$

$\therefore$The total Capacitanae of m rous $C = mc'$ OR $m = \frac{C}{C'} = \frac{2}{\frac{1}{3}} = 6$

(94) Common Potential $V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2} = \frac{4\pi\varepsilon_o r_1 v_1 + 4\pi\varepsilon_o r_2 v_2}{4\pi\varepsilon_o r_1 + 4\pi\varepsilon_o r_2}$

$V = \frac{r_1v_1 + r_2v_2}{r_1 + r_2}$

(95) $U_1 = \frac{1}{2}C_1V_1^2 = 0.121J$

Total charge on the two Capacitors $q = C_1V_1 + C_2V_2 = 11 \times 10^{-4}\,C$

$C = C_1V_2\ = 5l_1F + 2.5l_1F = 7.5l_1F = 7.5 \times 10^{-6}\,F$

$V = \frac{9}{C} = \frac{440}{3}V$

$U_2 = \frac{1}{2}CV^2 = 0.081J$ Now, energy lost = $U_1 - U_2$ find it.

(96) Apply wheatstone bridge law.

(97) $Cair = \frac{A\varepsilon_o}{d}$ with dielectlic slab $C^{1} = \frac{A\varepsilon_o}{(d - t + t/k)}$ Now C′ = $\frac{4}{3}$C

$= \frac{A\varepsilon_o}{\left(d - t + \frac{t}{k}\right)} = \frac{4}{3}\frac{A\varepsilon_o}{d}$

= K = $\frac{4t}{4t - d} = \frac{4(q/2)}{4(d/2 - d)}$ = 2

(98) As P.d. remains the same, Capacity must remain the same,

$\therefore\ x = t\left(1 - \frac{1}{k}\right)$ x = 1.5 mm t= 2mm Now find out k.

(99) $C_o = \frac{A\varepsilon_o}{d} = 18\ C = \frac{k}{3}\frac{A\varepsilon_o}{d} = 72$ Now, find out K.

(100) C = $4\pi\varepsilon_o R = \frac{R}{k}$. Put the Values of R&K.

(101) $C_1 = \frac{k_1\varepsilon_0 A}{d/2} = \frac{2k_1\varepsilon_0 A}{d}$

$C_2 = \frac{2k_2\varepsilon_o d}{2}$ Now, $\frac{1}{C_3} = \frac{1}{C_1} + \frac{1}{C_2}$

(102) Apply lawa of series & Parallel conection of Capacitor

(103) $\frac{1}{C_5} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

$C_p = C_1 + C_2 + C_3$ Now fiud out $\frac{C_s}{C_p}$

(104) Number of Capacitors to be connected in series

$V = \frac{voltage\ rat\text{int}\ g\ required}{voltage\ rat\text{int}\ g\ required} = \frac{700}{200} = 3.5$

$i.e$ 4.

$C_{eq} = \frac{10}{4} = 2.5 r_1 F$

Number of rows required = $\frac{Capacity required}{Capacity of\ each row} = \frac{10}{2.5} = 4$

$\therefore$ Total number of Capacitor 4×4 = 16

(105) $U_1 = \frac{1}{2}C_1V^2 \& U_2 = \frac{1}{2}C_2V^2$ $\in_o \Delta U = U_1 - U_2$

(106) Work done = $U_2 - U_1 = \frac{q_1 q_2}{4\pi \in_o}\left[\frac{1}{r_2} - \frac{1}{r_1}\right]$ Now Calculate.

(107) Volume of big drop = $1000\times$ volume of small drop $\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi l^2$

Now find out initial energy

$U_1 = 1000 \times \frac{q^2}{2C}$ then final energy $U_2 = \frac{(1000q)^2}{2C^1}$ then $\frac{U_2}{U_1}$

R = 10r so, $C^1 = 10C$

(108) minimum capacity is C_s (Series Connection) & maximum Capaciy is C_P (Parallel Connection) Now $\frac{C_p}{C_s}$ Calculate

(109) Theory base question

(110) With $S_1 \& S_3$ Closed , the Capacitors $C_1 \& C_2$ are in series arrangenent. So $\frac{V_1^1}{V_2^1} = \frac{C_2}{C_1} = \frac{3}{2} \& V_2^{\,1} + V_2^{\,1} = V_1 + V_2 = 30 + 20 = 50V$ Now simplify.

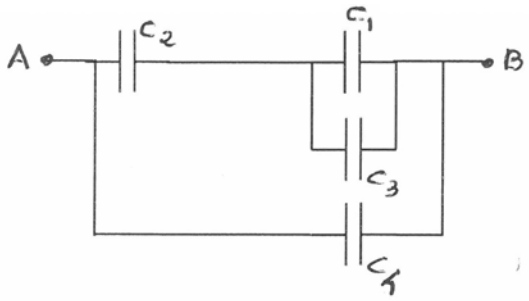
(111) $E = \frac{\sigma_1}{2 \in_o} = \frac{\sigma_2}{2 \in_o} = \frac{Q_1}{2A \in_o} - \frac{Q_1}{2A \in_o}$ Now Simplify.

(112) $C_{23} = 30 + 30 = 60F$ Now, $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_{23}} + \frac{1}{C_4}$

$\Rightarrow C = \frac{120}{7} pf$ Total charge Q =CV

$\Rightarrow V_1 = \frac{Q}{C_1},\ V_2 = V_3 = V_{23} = \frac{Q}{C_{23}} \& V_4 = \frac{Q}{C_4}$

(113) The arrangement can be redrawn as shown in figule Now find out.



(114) $V = 4x^2 \Rightarrow \vec{E} = -\frac{dv}{dr} = -8x$ Now Put the value of x.

(115) $\frac{1}{C_5} = \frac{1}{C_1} + \frac{1}{C_2}$ and $q = C_5 V$

(116) $C = \frac{A \in_o}{d} = 9PF$ $\quad C_1 = \frac{3k_1 \in_0 A}{d}$ and $C_2 = \frac{3k_2 \in_0 A}{2}$

Now $\frac{1}{C_3} = \frac{1}{C_4} + \frac{1}{C_2}$

(117) Knowledge base question.

(118) Knowledge base question.

(119) Knowledge base question.

(120) Use equation $E = \frac{-dV}{dx}$

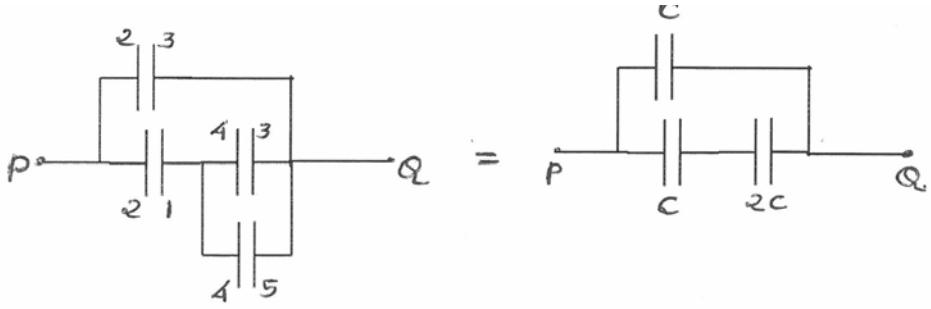
(121) $W = QV = (CV)V = CV^2$ and $\frac{1}{2}CV^2$ Now find $\frac{U}{w}$

(122) When Spheres are connected by a conducting wire their Potentials become equal.

$$\frac{C_1}{C_2} = \frac{r_1}{r_2} = \frac{1}{2} \quad \frac{q_1}{q_2} = \frac{C_1 V}{C_2 V} = \frac{C_1}{C_2} = \frac{1}{2}$$

Now, $\frac{E_1}{E_2} = \frac{\frac{Kq_1}{r_1^2}}{\frac{Kq_2}{r_2^2}}$ Now find out rativ.

(123)



(124) $Q = CV = 5\mu c$

$$= C^1 = \frac{A\varepsilon_o}{d - \left(t - \frac{t}{k}\right)} = \frac{A\varepsilon_o / d}{1 - \left(\frac{t - t/k}{d}\right)}$$ Now Put the value.

(125) Knowledge base question.

(126) $\frac{q}{c} + \frac{q}{c} + \frac{q}{c} + \frac{q}{c} = 10$ or $\frac{\Sigma}{C} = \frac{10}{4} = 2.5V$

Now $\therefore V_A - 0 = 7.5 \Rightarrow V_A = 7.5V$

Again $V_N - V_B = \ldots\ldots \quad O - V_B = 2.5 \quad V_B = -25V$

(127) $V = \frac{64q}{4\pi \in_o R} = \frac{64q}{4\pi \in_o (4r)}$

$C = A\pi \in_o / d$

(128) $C^1 = \frac{A \in_o}{d - t\left(1 - \frac{1}{k}\right)}$ Use this equation

(129) $C_1 = \frac{A_1 \in_o}{d_1} = \frac{5}{4}\pi \in_o$ and $C_2 = \frac{6}{5}\pi \in_o$

Now $C = \frac{C_1 C_2}{C_1 + C_2}$

(130) C = $\frac{3}{2}\frac{A \in_o}{d}$ and $V = \frac{Q}{C}$

(131) Use wheatston's bridge equation $C_{eq} = C$ and $Q = CV$

Charge on Capacitor between the terminals

A and B is $\frac{Q}{2} = \frac{CV}{2}$ = Enegy stored in capacitor = $\frac{\left(\frac{Q}{2}\right)^2}{2C}$

(132) Theory base question.

(133) Work done = Final energy - initial energy = $\frac{Q^2}{2C^1} - \frac{Q^2}{2C}$

(134) $U_i = \frac{1}{2}C\left(V_1^2 + V_2^2\right)$ and $V = \frac{q_1 + q_2}{c_1 + c_2} = \frac{V_1 + V_2}{2}$

$U_f = \frac{1}{2}(2C)V^2$ Now find $U_i U_f$

(135) $C_1 = \frac{A\varepsilon_O}{d/10}$ $C_2 = {A\varepsilon_O}/{d/2} \Rightarrow$

$C_5 = \frac{5C}{3}$

$\therefore$ Percentage increase in Capacitance = $\frac{C_5 - C}{C} \times 100\%$

(136) Use Series and Parallel Connection law.

(137) Theory base question.

(138) Theory base question.

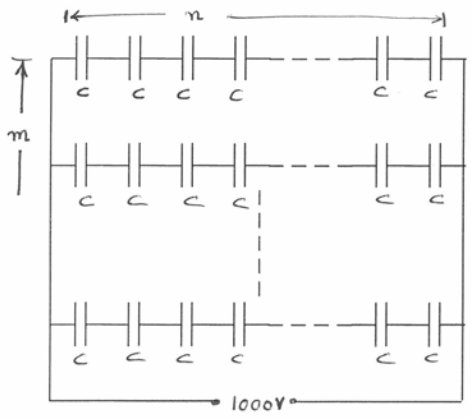
(139) $n = \frac{1000}{350} = 2.8 = 3$

$C_S = \frac{1}{3}$ F $= \frac{m}{3}$ F

$\frac{m}{3}$ F = 2 F

m = 6

total no = mn = $3 \times 6 = 18$



Unit - 12
Curreent Electricity

SOME IMPORTANT POINTS

1. Current $I = \frac{dQ}{dt}$

   If current is steady, then $I = \frac{Q}{t} = \frac{ne}{t}$

2. If a point charge q is moving in circle with constant speed and frequency f, then corrosponding current.

   $I = fq = \frac{\omega}{2\pi} q$

3. Current density at any point of conductor.

   $J = \frac{dI}{da\cos\theta}$

   $\therefore dI = \vec{J}.\vec{da} \Rightarrow I = \int \vec{J}.\vec{da}$

   If the cross-sectional area is perpendicular to the current and if J is constant over the entire cross-section, then

   $J = \frac{I}{A}$

4. Ohm's law $R = \frac{V}{I}$ where R= resistance

   1 Ohm $1 = \frac{volt}{Ampere}$ V= potential difference

   I= current flowing through the conductor

   $\frac{1}{R}$ is called the conductance of material.

   Its unit is Ω^{-1} or mho or seimen(s)

5. Resistivity $R = \rho\frac{\ell}{A}$ $\rho \rightarrow$ resistivity

   $\rho = \frac{RA}{\ell}$ $\rightarrow$ unit $\rightarrow$ Ohm.m (Ω m)

   $\rightarrow$ Dimension formula $\rightarrow M^1L^3T^{-3}A^{-2}$

6. Conductivity $\sigma = \frac{1}{\rho}$ $\rightarrow$ unit $\rightarrow$ mho.m^{-1}

   $\rightarrow$ Dimension $\rightarrow M^{-1}L^{-3}T^3A^2$

7. Drift velocity $V_d = \frac{eE}{m}\tau$ and $I = neAV_d$

$$V_d = \frac{I}{neA} = \frac{J}{ne} = \frac{\sigma E}{ne} = \frac{E}{\rho ne} = \frac{V}{\rho \ell ne}$$

where, η = Number of electrons per unit volume of the conductor

A = Area of cross-section

V = Potential difference across the conductor

E = electric field inside the conductor

I = Current

J = Current density

ρ = Specific resistance

σ = Conductivity $\left(\sigma = \frac{1}{\rho}\right)$

τ = relxation time between cons. collisan

8. Resistivity $\rho = \frac{m}{ne^2\tau}$

9. Mobility $\mu = \frac{v_d}{E}$

$$\mu = \frac{\sigma}{ne} \qquad \text{Unit} = \frac{m^2}{\text{volt.sec}}$$

(1) For conductor $\sigma = n_e e\mu_e$

(2) For Semiconductor $\sigma = n_e e\mu_e + n_h e\mu_h$

10. Temperature Dependence of Resisitivity

$\rho_\theta = \rho_{\theta_0}\left[1 + \alpha(\theta - \theta_0)\right]$ where, ρ_θ = resistivity at a temperature θ

ρ_{θ_0} = resistivity at a proper reference temperature θ_0

α = temperature co-efficient of resistivity $(.^0C^{-1})$

$$R_\theta = R_{\theta_0}\left[1 + \alpha(\theta - \theta_0)\right]$$

If R_1 and R_2 are the resistance at t_1℃ and t_2℃ respectively then $\frac{R_1}{R_2} = \frac{1+\alpha t_1}{1+\alpha t_2}$

and $\alpha = \frac{R_2 - R_1}{R_1(t_2 - t_1)}$

11. **The emf of a Cell and Terminal Voltage:** when unit positive charge is driven form negative terminal to the positive terminal due to non-elecrical forces, the energy gained by the charge (or work done by the non-electrical forces) is called an emf (ε) of a battery.

    The net potential difference between the two terminals of a battery is called the terminal voltage (V).

    The terminal voltage of a battery is, $V = \varepsilon - Ir$

12. **Secondary Cell:** The cell which can be restored to original condition by reversing chemical processes (i.e. by recharging) are called secondary cells. e. g. lead accumulator.

13. **Charging:** If the secondary cell is connected to some other external d.c. source of larger emf, current may enter the cell through the positive terminal and leave it at the negative terminal. The electrical energy is then converted into chemical energy. This is called charging of the cell.

    For the charging of a laed storage cell (lead accumulator),

    $VIt = \varepsilon It + I^2Rt + I^2rt$ and $I = \dfrac{V - \varepsilon}{r + R}$ where I = charging current

14. **Junction or branch Point:** It is the point in a network at which more then two conductors (minimum three) meet.

15. **Loop:** A closed circuit formed by conductors is known as loop.

16. **Kirchhoff's Rules:**

    First rule: `` The algebraic sum of all the electric currents meeting at the junction is zero."

    $\therefore \sum I = 0$

    Second Rule: `` For any closed loop the algebraic sum of the products of resistances and the respective currents flowing through them is equal to the algebraic sum of the emfs applied along the loop."

    $\sum IR = \sum \varepsilon$

17. **Connections of Resitors:**

    Series Connection:

    $R_S = R_1 + R_2 + R_3 + ... + R_n$

    where, R_S = Equivalent resistance of n resistors connected in series.

    Parallel connection:

    $$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} ..... + \frac{1}{R_n}$$

    where, R_p = Equivalent resistance of n resistors connected in parallel.

18. **Series Connection of Cells:** For the series connection of two cells of emfs ε_1 and ε_2 and internal resistances r_1 and r_2,

    $$I = \frac{\varepsilon_1 + \varepsilon_2}{R + (r_1 + r_2)} = \frac{\varepsilon_{eq}}{R + r_{eq}}$$ (for helping condition)

    where, I= Current flowing through the external resistance R connected across the series connection.

Equivalent emf $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$

Equivalent internal resistance $r_{eq} = r_1 + r_2$

19. **Parallel Connection of cells:** When n cells of equal emf E and internal resistance r are connected in series in helping condition.

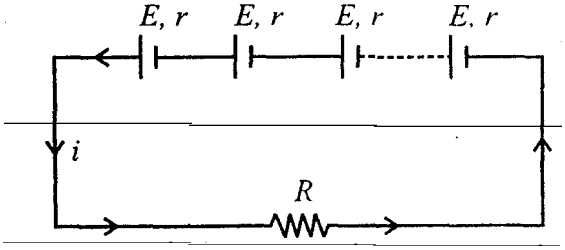
$$I = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}}{1 + \frac{R}{r_1} + \frac{R}{r_2}} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{R(r_1 + r_2) + r_1 r_2}$$

$$\therefore I = \frac{\frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{(r_1 + r_2)}}{R + \frac{r_1 r_2}{(r_1 + r_2)}} = \frac{\varepsilon_{eq}}{R + r_{eq}}$$

Equivalent emf $\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$

Equivalent internal resistance $r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$

(1) **Series grouping:** In series grouping of one cell is connected to cathode of other cell and so on, If n identical cells are connected in series.



(i) Equivalent emf of the combination $E_{eq} = nE$

(ii) Equivalent internal resistance $r_{eq} = nr$

(iii) main current = Current from each cell $= i = \frac{nE}{R + nr}$

(iv) Potential difference across external resistance $V = iR$

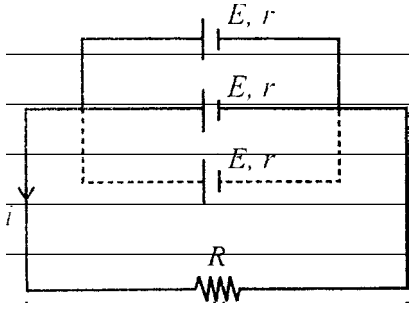
(v) Potential difference across each cell $V' = \frac{V}{n}$

(vi) Power dissipated in the external circuit $= \left(\frac{nE}{R + nr}\right)^2 R$

(vii) Condition for maximum power : $R = nr$ and $P_{max} = n\left(\frac{E^2}{4r}\right)$

(viii) This type of combination is used when $nr << R$.

(2) **Parallel grouping:** In parallel grouping all anodes are connected at one point and all cathodes are connected together at other point. of n identical cells are connected in parallel.



(i) Equivalent emf $E_{eq} = E$

(ii) Equivalent internal resistance $R_{eq} = \frac{r}{n}$

(iii) Main current $i = \frac{E}{R + r/n}$

(iv) Potential difference across external resistance = p.d across each cell = V= iR

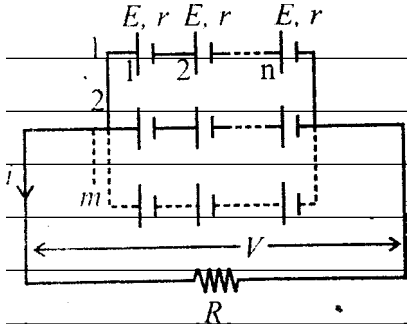
(v) Current form each cell $i' = \frac{i}{n}$

(vi) Power dissipated in the circuit $P = \left(\frac{E}{R + r/n}\right)^2 R$

(vii) Condition for max. power is $R = \frac{r}{n}$ and $P_{max} = n\left(\frac{E^2}{4r}\right)$

(viii) This type of combination is used when $r >> nR$

(3) **Mixed Grouping:** If n identical cells are connected in a row and such m rows are connected in parallel as shown, then



(i) Equivalent emf of the combination $E_{eq} = nE$

(ii) Equivalent internal resistance of the combination $r_{eq} = \frac{nr}{m}$

(iii) main current flowing through the load $i = \frac{nE}{R + \frac{nr}{m}} = \frac{mnE}{mR + nr}$

(iv) Potential difference across load $V = iR$

(v) Potential difference across each cell $V' = \frac{V}{n}$

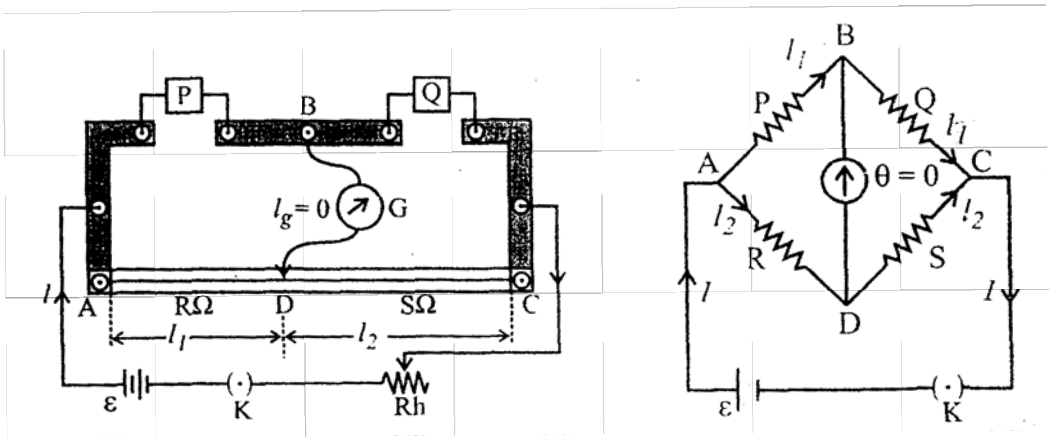
(vi) Current form each cell $i' = \frac{i}{n}$

(vii) Condition for maximum power $R = \frac{nr}{m}$ and $P_{max} = (mn)\frac{E^2}{4r}$

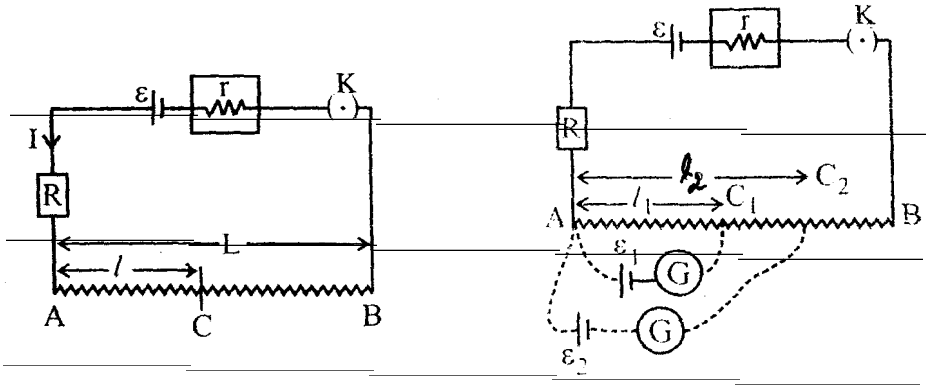
(viii) Total number of cells = mn

20. **Wheatsone Bridge:** For a balanced wheatstone bridge, $\frac{P}{Q} = \frac{R}{S}$ or $\frac{P}{R} = \frac{Q}{S}$

For Practical circuit $\frac{P}{\ell_1} = \frac{Q}{\ell_2}$ or $\frac{P}{Q} = \frac{\ell_1}{\ell_2}$



21. **Potentiometer:**



Current $I = \frac{\varepsilon}{R + \rho L + r}$ where, r = internal resistance of battery

L = length of potentiometer wire

ρ = resistance per unit length of potentiometer wire

ρL = resistance of potentiometer wire

ε = emf of battery

R = resistance connected in series

$\therefore$ Potential difference between two points on wire separated by distance

ℓ will be, $V_\ell = I(\rho\ell) = \left(\frac{\varepsilon\rho}{R + \rho L + r}\right)\ell$

$\therefore$ Potential gardient on wire will be σ, where $\sigma = \frac{V_1}{\ell} = \frac{\varepsilon\rho}{R + L\rho + r}$ OR $\sigma = \ell\rho$

(i) If the length of a Potentiometer wire required to balance the cell of emf ε_1 is ℓ_1, then $\varepsilon_1 = \sigma\ell_1$

(ii) If the length of a potentiometer wire required to balance the cell of emf ε_2 is ℓ_2, then $\varepsilon_2 = \sigma\ell_2$

$\therefore \frac{\varepsilon_2}{\varepsilon_1} = \frac{\ell_2}{\ell_1}$

22. **On Passing electric current in a conductor:**

Electric energy consumed = Heat enrgy generated (in joule)

$\therefore W = VQ = Vlt = I^2Rt = \frac{V^2t}{R}$, where

V = Potential difference between two ends of a conductor

Q = electric charge

I = electric current

R = ohmic resistance

t = time in seconds

23. **Heat or thermal energy:**

Heat (calorie) = $\frac{I^2Rt}{J}$, where J= Joule's constant = 4.2 J/cal

Heat or thermal energy:

H (joule) = I^2Rt

$\therefore$ Heat (H) per unit time α I^2

24. **Electric power (or electrical energy consumed in unit time):**

$$P = \frac{W}{t} = VI = I^2R = \frac{V^2}{R}$$

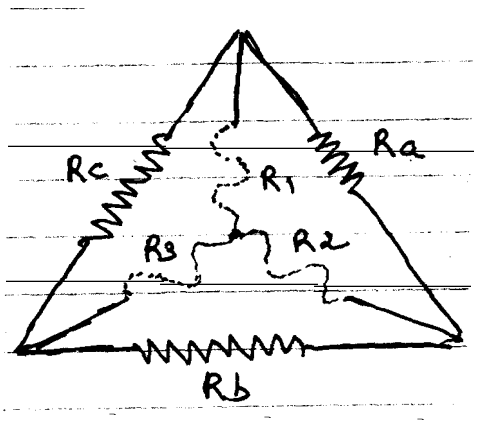
$\therefore P \alpha I^2 \rightarrow$ (Joule's Law)

25. **Star (Y) Delta (Δ) arrangment:** Here three ressistances Ra, Rb, Rc are replaced by R_1 R_2 and R_3 as shown, then

$$R_1 = \frac{RaRc}{Ra + Rb + Rc}$$

$$R_2 = \frac{RaRb}{Ra + Rb + Rc}$$

$$R_3 = \frac{RbRc}{Ra + Rb + Rc}$$



Current Electricity

Electric Current: The rate of flow of electric charge, $I = \frac{q}{t}$

Kirchhoff's Law

First Law: The algebraic sum of the currents meeting at a junction is zero
i.e., $\Sigma I = 0$

Second Law: In a closed loop, the algebraic sum of all the potential differences is zero, *i.e.*, $\Sigma \Delta V = 0$

Electric Power:
$P = \frac{\text{Electric workdone}}{\text{time taken}}$
$P = \frac{W}{t} = VI = I^2R = \frac{V^2}{R}$

Electric Energy: $E = p \times t$

Drift Velocity: $\vec{v}_d = \frac{-e\vec{E}}{m}\tau$

Mobility: It is the drift velocity per unit electric field applied $\mu_e = \frac{v_d}{E}$

Relation between drift velocity and mobility
$I = An_d e\mu_e E$

Ohm's Law: Current flowing through a conductor is directly proportional to the potential difference across its ends
$V = IR$

Electric Resistance: It is the obstruction posed by the conductor to the flow of current through it. $R = \rho\frac{l}{A}$

Internal Resistance of a Cell: Resistance offered by the electrolyte and electrodes of a cell when current flows through it.

Resistance in Series: $R_s = R_1 + R_2 + \ldots + R_n$

Resistance in Parallel: $\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$

Wheatstone Bridge : Its principle states that if four resistances P, Q, R, S are arranged to form a bridge and if the galvanometer G shows no deflection then the bridge is said to the balanced.
$\frac{P}{Q} = \frac{R}{S}$

Metre Bridge : It is based on the principle of Wheatstone's bridge. The unknown resistance,
$R = \frac{Sl}{100 - l}$

Potentiometer: Its principle is based on the facts that the fall of potential across any portion of the wire is directly proportional to the length of the wire
$V = Kl$

Determination of internal Resistance of a Cell:
$r = \left(\frac{l_1 - l_2}{l_2}\right)R$

Comparison of Emf of Cell:
$\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$

Electric Conductivity: It is the increase of resistivity.

Effect of Temperature on Resistance: The resistance of a metal conductor at a temperature t°C is given by $R_t = R_0(1 + \alpha t)$

Potential Difference of a Cell: It is the potential difference between the two electrodes of a cell in a closed circuit.

Emf of a Cell: It is the potential difference between the two terminals of a cell in an open circuit

Grouping of Cells

Series Grouping:
$I = \frac{n\varepsilon}{R + nr}$
- If $R << nr$, then $I = \frac{\varepsilon}{r}$
- If $R >> nr$, then $I = \frac{\varepsilon}{r}$

Mixed Grouping:
$I = \frac{n\varepsilon}{R + \left(\frac{r}{m}\right)}$
current in the circuit is maximum

Parallel Grouping:
$I = \frac{\varepsilon}{R + \left(\frac{r}{m}\right)}$
- If $\frac{r}{m} << R$, then $I = \frac{\varepsilon}{R}$
- If $\frac{r}{m} >> R$, then $I = m\frac{\varepsilon}{R}$

Symbols used:
I = electric current
v_d = drift velocity
ρ = resistivity
K = potential gradient
q = electric charge
μ_e = mobility
ε = emf
R_S, R_P = effective resistance
n, m = number of cells in one row and number of rows of cells respectively
t = time taken
E = electric energy
t = time of relaxation
l = length of the conductor
R, r, s = resistance
P = electric power
E = electric field
n_d = number density of electrons
A = Area of cross section of conductor

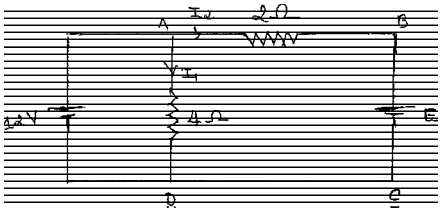
Question

For the answer of the following questions choose the correct alternative from among the given ones.

1. Two wires of equal lenghts, equal diameters and having resistivities ρ_1 and ρ_2 are connected in series The equivalent resistivity of the combination is....

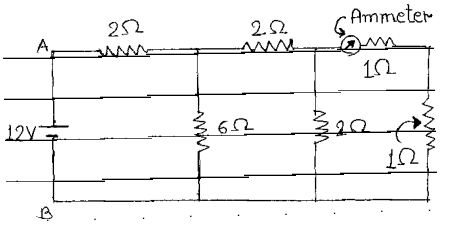
   (A) $(\rho_1 + \rho_2)$ (B) $\frac{1}{2}(\rho_1 + \rho_2)$ (C) $\frac{\rho_1\rho_2}{(\rho_1 + \rho_2)}$ (D) $\sqrt{\rho_1 \rho_2}$

2. In the circuit shown in fig, current $I_2 = 0$ The value of E is....



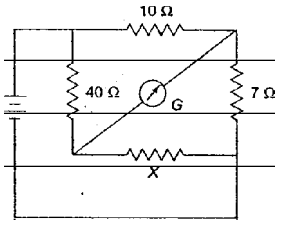
   (A) 3V (B) 6V
   (C) 9V (D) 12V

3. In the circuit shown in fig, the reading of ammetre is....



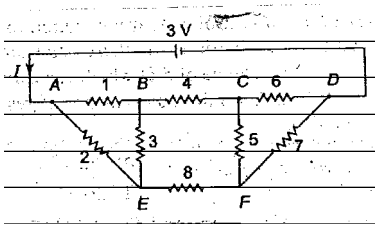
   (A) 1A (B) 2A
   (C) 3A (D) 4A

4. In Fig, the galvanometer shows no deflection. what is the resigtance X?



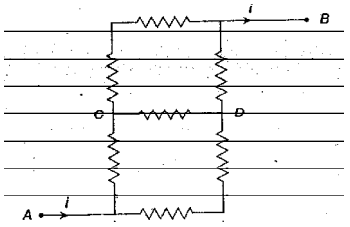
   (A) 7Ω (B) 14Ω
   (C) 21Ω (D) 28Ω

5. Figure, shows a network of eight resistors numbered 1 To 8, each equal to 2Ω, connected to a 3V battry of negligible internal resistance The current I in the circuit is....



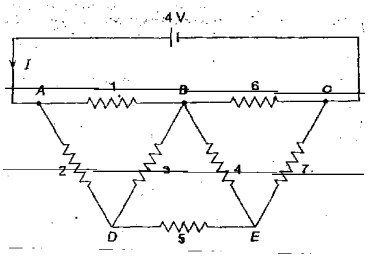
   (A) 0.25A (B) 0.5A
   (C) 0.75A (D) 1.0 A

6. Seven resistors, each of resistance 5Ω, are connected as shown in fig, The equiualent resistance between points A and B is....



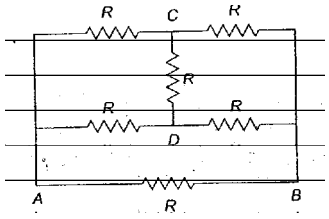
   (A) 1Ω (B) 7Ω
   (C) 35Ω (D) 49Ω

7. Figure, shows a network of seven resistors number 1 to 7, each equal to $1\Omega_3$ connecteal to a 4 V battery of negligible internal resistance The current I in the circuit is....



   (A) 0.5A (B) 1.5A
   (C) 2.0A (D) 3.5A

8. In the circuit shown in fig, the effective resistance between A and B is....



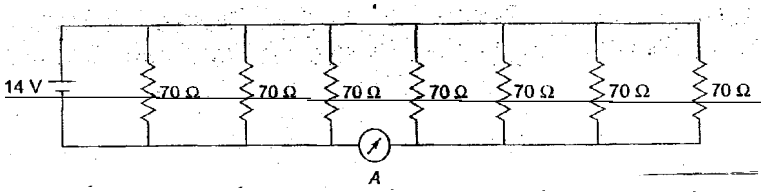
(A) $R/2$ (B) R

(C) 2 R (D) 4R

9. The effective resistance of a n number of resistors connected in parallel in x ohm. When one of the resistors is removed, the effective resistance becomes y ohm. The resistance of the resistor that is removed is....

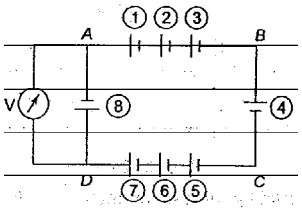
(A) $\frac{xy}{(x+y)}$ (B) $\frac{xy}{(y-x)}$ (C) (y – x) (D) $\sqrt{xy}$

10. The reading of the ammeter in the circuit in Fig, is ....



(A) $\frac{3}{5}$ A (B) $\frac{4}{5}$ A (C) $\frac{6}{5}$ A (D) $\frac{7}{5}$ A

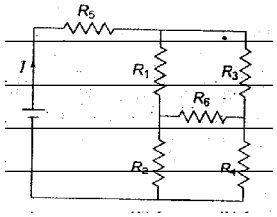
11. Eight cells marked 1 to 8, each of emt 5V and internal resistance 0.2Ω are connected as shown in Fig, what is the reading of the ideal voltmeter V?



(A) 40 V (B) 20 V

(C) 5V (D) zero

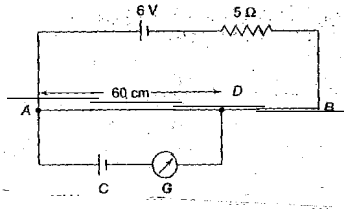
12. In the given circuit it is observed that the current I is independent of the value of the resistance R_6 Then the resistance values must satisfy (see fig)



(A) $R_1R_2R_5=R_3R_4R_6$ (B) $\frac{1}{R_5}+\frac{1}{R_6}=\frac{1}{R_1+R_2}+\frac{1}{R_3+R_4}$

(C) $R_1R_4=R_2R_3$ (D) $R_1R_3=R_2R_4=R_5R_6$

13. In the potentiometer circuit shown in fig, the internal resistance of the 6V battery is 1Ω and the length of the wire is 100 cm. when AD=60 cm, the galvanometer shows no deflection.

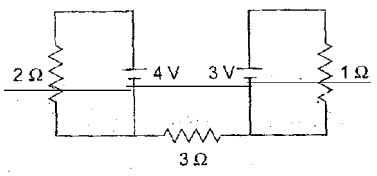


The emf of cell c is (the resistance of wire AB is 2Ω)

(A) 0.7 V (B) 0.8 V

(C) 0.9 V (D) 1.0V

14. The potential difference through the 3Ω resistor shown in fig is....



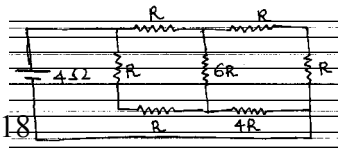
(A) Zero (B) 1V

(C) 3.5V (D) 7V

15. when a cell is connected to a resistance R_1 the rate at which heat is generated in it is the same as when the cell is connected to a resistance $R_2 (< R_1)$ the internal resistance of the cell is....

(A) $(R_1 - R_2)$ (B) $\frac{1}{2}(R_1 - R_2)$ (C) $\frac{R_1 R_2}{(R_1 + R_2)}$ (D) $\sqrt{R_1 R_2}$

16. A battary of internal resistance 4Ω is connected to the network of resistances as shown in fig, in order that maximun power can be delivered to the network, the value of R in ohm should be.



(A) $\frac{4}{9}$ (B) 2 (C) $\frac{8}{3}$ (D) 18

17. Length of a wire of resistance $R\Omega$ is increased to 10 times, so its resistance becomes 1000Ω, therfore R=.... (The volume of the wire remains same during increase in length)

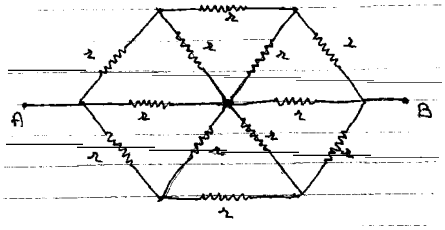
(A) 0.01 Ω (B) 0.1Ω (C) 1Ω (D) 10 Ω

18. On applying an electric field of 5×10^{-8} Vm^{-1} across a conductor, current density through it is 2.5 Am^{-2} The resistivity of the conductor is ....

(A) 1×10^{-8} Ω m (B) 2×10^{-8} Ω m

(C) 0.5×10^{-8} Ω m (D) 12.5×10^{-8} Ω m

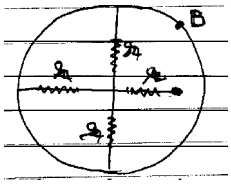
19. Calculate net resistance between A and B



(A) $4r/5$ (B) $5r/2$

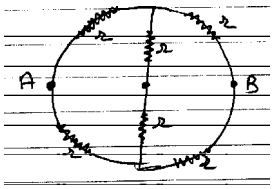
(C) 4r (D) $\frac{5r}{4}$

20. Calculate net resistance between A and B



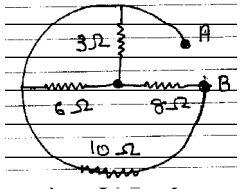
(A) r (B) $\frac{4r}{3}$ (C) $\frac{3r}{4}$ (D) 2 r

21. Calculate net resistance beween A and B



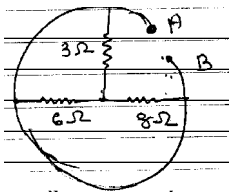
(A) 2 r (B) 3 r (C) $\frac{r}{2}$ (D) r

22. Calculate net resistance beween A and B



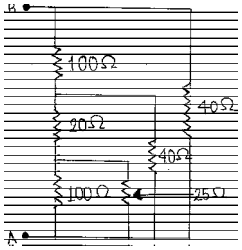
(A) 3 Ω (B) $\frac{1}{3}$ Ω (C) 5 Ω (D) $\frac{1}{5}$ Ω

23. Calculate net resistance beween A and B



(A) 6 Ω (B) 10 Ω (C) 9 Ω (D) 4 Ω

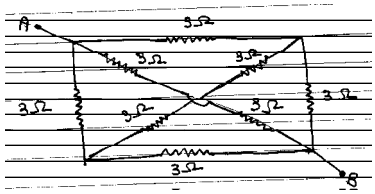
24. Calculate net resistance beween A and B



(A) 10 Ω (B) 30 Ω

(C) 25 Ω (D) $\frac{1}{25}$ Ω

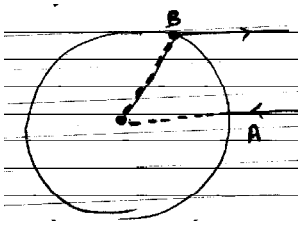
25. Calculate net resistance beween A and B



(A) 3 Ω (B) $\frac{1}{3}$ Ω

(C) 2 Ω (D) $\frac{1}{2}$ Ω

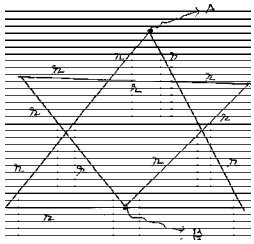
26. A wire in a circular shape has 10 Ω resistance. The resistance per one meter is 1Ω The resultant between A & B is equal to 2.4Ω, then the length of the chord AB will be equal to



(A) 2.4 (B) 4

(C) 4.8 (D) 6

27. Calculate net resistance beween A and B



(A) $\frac{6r}{7}$ (B) $\frac{3r}{4}$

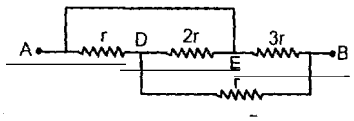
(C) $\frac{7r}{6}$ (D) $\frac{4r}{3}$

28. Area of cross-section of a copper wire is equal to area of a squre of 2mm length It carries a current of 8A Find drift velocity of electrons (Density of free electrons in copper = 8×10^{28} m^{-3})

(A) 1.56×10^{-2} ms^{-1} (B) 1.56×10^{-4} ms^{-1}

(C) 3.12×10^{-2} ms^{-1} (D) 3.12×10^{-3} ms^{-1}

29. What is the equivalent resistance across the teminals A and B?



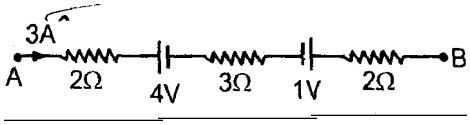
(A) $\frac{15r}{7}$ (B) $\frac{7r}{15}$

(C) $\frac{15r}{14}$ (D) $\frac{8r}{15}$

30. Two batteries each of emf 2V and internal resistance 1 Ω are connected in series to a resistor R. Maximum Possible power consumed by the resistor = ...

(A) 3.2 W (B) $\frac{16}{9}$ W (C) $\frac{8}{9}$ W (D) 2W

31. What is the p.d between the terminals A and B?



(A) 12 V (B) 24 V

(C) 36 V (D) 48V

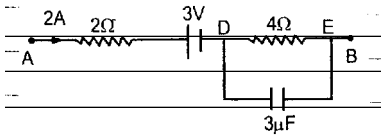
32. In an experiment to measure the intenal resistance of a cell by a potentiometer it is found that all the balance points at a length of 2m when the cell is shunted by a 5 ohm resistence and is at a length of 3m when the cell is shunted by a 10 ohm resistance, the internal resistance of the cell is then :

(A) 1.5 Ω (B) 10 Ω (C) 15 Ω (D) 1 Ω

33. Two wires of the metal have the same length but their cross-sections are in the ratio 3:1 They are joined in series: The resistance of the thicker wire is 10Ω. The total tesistance of the combination will be

(A) 40 (B) $\frac{40}{3}$ (C) $\frac{5}{2}$ (D) 100

34. What is the enrergy stored in the capacitor?



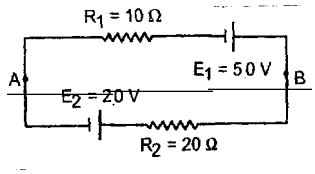
(A) 72 μJ (B) 96 μJ

(C) 96 mJ (D) 96 MJ

35. A wire of length L is drawn such that its diameter is reduced to half of its original diameter. If the resistance of the wire were 10Ω, its new resistance would be.

(A) 40 Ω (B) 60 Ω (C) 120 Ω (D) 160 Ω

36. In the circuit shown, the current sources are of negligible internal resistances. What is the potential difference between the points B and A ?



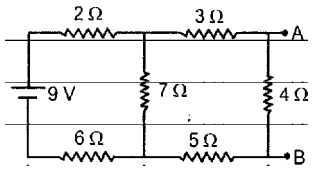
(A) –4.0 V (B) 4.0 V

(C) –8.0 V (D) 8.0V

37. Which of the following has negative temperature coefficient of resistance?

(A) Fe (B) C (C) Mn (D) Ag

38. what is the potential aross the points A and B?



(A) 0.9 V (B) 1.1 V

(C) 1.3 V (D) 0.7 V

39. A wire 50cm long and 1 mm^2 in cross-section carries a curent of 4A when connected to a 2V battery. The resistivity of the wire is:

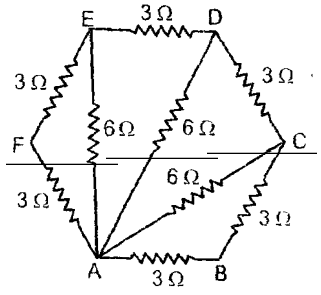
(A) 2×10^{-7} Ω m (B) 5×10^{-7} Ω m

(C) 4×10^{-6} Ω m (D) 1×10^{-6} Ω m

40. A parallel combination of three resistors takes a current of 7.5 A form a 30 V supply, It the two resistors are 10Ω and 12Ω find which is the third one?

(A) 4 Ω (B) 15 Ω (C) 12 Ω (D) 22 Ω

41. Six resistors of 3Ω each are connected along the sides of a hexagon and three resistors of 6Ω each are connected along AC, AD, and AE as shown in the figiure. The equivalent resistance between A and B is equal to:



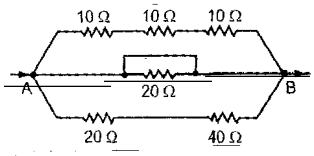
(A) 3 Ω

(B) 9 Ω

(C) 2 Ω

(D) 16 Ω

42. The total electrial resistance between the points A and B for the circuit figure shown below is:



(A) 0 Ω (B) 15 Ω

(C) 30 Ω (D) 100 Ω

43. A potentiometer wire has length 10 m and resitance 20Ω. A 2.5V battery of negligble internal resistance is connected across the wire with an 80Ω series resistance. The potential gradient on the wire will be:

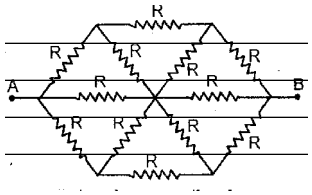
(A) 2.5×10^{-4} V/cm (B) 0.62×10^{-4} V/mm

(C) 1×10^{-5} V/mm (D) 5×10^{-5} V/mm

44. The drift velocity of free electrons through a conducting wire of radius r, carrying current I, is if the same current is passed through a conductor of radius 2r what will be the drift velocity?

(A) $\frac{V_d}{4}$ (B) V_d (C) $2V_d$ (D) $24V_d$

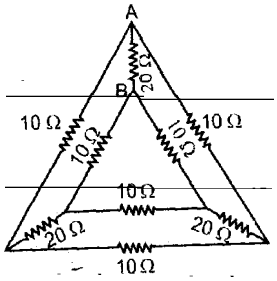
45. Net resistance between A and B in the given network is ...



(A) $\frac{5R}{7}$ (B) $\frac{7R}{6}$

(C) $\frac{4R}{5}$ (D) $\frac{6R}{7}$

46. Net resistance between A and B in the given network is:



(A) 10Ω (B) 40Ω

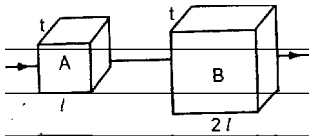
(C) $\frac{40}{7}\Omega$ (D) $\frac{60}{7}\Omega$

47. A carbon resistor has a set of coaxial coloured rings in the order brown, violet brown and silver. The value of resistance (in ohms) is.

(A) $(27 \times 10) \pm 5\%$ (B) $(27 \times 10) \pm 10\%$

(C) $(17 \times 10) \pm 5\%$ (D) $(17 \times 10) \pm 10\%$

48. The figure shows two metal plates A and B which are square in shape and have same thickness t. The side of B is twice that of A. Current flows through them in the direction as shown by the arrow marks. The ratio of resistance of A to that of B is.



(A) 1:2 (B) 2:1

(C) 1:1 (D) 1:4

49. A cross a wire of length l and thickness d, a p.d of V is applied. If the p.d is doubled the dirft velocity becomes....

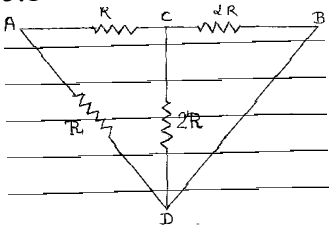
(A) becomes double (B) becomes half

(C) Close not change (D) becomes Zero

50. The masses of three wires of copper are in the ratio of 1:3:5 and their lengths are in the ratio of 5:3:1. The ratio of their electrical resistance is:

(A) 1:1:1 (B) 1:3:5 (C) 5:3:1 (D) 125:15:1

51. The effective resistance between points A and B is....



(A) R (B) $\frac{R}{3}$

(C) $2\frac{R}{3}$ (D) $3\frac{R}{5}$

52. Two resistors when connected in parallel have an equivalent of 2Ω and when in series of 9Ω The values of the two resistors are.

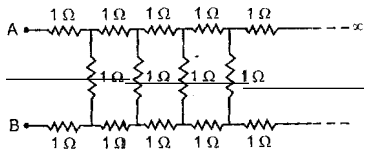
(A) 2Ω and 9Ω (B) 3Ω and 6Ω (C) 4Ω and 5Ω (D) 2Ω and7Ω

53. Which is the dimensional formula for condutance from the give below?

(A) $M^1L^2T^{-3}A^2$ (B) $M^1L^{-2}T^{-3}A^2$

(C) $M^1L^{-3}T^{-3}A^{-2}$ (D) $M^1L^{-3}T^3A^2$

54. The given figure shows an infinite ladder network of resistances The equivalent resistance between points A and B is.



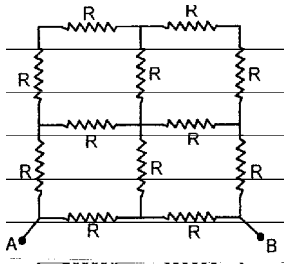
(A) Infinite (B) 3.73Ω

(C) 2.73 Ω (D) $\frac{2}{3}\Omega$

55. Resistivity of material of a conducting wire is 4×10^{-8} Ωm volume of the wire is $4m^3$ and its resistance is 4Ω Therefore its length will be.

(A) 500 m (B) 5000 m (C) 20,000 m (D) 4×10^{-5} m

56. Net resistance between A and B in the given network is:



(A) $\frac{5R}{7}$ (B) $\frac{7R}{6}$

(C) $\frac{4R}{5}$ (D) $\frac{5R}{4}$

57. How would you arrange 48 cells each of e.m.f 2V and inteanal resistance 1.5Ω so as to pass maximum current through the external resistance of 2Ω?

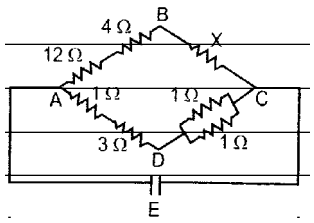
(A) 2 cells in 24 grounps (B) 4 cells in 12 groups

(C) 8 cells in 6 groups (D) 3 cells in 16 groups

58. How many dry cells, each of emf 1.5V and internal resistance 0.5Ω, much be joined in series with a resistor of 20Ω to give a current of 0.6A in the circuit ?

(A) 2 (B) 8 (C) 10 (D) 12

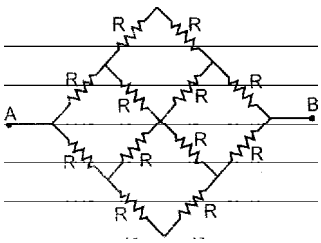
59. In the arrangement of resistances shown in the figure, the potential differenc between B and D will be zero when the unknown resistance X is



(A) 4Ω (B) 2Ω

(C) 3Ω (D) 6Ω

60. Net resistance between A and B in the given network is:



(A) $5\frac{R}{7}$ (B) $7\frac{R}{6}$

(C) $3\frac{R}{2}$ (D) $5\frac{R}{4}$

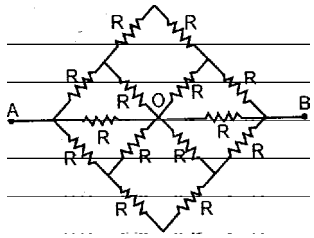
61. Two electirc bulbs whose resistances are in the ratio of 1:2 are connected in parallel to a constant voltage source the power dissipated in them have the ratio.

(A) 1:2 (B) 1:1 (C) 2:1 (D) 1:4

62. If the above two bulbs are connected in series, the power dissipatd in them have the ratio:

(A) 1:2 (B) 1:1 (C) 2:1 (D) 1:4

63. Net resistance between A and B in the given network is....



(A) $5\ \frac{R}{7}$ (B) $7\ \frac{R}{6}$

(C) $\frac{4R}{5}$ (D) $\frac{6R}{7}$

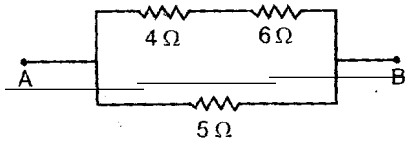
64. An electric kettle has two coils. when onc of them is switched on, the water in the kettle boils in 6 minutes. When the other coil is switched on, the water boils in 3 minutes If the two coils are connected in series the time taken to boil water in the kettle is:

(A) 3 minutes (B) 6 minutes (C) 2 minutes (D) 9 minutes

65. An electric kettle has two coils when one of these is switched on. the water in the kettle boils in 6 minutes. When the other coil is switched on, boils in 3 minutes If the two coils are connected in parallel, the time taken to boil water in the kettle is.

(A) 3 minutes (B) 6 minutes (C) 2 minutes (D) 9 minutes

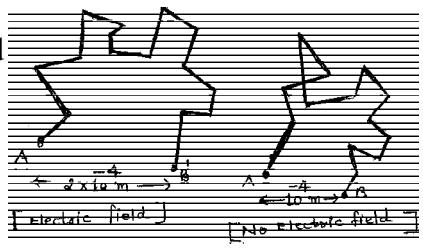
66. In the circuit shown, the heat produced in the 5 ohm resistor due to current flowing through it, is 10 calories per second. Then the heat generated in the 4 ohm resistor is:



(A) 1 calorie per sec (B) 2 colorie per sec

(C) 4 calorie per sec (D) 3 calorie per sec

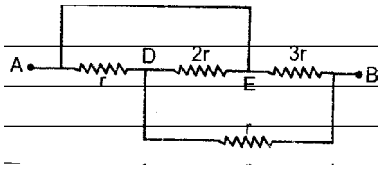
67. The figures below show the motion of electron is the absence and presence of eletric field in 10 sec. The drift velocity is....



(A) $10^{-5}\ ms^{-1}$ (B) $2 \times 10^{-5}\ ms^{-1}$

(C) $2 \times 10^{-4}\ ms^{-1}$ (D) $10^{-4}\ ms^{-1}$

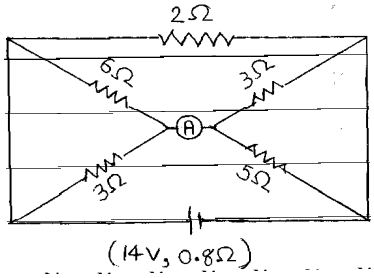
68. What is three equivelent resistance areoss the terminals A and B?



(A) $\frac{15r}{7}$ (B) $\frac{7r}{15}$

(C) $\frac{15r}{14}$ (D) $\frac{8r}{15}$

69. The reading of ammeter shown in figure is....



(A) 2.18 A (B) 3.28 A

(C) 6.56 A (D) 1.09 A

70. The potential difference between the terminals of a battery is 10V and internal resistance 1Ω drops to 8V when connected across an external resistor find the resistance of the external resistor.

(A) 40 Ω (B) 0.4 Ω (C) 4M Ω (D) 4 Ω

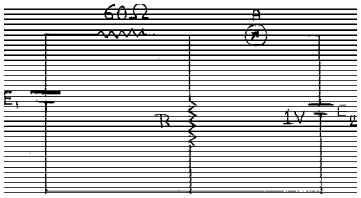
71. Two heater wires of equal length are first connected in sreies and then in parallel The ratio of heat produced in the two cases is....

(A) 2:1 (B) 1:2 (C) 4:1 (D) 1:4

72. A heater boils 1kg of water in time t_1 and another heater boils the same water in time t_2 If both are connected in series, the combination will boil the same water in time.

(A) $\frac{t_1 t_2}{t_2 + t_1}$ (B) $\frac{t_1 t_2}{t_1 - t_2}$ (C) $t_1 + t_2$ (D) $2(t_1 + t_2)$

73. In the circuit shown in fig, the ammeter A reads zero, If the batteries have negligible internal resistance, the value of R is.



(A) 20 Ω (B) 10 Ω

(C) 30 Ω (D) 40 Ω

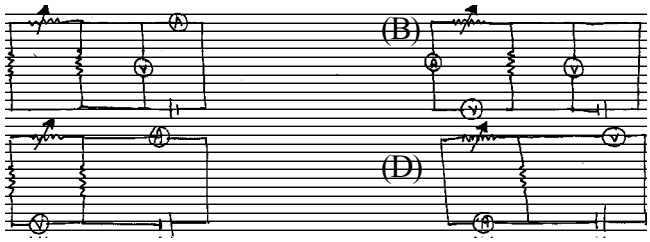
74. The drift velocity of free electrons in a conductor is v, when a current. I is flowing in it If both the radius and current are doubled, then drift velocity will be.

(A) $\frac{V}{4}$ (B) $\frac{V}{2}$ (C) 4V (D) 2V

75. Which of the follwing set up can be uesd to verify the ohm's law?

(A) (B)

(C) (D)



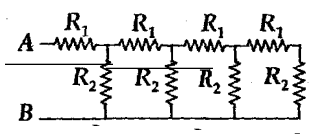
76. At what tempreature will the resistance of a copper wire be three times its value at $0^\circ C$? (Given: temerature coefficient of resistance for copper = $4 \times 10^{-3}\ ^\circ C^{-1}$)

(A) $400^\circ C$ (B) $450^\circ C$ (C) $500^\circ C$ (D) $550^\circ C$

77. The resistance of a coppre coil is 4.64Ω at 40° C and 5.6Ω at 100° C Its resistcnce at 0° C will be

(A) 5 Ω (B) 4 Ω (C) 3 Ω (D) 2 Ω

78. A circuit with an infinite no of resistance is shown in fig. the resultant resistance between A and B, when $R_1 = 1\Omega$ and $R_2 = 2\Omega$ will be



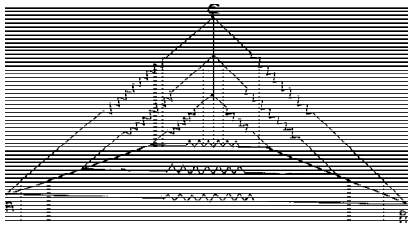
(A) 4 Ω (B) 1 Ω

(C) 2 Ω (D) 3 Ω

79. There are n resistors having equal value of resistance r. First they are connected in such a way that the possible minimum value of resistance is obtained. Then they are connected in such a way that possible maximum value of resistance is obtained the ratio of minimum and maximum values of resistances obtained in these way is....

(A) $\frac{1}{n}$ (B) n (C) n^2 (D) $\frac{1}{n^2}$

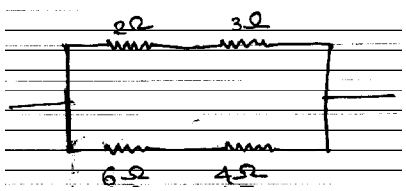
80. Nine resistors each of resistance R are connected as shown in fig. The effective resistance between A and B is.



(A) $\frac{7}{6}$ R (B) R

(C) $\frac{3}{5}$ R (D) $\frac{2}{9}$ R

81. 60 cal heat is produced per second in a 6Ω resistance on passing electric current through the circuit as shown in the figure. The amount of heat produced per second through 3Ω resistance is.... cal



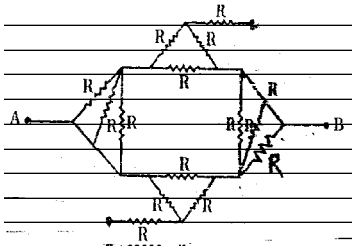
(A) 30 (B) 60

(C) 100 (D) 120

82. Temperature of a conductor increases by $5^{\circ}C$ passing electric current for some time. The increase in its temperature when double current is passed through the same conductor for the same time is.... $^{\circ}C$

(A) 10 (B) 12 (C) 16 (D) 20

83. Find equivalent resistance between A and B



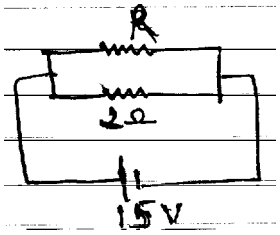
(A) R (B) $\frac{3R}{4}$

(C) $\frac{R}{2}$ (D) 2R

84. Area of cross-section of two wires of same length carrying same current is in the ratio of 1 : 2. Then the ratio of heat generated per second in the wires = ...

(A) 1: $\sqrt{2}$ (B) 1:1 (C) 1:4 (D) 2:1

85. In given circuit total power consumed is 150W. Then value of R =....



(A) 2 Ω (B) 6 Ω

(C) 5 Ω (D) 4 Ω

86. If σ_1, σ_2, and σ_3 are the coductances of three conductor then equivalent conductance when they are joined in series, will be.

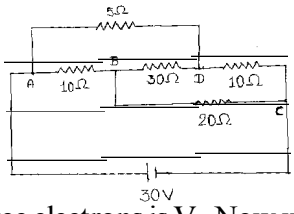
(A) $\sigma_1 + \sigma_2 + \sigma_3$ (B) $\frac{1}{\sigma_1} + \frac{1}{\sigma_2} + \frac{1}{\sigma_3}$

(C) $\frac{\sigma_1 \sigma_2 \sigma_3}{\sigma_1 + \sigma_2 + \sigma_3}$ (D) None of these.

87. Formula for current flowing through a wire is $I = 6t^2 + 4t - 2$ here t is in second and I is an ampere. In this wire, what is the quantity of electric charge passing in time interval 1 sec to 2 sec?

(A) 8C (B) 18C (C) 20C (D) 24C

88. What is current flowing through 5Ω resistor in the circuit given below?



(A) 1 A (B) 2 A
(C) 3A (D) 4A

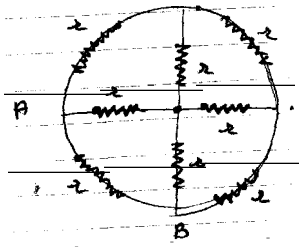
89. when current flowing through a conductor is I, average drift velocity of free electrons is V_d. Now when 6I current is flowing through a conductor having 3 times cross sectional are of same material what will be average dirft velocity of free electrons?

(A) $2\ V_d$ (B) $\frac{Vd}{2}$ (C) $3\ V_d$ (D) $18\ V_d$

90. n resistors each of resistance r are connected to a battery of emf E and intrnal resistance r. Then the ratio of terminal voltage to emf of battery =....

(A) n (B) $\frac{n}{n+1}$ (C) $\frac{1}{n+1}$ (D) $\frac{n+1}{n}$

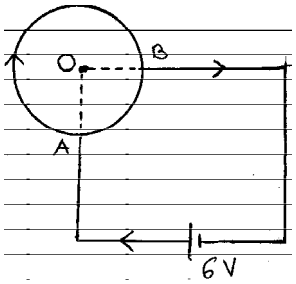
91. In a given circuit, resistance of each resistor is r. Then equivalent resistance between A and B=....



(A) $\frac{3}{4}$ r (B) $\frac{2}{3}$ r

(C) $\frac{8}{15}$ r (D) $\frac{8}{7}$ r

92. A wire is bent in the form of a circle of radius 4m Resistance per unit length of wire is $\frac{1}{\pi}\ \frac{\Omega}{m}$ battery of 6V is connected between A and B $\angle AOB = 90^\circ$ Find the curent through the battery



(A) 8A (B) 4A
(C) 3A (D) 9A

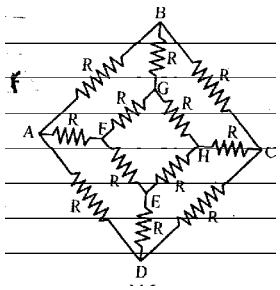
93. Masses of three conductors of same material are in the proportion of 1:2:3 their lengths are in the proportion of 3:2:1 then their resistance will be in the proportion of....

(A) 1:1:1 (B) 1:2:3 (C) 9:4:1 (D) 27:6:1

94. Resistance of a wire at 50° C is 5Ω, and at 100° C it is 6Ω find its resistance at 0° C

(A) $4\ \Omega$ (B) $3\ \Omega$ (C) $2\ \Omega$ (D) $1\ \Omega$

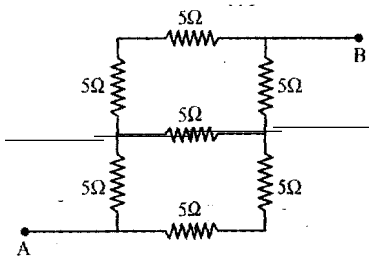
95. Twelve resistance cach of resistance R are connected in the circuit as shown in figure. Net resistance between points A and C would be.



(A) $\frac{6R}{3}$ (B) $\frac{7R}{6}$

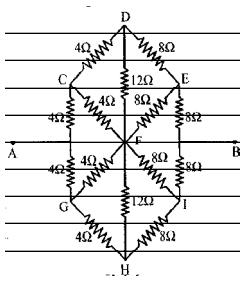
(C) R (D) $\frac{3R}{4}$

96. In the figure all the seven resistances joined in the circuit have a value of 5Ω each the equivalent resistance of AB is....



(A) $35\ \Omega$ (B) $25\ \Omega$
(C) $7\ \Omega$ (D) $15\ \Omega$

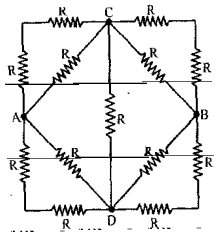
97. The effective resistance between the points A and B in the given network shown in figure will be.



(A) 9 Ω (B) 12 Ω

(C) 18 Ω (D) 7.5 Ω

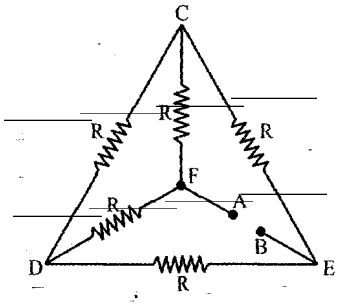
98. Thirteen resistances each of resistance $R\Omega$ are connected in the circuit as shown in the figure the effective resistance between A and B is....



(A) (2R) Ω (B) $\left(\frac{4R}{3}\right)$ Ω

(C) $\left(\frac{2R}{3}\right)$ Ω (D) R

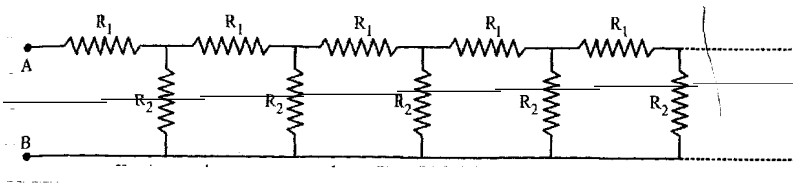
99. Five equal resistances each of resistances R are connected as shown in the figure A battery of V volt is connected between A and B. The current flowing in AFCEB will be.



(A) $\frac{3V}{R}$ (B) $\frac{V}{R}$

(C) $\frac{V}{2R}$ (D) $\frac{2V}{R}$

100. An infinit sequence of resistances is shown in the figure. The resultant resistance between A and B will be, when $R_1 = 1$ ohm and $R_2 = 2$ ohm



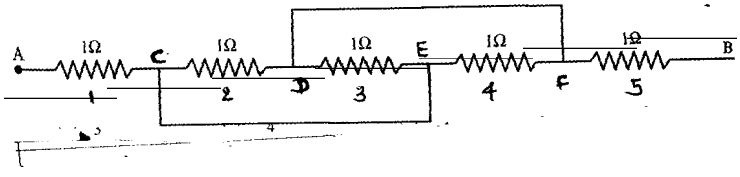
(A) 3 Ω (B) 2 Ω

(C) 1 Ω (D) 1.5 Ω

101. Two wires of equal dimeters of resistivities ρ_1 and ρ_2 are joined in series. The equivalent resistivety of the combination is....

(A) $\frac{\rho_1\ell_1+\rho_2\ell_2}{\ell_1+\ell_2}$ (B) $\frac{\rho_1\ell_2+\rho_2\ell_1}{\ell_1-\ell_2}$ (C) $\frac{\rho_1\ell_2+\rho_2\ell_1}{\ell_1+\ell_2}$ (D) $\frac{\rho_1\ell_1-\rho_2\ell_2}{\ell_1-\ell_2}$

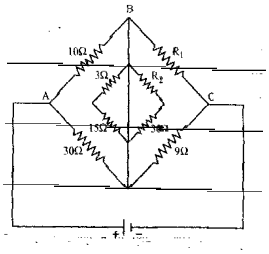
102. Equivalent resistance between the points A and B is (in Ω)



(A) $\frac{1}{5}$ Ω (B) $1\frac{1}{4}$ Ω

(C) $2\frac{1}{3}\Omega$ (D) $3\frac{1}{2}$ Ω

103. In the wheastone bridge shown below, in order to balance the bridge we must have



(A) $R_1 = 3\ \Omega,\ R_2 = 3\ \Omega$

(B) $R_1 = 6\ \Omega,\ R_2 = 15\ \Omega$

(C) $R_1 = 1.5\ \Omega,\ R_2$ = any finite value

(D) $R_1 = 3\ \Omega,\ R_2$ = any finite value

104. A bulb of 300W and 220V is connected with a source of 110V. What is the % decrease in power?

(A) 100.% (B) 75 % (C) 70 % (D) 25 %

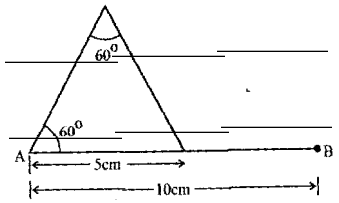
105. Length of a heating filament is reduced by 20% its power will....

(A) decrease by 20% (B) Increaseby 20% (C) Increase by 25% (D) Incease by 40%

106. What maximum power can be obtained from a battery of emf ε and internal resistance r connected with an external resistance R?

(A) $\frac{\varepsilon^2}{4r}$ (B) $\frac{\varepsilon^2}{3r}$ (C) $\frac{\varepsilon^2}{2r}$ (D) $\frac{\varepsilon^2}{r}$

107. A wire has resistance of 24Ω is bent in the following shape. The effective resistance between A and B is



(A) 24 Ω (B) 10 Ω

(C) $\frac{16}{3}\Omega$ (D) None of these

108. The tungsten filament of bulb has resistance equal to 18Ω at 27° C tempreature 0.25 A of current flows, when 45V is connected to it If $\alpha = 4.5\times10^{-3}\,K^{-1}$ for a tungsten then find the temperature of the filament.

(A) 2160 K (B) 1800.K (C) 2070 K (D) 2300 K

109. The resistance of the wire made of silver at 27° C temperature is equal to 2.1Ω while at 100° C it is 2.7Ω calculate the temprature eoefficient of the resistivity of silver. Take the reference temperature equal to 20° C

(A) $4.02 \times 10^{-3}\ {}^{o}C^{-1}$ (B) $0.402 \times 10^{-3}\ {}^{o}C^{-1}$

(C) $40.2 \times 10^{-4}\ {}^{o}C^{-1}$ (D) $4.02 \times 10^{-4}\ {}^{o}C^{-1}$

110. The temperature co-efficient of resistance of a wire is $0.00125^{\circ}k^{-1}$ Its resistance is 1Ω at 300K. Its resistance will be 2Ω at.

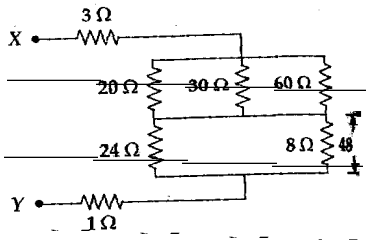
(A) 1400 K (B) 1200.K (C) 1000 K (D) 800 K

111. Two resistances R_1 and R_2 have effective resistance R_s when connected in sries combination and R_p when connected in parallel combination if $R_s R_p = 16$ and $R_1/R_2 = 4$ the values of R_1 and R_2 are

(A) 2 Ω and 0.5 Ω (B) 1 Ω and 0.25 Ω

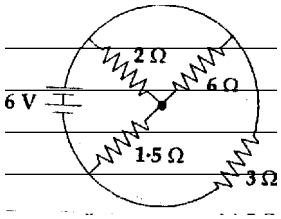
(C) 8 Ω and 2 Ω (D) 4 Ω and 1 Ω

112. The potential difference across 8Ω resistance is 48V as shown in the figure. The value of potential differences across X and Y will be.



(A) 180 V (B) 160 V
(C) 140 V (D) 120 V

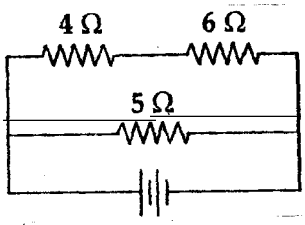
113. The total current supplied to the circuit by the battery is....



(A) 1 A (B) 2 A
(C) 4 A (D) 6A

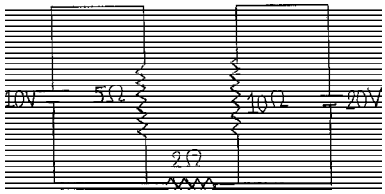
114. Three identical resistors connected in series with a battery, together dissipate 10W of power. What will be the power dissipated, if the same resistors are connected in parallel across the same battery?
(A) 60W (B) 30 W (C) 90 W (D) 120 W

115. If power dissipated in 5Ω resistor in 20W, then power dissipated across 4Ω resistor will be.



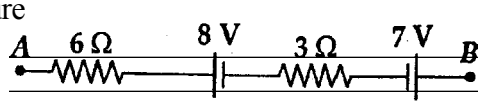
(A) 1 W (B) 2 W
(C) 3 W (D) 4 W

116. In the given curcuit, the value of current through 2Ω resistor is....



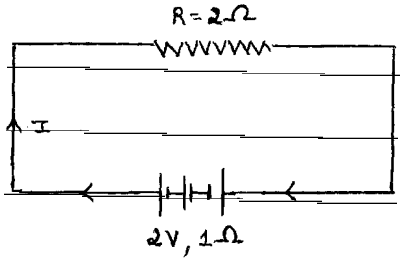
(A) 2 A (B) 4 A
(C) Zero (D) 5A

117. A curent of 3A flows from A to B through the wire shown in figure If the potential at A is 45V, then the potential at B will be.



(A) 17 V (B) 9 V
(C) 12 V (D) 6 V

118. In the electric circuit shown, each cell has an e.m.f of 2V and internal resistance of 1Ω. The external resistance is 2Ω the value of the current (I) is.

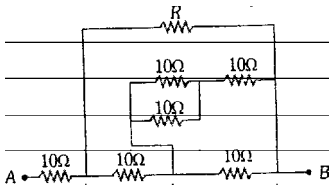


(A) 0.8 A (B) 0.6 A
(C) 0.4 A (D) 0.1A

119. A potentiometer wire of length 1 m and resistance 10Ω is connected in series with a cell of e.m.f 2V with internal resistance 1Ω and a resistance box of a resistance R if potential difference between ends of the wire is 1V the value of R is.
(A) 4.5 Ω (B) 9 Ω (C) 15 Ω (D) 20 Ω

120. For a cell of e.m.f 2V, a balance is obtained for 50 cm of the potentiometer wire If the cell is shunted by a 2Ω resistor and the balance is obtained across 40 cm of the wire, then the internal resistance of the cell is.
(A) 1 Ω (B) 0.5 Ω (C) 1.2 Ω (D) 2.5 Ω

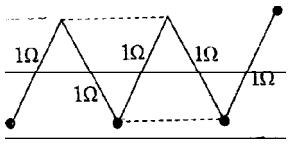
121. For what value of R the net resistance of the circuit will be 18 ohms.



(A) 8 Ω (B) 10 Ω

(C) 16 Ω (D) 24 Ω

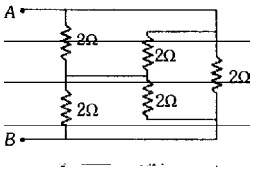
122. A circuit consists of five identical conductors as shown in figure the two similar conductors are added as indicated by the dotted lines. The ratio of resistances before and after addition will be....



(A) $\frac{7}{5}$ (B) $\frac{3}{5}$

(C) $\frac{5}{3}$ (D) $\frac{6}{5}$

123. Find the equivalent resistance a cross AB



(A) 1 Ω (B) 2 Ω

(C) 3 Ω (D) 4 Ω

124. n idential cells each of e.m.f E and internal resistance r are connected in series An external resistance R is connected in series to this combination the current through R is.

(A) $\frac{nE}{R+nr}$ (B) $\frac{nE}{nR+r}$ (C) $\frac{E}{R+nr}$ (D) $\frac{nE}{R+r}$

125. 4 cell each of emf 2v and internal resistance of 1Ω are connected in parallel to a load resistor of 2Ω Then the current through the load resistor is....

(A) 2A (B) 1.5 A (C) 1A (D) 0.888A

126. A Potentiometer wire, 10m long, has a resistance of 40Ω It is connected in series with a resislance box and a 2V storage cell If the potential gradient along the wire is 0.1mv/cm, the resistance unplugged in the box is.

(A) 260 Ω (B) 760 Ω (C) 960 Ω (D) 1060 Ω

127. The resistivity of a potentiometer wire is 40×10^{-8} ohm m and its area of cross-section is $8\times10^{-6}m^2$ If 0.2 amp current is flowing through the wire, the potential gradient will be.

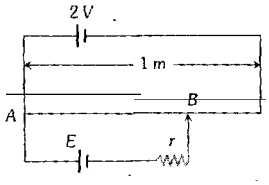
(A) 10^{-2} Volt / m (B) 10^{-1} Volt / m

(C) 3.2×10^{-2} Volt / m (D) 1 Volt / m

128. Potentiometer wire of length 1m is connected in series with 490Ω resistance and 2V battery If 0.2 mv/cm is the potential gradient, then resistance of the potentiometer wire is.

(A) 4.9 Ω (B) 7.9 Ω (C) 5.9 Ω (D) 6.9 Ω

129. In the given figure, battery E is balanced on 55 cm length of potentiometer wire but when a resistance of 10Ω is connected in parallel with the battery then it balances on 50cm length of the potenitometer wire then internal resistance r of the battery is.



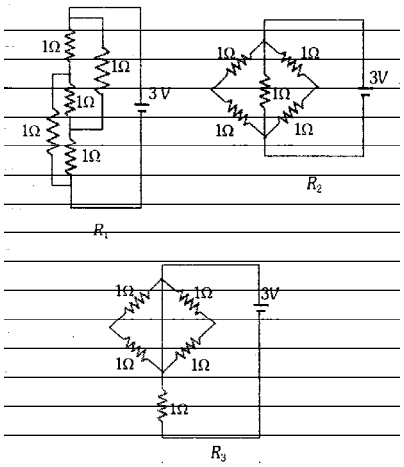
(A) 1 Ω (B) 3 Ω

(C) 10 Ω (D) 5 Ω

130. Figure shows three resistor configurations R_1, R_2 and R_3 connected to 3V battery If the power dissipated by the configuration R_1, R_2 and R_3 is P_1, P_2 and P_3 respectively then

(A) $P_1 > P_2 > P_3$ (B) $P_1 > P_3 > P_2$

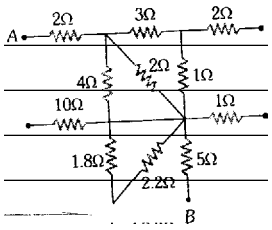
(C) $P_2 > P_1 > P_3$ (D) $P_3 > P_1 > P_1$



131. What is the equivalent resistance between the points A and B of the network

(A) $\frac{57\Omega}{7}$ (B) 8Ω

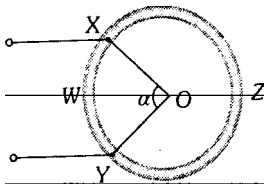
(C) 6Ω (D) $\frac{57\Omega}{5}$



132. A wire of resistor R is bent into a circular ring a circular ring of radius r Equivalent resistance between two points X and Y on its circumference, when angle xoy is α, can be given by

(A) $\frac{R\alpha}{4\pi^2}(2\pi - \alpha)$ (C) $R(2\pi - \alpha)$

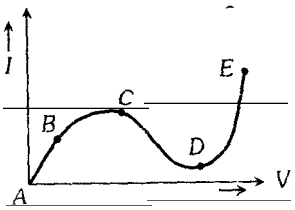
(B) $\frac{R}{2\pi}(2\pi - \alpha)$ (D) $\frac{4\pi}{R\alpha}(2\pi - \alpha)$



133. Form the graph between current I and voltage V shown below, identity the portion corresponding to negative resistance.

(A) AB (B) BC

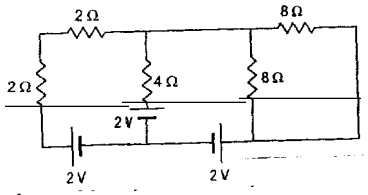
(C) CD (D) DE



134. Find the current in the differrent resistors shown in fingure

(A) Zero (B) 2 Amp.

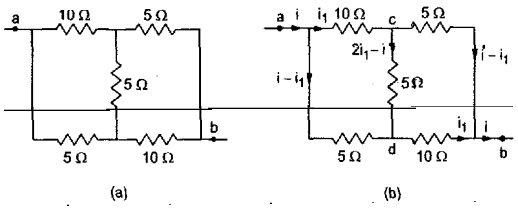
(C) 2.2 Amp. (D) 4 Amp.



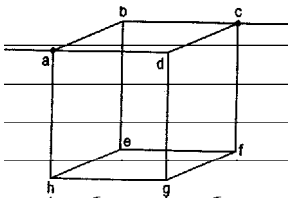
135. Find the equivalent resistance between the points a and b of the circuit shown in figure.

(A) 7 Ω (B) 9 Ω

(C) 7.5 Ω (D) 5 Ω



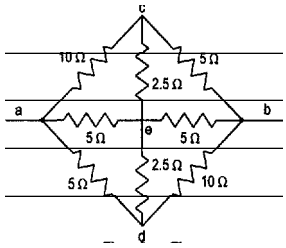
136. Twelve wires, each having resistance r, are joined to form a cube as shown in figure find the equivalent resistance between the ends of a face diagonal such as a and c.



(A) $\frac{5r}{7}$ (B) $\frac{3r}{4}$

(C) $\frac{12r}{7}$ (D) $\frac{7r}{12}$

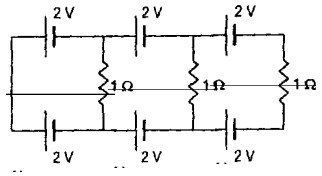
137. Find the equivalent resistance of the network shown in figure between the points a and b.



(A) 4.1 (B) 1.4

(C) 2 (D) 4

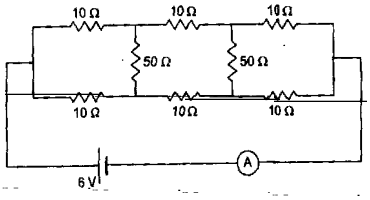
138. Find the current in the three resistors shown in figure.



(A) Zero (B) 2 Amp.

(C) 1 Amp. (D) 4 Amp.

139. Find the current measureal by the ammeter in the circuit shown in figure.



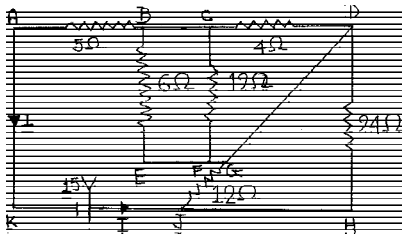
(A) Zero (B) 0.4 A.

(C) 4 A (D) 2 A

140. A and B are two points on a uniform ring of resistance R the $\angle AOB = \theta_3$ where O is the centre of the ring The equivent resistance between A and B is.

(A) $\frac{R\theta}{2\pi}$ (B) $\frac{R(2\pi - \theta)}{4\pi}$ (C) $R\left(1 - \frac{\theta}{2\pi}\right)$ (D) $\frac{R}{4\pi^2}(2\pi - \theta)\theta$

141. The equivalent resistance between points A and J and current I in the following circuit will be.



(A) 15 Ω, 0.5 A (B) 15 Ω, 1 A

(C) 12 Ω, 0.5 A (D) 12 Ω, 1 A

142. A cell supplies a current I, through aresistance R_1 and a current I_2 through a resistance R_2 the internal resistance of a cell is....

(A) $R_2 - R_1$ (B) $\frac{(I_1 + I_2)}{I_1 - I_2} R_1R_2$ (C) $\frac{I_1R_2 - I_2R_1}{I_1 - I_2}$ (D) $\frac{I_2R_2 - I_1R_1}{I_1 - I_2}$

143. Two wires of resistances R_1 and R_2 have temperature coeffcient of resistances α_1 and α_2 respectively they are joined in series the effective tempercture coefficient of resistance is ....

(A) $\frac{\alpha_1 + \alpha_2}{2}$ (B) $\sqrt{\alpha_1 \alpha_2}$ (C) $\frac{\alpha_1R_1 + \alpha_2R_2}{R_1 + R_2}$ (D) $\frac{\sqrt{R_1R_2\alpha_1\alpha_2}}{\sqrt{R_1^2 + R_2^2}}$

144. The resistance of the series combination of two resistances is S, when they are joined in parallel the total resistance is P If S=n P, then the minimum possible valueofn is....

(A) 4 (B) 3 (C) 2 (D) 1

145. Two sources of equal emf are connected to an external resistance R the internal resistance of the two soureces are R_1 and R_2 $(R_2 > R_1)$ if the potential difference across the source having internal resistance R_2 is Zero, then

(A) $R = R_1R_2 / (R_2 - R_1)$ (B) $R = R_1R_2 / (R_1 + R_2)$

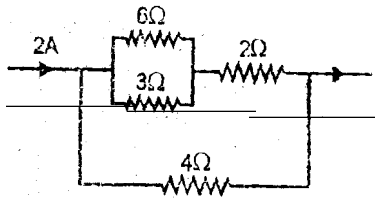
(C) $R = R_2 - R_1$ (D) $R = R_2 (R_1 + R_2) / (R_2 - R_1)$

146. In a wheatstone's bridge, three resistance P, Q and R connected in three are a and the fourth arm is formed by two resistances S_1 and S_2 connected in paralled The condifion for bridge to be balanced will be.

(A) $\frac{P}{Q} = \frac{R}{S_1 + S_2}$ (B) $\frac{P}{Q} = \frac{2R}{S_1 + S_2}$

(C) $\frac{P}{Q} = \frac{R (S_1 + S_2)}{\rho_1 S_2}$ (D) $\frac{P}{Q} = \frac{R (S_1 + S_2)}{2 S_1 S_2}$

147. In the circuit shown in fig the potential difference across 3Ω is.



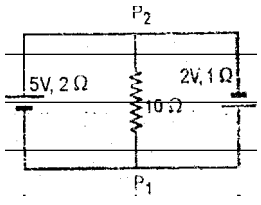
(A) 2 V (B) 4 V

(C) 8 V (D) 16 V

148. The resistance of a wire is 5Ω at $50^\circ C$ and 6Ω at $100^\circ C$ The resistance of the wire at $0^\circ C$ will be.

(A) 3 Ω (B) 2 (C) 1 Ω (D) 4 Ω

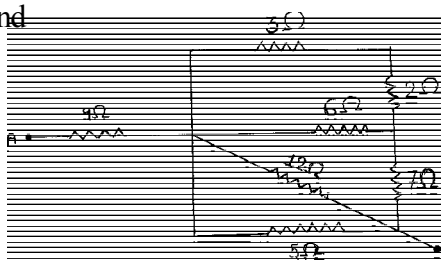
149. A 5V battery with internal resistance 2Ω and 2v battery with internal resistance 1Ω are connected to 10Ω resistor as shown in fig the current in 10Ω resistor is....



(A) 0.27 A, P_1 to P_2 (B) 0.27 A, P_2 to P_1

(C) 0.03 A, P_1 to P_2 (D) 0.03 A, P_2 to P_1

150. In the given circuit the equivalent resistance between the poins A and B in ohm is.



(A) 9 (B) 11.6

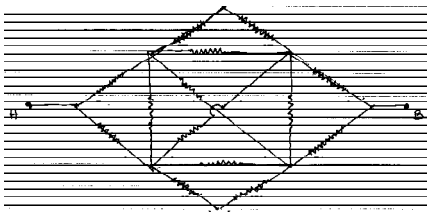
(C) 14.5 (D) 21.2

151. Resistors P and Q connected in the gaps of the meter bridge. the balancing point is obtained 1/3 m from the zero end If a 6Ω resistance is connected in series with p the balance point shifts to 2/3m form same end P and Q are.

(A) 4, 2 (B) 2, 4

(C) both (a) and (b) (D) neither (a) nor (b)

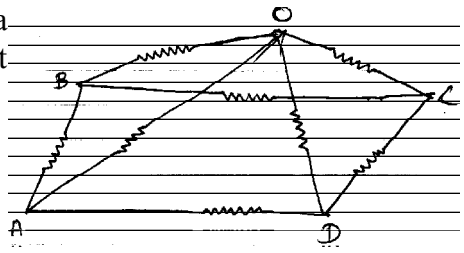
152. Fourteen identical resistors each of resistance r are connected as shown calculate equivalent resistance between A and B.



(A) 1.2 r (B) 2 r

(C) 2.1 r (D) r

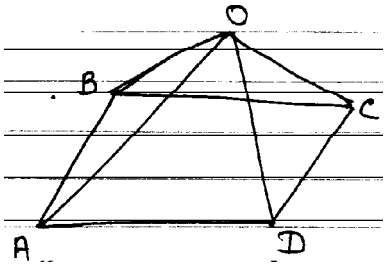
153. Eight identical resisitances r each are connected along edges of a pyramid having square base ABCD as shown calculate equivalent resistance between A and O.



(A) $\frac{15r}{7}$ (B) $\frac{7}{5r}$

(C) $\frac{7r}{15}$ (D) $\frac{5r}{7}$

154. Eight identical resistances r each are connected as shown find equivalent resistance between A and D



(A) $\frac{15r}{8}$ (B) $\frac{8r}{15}$

(C) $\frac{15}{7r}$ (D) $\frac{7r}{15}$

155. Two conductors have the same resistance at $0^\circ C$ but their temperture coefficients of resistonces are α_1 and α_2 The respective temperture coefficients of their series and parallel combinations are nearly....

(A) $\alpha_1 + \alpha_2 \times \frac{\alpha_1\alpha_2}{\alpha_1 + \alpha_2}$ (B) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$

(C) $\frac{\alpha_1 + \alpha_2}{2} . \alpha_1 + \alpha_2$ (D) $\alpha_1 + \alpha_2 . \frac{\alpha_1 + \alpha_2}{2}$

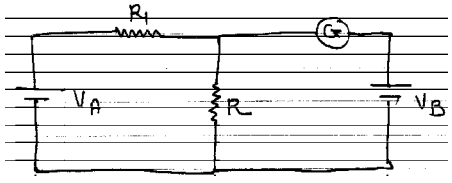
156. Two electric bulbs marked 25W-220V and 100W-220V are connected in series to a 440v supply which of the bulbs will fuse?

(A) 100 W (B) 25 W (C) None of this (D) Both

157. 2 A current is obtained when a 2Ω resistor is connectd with battery having r Ω as internal resistance 0.5A current is obtained if the above battery is connected to 9Ω resistor. Culculate the internal resistance of the battery.

(A) 0.5 Ω (B) $\frac{1}{3}\Omega$ (C) $\frac{1}{4}$ Ω (D) 1 Ω

158. Figure shown below the internal resistance of battery of A and B are negligible for $V_A = 12$ val, $R_1 = 500\Omega$ and $R = 100\Omega$ when the Galvenometer shows zero diflection then the value of V_B = ...



(A) 4 V (B) 2 V

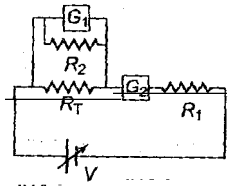
(C) 12 V (D) 6 V

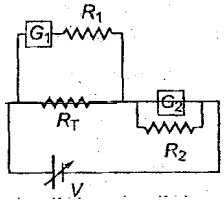
159. Incandescent bulbs are designed by keeping in mind that the resistance of their filament increases with the increase in temperature It at room temperature, 100w, 60w and 40w bulbs have filament resistances R_{100}, R_{60} and R_{40} respectively the relation between these resistances is

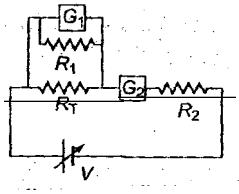
(A) $\frac{1}{R_{100}} = \frac{1}{R_{40}} + \frac{1}{R_{60}}$ (B) $R_{100} = R_{40} + R_{60}$

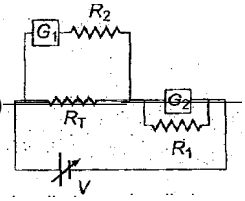
(C) $R_{100} > R_{40} > R_{60}$ (D) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

160. To verify ohm's law, a student is provided with a test resistor R_T a high resistance R_1 a small resistance R_2 two identical galvanometers G, and $G_2$and a variable voltage source V: the correct circuit to carry out the experiment is

(A) 

(B) 

(C) 

(D) 

In each of the follwing questions, match column i and column II and select the correct match out of the four given choices.

161.

	Column I		Column II
(a)	The series combination of cells is for	(p)	More current
(b)	The parallel combination of cell is for	(q)	More voltage
(c)	In series combination of n cells, each of emf ε the effective voltage is	(r)	ε
(d)	In parallel combination of n cells, each of emf ε the effective voltage is	(s)	$n\varepsilon$

(A) a - p, b - q, c - r, d - s (B) a - q, b - p, c - r, d - s

(C) a - q, b - p, c - s, d - r (D) a - p, b - q, c - s, d - r

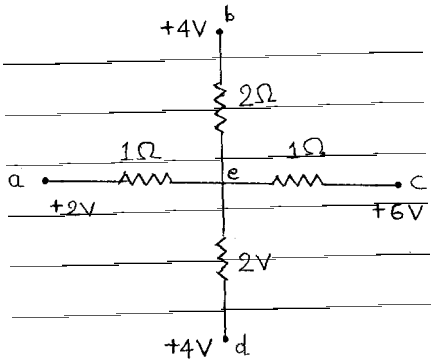
162.

	Column I		Column II
(a)	The unit of electrical resistivity is	(p)	$m^2S^{-1}V^{-1}$
(b)	The unit of current density is	(q)	$\Omega^{-1}m^{-1}$
(c)	The unit of electrical conductivity is	(r)	Am^{-2}
(d)	The unit of electric mobility is	(s)	Ωm

(A) a - p, b - q, c - r, d - s (B) a - s, b - r, c - q, d - p

(C) a - r, b - q, c - p, d - s (D) a - q, b - r, c - s, d - p

163. For the circuit shown in figure, match the two columns.

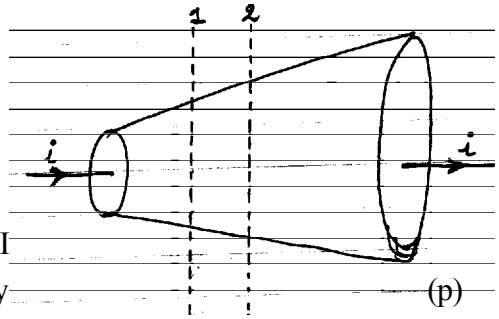


	Column I		Column II
(a)	Current in wire ae	(p)	1 A
(b)	Current in wire be	(q)	2 A
(c)	Current in wire ce	(r)	0.5Ω
(d)	Current in wire cle	(s)	None of these

(A) a - p, b - s, c - q, d - r (B) a - s, b - r, c - q, d - p

(C) a - q, b - s, c - q, d - s (D) a - s, b - q, c - p, d - r

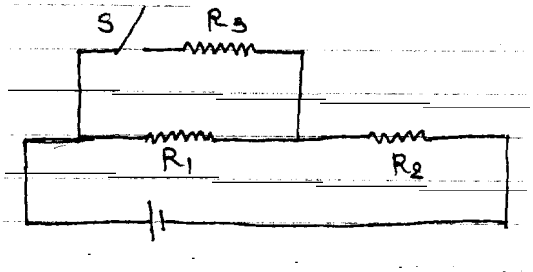
164. Current i is flowing through a wire of nonuniform cross section as shown match the follwing two columns.



Column I		Column II	
(a)	Current density	(p)	Is more at 1
(b)	Electric field	(q)	Is more at 2
(c)	Resistance per unit length	(r)	Is same at both sectinos l and 2
(d)	Potential cifference per unit length	(s)	elata insufficient

(A) a - p, b - p, c - p, d - p
(B) a - q, b - r, c - s, d - p
(C) a - q, b - q, c - p, d - p
(D) a - p, b - q, c - r, d - s

165. In the circuit shown in figure, after closing the switch S, match the follwing two columns.



Column I		Column II	
(a)	Current through R_1	(p)	Will increase
(b)	Current through R_2	(q)	Will decrease
(c)	Potential difference across R_1	(r)	Will remain same
(d)	Potential difference across R_2	(s)	Data insufficient

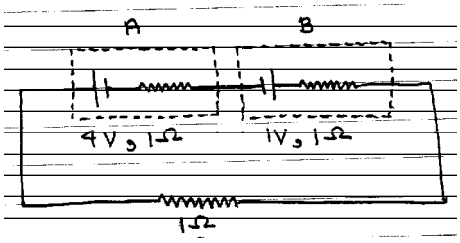
(A) a – p, b – q, c – r, d – s
(B) a – p, b – r, c – s, d – q
(C) a – p, b – r, c – s, d – q
(D) a – q, b – p, c – q, d – p

166. Match the following two columns.

Column I		Column II	
(a)	Electrical resistance	(p)	$[M\,L\,T^{-2}A^{2}]$
(b)	Electric potential	(q)	$[M\,L^{2}T^{-3}A^{-2}]$
(c)	Specific resistance	(r)	$[M\,L^{2}T^{-3}A^{-1}]$
(d)	Specific conductance	(s)	None of these.

(A) a – q, b – s, c – r, d – p
(B) a – q, b – r, c – s, d – s
(C) a – p, b – q, c – s, d – r
(D) a – p, b – r, c – q, d – s

167. In the circuit show in figure, match the following two colums:-



Column I		Column II	
(a)	Potential difference across battery A	(p)	Zero
(b)	Potential difference across battery B	(q)	1
(c)	net power supplied/consumed by A	(r)	2
(d)	net power supplied/consumed by B	(s)	3

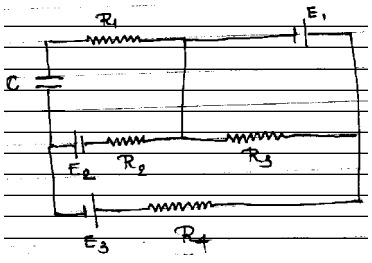
(A) a – p, b – q, c – r, d – s (B) a – s, b – q, c – r, d – p
(C) a – s, b – r, c – s, d – r (D) a – q, b – r, c – s, d – s

168. Match the physical quantities given in column I with their dimensional formulae given in column II -I stands for the dimesion of current.

Column I		Column II	
(a)	Electromotive force (emf)	(p)	$M \cdot L^2T^{-3}A^{-2}$
(b)	Resistance	(q)	$M\ L^3T^{-3}A^{-2}$
(c)	Resistivity	(r)	$M^{-1}L^{-3}T^3A^2$
(d)	Conductivity	(s)	$M\ L^2T^{-3}A^{-1}$

(A) a – s, b – p, c – q, d – r (B) a – p, b – s, c – r, d – q
(C) a – p, b – s, c – r, d – q (D) a – r, b – p, c – q, d – s

- **Questions 169 to 181 are based on the following passage.**



Passage-1 The circuit shown in fig consists of the following
$E_1 = 3, E_2 = 2, E_3 = 6$ Volt
$R_1 = 2, R_4 = 6$ Ohm
$R_3 = 2, R_2 = 4$ Ohm
$C = 5\mu F$

169. The current in resistance R_1 is.

(A) 0.5 A (B) 1.0 A (C) 1.5 A (D) Zero

170. The current through resistance R_3 is.

(A) 1.5 A (B) 1.2 A (C) 0.9 A (D) 0.6 A

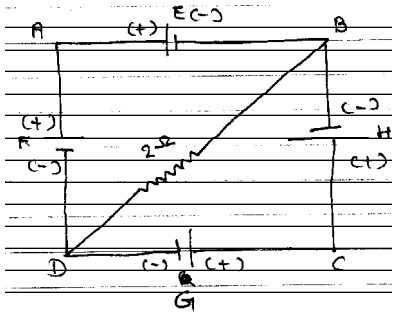
171. The current through resistance R_4 is

(A) 0.3 A (B) 0.25 A (C) 0.2 A (D) Zero

172. The energy stored in the capacitor is.

(A) 4.8×10^{-6} J (B) 9.6×10^{-6} J
(C) 1.44×10^{-5} J (D) 1.92×10^{-5} J

Passage-2 Figure shows four cell E, F, G and H of emfs 2V, 1V, 3V and 1V and internal resistances $2\Omega, 1\Omega, 3\Omega$ and 1Ω respectively



173. The current flowing in the 2Ω resistor is.

(A) $\frac{1}{7}$ A (B) $\frac{1}{9}$ A

(C) $\frac{1}{11}$ A (D) $\frac{1}{13}$ A

174. The potential difference between points B and D is.

(A) $\frac{2}{7}$ V (B) $\frac{2}{9}$ V (C) $\frac{2}{11}$ V (D) $\frac{2}{13}$ V

175. The Potential difference between the terminals of celi G is.

(A) Equal to 1V (B) More than 2V

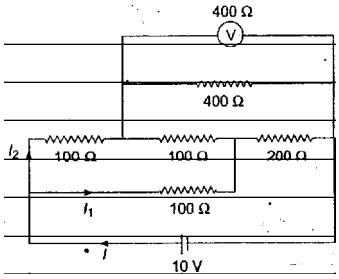
(C) Between 1.5V and 2V (D) Between 1V and 1.5V

176. The Potential difference between the terminals of cell is

(A) Equal to 1V (B) More than 2V

(C) Between 1.5V and 2V (D) Between 1v and 1.5V

Passage-3 An electrical circuit is shown in fig the values of resistasnces and the directions of the currents are shown A voltmeter of resistance 400Ω is connected across the 400Ω resister the battery has negligible internal resistance.



177. The value of current i_1 is

(A) $\frac{1}{10}$ (B) $\frac{1}{20}$ (C) $\frac{1}{30}$ (D) $\frac{1}{40}$

178. The value of current i_2 is

(A) $\frac{1}{30}$ A (B) $\frac{1}{15}$ A (C) $\frac{1}{10}$ A (D) $\frac{2}{15}$ A

179. The resding of the voltmeter is.

(A) $\frac{10}{3}$ V (B) 5 V (C) $\frac{20}{3}$ V (D) 4 V

Passage-4 The length of a potentiometer wire is 600cm and it carries a current of 40m A for cell of emf 2V and internal resistance 10Ω, the null point is found to be at 500cm on connecting a voltmeter across the cell, the balancing length is decreased by 10 cm.

180. The voltmeter reading will be.

(A) 1.96 V (B) 1.8 V (C) 1.64 V (D) 0.96 V

181. The resistance of the voltmeter is

(A) 500Ω (B) 290Ω (C) 490Ω (D) 20Ω

- **Assertion and reason typc question:**

Assertion and reason are given in follwing questions each question has four options one of them is correct select it.

(a) Both assertion and reason are true and the reason is correct ercplanation of the assertion.

(b) Both assertion and reason are true, but reason is not correct explanation of the assertion.

(c) Assertion is true, but the reason is false.

(d) Both, assertion and reason are false.

182. **Assertion:** There is no current in the metals in the absence of electric field.

Reason: Motion of free electrons is random.

(A) a (B) b (C) c (D) d

183. **Assertion:** the drift velocity of electrons in a metallic wire will decrease, if the tempreature of the wire is increased

Reason: On increasing temperature, conductivity of metallic wire decreases.

(A) a (B) b (C) c (D) d

184. **Assertion:** A potentiometer of longer length is used for acaurate measurement.

Reason: The potential gardient for a potentiometer of longer length with a given source of e.m.f become small.

(A) a (B) b (C) c (D) d

185. **Assertion:** The 200w bulbs glow with more brightness than 100w bulbs.

Reason: A 100w bulb has more resistance than a 200w bulb.

(A) a (B) b (C) c (D) d

186. **Assertion:** A series combination of cells is used when their internal resistance is much smaller than the external resistance.

Reason: It follows from the relation $I = \frac{nE}{R + n}$. Where the symbols have their standard meaning.

(A) a (B) b (C) c (D) d

187. **Assertion:** When a wire is stretched to three times its lenght, its resistance becomes a times.

Reason: $R = \frac{\rho \ell}{A}$

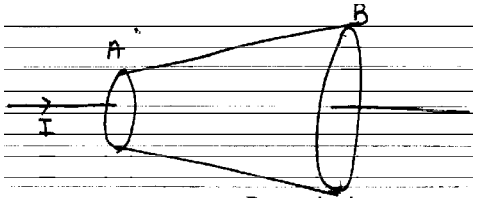
(A) a (B) b (C) c (D) d

188. **Assertion:** In balanced standard wheastone bridge, $R_{AC} = \frac{(P + Q)(R + S)}{(P + Q + R + S)}$

Reason: This is because B and D are at the same potential.

(A) a (B) b (C) c (D) d

189. **Assertion:** Current I is flowing through a cylindrical wire of non-uniform cross-section as shown section of wire near A will be more heated compared to the section near B.



Reason: Current density near A is more

(A) a (B) b

(C) c (D) d

190. **Assertion:** In the part of a circuit show in figure, given that $V_b > V_a$ the current should flow from b to a

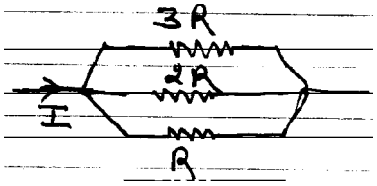


Reason: Direction of current inside a battery is always form negative teminal to positive terminal.

(A) a (B) b (C) c (D) d

191. **Assertion:** when temperature of a conductor is increased its resistance increses.
Reason: Free electorns collide collide frequently
(A) a (B) b (C) c (D) d

192. **Assertion:** In the part of the circuit shown in fig maximum power is produced across R.
Reason: Power $P = \frac{V^2}{R}$
(A) a (B) b (C) c (D) d



KEY NOTE

1	B	37	B	73	A	109	A	145	C	181	C
2	D	38	B	74	B	110	A	146	C	182	A
3	A	39	D	75	A	111	C	147	A	183	B
4	D	40	B	76	C	112	B	148	D	184	A
5	D	41	C	77	B	113	C	149	D	185	A
6	B	42	A	78	C	114	C	150	B	186	A
7	D	43	D	79	D	115	D	151	B	187	B
8	A	44	A	80	D	116	C	152	A	188	B
9	B	45	C	81	D	117	A	153	C	189	B
10	B	46	A	82	D	118	C	154	B	190	C
11	D	47	D	83	C	119	B	155	B	191	A
12	C	48	C	84	D	120	B	156	B	192	A & B
13	C	49	A	85	B	121	C	157	B		
14	A	50	D	86	D	122	C	158	B		
15	D	51	C	87	B	123	A	159	D		
16	B	52	B	88	B	124	A	160	C		
17	D	53	B	89	A	125	D	161	C		
18	B	54	C	90	B	126	B	162	B		
19	A	55	C	91	C	127	A	163	C		
20	B	56	D	92	B	128	A	164	A		
21	D	57	C	93	D	129	A	165	D		
22	C	58	C	94	A	130	C	166	B		
23	B	59	B	95	D	131	B	167	C		
24	B	60	B	96	C	132	A	168	A		
25	C	61	C	97	A	133	C	169	D		
26	B	62	A	98	C	134	A	170	A		
27	C	63	C	99	C	135	A	171	C		
28	B	64	D	100	B	136	B	172	C		
29	C	65	C	101	A	137	A	173	D		
30	D	66	B	102	C	138	A	174	D		
31	B	67	A	103	D	139	B	175	D		
32	B	68	C	104	B	140	D	176	D		
33	A	69	A	105	C	141	B	177	C		
34	B	70	D	106	A	142	D	178	A		
35	D	71	D	107	B	143	C	179	C		
36	B	72	C	108	D	144	A	180	A		

Solution

1. $R = R_1 + R_2 = \frac{\rho_1 L}{A} + \frac{\rho_2 L}{A} = \frac{L}{A}(\rho_1 + \rho_2)$ .....(1)

   $R = \frac{\rho(2L)}{A}$ .....(2)

   Equations (1) and (2) giveis $= \rho = \frac{1}{2}(\rho_1 + \rho_2)$

2. $I_1 = \frac{12V}{4\Omega} = 3A$

   Applying kirchhoff's loop rule to loop ABCDE

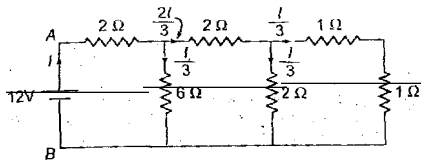
   $2L_2 + E - 4I_1 = 0$

   Putting $I_1 = 3A$ and $I_2 = 0$ we get $E = 12V$

3. The equivalent resistance between A and B $= 4\Omega$

   Current $I = \frac{12V}{4\Omega} = 3A$

   $\frac{I}{3} = \frac{3A}{3} = 1A$



4. This is a balanced wheatstone's bridge there fore $\frac{10}{40} = \frac{7}{x}$

   Which given $x = 28\ \Omega$ Hence the correct choice is (d)

5. $\frac{1}{R} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \therefore R = 3\Omega$

8. $\frac{1}{R'} = \frac{1}{R} + \frac{1}{2R} + \frac{1}{2R} = \frac{2}{R} \therefore R' = \frac{R}{2}$

9. $\frac{1}{R_1} + \frac{1}{R_2} + \ldots\ldots + \frac{1}{R_{n-1}} + \frac{1}{R_n} = \frac{1}{x}$ .....(1)

   $\Rightarrow$ If the nth resistor is removed, then

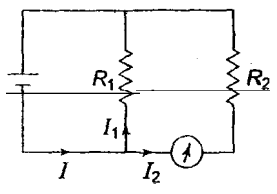
   $\frac{1}{R_1} + \frac{1}{R_2} + \ldots\ldots + \frac{1}{R_n - 1} = \frac{1}{y}$ .....(2)

   Subtracting (2) form (1), we have

   $\frac{1}{R_n} = \frac{1}{x} - \frac{1}{y}$

   Which given $R_n = \frac{xy}{(y - x)}$ which is choice (B)

10. Since the seven resistance are in parallel, the effective resistance is $R = \frac{70}{7} = 10\Omega$ There fore, the current in the circuit is $I = \frac{14}{10} = \frac{7}{5}$ A The given circuit can be redrawn as shown in fig where $R_1 = 70/3$ and $R_2 = 70 / 4\Omega$ The current I_2 is given by $I_2 = I \times \frac{R}{R_2} = \frac{7}{5} \times \frac{10}{70/4} = \frac{4}{5}$ A



12. Since no current flows through R_6 resistance R_1, R_2 R_3 and R_4 constitute the four arms of a balanced wheatestone's bridge hence

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

13. $I = \frac{6}{1+5+2} = \frac{3}{4}$ A

Now emf of cell C = Potential difference across AD

$$= \frac{3}{4} \times \frac{2 \times 60}{100} = 0.9V$$

15. $I = \frac{E}{(R_1 + r)} \Rightarrow Q_1 = I^2R_1 = \left[\frac{E}{(R_1 + r)}\right]^2 R_1$

$$Q_2 = \left[\frac{E}{(R_2 + r)}\right]^2 R_2$$

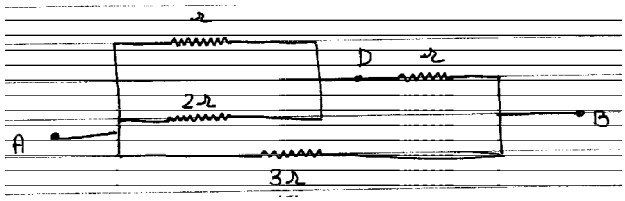
Equating Q_1 and Q_2 simplifying, we get $r = \sqrt{R_1R_2}$

16. $R_1 = R + R + R = 3R$, $R_2 = R + R + 4R = 6R$,

$$R^1 = \frac{R_1R_2}{R_1 + R_2} = \frac{3R \times 6R}{3R + 6R} = 2R,$$

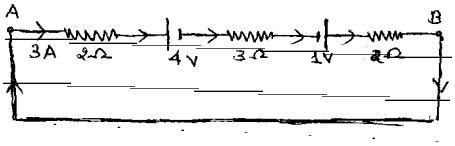
$\Rightarrow$ $4\Omega = R^1 = = 2R$

$R = 2\Omega$

29. 

$R_{net} = \frac{15}{14}$ r

30. Max power = $n\left(\frac{E^2}{4r}\right)$

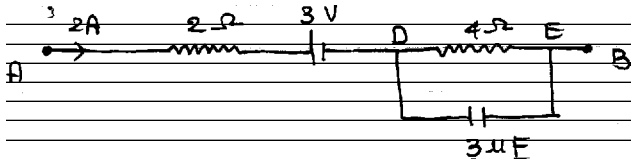
31. 

$V_A - V_B = (3 \times 2) + 4 + (3 \times 3) - 1 + (3 \times 2) = 6 + 4 + 9 - 1 + 6 = 24V$

32. $r = \left(\frac{l_1}{l_2} - 1\right) R$

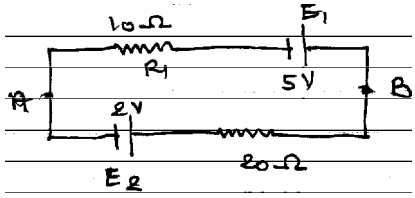
33. $R_1 = \frac{\delta l}{3} = 10\Omega,\ R_2 = \frac{\delta l}{1} = 30\Omega$

$Rs = R_1 + R_2 = 10\Omega + 30\Omega = 40\Omega$

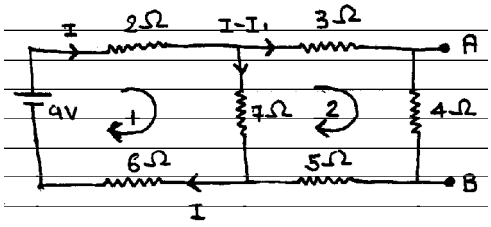
34. 

$V_D - V_C = IR = 2 \times 4 = 8V$

$\Rightarrow E = \frac{1}{2} CV^2 = \frac{1}{2} \times 3 \times 10^{-6} \times (8)^2 = 96\ \mu J$

36. 

$V_B - V_A = \frac{E_1R_2 + E_2R_1}{R_1 + R_2}$

38. 

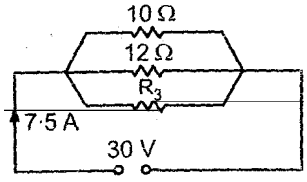
Inloop (1) $9 = 8I + 7I_1$

Inlong (2), $0 = 12 (I - I_1) - 7I_1$

$I = \frac{171}{236}$ A and $I - I_1 = \frac{63A}{236}$

$V_A - V_B = (I - I_1) R = \frac{63}{236} \times 4 \approx 1.1V$

39. $I = \frac{V}{R} = \frac{V.A}{\rho.l}$

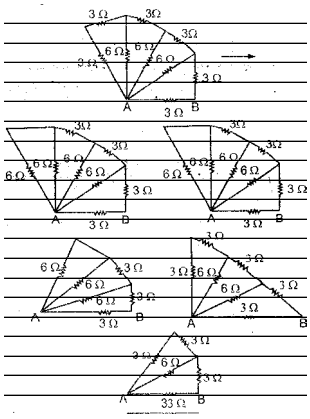


40. $I = 7.5\ A,\ V = 30V,\ R_1 = 10\ \Omega, R_2 = 12\ \Omega, R_3 = ?$

$$R_{net} = \frac{V}{I}$$

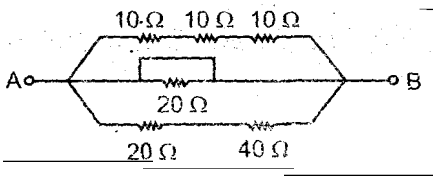
$$\Rightarrow \frac{1}{Rnet} = \frac{I}{V} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

41. Start reducing the circuit from opposite end of A and B



This way when we keep on reducing the circuit then at the end $R_{net} = 2\Omega$

42. R_{net} = Zero because points A and B are short circuited



43. $$V = \left(\frac{E}{R_S + R_l}\right) R\ell$$

$$= \left(\frac{2.5}{80 + 20}\right) \times 20 = 0.5 \text{ Volt}$$

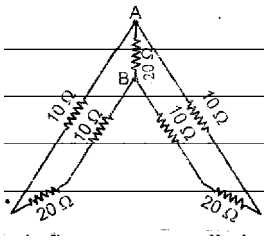
Potential gradient $= \frac{V}{l} = \frac{0.5\,\text{Volt}}{10\,\text{metre}} = 5 \times 10^{-2}\ \text{Vm}^{-1} = 5 \times 10^{-5}\ \text{V / mm}$

44. $I = nAV_dq$

$V_d\ \alpha\ \frac{1}{r^2}\ \therefore\ V^1_d = \frac{Vd}{4}$

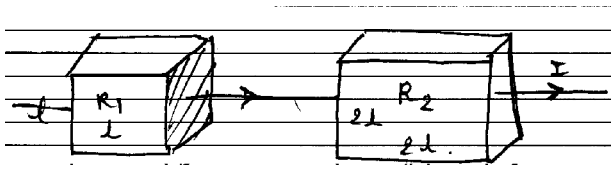
46. $$\frac{1}{R_{AB}} = \frac{1}{20} + \frac{1}{40} + \frac{1}{40} = \frac{1}{10}$$

$R_{AB} = 10\ \Omega$



48. $R_1 = \rho\ \frac{\ell}{A} = \rho \times \frac{\ell}{\ell \times t} = \frac{\rho}{t}$

$R_2 = \rho\ \frac{2\ell}{A} = \rho = \frac{\rho \cdot 2\ell}{2\ell \times t} = \frac{\rho}{t}\ \therefore\ \frac{R_1}{R_2} = \frac{1}{1}$



49. $I = \frac{V}{R} = n A V_d q$

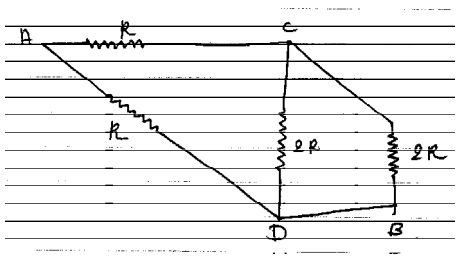
$\therefore V_d \alpha V$

50. $R \alpha \frac{l^2}{m}$ $\therefore R_1 : R_2 : R_3 = \frac{25}{1} : \frac{9}{3} : \frac{1}{5} = 125:15:1$

51. $R' = \frac{2R \times 2R}{2R + 2R} = \frac{4R^2}{4R} = R$

A–C–R resistance is = 2R

$R_{AB} = \frac{R \times 2R}{R + 2R} = \frac{2A^2}{3A} = \frac{2}{3} = R$



52. $R_P = \frac{R_1 R_2}{R_1 + R_2} = 2\Omega$ .....(1)

$R_S = R_1 + R_2 = 9\Omega$ .....(2)

Form (1) and (2)

$R_1 R_2 = 18\Omega$

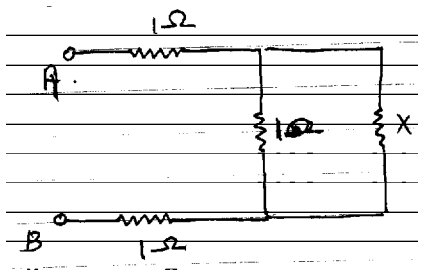
$R_1 (9 - R_1) = 18$

$R_1^2 - 9R_1 + 18 = 0$

$R_1 = 3\Omega$ OR 6Ω

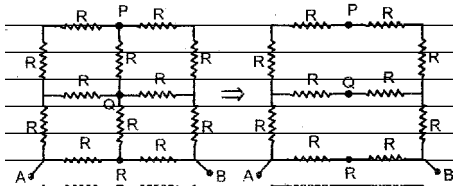
Putting $R_1 = 3$ in eq (2) we get $\therefore R_2 = 6\Omega$

54. $R_{AB} = 1 + \frac{1 \times x}{1 + x} + 1$



55. $\ell^2 = \frac{RV}{\rho}$

56. Rnet $= \frac{5R}{4}$

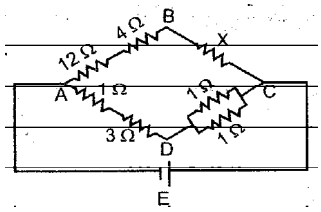


58. $I = \frac{n\varepsilon}{R + nr} \Leftrightarrow 0.6 = \frac{n \times 1.5}{20 \times 0.5n} \Leftrightarrow 12 + 0.3\,n = 1.5\,n \therefore 12 = 1.2n$

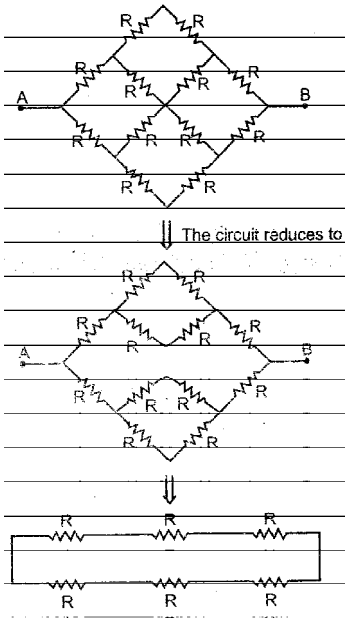
$\therefore n = 10$ cells to be connected in series.

59. $\frac{P}{Q} = \frac{R}{S} \quad \therefore \frac{16\Omega}{x} = \frac{4\Omega}{0.5\Omega}$

$\therefore 16 = 8x \qquad \therefore x = 2\Omega$



60.

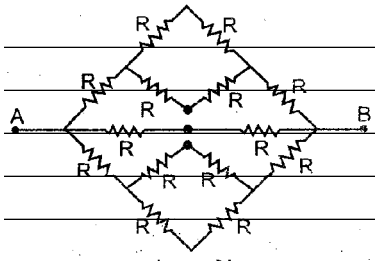


Rnet $= \frac{3R}{2}$

61. $\frac{P_1}{P_2} = \frac{\frac{V^2}{R_1}}{\frac{V^2}{R_2}} = \frac{R_2}{R_1} = \frac{2}{1}$

62. $\frac{P_1}{P_2} = \frac{I^2R_1}{I^2R_2} = \frac{R_1}{R_2} = \frac{1}{2}$

63. $P_{AB} = \frac{4R}{5}$



64. $H_1 = \frac{V^2}{R_1} t_1$

$R_1 = \frac{V^2}{H_1} t_1$ and $R_2 = \frac{V^2}{H_2} t_2$

$\frac{V^2}{H_1} = \frac{V^2}{H_2} = \text{const} \therefore R \alpha t$

Now $R_S = R_1 + R_2$

$t = 6 + 3 = 9$ Minutes

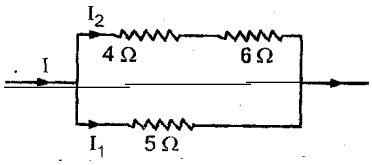
65. $R_P = \dfrac{R_1 R_2}{R_1 + R_2}$

$t = \dfrac{6 \times 3}{6 + 3} = 2$ minutes.

66. $I_2^2 R = 10$

$I_1^2 \times 5 = 10$

$I_1 = \sqrt{2}$ A



Potential difference across 5Ω resistance $= I_1 R - 5\sqrt{2}$ volt

$I_2 = \dfrac{V}{R} = \dfrac{5\sqrt{2}}{10} = \dfrac{1}{\sqrt{2}} =$ A

Produced per secin resistance 4Ω

$= I_2^2 R$

$= \left(\dfrac{1}{\sqrt{2}}\right)^2 \times 4 = 2$ calories per second

67. $V_d = \dfrac{\Delta r}{\Delta t} = \dfrac{10^{-4}}{10} = 10^{-5}$ m/s

$\Delta r = AB' - AB = 2 \times 10^{-4} - 10^{-4} = 10^{-4}$ m

70. $V = IR$ in $I = \dfrac{E}{R + r}$

$\therefore V = \dfrac{ER}{R + r}$

74. $\dfrac{V_{d1}}{V_{d2}} = \dfrac{I_1}{I_2} \times \left(\dfrac{r_2}{r_1}\right)^2 = \dfrac{I}{2I} \times \left(\dfrac{2r}{r}\right)^2 = 2$

$\therefore vd_2 = \dfrac{vd_1}{2} = \dfrac{v}{2}$

76. $R_\theta = R_{\theta o}\left[1 + \alpha(\theta_2 - \theta_1)\right]$

$3 = 1 + \left(4 \times 10^{-3}\right) T \therefore T = \dfrac{2}{4 \times 10^{-3}} = 500^\circ C$

77. $R_{40}\left[1 + \alpha(\theta_2 - \theta_1)\right]$

$5.6 = 4.64\left[1 + a(100 - 40)\right] = 4.64 + 278.4\alpha$

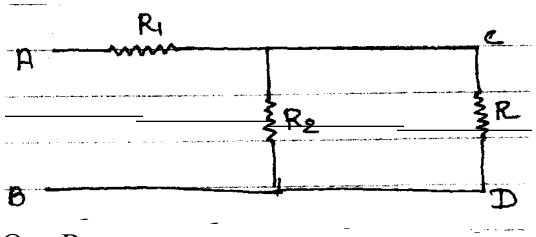
$a = \dfrac{5.6 - 4.64}{278.4} = 0.0035^\circ C^{-1}$

$\Rightarrow R_{100} = R_0(1 + \alpha T_2) = R_o\left[1 + (0.0035 \times 100)\right]$

$5.6 = R_o \times 1.35$

$Ro = 4\Omega$

78. $R_{eq} = \dfrac{R \times R_2}{R + R_2} = \dfrac{R \times 2}{R + 2} = \dfrac{2R}{R + 2}$



$R = R_1 + R_{eq} = 1 + \dfrac{2R}{R+2} = \dfrac{R+2+2R}{R+2} = \dfrac{3R+2}{R+2}$ oR

$R(R+2) = 3R+2$ oR $R^2 + 2R = 3R + 2$ oR $R^2 - R - 2 = O$ oR

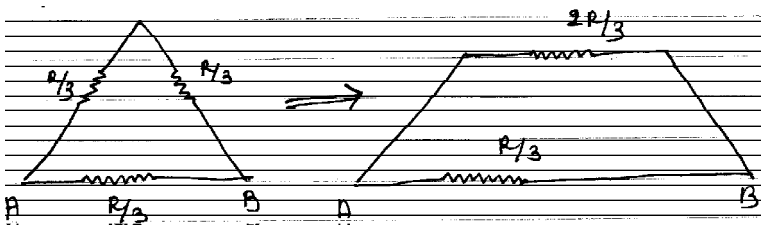
$(R-2)(R+1) = 0$ oR $R = 2\Omega$ oR $R = -1\Omega$

79. In parallal connection R minimum $= \dfrac{r}{n}$

In Series connection R maximum = nr

$\therefore \dfrac{R\,min}{R\,max} = \dfrac{r}{n(n-1)} = \dfrac{1}{n^2}$

80.



$R_{AB} = \dfrac{\left(\dfrac{2R}{3}\right)\left(\dfrac{R}{3}\right)}{2R/3 + R/3}$

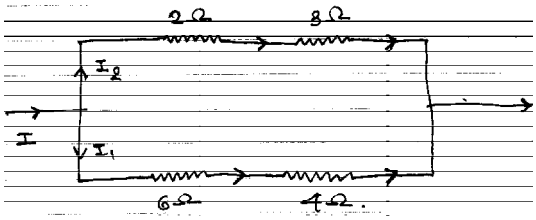
$= \quad \frac{2}{9}\ R$

81. $10I_1 = 5I_2$

$I_2 = 2T_1$

$H = \dfrac{6I_1^2}{J} = 60$

$\therefore\ I_1^2 = 10J$



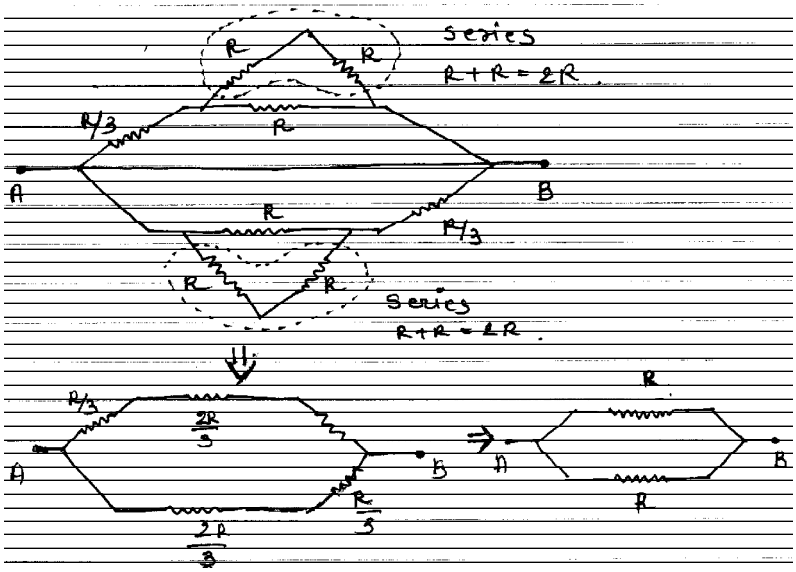
$H^1 = \dfrac{3I_2^2}{J} = 3\ \dfrac{(4I_2^2)}{J} = \dfrac{12I_1^2}{J} = \dfrac{12 \times 10J}{J} = 120$ cal

82. $\Delta Q\ \alpha\ I^2$

V is same so $P_1 = \dfrac{V^2}{R_1}$ and $P_2 = \dfrac{V^2}{R_2}$

83. $R_{eq} = \frac{R}{2}$

series $R + R = 2R$



84. I, ρ, ℓ are equal so $H \alpha \frac{1}{A} \therefore \frac{H_1}{H_2} = \frac{A_2}{A_1} = \frac{2}{1}$

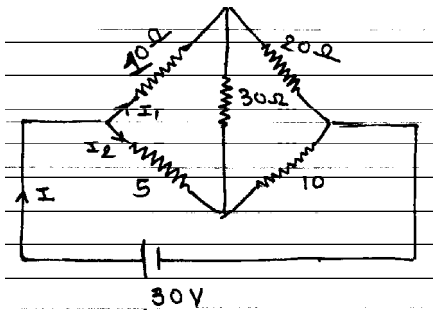
85. Equivalent resistance $R^1 = \frac{2R}{2 + R}$

Total power $P = \frac{R}{R^1} \therefore 150 = \frac{(15)^2}{2R}(2 + R) \therefore R = 6\Omega$

86. $Reff = R_1 + R_2 + R_3 \Rightarrow \frac{1}{\sigma_{eff}} = \frac{1}{\sigma_1} + \frac{1}{\sigma_2} + \frac{1}{\sigma_3}$

87. $Q = \int_1^2 Idt$

88. Wheatstone bridge is in balanced condition



so, $\frac{1}{R} = \frac{1}{30} + \frac{1}{15}$

$R = 10\Omega$

Now $I = \frac{V}{R} = \frac{30}{10} = 3$ Amp.

Now $I_2 = I \times \left(\frac{30}{30 + 15}\right) = 2$ Amp.

89. $I \alpha V_d \times A (\therefore I = neAV_d)$

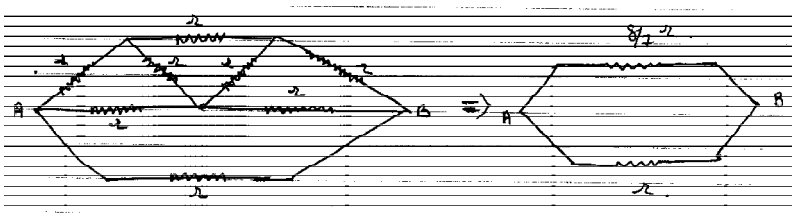
$\therefore \frac{I_1}{I_2} = \frac{(V_d)_1 A_1}{(V_d)_2 A_2}$

$\therefore (V_d)_2 = 2 (V_d)_1$

90. $V = \varepsilon - Ir, \ \varepsilon = V + Ir,$ But $V = nIr, \ \varepsilon = nIr + Ir = (n + 1) Ir$

$\therefore \frac{V}{\varepsilon} = \frac{n}{n + 1}$

91. $\therefore R_{AB} = \dfrac{\frac{8}{7}.r.r}{\frac{8}{7}r + r} = \dfrac{8}{15}r$



92. Length of wire $= 2\pi r = 8\pi m$

Total resistance of wire $= 8\pi \times \frac{1}{\pi} = 8\Omega$

$R = \dfrac{2 \times 6}{2 + 6} = \frac{12}{8} = 3/2\Omega$

$\therefore$ current $I = \frac{\varepsilon}{R} = \frac{6}{3/2} = 4A$

93. Mass m= density $\times$ volume = dAl

$\therefore A = \frac{m}{dl}$

Now , $R = \dfrac{\rho l}{A} = \dfrac{\rho \ell d\ell}{m} = \dfrac{\rho d\ell^2}{m}$

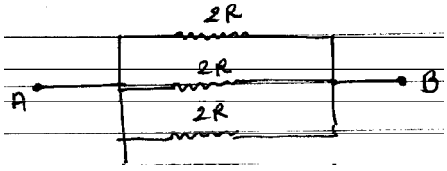
$\therefore R \alpha \frac{l^2}{m}$ ($\therefore$ p and d = constant) $\therefore R_1 : R_2 : R_3 = \dfrac{9}{1} : \dfrac{4}{2} : \dfrac{1}{3} = 27 : 6 : 1$

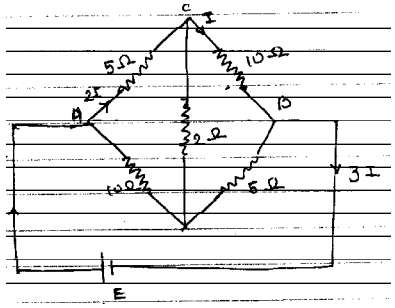
94. $R_\theta = Ro[1 + \alpha\theta]$

$5 = Ro\ (1 + 50\ \alpha)$ and

$6 = Ro\ (1 + 100\ \alpha)$

95. $\frac{1}{Reff} = \frac{1}{2R} + \frac{1}{3R} + \frac{1}{2R} = \frac{3R}{4}$





96. In closed circuit EADBE

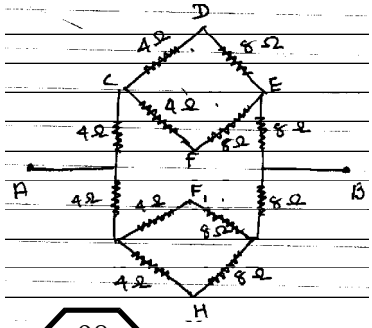
$E = 10I \times 3 \times 2I = 20I$ .....(i)

If R is the effetive resistance between A and B then

$E = 3IR$ .....(ii)

From (1) and (2) $3IR = 20I$

$R = 7$ ohms

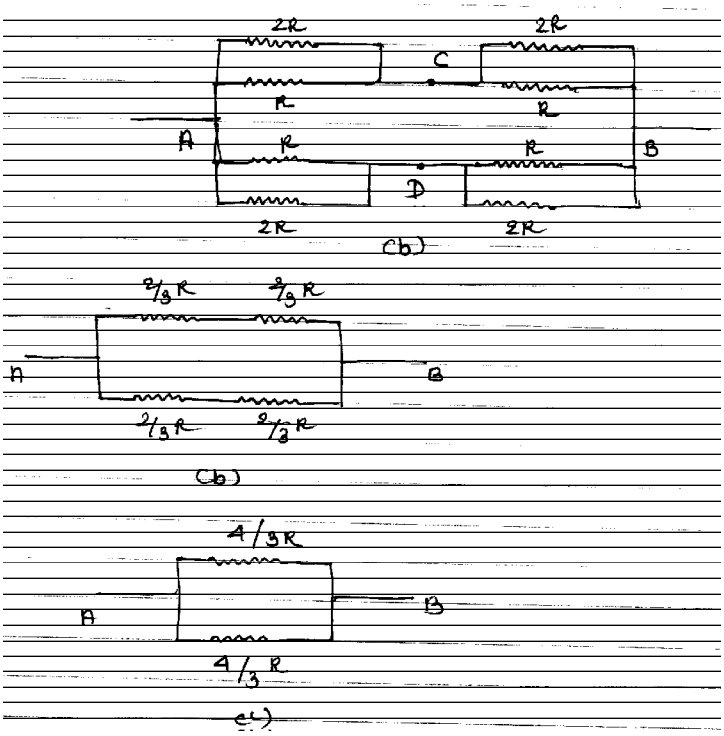


97. $R^1 = \dfrac{12 \times 12}{12 + 12} = 6\Omega$

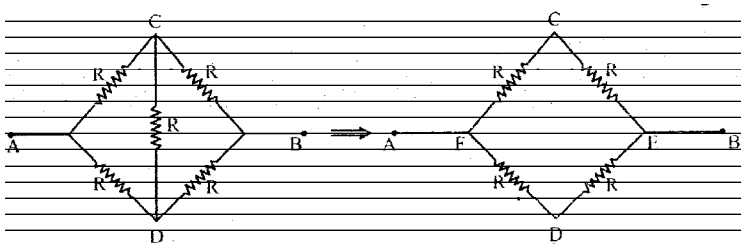
$R_1 = 4 + R^1 + 8 = 18\Omega$

$R_{eff} = \dfrac{18 \times 18}{18 + 18} = 9\Omega$

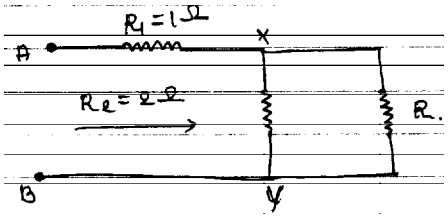
98. Total resistance $= \dfrac{\frac{4}{3}R \times \frac{4}{3}R}{\frac{4}{3}R + \frac{4}{3}R} = \dfrac{2}{3}\ R$



99. The resistance of AFCEB= R+R=2R, $I = \dfrac{V}{2R}$



100. Let the resultant resistance be R. If we add one more branch, then the resultant resistance would be the same because this is an infinite sequence.



$$\therefore \frac{R_1R_2}{R_1 + R_2} + R_1 = R \Rightarrow 2R + R + 2 = \Rightarrow R + 2R$$

$$\Rightarrow R^2 - R - 2 = 0 \Rightarrow R = -1 \text{ OR } R = 2\Omega$$

102. $R_{AB} = 2 + \dfrac{1}{3} = 2\dfrac{1}{3}\Omega$



103. The bridge ABCD is balanced if $\frac{10}{R_1} = \frac{30}{9} \Rightarrow R_1 = 3\Omega$

R_2 can have any finite value.

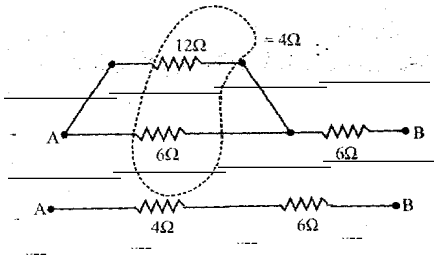
105. $P = \frac{v^2}{R} = \frac{V^2 A}{\rho l} \Rightarrow P \ \alpha \ \frac{I}{\ell} \ \therefore \ \frac{P_1}{P_2} = \frac{\ell_1}{\ell_2}$

106. Power obtained from abattery becomes maximum when r = R

$\therefore$ Power $P = I^2 r$

$= \left(\frac{\varepsilon}{R + r}\right)^2 \quad r = \frac{\varepsilon^2}{4r}$

107. $R_{eq} = 10\Omega$



108. $I = \frac{V}{R} \ \therefore \ R = 180\Omega$

$R_\theta = Ro\ [1 + \alpha\ (\theta - \theta o)$

109. $R_{27} = R_{20}\ [(I + \alpha\ (27 - 20)] = R_{20}\ [1 + \alpha\ (7)]$

$R_{100} = R_{20}\ [(I + 2\ (100 - 20)] = R_{20}\ [1 + \alpha\ (80)]$

110. $\frac{R_{300}}{R_T} = \frac{1 \times 2 \times 300}{1 + 2T} \ \therefore \ \frac{1}{2} = \frac{1 + 2 \times 300}{1 \times 2T}$

$\therefore 1 + \alpha\ T = 2 + 600\,\alpha$

$\therefore T = \frac{1 + 600\alpha}{\alpha} = \frac{1 + (600 \times 0.00125)}{(0.00125)} = \frac{1.75}{0.00125} = 1400\ K$

111. $R_P = \frac{R_1 R_2}{R_1 + R_2}$

$Rs = R_1 + R_2$

Use the $\frac{R_1}{R_2} = 4$

112. $R' = \frac{24 \times 8}{24 + 8} = 6\ \Omega$

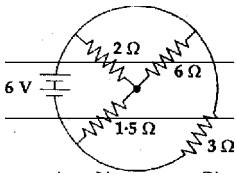
$\frac{1}{R''} = \frac{1}{20} + \frac{1}{30} + \frac{1}{60} = \frac{1}{10}$ OR $R'' = 10\Omega$

$(R_{xy}) = 3 + R' + R'' + 1 = 3 + 6 + 10 + 1 = 20\Omega$

$I = \frac{V}{R^1} = \frac{48}{6} = 8A$

$Vxy = I \times Rxy = 8 \times 20 = 160V$

113. $R^1 = \frac{R_1 R_2}{R_1 + R_2} = \frac{2 \times 6}{2 + 6}$



$R^{11} = R^1 + R_3 = 1.5 + 1.5 = 3\Omega$

$I = \frac{V}{R} = \frac{6}{1.5} = 4A$

114. $R_S = R_1 + R_2 + R_3 = R + R + R = 3R$

$R_p = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R}$ OR $R_p = \frac{R}{3}$

$P = \frac{V^2}{R} \alpha \frac{1}{R}$

$\frac{P_s}{P_p} = \frac{R_p}{R_s}$ OR $P_P = \frac{R_s}{R_p} \times P_S = \frac{3R}{R/3} \times 10 = 90W$

115. Use the equn power $p = \frac{V^2}{R}$

117. $V_B = V_A + (-I \times R_1) - V_1 - (-I \times R_2) + V_2$
$= 45 - (3 \times 6) - 8 - (3 \times 3) + 7$
$= 45 - 18 - 8 - 9 + 7 = 17V$

118. $E = E_1 + E_2 - E_3 = 2 + 2 - 2 = 2V$

$I = \frac{E}{nr + R} = \frac{2}{(3 \times 1) + 2} = \frac{2}{5} = 0.4A$

119. $I = \frac{E}{x + R + r} = \frac{2}{10 + R + 1} = \frac{2}{11 + R}$

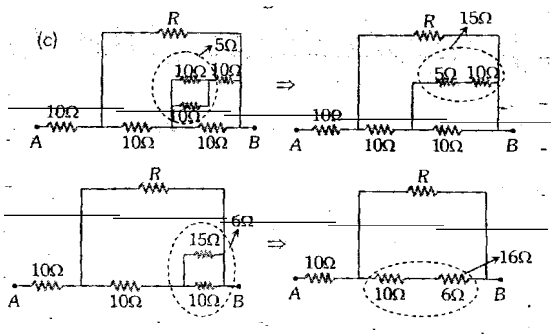
$\Rightarrow V = I \times X$ OR $I = \frac{2}{11 + R} \times 10 = \frac{20}{11 + R}$

OR $11 + R = 20$

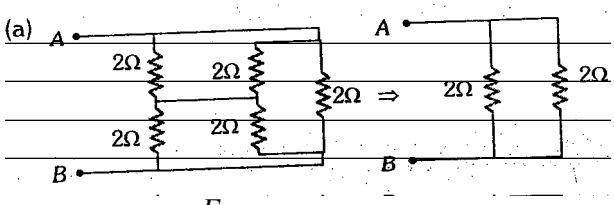
$R = 20 - 11 = 9\Omega$

120. $r = \frac{I_1 - I_2}{I_2} \times R = 0.5\ \Omega$

121. $\frac{R \times 16}{R + 16} + 10 = 18$, on solving we get, $R = 16\Omega$



123. $R_{AB} = \frac{2 \times 2}{2 + 2} = 1\Omega$



124. Total e.m.f = nE Total resistance R + nr $\Rightarrow$ $i = \frac{nE}{R + nr}$

125. $I = \frac{E}{R + \frac{r}{4}} = \frac{2}{2 + \frac{1}{4}} = \frac{2}{2.25} = 0.888A$

126. Potential gardient along wire

= Potential difference along wire / length of wire

$= 0.1 \times 10^{-3} = \frac{I \times 40}{1000}$ V/cm

$\Rightarrow$ Current in wire , $I = \frac{1}{400}$ A

$\frac{2}{40 + R} = \frac{1}{400}$ OR $R = 800 - 40 = 760\Omega$

127. Potential gardient = $\frac{V}{L} = \frac{iR}{L} = \frac{i\rho\ell}{AL} = \frac{i\rho}{A}$

128. Potential garadient $x = \frac{e}{(R + Rh + r)} \frac{R}{L}$

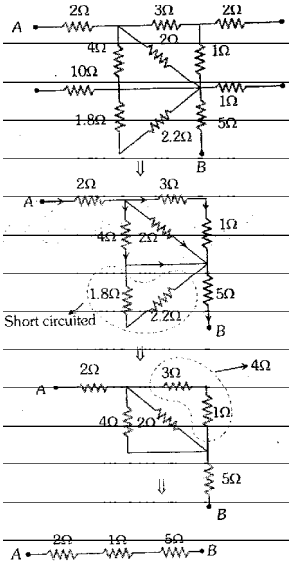
129. $r = \left(\frac{I_1 - I_2}{I_2}\right) \times R^1 \Rightarrow r = \left(\frac{55 - 50}{50}\right) \times 10 = 1\ \Omega$

130. Form the figure net resistance

$R_1 = 1$ Ohm, $R_2 = \frac{1}{2}$ Ohm, $R_3 = 2$ Ohm

It is clear that $R_3 > R_1 > R_2$ $\therefore P_3 < P_1 < P_2 \left(P = \frac{V^2}{R}\right)$

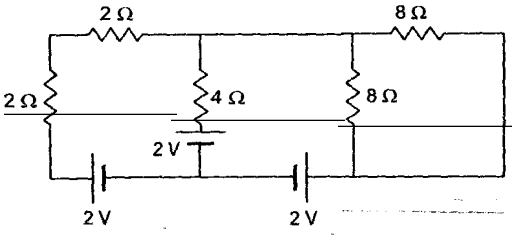
131.



132. Here Rxwy = $\frac{R}{2\pi} \times r\,\alpha = \frac{R\alpha}{2\pi}$ $(\therefore \alpha = \frac{I}{r})$

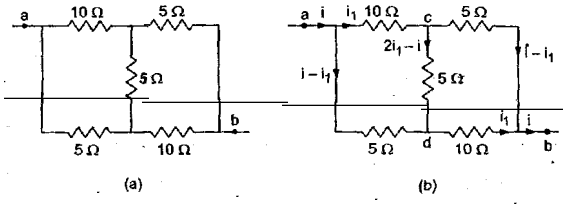
and Rxzy = $\frac{R}{2\pi r} \times r\,(2\pi - \alpha) = \frac{R}{2\pi} \times (2\pi - \alpha)$

134. Ans: Zero

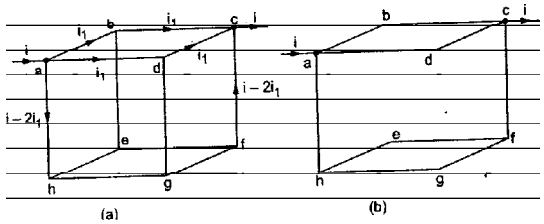


135. $V_a - V_b = (V_a - V_c) + (V_c - V_b)$

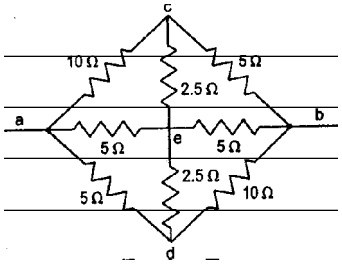
Also, $V_a - V_b = (V_a - V_c) + (V_c - V_d) + (V_d - V_b)$



136. $\frac{1}{R} = \frac{1}{3r} + \frac{1}{3r} + \frac{1}{2r}$ $\quad \therefore R = \frac{3}{4}\,r$



137. $R = \frac{(7\,\Omega) \times (10\,\Omega)}{7\,\Omega \times (10\,\Omega} = 4.1\ \Omega$



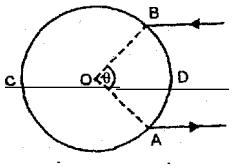
140. Resistance of section ADB

$R_1 = \rho r\theta = \frac{R}{2\pi r} r\theta = \frac{R\theta}{2\pi}$

Resistance of section ACB_1

$R_2 = \frac{R}{2\pi r} r\,(2\pi - \theta) = R\,\frac{(2\pi - Q)}{2\pi}$

$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$



141. $R_{EF} = \frac{12 \times 6}{12 + 6} = 4\ \Omega$, $R_{GD} = \frac{4 \times 4}{4 + 4} = 2\ \Omega$

$R_{DJ} = \frac{12 \times 24}{12 \times 24} = 8\ \Omega$, $R_{AJ} = 5 + 2 + 8 = 15\ \Omega$

$I = \frac{V}{R_{AJ}} = \frac{15}{15} = 1A$

142. $E = I_1 (R_1 + r) = I_2 (R_2 + r)$

143. $R_{it} = R_1 (1 + \alpha it)\ R_2 = R_2 (1 + \alpha_2 t)$

$R_{st} = R_1 t + R_2 t$

$= R_1 [1 + \alpha_1 t] + R_2 [1 + \alpha_2 t]$

$= R_S \left[1 + \left(\frac{(R_1\alpha_1 + R_2\alpha_2 t)}{R_1 + R_2}\right)\right]$

Cap comparing with $Rst = R_S [1 + \alpha st]$

$\alpha_S = \frac{R_1\alpha_1 + R_2\alpha_2}{R_L + R_2}$

144. $S = R_1 + R_2;\ P = \frac{R_1R_2}{R_1 + R_2};\ S = nP$

$R_1 + R_2 = \frac{nR_1R_2}{R_1 + R_2}$

Or $R_1^2 + R_2^2 + 2R_1R_2 = nR_1R_2$

Or $(R_1 - R_2)^2 + 4R_1R_2 = nR_1R_2$

Or $(R_1 - R_2)^2 - R_1R_2 (n - 4)$

If $R_1 = R_2$, Then $(n - 4) = 0$ or $n = 4$

145. $I = \frac{E + E}{R_1 + R_2 + R}$

$R_2 = \frac{2E \times R_2}{(R_1 + R_2 + R)}$

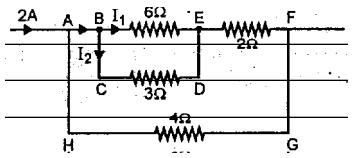
$\Rightarrow E - \frac{2ER_2}{R_1 + R_2 + R} = 0$ or $E = \frac{2ER_2}{R_1 + R_2 + R}$

$R = R_2 - R_1$

146. $S = \frac{S_1S_2}{S_1 + S_2}$

$\frac{P}{Q} = \frac{R}{S}$

149. $R = \frac{6 \times 3}{6 + 3} + 2 = 4\ \Omega$



$I = I_1 + I_2$ ..... (i)

$E = 6I_1 = 3I_2$

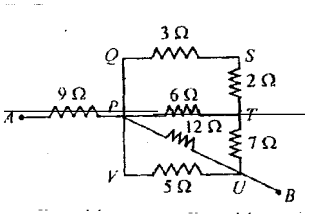
or $I_2 = 2I_1$

Solving (i) and (ii), we get

$I_1 = \frac{2}{3}$ A, $I_2 = \frac{1}{3}$ A

Pot diff current $3\Omega = \frac{2}{3}$ A $\times$ $3\Omega = 2V$

150. $x = \frac{5 \times 6}{5 + 6} = \frac{30}{11}\ \Omega$



$y = x + 7 = \frac{30}{11} + 7 = \frac{107}{11}\ \Omega$

$\therefore \frac{1}{Z} = \frac{11}{107} + \frac{1}{5} + \frac{1}{12} = 0.1 + 0.2 + 0.083 = 0.38$

$\therefore$ Z $= \frac{1}{0.38} = 2.6\ \Omega$ Now 2.6 Ω and 9.0 Ω are in series

$\therefore$ Resistance between A and B = 9.W + 2.6 Ω

= 11.6.Ω

$\therefore$ Required resistance between A and B = 11.6.Ω

151. $\frac{P}{1/3} = \frac{Q}{1 - (1/3)}$, l = 1m

Or 3P = 3/2 Q Or P = Q/2

$\frac{P + 6}{2/3} = \frac{Q}{1/3}$

P + 6 = 2Q

$6 = 2Q - \frac{Q}{2} = \frac{3Q}{2}$

So Q = 4 and P = 2

155. $R = R_o\ [1 + 2t]$

For series connection $R_S = R_1 + R_2$

at O^oC tap $R_S = R_o + R_o + 2R_o$

$2R_o\ [1 + \alpha st] = R_o\ [1 + \alpha_1 t] + R_o\ [1 + \alpha_2 t]$

$\therefore\ \alpha_3 = \frac{1}{\alpha}\ (\alpha_1 + \alpha_2)$ ..... (i)

For parallal connection $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$

at O^oC temp RP = Ro/2

$\therefore\ \frac{1}{Ro/2\left(1+\alpha_p t\right)} = \frac{1}{R_o\left[1+\alpha_1 t\right]} + \frac{1}{R_o\left[1+\alpha_2 t\right]}$

$\therefore\quad \alpha_P = \frac{1}{\alpha}\ (\alpha_1 + \alpha_2)$

157. $\varepsilon = I_1 (R_1 + r) = I_2 (R_2 + r)$

$\therefore \quad r = \frac{1}{3}\ \Omega$

158. Not flowing current in Golvememeter

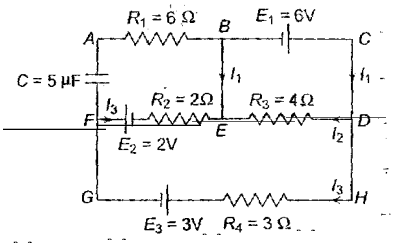
$$\text{So } I = \frac{V_A}{R_1 + R} = 12/600$$

$V_B = I_R = 12/600 \times 100$

$= 2V$

159. Wattage $P \alpha \frac{1}{R}$ Hence $R_{40} < R_{60} < R_{100}$

169.



170. $I_2 R_3 = E_1$ Or $I_2 = \frac{E_1}{R_3} = 1.5\ A$

171. $I_3R_2 - I_2R_3 + I_3R_4 = -E_2 - E_3$

Or $I_3 (R_2 + R_4) = I_2 R_3 - E_2 - E_3$

Or $I_3 (2 + 3) = 1.5 \times 4 - 2 - 3$

Or $5I_3 = I$ Or $I_3 = 0.2\ A$

172. $V = E_2 + I_3R_2 = 2 + 0.2 \times 2 = 2.4\ V$

Energy stored in the capacitor $= \frac{1}{2} CV^2$

174. Potential difference between B and D $= 2 \times (1/3) = 2/13V$

175. Potential difference between the terminals of cell $G = 3 - \left(\frac{6}{13}\right) \times 3 = \frac{21}{13}\ 1.6\ V$

176. Potential difference between the terminals of cell $H = I + \frac{6}{13} = \frac{19}{13} \approx 1.46$

177. Solve by kirchhof's second law

179. Voltmeter reding = potential difference across $R = (200\Omega)$

$= I_3R = \frac{1}{30} \times 200 = \frac{20}{3}\ V.$

●●●

Unit - 13
Magnetic Effects of Electric Current and Magnetism

SUMMARY

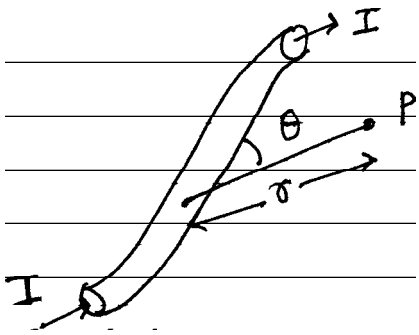
Important tips of each topic

1. Biot-Savart;s law :

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{1\ d\vec{l} \times \hat{r}}{r^2}$$

$$dB = \frac{\mu_0}{4\pi}\frac{1\ dl\ \sin\theta}{r^2}$$

} In Vaccum or AIR



$$d\vec{B} = \frac{\mu_r\ \mu_0}{4\pi}\frac{1\ d\vec{l} \times \hat{r}}{r^2}$$

$$dB = \frac{\mu_r\ \mu_0}{4\pi}\frac{1\ dl\ \sin\theta}{r^2}$$

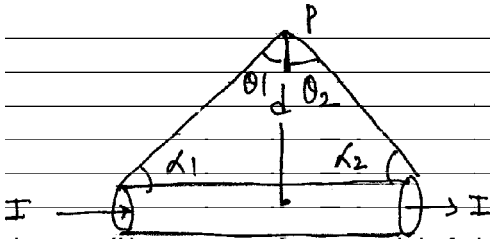
} In any Medium

2. For a WIRE

(A) Finite length of a wire

$$B = \frac{\mu_0}{4\pi}\frac{I}{d}\left[\sin\theta_1 + \sin\theta_2\right]$$

$$= \frac{\mu_0}{4\pi}\frac{I}{d}\left[\cos\alpha_1 + \cos\alpha_2\right]$$



(B) Infinite length of a wire

$\theta_1 = \theta_2 = 90^o$ OR $\alpha_1 = \alpha_2 = 0^o$

$$B = \frac{\mu_0}{4\pi}\frac{I}{d}\left[1 + 1\right]$$

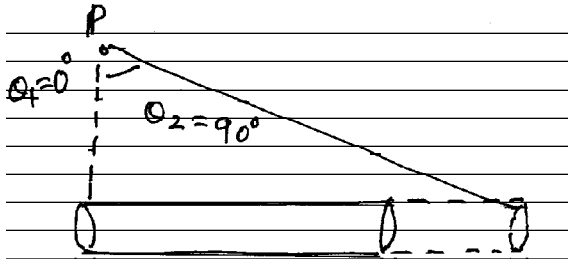
$$\frac{\mu_0}{4\pi}\frac{I}{d}\left[2\right] = \frac{\mu_0}{2\pi}\frac{I}{d}$$

(C) Semi - infinite length of a wire

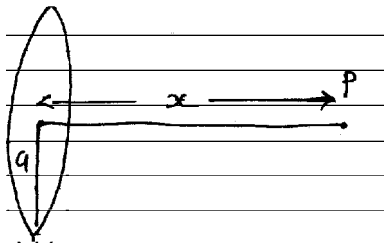
$\theta_1 = 0^o$; $\theta_2 = 90^o$

$$B = \frac{\mu_0}{4\pi}\frac{1}{d}\left[0 + 1\right]$$

$$= \frac{\mu_0}{4\pi}\frac{1}{d}$$



3. For a RING :



(A) For N = 1 turn

$$B = \frac{{}_0 I a^2}{2\left(a^2 + x^2\right)^{\frac{3}{2}}}$$

(B) For N = N turns

$$B = N\left[\frac{{}_0 I a^2}{2\left(a^2 + x^2\right)^{\frac{3}{2}}}\right]$$

(C) At the centre (x = 0)

$$B = N\left[\frac{\mu_0 I}{2a}\right]$$

(D) At $x >> a$

$$B = \frac{{}_0}{4\pi}\frac{2M}{x^3}$$

[Just as, mag. field on the axis of a Bar-magnet]

Where m = magnetic moment

4. For Solenoid

(A) Finite length solenoid



$$B = \frac{{}_0 nI}{2}\left[\sin\alpha + \sin\beta\right] \quad \text{Where } n = \frac{N}{\ell}$$

Where α and β are angles mode at the either end of the solenoids.

n = no. of turns per unit length ; N = total no. of turns.

(B) Infinite length solenoid

$\alpha = \beta = 90^o$

$$B = \frac{{}_0 nI}{2}\left[1 + 1\right]$$

$$B = {}_0 nI \qquad \text{Where } n = \frac{N}{\ell}$$

(C) Mag. field at either end

$\alpha = 0$ and $\beta = 90^o$

$$\text{Bend Point} = \frac{{}_0 nI}{2}\left[0 + 1\right]$$

$= \frac{1}{2} \mu_0 nI$

$\text{Bend} = \frac{1}{2} \text{Binside}$

(D) Toroid

$B = N\left(\frac{\mu_0}{2\pi} \frac{I}{r} \right)$

5. Force on a charged particle in magnetic field.

(A) $\vec{F} = q\left(\vec{V} \times \vec{B}\right)$

$F = q V B \sin\theta$

Direction of force can be determine by using

(B) Fleming's Left and rule

First finger indicates $\Rightarrow$ direction of magnetic field.

Middle finger indicates $\Rightarrow$ direction of motion of POSITIVE charge particle

Thumb indicates $\Rightarrow$ direction of force

(C) If $\theta = 0^o$ or 180°

charged particle moves on straight line.

(D) If $\theta = 90^o$ ie $\vec{V} \perp \vec{B} \Rightarrow F = q V B$

charged particle moves on circular path of radius r

$r = \frac{mv}{qB} = \frac{p}{qB} = \sqrt{\frac{2mK}{qB}} = \frac{1}{B}\sqrt{\frac{2mV}{q}}$

(E) If θ is neither zero nor perpendicular it performs Helical path.

- radius of helical path $r = \frac{m(V\sin\theta)}{qB}$
- periodic time $T = \frac{2\pi m}{qB}$
- pitch of the helix $= T(v\cos\theta) = \frac{2\pi m v\cos\theta}{qB} = \frac{2\pi r}{\tan\theta}$
- No. of pitches $= \frac{\ell}{\text{Pitch distance}}$

6. Lorentz's force

$$\vec{F} = q\left[\vec{E} + \left(\vec{V} \times \vec{B}\right)\right]$$

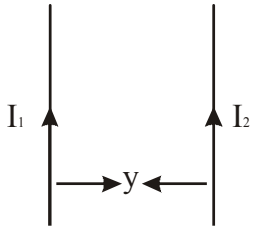
7. Cyclotorn

Frequency $f = \frac{1}{T} = \frac{Bq}{2\ \pi\ m}$

8. Force between two parallel current carrying wires.

$$F = \frac{0}{2\pi}\ \frac{I_1\ I_2}{Y}\ \ell$$

$$\frac{F}{\ell} = \frac{0}{2\pi}\ \frac{I_1\ I_2}{Y}$$



Case (i) If I_1 and I_2 are flowing in same direction Attraction.

Case (ii) If I_1 and I_2 are flowing in opposite direction Repulsion.

9. Torque acting on a rectangle frame

$\tau = BINA \sin\theta$

(i) If frame is parallel to the field $\theta = 0^o$ $\tau = 0$

(ii) If frame is perpendicular to the field $\theta = 90^o$ $\tau = BINA$

10. Moving coil Galvano meter.

(i) $\tau = BINA$

τrestoning $= K\phi$ Where ϕ = deflection in galvanometer

$BINA = K\phi$

$$I = \left(\frac{K}{BNA}\right)\varphi$$

(ii) Current sensitivity (S_I) :

The deflection produced in the Galvanometer per unit current flowing throught it.

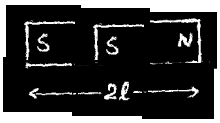
$$S_I = \frac{\phi}{I} = \frac{BNA}{K}$$

(iii) Voltage sensitivity (S_V) :

The deflection produced in the Galvanometer per unit voltage applied to it.

$$S_V = \frac{\phi}{V} = \frac{\phi}{IR} = \frac{SI}{R} = \frac{BNA}{KR}$$

11. Bar magnet and its pole strength (m)



Pole strength :

- The strength of a magnetic pole to atiract magnetic material towards itself.
- Unit is $Amp \times meter = \frac{Newton}{Tesla}$
- Pole strength of the magnet depends on the nature of material of magnet and area of cross- section.
- m does not depend upon length.

12. Magnetic dipole moment (M) :

$\vec{M} = m \times \left(2\vec{\ell}\right)$

- dir^{n} is from south pole to North pole
- unit is $\text{Amp} \times \text{meter}^2 = \frac{\text{Newton} - \text{meter}}{\text{Tesla}}$

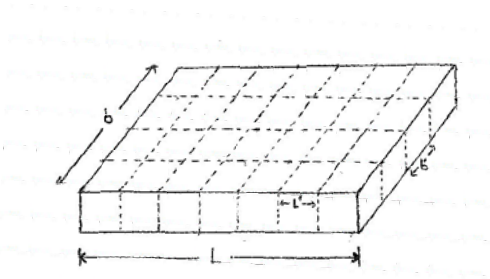
13. Cutting of a rectangular bar-magnet.

If a bar-magnet of length L and breadth b is cut into n equal parts then

(i) Length of each part $L' = \frac{L}{\sqrt{n}}$

(ii) Breadth of each part $b' = \frac{b}{\sqrt{n}}$

(iii) Mass of each part $w' = \frac{w}{n}$



(iv) Pole-strength (m) of each part $m' = \frac{m}{\sqrt{n}}$

(v) Magnetic moment (M) of each part $M' = \frac{M}{n}$

(vi) Initial (Original) moment of inertia of a bar $I = \frac{1}{12} W\left(L^2 + b^2\right)$

(vii) After cutting new moment of inertia $I' = \frac{I}{n^2}$

14. Cutting of a thin bar-magnet for thin bar magnet b = 0

$$L' = \frac{L}{n};\ w' = \frac{w}{n};\ m' = \frac{m}{n};\ I' = \frac{I}{n^2}$$

15. Magnetic field and Magnetic flux :

(i) Magnetic field is denoted by B and its units are

$$Tesla = \frac{Weber}{m^2} = \frac{Newton}{Amp \times meter} = \frac{Joule}{Amp \times m^2} = \frac{Volt.\ see}{m^2}$$

(G) unit is Gauss 1 Tesla = 10^4 Gauss

16. Magnetic permeability : $-(\mu)$

$\mu o =$ Absolute permeability of air or vaccum

$$= 4\pi \times 10^{-7}\ \frac{\text{tesla} \times \text{meter}}{\text{Amp}}$$

μ_r = relative permeability

$$_r = \frac{}{_0} = \frac{B}{B_0} = \frac{\text{mag. Flux density in material}}{\text{mag. Flux density in vaccum}}$$

17. Intansity of magnetising field (H^{-1}) :

It is the degree or extent to which a magnetic field can magnetise a substance.

$$H = \frac{B}{\mu}$$

$$\text{unit} = \frac{\text{Ampere}}{\text{meter}}$$

$$= \frac{A}{m} = \frac{N}{m^2 \times \text{tesla}} = \frac{N}{wb} = \frac{J}{m^3 \times \text{tesla}} = \frac{J}{m \times wb}$$

CGS unit : Oersted

$$1\ \text{Oersted} = \frac{80\ \text{Amp}}{\text{meter}}$$

18. Intensity of magnetisation (I)

(i) It is the degree to which a substance is magnetised when placed in a magnetic field.

(ii) It is also defined as the pole strength per unit cross-sectional area of the substance.

(iii) It is also defined as Induced dipole moment per unit volume.

$$I = \frac{m}{A} = \frac{M}{\text{Volume}}$$

$$\text{unit is } \frac{\text{Ampere}}{\text{meter}}$$

19. Magnetic susceptibility (χ_m) and permeability

$B = B_o + B_m$

$= {}_0H + {}_0I$

$= {}_0 (H + I)$

$B = \mu_0 H(1 + \chi_m)$

$\mu_r = 1 + \chi_m$

20. Coulomb's law in magnetism.

$F = \frac{K\, m_1\, m_2}{r^2}$ where m_1, m_2 = pole strength

where $K = \frac{\mu_0}{4\pi} = 10^{-7}$ in SI unit

= 1 in CGS unit

21. Magnetic field due to bar-magnet

(i) On axis of a bar-magnet

$B_{axis} = \frac{\mu_0}{4\pi} \frac{2M}{r^3}$

(ii) On equator of a bar-magnet

$B_{equator} = \frac{\mu_0}{4\pi} \frac{M}{r^3}$

22. Bar-magnet in magnetic field.

(i) Torque $\tau = MB \sin\theta$

(ii) Work $W = MB(\cos\theta_1 - \cos\theta_2)$

(iii) Potential energy $U = -\vec{M} \cdot \vec{B} = -MB\cos\theta$

23. Tangent Galvanometer :

In equilibrium

$B = B_H \tan\theta$

Where $B = \frac{\mu_0 nI}{2r}$

n = no. of turns

r = radius of the coil

I = Current to be measured

θ = angle made by needle from the direction of B_H in equilibrium.

24. Deflection magnetometer :

It works on principle of tangent law

(i) A-Position :

The magnetometer is set perpendicular to magnetic meridian so that magnetic field due to magnet is in AXIAL position.

$$B = B_H \tan\theta = \frac{\mu_0}{4\pi}\frac{2M}{r^3}$$

(ii) B-position :

The arms of magneto meter are set in magnetic meridian so that the magnetic field due to magnet is at its equatorial position.

$$B = B_H \tan\theta = \frac{\mu}{4\pi}\frac{M}{r^3}$$

(iii) Comparison :

$$\frac{M_1}{M_2} = \frac{\tan\theta_1}{\tan\theta_2}$$

$$= \left(\frac{r_1}{r_2}\right)^3$$

25. Vibration Magnetometer :

Periodic time $T = 2\pi\sqrt{\frac{I}{MB_H}}$

$$\therefore M = \frac{4\pi^2 I}{B_H \cdot T^2}$$

(i) Comparison of horizontal components of earth's magnetic field at two places.

$$T = 2\pi\sqrt{\frac{I}{MB_H}}$$

but I and M are constant

$$\therefore T^2 \alpha \frac{1}{B_H} \Rightarrow \frac{(B_H)_1}{(B_H)_2} = \frac{T_2^2}{T_1^2}$$

(ii) Comparison of magnetic moment of two magnets of same size and mass

$$T = 2\pi \sqrt{\frac{1}{MB_H}}$$

but I and B_H are constant.

$$\therefore\ T^2 \ \alpha\ \frac{1}{M} \Rightarrow \frac{M_1}{M_2} = \frac{T_2^{\,2}}{T_1^{\,2}}$$

26. Diamagnetic material :
- magnetic dipole moment M = 0
- experience fore towards weak mag. field.
- magnetic susceptibility $\chi_m = -\text{Ve}.$

27. Paramagnetic material :
magnetic dipole moment = M = 0
experience force towrards strong mag. field.
magnetic susceptibility $\chi_m = +\text{Ve}.$

28. Curie Law :

$$\chi\ \alpha\ \frac{1}{T}$$

$$\chi = \frac{C}{T}$$

29. Curie - weiss law :
At temperature above curie temperature the magnetic susceptibility of force magnetic material is inversely proportional to $(T - T_C)$

$$\chi\ \alpha\ \frac{1}{T - T_C}$$

$$\chi\ = \frac{C}{T - T_C}$$

MCQ

For the answer of the following questions choose the correct alternative from among the given ones.

1. An element $\overrightarrow{d\ell} = dx \uparrow$ (where dx = 1 cm) is placed at the origin and carries a large current I = 10 Amp. What is the mag. field on the Y-axis at a distance of 0.5 meter ?

   (a) $2 \times 10^{-8}\ \hat{k}$ T (b) $4 \times 10^{8}\ \hat{k}$ T

   (c) $-2 \times 10^{-8}\ \hat{k}$ T (d) $-4 \times 10^{-8}\ \hat{k}$ T

2. Two straight long conductors AOB and COD are perpendicular to each other and carry currents I_1 and I_2. The magnitude of the mag. field at a point "P" at a distance "a" from the point "O" in a direction perpendicular to the plane ABCD is

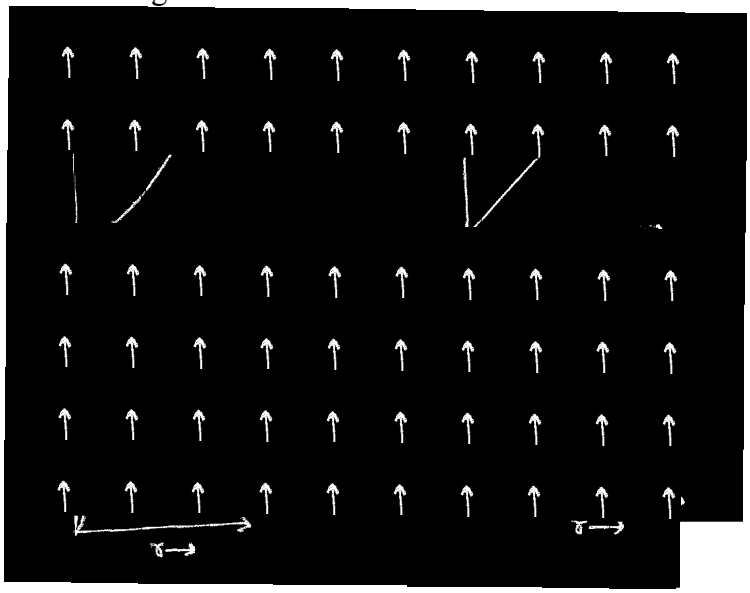
   (a) $\frac{\mu_0}{2\pi a}\ (I_1 + I_2)$ (b) $\frac{\mu_0}{2\ \pi a}\ (I_1 - I_2)$

   (c) $\frac{\mu_0}{2\ \pi a}\ (I_1^2 + I_2^2)^{\frac{1}{2}}$ (d) $\frac{\mu_0}{2\ \pi a}\ \frac{I_1\ I_2}{(I_1 + I_2)}$

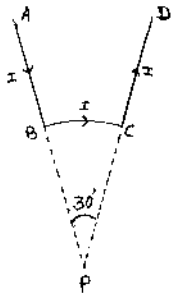
3. B $\rightarrow$ R graph. The mag. field B at a distance r from a long straight wire carrying a current varies with r as shown in Fig.

   (a)

   (c)



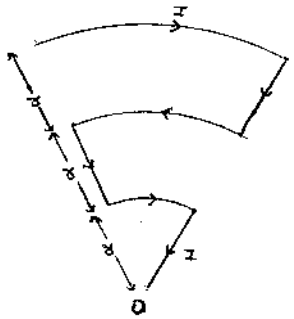
4. A current path shaped as shown in figure produces a mag. field at point "P", the centre of the arc BC. If the arc subtends an angle of 30^0 and the radius of the arc is 0.6 meter. What is the magnitude of the field at point P if the current is 3 AMP ?



   (a) 2.62×10^{-6} T (b) 2.62×10^{-7} T

   (c) 3.62×10^{-7} T (d) 2.62×10^{-8} T

5. As shown in figure a conductor carrying a current I. Find the magnetic field intensity at the point "O".



(a) $\frac{5}{24}\ \frac{\pi_o}{\pi} \times \frac{I}{r}\ \theta\ \hat{k}$

(b) $\frac{-5}{26}\ \frac{\pi_o}{\pi}\ \frac{I}{r}\ \theta\ \hat{k}$

(c) $\frac{-7}{24}\ \frac{\mu o}{\pi}\ \frac{1}{r}\theta\ \hat{k}$

(d) $-\frac{5}{24}\ \frac{o}{\pi}\ \frac{I}{r}\ \theta\ \hat{k}$

6. A length L of wire carries a steady current 1. It is bent first to form a coil of 1 turn. The same length is now bent more sharply to give a double loop of smaller radius. The magnetic field at the centre caused by the same current is.....

(a) A quater of its first value (b) Un changed

(c) Four times of its first value (d) A half of its first value

7. If a long hollow copper pipe carries a direct current, the magnetic field associated with the current will be .....

(a) Only inside the pipe (b) Only outside the pipe

(c) Neither inside nor outside the pipe (d) Both inside and outside the pipe

8. The magnetic induction at a point P which is at a distance 4 cm from a long current carrying wire is 10^{-8} tesla. The field of induction at a distance 12 cm from the same current would be .....tesla.

(a) 3.33×10^{-9} (b) 1.11×10^{-4} (c) 3×10^{-3} (d) 9×10^{-2}

9. The strength of the magnetic field at a point y near a long straight current carrying wire is B. The field at a distance y/2 will be.....

(a) $\frac{B}{2}$ (b) $\frac{B}{4}$ (c) 2B (d) 4B

10. The mag. field (B) at the centre of a circular coil of radius "a", through which a current I flows is.....

(a) $B \alpha a$ (b) $B \alpha \frac{1}{I}$ (c) $B \alpha I$ (d) $B \alpha I^2$

11. A current of a 1 Amp is passed through a straight wire of length 2 meter. The magnetic field at a point in air at a distance of 3 meters from either end of wire and lying on the axis of wire will be.....

(a) $\frac{\mu_o}{2\pi}$ (b) $\frac{\mu_o}{4\pi}$ (c) $\frac{\mu_o}{8\pi}$ (d) zero

12. If the strangth of the magnetic field produced at 10 cm away from a infinilely long straight conductor is 10^{-5} tesla. The value of the current flowing in the conductor will be..... Ampere.

(a) 5 (b) 10 (c) 500 (d) 1000

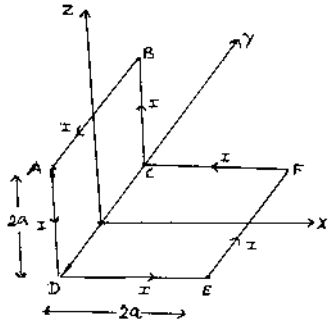
13. A long straight wire of radius "a" carries a steady current I the current is uniformly distributed across its cross-section. The ratio of the magnetic field at a/2 and 2a is .....

(a) $\frac{1}{4}$ (b) 4 (c) 1 (d) $\frac{1}{2}$

14. At a distance of 10 cm from a long straight wire carrying current, the magnetic field is 4×10^{-2}. At the distance of 40 cm, the magnetic field will be .....Tesla.

(a) 1×10^{-2} (b) 2×10^{-2} (c) 8×10^{-2} (d) 16×10^{-2}

15. As shown in figure ABCD and CDEF planes are kept carrying current I. Each side of the plane is having length "2a". The magnetic field due to ABCD and CDEF planes at the point P(a, 0, a) is in the direction.



(a) $\frac{-\hat{i} + \hat{k}}{\sqrt{2}}$ (b) $\frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$

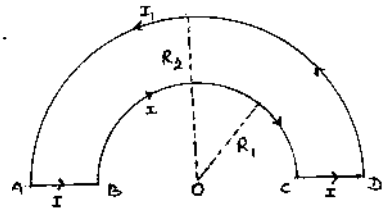
(c) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$ (d) $\frac{\hat{i} + \hat{k}}{\sqrt{2}}$

16. A He nucleus makes a full rotation in a circle of radius 0.8 meter in 2 sec. The value of the mag. field B at the centre of the circle will be ..... Tesla.

(a) $\frac{10^{-19}}{\mu_o}$ (b) $10^{-19}\ \mu_o$

(c) $2 \times 10^{-10}\ \mu_o$ (d) $\frac{2 \times 10^{-10}}{\mu_o}$

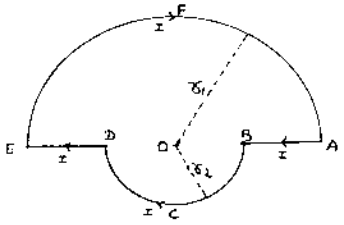
17. The magnetic field at pt. "O" in the figure shown is .....
Where AB = CD = 2 cm
R_1 = 10 cm
R_2 = 12 cm
I = 4 Amp



(a) $\frac{5}{3}\ \mu_o$ going inside (b) $\frac{5}{3}\ \mu_o$ going outside

(c) $\frac{3}{5}\ \mu_o$ going inside (d) $\frac{5}{3}\ \mu_o$ going outside

18. As shown in Fig. there are two semicircles of radii r_1 = 12 cm and r_2 = 10 cm in which 4 Amp current is flowing The mag. field at the centre "O" is .....



(a) $\frac{55}{3}\ \mu_o$ going inside (b) $\frac{3}{55}\ \mu_o$ going outside

(c) $\frac{6}{55}\ \mu_o$ going inside (d) $\frac{12}{55}\ \mu_o$ going inside

19. The direction of mag. field lines close to a straight conductor carrying current will be .....
    (a) Along the length of the conductor
    (b) Radially outward
    (c) Circular in a plane perpendicular to the conductor
    (d) Helical

20. Due to 10 Amp of current flowing in a circular coil of 10 cm radius, the mag. field produced at its centre is $\pi \times 10^{-3}$ Tesla. The number of turns in the coil will be

    (a) 5000 (b) 100 (c) 50 (d) 25

21. There are 50 turns / cm in a long solenoid. If 4 Amp current is flowing in the solenoid, the approximate value of mag. field along its axis at an internal point and one end will be respectively.

    (a) 12.6×10^{-3} Tesla ; 6.3×10^{-3} tesla

    (b) 12.6×10^{-3} Tesla ; 25.1×10^{-3} tesla

    (c) 25.1×10^{-3} Tesla ; 12.6×10^{-3} tesla

    (d) 25.1×10^{-5} Tesla ; 6.3×10^{-5} tesla

22. The distance at which the magnetic field on axis as compared to the mag. field at the centre of the coil carrying current I and radius R is $\frac{1}{8}$, would be

    (a) R (b) $\sqrt{2}$ R (c) 2R (d) $\sqrt{3}$ R

23. In a H-atom, an electron moves in a circular orbit of radius 5.2×10^{-11} meter and produces a mag. field of 12.56 Tesla at its nucleus. The current produced by the motion of the electron will be ..... Amp.

    (a) 6.53×10^{-3} (b) 13.25×10^{-10}

    (c) 9.6×10^{6} (d) 1.04×10^{-3}

24. A conducting rod of 1 meter length and 1 kg mass is suspended by two vertical wires through its ends. An external magnetic field of 2 Tesla is applied normal to the rod. Now the current to be passed through the rod so as to make the tension in the wires zero is [take g = 10 ms^{-2}]

    (a) 0.5 Amp (b) 15 Amp (c) 5 Amp (d) 1.5 Amp

25. A straight wire of mass 200 gm and length 1.5 meter carries a current of 2 Amp. It is suspended in mid-air by a uniform horizontal magnetic field B. [take g = 10 m/s^2]. The B is .....

    (a) $\frac{2}{3}$ tesla (b) $\frac{3}{2}$ tesla (c) $\frac{20}{3}$ tesla (d) $\frac{3}{20}$ tesla

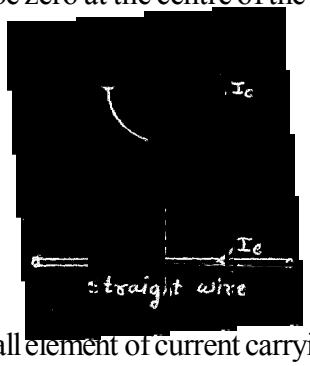
26. A long solenoid has 200 turns per cm and carries a current of 2.5 Amp. The mag. field at its centre is ..... tesla.

    (a) $\pi \times 10^{-2}$ (b) $2\pi \times 10^{-2}$ (c) $3\pi \times 10^{-2}$ (d) $4\pi \times 10^{-2}$

27. Two concentric co-planar circular Loops of radii r_1 and r_2 carry currents of respectively I_1 and I_2 in opposite directions. The magnetic induction at the centre of the Loops is half that due to I_1 alone at the centre. If $r_2 = 2r_1$ the value of $\frac{I_2}{I_1}$ is

(a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) 1

28. Circular loop of a wire and a long straight I_C, I_E respectively as shown in fig. Assuming that these are placed in the same plane. The mag. field will be zero at the centre of the Loop when separation H is .....



(a) $\frac{I_e R}{I_c \pi}$ (b) $\frac{I_c R}{I_e \pi}$

(c) $\frac{\pi I_c}{I_e R}$ (d) $\frac{I_e . \pi}{I_c R}$

29. For the mag. field to be maximum due to a small element of current carrying conductor at a point, the angle between the element and the line joining the element to the given point must be

(a) 0° (b) 90° (c) 180° (d) 45°

30. When a certain length of wire is turned into one circular Loop, the magnetic induction at the centre of coil due to some current flowing is B_0. If the same wire is turned into three Loops to make a circular coil, the magnetic induction at the centre of this coil for the

(a) B_0 (b) $9 B_0$ (c) $3 B_0$ (d) $27 B_0$

31. A long straight wire carrying current of 30 Amp is placed in an external uniform mag. field of induction 4×10^{-4} tesla. The mag. field is acting parallel to the directon of current. The magnitude of the resultant magnetic induction in tesla at a point 2 cm away from the wire is ..... tesla.

(a) 10^{-4} (b) 3×10^{-4} (c) 5×10^{-4} (d) 6×10^{-4}

32. Two similar coils are kept mutually perpendicular such that their centres coinside. At the centre, find the ratio of the mag. field due to one coil and the resultant magnetic field by both coils, if the same current is flown.

(a) $1 : \sqrt{2}$ (b) 1 : 2 (c) 2 : 1 (d) $\sqrt{3} : 1$

33. A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the centre of the coil is B. It is then bent into a circular Loop of n turns. The magnetic field at the centre of the coil for same current will be.

(a) nB (b) n^2B (c) 2nB (d) $2n^2B$

34. The mag. field due to a current carrying circular Loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is $54\,\mu T$ what will be its value at the centre of the LOOP.

(a) $250\,\mu T$ (b) $150\,\mu T$ (c) $125\,\mu T$ (d) $75\,\mu T$

35. When the current flowing in a circular coil is doubled and the number of turns of the coil in it is halved, the magnetic field at its centre will become

(a) Four times (b) Same (c) Half (d) Double

36. Two wires of same length are shaped into a square and a circle. If they carry same current, ratio of the magnetic moment is

(a) $2 : \pi$ (b) $\pi : 2$ (c) $\pi : 4$ (d) $4 : \pi$

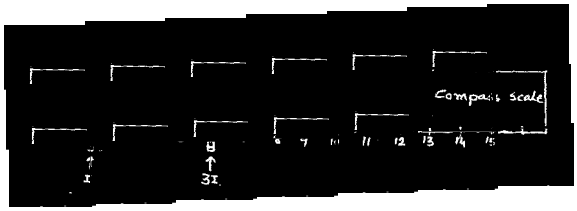
37. Two concentric coils each of radius equal to 2π cm are placed at right angles to each other. 3 Amp and 4 Amp are the currents flowing in each coil respectively. The magnetic field intensity at the centre of the coils will be ..... Tesla.

(a) 5×10^{-5} (b) 7×10^{-5} (c) 12×10^{-5} (d) 10^{-5}

38. Two parallel long wires A and B carry currents I_1 and I_2. ($I_2 < I_1$) when I_1 and I_2 are in the same direction the mag. field at a point mid way between the wires is 10 T. If I_2 is reversed, the field becomes $30\mu T$. The ratio $\frac{I_1}{I_2}$ is .....

(a) 1 (b) 2 (c) 3 (d) 4

39. Two parallel long straight conductors are placed at right angle to the meter scale at the 2 cm and 6 cm marks as shown in the figure. If they carry currents I and 3I respectively in the upward direction, then will produce zero magnetic field at



(a) Zero mark (b) 9 cm mark (c) 3 cm mark (d) 7 cm mark

40. A long solenoid has 800 turns per meter length of solenoid. A current of 1.6 Amp flows through it. The magnetic induction at the end of the solenoid on its axis is .....tesla.

(a) 16×10^{-4} (b) 8×10^{-4}

(c) 32×10^{-4} (d) 4×10^{-4}

41. A solenoid of 1.5 meter length and 4 cm diameter possesses 10 turn per cm. A current of 5 Amp is flowing through it. The magnetci induction at axis inside the solenoid is

(a) $2\pi \times 10^{-3}$ T (b) $2\pi \times 10^{-5}$ T

(c) $2\pi \times 10^{-2}$ G (d) $2\pi \times 10^{-5}$ G

42. A straight wire of length 30 cm and mass 60 milligrawm lies in a direction 30^0 east of north. The earth's magnetic field at this site is horizontal and has a magnitude of 0.8 G. What current must be passed through the wire so that it may float in air ? $\left(g = 10 \frac{m}{s^2}\right)$

(a) 10 Amp (b) 20 Amp (c) 40 Amp (d) 50 Amp

43. A long horizontal wire "A" carries a current of 50 Amp. It is rigidly fixed. Another small wire "B" is placed just above and parallel to "A". The weight of wire-B per unit length is $75X10^{-3}$ Newton/meter and carries a current of 25 Amp. Find the position of wire B from A so that wire B remains suspended due to magnetic repulsion. Also indicate the direction of current in B w.r.t. to A.

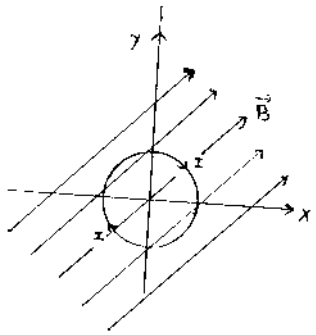
(a) $\frac{1}{2} \times 10^{-2}$ m; in sume direction

(b) $\frac{1}{3} \times 10^{-2}$ m; in mutually opposite direction

(c) $\frac{1}{4} \times 10^{-2}$ m; in same direction

(d) $\frac{1}{5} \times 10^{-2}$ m ; in mutally opposite direction

44. A circular loop of radius R = 20 cm is placed in a uniform magnetic field B = 2 Tesla in xy - Plane as shown in figure. The loop carries a current I = 1 Amp in the direction shown in fig. Find the magnitude of torque acting on the Loop.



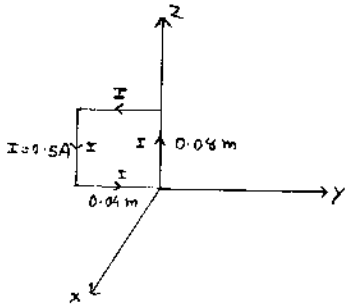
(a) 0.15 N - m

(b) 0.25 N - m

(c) 0.55 N - m

(d) 0.35 N - m

45. The rectangular coil having 100 turns is turned in a uniform mag. field of $\frac{0.05}{\sqrt{2}}\hat{j}$ as shown in the fig. The torque acting on the Loop is .....



(a) 11.32×10^{-4} N.m.$\hat{k}$ (b) 22.64×10^{-4} N.m.$\hat{k}$

(c) 5.66×10^{-5} N.m.$\hat{k}$ (d) zero

46. Two particles X and Y having equal charges, after being accelerated through the same potential difference, enter a region of uniform mag. field and desoribe circular path of radius R_1 and R_2 respectively. The ratio of mass of X to that of Y is .....

(a) $\sqrt{\frac{R_1}{R_2}}$ (b) $\frac{R_2}{R_1}$

(c) $\left(\frac{R_1}{R_2}\right)^2$ (d) $\frac{R_1}{R_2}$

47. An electron having mass 9×10^{-31} kg, charge 1.6×10^{-19} C and moving with a velocity of 10^6 m/s enters a region where mag. field exists. If it describes a circle of radius 0.10 m, the intensity of magnetic field must be ..... Tesla

(a) 1.8×10^{-4} (b) 5.6×10^{-5}

(c) 14.4×10^{-5} (d) 1.3×10^{-6}

48. A proton and an particle are projected with the same kinetic energy at right angles to the uniform mag. field. Which one of the following statements will be true.

(a) The $\alpha-$ particle will be bent in a circular path with a small radius that for the proton.

(b) The radius of the path of the $\alpha-$ particle will be greater than that of the proton.

(c) The $\alpha-$ particle and the proton will be bent in a circular path with the same radius.

(d) The $\alpha-$ particle and the proton will go through the field in a straight line.

49. A 2 Mev proton is moving perpendicular to a uniform magnetic field of 2.5 tesla. The force on the proton is

(a) 3×10^{-10} N (b) 70.8×10^{-11} N

(c) 3×10^{-11} N (d) 7.68×10^{-12} N

50. A proton is projected with a speed of $2 \times 10^6 \frac{m}{s}$ at an angle of 60^0 to the X-axis. If a uniform mag. field of 0.104 tesla is applied along Y-axis, the path of proton is .....

(a) A circle of r = 0.2 m and time period $\pi \times 10^{-7}$ sec

(b) A circle of r = 0.1 m and time period $2\pi \times 10^{-7}$ sec

(c) A helix of r = 0.1 m and time period $2\pi \times 10^{-7}$ sec

(d) A helix of r = 0.2 m and time period $4\pi \times 10^{-7}$ sec

51. A charged particle moves in a uniform mag. field. The velocity of the particle at some instant makes an acute angle with the mag. field. The path of the particle will be

(a) A straight line (b) A circle

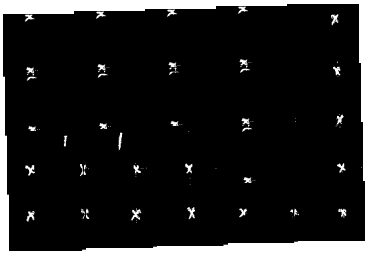
(c) A helix with uniform pitch (d) A helix with non-uniform pitch

52. A, proton, a deuteron and α - an particle having the same kinetic energy are moving in circular trajectories in a constant magnetic field. If r_p, r_d and denote respectively the radii of trajectories of these particles, then

(a) $r_\alpha = r_p < r_d$ (b) $r_\alpha > r_d > r_p$

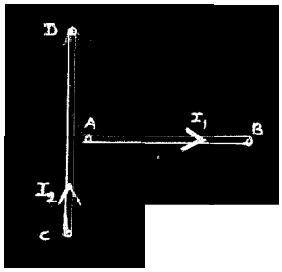
(c) $r_\alpha = r_d > r_p$ (d) $r_p = r_d = r_\alpha$

53. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform mag. field exists perpendicular to this plane. The speeds of the particles are V_A and V_B respectively and the trajectories are as shown in the figure, then.



54. An electron and a proton with equal momentum enter perpendicularly into a uniform magnetic then.
    (a) The path of proton shall be more curved than that of electron.
    (b) The path of proton shall be less curved turn that of electron.
    (c) Both are equally curved.
    (d) Path of both will be straight line.

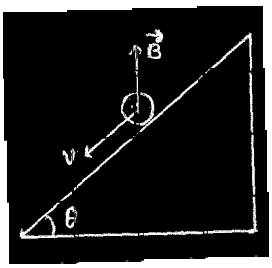
55. A current I, carrying wire AB is placed near an another long wire CD carrying current I_2 As shown in Fig. If free to move, wire AB will have
    (a) rotational motion only
    (b) translational motion only
    (c) rotational as well as translational motion
    (d) neither rotational nor translationed motion



56. A conducting rod of length ? [cross-section is shown] and mass m is moving down on a smooth inclined plane of inclination θ with constant speed v. A vertically upward mag. field $\vec{B}$ exists in upward direction. The magnitude of mag. field $\vec{B}$ is .....

    (a) $\frac{mg \sin\theta}{I\ell}$ (b) $\frac{mg \cos\theta}{I\ell}$

    (c) $\frac{mg \tan\theta}{I\ell}$ (d) $\frac{mg}{I\ell \sin\theta}$



57. A deutron of K. E. 50 kev is describing a circular orbit of radius 0.5 m in a plane perpendicular to magnetic field $\vec{B}$. The K.E. of the proton that describe a circular orbit of radius 0.5 m in the same plane with the same $\vec{B}$ is .....

    (a) 200 kev (b) 100 kev
    (c) 50 kev (d) 25 kev

58. A magnetic field existing in a region is given by $\vec{B} = Bo\left[1 + \frac{x}{\ell}\right]\hat{k}$. A square Loop of side *l* and carrying current I is placed with edges (sides) parallel to X-Y axis. The magnitude of the net magnetic force experienced by the Loop is .....

    (a) $2 Bo I\ell$ (b) $\frac{1}{2} B_o I\ell$

    (c) $Bo I\ell$ (d) $B I\ell$

59. The forces existing between two parallel current carrying conductors is F. If the current in each conductor is doubled, then the value of force will be

(a) 2F (b) 4F (c) 5F (d) $\frac{F}{2}$

60. At a given place the horizontal componnent of earth's field is 0.2 G. If a vertical wire carries a current of 30 Amp upward, what is the magnitude and direction of the force on 1 meter of wire ?

(a) 6 E to W (b) 6×10^{-3} E to W

(c) 6×10^{-3} E to W (d) 6×10^{-4} E to W

61. A Galvanometer has a resistance G and 9 current I_G flowing in it produces full scale deflection. S_1 is the value of the shunt which converts it into an ammeter of range 0 to I and S_2 is the value of the shunt for the range 0 to 2I. The ratio $\frac{S_1}{S_2}$ is .....

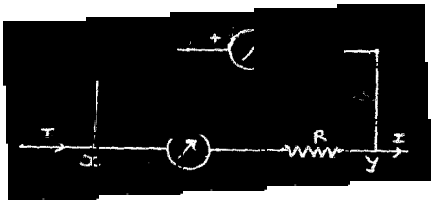
(a) $\frac{2I - I_G}{I - I_G}$ (b) $\frac{1}{2}\left(\frac{I - I_G}{2I - I_G}\right)$

(c) 2 (d) 1

62. A student connect a moving coil voltmeter V and 9 moving coil Ammeter A and resistor R as shown in figure ? If the voltmeter reads 10 volt and the ammeter reads 2 Amp then R is .....

(a) $= 5\ \Omega$ (b) $> 5\ \Omega$

(c) $< 5\Omega$ (d) 10Ω



63. The deflection in a Galvanometer falls from 50 division to 20 when 12Ω shunt is applied. The Galvanometer resistance is .....

(a) 18Ω (b) 36Ω (c) 24Ω (d) 30Ω

64. In a mass spectrometer used for measuring the masses of ions, the ions are initially accelerated by an ele. potential V and then made to describe semicircular paths of radius R using *a* magnetic field B.If V and B are kept constant, the ratio $\frac{\text{Charge on the ion}}{\text{mass of the ion}}$ will be proportional to.

(a) $\frac{1}{R^2}$ (b) R^2 (c) R (d) $\frac{1}{R}$

65. A Galvanometer of resistance 50Ω is connected to a battery of 3 volt along with a resistance of 2950Ω in series. A full scale deflection of 30 divisions is obtained in the galvanometer. In order to reduce this deflection to 20 divisions, the resistance in series should be

(a) 6050Ω (b) 4450Ω (c) 5050Ω (d) 5550Ω

66. A Galvanometer coil has a resistance of 15Ω and gives full scale deflection for a current of 4 mA. To convert it to an ammeter of range 0 to 6 Amp.....

(a) $10\,m\Omega$ resistance is to be connected in parallel to the galvanometer.

(b) $10\,m\Omega$ resistance is to be connected in series with the galvanometer.

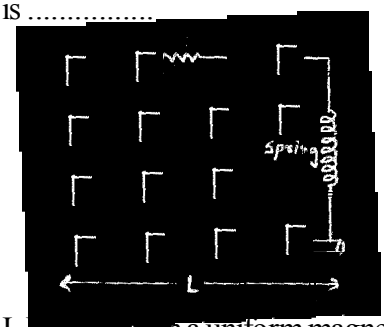
(c) $0.1\,\Omega$ resistance is to be connected in parallel to the galvanometer.

(d) $0.1\,\Omega$ resistance is to be connected in series with the galvanometer.

67. The deflection in moving coil Galvanometer is reduced to half when it is shunted with a 40Ω coil. The resistance of the Galvanometer is .....

(a) 60Ω (b) 10Ω (c) 40Ω (d) 20Ω

68. A straight rod of mass m and length L is suspended from the two idential springs as shown in figure. The spring is streched a distawnce y_0 due to the weight of the wire. The circuit has total resistance R. when the magnetic field perpendicular to the plane of paper is switched on, springs are observed to extend further by the same distance y_0 the magnetic strength is .....



(a) $\frac{2mgR}{LV}$ (b) $\frac{mgR}{LV}$

(c) $\frac{mgR}{2LV}$ (d) $\frac{mgR}{V}$

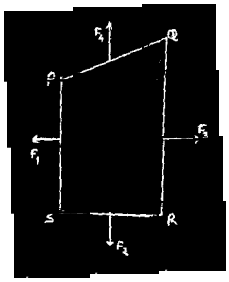
69. A conducting circular loop of radius a carries a constant current I. It is placed in a uniform magnetic field $\vec{B}$, such that $\vec{B}$ is perpendicular to the plane of the Loop. The magnetic force acting on the Loop is .....

(a) $\vec{B}Ir$ (b) $\vec{B}I\pi r^2$ (c) Zero (d) $BI(2\pi r)$

70. Two thin long parallel wires separated by a distance Y are carrying a current I Amp each. The magnitude of the force per unit length exerted by one wire on their is .....

(a) $\frac{\mu o I^2}{Y^2}$ (b) $\frac{\mu o I^2}{2\pi Y}$ (c) $\frac{\mu o}{2\ \pi}\ \frac{I}{Y}$ (d) $\frac{\mu o}{2\pi}\ \frac{I}{Y^2}$

71. A closed Loop PQRS carrying a current is placed in a uniform magnetic field. If the magnetic forces on segment PS, SR and RQ are F_1, F_2 and F_3 respectivley and are in the plane of the paper and along the directions shown, the force on the segment QP is .....



(a) $\sqrt{(F_3 - F_1)^2 - F_2^{\,2}}$ (b) $F_1 - F_2 + F_3$

(c) $-F_1 + F_2 + F_3$ (d) $\sqrt{(F_3 - F_1)^2 + F_2^{\,2}}$

72. If two streams of protons moive parallel to each other in the same direction, then they
(a) Do not exert any force on each other
(b) Repel each other
(c) Attract each other
(d) Get rotated to be perpendicular to each other.

73. A coil in the shape of an equilateral triangle of side l is suspended between the pole pieces of a permanent magnet such that $\vec{B}$ is in plane of the coil. If due to a current I in the triangle a torque τ acts on it, the side l of the triangle is .....

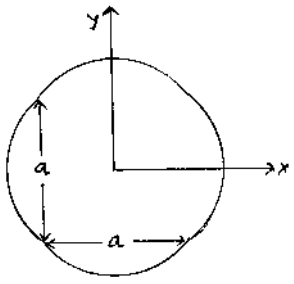
(a) $\frac{2}{\sqrt{3}}\left(\frac{\tau}{BI}\right)^{\frac{1}{2}}$ (b) $\frac{2}{3}\left(\frac{\tau}{BI}\right)$

(c) $2\left(\frac{\tau}{\sqrt{3}BI}\right)^{\frac{1}{2}}$ (d) $\frac{1}{\sqrt{3}}\ \frac{\tau}{BI}$

74. In a moving coil galvanometer, the deflection of the coil θ is related to ele. current I by the relation.
(a) $I \alpha \tan\theta$ (b) $I \alpha \theta$
(c) $I \alpha \theta^2$ (d) $I \alpha \sqrt{\theta}$

75. The unit of ele. current "AMPEAR" is the current which when flowing through each of two parallel wires spaced 1 meter apart in vaccum and of infinite length will give rise to a force between them equal to ..... N/m.
(a) 1 (b) 2×10^{-7}
(c) 1×10^{-2} (d) $4\pi \times 10^{-7}$

76. A Loop carrying current I lies in XY - plane as shown in the figure. The unit vector $\hat{k}$ is comming out of the plane of the paper. The magnetic moment of the current Loop is.....



(a) $Ia^2\hat{k}$ (b) $\left(\frac{\pi}{2}+1\right)a^2 I\,\hat{k}$
(c) $-\left(\frac{\pi}{2}+1\right)a^2 I\,\hat{k}$ (d) $(2\pi+1)a^2 I\hat{k}$

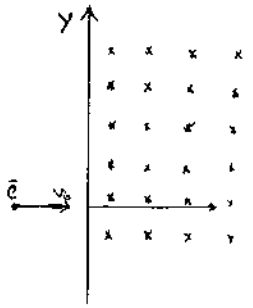
77. A coil having N turns is wound tightly in the form of a spiral with inner and outer radii "a" and "b" respectively. When a current I passes through the coil, the magnetic field at the centre is .....
(a) $\frac{\mu_o NI}{b}$ (b) $\frac{2\mu_o NI}{a}$
(c) $\frac{{}_0 NI}{2\ (b-a)}\ ln\left(\frac{b}{a}\right)$ (d) $\frac{{}^0}{2NI(b-a)}\ ln\,(ab)$

78. A particle of mass m and charge q moves with a constant velocity v along the positive x - direction. It enters a region containing a uniform magnetic field B directed along the negative z-direction, extending from x = a to x = b. The minimum value of required so that the particle can just enter the region x > b is

(a) $\frac{qbB}{m}$ (b) $q(b-a)\frac{B}{m}$

(c) $\frac{qaB}{m}$ (d) $q(b+a)\frac{B}{2m}$

79. An electron moving with a speed along the positive x-axis at y = 0 enters a region of uniform magnetic field $\vec{B} = -B_o\hat{k}$ which exists to the right of y - axis. The electron exits from the region after some time with the speed at co-ordinate y then.



(a) $v > v_o, y < 0$

(b) $v = v_o, y > 0$

(c) $v > v_o, y > 0$

(d) $v = v_o, y < 0$

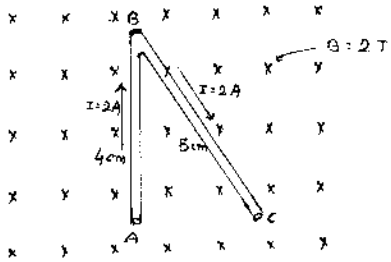
80. A uniform conducting wire ABC has a mass 10 gram. A 2 Amp current is flowing through it. The wire is kept in uniform magnetic field B = 2 tesla the acceleration of the wire will be

(a) zero

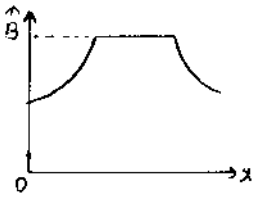
(b) 12 m/s^2 along y - axis

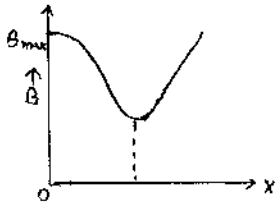
(c) $1.2 \times 10^{-3}\ \frac{m}{s^2}$ along y - axis

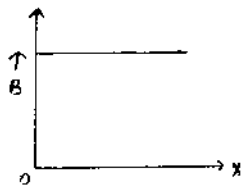
(d) $0.6 \times 10^{-3}\ \frac{m}{s^2}$ along y - axis

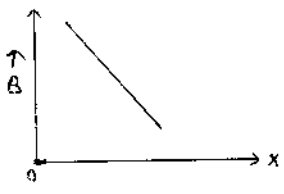


81. The correct curve between the magnetic induction (B) along the axis of a long solenoid due to current flow 1 in it and distance x from one end is

(a) 

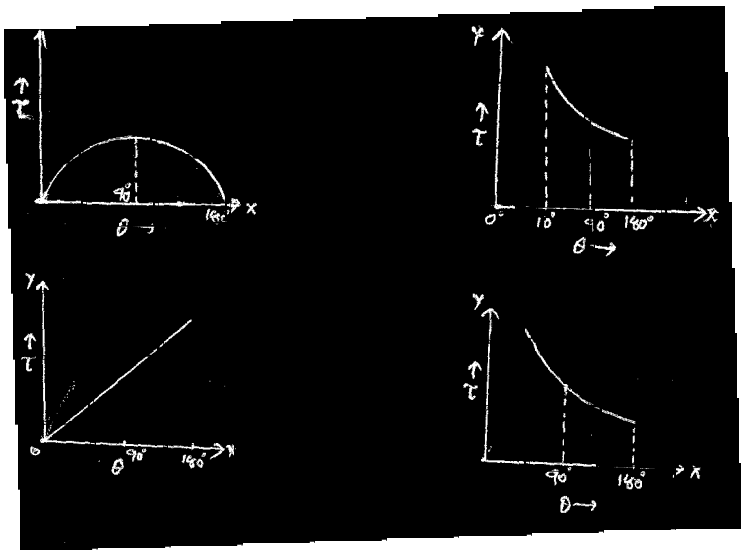
(b) 

(c) 

(d) 

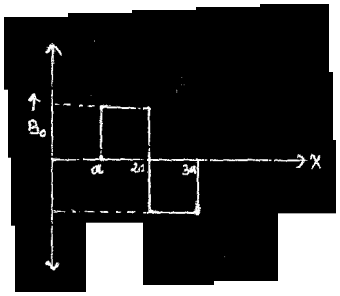
82. When any coil is placed in a uniform mag. field torque is acting on it. The graph of $\tau \rightarrow \theta$ is

(a)



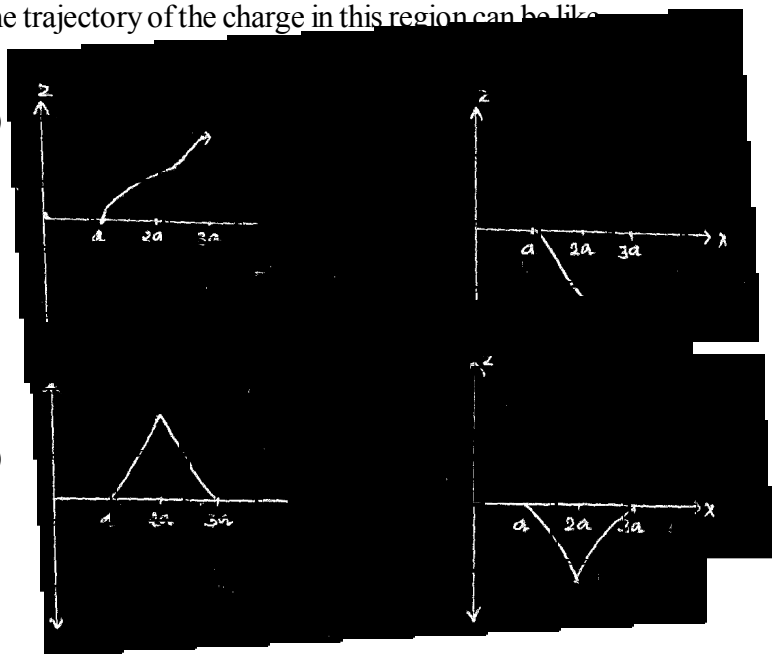
(c)

83. A magnetic field $\vec{B} = Bo\,\hat{j}$ exists in the region $a < x < 2a$ and $\vec{B} = -Bo\,\hat{j}$ in the region $2a < x < 3a$ where Bo is a positive constant. A positive point charge moving with a velocity $\vec{V} = Vo\uparrow$, where Vo is a positive constant, enters the magnetic field at x = a. The trajectory of the charge in this region can be like



(a)

(c)

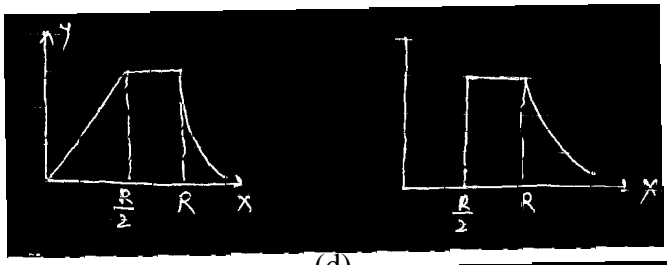


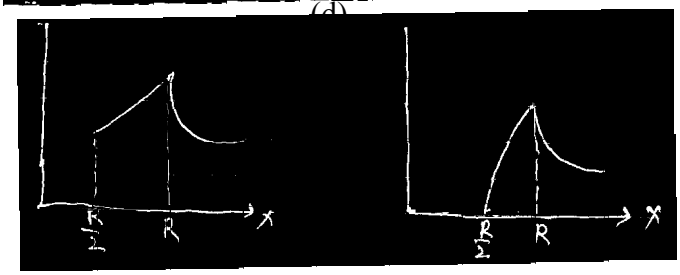
84. Graph of force per unit length between two long parallel current carrying conductors and the distance between them is .....

(a) Straight line (b) Parabola

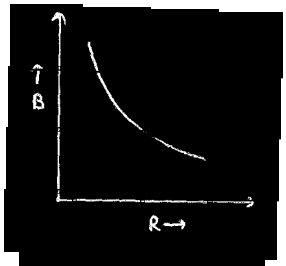
(c) Ellipse (d) Rectangular hyperbola

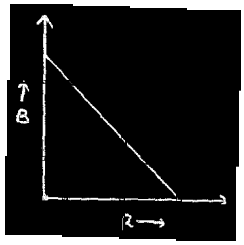
85. An infinitely long hollow conducting cylinder with inner radius $\frac{R}{2}$ and outer radius R carries a uniform current density along its length. The magnitude of the magnetic field $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by

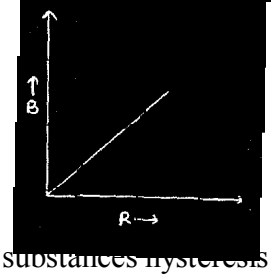
(a) 

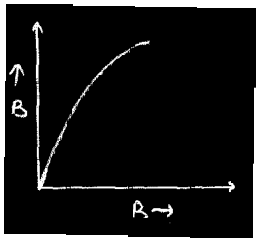
(c) 

86. A charge Q is uniformly distributed over the surface of non-conducting disc of radius R. The disc rotates about an axis perpendicular to its plane and passing tarough its centre with an angular velocity co. As a result of this rotation a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction at the centre of the disc will be represented by the figure.

(a) 

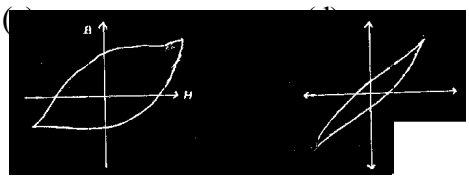
(b) 

(c) 

(d) 

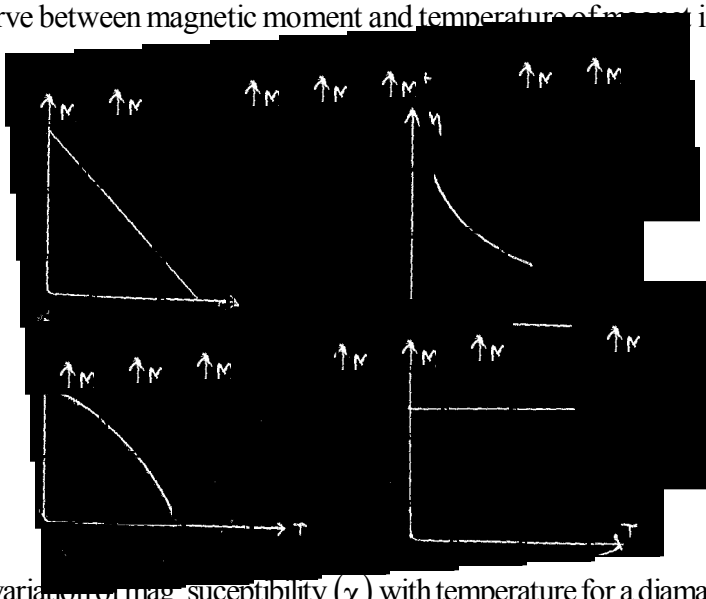
87. For substances hysteresis B-H curves are given as shown in the figure. For making temporary magnet which of the following group is best.

(a) 



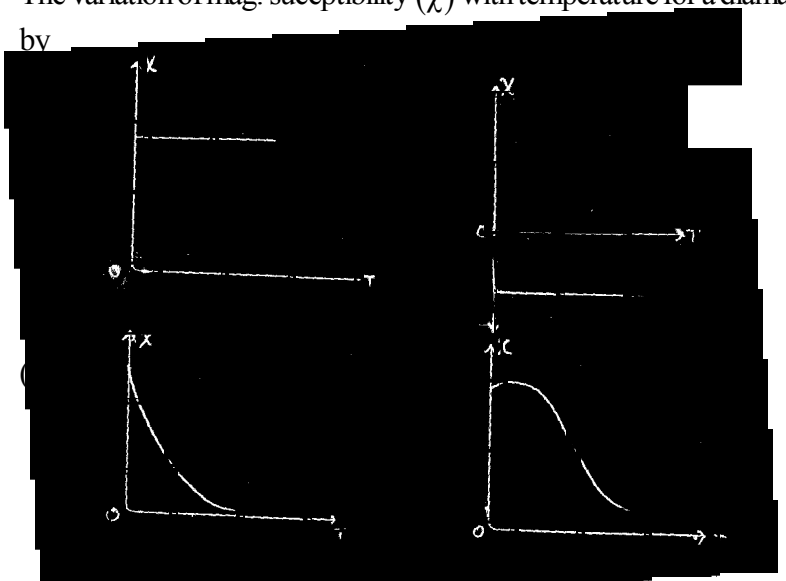
88. A curve between magnetic moment and temperature of magnet is

(a)



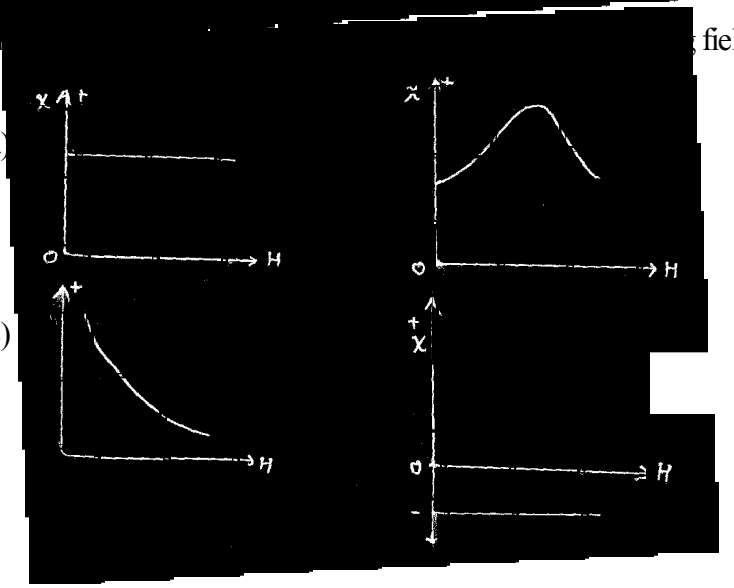
(c)

89. The variation of mag. suceptibility (χ) with temperature for a diamagnetic substance is best represented by



90. T field for a paramagnetic substance is

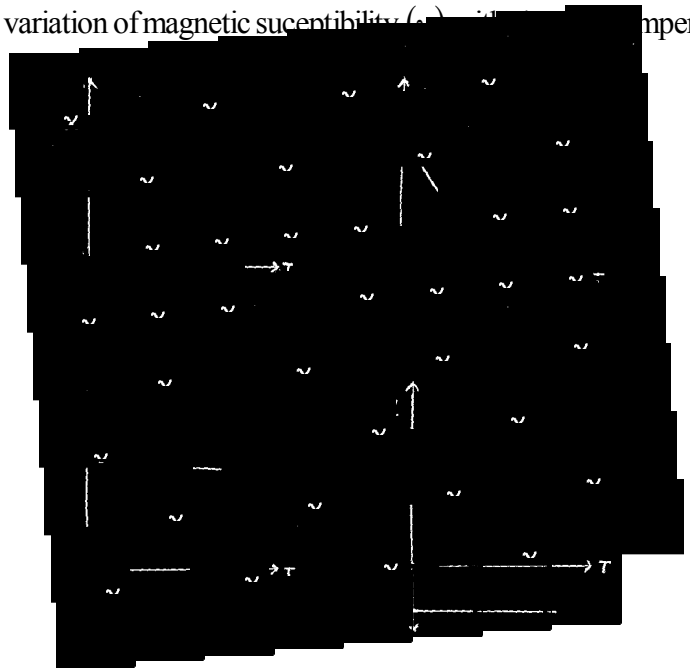
(a)



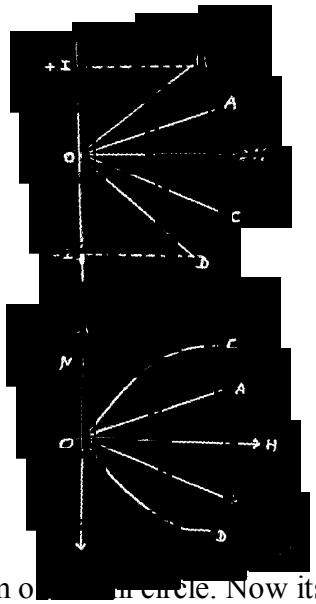
(c)

91. The variation of magnetic suceptibility (χ) with temperature T for a termomagnetic material is

(a)

(c) 

92. The variation of the intensity of magnetisation (I) with respect to the magnetising field (H) in a diamagnetic substance is described by the graph



(a) OD (b) OC

(c) OB (d) OA

93. The most appropriate magnetization M $\rightarrow$ magnetising field H curve for a paramagnetic substance is

(a) A (b) B

(c) C (d) D

94. An iron rod of length L and magnetic moment M is bent in the form of a semicircle. Now its magnetic moment will be

(a) M (b) $\frac{2M}{\pi}$

(c) $\frac{M}{\pi}$ (d) $M\pi$

95. Unit of magnetic Flux density is .....

(a) Tesla (b) $\frac{\text{Weber}}{\text{meter}^2}$

(c) $\frac{\text{Newton}}{\text{Amp}-\text{meter}}$ (d) All of the above

96. Magnetic intensity for an axial point due to a short bar magnet of magnetic moment M is given by

(a) $\frac{\mu_o}{4\pi}\frac{M}{d^3}$ (b) $\frac{\mu_o}{4\pi}\frac{M}{d^2}$

(c) $\frac{\mu_o}{2\pi}\frac{M}{d^3}$ (d) $\frac{\mu_o}{2\pi}\frac{M}{d^2}$

97. A magnet of magnetic moment M and pole strength m is divided in two equal parts, then magnetic moment of each part will be

(a) M (b) $\frac{M}{2}$

(c) $\frac{M}{4}$ (d) 2 M

98. If a magnet of pole strength m is divided into four parts such that the length and width of each part is half that of initial one, then the pole strength of each part will be

(a) $\frac{m}{4}$ (b) $\frac{m}{2}$

(c) $\frac{m}{8}$ (d) 4m

99. The magnetism of magnet is due to
(a) The spin motion of electron
(b) Earth
(c) Pressure inside the earth core region
(d) Cosmic rays

100. The magnetic field at a point x on the axis of a small bar magnet is equal to the at a point y on the equator of the same magnet. The ratio of the distances of x and y from the centre of the magnet is

(a) 2^{-3} (b) $2^{-\frac{1}{3}}$

(c) 2^{3} (d) $2^{+\frac{1}{3}}$

101. The magnetic field due to a short magnet at a point on its axis at a distance x cm from the middle point of the magnet is 200 gauss. The magnetic field at a point on the neutral axis at a distance x cm from the middle of the magnet is ..... gauss.

(a) 100 (b) 400

(c) 50 (d) 200

102. A bar magnet having a magnetic moment of 2×10^4 J T^{-1} is free to rotate in a horizontal plane. A horizontal magnetic field $B = 6 \times 10^{-4}$ Tesla exists in the space. The work done in taking the magnet slowly from a direction parallel to the field to a direction 60^0 from the field is

(a) 0.6 J (b) 12 J

(c) 6 J (d) 2 J

103. A magnet of length 0.1 m and pole strength 10^{-4} A.m. is kept in a magnetic field of 30 tesla at an angle of 30^0. The couple acting on it is ..... $\times 10^{-4}$ Joule.

(a) 7.5 (b) 3

(c) 1.5 (d) 6

104. In the case of bar magnet, lines of magnetic induction

(a) Start from the North pole and end at the South pole

(b) Run continuously through the bar and outside

(c) Emerge in circular paths from the middle of the bar

(d) Are produced only at the North pole like rays of light from a bulb.

105. A small bar magnet of moment M is placed in a uniform field of H. If magnet makes an angle of 30^0 with field, the torque acting on the magnet is

(a) MH (b) $\frac{MH}{2}$

(c) $\frac{MH}{3}$ (d) $\frac{MH}{4}$

106. The effective length of a magnet is 31.4 cm and its pole strength is 0.5 A.m. The magnetic moment, if it is bent in the form of a semicircle will be .....Amp.m^2.

(a) 0.1 (b) 0.01

(c) 0.2 (d) 1.2

107. A bar magnet of length 10 cm and having the pole strength equal 00.1×10^{-3} to is kept in a magnetic field having magnetic induction (B) equal to $4\pi \times 10^{-3}$ tesla. It makes an angle of 30^0 with the direction of magnefic induction. The value of the torque acting on the magnet is ..... Joule.

(a) $2\pi \times 10^{-7}$ (b) $2\pi \times 10^{-5}$

(c) 0.5 (d) 0.5×10^2

108. A small bar magnet has a magnetic moment 1.2 A.m^2. The magnetic field at a distance 0.1 m on its axis will be ..... tesla.

(a) 1.2×10^{-4} (b) 2.4×10^{-4}

(c) 2.4×10^4 (d) 1.2×10^4

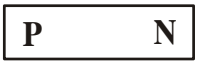
109. Force between two idential bur magnets whose centres are r meter apart is 4.8 N, when their axes are in the same line. If separation is increased to 2r, the force between them is reduced to

(a) 2.4 N (b) 1.2 N

(c) 0.6 N (d) 0.3 N

110. Two equal bar magnets are kept as shown in the figure. The direction of resultant mag. field indicated by arrow head at the point P is approximately

(1) 

P. 

(2)

(a) 

(b) 

(c) 

(d) 

111. A magnet of magnetic moment $50\uparrow$ A.m^2 is placed along the X-axis in a mag. field $\vec{B} = \left(0.5\uparrow + 3.0\hat{j}\right)$ Tesla. The torque acting on the magnet is ..... N.m.

(a) $175\hat{k}$ (b) $150\hat{k}$

(c) $75\hat{k}$ (d) $25\sqrt{5}\ \hat{k}$

112. A straight wire currying current I is turned into a circular Loop. If the magnitude of magnetic moment associated with it in MKs unit is M, the length of wire will be

(a) $4\pi MI$ (b) $\sqrt{\frac{4\pi M}{I}}$

(c) $\sqrt{\frac{4\pi I}{M}}$ (d) $\frac{M\pi}{4I}$

113. A bar magnet is 10 cm long and is kept with its North (N) pole pointing North. A neutral point is formed at a distance of 15 cm from each pole. Given the horizontal component of earth's field to be 0.4 Gauss. The pole strength of the magnet is ..... A.m.

(a) 9 (b) 6.75

(c) 27 (d) 1.35

114. The true value of angle of dip at a place is 60^0, the apparent dip in a plane inclined at an angle of 30^0 with magnetic meridian is.

(a) $\tan^{-1}\left(\frac{1}{2}\right)$ (b) $\tan^{-1}(2)$

(c) $\tan^{-1}\left(\frac{2}{3}\right)$ (d) None of these

115. A dip needle lies initially in the magnetic meridian when it shows an angle of dip at a place. The dip circle is roated through an angle x in the horizontal plane and then it shows an angle of dip θ'. Then $\frac{\tan\theta'}{\tan\theta}$ is .....

(a) $\frac{1}{\cos x}$ (b) $\frac{1}{\sin x}$

(c) $\frac{1}{\tan x}$ (d) $\cos x$

116. A dip needle vibrates in the vertical plane perpendicular to the magnetic meridian. The time period of vibration is found to be 2 sec. The same needle is then allowed to vibrate in the horizontal plane and the time period is again found to be 2 sec. Then the angle of dip is

(a) 0^0 (b) 30^0

(c) 45^0 (d) 90^0

117. Two identical short bar magnets, each having magnetic moment M are placed a distance of 2d apart with axes perpendicular to each other in a horizontal plane. The magnetic induction at a point midway between them is.

(a) $\sqrt{2}\ \frac{\mu_o}{4\pi}\frac{M}{d^3}$ (b) $\sqrt{3}\ \frac{\mu_o}{4\pi}\ \frac{M}{d^3}$

(c) $\sqrt{4}\ \frac{\mu_o}{4\pi}\ \frac{M}{d^3}$ (d) $\sqrt{5}\ \frac{\mu_o}{4\pi}\ \frac{M}{d^3}$

118. The magnetic suceptibility of a paramagnetic substance at $-73^\circ C$ is 0.0060, then its value at $-173^\circ C$ will be

(a) 0.0030 (b) 0.0120

(c) 0.0180 (d) 0.0045

119. Needles N_1, N_2 and N_3 are made of a ferrowmagnetic, a paramagnetic and a dia-magnetic substance respectively. A magnet when brought close to them will

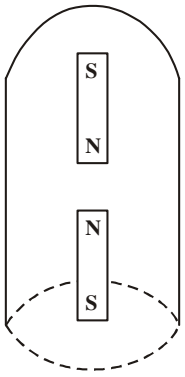
(a) Attract N_1 strongly, N_2 weakly and repd N_3 weakly

(b) Attract N_1 strongly, but repd N_2 and N_3 weakly

(c) Attract all three of them

(d) Attract N_1 and N_2 strongly but repd N_2

120. Two indential bar magnets with a length 10 cm and weight 50 gm weight are arranged freely with their like pole facing in a inverted vertical glass tube. The upper magnet hangs in the between the nearest pole of the magnet is 3 mm. Pole strength of the poles of earth magnet will be ..... Amp. meter



(a) 6.64 (b) 2

(c) 10.25 (d) None

121. Susceptibility of one material at 300 k is 1.2×10^{-5}. The temperature at which susceptibility will be 1.8×10^{-5} is ..... kelvin.

(a) 450 (b) 200

(c) 375 (d) None

122. Due to a small magnet, intensity at a distance x in the end on position is 9 Gauss. what will be the intensity at a distance $\frac{x}{2}$ on broad side on position.

(a) 9 Gauss (b) 4 Gauss

(c) 36 Gauss (d) 4.5 Gauss

123. A domain in a ferro magnetic substance is in the form of a cube of side length 1 μm. If it contains 8×10^{10} atoms and each atomic dipole has a dipole moment of 9×10^{-24} A.m^2 then magnetization of the domain is ..... A.m^{-1}.

(a) 7.2×10^{5} (b) 7.2×10^{3}

(c) 7.2×10^{-5} (d) 7.2×10^{-3}

124. The magnetic susceptibility is negative for

(a) Paramagnetic materials (b) Diamagnetic materials

(c) Ferromagnetic materials (d) Paramagnetic and ferromagnetic materials

125. When 2 Amp current is passed tarough a tangent galvanometer. It gives a deflection of 30^0. For 60^0 deflaction, the current must be

(a) 1 Amp (b) $2\sqrt{3}$ amp

(c) 4 amp (d) 6 Amp

126. The time period of a freely suspended magnet is a 4 seconds. If it is broken in length into two equal parts and one part is suspended in the same way, then its time period will be

(a) 4 sec (b) 2 sec

(c) 0.5 sec (d) 0.25 sec

127. A thin magnetic needle oscillates in a horizontal plane with a period T. It is broken into n equal parts. The time period of each part will be

(a) T (b) n^2 T

(c) $\frac{T}{n}$ (d) $\frac{T}{n^2}$

128. The plane of a dip circle is set in the geographic meridian and the apparent dip is δ_1. It is then set in a vertical plane per pendicular to the geographic meridian. The appartent dip angle is δ_2. The declination at the place is

(a) $\theta = \tan^{-1}\left(\tan\sqrt{1} \text{ X } \tan\sqrt{2}\right)$ (b) $\theta = \tan^{-1}\left(\tan\sqrt{1} + \tan\sqrt{2}\right)$

(c) $\theta = \tan^{-1}\left(\frac{\tan\sqrt{1}}{\tan\sqrt{2}}\right)$ (d) $\theta = \tan^{-1}\left(\frac{\tan\sqrt{2}}{\tan\sqrt{1}}\right)$

129. The coercivity of a bar magnet is 100 A/m. It is to be demagnetised by placing it inside a solenoid of length 100 cm and number of turms 50. The current flowing through the solenoid will be

(a) 4 A (b) 2 A

(c) 1 A (d) Zero

130. The angles of dip at two places are 30^0 and 45^0. The ratio of horizontal components of earth's magnetic field at the two places will be

(a) $\sqrt{3} : \sqrt{2}$ (b) $1 : \sqrt{2}$

(c) 1 : 2 (d) $1 : \sqrt{3}$

ASSERTION - REASON TYPE

Questions (Neet)

Read the assertion and reason carefully to mark the correct option out of the options given below.

(A) If both assertion and reason are true and the reason is the correct explanation of the asseration.

(B) If both assertion and reason are true but reason is not the correct explaination of the asseration,

(C) If assertion is true but reason is false.

(D) If the assertion and reason both the false.

(E) If assertion is false but reason is true.

131. Assertion : We cannot think of magnetic field configuration with three poles.

Reason : A bar magnet does exert a torque on itself due to its own field.

132. Assertion : If a compass needle be kept at magnetic north pole of the earth, the compass needle may stay in any direction.

Reason : Dip needle will stay vertical at the north pole.

133. Assertion : Dia-magnetic materials can exhibit magnetism.

Reason : Dia-magnetic materials have permanent magnetic dipole moment.

134. Assertion : A paramagnetic sample displays greater magnetisation when it is cooled.

Reason : The magnetisaution does not depend on temperature.

135. Assertion : Two short magnets are placed on a cork which floats on water. The magnets are placed such that the axis of one produced bisects the axis of other at right angles. Then the cork has neither translational nor rotational motion.

Reason : Not force on the cork is zero.

136. Assertion : Cyclotorn does not accelerate electron.

Reason : Mass of the electron is very small.

137. Assertion : Cyclotron is a device which is used to accelerate the positive ion.

Reason : Cyclotron frequency depends upon the veolocity.

138. Assertion : The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area

Reason : The magnetic field inside the solenoid is uniform.

139. Asseration : Torque on the coil is the maximum, when coil is suspended in a radial magnetic field.

Reason : The torque tends to rotate the coil on its own axis.

Comprehension Type Questions (For JEE)

Passage - 1

Advanced countries are making use of powerful electro magnets to move trains at very high speed. These trains are called maglev trains. These trains float on a guideway and do not run on steel tail trucks.

Instead of uising an engine based on fossile fuels, they make use of magnetic field forces. The magnetized coil are arranged in the guideway which repel the strong magnets placed in the train's under carriage. This helps train move over the guideway, a technique called electrodynamics suspension. When current passes in the coils guideway, a typical magnetic field is set up between the under carriage of train and guideway which pushes and pulls the train along the guideway depending on the requirement.

The lack of friction and its aerodynamic style allows the train to move at very high speed.

140. The force which makes manglev move is

(a) Gravitational (b) Magnetic

(c) Nuclear forces (d) Air drag

141. The disadvantage of magtev train is

(a) More friction (b) Less pollution

(c) Less wear and tear (d) High initial cost

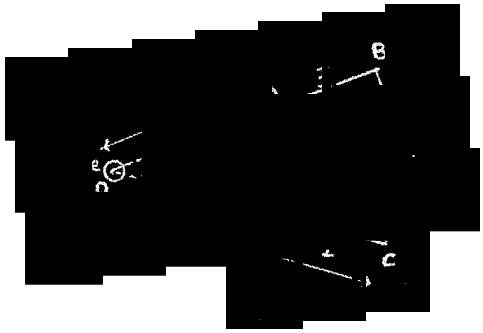
142. The levitation of the train is due to

(a) Mechnical tone

(b) Electros static attraction

(c) Electrostatic repulsion

(d) Magnetic repulsion



Passage - 2

A current Loop ABCD is held fixed on the plane of the paper as shown in the figure. The arcs BC (radius = b) and DA (radius = a) of the Loop are joined by two straight wires AB and CD. A steady current I is flowing in the Loop. angle made by AB and CD at the origin is 30^0. Another straight thin wire with steady current I_1 flowing out of the plane of the paper is kept at the origin.

143. The magnitude of the magnetic field (B) due to the Loop ABCD at the origin (0) is

(a) zero

(b) $\frac{\mu_0 I\ (b\ -\ a)}{24\ ab}$

(c) $\frac{\mu_0 I\ (b\ -\ a)}{4\ \pi\ ab}$

(d) $\frac{{}_0 I}{4\ \pi}\left[2\ (b - a) + \frac{\pi}{3}\ (a + b)\right]$

144. Due to the presence of the current I_1 at the origin

(a) The forces on AB and DC are zero.

(b) The forces on AD and BC are zero.

(c) The magnitude of the net force on the looop is given by $\frac{\mu o I\ I_1}{4\pi}\left[2\ (b - a) + \frac{\pi}{3}\ (a + b)\right]$

(d) The magnitude of the net force on the loop is given by $\frac{\mu o I\ I_1}{24\ ab}\ (b - a)$

Matching Type Questions

In each of the following questions, Match column-I and column-II and select the correct match out of the four given choices.

145.

Coulmn - I	**Coulmn - II**
(A) Biot-savart's law	(P) Direction of magnetic field induction
(B) Right hand thumb rule	(Q) Magnitude of magnetic field induction
(C) Fleming's left hand rule	(R) Direction of induced current
(D) Fleming's right and rule	(S) Direction of force due to a mag. field.

(a) $A \to Q;\ B \to P;\ C \to R;\ D \to S$

(b) $A \to Q;\ B \to P;\ C \to S;\ D \to R$

(c) $A \to P;\ B \to Q;\ C \to R;\ D \to S$

(d) $A \to P;\ B \to Q;\ C \to S;\ D \to R$

146.

	Column - I	Column - II
(A)	Magnetic field induction due to current 1 through straight conductor at a perpendicular distance r.	(P) $\frac{\mu_o I}{2r}$
(B)	Magnetic field induction at the centre of current (1) carrying Loop of radius (r)	(Q) $\frac{\mu_o I}{4\pi r}$
(C)	Magnetic field induction at the axis of curreint (1) carrying coil of radius (r) at a distance (r) from centre of coil.	(R) $\frac{_o}{4\pi}\ \frac{2I}{r}$
(D)	Magnetic field induction at the centre due to circular arc of length r and radius (r) carrying current (I).	(S) $\frac{\mu_o}{4\sqrt{2}}\ \frac{I}{r}$

(a) $A \rightarrow R;\ B \rightarrow S;\ C \rightarrow P;\ D \rightarrow Q$

(b) $A \rightarrow R;\ B \rightarrow P;\ C \rightarrow S;\ D \rightarrow Q$

(c) $A \rightarrow P;\ B \rightarrow Q;\ C \rightarrow S;\ D \rightarrow R$

(d) $A \rightarrow Q;\ B \rightarrow P;\ C \rightarrow R;\ D \rightarrow S$

147.

	Coulmn - I	Coulmn - II
(A)	Moving coil Galvanometer	(P) Low resistance
(B)	Ammeter	(Q) Moderate resistance
(C)	Voltmeter	(R) High, Low or moderate resistance
(D)	Avometer	(S) High resistance

(a) $A \rightarrow P;\ B \rightarrow Q;\ C \rightarrow R;\ D \rightarrow S$

(b) $A \rightarrow P;\ B \rightarrow Q;\ C \rightarrow S;\ D \rightarrow R$

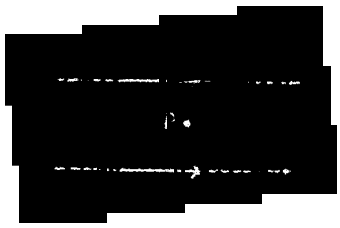
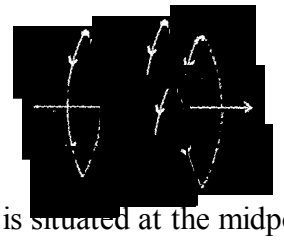
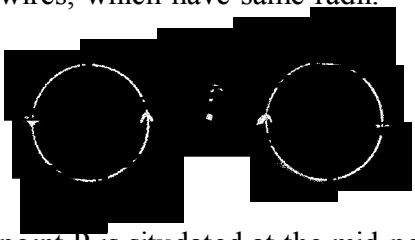
(c) $A \rightarrow Q;\ B \rightarrow P;\ C \rightarrow R;\ D \rightarrow S$

(d) $A \rightarrow Q;\ B \rightarrow P;\ C \rightarrow S;\ D \rightarrow R$

Matrix Match Type Questions

In this section each equation has some statements (A, B, C, D.....) given in column-I and some statements (P, Q, R, S, T,.....) in column-II. Any given statement in column-I can have correct matching with ONE OR MORE statements (s) in column-II.

148. Two wires each carrying a steady current - I are shown in four configureations in coulmn-I. Some of the resulting effects are described in coulmn-II. Match the statements in column-I with the statements in column-II.

Coulmn - I	**Coulmn - II**
(A)  point P is situated midway between the wires	(P) The mag. fields at P due to the currents in the wires are in the same direction.
(B)  point P is situated at the midpoint of the line joining the centres of the circular wires, which have same radii.	(Q) The magnetic fields at P due to the currents in the wires are in opposite directions.
(C)  point P is situdated at the mid-point of the two Loops.	(R) There is no magnetic field at P.
(D)  point P is situated at the common centre of the wires.	(S) The wires repel each other.

149. The physical quantities are given in column-I and their various related factors in column-II.

	Coulmn - I		**Coulmn - II**
(A)	Torque on a coil carrying current when held in a mag. field .	(P)	Restoring torque per unit twist of the suspension strip (K).
(B)	current sensitivity of galvanometer	(Q)	Number of turns in the coil (N)
(C)	voltage sensitivity of galvanometer	(R)	Magnetic field (B)
(D)	figure of merit ofgalvonometer	(S)	Area of the coil (A)

150.

	Coulmn - I		**Coulmn - II**
(A)	A chared particle moving parallel to direction of mag. field	(P)	undeflected
(B)	Acharged particle moving perpendicular to the direction of magnetic field	(Q)	circular path
(C)	A charged particle moving at an angle in a region of strong mag. field	(R)	Helical path
(D)	A charged particle moving in a strong and uniform electric field of large region	(S)	parabolic path

KEY NOTE

1	B	26	B	51	C	76	B	101	A	126	B
2	C	27	D	52	A	77	C	102	C	127	C
3	D	28	A	53	B	78	B	103	C	128	C
4	B	29	B	54	C	79	D	104	B	129	B
5	D	30	B	55	C	80	B	105	B	130	A
6	C	31	C	56	C	81	A	106	A	131	D
7	B	32	A	57	B	82	A	107	A	132	B
8	A	33	B	58	C	83	A	108	B	133	C
9	C	34	A	59	B	84	D	109	D	134	C
10	C	35	B	60	D	85	D	110	B	135	A
11	D	36	C	61	A	86	A	111	B	136	A
12	A	37	A	62	C	87	D	112	B	137	C
13	C	38	B	63	A	88	C	113	D	138	B
14	A	39	C	64	A	89	B	114	B	139	D
15	D	40	B	65	B	90	A	115	A	140	B
16	B	41	A	66	A	91	A	116	C	141	D
17	A	42	D	67	C	92	B	117	D	142	D
18	A	43	B	68	B	93	A	118	B	143	B
19	C	44	B	69	C	94	B	119	A	144	B
20	C	45	C	70	B	95	D	120	A	145	B
21	C	46	C	71	D	96	C	121	B	146	B
22	D	47	B	72	B	97	B	122	C	147	D
23	D	48	C	73	C	98	B	123	A	148	ઉકેલ જુઓ
24	C	49	D	74	B	99	A	124	B	149	ઉકેલ જુઓ
25	A	50	C	75	B	100	D	125	D	150	ઉકેલ જુઓ

HINT

1. $d\ell = dx = 10^{-2}$ m ; I = 10 Amp ; r = 0.5 m.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\ \vec{d\ell} \times \vec{r}}{r^3}$$

$= 4 \times 10^{-8}\ \hat{k}$ Tesla.

2. Point "P" is lying symmetrically w.r.t.the two long wires

$$B_1 = \frac{\mu_O}{2\pi} \frac{I_1}{a} \ ; \ B_2 = \frac{\mu_O}{2\pi} \frac{I_2}{a}$$

$$B = \sqrt{B_1^{\ 2} + B_2^{\ 2}}$$

$$= \frac{\mu_O}{2\pi\ a} \left(I_1^{\ 2} + I_2^{\ 2}\right)^{\frac{1}{2}}$$

3. mag. field inside the wire $B \alpha r$

mag. field outside the wire $B \alpha \frac{1}{r}$

4. $\frac{\mu_O}{4\pi} \frac{I}{R} (\theta) = \frac{\mu_O}{4\pi} \times \frac{3}{0.6} \times \frac{\pi}{6} = 2.6 \times 10^{-7}$ T

5. $B_3 = B_5 = B_7 = 0$

$\overrightarrow{B_2} = \frac{\mu_O}{4\pi} \frac{I}{3r}$ going inside

$\overrightarrow{B_4} = \frac{\mu_O}{4\pi} \frac{I}{2r}$ θ comming outside

$\overrightarrow{B_6} = \frac{\mu_O}{4\pi} \frac{I}{r}$ θ going inside

$\therefore$ total mag. field $\vec{B} = -\overrightarrow{B_2} + \overrightarrow{B_4} - \overrightarrow{B_6}$

6. $B = N \left(\frac{\mu_O\ I}{2\ \pi}\right) \Rightarrow B \ \alpha \ \frac{N}{r} \Rightarrow \frac{B_1}{B_2} = \frac{N_1}{r_1} \times \frac{r_2}{N_2}$

$B_2 = 4B_1$

techanique $B_2 = n^2\ Bl$

$= (2)^2\ Bl$

$= 4Bl$

7. only outside the pipe

Hollow copper pipe I = 0

$$\therefore \quad B = \frac{\mu_O}{2\pi}\frac{I}{r} = 0$$

i.e. Inside mag.fieldis ZERO

8. $$B = \frac{\mu_O}{2\pi}\frac{I}{y} \Rightarrow \quad \frac{B_1}{B_2} = \frac{y_2}{y_1}$$

$$\therefore B \alpha \frac{1}{y} \qquad \frac{10^{-8}}{B_2} = \frac{12\times10^{-2}}{4\times10^{-2}}$$

$$B_2 = 3.33\times10^{-9} \text{ tesla}$$

9. $$B = \frac{\mu_O}{2\pi}\frac{I}{y} \Rightarrow B \alpha \frac{1}{y} \Rightarrow \frac{B_1}{B_2} = \frac{y_2}{y_1}$$

$$\Rightarrow B_2 = 2B_1$$

10. $$B = \frac{\mu_O}{2}\frac{I}{a} \Rightarrow B \alpha I$$

11. $\theta_1 = \theta_2 = 0$

$$B = \frac{\mu_O}{4\pi}\frac{I}{y}\left[\sin\theta_1 + \sin\theta_2\right]$$

12. I = 5 Amp.

13. (c)

$$\left(B_{Wire}\right)_{r<9} = \left(B_{Wire}\right)_{r>9}$$

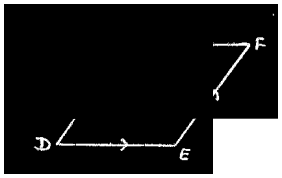
$$\left(\frac{\mu_O I}{2\pi\ 9^2}\right)\frac{9}{2} = \frac{\mu o}{2\pi}\frac{I}{29}$$

$$B_1 = B_2$$

$$\frac{B_1}{B_2} = 1$$

14. $$B = \frac{\mu o}{2\pi}\frac{I}{y} \Rightarrow B \alpha \frac{1}{y} \Rightarrow \frac{B_1}{B_2} = \frac{y_2}{y_1} \qquad \Rightarrow B_2 = 1\times10^{-2}$$

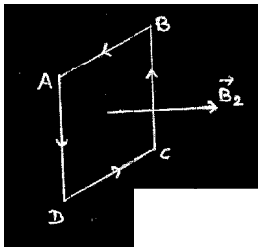
15. CDEF plane is in XY plane



As shown in fig. $\overrightarrow{B_1}$ is along +Z axis i.e. $\hat{k}$

ABCD plane is in YZ - plane

$\therefore$ $\overrightarrow{B_2}$ is along + x - axis i.e. $\uparrow$



$\therefore$ direction of resultant mag. field is in the direction of P(q, o, a) is along unit vector of $\hat{1}+\hat{k}$

$$\frac{\uparrow+\hat{k}}{\left|\uparrow+\hat{k}\right|}=\frac{\uparrow+\hat{k}}{\sqrt{(1)^2+(1)^2}}=\frac{\uparrow+\hat{k}}{\sqrt{2}}$$

16. $I=\frac{Q}{t}=\frac{2\times1.6\times10^{-19}}{2}=1.6\times10^{-19}$ Amp

$$B=\frac{\mu_O}{2a}=\mu_O\times10^{-19}\ \text{Tesla.}$$

17. Mag. field due to ABandCD wire ZERO.

$$B_{BC}=\frac{\mu_O}{4\pi}\frac{I}{R_1}\cdot\theta\quad\text{going inside}$$

$$B_{AD}=\frac{\mu_O}{4\pi}\frac{I}{R_2}\cdot\theta\quad\textit{Outside}$$

$\therefore$ $B = B_{BC} - B_{AD}$

$$=\frac{\mu_O}{4\pi}I\cdot\theta\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

18. Mag.fielddue to ABand DE wire is ZERO.

$$B_{BCD}=\frac{\mu_O}{4\pi}\frac{I}{R_1}\cdot\theta=\frac{\mu_O}{4\pi}\frac{I}{R_1}\times\pi=\frac{\mu_O I}{4\ R_1}$$

$$B_{EFA}=\frac{\mu_O I}{4\ R_2}$$

$\therefore$ $B = B_{BCD} + B_{EFA}$

$$=\frac{\mu_O I}{4}\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$$

20. $B = N\left(\frac{\mu_O I}{2a}\right) \Rightarrow N = 50$

21. $n = 50\ \frac{\text{turns}}{\text{cm}} = 5000\ \frac{\text{turns}}{\text{meter}}$

$B_{Inside} = \mu_O nI$

$= 25.1 \times 10^{-3}$ tesla

$Bend = \frac{\mu_O nI}{2}\left[\sin 0^o + \sin 90^o\right]$

$= \frac{\mu_O nI}{2}$

22. $\frac{B_1}{B_2} = \left(\frac{R^2}{x^2 + R^2}\right)^{\frac{3}{2}}$ $\qquad 2^{-1} = \left(\frac{R^2}{x^2 + R^2}\right)^{\frac{1}{2}}$

$\frac{1}{8} = \left(\frac{R^2}{x^2 + R^2}\right)^{\frac{3}{2}}$ $\qquad \frac{1}{4} = \frac{R^2}{x^2 + R^2}$

$2^{-3} = \left(\frac{R^2}{x^2 + R^2}\right)^{\frac{3}{2}}$ $\qquad x = \sqrt{3}\ R$

23. $B = \frac{\mu oI}{2a} \Rightarrow 12.56 = \frac{4\pi \times 10^{-7} I}{2 \times 5.2 \times 10^{-11}}$

24. $f_{mag} = f_{grav.}$

$BI\ell = mg \quad I = \frac{mg}{B\ell} \qquad (\theta = 90^o)$

$= 5$ Amp.

25. $f_{mag} = f_{gra.}$

$B = \frac{mg}{I\ell}$

$= \frac{2}{3}$ tesla

26. $B = \mu_O nI = 6.28 \times 10^{-2} = 2\pi \times 10^{-2}$ *tesla*

27. for smaller Loop $B_1 = \frac{\mu_O I_1}{2\ r_1}$ ....(1)

for Bigger Loop $B_2 = \frac{\mu_O I_2}{2\ r_2}$ ....(2)

but $r_1 < r_2 \Rightarrow B_1 > B_2$

$\therefore\ B = B_1 - B_2$ ....(3)

$= \frac{\mu_O}{2} \left[\frac{I_1}{r_1} - \frac{I_2}{r_2} \right]$

$B = \frac{\mu_O}{2} \left[\frac{I_1}{r_1} - \frac{I_2}{2r_1} \right] \qquad (\because\ r_2 = 2r_1)$

$B = \frac{1}{2} B_1$ From the data

$\frac{1}{2} B_1 = \frac{\mu_O}{2r_1} \left[I_1 - \frac{I_2}{2} \right]$

$\therefore\ \frac{I_2}{I_1} = 1$

28. $B_{Loop} = B_{wire}$

$\frac{\mu_O\ I_C}{2\ R} = \frac{\mu_O\ I_e}{2\ \pi\ H}$

$\therefore\ H = \frac{Ie \cdot R}{\pi I_C}$

29. 90°

$dB = \frac{\mu_O}{4\ \pi} \frac{I\ d\ell\ \sin\ \theta}{r^2}$

where $\theta = 90^o$; dB becomes maximum

30. $B = n^2 Bo$

$= (3)^2 Bo$

$= 9 Bo$

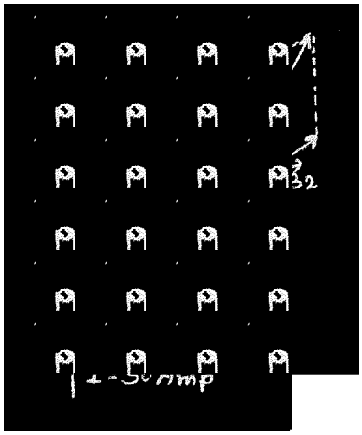
31. $B_1 = 4 \times 10^{-4}$ tesla

(parallel to wire)

$B_2 = \frac{\mu_O}{2\pi} \frac{I}{y}$

$= 3 \times 10^{-4}$ tesla

$= 5 \times 10^{-4}$ tesla.



32. $B = B_1 = B_2 = \frac{\mu_O}{2} \frac{I}{a}$

$B_{net} = \sqrt{B_1^2 + B_2^2}$

$= \sqrt{2B}$

$\therefore \frac{B}{B_{net}} = \frac{1}{\sqrt{2}}$

33. for 1turn $B = \frac{\mu_O}{2} \frac{I}{r}$ where $\ell = 2\pi r \Rightarrow r = \frac{\ell}{2\pi}$

for n turn $B_n = \frac{\mu o I}{2r'}$ where $\ell = n(2\pi r')$

$B_n = n \left(\frac{\mu_O}{2} \frac{I}{\frac{r}{n}} \right) \quad r' = \frac{r}{n}$

$B_n = n^2 B$

34. $\frac{B\,Centre}{B\,axis} = \left(1 + \frac{x^2}{a^2}\right)^{\frac{3}{2}}$

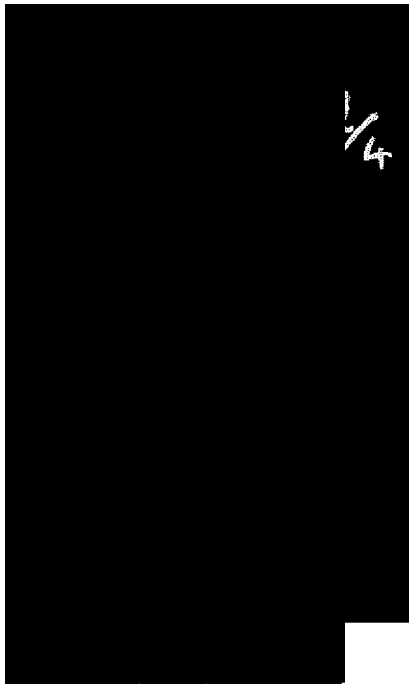
$Bcentre = 250\,\mu T$

35. $B = n\left(\frac{\mu_O I}{2a}\right)$

$B \alpha nI$

36. Suppose length of the wire is ℓ

$$\text{Asquare} = \left(\frac{\ell}{4}\right)\left(\frac{\ell}{4}\right) = \frac{\ell^2}{16}$$



$\therefore$ magnetic moment $M_{Square} = I A_{Square}$

$$= \frac{I\ell^2}{16}$$

$$\ell = 2\pi r \Rightarrow r = \frac{\ell}{2\pi}$$

$$\therefore \text{ Acircle } = \pi r^2 = \frac{\pi\ell^2}{4\pi^2}$$

$$= \frac{\ell^2}{4\pi}$$

$\therefore$ magnetic moment $M_{Circle} = 1\ A_{Circle}$

$$= \frac{I\ell^2}{4\pi}$$

$\text{eq} - (1) \div (2)$

$$\frac{M_{Square}}{M_{Circle}} = \frac{\pi}{4}$$

37. $B = \sqrt{B_1^2 + B_2^2}$

$$= \frac{\mu_O}{2\ r}\ \sqrt{I_1^2 + I_2^2}$$

$= 5 \times 10^{-5}$ tesla. suppose point "p" is at same r distance from the wires.

38. When I_1 and I_2 are in the same direction

$$\frac{\mu_O}{2\pi}\ \frac{I_1}{r} - \frac{\mu_O}{2\pi}\ \frac{I_2}{r} = 10\ \mu T$$

when I_1 and I_2 are in the opposite direction

$$\frac{\mu_O}{2\pi}\frac{I_1}{r} + \frac{\mu_O}{2\pi}\frac{I_2}{r} = 30\ \mu T$$

solve above two equations

$I_1 = 20$ Amp and $I_2 = 10$ Amp

$$\therefore \frac{I_1}{I_2} = 2$$

39. from the fig, the distance between two wires is = (6-2) = 4 cm

$|B_1| = |B_2|$

$$\frac{\mu_O}{2\pi}\frac{I_1}{x \times 10^{-2}} = \frac{\mu_O}{2\pi}\frac{I_2}{(4 - x) \times 10^{-2}}$$

$\therefore$ x = 10 m

$\therefore$ Location of point on sicle = 2 + 1 = 3 cm mark.

40. $$B = \frac{\mu_O nI}{2}\left[\sin 0^o + \sin\frac{\pi}{2}\right] = \frac{\mu onI}{2} = 8 \times 10^{-4} \text{ tesla.}$$

41. $$n = 10\ \frac{\text{turns}}{\text{m}} = 1000\ \frac{\text{turns}}{\text{metre}} \Rightarrow B = \mu_O nI = 2\pi \times 10^{-3} \text{ Tesla}$$

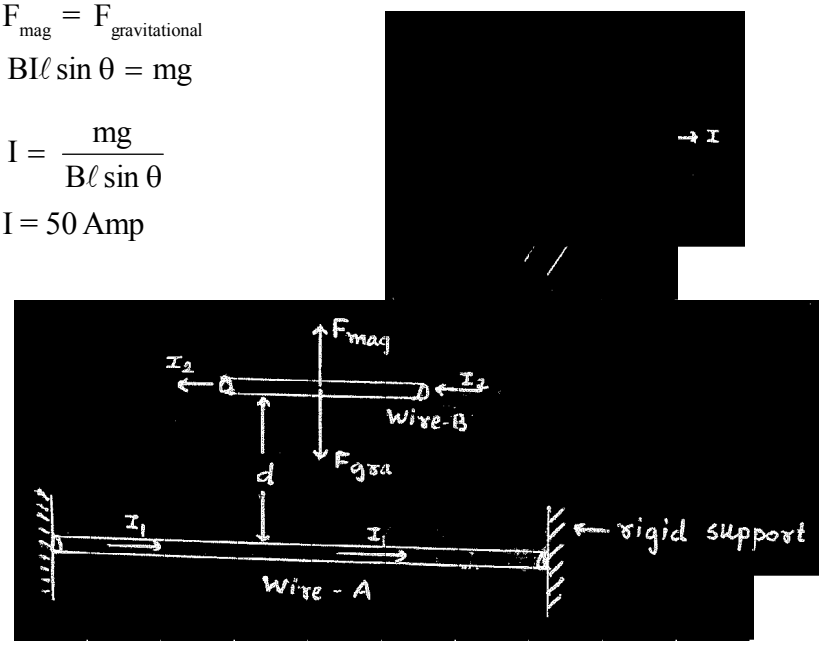
42. $F_{mag} = F_{gravitational}$

$BI\ell \sin\theta = mg$

$$I = \frac{mg}{B\ell \sin\theta}$$

$I = 50$ Amp

43.



$$\frac{F_{mag}}{L} = \frac{\mu_O}{2\pi}\frac{I_1 I_2}{d} \text{ (acting in upward dir}^{n}\text{)}$$

$$\frac{F_{mag}}{L} = \frac{Mg}{L}$$

$$\frac{\mu_O}{2\pi}\ \frac{I_1\ I_2}{d} = \frac{Mg}{L}$$

$$\frac{\mu_O}{2\pi}\ \frac{I_1\ I_2}{d} = 75 \times 10^{-3}$$

$$\therefore\ d = \frac{1}{3} \times 10^{-2} \text{ meter}$$

44. magnetic moment $M = IA$

$= (1)\ \pi\ (0.2)^2$

$= 0.04\pi$ Amp.m^2 $(\perp$ to the plane Inside$)$

$\therefore\ \left|\vec{\tau}\right| = MB\sin\theta$ $\qquad\qquad$ $\theta =$ Angle bet^{-n} M and

$= 0.25$ N.m

45. mag. moment $\vec{M} = N\ (IA)\ \hat{n}$

$= 16 \times 10^{-2} \uparrow$ Amp.m^2

$\therefore\ \vec{\tau} = \vec{M} \times \vec{B}$

$= 5.66 \times 10^{-5}\ \hat{k}$

46. $r = \sqrt{\frac{2mk}{qB}}$

$r\ \alpha\ \sqrt{m} \Rightarrow \frac{m_1}{m_2} = \left(\frac{R_1}{R_2}\right)^2$

47. $B = \frac{m\vartheta}{qr} \Rightarrow 5.6 \times 10^{-5}$ tesla

48. $r = \sqrt{\frac{2mk}{qB}}$

$r\ \alpha\ \frac{\sqrt{m}}{q}$ $\qquad\qquad$ $(\because$ k. E and B are same$)$

$\frac{rp}{r\alpha} = 1 \Rightarrow r_p = r_\alpha$

49. $F = Bqv$

$= Bq\sqrt{\frac{2E}{M}}$

$= 7.6 \times 10^{72}$ N

50. path of the proton will be a helix of radius $r = \frac{mv\sin\theta}{qB}$

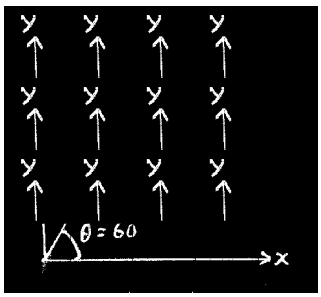
where

θ = Angle bet^{-n} $\vec{B}$ and $\vec{v}$

$\theta = 30^{o}$

r = 0.1 meter

* time period $T = \frac{2\pi m}{qB} = 2\pi \times 10^{-7}$ Sec



51. when particle entres at angles other than 0° or 90° or 180°, path followed is helix.

52. $r_p = \frac{\sqrt{2mk}}{e.B}$

$r_d = \sqrt{\frac{2(2m)k}{eB}} = \sqrt{2}\ r_p$

$r_\alpha = \sqrt{\frac{2\ (4m)k}{(2e)\ B}} = r_p$

$r_\alpha = r_p < r_d$

53. $r = \frac{mv}{qB} \Rightarrow r\ \alpha\ mv$

$\therefore\ r_A > r_B$

$\therefore\ m_A v_A > m_B v_B$

54. $r = \frac{mv}{qB}$

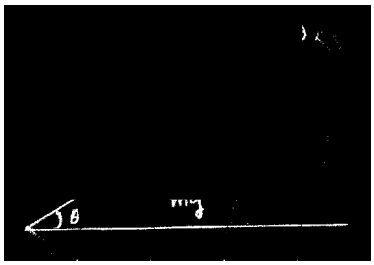
since both have same momentum,therefore the circularpath of both will have the same radius.

55. wire ABis placed in non-uniform mag. field generated by CD wire hence ABwire will perform translational and rotational motion.

56. The cross-section of a rod is appears as a circle.

The rod will move with a constant speed v if the net force on the rod is zero.

$BI\ell \cos\theta = mg \sin\theta$



$$B = \frac{mg}{I\ell}\frac{\sin\theta}{\cos\theta}$$

$$B = \frac{mg\ \tan\theta}{I\ell}$$

57. $$r = \frac{\sqrt{2m_1 E k_1}}{Bq_1} = \frac{\sqrt{2m_2 Ek_2}}{Bq_2}$$

$$Ek_2 = \frac{m_1}{m_2}\frac{q_2}{q_1} Ek_1$$

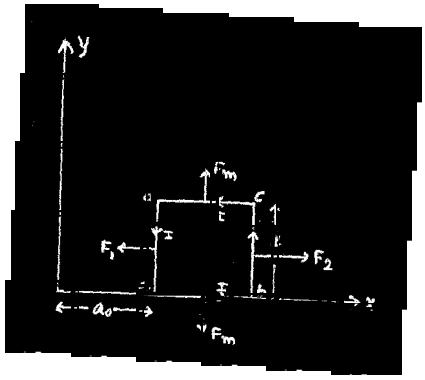
$$= \frac{2m}{m} \times \frac{q}{q} \times 50 \text{ keV}$$

$$= 2 \times 50$$

$$= 100 \text{ keV}$$

58. The mag. force (Fm) on wire ab and cd is equal and opposite, hencecencelled each other

$$F_1 = F_{ad} = BoI\left[1 + \frac{90}{\ell}\right]\hat{k}$$



$$F_2 = F_{cb} = BoI\left[1 + \frac{90 + \ell}{\ell}\right]\hat{k}$$

$$F = F_1 - F_2$$

$$= Bo.I.\ell$$

59. $$F = \frac{\mu_O}{2\pi}\frac{I_1\ I_2\ \ell}{y}$$

$F \alpha I_1\ I_2$

4 times

60. $F = BI\ell$

$= 6 \times 10^{-4}$ east to west

61. $S_1 = \dfrac{G.\ I_G}{I - I_G}$; $S_2 = \dfrac{G.I_G}{2I - I_G}$

$\therefore \dfrac{S_1}{S_2} = \dfrac{2I - I_G}{I - I_G}$

62. If x is the resistance of ammeter then

$10 = 2\left(x + R\right) \Rightarrow \dfrac{10}{2}\ x + R$

$V = I\left(R\right)\ \ 5 = x + R$

$x = 5 - R$

$\therefore$ x is less than 5Ω

63. $I = 50$ k ; $I_G = 20$ k

where K = figure of merit

$S = \dfrac{G.\ I_G}{I - I_G}$

$G = 18\Omega$

64. ACC to work-energy theorem

$qV = \dfrac{1}{2}\ mv^2$ (for ele. field)

$BqV = \dfrac{mv^2}{R}$ (for mag.field)

$v = \dfrac{BqR}{m}$ ...(2)

sub eq.(2) in (1)

$qV = \dfrac{1}{2}\ m\ \dfrac{B^2\ q^2\ R^2}{m^2}$

$V = \dfrac{B^2\ R^2}{2}\ \dfrac{q}{m}$

$$\frac{q}{m} = \frac{2V}{B^2 R^2}$$

$$\frac{q}{m} \alpha \frac{1}{A^2}$$

65. total initial resistance = G + R

= 50 + 2950

$= 3000\Omega$

$$\therefore\ I = \frac{V}{G + R} = \frac{3}{3000} = 0.001 \text{ Amp}$$

Let x be the effective resistance of the circuit

$$3 \text{ volt} = 3000 \times 0.001 = x \times \frac{20}{30} \times 0.001$$

$x = 4500\ \Omega$

$\therefore$ resistance to be added $= 4500 - 50$

$= 4450\ \Omega$

66. $$S = \frac{G\ I_G}{-I_G} = \frac{4 \times 10^{-3} \times 15}{6 - \left(4 \times 10^{-3}\right)} = 10 \times 10^{-3} = 10\ m\ \Omega$$

above shunt resistance should be connected in parallel

67. $I_G \cdot G = (I - I_G) \cdot S$

$$G = \frac{(I - I_G) \cdot S}{I_G} = S = 40\ \Omega$$

68. In absence of mag. field

mg = 2 Kyo .....(1)

From the cct $I = \frac{V}{R}$

$\therefore$ mag. force on the rod $F_m = BI\ell \sin 90^\circ$

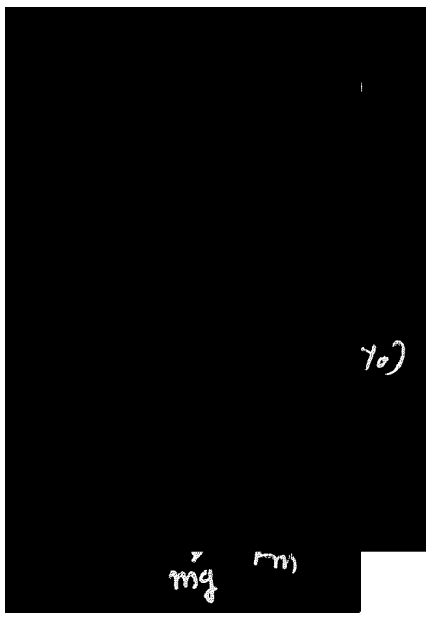
$$= \frac{B\ell V}{R}$$

In presence of mag. field $mg + f_m = 4$ Kyo .....(2)

Sub Eq. (1) in (2)

$2ky_o + F_m = 4ky_o$

$Fm = 2ky_o$



$$\frac{B\ell V}{R} = 2ky_o$$

$$B = \frac{2ky_o \cdot R}{LV}$$

Sub eq. (1)

$$B = \frac{mgR}{LV}$$

69. Net force on a current carrying closed Loop is alwys zero if it is placed in a uniform mag. field.

70. $$F = \frac{\mu o}{2\pi} \frac{I_1 I_2 \ell}{y}$$

$$\frac{F}{\ell} = \frac{\mu o}{2\pi} \frac{I.I}{y} = \frac{\mu o}{2\pi} \frac{I^2}{y}$$

71. since all the given forces are lying in plane, so the given Loop is in equilibrium.
From the fig.

$F_4 \cos\theta = F_2$

$F_4 \sin\theta = F_3 - F_1$

$\therefore F_4^2 = (F_4 \cos\theta)^2 + (F_4 \sin\theta)^2$

$F_4^2 = (F_2)^2 + (F_3 - F_1)^2$

$\therefore F_4 = \sqrt{F_2^2 + (F_3 - F_1)^2}$



72. For charge particles, if they are moving freely in space ver mag. force between them. Hence due to ele. force they repel each other.

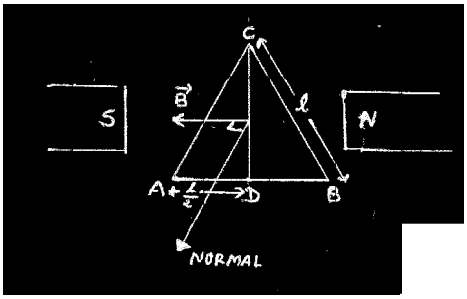
73. In ΔCAD

$AC^2 = AD^2 + DC^2$

$\ell^2 = \frac{\ell^2}{4} + DC^2$

$DC = \frac{\sqrt{3}}{2} \ell$

Area of ΔABC



$$A = \frac{1}{2}(\ell)\left(\frac{\sqrt{3}}{2}\ell\right)$$

$$A = \frac{1}{4}\sqrt{3}\,\ell^2$$

torque acting on ΔABC is

$\tau = IAB\sin\theta$

$$= I\left(\frac{1}{4}\sqrt{3}\ \ell^2\right) B \sin\theta$$

$\theta = 90^o$

$$\tau = \frac{\sqrt{3}}{4} I\ \ell^2 B$$

$$\therefore\ \ell^2 = \frac{4\tau}{\sqrt{3}\,IB}$$

$$\therefore \ell = 2\left(\frac{\tau}{\sqrt{3}\ I\ B}\right)^{\frac{1}{2}}$$

74. $\tau = NIAB$ and $\tau = k\theta$

$\therefore NIAB = k\theta$

$$I = \left(\frac{k}{NAB}\right)\theta$$

$\therefore I\ \alpha\ \theta$

75. $$F = \frac{\mu o}{2\pi}\frac{I_1 I_2 \ell}{y}$$

$$\frac{F}{\ell} = 2\times 10^{-7}\ \frac{N}{m}$$

76. Area of aceq square = $a \times 9$

$= 9^2$

Now area of 4 semi circles

$$= 4\times\frac{1}{2}\left(\pi\frac{a^2}{4}\right)$$

$= \frac{\pi}{2} a^2$

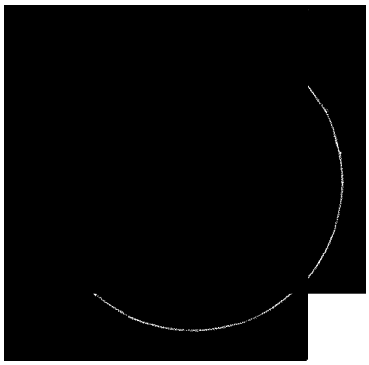
$\therefore$ total Area A = Area of + Area of 4 semi circles square

$= a^2 + \frac{\pi}{2} a^2 \quad = a^2 \left[1 + \frac{\pi}{2} \right]$

$\therefore M = IA$

$Ia^2 \left[1 + \frac{\pi}{2} \right] \hat{k}$

77. No. of turns per unit width $= \frac{N}{b-a}$



$\therefore$ the no. of turns in thickness dx is $dN = \left(\frac{N}{b-a} \right) dx$

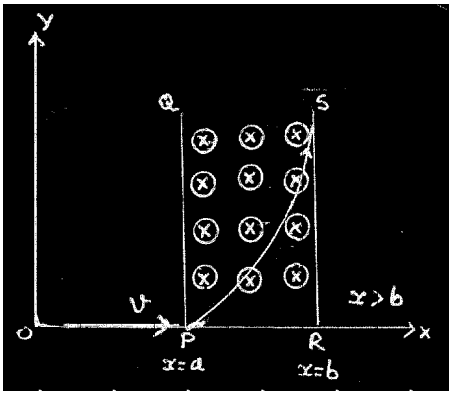
$\therefore$ mag. field at the centre is $dB = dN \left(\frac{\mu_0 I}{2x} \right)$

$dB = \left(\frac{N}{b-a} \right) \frac{\mu_0 I}{2x} . dx$

$\therefore$ total mag. field $B = \int dB$

$= \frac{\mu_0 NI}{2(b-a)} \; n \left(\frac{b}{a} \right)$

78. In the fig. the z-axis points out of the paper and mag. field is direced into the paper represented by



It is present between PQ and RS only

The particle moves in a circular path of radius r in the magnetic field.

It can just enter the region x > 9 for

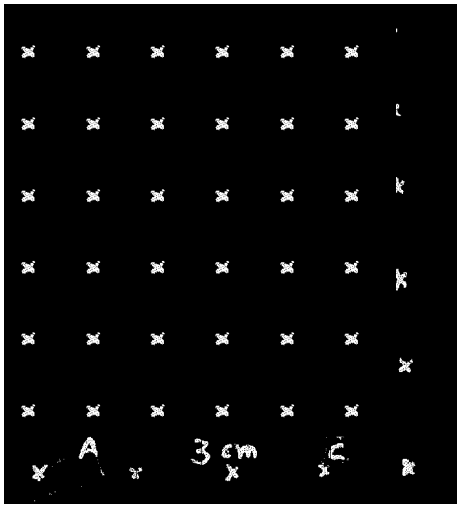
Now $r = \frac{mv}{qB} \geq (b-a)$

$v \geq \frac{(b-a)qB}{m}$

$\therefore v \min = \frac{qB(b-a)}{m}$

79. Electron is performing circular motion so according to work Energy theorem mag. field does not do any work.K.E. remains constant the force on $\bar{e}$ will act along negative y-axis initially in clockwise direction.

$\therefore v = v_0$



80. the AB and BC wire is equivalent to AC wire

$\therefore$ Force acting on AC wire $F = BI\ell \sin\theta$

$= 2 \times 2 \times 3 \times 10^{-2} \sin 90^0$

$F = 12 \times 10^{-2} N$ along y -axis

$\therefore$ acceleration $a = \frac{F}{m}$

$= 12\ m/s^2$

81. mag. field in the middle of the solenoid is maximum.

mag. field is half at the end compare to its value in the middle.

Bend $= \frac{1}{2}$ B centre

82. $\tau = BINA \sin\theta$

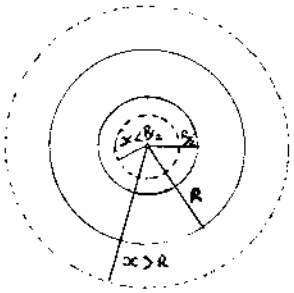
$\tau \to \sin\theta$

$\therefore \tau \to \theta$ is a sinusoidal graph

sinusoidal graph

85. Case - 1 $x < \frac{R}{2}$

$|\vec{B}| = 0$



Case - 2 $\frac{R}{2} \le x < R$

$\int \vec{B}.\vec{d\ell} = \mu_O I \quad \left[\because J = \frac{I}{A} \quad I = JA\right] oJ$

$|\vec{B}| 2\pi x = \mu_O JA$

$$|\vec{B}|2\pi x = \mu_O.J\left[\pi x^2 - \pi\left(\frac{R}{2}\right)^2\right]$$

$$|\vec{B}| = \frac{\mu_O J}{2x}\left[x^2 - \frac{R^2}{4}\right]$$

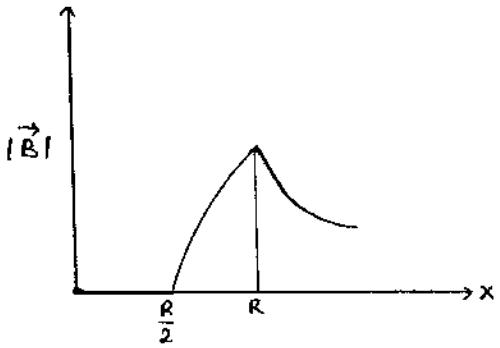
Case - 3 $x \geq R$

$$\int \vec{B}.\vec{dl} = \mu_O JA$$

$$|\vec{B}|2\pi x = \mu 0 J\left[\pi R^2 - \pi\left(\frac{R}{2}\right)^2\right]$$

$$= \frac{\mu_O J}{2x}\frac{3}{2}R^2$$

$$= \frac{3\mu_O J}{2x}\frac{3}{2}R^2$$



86. $dB = \frac{\mu_0 I}{2r}$ (mag. field at the centre of the ring)

$$= \frac{\mu_0}{2r}.dq.f$$

$$dB = \frac{\mu_0}{2r}\ dq.\frac{\omega}{2\pi}$$

$$B = \frac{\mu_0 \omega}{4\pi}\int \frac{dq}{r}$$

$$= \frac{\mu_0 \omega}{4\pi}\int \frac{Q}{\pi R^2}(2\pi r.dr)\frac{1}{r}$$

$$= \frac{\mu_0 \omega.Q}{2\pi R^2}\int_{r=0}^{r=R} dr$$

$$= \frac{\mu_0.\omega.Q.}{2\pi R}$$

$$B = \left(\frac{\mu_0 \omega.Q.}{2\pi}\right)\frac{1}{R}$$

$$B\alpha\frac{1}{R}$$

$\therefore$ Graph - A is true

87. For a temporarymagnet the hysteresis Loop should be long and narrow.

88. Magnetism of a magnet falls with rise of temp and becomes practically zero above curie temperature.

89. For a diamagnetic substance χ is small negative and independent of temperature.

90. For a paramagnetic substance χ is independent is magnetic field.

91. For a ferromagnetic substance sueptibility $\chi = \dfrac{C}{T - T_C}$

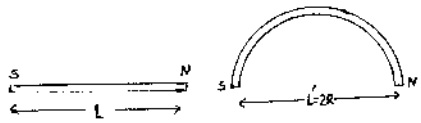
As temp T of substance isincrease its χ is decreasing

92. Intensity of magnetisation of diamagnetic substance is very small and negative

93. For a paramagnetic substance magnetization M is proportional to magnetising field H and M is positive.

94.

On bending a rod its pole strangth remains unchanged where as its magnetic moment changes.

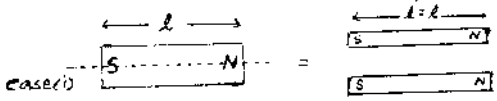


$M' = m(2R)$

$= m\left(2\dfrac{L}{\pi}\right) \quad = \dfrac{2}{\pi}mL \quad = \dfrac{2m}{\pi}$

95. All of the above

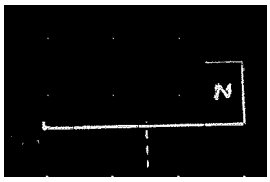
97. Case (i)



If cut along the axis of magnet of length ℓ, then new pole strength $m' = \dfrac{m}{2}$ and new length $\ell' = \ell$

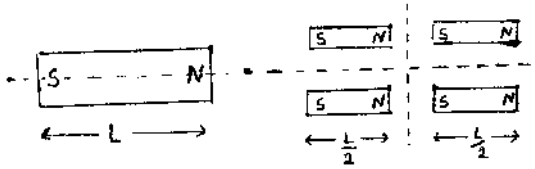
$\therefore$ New magnetic moment $M' = \dfrac{m}{2} \times \ell = \dfrac{m\ell}{2} = \dfrac{M}{2}$

Case (ii)



Ifcut perpendicular to the axis of magnet, then new pole strength $m' = m$ and new length $\ell' = \dfrac{\ell}{2}$

$\therefore$ New magnetic moment $M' = m \times \dfrac{\ell}{2} = \dfrac{m\ell}{2} = \dfrac{M}{2}$

98. 

For each part $m' = \frac{m}{2}$

99. The spin motion of electron

100. On the axis $B_1 = \frac{2M}{x^3}$

On the equator $B_2 = \frac{M}{y^3}$

As $B_1 = B_2$

$$\frac{2M}{x^3} = \frac{M}{y^3}$$

$$\frac{x^3}{y^3} = 2$$

$$\frac{x}{y} = 2^{\frac{1}{3}}$$

101. On the axis Baxis $= \frac{2M}{x^3}$

$$200 \text{ gauss} = \frac{2M}{x^3}$$

$$100 \text{ gauss} = \frac{M}{x^3}$$

100 gauss = Bequator

102. $W = MB(1-\cos\theta)$

$= 2\times10^4 \times 6\times10^{-4}(1-\cos 60^\circ)$

$= 6$ Joule

103. $\tau = MB\sin\theta$

$= m(2\ell)\times B\sin\theta$

$= 10^{-4}\times 0.1\times 30\sin 30^\circ$

$= 1.5\times10^{-4}$ Joule

105. $\tau = MH\sin\theta$

$= MH\sin 30^{o}$

$= \dfrac{MH}{2}$

106. 0.1 Amp m^2

107. torque $\tau = MH_H \sin\theta$

$= 2\pi \times 10^{-7}$ Joule

108. $B = \dfrac{\mu 0}{4\pi}\dfrac{2M}{d^3}$

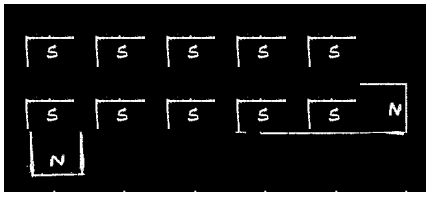
$= 2.4 \times 10^{-4}$ tesla

109. In magnetic dipole, force $\times \dfrac{1}{r^4}$

$\dfrac{F_2}{F_1} = \dfrac{r_1^4}{r_2^4}$

$\dfrac{F_2}{4.8} = \left(\dfrac{r_1}{2r_1}\right)^4$

$= 0.3$ Newton

110. 

111. $\vec{\tau} = \vec{M} \times \vec{B}$

$= 150\hat{K}$ N.m

112. Mag. moment of circular Loop carrying current is

$M = IA \quad = I(\pi R^2) \quad = I\pi\left(\dfrac{L}{2\pi}\right)^2$

$M = \dfrac{IL^2}{4\pi} \quad \Rightarrow L = \sqrt{\dfrac{4\pi M}{I}}$

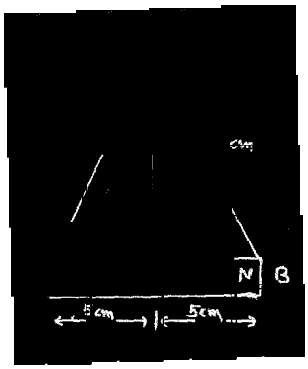
113. $L = 10 \times 10^{-2}\,m$

$r = 15 \times 10^{-2}\,m$

$OP = \sqrt{225 - 25}$

$= \sqrt{200}$ cm

since, at theneutral point, magnetic field due to the magnetic equal to B_H



$$B_H = \frac{\mu o}{4\pi} \frac{M}{\left(OP^2 + AO^2\right)^{\frac{3}{2}}}$$

= 1.35 Amp. meter

114. $\tan\phi' = \frac{\tan\phi}{\cos\beta}$ where ϕ' = Apparent angle of dip

$= \frac{\tan 60^o}{\cos 30^o}$ ϕ = true angle of dip

$\tan\phi' = 2$ β = Angle made by vertical plane with

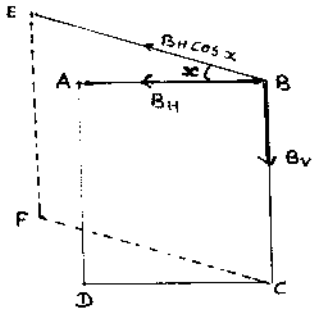
$\phi' = \tan^{-1}(2)$ magnetic meridian

115. for ABCD plane

$$\tan\theta = \frac{B_V}{B_H} \quad .....(1)$$

for BCFE plane

$$\tan\theta = \frac{B_V}{B_H \cos x} \quad .....(2)$$



solved the equation 1 and 2.

116. In vertical plane $T = 2\pi\sqrt{\frac{I}{MB_V}}$ ...(1)

In horizontal plane $T = 2\pi\sqrt{\frac{I}{MB_H}}$ ....(2)

but in both the cause T = 2 sec.

$$\sqrt{\frac{I}{MB_V}} = \sqrt{\frac{I}{MB_H}}$$

$$\frac{1}{B_V} = \frac{1}{B_H}$$

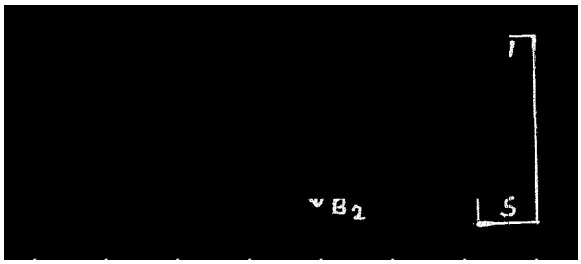
$$1 = \frac{B_V}{B_H}$$

$1 = \tan\phi$

$\phi = 45^o$

117. $B_{1\ axis} = \frac{\mu_O}{4\pi} \frac{2M}{d^3}$ .....(1)

$B_{2\ equator} = \frac{\mu_O}{4\pi} \frac{M}{d^3}$ .....(2)



At point P

$B_{resultant} = \sqrt{B_1^2 + B_2^2}$

$= \sqrt{5} \frac{\mu_O}{4\pi} \frac{M}{d^3}$

118. Magnetic suceptibility $\chi_m \alpha \frac{1}{T}$

$\frac{\chi_{m2}}{\chi_{m1}} = \frac{T_1}{T_2}$

$\frac{\chi_{m2}}{0.0060} = \frac{273-73}{273-173}$

$\frac{\chi_{m2}}{0.0060} = \frac{200}{100}$

$\chi_{m2} = 0.0120$

119. ferro magnetic substances, magnetised strongly in the direction of magnetic field.
Para magnetic substances magnetised weakly in the direction of magnetic field.
Diamagnetic substances is magnetised weakly in opposite direction of magnetic field.

120. The repulsive force between tab magnets = weight

$\frac{\mu o}{4\pi} \frac{m_1 m_2}{r^2}$ = 50 gm. weight

$10^{-7} \frac{m^2}{9 \times 10^{-6}} = 50 \times 10^{-3} \times 9.8$

$m^2 = \frac{9 \times 10^{-6} \times 50 \times 10^{-3} \times 9.8}{10^{-7}}$

m = 6.64 Amp. meter

121. $\frac{\chi_{m2}}{\chi_{m1}} = \frac{T_1}{T_2}$

$\frac{1.8 \times 10^{-5}}{1.2 \times 10^{-5}} = \frac{300}{T_2}$

$T_2 = 200$ kelvin

122. $\text{Baxis} = \dfrac{2M}{x^3} = 9 \ (\text{In CGS})$

$= \dfrac{M}{x^3} = \dfrac{9}{2}$

$\text{Bequater} = \dfrac{M}{\left(\dfrac{x}{2}\right)^3}$

$= 8\dfrac{M}{x^3}$

$= 8\left(\dfrac{9}{2}\right)$

$= 36$ Gauss

123. $\text{Volume of the domain} = \left(1\times10^{-6}\right)^3 \text{ m}^3$

$= 10^{-18} \text{ m}^3$

$\therefore$ New dipole moment mnet $= \text{Nm}$

$= 8\times10^{10}\times9\times10^{-24}$

$= 72\times10^{-14} \text{ A.m}^2$

$\therefore \text{Magnetization M} = \dfrac{m_{net}}{\text{vol}}$

$= \dfrac{72\times10^{-14}}{10^{-18}} \dfrac{\text{A.m}^2}{\text{m}^3}$

$= 72\times10^{+4}$

$= 7.2\times10^5$ Amp. meter^{-1}

124. Diamagnetic materials

125. $I \alpha \tan\phi$

$\dfrac{I_1}{I_2} = \dfrac{\tan\phi 1}{\tan\phi 2}$

$\dfrac{2}{I_2} = \dfrac{\tan 30^o}{\tan 60^o}$

$I_2 = 6$ Amp.

126. $T = 2\pi\sqrt{\dfrac{I}{MH_H}} = 4\sec$ where $I = \dfrac{1}{12}ML^2$

when magnet is cut into two equal havies, then

new magnet moment $M' = \dfrac{M}{2}$

moment of inertia $I' = \dfrac{1}{12}M'L'^2$

$= \dfrac{1}{12}\left(\dfrac{M}{2}\right)\left(\dfrac{L}{2}\right)^2$

$= \dfrac{1}{8}.\dfrac{1}{12}ML^2$

$I' = \dfrac{I}{8}$

$\therefore$ new time period $T' = 2\pi\sqrt{\dfrac{I'}{M'BH}}$

$= 2\pi\sqrt{\dfrac{\frac{I}{8}}{\left(\frac{M}{2}\right)BH}}$

$= \dfrac{1}{2}2\pi\sqrt{\dfrac{I}{MH}}$

$= \dfrac{T}{2}$

$= \dfrac{4}{2} \quad = 2$ sec.

127. Moment of inertio $I = \dfrac{\text{mass} \times (\text{length})^2}{12}$ $\left(\because I = \dfrac{1}{12}ML^2\right)$

$= \dfrac{1}{n} \times \left(\dfrac{1}{n}\right)^2$

$= \dfrac{1}{n^3}$ time

Magnetic moment $M =$ pole strength $\times$ length

$= \dfrac{1}{n}$ time

$$T = 2\pi\sqrt{\frac{I}{MH}}$$

$$= \sqrt{\frac{\frac{1}{n^3}}{\frac{1}{n}}}\,\text{time}$$

$$= \frac{1}{n}\,\text{time}$$

$$\therefore T^1 = \frac{T}{n}\,\text{sec.}$$

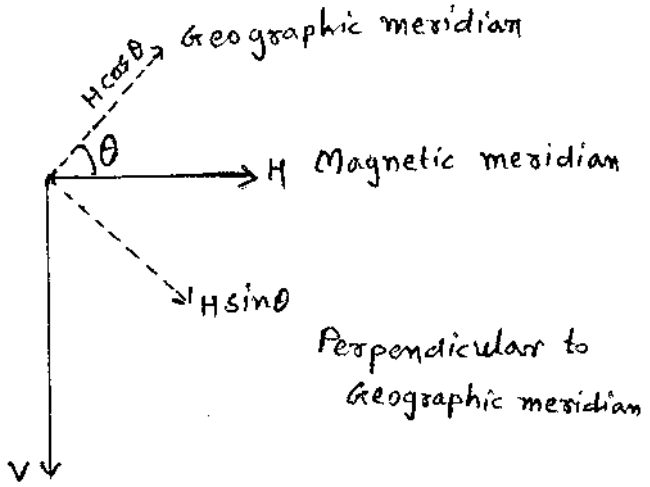
128. $\tan\delta_1 = \dfrac{V}{H\cos\theta}$

$$\tan\delta_2 = \frac{V}{H\cos(90^\circ - \theta)} = \frac{V}{H\sin\theta}$$

$$\frac{\tan\delta_1}{\tan\delta_2} = \frac{\sin\theta}{\cos\theta}$$

$$\frac{\tan\delta_1}{\tan\delta_2} = \tan\theta$$

$$\theta = \tan^{-1}\left(\frac{\tan\delta_1}{\tan\delta_2}\right)$$



129. coercivity $H = 100\dfrac{A}{m}; \ell = 100\,\text{cm} = 1\,\text{m}; n = 50$

As $H = nI \Rightarrow I = \dfrac{H}{n} = \dfrac{100}{50} = 2$ Amp.

130. $H_1 = B\cos\delta_1$; $H_2 = B\cos\delta_2$

$$\therefore \frac{H_1}{H_2} = \frac{\cos 30^\circ}{\cos 45^\circ}$$

$$= \frac{\sqrt{3}}{\sqrt{2}}$$

143. There will be nomagnetic field at "0" due to wire AB and CD carrying current "I".

Wire carrying I_1 is also produced zero magnetic field at "0".

Mag. field at "0 due to arc AD" $= \frac{\mu_O I}{24\ ab}(b-a)$

$= \frac{\mu_O}{4\pi}\ \frac{(I)\left(\frac{\pi}{6}\right)}{a}$ (coming out at pt "0")

Mag. field at "0" due to arc BC $B_2 = \frac{\mu_O}{4\pi}\ \frac{(I)\left(\frac{\pi}{6}\right)}{b}$ (going inside atpt "0")

$\therefore$ Net mag. field B $B = B_1 - B_2$ (comoing out)

$= \frac{\mu_O}{4\pi}\ \frac{\pi}{6}\ .\ I\left(\frac{1}{a}-\frac{1}{b}\right)$

$= \frac{\mu_O I}{24}\ \left(\frac{b-a}{ab}\right)$

$= \frac{\mu_O I}{24\ ab}\ (b-a)$

144. The forces on AD and BC are zero because mag.field d we to a straight wire on AD and BC is parallel to elementary length of the Loop and both the fields are in nuturally opposite direction.

148.

	P	Q	R	S
A	○	○	○	○
B	○	○	○	○
C	○	○	○	○
D	○	○	○	○

149. $A \rightarrow Q, R, S$

$B \rightarrow P, Q, R, S$

$C \rightarrow P, Q, R, S$

$D \rightarrow P, Q, R, S$

150. $A \rightarrow P$

$B \rightarrow Q$

$C \rightarrow R$

$D \rightarrow P, S$

Unit-14
Electromagnetic Induction and Alternating Current

SUMMERY

1. Net flux through the surface $\phi = \oint \vec{B} \cdot \vec{dA} = BA\cos\theta$

   θ = angle between area vector $\vec{A}$ & magnetic field $\vec{B}$

   • if $\theta = 0$ then $\phi max = BA$

   • if $\theta = 90$ then $\phi = 0$

2. Unit of magnatic flux is weber (wb) in SI unit and maxwell or Gauss x cm^2 in CGS unit where $1wb = 10^8$ maxwell

   • other units = tesla $\times$ m^2 = $\frac{Nm}{A} = \frac{Joule}{A}$ =

   $$\frac{Volt \times Coulomb}{A} = Volt \text{ x } sec = ohm \text{ x } coulomb = henery \text{ x } \mathbf{A}$$

   • Its dimensional formula = $M^1L^2T^{-2}A^{-1}$

3. **Faraday's Law**

   • The induced emf $\varepsilon = -N\frac{d\phi}{dt} = -N\frac{d}{dt}(AB\cos\omega t)$

   • If A alone is changing $\varepsilon = -NB\cos\omega t\frac{dA}{dt}$

   • If B alone is changing $\varepsilon = -NA\cos\omega t\frac{dB}{dt}$

   • If θ alone is changing $\varepsilon = +NAB\omega\sin\omega t$

4. Motional emf $\varepsilon = -Bvl\sin\theta$ where, θ = angle between $\vec{v}$ & $\vec{B}$

5. If conducting rod moves on two parallel conducting rails placed in magnetic field perpendicular to plane of close loop.

   • induced emf $\varepsilon = -Bvl$

   • induced current $I = \varepsilon = -\frac{Bvl}{R}$

   • magnetic force $F = IlB = -\frac{B^2vl^2}{R}$

   • power dessipated $P = \frac{B^2v^2l^2}{R} = F_m \cdot v$

   where l = distance between two rails

   R = Resistance of the loop

6. Self induction

• self inductance $L = \frac{N\phi}{I}$

• induced emf $\varepsilon = -L\frac{dI}{dt}$

• unit of L is $\frac{wb}{A}$ = ohm x sec = Henery

• dimensional formula of L is $M^1L^2T^{-2}A^{-2}$

• magnetic potential energy of inductor $U = \frac{1}{2}LI^2$

• magnetic energy density $S_m = \frac{U}{Al} = \frac{B^2}{2\mu_0}$

7. Mutual Induction

• Mutual inductance $M = \frac{N_1\phi_1}{I_2} = \frac{N_2\phi_2}{I_1}$

• induced emf = $\varepsilon_2 = -M\frac{dI_1}{dt}$ OR $\varepsilon 1 = -M\frac{dI_2}{dt}$

8. Relation between M, L, & L_2

For two magnetically coupled coils $M = k\sqrt{L_1L_2}$ where k = coefficient of coupling or coupling factor $(0 \le k \le 1)$

$$k = \frac{\text{magnetic flux linked in secondary}}{\text{magnetic flux linked in primary}}$$

9. Combination of inductances

• series $L_s = L_1 + L_2$ if they are situated close to each other then $L_s = L_1 + L_2 \pm 2M$

• Parellel $L_p = \frac{L_1L_2}{L_1 + L_2}$ OR $\frac{1}{L_p} = \frac{1}{L_1} + \frac{1}{L_2}$ if they are situated close to each other then

$$L_p = \frac{L_1L_2 - M^2}{L_1 + L_2 \pm 2M}$$

10. Growth & decay of current in LR circuit

• The value of current at time t after closing the circuit $I = I_0[1 - e^{\frac{-Rt}{L}}]$

where $I_0 = I_{max} = \frac{\varepsilon}{R}$

• The value of current at time t after opening from the steady state condition $I = I_0 \; e^{\frac{-Rt}{L}}$

• Time constant $\tau = \frac{L}{R}$

11. A.C. generator

• emf $V = NAB\omega \sin \omega t = V_m \sin \omega t$

• current $I = \frac{V_m}{R} \sin \omega t = I_m \sin \omega t$

• A.C. current & voltage are positive for half the cycle & negative for the rest half. So aweerage value of ac quantity is zero over a complete cycle.

• at $t = \frac{T}{4}$ from the begining I or V reach their maximum value.

12. Series LCR circuit

• current $I = I_m \cos(\omega t - \delta)$

where $I_m = \frac{V_m}{|z|}$

δ = phase difference between V & I

• Voltage $V = \sqrt{V_R{}^2 + (V_L - V_C)^2}$

• Impedence $|z| = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + (\omega L - \frac{1}{\omega c})^2}$

• Phase difference $\tan \delta = \frac{X_L - X_C}{R} = \frac{V_L - V_C}{V_R}$

13. At resonance

• $X_L = X_C$ then $|z| = R$ then $\omega_0 = \frac{1}{\sqrt{LC}}$ then $f_0 = \frac{1}{2\pi\sqrt{LC}}$

• Half power band width $\Delta\omega = \frac{R}{L}$

• Q-factor $Q = \frac{\omega_0}{\Delta\omega} = \frac{1}{R}\sqrt{\frac{L}{C}}$

14. Power

- $P = V_{rms}\ I_{rms} \cos \delta$

  where $\cos \delta$ = power factor

15. Parellel LC circuit in series with R

- $|z| = \sqrt{R^2 + \frac{1}{(\omega c - \frac{1}{\omega L})^2}}$

- $\tan \delta = \frac{1}{R(\omega c - \frac{1}{\omega L})}$

16. Transformer

- $\frac{\varepsilon_s}{\varepsilon_p} = \frac{I_p}{I_s} = \frac{N_s}{N_p} = r$ = transformation ratio

MCQ

For the answer of the following questions choose the correct alternative from among the given ones.

1. A coil having area $2m^2$ is placed in a magnetic field which changes from 1 wb /m^2 to 4 wb/m^2 in an interval of 2 second. The emf induced in the coil of single turn is....

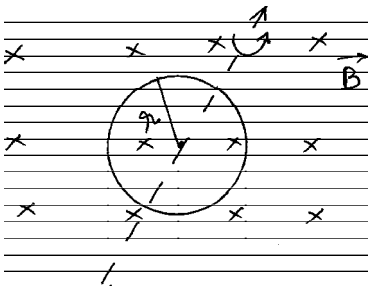
   (a) 4 v (b) 3 v (c) 1.5 v (d) 2 v

2. Two different loops are concentric & lie in the same plane. The current in outer loop is clockwise & increasing with time. The induced current in the inner loop then, is.....

   (a) clockwise (b) zero

   (c) counter clockwise (d) direction depends on the ratio of loop radii

3. A ring of radius r is rotating about its diameter with angular velocity ω in a perpendicular magnetic field $\vec{B}$ It has 20turns.The emf induced is



   (a) $20B\pi r^2 \text{Sin wt}$ (b) $20B\pi r^2 \cos wt$

   (c) $10\sqrt{2}B\pi r^2$ (d) $20B\pi r^2 w \sin wt$

4. A magnetic field $2\times10^{-2}T$ acts at right angles to a coil of area 200 cm^2 with 25 turns. The average emf induced in the coil is 0.1 v when it removes from the field in time t. The value of t is

   (a) 0.1 sec (b) 1 sec (c) 0.01 sec (d) 20 sec

5. The magnetic flux linked with a coil, in webers, is given by the equation $\phi = 4t^2 - 3t + 7$. Then the magnitude of induced emf at 2 sec will be ...

   (a) 15 v (b) 19 v (c) 17 v (d) 21 v

6. A solenoid of 1.5 m long with inner diamter of 4 cm has three layers of windings of 1000 turns each & carries a current of 2 A. The magnetic flux through a cross section of the solenoid is nearly.....

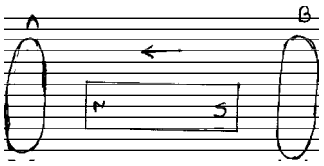
   (a) 2.5×10^{-7} wb (b) 6.31×10^{-6} wb

   (c) 2.1×10^{-6} wb (d) 4.1×10^{-5} wb

7. A coil has an area of 0.05 m^2 & it has 800 turns. It is placed perpendicularly in a magnetic field of strength , $4x10^{-5}$ Wb/m^{-2}. It is rotated through 90° in 0.1 sec. The average emf induced in the coil is...

   (a) 0.056 v (b) 0.046 v (c) 0.026 v (d) 0.016 v

8. A bar magnet is moving along the common axis of two coils A &B towards A. current is induced in



(a) Only A (b) Only B

(c) both A& B in same direction (d) both A & B in opposite direction

9. A coil having n turns & resistance $R\Omega$ is connected with a galvanometer of resistance $4R\Omega$. This combination is moved from a magnetic field W_1 Wb to W_2 Wb in t second. The induced current in the circut is....

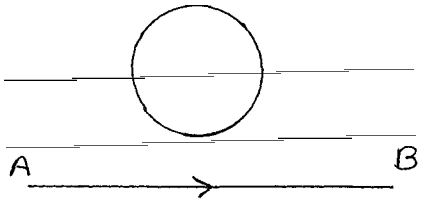
(a) $-\frac{(W_2 - W_1)}{5Rnt}$ (b) $-n\frac{(W_2 - W_1)}{5Rt}$

(c) $-\frac{(W_2 - W_1)}{Rnt}$ (d) $-n\frac{(W_2 - W_1)}{Rt}$

10. When a train travels with a speed 360 km h^{-1} along the track separated by 1 meter. The vertical component of earth's magnetic field is 0.1×10^{-4} T .What is the value of induced emf between the rails ?

(a) 10^{-2} V (b) 10^{-4} V (c) 10^{-3} V (d) 1V

11. An electron moves along the line AB,which lies in the same plane as a circular loop of conducting wires as shown in figure. What will be the direction of current induced if any, in the loop ?



(a) No current will be induced

(b) The current will be clockwise

(c) The current will be anticlockwise

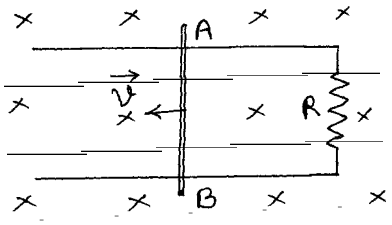
(d) The current will change direction as the electron passes by

12. A circular loop of radius R carrying current I lies in X-Y plane with its centre at origin. The total magnetic flux through x-y plane is.....

(a) Directly proportional to I (b) Directly proportional to R

(c) Directly proportional to R^2 (d) zero

13. Consider the situation shown in the figure. The wire AB is sliding on the fixed rails with a constant velocity $\vec{v}$. If the wire AB replaced by semi circular wire the magnitude of the induced current will.



(a) Increases (b) decreased

(c) Remain same (d) depending on whether the semicircle bulge is towards the resistance or away from it.

14. Two similar circular loops carry equal currents in the same direction. On moving the coils further apart, the electric current will

(a) Remain unchanged (b) Increasing in both

(c) Increasing in one decreasing in other (d) Decreasing in both

15. When the number of turns in a coil is make four time without any change in length of the coil, its self inductance becomes.....

(a) unchanged (b) two times

(c) four times (d) sixteen times

16. A metal road moves at a constant velocity in a direction perpendicular to its length & a constant uniform magnetic field too. Select the correct statement (s) from the following.

(a) The entire rod is at the same electrical potential

(b) There is an electric field in the rod

(c) The electric polential is highest at the centre of the rod.

(d) The electric polential is lowest at the centre of the rod.

17. Two co-axial solenoids are made by a pipe of cross sectional area 10 cm^2 and length 20 cm If one of the solenoid has 300 turns and the other 400 turns, their mutual inductance is.....

(a) $4.8\pi \times 10^{-4}$ H (b) $4.8\pi \times 10^{-5}$ H

(c) $2.4\pi \times 10^{-4}$ H (d) $4.8\pi \times 10^{4}$ H

18. The self inductance of a straight conductor is...

(a) zero (b) very large (c) very small (d) ∞

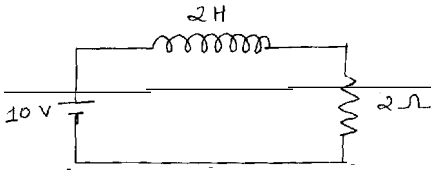
19. Two coils of self inductances 2 mH & 8 mH are placed so close togather that the effective flux in one coil is completely half with the other. The mutual inductance between these coils is.....

(a) 4 mH (b) 6 mH (c) 2 mH (d) 16 mH

20. In circular coil. when no. of turns is doubled & resistance becomes half of the initial then inductance becomes....

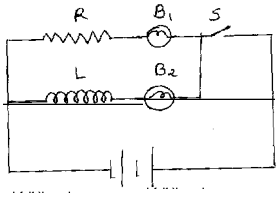
(a) 4 times (b) 2 times (c) 8 times (d) No change

21. In the figure the magnetic energy stored in the coil is



(a) zero (b) infinite (c) 25 J (d) 16 J

22. When the switch S turned off.



(a) Both B_1 and B_2 die out promptly

(b) Both B_1 and B_2 die out with some delay

(c) B_1 dies out promptly but B_2 with some delay

(d) B_2 dies out promptly but B_1 with some delay

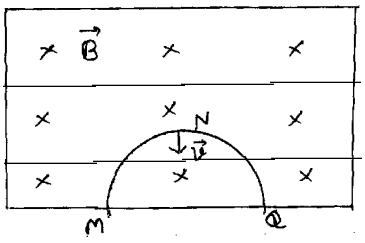
23. A transformer of efficiency 90% draws an input power of 4 kW. An electrical applience connected across the secondary draws a current of 6 A. The impedence of device is.....

(a) 60Ω (b) 50Ω (c) 80Ω (d) 100Ω

24. The armature of dc motor has 20 Ω resistance. It draws current of 1.5 A when run by 220 V dc suppy. The value of back induced in it will be

(a) 150 V (b) 190 V (c) 170 V (d) 180 V

25. A thin semicircular conducting ring of radius R is falling in a magnetic field B. Which is in limited area as shown in figure. At the position MNQ, the speed of ring is v and the potential difference developed across the ring is



(a) zero (b) $B v \pi \frac{R^2}{2}$ and M is at higher potential

(c) πRBv and Q is at higher potential (d) $2RBv$ and Q is at higher potential

26. Two identical circular loops of metal wire are lying on a table near to each other without touching. Loop A carries a current which increasing with time. In response the loop B.....

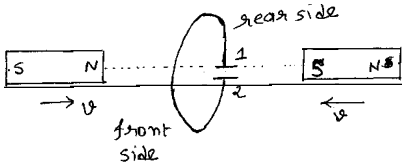
(a) Is repelled by loop A (b) Is attracted by loop A

(c) rotates about its centre of mass (d) remains stationary

27. A wire of length 2m is moving at a speed $2ms^{-1}$ keep its length perpendicular to uniform magnetic field of 0.5 T. The resistance of circuit joined with this wire is 6Ω. The rate at which wolrk is being done to keep the wire moving at constant speed is .....

(a) $\frac{1}{3}\omega$ (b) $\frac{2}{3}\omega$ (c) $\frac{1}{6}\omega$ (d) 2ω

28. Two identical magnets moving towards a coil, connecting a condenser at rear side shown in fig., with equal speed from opposite sides Then



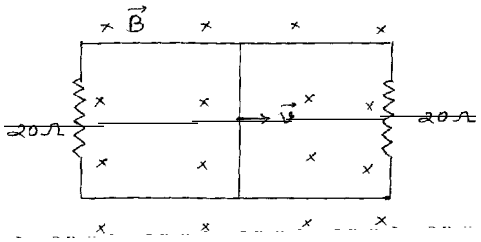
(a) Both plate will be positive

(b) There is no charging of condenser

(c) Plate 1 will be positive and 2 negative

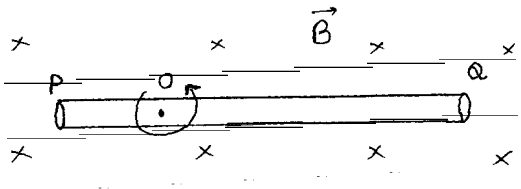
(d) Plate 2 will be positive and 1 negative

29. A rectangular loop with a sliding rod of length 2m & resistance 2Ω. It moves in a uniform magnetic field of 3T perpendicular to plane of loop. The external force required to keep the rod moving with constant velocity of $2ms^{-1}$ is



(a) 2N (b) 4N (c) 6N (d) 8N

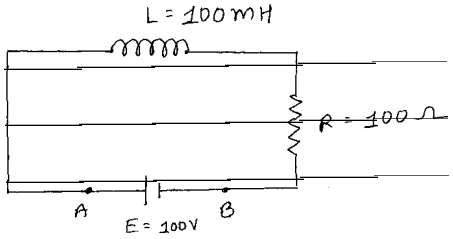
30. A conducting rod PQ of length $4l$ is rotated about a point O in a uniform magnetic field $\vec{B}$. PO = ℓ Then



(a) $V_Q - V_P = -\frac{B\omega\ell^2}{2}$ (b) $V_Q - V_O = \frac{5}{2}B\omega\ell^2$

(c) $V_Q - V_O = \frac{9}{2}B\omega\ell^2$ (d) $V_A - V_Q = 4\ B\omega\ell^2$

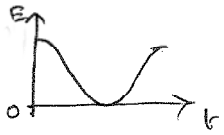
31. As shown in figure after a long time the battery is disconnected and short circuiting the points A & B. The current in circuit after 2ms after the short circuit is.

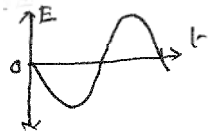


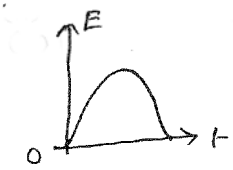
(a) $\frac{1}{e^2}$A (b) $\frac{1}{e}$A (c) 0.2A (d) e^2A

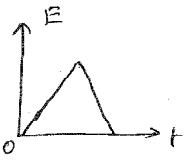
32. The variation of induced emf with time in a coil if a short bar magnet is moved along its axis with constant velocity is.



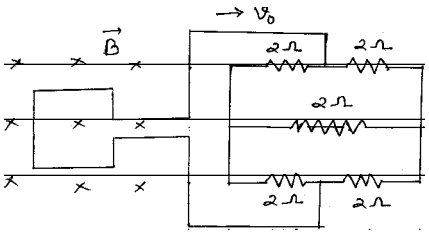
(a) 

(b) 

(c) 

(d) 

33. A square loop of side 10 cm and resistance 2 Ω is moved with velocity v_0 as shown in fig. The uniform magnetic field is 3T with what speed should the loop be moved so that a steady current of 1.5 mA flows in the loop.

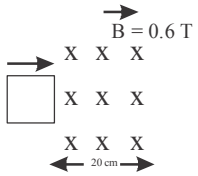


(a) 1 cms^{-1} (b) 2 cms^{-1} (c) 3 cms^{-1} (d) 4 cms^{-1}

34. An inductor-resistor-battery circuit is switched on at t = 0. If the emf of battry is ε find the charge passes through the battery in one time constant τ.

(a) $\frac{i_{max}\tau}{e}$ (b) $(\tau\text{-}1)e\ i_{max}$ (c) $\frac{i_{max}}{e}$ (d) $\tau\ i_{max}$

35. A square loop of side 5 cm being moved right at a constant speed $2\,cms^{-1}$. The front edge of it enters in magnetic field at t = 0. Fined the induced emf in the loop at t = 1s, 5s, 11s.



(a) 0.3 mv, 0.3 mv, 0.3mv (b) 0, 0.3mv, 0

(c) 0.3 mv, 0,0.3 mv (d) 0, 0, 0

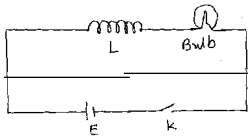
36. The time constant of a LR circuit is 20 ms The circuit is connected at t = 0 and the steady state current is found to be 4A. Find the current at 80 ms.

(a) 0.98 A (b) 1 A (c) 0.44 A (d) 0.88 A

37. A metal rod of length 2 m rotates vertically about one of its end with frequency 2 Hz. The horizontal component of earth's magnetic field is 3.14×10^{-5} T then emf developed between two ends of road is.....

(a) 78.87×10^{-4} V (b) 7.887×10^{-4} V (c) 78.87×10^{-6} V (d) 0V

38. In the following circuit the bulb will become suddenly bright if



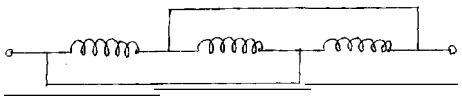
(a) contact is made (b) contanct is broken

(c) contact is made or broken (d) Won't become bright at all

39. The self inductance of a coil is 5H, a current of 1A changes to 2A within 5 sec. through the coil. The value of induced emf will be.....

(a) 10 V (b) 0.1 V (c) 1 V (d) 100 V

40. Pure inductance of 3H each are connected as shown below. The equivalent inductance of the circuit is



(a) 1 H (b) 2 H (c) 3 H (d) 9 H

41. A coil of inductance 300 mH and resistance 2 Ω is connected to a source of voltage 2V. The current reaches half of its steady state value in.....

(a) 0.15 sec (b) 0.3 sec (c) 0.05 sec (d) 0.1sec

42. If rotational velocity of a dynamo armature is doubled, then induced emf will become...

What is increased in step down transformer ?

(a) Half (b) Two times (c) Four times (d) unchanged

43. The core of a transformer is laminated so that.....
    (a) Ratio of I/p & O/p voltage increases
    (b) Rusting of core may be stopped
    (c) Energy loss due to eddy current may be reduced
    (d) Change in flux is increased

44. In transformer, core is made of soft iron to reduce.....
    (a) Hysterlsis losses (b) Eddy current losses
    (c) Force opposing current (d) the weight

45. An ideal transformer has 1:25 turn ratio. The peak value of the ac is 28 V. The rms ms secondary voltage is nearest to.....
    (a) 50 V (b) 70 V (c) 100 V (d) 40 V

46. A primary winding of transformer has 500 turns whereas its secondary has 5000 turns. Primary is connected to ac supply of 20V, 50Hz The secondary output of....
    (a) 200V, 25 Hz (b) 200 V, 50Hz (c) 2 V, 100 Hz (d) 2V, 50 Hz

47. A step down transformer is connected to main supply 200 V to operate a 6V, 30 w bulb. The current in primary is.....
    (a) 3A (b) 1.5 A (c) 0.3 A (d) 0.15 A

48. An emf of 15 V is applied in a circuit containing 5H inductance & 10 Ω resistance. The ratio of the currents at time $t = \infty$ and at t = 1sec.
    (a) $\frac{e^{\frac{1}{2}}}{e^{\frac{1}{2}}-1}$ (b) $1-e^{-1}$ (c) $\frac{e^2}{e^2-1}$ (d) e^{-1}

49. Two coils have a mutual inductance 0.005H The current changes in a coil according to equation $I = I_O \sin \omega t$ Where $I_O = 10\,A$ & $\omega = 100\ \pi$ rad s^{-1} The maximum value of emf in second coil is.....
    (a) 2π (b) 5π (c) π (d) 4π

50. A coil of inductance 8.4 mH and resistance 6Ω is connected to a 12 V battery. The current in the coil is 1A in the time.....
    (a) 500 sec (b) 20 sec (c) 35 ms (d) 1 ms

51. Alternating current can not be measured by dc ammeter because,
    (a) ac can not pass through dc ammeter
    (b) Average value of complete cycle is zero
    (c) ac is virtual
    (d) ac changes its direction

52. The resistance of a coil for dc is in ohms. In ac, the resistance
    (a) will remain same (b) will increase
    (c) will decrease (d) will be zero

53. An alternating current of rms valuve 10 A is passed through a $12\,\Omega$ resistance. The maximum potential difference across the resistor is,

(a) 20 V (b) 90 V (c) 169. 68 V (d) None of these

54. 220 V,50 Hz, ac is applied to a resistor. The instantancous value of voltage is

(a) $220\sqrt{2}\sin 100\pi t$ (b) $220\sin 100\pi t$

(c) $220\sqrt{2}\sin 50\pi t$ (d) $220\sin 50\pi t$

55. The rms value of an ac of 50 Hz is 10 amp.The time taken by the alternating current in reaching from zero to maximum value and the peak value of current will be,

(a) 2×10^{-2} sec and 14.14 amp (b) 1×10^{-2} sec and 7.07 amp

(c) 5×10^{-3} sec and 7.07 amp (d) 5×10^{-3} sec and 14.14 amp

56. If a current I given by $I_O \sin\left(\omega t - \frac{\pi}{2}\right)$ flows in an ac circuit across which an ac potential of $E = E_O$ sin ωt has been applied, then the power consumption p in the circuit will be,

(a) $P = \frac{E_0 I_0}{\sqrt{2}}$ (b) $P = \sqrt{2}E_0 I_0$

(c) $P = \frac{E_0 I_0}{2}$ (d) $P = 0$

57. In general in an alternating current circuit.

(a) The average value of current is 20w.

(b) The average value of square of current is zero.

(c) Average power dissipation is zero.

(d) The phase difference between voltage and current is zero.

58. An alternating current is given by the eq^n $I = I_1$ coswt + I_2 sinwb. The rms current is given by,

(a) $\frac{1}{\sqrt{2}}(I_1 + I_2)$ (b) $\frac{1}{\sqrt{2}}(I_1 + I_2)^2$

(c) $\frac{1}{\sqrt{2}}\left(I_1^2 + I_2^2\right)^{\frac{1}{2}}$ (d) $\frac{1}{2}\left(I_1^2 + I_2^2\right)^{\frac{1}{2}}$

59. In an ac circuit, the current is given by $I = 5\sin\left[100\ \ t - \frac{\pi}{2}\right]$ and the ac potential is V = 200 sin 100t. Then the power consumption is,

(a) 20 watts (b) 40 watts

(c) 1000 watts (d) 0 watts

60. In ac circuit with voltage V and current I, the power dissipaled is.

(a) VI (b) $\frac{1}{2}$VI

(c) $\frac{1}{\sqrt{2}}$VI (d) Depends on the phase between V and I

61. In the transmission of a.c. power through transmission lines, when the voltage is shapped up n times, the power loss in transmission,

(a) increase n times (b) Decrease n times

(c) Increase n^2 times (d) Decrease n^2 times

62. An alternating voltage is represented as E = 20 sin 300t. The average value of voltage over one cycle will be.

(a) zero (b) 10 volt

(c) $20\sqrt{2}$ volt (d) $\frac{20}{\sqrt{2}}$ volt

63. An ac source is rated at 220V, 50 Hz. The time taken for voltage to change from its peak value to zero is,

(a) 50 sec (b) 0.02 sec

(c) 5 sec (d) 5×10^{-3} sec

64. A lamp consumes only 50% of peak power in an ac circuit. What is the phase difference between the applied voltage and the circuit current.

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{2}$

65. The instantancous voltage through a device of impedance 20Ω is $\varepsilon = 80\sin 100\pi t$. The effective value of the current is,

(a) 3A (b) 2.828 A

(c) 1.732 A (d) 4A

66. A choke coil has.

(a) High inductance and low resistance (b) Low inductance and high resistance

(c) High inductance and high resistance (d) Low inductance and low resistance

67. A resistor and a capicitor are connected in series with an ac source. If the potential drop across the capacitor is 5 V and that across resistor is 12 V, the applied voltage is,

(a) 13 V (b) 17 V

(c) 5 V (d) 12 V

68. In an ac circuit the emf (e) and the current (i) at any instant core given respectively by $e = E_O \sin \omega t$, $I = I_0 \sin(\omega t - \varphi)$ The average power in the circuit over one cycle of ac is.

(a) $\frac{E_0 I_0}{2} \cos\phi$ (b) $E_0 I_0$

(c) $\frac{E_0 I_0}{2}$ (d) $\frac{E_0 I_0}{2} \sin\phi$

69. The instanteneous value of current in an A.C. circuit is $I = 2\sin\left(100\pi t + \frac{\pi}{3}\right) A$. The current will be maximum for the first time at,

(a) $t = \frac{1}{100} S$ (b) $t = \frac{1}{200} S$

(c) $t = \frac{1}{400} S$ (d) $t = \frac{1}{600} S$

70. An alternating current of frequency 'f' is flowing in a circuit containing a resistance R and a choke L in series. The impedance of this circuit is,

(a) $R + 2\pi fL$ (b) $\sqrt{R^2 + 4\pi^2 f^2 L^2}$

(c) $\sqrt{R^2 + L^2}$ (d) $\sqrt{R^2 + 2\pi fL}$

71. The resistance of an R-L circuit is $10\,\Omega$. An emf E_O applied across the circuit at $\omega = 20$ rad/s. If the current in the ckt is $\frac{I_0}{\sqrt{2}}$ what is the value of L.

(a) 1 H (b) 2 H (c) 3 H (d) 0.5 H

72. A resistor $30\,\Omega$, inductor of reactance $10\,\Omega$ and the capacitor of reactance $10\,\Omega$ are connected in series to an ac voltage source $e = 300\sqrt{2}\ \sin(\omega t)$ The current in the circuit is

(a) $10\sqrt{2}$A (b) 10 A (c) $30\sqrt{11}$A (d) $\frac{30}{\sqrt{11}}$A

73. Same current is flowing in two alternating circuits. The first circuits contains only inductance and the other contains only a capacitor. If the frequency of the emf of ac is increased the effect on the value of the current will be.

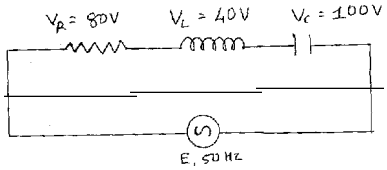
(a) Increase in the first circuit and dicrease in other

(b) Increase in both the circuit

(c) Decrease in both the circuit

(d) Decrease in the first and increase in other

74. The value of alternating emf E in the given ckt will be.



(a) 100 V (b) 20 V (c) 220 V (d) 140 V

75. A 20 volts ac is applied to a circuit consisting of a resistance and a coil with negligible resistance. If the voltage across the resistance is 12 V, the voltage across the coil is,

(a) 16 volts (b) 10 volts (c) 8 volts (d) 6 volts

76. An alternating voltage $E=200\sqrt{2}\sin(100t)$ is connected to 1 microfarad capacitor through an ac ammeter. The reading of the ammeter shall be.

(a) 10 m A (b) 20 m A (c) 40 m A (d) 80 m A

77. In a region of uniform magnetic induction $B = 10^{-2}$ tesla, a circular coil of radius 30 cm and resistance π^2 ohm is rotated about an axis which is perpendicular to the direction of B and which forms a diameter of the coil. If the coil rotates at 200 rpm the amplitude of the alternating current induced in the coil is,

(a) $4\pi^2$ mA (b) 30 mA (c) 6 mA (d) 200 mA

78. An LCR series circuit with R = 100Ω is connected to a 200 V, 50 Hz a.c source when only the capacitance is removed the current lies the voltage by 60^0 when only the inductance is removed, the current leads the voltage by 60^0. The current in the circuit is,

(a) 2A (b) 1A (c) $\frac{\sqrt{3}}{2}$ A (d) $\frac{2}{\sqrt{3}}$ A

79. The impedance of a circuit consists of 3Ω resistance and 4Ω reactance. The power factor of the circuit is.

(a) 0.4 (b) 0.6 (c) 0.8 (d) 1.0

80. The power factor of a good choke coil is

(a) Nearly zero (b) Exactly zero (c) Nearly one (d) Exactly one

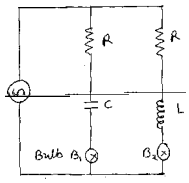
81. A coil of inductance L has an inductive reactance of X_L in an AC circuit in which the effective current is I. The coil is made from a super-conducting material and has no resistance. The rate at which power is dissipated in the coil is.

(a) 0 (b) IX_L (c) I^2X_L (d) Ix^2_L

82. When 100 volt dc is applied across a coil, a current of 1A flows through it. When 100 volt ac at 50 cyle s^{-1} is applied to the same coil, only 0.5 A current flows. The impedance of the coil is,

(a) 100 Ω (b) 200Ω (c) 300Ω (d) 400Ω

83. Two identical incandesant light bulbs are connected as shown in figure. when the circuit is an AC voltage source of frequency f, which of the following observations will be correct.



(a) both bulbs will glow alternatively

(b) Both bulbs will glow with same brightness provided $f = \frac{1}{2\pi}\sqrt{11LC}$

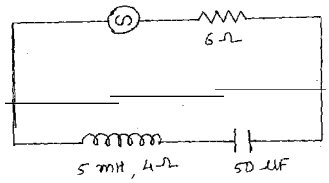
(c) Bulb b_1 will light up initially and goes off, bulb b2 will be ON constantly.

(d) Bulb b_1 will blink and bulb b_2 will be ON constantly.

84. An LC circuit contains a 20 mH inductor and a 50 μF capacitor with an initial charge of 10 mc. The resistance of the circuit is negligible. At the instant the circuit is closed be t = 0. At what time is the energy stored compeletly magnetic.

(a) t = 0 (b) t = 1.54 ms (c) t = 3.14 ms (d) t = 6.28 ms

85. In the circuit shown below, the ac source has voltage v = 20 cos ωt volts with ω = 2000 rad/sec. The amplitude of the current will be nearest to



(a) 2A (b) 3.3 A (c) $21\sqrt{5}$ A (d) $\sqrt{5}$ A

86. The quality factor of LCR circuit having resistance (R) and inductance (L) at resonance frequency (ω) is given by

(a) $\frac{\omega L}{R}$ (b) $\frac{R}{\omega L}$ (c) $\left(\frac{\omega L}{R}\right)^{\frac{1}{2}}$ (d) $\left(\frac{\omega}{L}\right)^{2}$

87. For high frequency, a capacitor offers

(a) More reactance (b) Less reactance (c) Zero reactance (d) Infinite reactance

88. The coil of a choke in a circuit

(a) increase the current (b) Decrease the current

(c) Does not change the current (d) Has high resistance to bc circuit

89. The power factor of an ac circuit having resistance (R) and inductance (L) connected in series and an angular velocity w is,

(a) $R/\omega L$ (b) $R/\left(R^2+\omega^2L^2\right)^{\frac{1}{2}}$

(c) $\omega L/R$ (d) $R/\left(R^2-\omega^2L^2\right)^{\frac{1}{2}}$

90. An inductor of inductance L and resistor of resistance R are joined in series and connected by a source of frequency ω power dissipated in the circuit is,

(a) $\frac{\left(R^2+\omega^2 L^2\right)}{V}$ (b) $\frac{V^2R}{\left(R^2+\omega^2 L^2\right)}$ (c) $\frac{V}{\left(R^2+\omega^2 L^2\right)}$ (d) $\frac{\sqrt{R^2+\omega^2 L^2}}{V^2}$

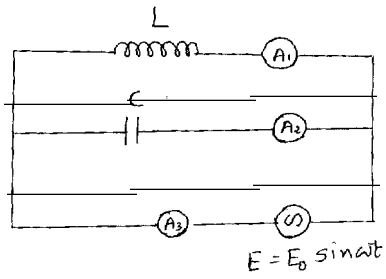
91. In a LCR circuit capacitance is changed from C to 2C. For the resonant fequency to remain unchanged, the inductance should be change from L to

(a) 4L (b) 2L (c) L/2 (d) L/4

92. In an LCR series ac circuit the voltage across each of the components L, C and R is 50 V. The voltage across the LC combination will be

(a) 50 V (b) $50\sqrt{2}$ V (c) 100 V (d) 0 V (zero)

93. An inductor L and capacitor C are connected in the circuit as shown in fig. The frequency of the power supply is equal to the resonant frequency of the circuit. Which ammeter will read zero ?



(a) A_1 (b) A_2

(c) A_3 (d) None of these

94. In a circuit L, C and R are connected in series with an alternating voltage source of frequency f. The current leads the voltage by 45^0. The value of c is,

(a) $\frac{1}{2\pi f\left(2\pi f L+R\right)}$ (b) $\frac{1}{\pi f\left(2\pi f L+R\right)}$ (c) $\frac{1}{2\pi f\left(2\pi f L-R\right)}$ (d) $\frac{1}{\pi f\left(2\pi f L-R\right)}$

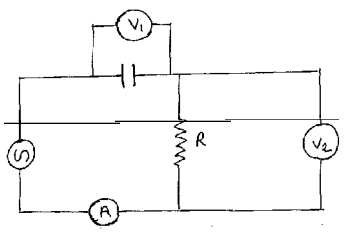
95. In a series resonant LCR circuit, the voltage across R is 100 V and R= 1kΩ with C =2 μF. The resonant frequency ω is 200 rad/s. At resonance the voltage across L is.

(a) 40V (b) 250 V (c) 4×10^{-3} V (d) 2.5×10^{-2} V

96. A coil of inductive reactance 31 Ω has a resistance of 8 Ω . It is placed in series with a condenser of copacitive reactance 25 Ω . The combination is connected to an a.c. source of 110 volt. The power factor of the circuit is.

(a) 0.80 (b) 0.33 (c) 0.56 (d) 0.64

97. The diagram shows a capacitor C and resistor R connected in series to an ac source. V_1 and V_2 are voltmeters and A is an ammeter, consider the following statements.



(1) Readings in A and V_2 are always in phase.

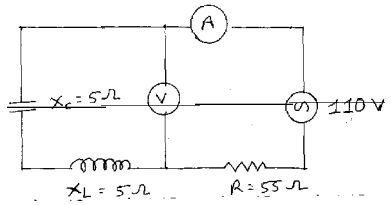
(2) Reading in V_1 is ahead in phase with reading in V_2.

(3) Reading in A and V_1 are always in phase.

Which of these statements are / is correct

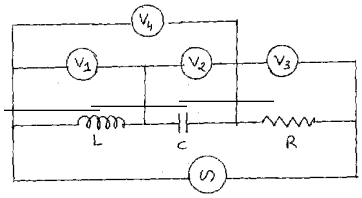
(a) 1) only (b) 2) only (c) 1) and 2) only (d) 2) and 3) only

98. The reading of ammeter in the circuit shown will be



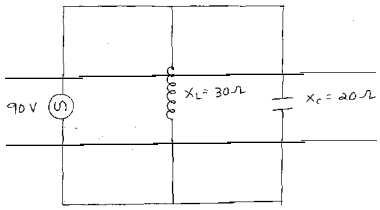
(a) 24 A (b) 2.4 A (c) Zero (d) 1.7 A

99. In the adjoining circuit the voltmeter whose reading will be zero at resonance is



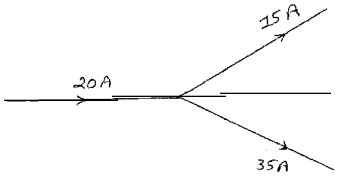
(a) V_1 (b) V_2 (c) V_3 (d) V_4

100. In the adjoining figure the impedance of the circuit will be



(a) 120Ω (b) 50Ω (c) 60Ω (d) 90Ω

101. Is it possible



(a) Yes (b) No

(c) can not be predicted (d) Insufficient data to reply

KEY NOTE

1	B	26	A	51	B	76	B
2	C	27	B	52	B	77	C
3	D	28	C	53	C	78	A
4	A	29	C	54	A	79	B
5	A	30	D	55	D	80	A
6	B	31	A	56	D	81	A
7	D	32	B	57	A	82	B
8	D	33	B	58	C	83	A
9	B	34	A	59	D	84	B
10	C	35	C	60	D	85	A
11	D	36	D	61	D	86	A
12	D	37	B	62	A	87	B
13	C	38	B	63	D	88	B
14	B	39	C	64	B	89	B
15	D	40	A	65	B	90	B
16	B	41	D	66	A	91	C
17	C	42	B	67	A	92	D
18	A	43	A	68	A	93	C
19	C	44	C	69	D	94	A
20	A	45	A	70	B	95	B
21	C	46	A	71	D	96	A
22	C	47	B	72	B	97	B
23	D	48	C	73	D	98	C
24	B	49	B	74	A	99	D
25	D	50	D	75	A	100	B
						101	A

Hints & Solution

13. $\varepsilon \alpha l$, distance bet n rails.

14. Oppose any change

19. $L = k\sqrt{L, L_2}$, $k = \frac{1}{2}$

20. $L \alpha N^2$

21. $\frac{1}{2} LI^2 \quad I = \frac{10}{2} 5A$

22. Current in B_1 promptly becomes zero while in B_2 becomes slowly zero.

25. Rate of decrease of area of semicircle ring $\frac{dA}{dt} = 2R\nu$

$$\varepsilon = \frac{d\theta}{dt} = B\frac{dA}{dt} = 2R\nu B$$

to oppose the change, Q at higher potential

27. $\rho = F.\nu. = \frac{B^2\ell^2\nu^2}{R} = \frac{1}{6}w$

28. Current will flow anticlock wise as seen from left side of the figure.

29. Consider rod as a bettary $I = 1A$ $F = I\ell b$

30. $\varepsilon_O - \varepsilon_P = \frac{1}{2} BW\ell^2$, $\varepsilon_O - \varepsilon_Q = \frac{1}{2} BW(3\ell)^2$

$$\varepsilon_P - \varepsilon_Q = 4\,BW\ell^2$$

31. $I = I_O e^{\frac{-Rt}{L}}$ where $I_O = \frac{E}{R} \Rightarrow I = E^{-\alpha} = \frac{1}{e^2}$

33. $I = \frac{B\nu\ell}{R}$, $R = 4\Omega$

34. $I = I_O(1 - e^{-t/T})$ $(I_O = I_{max})$

$$Q = \int_0^T (I_O - I_O e^{-t/T})\, dt = I_O T/e$$

36. $I = I_O(1 - e^{-t/T}) \Rightarrow 4(1 - e^{-\frac{1}{4}}) = 4\left(1 - \frac{1}{(2.718)}\frac{1}{4}\right) = 0.88\,A$

38. When the contact is broken current decrease in circuit, inductor release current to oppose it.

48. $I = I_O (1 - e^{-Rt/L})$

$I_\alpha = I_O (1 - e^{-\infty}) = 1.5 \quad I1 = 1.5 (1 - e{-}2) \Rightarrow \frac{I_\infty}{I_1} = \frac{1}{1 - e^{-2}}$

49. $\varepsilon = M\frac{dI}{dt} \Rightarrow 0.005 \frac{d}{dt} (10 \sin 100 \pi t) = \varepsilon_{max} = 5\pi$

50. $I_O = \frac{12}{6} = 2A$. Current becomes half in time $t = 0.693 \frac{L}{R} = 1$ ms

64. $\rho = \rho_m \cos \theta$

67. With help of phasor $\nu = \sqrt{\nu_R^2 + \nu_C^2}$

71. $I = \frac{\varepsilon_O}{\sqrt{R^2 + \omega^2 L^2}} = \frac{\varepsilon_O}{R\sqrt{1 + \left(\frac{\omega L}{R}\right)^2}} = \frac{I_O}{\sqrt{1 + \left(\frac{\omega L}{R}\right)^2}}$

72. $I_{rms} = \frac{\nu_{rms}}{|z|} = \frac{300}{30} = 10A$

77. $I_O = \frac{NAB\omega}{R}$

78. for inductor $\tan\theta = \frac{\omega L}{R}$

for capacitor $\tan\theta = \frac{1}{\omega CR}$

$|\theta|$ is same 50, $\omega L = \frac{1}{\omega C}$ means resoance

$I_{rms} = \frac{\nu_{rms}}{R}$

83. L.C. oscillator

84. $t = \frac{T}{4} = \frac{2\pi\sqrt{LC}}{4}$

95. $\nu_L = \nu_C = I_{x_C} = \frac{\nu}{R\omega C}$

97. In RC series circuit ν_C leads ν_R by $\frac{\pi}{2}$

98. $X_L = X_C = 5\ \Omega \Rightarrow V = 0 \Rightarrow I = 0$

100. $I_L = \frac{90}{30} = 3A$, $I_C = \frac{90}{20} = 4.5\ A$

$I = I_C - I_L = 1.5\ A$

$Z = \frac{\nu}{I} = \frac{90}{1.5} = 60\ \Omega$

101. in AC circuit AB has R, BC has C & BD has pure L

Unit 15
Electromagnetic Waves

SUMMARY

Type	Wavelength Range	Production	Detection
Radio	> 0.1 m	Rapid acceleration and decelerations of electrons in aerials.	Receiver's aerials (conducting wire)
Microwave	0.1 m to 1 mm	Klystron, magnetron valve, Gun diode.	Point contact diodes
Infra- red	1 mm to 700 nm	Vibration of atoms and molecules.	Thermopile, Bolometer, infrared photographic film
Visible Light	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level.	The eye, photocells, photographic film, photo diode (LDR), light dependent resistor
Ultraviolet	400 nm to 1 nm	Inner shell electrons in atoms moving from one energy level to a lower level.	solar cell, Photocells, photographic film
X-rays	1 nm to 10^{-3} nm	X- ray tubes or inner shell electrons of atom	Photographic film, Geiger tubes, Ionization chamber,
Gamma rays	< 10^{-3} nm	Radioactive decay of the nucleus.	– do –

- The oscillating charges are responsible for the generation of periodically varying electric field in the space. Further, the oscillating charges generate varying electric current which in turn is responsible for the generation of periodically varying magnetic field. This way the electromagnetic waves are generated.
- The frequency of generated electromagnetic waves is equal to the frequency of oscillation of the electric charges. In case of electromagnetic waves

  c (Velocity) = λ (Wavelength) × f (Frequency)
- In the region closer to the oscillating charges, the phase difference between $\vec{E}$ and $\vec{B}$ fields is $\frac{\pi}{2}$, and their magnitude quickly decreases as $\frac{1}{r^3}$ (where r = distance from the source). These components of the transmitted waves (or fields) are called inductive components.
- At large distances from the source, $\vec{E}$ and $\vec{B}$ are in phase and the decrease in their magnitude is comparatively slower with distance, as per $\frac{1}{r}$. These components of electromagnetic radiation are called radiated components.

- Electromagnetic waves are self sustaining oscillations of electric and magnetic fields in free space, or vacuum. No material medium is associated with vibrations of the electric and magnetic fields.
- The velocity of electromagnetic waves in vacuum (free space) is

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = 2.99792 \times 10^8 \text{ ms}^{-1}$$

- The velocity of the electromagnetic waves in any medium is given by $v = \frac{1}{\sqrt{\mu\varepsilon}}$

  where μ = Permeability of the medium, and

  ε = Permittivity of the medium.
- The velocity of light depends on electric and magnetic properties of the medium.
- The refractive index of a medium is $n = \frac{c}{v} = \sqrt{\mu_r \varepsilon_r} = \sqrt{\mu_r K}$.
- Electromagnetic waves exert pressure on a surface when they are incident on it, called radiation pressure.
- If ΔU is the energy of electromagnetic waves incident on a surface of unit area per unit time normal to the direction of flow of energy, then assuming that the energy is completely absorbed, the momentum of the electromagnetic radiation transferred to the surface is $\Delta p = \frac{\Delta U}{c}$

  which also represents radiation pressure (P_S).
- The electromagnetic energy per unit volume (energy density) in a region is given by

$$\rho = \rho_E + \rho_B = \frac{1}{2} \varepsilon_0 E^2 + \frac{B^2}{2\mu_0} = \varepsilon_0 E_{rms}^2$$

- The radiant energy passing through unit area normal to the direction of propagation in one second is called the intensity of radiation I.
- The energy of the electromagnetic waves is equal to the kinetic energy of the charges oscillating between the two spheres.

MCQ

For the answer of the following questions choose the correct alternative from among the given ones.

(1) Who produced the electromagnetic waves first ?

(A) Marconi (B) Maxwell (C) J.C. Bose (D) Hertz

(2) The dimensional formula of $\ell_o E_o$ is _____

(A) L^2T^{-2} (B) $L^{-2}T^2$ (C) L^1T^{-1} (D) $L^{-1}T^1$

(3) A plane electromagnetic wave is incident on a material surface. The wave delivers momentum P and energy E

(A) P=0, $E \neq 0$ (B) $P \neq 0$, E=0 (C) $P \neq 0$, $E \neq 0$ (D) P=0, E=0

(4) If V_γ, V_x and V_m are the velocity of the *v* rays, x rays, micro waves respectively in space, then

(A) $V_r < V_x < V_m$ (B) $V_r = V_x = V_m$ (C) $V_r > V_x > V_m$ (D) $V_r > V_x < V_m$

(5) If $\lambda_r, \lambda_x and \lambda_m$ are the wave lengths of the r-rays, x rays and micro waves respectively in space then

(A) $\lambda_r > \lambda_x > \lambda_m$ (B) $\lambda_r < \lambda_x < \lambda_m$ (C) $\lambda_r = \lambda_x = \lambda_m$ (D) $\lambda_r < \lambda_m < \lambda_x$

(6) According to Maxwell, a changing electric field produces

(A) emf (B) Electric current

(C) magnetic field (D) radiation pressure

(7) An electromagnetic wave going through vaccum is described by E= E_o sin(kx- ω t). Which of the following is independent of the wavelength?

(A) ω (B) k/ω (C) kω (D) k

(8) Which of the following have zero average value in a plane electromagnetic wave?

(A) Electric energy (B) Magnetic energy

(C) Electric field (D) None of these.

(9) If the relative permeability and dielectric constant of a given medium are equal to ℓ_r nd K respectively, then the refractive index of the medium is equal to_____

(A) $\sqrt{\ell_r K}$ (B) $\sqrt{\ell_r E_o}$ (C) $\sqrt{\ell_o E_o}$ (D) $1/\sqrt{\ell_r k}$

(10) Astraonomers have found that electromagnetic waves of wavelength 21cm are continuously reaching the Earth's surface . Calculate the frequency of this radiation. (c= 3×10^8 m/s)

(A) 14.28 GHz (B) 1.428 kHz (C) 1.428 MHz (D) 1.428 GHz

(11) Electric field in an electromagnetic wave is given by E=50 $\sin\omega$(t-x/c) NC^{-1}. Intensity of this wave is _____ Wm^{-2}

(A) 50 (B) 1.1×10^{8} (C) 3.3 (D) 5.5×10^{-19}

(12) The amplitude of the electric field in a parallel beam of light of intensity 2.0 Wm^{-2} is _____

(A) $38.8NC^{-1}$ (B) $19.4NC^{-1}$ (C) $9.7NC^{-1}$ (D) None of these.

(13) Speed of electromagnetic wave is the same

(A) for all wavelengths (B) in all media

(C) for all intensities (D) for all frequencies

(14) The maximum electric field in a plane electromagnetic wave is $900NC^{-1}$. The wave is going in the x direction and the electric field is in the y direction. The maximum magnetic field in the wave is _____T

(A) 3×10^{-8} (B) 3×10^{-6} (C) 27×10^{-6} (D) 27×10^{10}

(15) Electromagnetic waves are produced by

(A) a static charge (B) a moving charge

(C) an accelerating charge (D) chargeless particles

(16) Maxwells equations are derived from the laws of_____

(A) electricity (B) magnetism

(C) both electricity and magnetism (D) mechanics

(17) Which of the following electromagnetic waves has the longest wavelength?

(A) Radio waves (B) Infrared radiations

(C) x rays (D) visible rays

(18) Which of the following electromagnetic waves has the highest frequency?

(A) radio waves (B) micro waves (C) r rays (D) x rays

(19) Which of the following electromagnetic waves is used in telecommunication?

(A) radio waves (B) visible radiations

(C) ultraviolet rays (D) micro waves

(20) The maximum value of $\vec{E}$ in an electromagnetic waves in air is equal to 6.0×10^{-4} Vm^{-1}. The maximum value of $\vec{B}$ is _____

(A) 1.8×10^{5} T (B) 2.0×10^{4} T (C) 2.0×10^{-12} T (D) 1.8×10^{13} T

(21) Dimensional formula of intensity of radiation is _____

(A) $M^1L^2T^{-2}$ (B) $M^1L^0T^{-2}$ (C) $M^1L^2T^{-3}$ (D) $M^1L^0T^{-3}$

(22) The frequency of an electromagnetic wave in free space is 3MHz. When it passes through a medium of relative permeability ε_r=4.0 , then its frequency

(A) becomes half (B) become doubled

(C) remain same (D) become $\sqrt{2}$ times

(23) The frequency of electromagnetic wave having wavelength 25mm is _____ Hz

(A) 1.2×10^{10} (B) 7.5×10^{5} (C) 1.2×10^{8} (D) 7.5×10^{6}

(24) Unit of energy density of electromagnetic wave is _____

(A) Jm^{-3} (B) Jm^{-2} (C) wm^{-2} (D) None of these

(25) What is the ratio of velocities of light rays of wavelengths 4000^0 A and 8000^0 A in vaccum ?

(A) 1:2 (B) 1:1

(C) 2:1 (D) cannot be determined

(26) Which of the following rays are not electromagnetic waves?

(A) $\propto$ rays (B) γ rays (C) β rays (D) heat rays

(27) A new system of unit is evolved in which the values of ℓ_0 and ε_0 are 2 and 8 respectively. Then the speed of light in this system will be

(A) 0.25 (B) 0.5 (C) 0.75 (D) 1

(28) Our eyes respond to wavelength ranging from

(A) 400nm to 700nm (B) $-\infty$ to $+\infty$

(C) 1mm to 700nm (D)700nm to 800nm

(29) In microwave oven, we use electromagnetic oscillators which produce electromagnetic waves in the wavelength range

(A) 1mm to 10m (B) 0.7 μm to 1mm

(C) 0.1m to 1mm (D) 0.1 μm to 0.7 μm

(30) What is the direction of $\vec{E} \times \vec{B}$ in an electromagnetic wave?

(A) same as that of $\vec{E}$

(B) same as that of $\vec{B}$

(C) same as the direction of propagation of electromagnetic wave

(D) none of these

(31) The wavelength of x rays is of the order of

(A) 1cm (B) 1m (C) 1micron (D)1angstrom

(32) A plane electromagnetic wave of frequency 25MHz travels in free space along the x direction. At a particular point in space and time $\vec{E} = 6.3\,\hat{j}\ Vm^{-1}$ then $\vec{B}$ at this point is_____

(A) $2.1 \times 10^{-8}\,\hat{i}$ T (B) $2.1 \times 10^{-8}\,\hat{k}$ T (C) $1.89 \times 10^{9}\,\hat{k}$ T (D) $2.52 \times 10^{-7}\,\hat{k}$ T

(33) The magnetic field in a plane electromagnetic wave is given $B_y = 2 \times 10^{-7} \sin(0.5 \times 10^{3}x + 1.5 \times 10^{11}t)$T. The expression for electric field is

(A) $E_x = 60\sin(0.5\times10^{3}x+1.5\times10^{11}t)Vm^{-1}$

(B) $E_z = 60\sin(0.5\times10^{3}x+1.5\times10^{11}t)Vm^{-1}$

(C) $E_z = 60\sin(1.5\times10^{11}x+0.5\times10^{3}t)Vm^{-1}$

(D) $E_z = 60\times10^{15}\sin(1.5\times10^{11}x+0.5\times10^{3}t)Vm^{-1}$

(34) Light with an energy flux of w/m^3 or Wm^{-3} falls on a non-reflecting surface at normal to surface. If the surface has an area of $20m^2$. The average force exerted on the surface during 30 minutes is_____

(A) $6.48\times10^{5}N$ (B) $3.60\times10^{2}N$ (C) $1.2\times10^{-6}N$ (D) $2.16\times10^{-3}N$

(35) Energy density of an electromagnetic wave of intensity $0.02\ Wm^{-2}$ is _____

(A) $6.67\times10^{-11}Jm^{-3}$ (B) $6\times10^{6}Jm^{-3}$

(C) $1.5\times10^{10}Jm^{-3}$ (D) none of the above

(36) The waves used in communication are generally called

(A) γ rays (B) α rays (C) microwaves (D) radiowaves

(37) For an electromagnetic wave, the phase difference between vectors $\vec{E}$ and $\vec{B}$ (far away from the source)

(A) 0 (B) $\frac{\pi}{2}$ (C) π (D) $\frac{3\pi}{2}$

(38) In an electromagnetic wave, if the amplitude of magnetic field is $3\times10^{-10}T$, the amplitude of the associated electric field will be_____

(A) $9\times10^{-2}Vm^{-1}$ (B) $3\times10^{-10}Vm^{-1}$ (C) $3\times10^{-2}Vm^{-1}$ (D) $1\times10^{-18}Vm^{-1}$

(39) The electric and magnetic field of an electromagnetic wave are

(A) in phase and perpendicular to each other

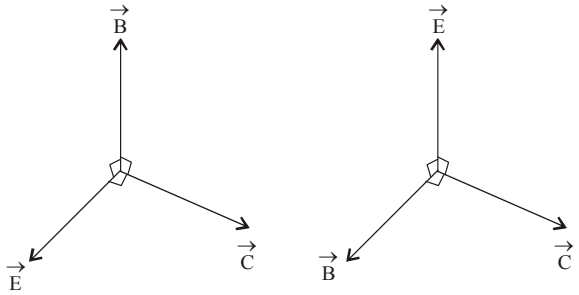
(B) in phase and parallel to each other

(C) in opposite phase and perpendicular to each other

(D) in opposite phase and parallel to each other

(40) The direction of electric and magnetic vector for an electromagnetic wave travelling along X-axis are shown in fig. A and B then

(A) A is correct, B is wrong
(B) B is correct, A is wrong
(C) Both A and B are correct
(D) Both A and B are wrong



(41) The dimensions of $\frac{1}{\mu\varepsilon}$ are

(A) $M^0L^1T^{-1}$ (B) $M^0L^{-1}T^1$ (C) $M^0L^2T^{-2}$ (D) $M^0L^1T^{-2}$

(42) The dimension of $\sqrt{\mu_r}k$ is

(A) $M^0L^{-1}T^{-1}$ (B) $M^0L^1T^{-1}$ (C) $M^0L^2T^{-2}$ (D) $M^0L^0T^0$

(43) The frequency of light wave of wavelength 5000^0A is _____Hz

(A) 6×10^{14} (B) 1.5×10^{-2} (C) 1.5 (D) 6×10^{1}

(44) Unit of μ_0C is same as that of

(A) current (B) resistance (C) electric charge (D) velocity

(45) The amplitude of the magnetic field part of an electromagnetic wave in vaccum is Bm = 510 nT. Then the amplitude of the electric part of the wave is_____

(A) $1.53\times10^{11}V/m$ (B) 1.53 V/m

(C) $1.53\times10^{2}V/m$ (D) $1.53\times10^{8}V/m$

(46) If the direction of magnetic field $\vec{B}$ at some instant is along +ve Z direction and the electromagnetic wave is propagating along +ve X direction, then the direction of electric field $\vec{E}$ at that instant is_____

(A) along -ve Y direction (B) along +ve Y direction

(C) along +ve X direction (B) along -ve X direction

(47) Relation between amplitudes of electric and Magnetic field is.....

(A) $E_0 = B_0$ (B) $E_0 = cB_0$ (C) $E_0 = \frac{B_0}{c}$ (D) $E_0 = \frac{c}{B_0}$

(48) If a source is transmitting electromagnetic wave of frequency 8.2×10^6Hz, then the wavelength of the electromagnetic waves transmitted from the source will be_____

(A) 36.6m (B) 42.3m (C)40.5m (D)50.9m

(49) The velocity of light in vaccum can be changed by changing____

(A) frequency (B)wavelength (C)amplitude (D) none of these

(50) An electromagnetic wave going through vaccum is described by E=Eo sin(kx-ωt) then B=Bo sin(kx-ωt) then

(A) $E_0B_0 = \omega k$ (B) $E_0k = B_0\omega$ (C) $E_0\omega = B_0k$ (D)none of these

(51) If the wavelength of light is 4000^0 A then the number of waves in 1mm length will be___

(A)2.5 (B)2500 (C)250 (D)25000

(52) The SI unit of displacement current is_____

(A) coulomb (B)henry (C) ampere (D)faraday

(53) The electromagnetic waves do not transport_____

(A) energy (B) charge (C)momentum (D)information

(54) An electric charge oscillating with a frequency of 1kilo cycles/s can radiates electromagnetic waves of wavelength

(A) 100km (B)200km (C) 300km (D)400km

(55) The frequency 1057MHz of radiation arising from two close energy levels in hydrogen belongs to_____

(A) radio waves (B)infrared waves (C)micro waves (D) γ rays

(56) Electromagnetic waves travelling in a medium which has relative permeability 1.3 and relative permeability 2.14 speed of electromagnetic waves in this medium will be_____

(A) $3.6\times10^8 m/s$ (B) $1.8\times10^8 m/s$ (C) $1.8\times10^6 m/s$ (D) $13.6\times10^6 m/s$

(57) A plane electromagnetic wave is incident on a material surface. If the wave delivers momentum p and energy E, then

(A) p=0,E=0 (B) $p \neq 0, E \neq 0$ (C) $p \neq 0, E = 0$ (D) $p = 0, E \neq 0$

(58) Maxwell's modified form of Ampere's circuital law is____

(A) $\oint \vec{B} \cdot d\vec{S}$ (B) $\oint \vec{B} \cdot d\vec{S} = \mu_o i$

(C) $\oint \vec{B} \cdot d\vec{l} = \mu_o i + \mu_o \in_o \frac{d\phi_E}{dt}$ (D) $\oint \vec{B} \cdot d\vec{l} = \mu_o i + \frac{1}{\in_o}\frac{dq}{dt}$

(59) The wavelength of x rays is of the order of

(A) 10^{-3}m (B) 10^{-5}m (C) 10^{-10}m (D) 10^{-12}m

(60) A point source of electromagnetic radiation has an average output power of 800W. The maximum value of electric field at a distance of 4.0m from the source is

(A) $64.7 Vm^{-1}$ (B) $57.8 Vm^{-1}$ (C) $56.72 Vm^{-1}$ (D) $54.77 Vm^{-1}$

(61) A plane electromagnetic wave $E_z = 100\cos(6\times10^8 t + 4x)Vm^{-1}$ propagate in a medium of refractive index

(A)1.5 (B)2.0 (C)2.4 (D)4.0

(62) A plane electromagnetic wave of wave intensity $10\omega m^{-2}$ strikes a small mirror of area $20\,cm^2$, held perpendicular to the approaching wave. The radiation force on the mirror will be

(A) $6.6\times10^{-11}N$ (B) $1.33\times10^{-11}N$ (C) $1.33\times10^{-10}N$ (D) $6.6\times10^{-10}N$

(63) An observer is at 2m from an isotropic point source of light emitting 40w power. The rm.s value of electric due to the source at the position of the observer is_____

(A) $5.77\times10^{-8}Vm^{-1}$ (B) $17.3Vm^{-1}$ (C) $57.7\times10^{-8}Vm^{-1}$ (D) $1.73Vm^{-1}$

(64) Electromagnetic waves used in medicine to destroy cancer cells

(A) radio waves (B)infrared rays

(C)gamma rays (D)ultraviolet rays

(65) What is the name associated with the equation $\phi\,\vec{E}\,.\,\overrightarrow{dt} = -\dfrac{d\phi\beta}{dt}$

(A) Gauss law for electricity (B) Gauss law for magnetism

(C) ampere's law (D)faraday's law

(66) What oscillates in an electromagnetic wave?

(A) $\vec{E}$ and $\vec{B}$ (B) $\vec{B}$ (C) $\vec{E}$ (D)none of these

(67) Which of the following rays are not electromagnetic waves?

(A) α rays (B) γ rays (C) β rays (D)heat rays

(68) The rms value of the electric field of the light coming from the sun is 720 N/c. The average tatal energy density of the electromagnetic wave is

(A) $4.58\times10^{-6}Jm^{-3}$ (B) $6.3\times10^{-9}Jm^{-3}$ (C) $81.35\times10^{-12}Jm^{-3}$ (D) $3.3\times10^{-3}Jm^{-3}$

(69) What is the wave length of range of electromagnetic waves?

(A) 10^{-8}m to 10^{15}m (B) 10^{-15}m to 10^{8}m

(C) 10^{-15}m to 10^{15}m (D) 10^{8}m to 10^{15}m

(70) What is the wavelength range of visible light?

(A) 10^0 A to 100^0 A (B) 4000^0 A to 7000^0 A

(C) 8000^0 A to 10000^0 A (D) 10000^0 A to 15000^0 A

(71) Unit of ℓ oC is same as that of

(A)current (B)resistance (C)electric charge (D) velocity

(72) In electromagnetic spectrum, the visible light lie between

(A) radiowaves and microwaves

(B) ultraviolet rays and infrared rays

(C) ultraviolet rays and x rays

(D) infrared rays and microwaves

(73) Which of the following statements is not true in case of electromagnetic waves?

(A) they are light waves (B)theay are transverse waves

(C) propagates through space (D) they are longitudinal waves

(74) The oscillating electric and magnetic field vectors of an electromagnetic waves far away from source are oriented along

(A) Mutually perpendicular direction and differ in phase by 90^0

(B) Mutually perpendicalar and in same phase

(C) In same direcfi on and in same phase

(D) In same divecfi on and differ in phaseby 90°

(75) Which of the following option of electromagnetic waves is in order of increasing frequency ?

(A) microwaves, ultraviolet rays, x rays

(B) gamma rays, ultraviolet rays, radiowaves

(C) radiowaves, visible light, infrared rays

(D) gamma rays, visible light, ultraviolet rays

(77) The sum delivers $_{10^3}$ Wm^{-2} of electromagnetic flux to earth's surface. The total power that is incident on a roof of dimension 8m x 20m will be_____

(A) $4\times10^5 w$ (B) $2.56\times10^4 w$ (C) $6.4\times10^5 w$ (D) $1.6\times10^5 w$

(77) Bolometer is used to detect _____.

(A) infrared rays (B) ultraviolet rays (C) x rays (D) γ rays

(78) Range of frequency of microwaves is about_____.

(A)530kHz to 1710kHz (B)54MHz to 890MHz

(C) 3GHz to 300GHz (D) $4\times10^{14} Hz$ to $7\times10^{14} Hz$

(79) SI unit of displacement current is

(A) coulomb (B)ampere (C)faraday (D)

(80) The frequencies of x rays, γ rays and ultraviolet rays are respectively p, q and r then

(A)p<q, q>r (B) p>q, q>r (C)p>q, q<r (D)p<q, q<r

(81) At room temperature, if the relative permitivity of water is 80 and the realtive permiability be 0.0222 then the velocity of light in water is _____m/s

(A) 2.5×10^{8} (B) 2.25×10^{8} (C) 3.5×10^{8} (D) 3×10^{8}

(82) If the electric field associated with a radiation of frequency 10MHz is E=10sin(kx-ωt)mV/m then its energy density is_____ Jm^{-3} ($\varepsilon o=8.85\times10^{-12}C^{2}N^{-1}m^{-2}$)

(A) 4.425×10^{-16} (B) 6.26×10^{-14} (C) 8.85×10^{-16} (D) 8.85×10^{-14}

(82) In an electromagnetic wave in free space, the direction of electric field vector $\vec{E}$ is along y axis and magnetic field vector $\vec{B}$ is along z axis then which of the following is true

(A) $\left(\vec{E}\times\vec{B}\right)\times\vec{E}=1$ (B) $\left(\vec{E}\times\vec{B}\right)\times\vec{B}=1$

(C) $\left(\vec{E}\times\vec{B}\right)\times\vec{B}=0$ (D) none of these

(84) When a plane electromagnetic wave travels in vaccum, the average electric energy density is given by (Eo is the amplitude of the electric field)

(A) $\frac{1}{4}\varepsilon_{o}Eo^{2}$ (B) $\frac{1}{2}\varepsilon_{o}Eo^{2}$ (C) $2\varepsilon_{o}Eo^{2}$ (D) $4\varepsilon_{o}Eo^{2}$

(85) In a plane electromagnetic wave , the electric field oscillates sinusoidaly at a frequency of $2.0\times10^{10}Hz.$ if the peak value of electric field is $60\,Vm^{-1}$ the average energy density (in Jm^{-3}) of the magnetic field of the wave will be (given $\mu o=4\pi\times10^{-7}$ Tm/A)

(A) $2\pi\times10^{-7}$ (B) $\frac{1}{2\pi}\times10^{-7}$ (C) $4\pi\times10^{-7}$ (D) $\frac{1}{4\pi}\times10^{-7}$

(86) Which of the following pairs of the component of space and time varying $\vec{E}=\left(E_x\hat{i}+Ey\hat{j}+Ez\hat{k}\right)$ and $\vec{B}=\left(B_x\hat{i}+By\hat{j}+Bz\hat{k}\right)$ would generate a plane electromagnetic wave travelling in +ve y direction

(A) $B_x\hat{i},\ E_z\hat{i}$ (B) $E_y\hat{i},B_z\hat{i}$ (C) $E_x\hat{i},B_x\hat{i}$ (D) $E_z\hat{i},B_x\hat{i}$

(87) Infrared radiations are detected by

(A) spectrometer (B)bolometer (C)photocells (D)geiger tubes

(88) Electromagnetic wave is produced by oscillating electric and magnetic fields $\vec{E}$ and $\vec{B}$. Choose only the incorrect statement from the following

(A) $\vec{E}$ is perpendicular to $\vec{B}$.

(B) $\vec{E}$ is perpendicular to the direction of propagation of the wave

(C) $\vec{B}$ is perpendicular to the direction of propagation of the wave

(D) $\vec{E}$ is parallel to $\vec{B}$

(89) The potential difference between the plates of a parallel plate capacitor is charging at the rate of $10^6 Vs^{-1}$. If the capatance is $2\mu F$. The displacement current in the dielectric of the capacitor will be

(A) 4A (B)3A (C)2A (D)1A

(90) Which of the following electromagnetic wave has the least frequency?

(A) radiowave (B) infrared radiation (C)microwave (D)x rays

(91) Which of the following electromagnetic wave has the least wavelength?

(A) rradiowave (B)visible wave (C) ultraviolet rays (D) microwaves

(92) Which of the following waves are not transverse in nature?

(A) light emitted from a sodium lamp

(B) sound waves travelling in air

(C) xrays from an x ray machine

(D) microwaves used in radar

(93) An electromagnetic wave

(A) can be deflected by electric field

(B) can be deflected by magnetic field

(C) can be deflected bye both electric and magnetic field

(D) none of these

(94) When an electromagnetic wave encounters a dielectric medium, the transmitted wave has

(A) same frquency but different amplitude

(B) same amplitude but different frequency

(C) same frequency and amplitude

(D) different frequency and amplitude

(95) According to Maxwell, a changing electric field produces_____

(A) emf (B)radiation pressure

(C) electric current (D)magnetic field

(96) _____was the first to predict the existence of electromagnetic waves

(A) Maxwel (B) Faraday (C)Ampere (D) hertz

(97) If the earth were not having atmosphere, its temperature_____.

(A) would have been low (B) would have been high

(C)would hav remain constant (D) none of these

(98) _____ is responsible for the green house effect

(A) infrared rays (B)ultraviolet rays (C)x raya (D)radiowaves

(99) The dimensional formula of energy density is

(A) $M^1L^0T^{-2}$ (B) $M^1L^{-1}T^{-2}$ (C) $M^1L^{-1}T^{-3}$ (D) $M^1L^0T^{-3}$

KEY NOTES

1	D	18	C	35	A	52	C	69	B	86	A
2	D	19	D	36	D	53	B	70	B	87	B
3	C	20	C	37	A	54	C	71	B	88	D
4	B	21	D	38	A	55	A	72	B	89	C
5	B	22	C	39	A	56	B	73	D	90	A
6	C	23	A	40	B	57	B	74	B	91	C
7	B	24	A	41	C	58	C	75	A	92	B
8	C	25	B	42	D	59	C	76	D	93	D
9	A	26	C	43	A	60	D	77	A	94	A
10	D	27	A	44	B	61	B	78	C	95	D
11	C	28	A	45	C	62	C	79	B	96	A
12	A	29	C	46	B	63	B	80	A	97	A
13	C	30	C	47	B	64	C	81	B	98	A
14	B	31	D	48	A	65	D	82	A	99	B
15	C	32	B	49	D	66	A	83	C		
16	C	33	B	50	B	67	C	84	A		
17	A	34	A	51	B	68	A	85	B		

Unit – 16

Rays Optics

Wave Optics

SUMMARY

- The path of the light propagation is called ray, but a bundald of such rays is called beam of light.
- The relation between focal length and radius of curvature is $f = \frac{R}{2}$ (for both the mirror) or R = 2f.
- In the case of plane mirror, R is infinite and therefore its focal length is also infinite.
- For mirrors, Gauss' equation is $\frac{1}{u} + \frac{1}{v} = \frac{1}{f} = \frac{2}{R}$, where, u = object distance, v = image distance, f = focal length, R = Radius of Curvature.
- Lateral magnification for mirrors is given by $m = \frac{h'}{h} = -\frac{v}{u}$, where, h' = height of image, h = height of object.
- The ratio of the sine of the angle of incidence to the sine of the angle of refraction for the given two media is constant, i.e., $\frac{\sin\theta_1}{\sin\theta_2} = n_{21}$ = Constant.

  where n_{21} is known as the refraative index of medium–2 with respect to medium–1.
- For a compound slab of different transparent media general form of Snell's Law is written as:

  $n_1 \sin\theta_1 = n_2 \sin\theta_2$
- Lateral shift x = t . $\theta_1\left(1 - \frac{n_1}{n_2}\right)$
- If two plane mirros M_1 and M_2 are inclined at angle θ, then no. of images form $n = \left(\frac{360}{\theta} - 1\right)$
- In concave mirror, when objects is between P and F, image formed is virtual, ereect and magnified. $\therefore$ m is positive. However, when object lies beyond F, image formed is real and inverted. $\therefore$ m is negative.
- In convex mirror, image is always virtual and erect, whatever be the position of the object.
- Critical angle $C = \sin^{-1}\left(\frac{1}{n}\right)$ where, n = refractive index medium.

  As $n_v > n_r$ $\therefore$ $C_v < C_r$. 'C' increases with temperature.

- Refraction from a spherical surface (for lens)

  (1) For refrection from rarer to denser medium : $\frac{-n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{R}$

  (2) Form denser to rarer medium : $\frac{-n_1}{v} + \frac{n_2}{u} = \frac{n_2 - n_1}{R}$

- Lens Maker's Formula is :

  $$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

  For Convex lens : R_1 = +, R_2 = –, f = +

  Convave lens : R_1 = –, R_2 = +, f = –

- Power of lens $P = \frac{1}{f}$ when f = 1m, $\therefore$ P = 1 diopter (D)

  For Convex lens P = +,

  Convave lens P = –

- If two lenses are in contact coaxilly,

  (i) $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$ (ii) $P = P_1 + P_2$ (iii) $m = m_1 \times m_2$

- For Prism equation is given by δ = i + e – A or A+δ = i + e

  At minimum angle of deviation, δ_m = 2i–A. $(\because i = e)$

  For thin prism $\delta_m = A(n-1)$

  $$n = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

- Newton's formula $f^2 = x_1 . x_2$
- The relation between δ, A and n is : δ = A(n– 1)

  Angular disperson, $\theta = \delta_V - \delta_r = (n_V - n_r)A$

  Dispersive power $\omega = \frac{\delta_V - \delta_r}{\delta} = \frac{n_V - n_r}{n-1}$, Where, $n = \frac{n_V + n_r}{2}$

- (i) Resolving power of human eye = 1'

  (ii) R.P. of Microscope = $\frac{2n\sin\theta}{\lambda}$

  (iii) R.P. of Telescope = $\frac{D}{1.22.\lambda}$

WAVE OPTICS

● Problem Solving Skills :

(1) Resultant intensity $I = R^2 = a^2 + b^2 + 2ab\cos\phi$

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

If $I_1 = I_2 = I_0$ तो $I_1 = I_0 + I_0 + 2I_0$ $\cos\phi = 4\, I_0 \cos^2\left(\frac{\phi}{2}\right)$

(2) Phase differance $\phi = \frac{2\pi}{\lambda}(\Delta x)$

$\Rightarrow$ Form $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

When $\cos\phi = +1$ $\therefore$ $I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

$\cos\phi = -1$ $\therefore$ $I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$

$$\therefore \frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2}$$

$\Rightarrow$ If the Sources are incoherent, $I = I_1 + I_2$

$\Rightarrow$ If W_1 and W_2 are widths of two Slis then , $\frac{W_1}{W_2} = \frac{I_1}{I_2} = \frac{a^2}{b^2}$

$\Rightarrow$ In the interfernce pattern $\frac{I_{max}}{I_{min}} = \frac{(a+b)^2}{(a-b)^2}$ where a, b = Amplitude.

$\Rightarrow$ In young's double slit experiment

(a) Position of bright fringes $x = n\lambda\frac{D}{d}$ (where n = 0 for centeral fringe)

(b) Positiion of dark fringes $x = \frac{(2n-1)\lambda D}{2d}$

(c) Width of each bright fringes = width of each darkfringes $\beta = \bar{x} = \frac{\lambda D}{d}$

(d) when entire apparatus is immersed in a medium of refractve index n, fringe width becomes

$$\bar{x}' = \beta' = \frac{\lambda' D}{d} = \frac{\lambda D}{nd} = \frac{\bar{x}}{n} = \frac{\beta}{n}$$

(e) Angular fringe width $= \frac{\beta}{D} = \frac{\lambda}{d}$

(f) fringe visibility is $V = I_{max} - I_{min} / I_{max} + I_{min}$

$\Rightarrow$ A thickness **t** of a medium of refractive index **n** is equivalent to a length **nt** in vacuum or air**.** This is called optical path length**.**

$\Rightarrow$ When a thin transperent plate of thickness **t** and refractive index **n** is placed in the path of one of the interfering waves, fringe width remains unaffected but the entire patern shiffts by

$$\Delta x = (n-1)\ t\ .\ \frac{D}{d} = (n-1)t\frac{\beta}{\lambda}$$

$\Rightarrow$ Law of Malus : $I = I_0 \cos^2\theta$

$\Rightarrow$ Brewster's law : $n = \tan\theta_P$

$\Rightarrow$ the intensity of porlarised light : $I = \frac{I_0}{2}$ where I_0 = intensity of unpolarsed light

$\Rightarrow$ Accoding to Doppler's effect for light waves

$$f' = f\left(1 \pm \frac{V}{C}\right)$$ where f' = apparent frequency of light

f = true frequency of light

$\Rightarrow$ **A symbol of refractive index is also denoted by**

MCQ Questions

(1). The velocity of light is maximum in a medium of_____.

(A) diamond (B) water (C) glass (D) vaccum

(2). A light of wavelength 320 nm enters in a medium of refractive index 1.6 from the air of refractive index 1.0 The new wavelength of light in the medium will be_____nm.

(A) 520 (B) 400 (C) 320 (D) 220

(3). "Bhautik" runs towards a plane mirror with a speed of 20 ms^{-1}, what is the speed of his image ?

(A) 45 ms^{-1} (B) 20 ms^{-1} (C) 15 ms^{-1} (D) 7.5 ms^{-1}

(4). A ray of light is incident at an angle 30^0 on a mirror, The angle between normal and reflected ray is_____.

(A) 15° (B) 30° (C) 45° (D) 60°

(5). The no. of images formed between two parallel plane mirror are _____.

(A) ∞ (B) 0 (C) 180 (D) 360

(6). To get five images of a single object one shold have two plane mirrors at an angle of_____.

(A) 36° (B) 72° (C) 180° (D) 302°

(7). If a glass rod is immersed in a liquid of the same refractive index, then it will_____.

(A) appear bent (B) appear longer (C) disappear (D) appear shorter

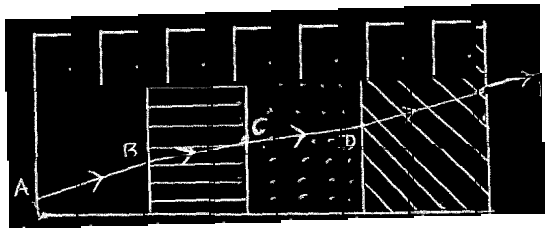
(8). For four different transperent medium $n_{41} \times n_{32} \times n_{21} =$ _____.

(A) $\frac{1}{n_{41}}$ (B) n_{41} (C) n_{14} (D) $\frac{1}{n_{14}}$

(9). A Plane mirror produces a magnification of_____.

(A) 0 (B) +1 (C) -1 (D) ∞

(10). A ray light passes through four transperent media with refractive indices n_1, n_2, n_3, n_4 as shown in figure The surfaces of all media are parallel, if the emergent ray DE is parallel of the incident ray AB we must have_____

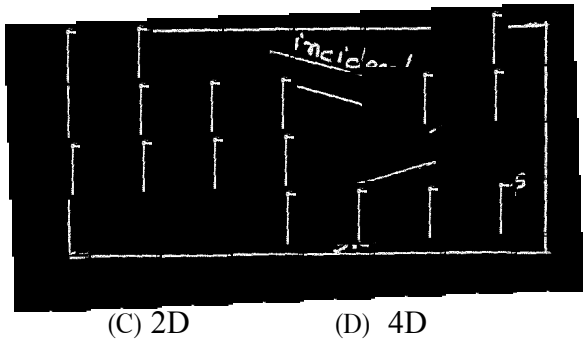


(A) $n_1 = n_2$ (B) $n_3 = n_4$ (C) $n_1 = n_4$ (D) $n_2 = n_3$

(11). A convex lens forms a real image of an object for it s two different positons on a screen if hight of the image in both cases be 16 cm and 4 cm then height of the object is_____cm.

(A) -4 (B) 4 (C) -8 (D) 8

(12). In shown figure two parallel rays incident on a mirror they are reflected as paraller rays as shown in the same figure what is the nature of the mirror ?



(A) plane mirror (B) concave

(C) convex (D) plano–concave

(13). The power of plane glass is_____.

(A) ∞ (B) 0 (C) 2D (D) 4D

(14). A convex lens is made up of three diffrent materials as shown in figure, for point object placed on its axis, the no. of imges formed are_____.



(A) 4 (B) 2

(C) 3 (D) 1

(15). Two thin lenses of focal lengths f_1 and f_2 are coaxially placed incontact with eachother then, the power of combination is_____.

(A) $\frac{f_1 + f_2}{2}$ (B) $\sqrt{\frac{f_1}{f_2}}$ (C) $\frac{f_1 f_2}{f_1 + f_2}$ (D) $\frac{f_1 + f_2}{f_1 f_2}$

(16). If thin prism of 5° gives a deviation of 2° then the refractive index of material of prism is_____.

(A) 1.4 (B) 1.5 (C) 1.6 (D) 1.0

(17). It is difficult to see through the fog because _____

(A) light is seattered by the doplets in the fog.

(B) fog absorbs light.

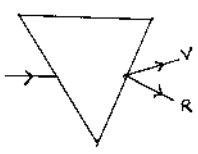
(C) rafractive index of fog is infinity.

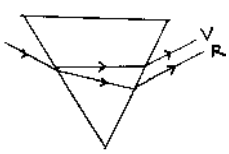
(D) light suffers totl internal reffection.

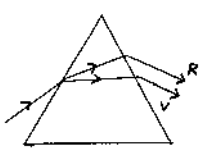
(18). what is the time taken in seconds to cross a glass plate of thikness 6 mm and $=2.0$ by light ?

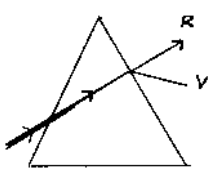
(A) 8×10^{-11} (B) 4×10^{-11} (C) 2×10^{11} (D) 16×10^{-11}

(19). Which of the following diagrams shows correctly the dispersion of white light by a prism ?

(A) 

(B) 

(C) 

(D) 

(20). Read the following quetions and choose if_____.

(A) both assertion and reason are true and the reason is correct explanation of the assertion.

(B) both assertion and reason are true but reason do not explain the assertion.

(C) Assertion is true but the reason are false.

(D) both assertion and reason are false.

(1) Assertion : Focal lengh of a lens for red colour is smaller than its focal length for violet colour

Reason : is becuse $n_r > n_V$

(A) B (B) A (C) C (D) D

(21). In which of the following cases a man will not see image grater than himself.

(A) convex mirror (B) concave mirror (C) plane mirror (D) none of these

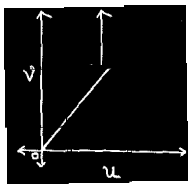
(22). A glass slab $(n = 1.5)$ of thikness 9 cm is placed over a written paper what is the Shift in the latters ?

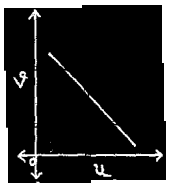
(A) 6 cm (B) 3 cm (C) 2 cm (D) 0 cm

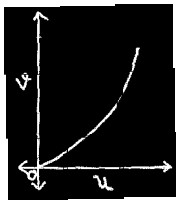
(23). A concave mirror of focal length 20 cm forms an virtual image having twice the linear dimensions of the object, the position of the object will be _____cm

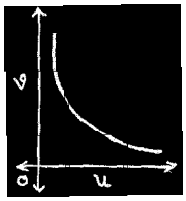
(A) 7.5 (B) -10 (C) 10 (D) -7.5

(24). In experiment to find focal length of a concave mirror a graph is drawn between the magnitude of u and v. The graph looks like_____.

(A) 

(B) 

(C) 

(D) 

(25). A mark at the bottom of the liquid appears to rise by 0.2 m, If depth of the liquid is 2.0 m then refractive index of the liquid is_____.

(A) 1.80 (B) 1.60 (C) 1.33 (D) 1.11

(26). A Sound wave travels from air to water. the angle of incidence is α_1 and the angle of reflection is α_2 If the snell's Law is valid then,_____.

(A) $\alpha_1 \geq \alpha_2$ (B) $\alpha_1 = \alpha_2$ (C) $\alpha_1 > \alpha_2$ (D) $\alpha_1 < \alpha_2$

(27). 1.6 is a refractive index of plano-convex lens, then the redius of curvature of the curved sunface is 60 cm. The focal length of the lens is_____ cm

(A) 50 (B) 100 (C) -50 (D) -100

(28). One convex lens and one concave lens placed is contect with eachother. If the ratio of their power is $2/3$ and focal length of the combination is 30 cm, then indivedual focal lengths are_____.

(A) 15 cm and -10 cm (C) 30 cm and -20 cm

(B) -15 cm and 10 cm (D) -30 cm and -30 cm

(29). A thin prism of 3°, angle made from glass of refractive index 1.5 is combined with another thin prism made from glass of refractive index 1.3 to produce dispersion without deveation. what is the angle of Prism of second prism.

(A) 3° (B) -3° (C) -5° (D) 5°

(30). If a ray of light is incident on a plane mirror at an angle of 30° then deviation produced by a plane mirror is_____.

(A) 60° (B) 90° (C) 120° (D) 150°

(31). The frequency of a light wave in a material is $4\times10^{14}\ H_z$ and wavelensth is 5000 A°. The refractive index of material will be_____ (take $c = 3\times10^{8}\ ms^{-1}$)

(A) 1.5 (B) 1.7 (C) 1.33 (D) None of thense

32. Mono chromatic light of wavelength 399 nm is incident from air on a water $(n = 1.33)$ Surface. The wavelength of refracted light is _____nm

(A) 300 (B) 600 (C) 333 (D) 443

33. If the refractive index of a material of an equilateral Prism is $\sqrt{3}$, then angle of minimum deviation will be_____.

(A) 50° (B) 60° (C) 39° (D) 49°

34. If the critical angle for total internal reflection from a medium to vacuum is 30° then velocity of light in the medium is _____ms^{-1} (take $c = 3.0\times10^{8}\ ms^{-1}$)

(A) 2.0×10^{8} (B) 1.5×10^{8} (C) 10^{8} (D) 1.5×10^{-8}

35. A ray of light passes from glass $(n = 1.5)$ to medium $(n = 1.60)$ The value of the critical angle of glass is_____.

(A) $\sin^{-1}\left(\frac{16}{15}\right)$ (B) $\sin^{-1}\sqrt{\frac{16}{15}}$ (C) $\sin^{-1}\left(\frac{1}{2}\right)$ (D) $\sin^{-1}\left(\frac{15}{16}\right)$

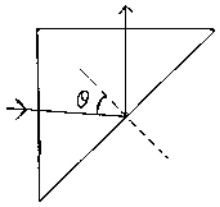
36. A double convex lens of focal length 6 cm is made of glass of refractive index 1.5, The radius of curvature of one surface is double than that of the other surface. The value of small radius of curvature is_____.

(A) 6 (B) 9 (C) 12 (D) 4.5

37. When a ray of light enters in a transperent medium of refractive index n, then it is observed that the angle of refraction is half of the angle of incidence. The value of angle of incidence will be _____.

(A) $2\cos^{-1}\left(\frac{n}{2}\right)$ (B) $\cos^{-1}\left(\frac{n}{2}\right)$ (C) $2\sin^{-1}\left(\frac{n}{2}\right)$ (D) $\sin^{-1}\left(\frac{n}{2}\right)$

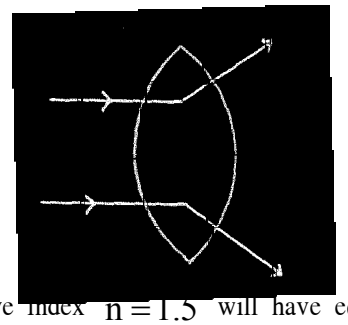
38. A prsim of glass is shown in figure A, ray incident normally on one face is totally reflected. If θ is 45°, the index of refraction of glass is_____.



(A) $< \sqrt{2}$ (B) $> \sqrt{2}$

(C) = 2 (D) None of these

39. A convex lens made up of a material of refractive index n_1 is immersed in a medium of refractive index n_2 as shown in the figure. The relation between n_1 and n_2 is_____.



(A) $n_1 = \sqrt{n_2}$ (B) $n_1 = n_2$

(C) $n_1 < n_2$ (D) $n_1 > n_2$

40. Two plano–convex lenses of radius of curvature R and refractive index $n = 1.5$ will have equivalent focal length equal to R, when they are placed_____.

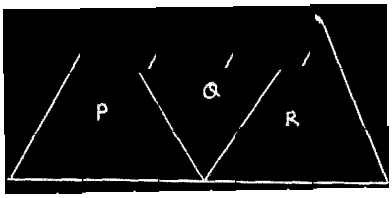
(A) at distance R (C) at distance $R/4$

(B) at distance $R/2$ (D) in contact with each other

41. A double convex lens made of glass of refractive index 1.6 has radius of curvature 15 cm each. The focal length of this lens when immersed in a fluid of refractive index 1.63 is_____.

(A) -40.75 (B) -407.5 (C) -125 (D) 12.5

42. One ray of light suffers minimum deviation in an equilateral prism P additlonal prism Q and R of indentical shown in fiqure. The ray will now suffer_____.



(A) greater devaition (C) total internal reflection

(B) same divaition as before (D) no devaition

43. Which of the following colours is scattered minimum ?

(A) Violet (B) red (C) blue (D) yellow

44. Angle of minimum devaition for a prism refractive index 1.5 is equal to the angle of the prism Then the angle of prism_____ $\left(given,\ \sin 48^\circ\ 36' = 0.75\right)$

(A) 62° (B) 82° (C) 60° (D) 41°

45. In a thin prism of glass ($a_{ng} = 1.5$) which of the following relation between the angle of minimum deviation δm and the angle of refraction r will be correct ?

(A) $\delta_m = \frac{r}{2}$ (B) $\frac{\delta_m}{2} = r$ (C) $\delta_m = 1.5\, r$ (D) $\delta_m = r$

46. An observer look at a tree of height 10 meters away with a telescope of magnifying power 10. To him, the tree appears_____.

(A) 10 times taller (B) 10 times smaler (C) 10 times nearer (D) 20 times nearer

47. A normal person wants to see two pillars at a distant 11 km away separately. The distance between two pillars should be approximately_____.

(A) 1 m (B) 3.2 m (C) 0.5 m (D) 1.6 m

48. When the length of microscope tube increases, its magnifying power_____.

(A) decreases (B) increaes (C) does not change (D) none of these

49. The focal lengths of objective and the eye–piece of a compound microscpe are fo and fe raspectively. Then_____.

(A) $F_o > F_e$ (B) $F_o < F_e$ (C) $F_o = F_e$ (D) none of these

50. The magnifying power of a telescope is 9.0 when it is focussed for parallel rays, then the dictance between its objective and eye–piece is 20 cm The focal lengths of lenses will be_____.

(A) 15 cm, 5 cm (B) 18 cm, 2 cm (C) 10 cm, 5 cm (D) 11 cm, 9 cm

51. A plano convex lens of f = 20 cm is silvered at plane surface New f will be_____cm

(A) 20 (B) 40 (C) 30 (D) 60

52. A ray of light from denser medium strikes a rarer medium at angle of incidence i. The reflected and refracted rays make an angle of 90° with each other The angle of reflection and refration are r and r' respectively. The crictical angle is _____.

(A) $\sin^{-1}$ (tan p) (B) $\tan^{-1}$ (tan r) (C) $\tan^{-1}$ (sin i) (D) $\sin^{-1}$ (tan r)

53. Relation between critical angle of water C_w and that of the glass C_g is _____. (given, $n_w = 4/3$, $n_g = 1.5$)

(A) $C_w < C_g$ (B) $C_w = C_g$ (C) $C_w > C_g$ (D) $C_w = C_g = O$

54. The radius of curvature of convex surface of a thin plano–convex lens is 15 cm and refractve index of its material is 1.6 The power of the lens will be_____.

(A) 6 D (B) 5 D (C) 4 D (D) 3 D

55. A ray of light passes through a prism having refractive index $\left(n=\sqrt{2}\right)$, Suffers minimum deviation If angle of incident is double the angle of refration within prism then angle of prism is_____.

(A) 30° (B) 60° (C) 90° (D) 180°

56. An air bubble inside glass slab $(n=1.5)$ appear from one side at 6 cm and from other side at 4 cm. Then the thikness of glass slab is_____cm

(A) 5 (B) 10 (C) 15 (D) 20

57. The magnifying power of objective of a compound microscope is 5.0 If the maginfying power of microscope is 30, then magnifying power of eye–piece will be_____.

(A) 3 (B) 6 (C) 9 (D) 12

58. Light of certain colour contain 2000 waves in the length of 1 mm in air. What will be the wavelength of this light in medium of refractive index 1.25 ?

(A) $1000\ \overset{o}{A}$ (B) $2000\ \overset{o}{A}$ (C) $3000\ \overset{o}{A}$ (D) $4000\ \overset{o}{A}$

59. In each of the following questions match columm-I and columm-II and select the correct match out of the four given choices.

Columm : I	Columm : II
(i) Snell's Law	(a) Frequency remais unaffected
(ii) In vacum	(b) n = sini/sinr
(iii) In glass	(c) $v_{violet} = v_{red}$
(iv) In going form one medium	(d) $v_{violet} < v_{red}$

(A) i–a, ii–b, iii–c, iv–d (C) i–b, ii–c, iii–d, iv–a

(B) i–d, ii–c, iii–b, iv–a (D) i–c, ii–b, iii–d, iv–a

60.

Columm – I	Columm – II
(i) While going from rarer to denser medium	(a) wavelength charger
(ii) While going from denser to rarer medium	(b) $= \frac{C}{V}$
(iii) While going to one medium to another	(c) Ray bends towards normal
(iv) Refractive index of medium	(d) Ray bends away from normal

(A) i–c, ii–d, iii–b, iv–a (C) i–d, ii–c, iii–b, iv–a

(B) i–a, ii–b, iii–c, iv–d (D) i–b, ii–c, iii–a, iv–d

61.

Columm – I	Columm – II
(i) Mean deviation	(a) $(n-1)A+(n'-1)A'=0$
(ii) Angular dispersion	(b) $\frac{n_v - n_r}{n-1}$
(iii) Dispersive power	(c) $(n_v - n_r)A$
(iv) Condition for no deviation	(d) $(n-1)A$

(A) i–c, ii–d, iii–b, iv–a (C) i–c, ii–b, iii–a, iv–d

(B) i–a, ii–b, iii–c, iv–d (D) i–d, ii–c, iii–b, iv–d

62. A convex lens of glass $(n=1.5)$ has focal lergth 0.2 m The lens is immersed in water of refractive index 1.33. The change in the power of convex lens is _____.

(A) 3.72 D (B) 4.62 D (C) 6.44 D (D) 1.86 D

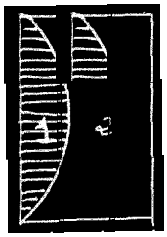
63. For a prism of refractive index $\sqrt{3}$, the angle of minimum deviation is eqviation is equal to the angle of prism, then angle of the prism is_____.

(A) 60° (B) 90° (C) 45° (D) 180°

64. A ray of light is incident normally on one of the faces of a solid prism of angle 30° and refractive index $\sqrt{2}$. The angle of minimum deviation is_____.

(A) 39° (B) 42° (C) 52° (D) 15°

65. A plano–convex lens has been fixed exactly into a plano–concave lens as shown in figure. Thier plane surface are parallel to each other. If both the lenses are made of different materials of refractive indices n_1 and n_2, R is the radius of curvature of the curved suface of the lens, their focal length of the combination will be_____.



(A) $\frac{R}{2(n_1+n_2)}$ (B) $\frac{R}{n_1-n_2}$

(C) $\frac{R}{n_1+n_2}$ (D) $\frac{R}{2(n_1-n_2)}$

66. A concave mirror has a focal langth 30 cm The distance between the two position of the object for whi ch image size is double of the object is_____.

(A) 30 cm (B) 15 cm (C) –25 cm (D) –15 cm

67. A concave lens forms the image of an object such that the distance between the object and the image is 10 cm and the magnification produced is $\frac{1}{4}$, the focal length of lens will be_____cm

(A) –6.2 (B) –12.4 (C) –4.4 (D) –8.8

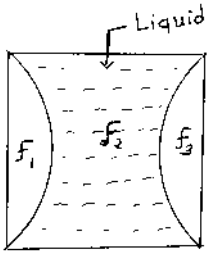
68. A prism of certain angle deviates the red and blue rays by 8° and 12° respectively. Another prism of the same prism angle deviates the red ligst at small angle and made of diffeunt materials The dispersive powers of the materials of the prisms are in the ratio_____.

(A) 5:6 (B) 9:11 (C) 6:5 (D) 11:9

69. The head light of a jeep are 1.2 m apart. If the pupil of the eye of an objerver has a diameter of 2 mm and light of wavelength $5836\ A^\circ$ is used what should be the maximum distance of the jeep from the observer if two head lights are just seem to be separated apart ?

(A) 30.9 km (B) 33.4 km (C) 3.34 km (D) 30.9 km

70. The effective focal length of the lens combination shown in the curved surface of the plano convex lenses are 12 cm each and refractive index of the material of the lens is 1.5. the refractive index of liquid is_____.



(A) 1.33 (B) 1.42

(C) 1.53 (D) 1.60

71. Interference is possible in_____.

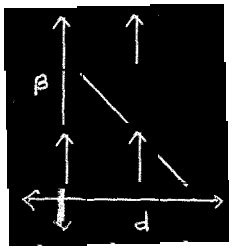
(A) light waves only (C) both light and Sound waves

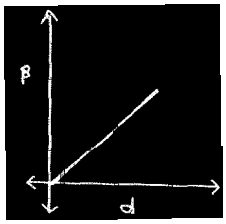
(B) Sound waves only (D) nono of these

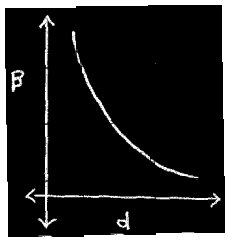
72. Huygin's wave theory of light can not explain_____phenomina.

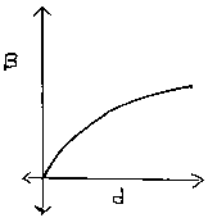
(A) Diffraction (B) Interference (C) palarization (D) Photoelectric effect

73. The correct curve between fring width β and distance between the slits d is shown below figure is_____.

(A) 

(B) 

(C) 

(D) 

74. The fringe width for red β_r $\left(\lambda_r = 8000\,A^\circ\right)$ and the fringe width for violct β_υ $\left(\lambda_\upsilon = 4000\,A^\circ\right)$ then $\frac{\beta_r}{\beta_\upsilon} =$ _____.

(A) 2:1 (B) 1:2 (C) 1:1 (D) $\sqrt{2}:1$

75. Wave ligth travels from an optically rarer medium to an optically denser medium its velocity decreaes because of change in_____.

(A) frequency (B) wavelensth (C) amplitude (D) phase

76. In young's double slite experiment if the width of 3rd fringe is 10^{-2} cm, then the width of 5th fringe will be_____cm

(A) 10^{-2} (B) 5×10^{-2} (C) 2×10^{-2} (D) 10^{+2}

77. The young's double slit experiment is performed with blue and with green light of wavelensths $4360\,A^\circ$ and $5460\,A^\circ$ respectively. If x is the distance of 4th maxima from the central one, then_____.

(A) $X_{blue} = X_{green}$ (B) $X_{blue} > X_{green}$ (C) $X_{blue} < X_{green}$ (D) none of these

78. The light waves from two coherent sources of same intensity interfere eachother. Then what will be maximam intensity when minimum intensity is zero ?

(A) 4 I (B) I (C) 4 I^2 (D) I^2

79. In young's doble Slit experiment the seventh maxima with wavelength λ_1, is at a distance d_1 and the same maxima with wavelength λ_2, is at a distance d_2 Then $\frac{d_1}{d_2} =$ _____.

(A) $\frac{\lambda_2}{\lambda_1}$ (B) $\frac{\lambda_1^{\,2}}{\lambda_2^{\,2}}$ (C) $\frac{\lambda_2^{\,2}}{\lambda_1^{\,2}}$ (D) $\frac{\lambda_1}{\lambda_2}$

80. The wave length corressponding to photon is $0.016\ A^\circ$. Its K.E._____ J.

($h = 6.66 \times 10^{-34}$ SI, $c = 3.0 \times 10^8 ms^{-1}$)

(A) 1.237×10^{-13} (B) 1.237×10^{13} (C) 12.37×10^{-13} (D) $12.37 \times 10^{+13}$

81. In young's double slit experiment, phase diffrence between light waves reaching 3rd bright fringe from central fringe with, is_____.($\lambda = 5000\ A^\circ$)

(A) zero (B) 2π (C) 4π (D) 6π

82. n^{th} bright fringe of red light $(\lambda_1 = 7500\ A^\circ)$ Coincides with $(n+1)^{th}$ bright fringe of green light $(\lambda_2 = 6000\ A^\circ)$. The value of n = _____.

(A) 8 (B) 4 (C) 2 (D) I

83. Which of the following will undergo maximum diffration ?

(A) α - particle (B) γ - rays (C) radio waves (D) light waves

84. A Slit of width $12 \times 10^{-7} m$ is illminated by light of wavelenth $6000\ A^\circ$. The angular width of the central maxima is appoximately_____.

(A) 30° (B) 60° (C) 90° (D) 0°

85. The distance between the first and sixth minima in the diffraction pattern of a single slit, it is 0.5 mm. The screen is 0.5 m away from the Slit. If the wavelength of light is $5000\ A^\circ$, then the width of the slit will be_____mm

(A) 5 (B) 2.5 (C) 1.25 (D) 1.0

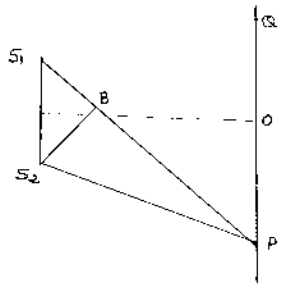
86. _____change in the polarization phynomina of ligst ?

(A) intensity (B) wavelength (C) phase (D) frequency

87. In yong's double slit experiment the phase diffrence is constant between two sources is $\pi/2$. The intensity at a point equi distant from the slits in terms of max. intensity I_0 is_____.

(A) $3\ I_0$ (B) $I_0/2$ (C) I_0 (D) $3I_0/4$

88. In figure young's double slit experiment Q is the position of the first bright fringes on the right side of O, p is the 11th fringe on the other side as measured from Q If $\lambda = 6000\ A^0$ then S_1B will be_____m



(A) 6.6×10^{-6} (B) 3.3×10^{-6}

(C) 6×10^{-6} (D) $6 \times 10^{+6}$

89. The two coherent sources of intensity β produce interference. The fringe visibeility will be_____

(A) $2\ \beta$ (B) $\frac{\beta}{2}$ (C) $\sqrt{\beta}/1+\beta$ (D) $2\sqrt{\beta}/1+\beta$

90. Light of wave–length λ is incident on a slit of width d. The resulting diffraction pattern is observed on a screen placed at a distance D. The linear width of the principal maximum is equal to the width of the slit, then D = _____.

(A) $d^2/2\lambda$ (B) $2\lambda^2/d$ (C) d/λ (D) $\frac{2\lambda}{d}$

91. A polariser is used for_____.

(A) producp polarised light (C) produced unpolarised light

(B) produced unpolarised light (D) none of these

92. Read the paragraph and chose the correct answer of the following quetions

In young expriment position of bright fringes is given by $x = n\lambda\frac{D}{d}$ and the positon of dark fringes is given by $x = (2n-1)\frac{\lambda}{2}\frac{D}{d}$ where n = 1,2,3..... for first, second, third bright / dark fringe. The center of the fringe pattern is bright (for n = 0). The width of each briht/dark fringe is $\beta = \frac{\lambda D}{d}$, Where λ = 5000 A^0.

(i) If light of wavelength $6000\ A^\circ$ be used in the above experiment the fringe width would be_____mm

(A) 0.36 (B) 3 (C) 0.6 (D) 6

(ii) with the light of wavelength $5000\ A^\circ$, If experiment were carried out under water of a $n = 4/3$ the fringe width would be_____

(A) zero (B) $\frac{4}{3}$ times (C) $\frac{3}{4}$ times (D) none of these

93. The width of a single slit, if the first minimum is observed at an angle of 2° with a wavelength of light $6980\ A^\circ$ is_____mm

(A) 0.2 (B) 2×10^{-5} (C) 2×10^{5} (D) 0.02

94. In a fraunhofer diffraction by single slit of width d with incident light of wavelength $5500\ A^\circ$ the first minimum is observed at angle of 30°. The first secondary maximum is observed at an angle θ = _____.

(A) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (B) $\sin^{-1}\left(\frac{3}{4}\right)$ (C) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (D) $\sin^{-1}\left(\frac{1}{4}\right)$

95. The phenomenon of polarisation of electromagnetic waves proves that the electromagnetic waves are_____.

(A) mechanical (B) longitudial (C) transverse (D) nono of these

96. Light from two coherent Sources of the same amplitude A and wavelength λ, illuminates the Screen. The intensity of the central maximum is Io. If the sources were incoherent, the intensity at the same point will be_____.

(A) $I_0/2$ (B) $I_0/4$ (C) $4I_0$ (D) $2I_0$

97. When the angle of incidence is 60° on the Surface of a glass slab, it is found that the reflected ray is completely palarised. Then the velocity of light in glass is_____ ms^{-1}

(A) $\sqrt{2}$ (B) $\sqrt{3}\times10^8$ (C) $\sqrt{3}$ (D) $\sqrt{2}\times10^8$

98. Two beams of Light of intensity I_1 and I_2 interfere to give an interference pattern. If the ratio of maximum intersity to that of minimum intensity is $\frac{16}{4}$ then I_1/I_2 = _____

(A) 1:9 (B) 1:4 (C) 4:1 (D) 9:1

99. Which of the following phenomenon is used in optical fibres ?

(A) Reflection (B) Scatterting

(C) Total internal reflection (D) Interference

100. Two beams of light having intensities I and 4I interfere to produce a fringe pattern on a screen. The phase diffrence between the resultant intensities at A and B is_____.

(A) I (B) 4I (C) 2I (D) 6I

101. A sound source emits sound of 600 Hz frequency, this sound enters by opened door of width 0.75 m. Find the angle on one side at which fitst minimum is formed. The speed of sound = $300\,ms^{-1}$.

(A) 84.4° (B) 90° (C) 74.2° (D) 47.2°

102. A plane polarised light is incident normally on the tourmaline plate. its $\vec{E}$ vectors make an agnle of 45° with the optical axis of the plate. find the percentage difference between intial and final maximum values of $\vec{E}$ vectors.

(A) 19% (B) 92% (C) 50% (D) 29%

103. In Yong's double slit experiment, the inensity on screen at a point where path difference is λ, is K , What will be intensity at the point where path differnce is $\frac{\lambda}{4}$

(A) $\frac{K}{2}$ (B) 2 K (C) 4 K (D) zero

104. Ordinary light incident on a glass slab at the polarising angle, suffers a deviation of 22°. The value of angle of refracion in this case is_____.

(A) 44° (B) 34° (C) 22° (D) 11°

105. The ratio of inensities of rays emitted from two different coherent Sourees is α. For the inerference pattern by them, $\frac{Imax+Imin}{Imax-Imin}$ will be equal to _____.

(A) $\frac{1+\sqrt{\alpha}}{2\alpha}$ (B) $\frac{1+\alpha}{2\alpha}$ (C) $\frac{1+\sqrt{\alpha}}{2}$ (D) $\frac{1+\alpha}{2\sqrt{\alpha}}$

KEY NOTES

1	D	26	D	51	C	76	A	100	B
2	C	27	B	52	D	77	C	101	D
3	B	28	B	53	C	78	A	102	D
4	D	29	C	54	C	79	D	103	A
5	A	30	C	55	C	80	A	104	C
6	B	31	A	56	C	81	D	105	D
7	C	32	A	57	B	82	B		
8	D	33	B	58	D	83	C		
9	B	34	B	59	C	84	B		
10	C	35	D	60	A	85	B		
11	D	36	D	61	D	86	A		
12	A	37	A	62	A	87	B		
13	B	38	B	63	A	88	C		
14	C	39	C	64	D	89	D		
15	D	40	D	65	B	90	A		
16	A	41	B	66	A	91	A		
17	A	42	B	67	C	92(i)	A		
18	B	43	B	68	C	92(ii)	C		
19	C	44	B	69	C	93	D		
20	D	45	D	70	D	94	B		
21	A	46	C	71	C	95	C		
22	B	47	B	72	D	96	A		
23	C	48	A	73	C	97	B		
24	D	49	A	74	A	98	D		
25	D	50	B	75	B	99	C		

HINTS

2. use $n = \frac{\lambda}{\lambda'}$.

5. No of image $= \frac{360^0}{\theta} = \frac{360^0}{0^0} = \infty$

11. Use $ho = \sqrt{h_1 h_2}$

22. Use, shift $= x\left(1 - \frac{1}{n}\right)$

25. Here, $n = \frac{h}{h'}$ $\therefore$ $h' = 2.0 - 0.2 = 1.8$ $\therefore$ $\frac{2}{1.5} = \frac{20}{18} = 1.11 = n$

26. Here, $i = \alpha_1$ $r = \alpha_2$ and $a\mu_w = \frac{V_a}{V_w}$ _____(*i*)

Now $a\mu_w = \frac{\sin i}{\sin r} = \frac{\sin \alpha_1}{\sin \alpha_2}$ _____(*ii*)

But $\frac{Va}{Vw} < 1$ $\therefore$ $\alpha_1 < \alpha_2$

27. Here, $\frac{1}{f} = (1.6 - 1)\left(\frac{1}{60} - \frac{1}{\infty}\right) = \frac{1}{100}$ $\therefore$ $f = 100$ cm

28. Here, $\frac{P_1}{P_2} = \frac{2}{3}$ $\therefore$ $\frac{f_2}{f_1} = \frac{2}{3}$ $\therefore$ $f_2 = \frac{2f_1}{3}$ _____(i)

Now, $\frac{1}{30} = \frac{-1}{f_1} + \frac{1}{\frac{2f_1}{3}}$ $\therefore$ $f_1 = -15$ cm and $f_2 = 10$ cm

29. Here, $(n_1 - 1)A_1 + (n_2 - 1)A_2 = 0$

$\therefore$ $0.5A_1 + 0.3A_2 = 0$ $\therefore$ $A_2 = \frac{-1.5}{0.3} = -5^o$

30. Here, $180 - 60^\circ = 120^\circ$

31. Use, $v = f\lambda$ and $n = \frac{c}{v}$

36. Use, $\therefore \ \frac{1}{f} = (n-1)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$ take $R_1 = R_1,\ R_2 = -2R$

R = 4.5 cm

37. Here, $n = \sin/\sin r = \frac{\sin(i)}{\sin\left(\frac{i}{2}\right)} = \frac{2\ \sin\left(\frac{i}{2}\right).\cos\left(\frac{i}{2}\right)}{\sin\left(\frac{i}{2}\right)}$

$\therefore \ n = \cos\left(\frac{i}{2}\right) \qquad \therefore \ \frac{n}{2} = \cos\left(\frac{i}{2}\right) \qquad \therefore \ i = 2\cos^{-1}\left(\frac{n}{2}\right)$

38. Here, $i > \theta\ (=c) \ = 45^0 \qquad \therefore \ n = \frac{1}{\sin c} = \sqrt{2}$

The value of critical angle (c) is minimum for 45°

$\therefore \ n > \sqrt{2}$

39. its divers the rays, $n_1 < n_2$

41. Here, $\frac{1}{fa} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (1.6-1)\left(\frac{1}{15} \times \frac{1}{15}\right) = \frac{1.2}{15}$

$\therefore \ fa = 12.5cm$

Now, $\frac{f_2}{f_a} = \frac{(-1)}{\left(\frac{}{2}-1\right)} = \frac{1.6-1}{\left(\frac{1.6}{1.63}-1\right)} = \frac{0.6\times1.63}{-0.03}$

$\therefore \ f_\ell = -20 \times 1.63 \times 12.5 = -407.5 \ cm$

43. wavelegth maximum, Scattering is minimum

44. Here, $n = \frac{\sin\left(\frac{A+A}{2}\right)}{\sin\left(A/2\right)} = \frac{\sin(A)}{\sin\left(\frac{A}{2}\right)} = \frac{2\sin\left(\frac{A}{2}\right).\cos\left(\frac{A}{2}\right)}{\sin\left(\frac{A}{2}\right)}$

$\frac{3}{4} = 2\cos\left(\frac{A}{2}\right), \ \frac{A}{2} = \cos^{-1}(0.75) = 41^o, \ \therefore \ A = 82^o$

45. Here $\delta m = r$ and $\delta = i_1 + i_2 - (r_1 - r_2)$

when $\delta = \delta m$ then $i_1 = i_2 = i \qquad r_1 = r_2 = r$

$$\therefore\ \delta m = 2i - 2r = 2nr - 2r \qquad \left(\therefore n = \frac{\sin i}{\sin p} = \frac{i}{r} \ \therefore\ i = nr\right)$$

$$= 2r\,(n-1) = 2r\left(\frac{3}{2} - 1\right)$$

$$\therefore \delta m = r$$

46. Here magnifying power is 10 there it can be seen 10 times near.

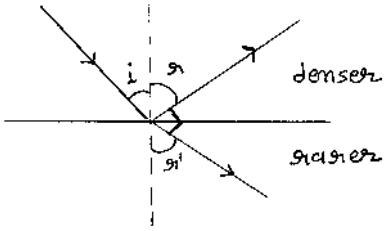
47. Use : $d = \dfrac{\pi D}{180^\circ \times 60^\circ} = 3.2$ m

51. Use $\dfrac{1}{f} = (n-1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right) \qquad \therefore\ R = 10$ cm

for rarer medium to denser, $-\dfrac{n_1}{u} + \dfrac{n_2}{v} = \dfrac{n_2 - n_1}{R} \quad (\therefore u = \infty \ \ v = f)$

$$\therefore \quad \frac{0 + 1.5}{f} = \frac{1.5 - 1}{10} \qquad \therefore f = 30 \text{ cm}$$

52.



from fig : $90^\circ - r + 90^\circ - r' = 90^\circ$

$\therefore r' = 90 - r$

$\therefore$ now $\sin c = \sin i \,/\, \sin(90 - r) \qquad = \dfrac{\sin i}{\cos(i)}$

$\therefore \sin c = \tan(i) = \tan(r) \qquad (\therefore i = r)$

$\therefore C = \sin^{-1}(\tan r)$

53. Here $Cw = \sin^{-1}\left(\dfrac{1}{n_w}\right) = \sin^{-1}\left(\dfrac{3}{4}\right) = 48.6^\circ$

$Cg = \sin^{-1}\left(\dfrac{1}{ng}\right) = 42^\circ \qquad \therefore \quad Cw > Cg$

54. Use $\dfrac{1}{f} = (n-1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right) \qquad \therefore\ f = 0.25\,m\,, \quad P = 4D$

55. Use $\mu = \dfrac{\sin i}{\sin p} = \dfrac{\sin 2r}{\sin r} = \dfrac{2 \sin r . \cos r}{\sin r} = 2\cos r \qquad \therefore \cos = \dfrac{\sqrt{2}}{2} = \dfrac{1}{\sqrt{2}}$

$r = 45^\circ \quad \therefore\ A = 90^\circ$

56. Use $n = \dfrac{\text{Real depth}}{\text{apperntdepth}} = \dfrac{x}{y}$ $\quad \therefore\ x = ny = 15$ cm

($\therefore$ apperent depth = 6+4=10)

58. Use = $\dfrac{\text{dis an t}}{\text{wave No.}} = 5000\ A^\circ$

Now $\lambda' = \dfrac{\lambda}{n} = 4000\ A^\circ$

62. Here, $wn_g = {}^{an_g}\!/_{an_g} = 1.128$

Now, $\dfrac{1}{fa} = (an_g - 1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right) \quad \therefore\ \dfrac{1}{R_1} - \dfrac{1}{R_2} = 10$

and $\dfrac{1}{fw} = \left(w n_g\text{-}1\right)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right) = (1.128 - 1) \times 10 = 1.28$

then $\therefore\ Pa = \dfrac{1}{fa} = 5D$ and $Pw = 01.28 \quad \therefore\ Pa - Pw = 3.72$ D

63. Use, $n = \dfrac{\sin\left(\dfrac{A + \delta m}{2}\right)/2}{\sin\left({}^{A}\!/_{2}\right)} = 2\cos\left(\dfrac{A}{2}\right) \qquad (\delta m = A)$

$\therefore \cos\left(\dfrac{A}{2}\right) = \dfrac{\sqrt{3}}{2} \quad \therefore\ A = 60^\circ$

64. Here, $i = 90^\circ \quad r_1 = 0, \quad r_1 + r_2 = A, \quad r_2 = 30^\circ$

Now, $n = \dfrac{\sin(i_2)}{\sin(r_2)} \qquad \therefore\ i_2 = 45^\circ$

$i + e = A + \delta m \qquad \therefore\ \delta m = 15^0$

65. Use $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{n_1-1}\left(\frac{1}{\infty}+\frac{1}{R}\right)+(n_2-1)\left(\frac{1}{R}-\frac{1}{\infty}\right)$

$\therefore F = R/n_1-n_2$

66. Here : For Real image $u=-v_1,\ v=2v_1\quad f=-30cm$

$\therefore\ \frac{-1}{-2u_1}-\frac{1}{u_1}=\frac{1}{-30}\quad \therefore\ u_1 = 45\ cm$

For virtul image $u=-u_2\quad v=+2v_2\quad f=-30\ cm$

$\therefore\ \frac{-1}{u_2}+\frac{1}{2u_2}=-\frac{1}{30}\quad u_2=15\ cm$

$u_1 - u_2 = 30\ cm$

67.



Here, $m=\frac{1}{4}=\frac{v}{u}\quad \therefore\ u=4v$

if $v=-x\quad u=-4x$ then

from figure, $|0\ I|$ = 4x-x , 3x = 10 cm $\therefore\ x=\frac{10}{3}\ cm$

Now, $u=4v=4x\quad (\therefore\ \text{formfig}: v=x)$

$=+40/3$ and $v=\frac{u}{4}=\frac{40}{3\times4}=\frac{-10}{3}cm$

$\therefore$ from $\frac{1}{f}=\frac{1}{u}-\frac{1}{v}\quad \therefore\ f = -4.4\,cm$

68. Use $\omega=\frac{\delta_V-\delta_V}{\delta}=\frac{2}{5}$ and $\omega'=\frac{\delta' v-\delta' v}{\delta'}=\frac{1}{3}$

$\therefore\ \frac{\omega}{\omega'}=6:5$

69. Use $d\theta=\frac{1.22\lambda}{D}=\frac{x}{r}$ where r = distant of jeep car $\therefore$ r = 3.34 km

70. For, plano-convex lens $\frac{1}{f_1}=\frac{1}{f_3}=(n-1)\left(\frac{1}{\infty}\times\frac{1}{R}\right)=\frac{1}{24}$

For, doble convex lens $\therefore\ \frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}=\frac{-1}{60}\quad \therefore\ \frac{1}{f_2}=\frac{-1}{10}$

Now $\therefore\ \frac{1}{f_2}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)\quad n=1.6$

80. A. Use K-E = $\frac{hc}{\lambda}$

82. Use $n\lambda_1 = (n+1)\frac{\lambda}{2}$ $\therefore$ n = 4

97. Here, n = tan $\theta_p = \sqrt{3}$, n = $\frac{c}{v}$ $\therefore$ v = $\sqrt{3} \times 10^8$

98. From $\frac{I_{max}}{I_{min}} = \frac{(a+b)^2}{(a-b)^2}$ 3 b = 9

Now $\therefore \frac{I_1}{I_2} = \frac{a^2}{b^2} = 9:1$

100. Here, $I_A = I_1 + I_2 + 2\sqrt{I_1 I_2}$ $\cos\frac{\pi}{2} = I \times 4I \times 2\sqrt{I \times 4I} \times \cos 90^\circ$

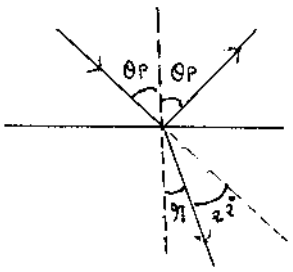
$I_A = 5I$

and, $I_B = 5I + 2\sqrt{I \times 4I} \times \cos\pi =$ 5I-4I=I

$\therefore I_A - I_B = 4I$

102. $I = I_0 \cos^2\theta = \frac{I_0}{2}$ and $\frac{E^2}{E_0{}^2} = \frac{1}{2}$, $\frac{E}{E_0} = \frac{1}{\sqrt{2}}$

$\therefore \frac{|E - E_0|}{E_0} = 0.29 = 29\%$

104. 

from Fig, $\theta p + 90^\circ + r = 180^\circ$

$\therefore$ $\theta p + r = 90^\circ$ and $\theta p - r = 22^\circ$

$\therefore r = 34^\circ$

105. Here $\frac{I_1}{I_2} = \alpha$ $\therefore \frac{E_1}{E_2} = \sqrt{\alpha}$

and $\frac{E_1 + E_2}{E_1 - E_2} = \frac{\sqrt{\alpha}+1}{\sqrt{\alpha}-1}$ $\therefore \frac{I_{max}}{I_{min}} = \frac{(\sqrt{\alpha}+1)^2}{(\sqrt{\alpha}-1)^2}$

$\therefore \frac{I_{max} + I_{min}}{I_{max} - I_{min}} = \frac{2(\alpha+1)}{4\sqrt{\alpha}} = \frac{\alpha+1}{2\sqrt{\alpha}}$

Unit - 17
Important Formula

SUMMARY

1. Photon energy $E = hf$ or $E = nhf = \frac{nhc}{\lambda}$

2. Number of photon emitted in one second $N = \frac{P}{E}$

3. Mamentum of Photon $= p = mv = \frac{hc}{\lambda}$

4. Mass of Photon $= m = \frac{hf}{c^2}$

5. Work function $W_0 = \phi_0 = hf_0 = \frac{hc}{\lambda_0}$

$$K_{max} = \frac{1}{2} mv^2_{max} = hf - w_0 = h\left(f - f_0\right) = h\left(\frac{c}{\lambda} - \frac{c}{\lambda_0}\right) = hc\left(\frac{\lambda_0 - \lambda}{\lambda_0 \lambda}\right)$$

OR

$$eV_0 = hc\left(\frac{\lambda_0 - \lambda}{\lambda_0 \lambda}\right) = \frac{hc}{e}\left(\frac{\lambda_0 - \lambda}{\lambda_0 \lambda}\right)$$

$$\text{Intensity} = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = (I) = \frac{E}{A \times t} = \frac{p}{A} \therefore P = IA$$

$$\text{Kinetic Energy}\,(K) = \frac{1}{2}mv^2 = \frac{p^2}{2m} \therefore p = \sqrt{2mk}$$

6. De-Brogli wave length equation

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mk}} = \frac{h}{\sqrt{3mkT}} \qquad \therefore \lambda = \frac{h}{\sqrt{2meV}} = \frac{1.23}{\sqrt{V}} nm$$

7. Braga's law for crlstal diffraction is $2d \sin\theta = n\lambda$

Thomsan Equation $\frac{1}{2} mv^2 = eV$ OR $v^2 = \frac{2eV}{m}$ ...(i)

$\therefore$ Centripetal force $= evB = \frac{mv^2}{r}$ or $eB = \frac{mv}{r}$ ...(ii)

form Eq. (i) and (ii) $F = evB = \frac{mv^2}{r}$ or $eB = \frac{mv}{r}$ ...(ii)

8. According to Einstein's theory of relativity, relation between energy of particle and momentum.

$E = \sqrt{p^2 c^2 + m_0^2 c^4}$

m_0 = rest mass of photon (particle)

If we take $m_0 = 0$

$E = pc$

9. Mass of particle moving with velocity v is given by :

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where v = velocity of moving particle

m_0 = rest mass of particle

c = Velocity of light

MCQ QUESTIONS

For the answer of the following questions choose the correct alternative from among the given ones.

1. Particle A and B have electric charge + q and + 4 q. Both have mass m. If both are allowed to fall under the same p.d., ratio of velocities $\frac{v_A}{v_B}$ = ...

(A) 2:1 (B) 1:2 (C) 1:4 (D) 4:1

2. Energy of photon having wavelength λ is 2 eV. This photon when incident on metal. maximum velocity of emitted is v. If λ is decreased 25% and maximumu velocity is made double, work function of metal is .....ev

(A) 1.2 (B) 1.5 (C) 1.6 (D) 1.8

3. When elctric bulb having 100 W efficiency emits photon having wavelength 410 mm every second, numbers of photons will be..... $\left(h = 6 \times 10^{-34} \text{ J.s, } c = 3 \times 10^8 \text{ ms}^{-1}\right)$

(A) 100 (B) 1000 (C) 3×10^{20} (D) 3×10^{18}

4. If velocity of free election is made double, change in its de-Broglie wavelength will be.....

(A) increase by $\frac{\lambda}{2}$ (B) decrease by $\frac{\lambda}{2}$ (C) increase by 2λ (D) decrease 2λ

5. de-Broglie wavelength of proton accelerated under 100V electric potential difference is λ_0 . If de - wave length α - particle accelerated by the same electric potential differencewill its bouglie .....

(A) $2\sqrt{2\lambda_0}$ (B) $\frac{\lambda_0}{2\sqrt{2}}$ (C) $\frac{\lambda_0}{2\sqrt{2}}$ (D) $\frac{\lambda_0}{2}$

6. Work function of a body is 4.0 eV. For emission of photoelectron from body, maximum wavelength of light = ...

(A) 540 nm (B) 400 nm (C) 310 nm (D) 220 nm

7. A body of mass 200 g moves at the speed of 5 m/hr. So de-Broglie wavelength related to it is of the order..... $\left(h = 6.626 \times 10^{-34} J - s\right)$

(A) 10^{-10} m (B) 10^{-20} m (C) 10^{-30} m (D) 10^{-40} m

8. Photo electric effect on surface is found for frequencies 5.5×10^8 MHz and 4.5×10^8 MHz If ratio of maximum kinetic energies of emitted photo electrons is 1 : 5, threshold frequency for metal surface is.....

(A) 7.55×10^8 MHz (B) 4.57×10^8 MHz

(C) 9.35×10^8 MHz (D) 5.75×10^8 MHz

9. For wave concerned with proton, de-Broglie wavelength change by 0.25% . If its momentum changes by P_O initial momentum = ...

(A) $100\,P_O$ (B) $\frac{P_O}{400}$ (C) $401\,P_O$ (D) $\frac{P_O}{100}$

10. According to Einstein's photoelectric equation, graph of kinetic energy of emitted photo electrons from metal versus frequency of incident radation is linear. Its shope.....

(A) depends on type of metal used

(B) depends on intensity of radiation

(C) depends on both metal used and intensity of radiation.

(D) is same for all metals and free from intensity of radiation.

11. Photocell cell is enlightended by small bright source 1 m away. If the same light source is placed $\frac{1}{2}$ m away, number of electrons emitted by cathode will be.....

(A) increases twice (B) decreases twice

(C) increases 4 times (D) decreases 4 times

12. If kinetic energy of free electron is made double, change in de-Broglie wavelength will be.....

(A) $\sqrt{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) 2 (D) $\frac{1}{2}$

13. Energy corresponding to threshod frequency of metatl is 6.2 eV. If stopping potential correspionding to radiation incident on srurface is 5V, incident radiation will be in the.....region.

(A) X-ray (B) Ultraviolet (C) infrared (D) Visible

14. At $10^{o}C$ temperature, de-Broglie wave length of atom is $0.4\,\mathring{A}$. If temperature of atom is increased by $30^{o}C$, what will be change in de-Broglie wavelength of atom ?

(A) decreases $10^{-2}\,\mathring{A}$ (B) decreases $2\times10^{-2}\,\mathring{A}$

(C) increases $10^{-2}\,\mathring{A}$ (D) increases $2\times10^{-2}\,\mathring{A}$

15. Wavelength of incident radiation on photo sensitive surface is changed from $4000\,\mathring{A}$ to $3000\,\mathring{A}$, so change in stopping potential will be.....

(A) 0.345 V (B) 0.435 V (C) 0.543 V (D) 0.534 V

16. Wavelength of incident radiation on photo sensitive surface is $4000\,\mathring{A}$ If wavelength of this light is $3600\,\mathring{A}$, what will be change in kinetic energy of emitted photo electron ?

$\left(h = 6.625\times10^{-34}\ J.s,\ c = 3\times10^{8}\ ms^{-1},\ 1ev = 1.6\times10^{-19}\ J\right)$

(A) 0.52 eV (B) 1.04 eV (C) 2.08 eV (D) 4.16 eV

17. A hollow metalic cubodial has mass 10 kg and length 10 cm. At what speed this cubodial is moved in X-direction so that its de-Broglie wavelength exactly trapped in the cubodial ?

(A) $1.1\times10^{-34}\,\frac{m}{s}$ (B) $2.2\times10^{-34}\,\frac{m}{s}$ (C) $3.3\times10^{-34}\,\frac{m}{s}$ (D) $4.4\times10^{-34}\,\frac{m}{s}$

18. If we take accelerating voltage V = 50 V, electric charge of electron $e = 1.6\times10^{-19}\ C$ and mass of electron $m = 9.1\times10^{-31}\ kg$ find the wavelength of concerned electrion.

(A) $0.1735\,\mathring{A}$ (B) $1.735\,\mathring{A}$ (C) $17.35\,\mathring{A}$ (D) $1735\,\mathring{A}$

19. What will be energy in eV of photons of λ - rays having length $0.1\,\mathring{A}$ coming out of excited nucleus of radium? $\left(c = 3\times10^{8}\,\frac{m}{s},\ h = 6.625\times10^{-34}\,\frac{m}{s}\right)$

(A) 42.12 (B) 12.42 (C) 22.41 (D) 24.21

20. How many photons of red coloured light having wavelength $8000\,\mathring{A}$ will have same energy as one photon of violet coloured light of wavelength $4000\,\mathring{A}$?

(A) 2 (B) 4 (C) 6 (D) 8

21. When a radiation of wavelength 3000 $\mathring{A}$ is incident on metal, 1.85 V stopping potential is obtained. What will be threshod wave length of metal ? $\left(h = 6.6 \times 10^{-34} \text{ J.s, } c = 3 \times 10^{8} \frac{m}{s} \right)$

(A) 4539 $\mathring{A}$ (B) 3954 $\mathring{A}$ (C) 5439 $\mathring{A}$ (D) 4395 $\mathring{A}$

22. Output power of He-Ne LASER of low energy is 1.00 mW. Wavelength of the ligth is 632.8 nm. What will be the number of photons emitted per second from this LASER ?

(A) $8.31 \times 10^{15} s^{-1}$ (B) $5.38 \times 10^{15} s^{-1}$ (C) $1.83 \times 10^{15} s^{-1}$ (D) $3.18 \times 10^{15} s^{-1}$

23. A star which can be seen with naked eye from Earth has intensity $1.6 \times 10^{-9} \text{ Wm}^{-2}$ on Earth. If the corresponding wavelength is 560 nm, and the diameter of the human eye is 2.5×10^{-3} m , the number of photons entering in our in 1 s is.....

(A) $7.8 \times 10^{4} s^{-1}$ (B) $8.85 \times 10^{4} s^{-1}$ (C) $7.85 \times 10^{5} s^{-1}$ (D) $8.85 \times 10^{5} s^{-1}$

24. Find the velocity at which mass of a proton becomes 1.1 times its rest mass, $m_g = 1.6 \times 10^{-27}$ kg Also, calculate corresponding temperature. For simplicity, consider a proton as non-interacting ideal-gas particle at 1 atm pressure.

$\left(2eV = 1.6 \times 10^{-19} \text{ J. } h = 6.63 \times 10^{-34} \text{ J.s}, c = 3 \times 10^{8} \text{ ms}^{-1} \right)$

(A) $v = 126 \times 10^{8} \frac{m}{s}$, $T = 6.75 \times 10^{11} K$ (B) $v = 12.6 \times 10^{8} \frac{m}{s}$, $T = 7.65 \times 10^{11} K$

(C) $v = 1.26 \times 10^{7} \frac{m}{s}$, $T = 5.76 \times 10^{11} K$ (D) $v = 12.6 \times 10^{7} \frac{m}{s}$, $T = 7.56 \times 10^{11} K$

25. Power produced by a star is 4×10^{28} W . If the average wavelength of the emitted radiations is considered to be 4500 $\mathring{A}$ the number of photons emitted in 1 s is.....

(A) 1×10^{45} (B) 9×10^{45} (C) 8×10^{45} (D) 12×10^{46}

26. What should be the ratio of de-Broglie wavelength of an atom of nitrogen gas at 300 K and 1000 K. Mass of nitrogen atom is 4.7×10^{-26} kg and it is at 1 atm pressure Consider it as an idecal gas.

(A) 2.861 (B) 8.216 (C) 6.281 (D) 1.826

27. Wavelength of light incident on a photo - sensitive surface is reduced form 3500 $\mathring{A}$ to 290 mm. The change in stopping potenital is..... $\left(h = 6.625 \times 10^{-24} \text{ J.s} \right)$

(A) 42.73×10^{-2} V (B) 27.34×10^{-2} V (C) 73.42×10^{-2} V (D) 43.27×10^{-2} V

28. An electric bulb of 100 W converts 3% of electrical energy into light energy. If the wavelength of light emitted is $6625\,\text{Å}$, the number of photons emitted is 1 s is.....$\left(h = 6.625 \times 10^{-34}\ \text{J.s}\right)$

(A) 10^{17} (B) 10^{19} (C) 10^{21} (D) 10^{15}

29. Work function of Zn is 3.74 eV. If the sphere of Zn is illuminated by the X-ray of wavelength $12\,\text{Å}$ the maximum potential produced on the sphere is.....$\left(h - 6.625 \times 10^{-34\ \text{J-s}}\right)$

(A) 10.314 V (B) 103.14 V (C) 1031.4 V (D) 10314 V

30. Consider the radius of a nucleus to be $10^{-15}\,\text{m}$. If an electron is assumed to be in such nucleus, what ill be its energy ? $\left(\text{me} = 9.1 \times 10^{-31}\ \text{kg. h} = 6.625 \times 10^{-34}\ \text{J.s}\right)$

(A) 5.59×10^3 (B) 9.55×10^3 (C) 5.95×10^3 (D) 7.45×10^3

31. A proton falls freely under gravity of Earth. Its de Broglie wavelength after 10 s of its mortion is....., Neglect the forces other than gravitational force.

$\left(g = 10\ \frac{m}{s^2},\ m_p = 1.6 \times 10^{-27}\ \text{kg},\ h = 6.625 \times 10^{-34}\ \text{J.s}\right)$

(A) $3.96\,\text{Å}$ (B) $39.6\,\text{Å}$ (C) $6.93\,\text{Å}$ (D) $69.3\,\text{Å}$

32. Compare energy of a photon of X-rays having $1\,\text{Å}$ wavelength with the energy of an electron having same de Broglie wavelength.

$\left(h = 6.625 \times 10^{-34}\ \text{J.s},\ c = 3 \times 10^8\ \text{ms}^{-1},\ 1\text{ev} = 1.6 \times 10^{-19}\ \text{J}\right)$

(A) 8.24 (B) 2.48 (C) 82.4 (D) 24.8

33. An electron is at a distance of 10 m form a charge of 10 C. Its total energy is 15.6×10^{-10} J. Its de Broglie wavelength at this point is.....

$\left(h = 6.625 \times 10^{-34}\ \text{J.s},\ m_e = 9.1 \times 10^{-31}\ \text{kg. K} = 9 \times 10^9\ \text{SI}\right)$

(A) $9.87\,\text{Å}$ (B) 9.87 Fermi (C) $8.97\,\text{Å}$ (D) 8.97 Fermi

34. Wavelength of an electron having energy E is $\lambda_0 = \frac{h}{\sqrt{2mE}}$, where m is the mass of an electron. Find the wavelength of the elecrton when it centers in X-direction in the region having potential X-$V_{(X)}$ If we imaging that due to the potential, electrion enters from one medium to another, what is the refractive index of the medium ?

(A) $\lambda = \frac{2mh}{eVx^{\frac{1}{2}}} \cdot \mu = \left[\frac{eVx}{E}\right]^{\frac{1}{2}}$ (B) $\lambda = \frac{h}{2m\left[E + eV_{(X)}\right]^{\frac{1}{2}}} \cdot \mu = \left[\frac{E + eV_{(x)}}{E}\right]^{\frac{1}{2}}$

(C) $\lambda = \frac{\left(E + eV_{(x)}\right)^{\frac{1}{2}} h}{2m} \cdot \mu = \left[\frac{E}{E + eV_x}\right]^{\frac{1}{2}}$ (D) $\lambda = \frac{Eh}{2m\left(eV_{(x)}\right)^{\frac{1}{2}}} \cdot \mu = \frac{\left(eV_{(x)}\right)^{\frac{1}{2}}}{E}$

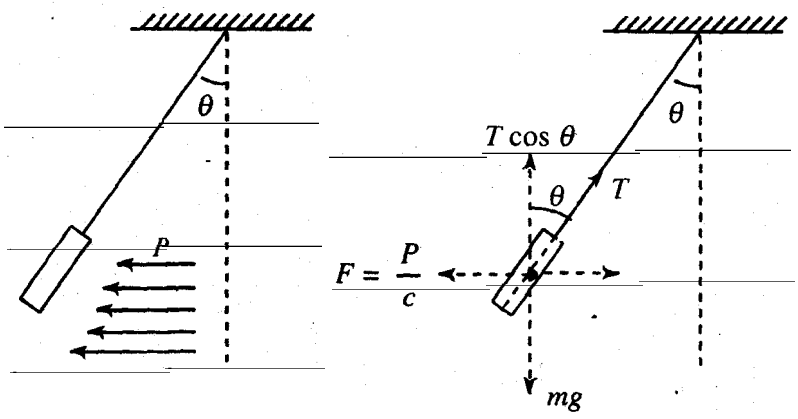
35. As shown in the figure, light of energy P (joule) is incident on a small, flat strip of metal of mass m, suspended with the help of weightless string of length l in 1s. All the energy incident on it is absorbed and the strip remains in equillibrium at an angle θ with respect to vertical. If the light is monochromatic, angle θ is.....

(A) $\theta = \tan^{-1}\left(\frac{cmg}{p}\right)$

(B) $\theta = \tan^{-1}\left(\frac{p}{cmg}\right)$

(C) $\theta = \sin^{-1}\left(\frac{cmg}{p}\right)$

(D) $\theta = \sin^{-1}\left(\frac{p}{cmg}\right)$



36. U. V. light of wavelength 200 mm is incidenet on polished surface of Fe. work function of the surface is 4.5 eV. Find maximum speed of phote electrons.

$\left(h = 6.625 \times 10^{-34}\ \text{J.s},\ c = 3 \times 10^{8}\ ms^{-1},\ 1\ eV = 1.6 \times 10^{-19}\ J\right)$

(A) $7.75 \times 10^{4}\ \frac{m}{s}$ (B) $875 \times 10^{5}\ \frac{m}{s}$ (C) $8.75 \times 10^{4}\ \frac{m}{s}$ (D) $7.75 \times 10^{5}\ \frac{m}{s}$

37. Light of $4560\ \text{Å}$ 1 mW is incident on photo-sensitive surface of Cs (Cesium). If the quantum efficiency of the surface is 0.5% what is the amount of photoelectric current produced ?

(A) 1.84 mA (B) 4.18 μA (C) 4.18 mA (D) 1.84 μA

38. Work function of metal is 2 eV. Light of intensity $10^{-5}\ Wm^{-2}$ is incident on $2\ cm^{2}$ area of it. If 10^{17} electrons of these metals absorb the light, in how much time does the photo electric effectc start ? Consider the waveform of incident light.

(A) 1.4×10^{7} s (B) 1.5×10^{7} s (C) 1.6×10^{7} s (D) 1.7×10^{7} s

39. Radius of a beam of radiation of wavelength $5000\ \text{Å}$ is 10^{-3} m . Power of the beam is 10^{-3} W . This beam is normally incident on a metal of work function 1.9 eV. The charge emitted by the metal per unit area in unit time is.....Assume that each incident photoen emits one elctron. $\left(h = 6.625 \times 10^{-34}\ \text{J. s}\right)$

(A) 1.282 C (B) 12.82 C (C) 128.2 C (D) 1282 C

40. 11×10^{11} Photons are icident on a surface in 10 s. These photons correspond to a wavelength of $10\,\mathring{A}$. If the surface area of the given surface is 0.01 m^2, the intensity of given radiations is.....

$\left(h = 6.625 \times 10^{-34}\ J.\ s,\ c = 3 \times 10^{8}\ \frac{m}{s}\right)$

(A) $21.86 \times 10^{-3}\ \frac{W}{m^2}$ (B) $2.186 \times 10^{-3}\ \frac{W}{m^2}$ (C) $218.6 \times 10^{-3}\ \frac{W}{m^2}$ (D) $2186 \times 10^{-3}\ \frac{W}{m^2}$

41. If alpha particale and duetron move with velocity v and 2v, the ratio of their de-Brogle wavelength will be.....

(A) $1 : \sqrt{2}$ (B) 2 :1 (C) 1 : 1 (D) $\sqrt{2} : 1$

42. Uncertainty in position of electron is found of the order of de-Broglie wavelength. Using Heisemberg's uncertainty principle, it is found that order of uncertainty in its velocity = ...

(A) 1 v (B) 2 v (C) $\frac{v}{2\pi}$ (D) $2\pi v$

43. Photoelectric effect is obtained on metal surface for a light having frequencies f_1 & f_2 where $f_1 > f_2$. If ratio of maximum kinetic energy of emitted photo electrons is 1 : K , so threshold frequency for metal surface is.....

(A) $\frac{f_1 - f_2}{K - 1}$ (B) $\frac{K f_1 - f_2}{K - 1}$ (C) $\frac{K\ f_2 - f_1}{K - 1}$ (D) $\frac{f_2 - f_1}{K}$

44. Frequency of incident light on body is f. Threshold frequency of body is f_0. Maximum velocity of electron = ...where m is mass of electron.

(A) $\left[\frac{2h(f - f_0)}{m}\right]^{\frac{1}{2}}$ (B) $\frac{2h(f - f_0)}{m}$ (C) $\left[\frac{2hf}{m}\right]^{\frac{1}{2}}$ (D) $h(f - f_0)$

45. If electron is accelerated under 50 KV in microscope, find its de-Broglie wavelength.

(A) 5.485×10^{-12} m (B) 8.545×10^{-12} m (C) 4.585×10^{-12} m (D) 5.845×10^{-12} m

46. Energy of photon having wavelenth is 2 eV. Maximum velocity of emitted photo electron after incidence of photon is v. If value of λ is decreased by 25% and maximum velocity is made double, work function metal will be.....eV.

(A) 1.2 (B) 1.5 (C) 1.6 (D) 1.8

47. Energys of photon of light having two different frequencies are 2 eV and 10 eV respectively. If both are incident on the metal having work function 1 eV, ratio of maximum velocities of emitted electron is.....

(A) 1 : 5 (B) 3 : 11 (C) 2 : 9 (D) 1 : 3

48. What will be velocity of particle having mass 3 times the rest mass ? $\left(c = 3\times10^8 \frac{m}{s}\right)$

(A) $1.83 \times 10^8 \frac{m}{s}$ (B) $0.92 \times 10^8 \frac{m}{s}$

(C) $2.83 \times 10^8 \frac{m}{s}$ (D) $5.66 \times 10^8 \frac{m}{s}$

49. De-Broglie wavelength of particle moving at a $\frac{1}{4}$ th of speed of light having rest mass m_0 is.....

(A) $\frac{3.87h}{m_0C}$ (B) $\frac{4.92h}{m_0C}$ (C) $\frac{7.57h}{m_0C}$ (D) $\frac{9.46h}{m_0C}$

50. Work function of metal is 2.5 eV. If wave length of light incident on metal plate is $3000\ \text{Å}$, stopping potential of emitted electron will be.....

$\left(h = 6.62 \times 10^{-34}\ \text{J.s}, c = 3 \times 10^8 \frac{m}{s}\right)$

(A) 0.82 V (B) 0.41 V (C) 1.64 V (D) 3.28 V

51. An electron enters perpendicularly into uniform magnetic field having magnitude 0.5×10^{-4} T . If it moves on a circular path of radius 2 mm, its de - Broglie wavelength is ..... Å

(A) 3410 (B) 4140 (C) 2070 (D) 2785

52. In photoelectric effect, work function of mateal is 3.5 eV. By applying - 1.2V potental, photo electric current becomes zero, so.....

(A) energy of incident photon is 4.7 eV.

(B) energy of incident photon is 2.3 eV

(C) If photon having higher frequency is used, photo electric current is produced.

(D) When energy of photon is 2.3 eV, photo electric current becomes maximum

53. Work function of tungsten and sodium are 4.5 eV and 2.3 eV respectively. If threshold wavelength for sodium is 5460A, threshold frequnecy for tungsten will be.....

(A) 10^{15} Hz (B) 1.1×10^{15} Hz (C) 1.2×10^{15} Hz (D) 1.4×10^{15} Hz

54. Ration of momentum of photons having wavelenghth $4000\ \text{Å}$ & $8000\ \text{Å}$ is.....

(A) 2 : 1 (B) 1 : 2 (C) 20 : 1 (D) 1 : 20

55. Work function of metal is 4.2 eV. If ultraviolet radiation (photon) having energy 6.2 eV, stopping potential will be.....

(A) 2 eV (B) 2 V (C) શૂન્ય (D) 10.4 V

56. If intensity of incident light is increased,.....of photo elctrons will increase.

(A) number (B) Frequency (C) energy (D) wavelength

57. A photon having energy 5.5 eV is incident on metal suface and emits photo elctrons having maximum kinetic energy 0.4 eV. Then stopping potential of this electron is.....

(A) 5.5 eV (B) 5.1 eV (C) 5.9 eV (D) 4.0 eV

58. Frequency of photon having energy 66 eV is.....$\left(h = 6.6 \times 10^{-34}\ J.s\right)$

(A) 8×10^{-15} Hz (B) 12×10^{-15} Hz (C) 16×10^{-15} Hz (D) 24×10^{-15} Hz

59. Kinetic energy of proton accelerated under p.d. 1 V will be.....

(A) 1840 eV (B) 13.6 eV (C) 1 eV (D) 0.54 eV

60. Which of the following phenomenon can not be explained by quantum theroy of light ?

(A) Emission of radiation from black body

(B) Photo electric effect

(C) Polarisation

(D) Crompton effect

61. What will be velocity of light of particle having mass double than its rest mass ?

(A) $\frac{2c}{3}$ (B) $\frac{c}{2}$ (C) $\frac{\sqrt{3}c}{2}$ (D) $\frac{3c}{4}$

62. If de-Broglie wavelength of electron is increassed by 1 % its momentum.....

(A) increases by 1% (B) decreases by 1%

(C) increased by 2% (D) decreases by 2%

63. With how much p.d. should an electron be accelerated, so that its de-Broglie wavelength is $0.4\ \text{Å}$

(A) 9410 V (B) 94.10 V (C) 9.140 V (D) 941.0 V

64. de-Broglie wavelength of atom at T K absolute temperature will be.....

(A) $\frac{h}{mkT}$ (B) $\frac{h}{\sqrt{3mKT}}$ (C) $\frac{\sqrt{2mKT}}{h}$ (D) $\sqrt{2mKT}$

65. Uncertainty of momentum of particle is 10^{-30} kg ms^{-1} so minimum uncertainty in its position is.....m.

(A) 10^{-8} m (B) 10^{-12} m (C) 10^{-16} m (D) 10^{-4} m

66. An electron is accelerated between two points having potential 20 V and 40 V, de- Broglic wavelength of electron is.....

(A) $0.75\ \text{Å}$ (B) $7.5\ \text{Å}$ (C) $2.75\ \text{Å}$ (D) 0.75 nm

67. Energy of α particle having de - Broglie wavelength $0.004\ \text{Å}$ is.....

(A) 1280 eV (B) 1200 eV (C) 1200 MeV (D) 1200GeV

68. de - Broglic wavelength of electron in n^{th} Bohr orbit is.....

(A) $\frac{2\pi r}{n}\,\text{Å}$ (B) $2\pi rn\,\text{Å}$ (C) $\frac{1}{n}\,\text{Å}$ (D) $n\,\text{Å}$

69. If work function of potesium is 3 eV. its threshold wavelength = ...

$\left(h = 6.6 \times 10^{-34}\ \text{J.s.}\ c = 3 \times 10^{8}\ \frac{m}{s},\ 1\ \text{eV} - 1.6 \times 10^{-19}\ \text{J.s}\right)$

(A) 4125 Å (B) 1860 Å (C) 1860 nm (D) 4125 nm

70. In photo electric effect, if threshold wave length of a metal is 5000 Å work function of this metal is.....eV. $\left(h = 6.62 \times 10^{-34}\ \text{J.s},\ c = 3 \times 10^{8}\ \text{m/s},\ 1\text{eV} = 1.6 \times 10^{-19}\ \text{J}\right)$

(A) 1.24 (B) 2.48 (C) 4.96 (D) 3.72

71. By four photons having frequency $f < f_0$,.....

(A) શૂન્ય (B) એક (C) બે (D) એકપણ નહિ

72. Work functions for tungsten and sodium are 4.5 eV and 2.3 eV respectively. If threshold wavelength of sodium is 5460 Å , threshold wavelength for tungsten will be.....Å

(A) 528 (B) 10683 (C) 2791 (D) 5893

73. Photo senstive surface is incident by light having frequecy 3 times its threshold frequency. In this condition, if frequency of light is made half and intensity of light is made doubel, magnitude of photo electric current becoms.....

(A) Fourth (B) double (C) half (D) zero

74. If ratio of threshold frequencies of two metals is 1 : 3, ratio of their work functions is.....

(A) 1 : 3 (B) 3 : 1 (C) 4 : 16 (D) 16 : 4

75. It work function of Na and Fe are 2.5 eV and 5eV respectively ratio of their threshold frequencies.....

(A) 2 :1 (B) 1 : 2 (C) 125 : 1 (D) 1 : 1

76. If electrion is accelerated under the effect of 200V p.d., its kinetic energy = ...

(A) 1.6×10^{-17} J (B) 23×10^{-17} J (C) 4.8×10^{-17} J (D) 6.4×10^{-10} J

77. In quantum mechaincs, a particle.....

(A) can be regarded as group of harmonic waves.

(B) can be regarded as a single wave of definite wavelength only

(C) Can be regarded as only a pair of two harmonic waves

(D) is a point-like object with mass.

78. The de-Broglie wavelength of a proton and α - particle is same. The ratio of their velocities will be.....

(α particle is the He-nucleus, having two protons and two neutrons. Thus, its mass $M_\alpha = 4m_p$ where m_p is the mass of the proton.)

(A) 1 : 4 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1

79. The de-Broglie wavelength associated with a particle with rest mass m_0 and moving with speed of light in vacuum is.....

(A) $\frac{h}{m_0 c}$ (B) 0 (C) ∞ (D) $\frac{m_0 c}{h}$

80. A proton and electron are lying in a box having unpenetrable walls, the ratio of uncertainty in their velocities are.....(m_e = mass of electron and m_p = mass of proton.)

(A) $\frac{m_e}{m_p}$ (B) $m_e \cdot m_p$ (C) $\sqrt{m_e \cdot m_p}$ (D) $\sqrt{\frac{m_e}{m_p}}$

81. Photons of enerty 1 eV and 2.5 ev successively illuminate a metal, whose work function is 0.5 eV, the ratio of maximum speed of emitted electron is.....

(A) 1 : 2 (B) 2 : 1 (C) 3 : 1 (D) 1 : 3

82. A proton and an x-particle are passed through same potential difference. If their initial velocity is zero, the ratio of their de Broglie's wavelength after getting accelerated is.....

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) $2\sqrt{2}$: 1

83. The uncertainty in position of a particle is same as its de-Broglie wavelength, uncertainty in its momentum is.....

(A) $\frac{\hbar}{\lambda}$ (B) $\frac{2\hbar}{3\lambda}$ (C) $\frac{\lambda}{\hbar}$ (D) $\frac{3\lambda}{2\hbar}$

84. Wavelength of an electron having energy 10 keV is..... $\mathring{A}$

(A) 0.12 (B) 1.2 (C) 12 (D) 120

85. Wavelength λ_A and λ_B are incident on two identical metal plates and photo electrons are emitted. If $\lambda_A = 2\lambda_B$, the maximum kinetic energy of photo electrons is.....

(A) $2K_A = K_B$ (B) $K_A < \frac{K_B}{2}$ (C) $K_A = 2K_B$ (D) $K_A > \frac{K_B}{2}$

86. The de-Broglie wave length of a particle having velosity of 2.25×10^8 ms^{-1}, is the same value of a photon wavelength, then the ratio of kinetic energy and photon energy of the particle is ..... (take c = 3×10^8 ms^{-1})

(A) $\frac{1}{8}$ (B) $\frac{3}{8}$ (C) $\frac{5}{8}$ (D) $\frac{7}{8}$

87. An image of sun is formed by a lens of focal length 30 cm on the metal surface of a photo-electric cell and a photoelectic current (I) is produced. The lens forming the image is then replaced by another of the same diameter but of focal length of 15 cm. The photoelectric current in this case is.....

(A) $\frac{1}{2}$ (B) 1 (C) 2I (D) 4I

88. The rationof de - Begli wavelengths of molecules of hydrogen and helium which are at temperature 27° and $127^{o}C$ respectively is.....

(A) $\frac{1}{2}$ (B) $\sqrt{\frac{3}{8}}$ (C) $\sqrt{\frac{8}{3}}$ (D) 1

89. A photon, an electron and a uranium nucleus all have same wavelength. The one with the msot energy.....

(A) is the photon

(B) is the electron

(C) is the uranium nucleus

(D) depends upon the wavelength and properties of the particle.

90. A light of frequency 1.5 times the threshold frequency is incident on photo-sensitive material. If the frequencey is halved and intensity is doubled, the photoelectric current becomes.....

(A) zero (B) doubled (C) quadrpled (D) halved

91. An electron with rest mass m_0 moves with a speed of 0.8 C. Its mass, when it moves with this speed is.....

(A) m_0 (B) $\frac{m_0}{6}$ (C) $\frac{5m_0}{3}$ (D) $\frac{3m_0}{5}$

92. The cathode of a photoelectiric cell is changed such that the work function changes from W_1 to W_2 $(W_2 > W_1)$. If the currents before and after change are I_1 and I_2, all other conditions remaining unchanged, then assuming $hf > W_2$ .....

(A) $I_1 = I_2$ (B) $I_1 < I_2$ (C) $I_1 > I_2$ (D) $I_1 < I_2 < 2I_1$

93. The mass of a particle is 400 times than that of an electron and charge is double. The partcile is acceleratied by 5V. Initially the particle remained at rest, then its final kinetic energy is.....

(A) 5 eV (B) 10 eV (C) 100 eV (D) 2000 eV

94. The work function of a metal is 1 eV. Light of wavelength $3000\ \mathring{A}$ is incident on this metal surface. The maximum velocity of emitted photoelection will be.....

(A) $10\ ms^{-1}$ (B) $10^3\ ms^{-1}$ (C) $10^4\ ms^{-1}$ (D) $10^6\ ms^{-1}$

95. The work function for tungsten and solidum are 4.5 eV and 2.3 eV respectively. If the threshold wavelength λ_0 for sodium is $5460\,\text{Å}$ the value of λ_0 for tungsten is.....

(A) $528\,\text{Å}$ (B) $2791\,\text{Å}$ (C) $5893\,\text{Å}$ (D) $10683\,\text{Å}$

96. In the Davission and Germer's experiment the filament of electron gun is coated with.....

(A) Cotton (B) BaO (C) Oil (D) FeO

97. Suppose $\psi(x, y, z)$ represents a particle in three dimensional space, then probility of finding the particle in the unit volume at a given point x,y,z is .....

(A) inversely proportional to $\psi(x, y, z)$ (B) directly proportional ψ^*

(C) directly proportional to $|\psi\ \psi^*|$ (D) inversely proportional to $|\psi\ \psi^*|$

98. Select the correct statement from the following

(A) Radiantion and matter (particles) may not exhibit both the wave nature and particle nature simultaneously at the some moment.

(B) At some moment electromagnetic waves get divided in to small pieces named particles

(C) In a given circumstance a particle at one moment behaves like particle and the at the next moment as wave and so on

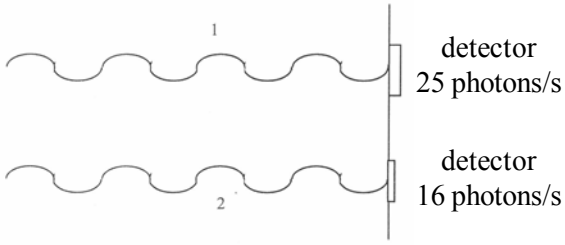
(D) Each microscopic particle is envloped by a wave

99. In which of the following phenomena the photon picture is required?

(A) Energy distribution in black body radiation (B) Compton scattering

(C) Photoelectric effect (D) all of the none

100. Two identical detect 25 and 16 photons per second as shown in the Fig. ..... the ratio of the amplitudes of these light waves.



(A) $\frac{26}{16}$ (B) $\frac{5}{4}$

(C) $\frac{4}{5}$ (D) $\frac{25}{16}$

101. Calculate the energy of a photon of radian wavelength $6000\,\text{Å}$ in eV.

(A) 20.6 eV (B) 2.06 eV (C) 1.03 eV (D) 4.12 eV

102. A 100 W bulb with 10% efficiency is placed at the centre of a sphere (hollow) of radius $\frac{1}{\sqrt{4\pi}}$ m. Find the number of photons arriving on unit area of the surface in the unit time.

(A) $4\pi \times 10^{20}$ (B) 1.51×10^{20} (C) $16\pi^2 \times 10^{19}$ (D) $\sqrt{4\pi} \times 10^{18}$

103. The work function of metal is 5.3 eV. What is threshold frequency ?

(A) 3.1×10^{15} Hz (B) 3.1×10^{45} Hz (C) 1.3×10^{15} Hz (D) 1.3×10^{45} Hz

104. An electron moving with velocity 0.6c, then de-brogly wavelength associated with is.....

(rest mars of electron, $m_0 = 9.1 \times 10^{-31} \frac{k}{s}$ $h = 6.63 \times 10^{-34}$ Js)

(A) 3.24×10^{-12} m (B) 32.4×10^{-12} m (C) 320×10^{-12} m (D) 3.29×10^{-14} m

105. In an experiment to determine photoelectric charactheristics for a metal the intensity of radiation is kept constant. Strating with threshold frequency. Now, frequency of incident radiation is increased. It is observed that .....

(A) the nuber of photoelectrons increases (B) the energy of photoelectrons decreases

(C) the number of photoelectrons decreases (D) the energy of photoelectrons increases.

106. An oscillator in the walls of cavity in which electromagnetic radiation, has energy equal to 5 h f. Then the oscillator is equivalent to .....

(A) 2 : 1 (B) 1 : 2 (C) 125 : 1 (D) 1 : 1

107. Valance electrons in metals .....

(A) are completely free to move anywhere in metal (B) move in the spave acfording to their wave function inside the metal.

(C) can move outside metal also (D) never escape from metal.

108. Rayleigh and jeans regarded the black body radiations as

(A) $E_\lambda = \lambda^5 f(\lambda T)$ (B) $E_\lambda = \lambda^3 f(kT)$ (C) $E_\lambda = \lambda^{-5} f(kT)$ (D) $E_\lambda = \lambda^{-4} f(\lambda T)$

109. According to Rayleigh and Jeans the black body radiation in the cavity is system of

(A) progressive electromagnetic waves (B) standing electromagnetic waves

(C) electromagnetic waves of discrete frequencies (D) standing waves in lattice

110. $\frac{e}{m}$ of electrons $1.76 \times 10^{11} \frac{C}{kg}$ and the stopping potential is 0.71 V, them the maximum velocity of photo electrons is .....

(A) 150 kms^{-1} (B) 200kms^{-1} (C) 500kms^{-1} (D) 250kms^{-1}

111. Radius of a nucleus 2×10^{-15} m . If we imagine an electron inside the nucleus then energy of electron will be = ... MeV.

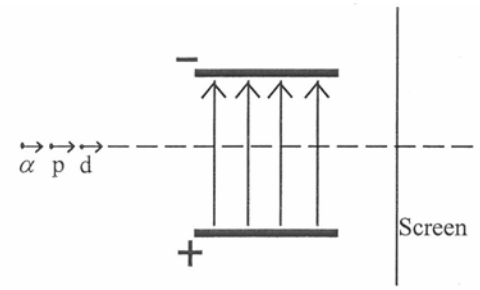
$m_e = 9.1 \times 10^{-31}$ kg, $h = 6.6 \times 10^{-34}$ Js

(A) 6.98×10^3 (B) 8.94×10^3 (C) 4.98×10^3 (D) 9.48×10^3

112. 2mW light of wave length $4400\,\mathring{A}$ is incident on photo sensitive surface of Cs. If quantum efficiency is 0.5 %, what will be the value of photoelectric current ?

(A) 1.56 μA (B) 2.56 μA (C) 4.56 μA (D) 3.56 μA

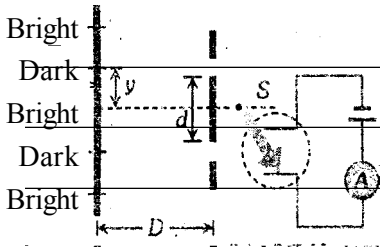
113. A proton, a deutron and an a-particle having the same momentum, enters a region of uniform electric field between the parallel plates of a capacitor. The electric field is perpendicular to the initial path of the particles. Then the ratio of deflections suffered by them is



(A) 1 : 2 : 8 (B) 1 : 2 : 4

(C) 1 : 1 : 2 (D) None of these

114. In the following aggrement y = 1.0 mm, d = 0.24 mm and D = 1.2 m. The work function of the material of the emitter is 2.2 eV. The stopping potential V needed to stop the photon is .....



(A) 0.9 V (B) 0.5 V

(C) 0.1 V (D) 0.4 V

115. The difference of kinetic energy of photoelectrons emitted from a surface wavelength $2500\,\mathring{A}$ and $5000\,\mathring{A}$ will be

(A) 1.98×10^{-19} J (B) 1.98×10^{-19} J (C) 3.96×10^{-19} eV (D) 3.96×10^{-19} J.

Assertion and Reason Type Questions :

Direction : Read the following questions choose if :

(a) Both Assertion and Reason are true and Reason is correct explanantion of Assertion.

(b) Both Assertion and Reason are true, but Reason is not correct explanation of Assertion.

(c) Assertion is true but the Reason is false.

(d) Both Assertion and Reason is false.

116. Assertion :

Stopping potential is a measure of K.E. of photo-electrons.

Reason : $W = eV_s = \frac{1}{2}mv^2 = K.E$

(A) a (B) b (C) c (D) d

117. Assertion :

Metals like Na or K, emit electrons even when visible lights fall on them .

Reason : This is because their work function is low.

(A) a (B) b (C) c (D) d

118. Assertion : The de Broglie wavelength of an electron accelerated through 941 volts is 0.4 $\overset{o}{A}$.
Reason : Higher the acceleration potentials of electron, smaller is the de Broglie wavelength.

(A) a (B) b (C) c (D) d

119. Assertion :
Photo-electric can take place only with an electron bound in the atom.
Reason : Electron is Fermion and photon is boson.

(A) a (B) b (C) c (D) d

120. Theshold-Frequency is equal to $= \ldots\ldots\ldots\ldots \times 10^{14}$ Hz.

(A) 4.0 (B) 4.2 (C) 8.0 (D) 8.4

121. An electron is accelerated under a potential difference of 64 V, the de Broglie wave length associated with electron is $= \ldots\ldots\ldots\ldots \overset{o}{A}$

(Use charge of electon $= 1.6 \times 10^{-19}$ C, $= 9.1 \times 10^{-31}$ Kg, mass of electron $h = 6.623 \times 10^{-34}$ J.sec)

(A) 4.54 (B) 3.53 (C) 2.53 (D) 1.534

122. A α particle and proton has same K.E. The ratio of de Broglie wavelength of a α particle and proton is....

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2

123. If α particle and proton have same momenta, the ratio of de Broglie wavelength of α-particle and proton is.....

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2

124. If α - particle and proton have same velocities, the ratio of de Broglie wavelength of α-particle and proton is ....

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 1 (D) 2

125. If α - particle and proton are accelerated through the same potential difference, then the ratio of de Brogile wavelength of α - particle and proton is .....

(A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{1}{2\sqrt{2}}$ (D) $2\sqrt{2}$

126. Read the paragraph carefully and select the proper choice from given multiple choices.
According to Einstein when a photon of light of frequeny f or wavelength λ is incident on a photo sensitive metal surface of work function ϕ. Where $\phi < hf$ (here *h* is Plank's constant) then the emission of photo-electronss place takes place. The maximum K.E. of emitted photo electrons is given by $K_{max} = hf - \phi$. If the therehold frequency of metal is f_0 then $hf_0 = \phi$.

(i) A metal of work function 3.3 eV is illuminated by light of wave length 300 nm. The maximum K.E> of photo-electrons is = ...eV. $\left(h = 6.6 \times 10^{-34} \text{ J.sec}\right)$

(A) 0.825 (B) 0.413 (C) 1.32 (D) 1.65

(ii) Stopping potential of emitted photo-electron is = ...V.

(A) 0.413 (B) 0.825 (C) 1.32 (D) 1.65

(iii) The threshold frequency fo = ...$\times 10^{14}$ Hz.

(A) 4.0 (B) 4.2 (C) 8.0 (D) 8.4

127. Matching type questions : (Match, Column-I and Column-II property)

Column-I		Column-II	
(A)	Planck's theory of quanta	(p)	Light energy = $h\nu$
(B)	Einstein's theory of quanta	(q)	Angular momentum of electron in an orbit.
(C)	Bohr's stationary orbit	(r)	Oscillator energies
(D)	D-Broglie waves	(s)	Electron microscope

(A) $(A-p), (B-q), (C-r), (D-s)$ (B) $(A-q), (B-r), (C-s), (D-p)$

(C) $(A-r), (B-p), (C-q), (D-s)$ (D) $(A-r), (B-p), (C-s), (D-q)$

128. Matching type questions : (Match, Column-I and Column-II property)

Column-I		Column-II	
(A)	Particle nature of light	(p)	Davisson and Germer
(B)	Wave nature of light	(q)	G. P. Thomson
(C)	Wave nature of slow moving electrons	(r)	Max. Planck
(D)	Wave nature of fast moving electrons	(s)	Muygen

(A) $(A-p), (B-q), (C-r), (D-s)$ (B) $(A-q), (B-r), (C-s), (D-p)$

(C) $(A-r), (B-s), (C-p), (D-q)$ (D) $(A-s), (B-r), (C-q), (D-p)$

129. Matching type questions : (Match, Column-I and Column-II property)

	Column-I		Column-II
(I)	Quantisation of charge	(P)	Diffraction of light
(II)	Wave nature of light	(Q)	de Broglie hypothesis
(III)	Dual nature of matter	(R)	Photo-electric effect
(IV)	Particle nature of light	(S)	Milikan's drop experiment

(A) I – P, II – Q, III – R, IV – S (B) I – S, II – P, III – Q, IV – R

(C) I – Q, II – R, III – S, IV – P (D) I – R, II – S, III – P, IV – O

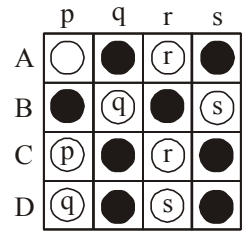
130. Matching type questions : (Match, Column-I and Column-II property)

	Column-I		Column-II
(I)	Energy of photon of wavelength λ is	(P)	$\frac{E}{p}$
(II)	The be Broglie wavelength associated with particle of momentum P is	(Q)	$\frac{hf}{c^2}$
(III)	Mass of photon in motion is	(R)	$\frac{hc}{\lambda}$
(IV)	The velocity of photon of energy E and momentum P is	(S)	$\frac{h}{p}$

(A) I – P, II – Q, III – R, IV – S (B) I – R, II – S, III – Q, IV – P

(C) I – R, II – S, III – P, IV – Q (D) I – S, II – R, III – Q, IV – P

131. Each of the following questions contains statement given in two columns, which have to be matched. The answers to these questions have to be appropriately dubboed. If the correct matches are A - q, s, B - p, r, C- q, s and D - s then the correctly dubbled matrix will look like the one shown here :



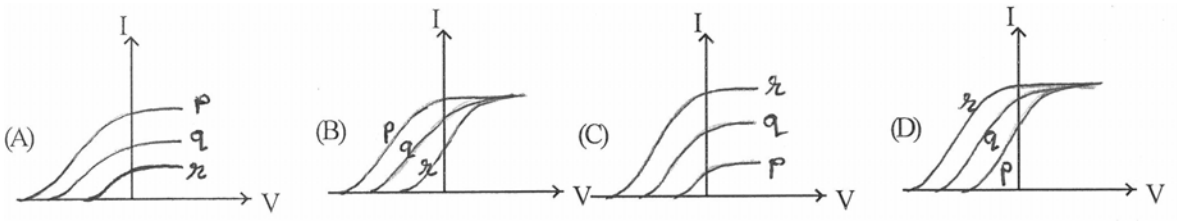
Match the statements of column I with that of column II.

	Column-I		Column-II
(A)	a-particle and proton have same K.E.	(p)	$\lambda_{p'} = \lambda_a$
(B)	*a*-particle has one quarter K.E. then that of protot.	(q)	$\lambda_{p'} = \lambda_a$
(C)	*a*-particle has one quarter K.E. than that of protot.	(r)	$\lambda_{p'} = p_a$
(D)	*a*-particle and proton has same velocity	(s)	$p_p > p_a$

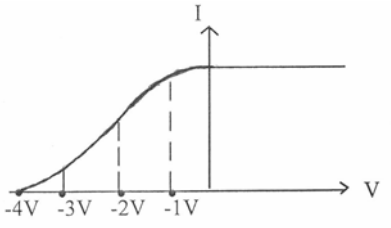
132. In photoelectric effect experiment match the statement of column I with that of column II :

	Column-I		Column-II
(A)	When intensity of incident light is increased on photo-cathode without changing its frequency.	(p)	The value of photo-electric current increases.
(B)	When the frequency of incident light is increased on photocathode without changing its intensity.	(q)	The value of stopping potential does not change.
(C)	When the wavelength of incident light is decreased without changing its intensity.	(r)	The value of stopping potential does not change.
(D)	When the potential of anode is increase without changing the frequency and intensity of the incident light	(s)	The value of photo-electric current does not change.

133. Photoelectric effect experiments are performed using three different metal plates p, q and r having work functions $\phi_p = 2eV$, $\phi_Q = 2.5eV$, $\phi_R = 3$ eV, respectively. A light beam containging wavelengths of 550 nm, 450 nm and 350 nm with equal Intensities illuminates each of the plates. The correct I-V graph for the experiments is.



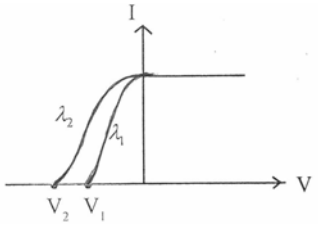
134. The value of stopping potential in the following diagram



(A) $-4V$ (B) $-3V$

(C) $-2V$ (D) $-1V$

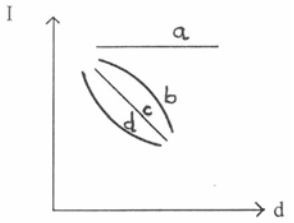
135. In the following diagram if $V_1 > V_1$ then.....



(A) $\lambda_1 = \sqrt{\lambda_2}$ (B) $\lambda_1 < \lambda_2$

(C) $\lambda_1 = \lambda_2$ (D) $\lambda_1 > \lambda_2$

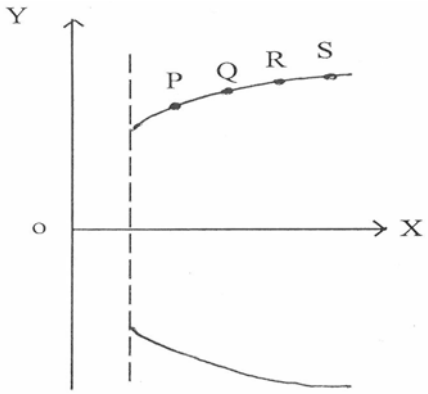
136. A point source of light is used n an experiment on photoelectric effect. Which of the following curves best represent the variation of photo current (i) with distance (d) of the source from the emitter.



(A) *a* (B) *b*

(C) *c* (D) *d*

137. In a parabola spectrograph, the velocities of four positive ions P, Q, R and S are u_1, u_2, u_3 and u_4 respectively. Then



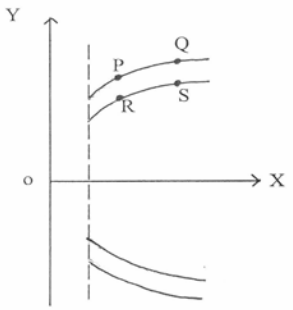
(A) $v_1 > v_2 > v_3 > v_4$

(B) $v_1 < v_2 < v_3 < v_4$

(C) $v_1 = v_2 = v_3 = v_4$

(D) $v_1 << v_2 > v_3 < v_4$

138. In Thomson spectrograph experiment, four positive ions P, Q, R and S are situated on Y-X curve a shown in the figure.



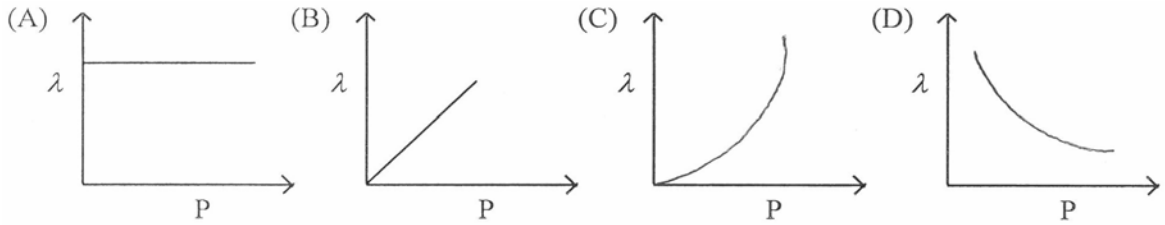
(A) The specific charge of R and S are same

(B) The masses of P and S are same

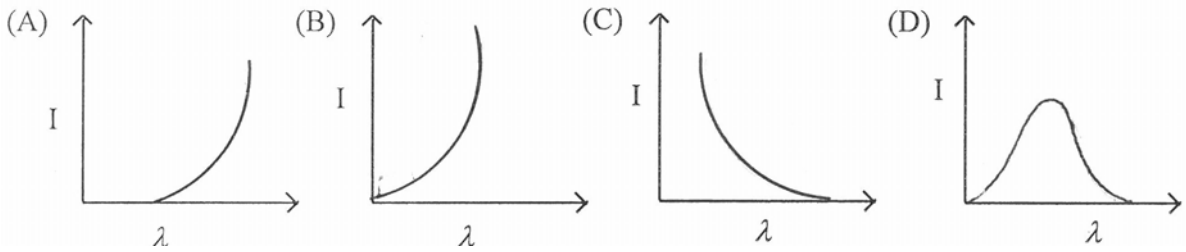
(C) The specific chrage of Q and R are same

(D) The velocities of R and S are same

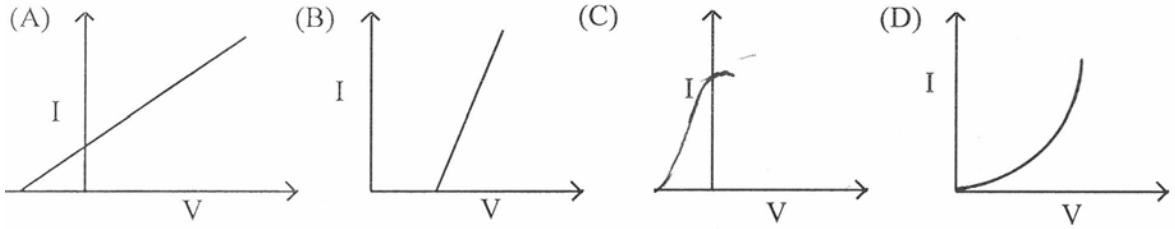
139. Kinetic energy of photo electron E_K changes with frequency (f) of light. Which of the following graph represents this emission ?



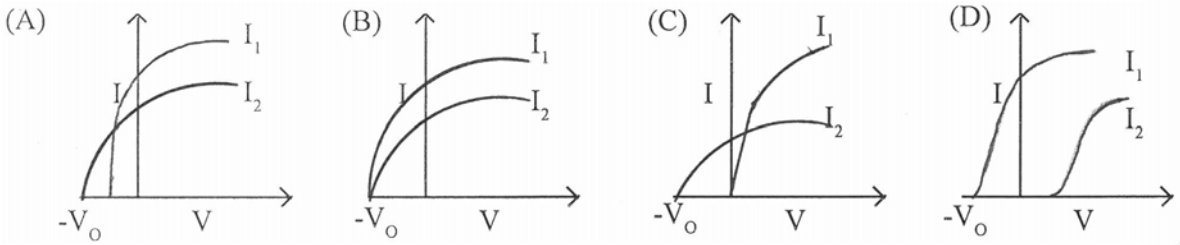
140. Which of the following graph shows change in momentum of particle which de-Broglie wavelength ?



141. In a photocell, the curve for current 1 versus potential difference V is best represented by Fig.



142. The curves (a), (b), (c) and (d) show the variation between the applied potential difference (V) and the photoelectric current (i), at two different intensities of light ($I_1 > I_2$) In which figure is the correct variation shown.



KEY NOTE

1	B	26	D	51	B	76	B	101	B	126(i)	A
2	D	27	C	52	A	77	A	102	C	(ii)	B
3	C	28	B	53	B	78	D	103	C	(iii)	C
4	B	29	C	54	A	79	B	104	A	127	C
5	B	30	B	55	B	80	A	105	D	128	C
6	C	31	B	56	A	81	A	106	D	129	B
7	C	32	C	57	D	82	D	107	B	130	B
8	D	33	D	58	C	83	A	108	A	131	A-q,s B-P,r C-q,s D-q,s
9	C	34	B	59	C	84	A	109	B	132	A-p,r B-q,s C-q,s D-r
10	D	35	B	60	C	85	B	110	C	133	A
11	C	36	D	61	C	86	B	111	D	134	A
12	B	37	D	62	B	87	B	112	D	135	D
13	B	38	C	63	D	88	C	113	A	136	D
14	B	39	C	64	B	89	A	114	A	137	A
15	A	40	B	65	D	90	A	115	D	138	A
16	B	41	C	66	C	91	C	116	D	139	D
17	B	42	C	67	A	92	A	117	B	140	C
18	B	43	B	68	A	93	B	118	C	141	C
19	B	44	A	69	B	94	D	119	C	142	B
20	A	45	A	70	B	95	B	120	C		
21	C	46	D	71	A	96	B	121	D		
22	D	47	D	72	A	97	C	122	B		
23	B	48	C	73	B	98	A	123	C		
24	A	49	A	74	A	99	D	124	A		
25	B	50	C	75	B	100	B	125	C		

HINT

1. $W = \frac{1}{2} mv^2 = q\ V$

$\therefore\ v = \sqrt{\frac{2qV}{m}}$

$\therefore\ \frac{v_A}{v_B} = \sqrt{\frac{q_A}{q_B}} = \sqrt{\frac{q}{4q}} = \frac{1}{2}$

$\therefore\ \frac{v_A}{v_B} = 1 : 2$

2. $\frac{1}{2} mv^2{}_{max} = \frac{hc}{\lambda} - \phi$ ...(1)

$\lambda' = \lambda - 0.25\lambda = 0.75\lambda$ & $v' = 2v$

$\therefore\ \frac{1}{2} m \left(2v_{max}\right)^2 = \frac{hc}{0.75\lambda} - \phi$ ...(2)

$4\left(\frac{\lambda c}{\lambda} - \phi\right) = \frac{4hc}{3\lambda} - \phi$

$\therefore\ \frac{8hc}{3\lambda} = 3\phi$

$\therefore\ \phi = \frac{8hc}{9\lambda}$

$\frac{hc}{\lambda} = 2eV$

$\phi = \frac{8}{9} \times 2\,eV = \frac{16}{9}$

$\therefore\ \phi = 1.8\ eV$

3. $p = \frac{nhc}{\lambda}$

$\therefore\ n = \frac{p\lambda}{hc} = \frac{100 \times 540 \times 10^{-9}}{6 \times 10^{-34} \times 3 \times 10^8}$

$= 3 \times 10^{20}$

4. $\frac{\lambda}{2}$ determent by

$v_2 = 2v_1$

K.E. of election $E_1 = \frac{1}{2} mv_1^2$

$E_2 = \frac{1}{2}mv_2^2 = \frac{1}{2} m (2v_1)^2 = \frac{1}{2}m(4v_1^2)$

$\therefore E_2 = 4\left[\frac{1}{2}mv_1^2\right]$

$\therefore E_2 = 4E_1$

$\therefore \lambda = \frac{h}{\sqrt{2mE}}$

$\therefore \lambda \propto \frac{1}{\sqrt{E}}$ $\quad \therefore \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{E_1}{E_2}} = \sqrt{\frac{E_1}{4E_1}} = \frac{1}{2}$ $\quad \because \lambda_2 = \frac{\lambda_1}{2}$

$\therefore \lambda_1 - \frac{\lambda_1}{2} = \frac{\lambda_1}{2}$

5. $\frac{1}{2} mv^2 = eV$

$\therefore m^2v^2 = p^2 = 2meV$

$\therefore p = \frac{h}{\sqrt{2meV}}$

$\therefore p \propto \frac{1}{\sqrt{me}}$

$\therefore \frac{\lambda_\infty}{\lambda_p} = \sqrt{\frac{m_p\ e_p}{m_\alpha e_\infty}}$

$m_\alpha = 4mp,\ e_\alpha = 2e_p$

$\therefore \frac{\lambda_\infty}{\lambda_p} = \sqrt{\frac{m_p e_p}{4m_p \times 2e_p}} = \frac{1}{\sqrt{8}}$ $\quad \therefore \frac{\lambda_\infty}{\lambda_p} = \frac{1}{2\sqrt{2}}$

$\therefore \frac{\lambda_\infty}{\lambda_0} = \frac{1}{2\sqrt{2}}$ $\quad \therefore \lambda_\infty = \frac{\lambda_0}{2\sqrt{2}}$

6. $\phi = 4eV = 4 \times 1.6 \times 10^{-19}$ J

$\therefore \phi = \dfrac{hc}{\lambda_0}$

$\therefore \lambda_0 = \dfrac{hc}{\phi}$

$= \dfrac{6.62 \times 10^{-34} \times 3 \times 10^8}{4 \times 1.6 \times 10^{-19}}$

$= 3.103 \times 10^{-7}$ m

$= 310.3 \times 10^{-9}$ m

$= 310$ m

7. $m = 200\,g = 0.2\,kg,\ v = 5\ \dfrac{m}{hr} = \dfrac{5}{3600}\ \dfrac{m}{s}$

$p = \dfrac{h}{\lambda} = mv$

$\therefore \lambda = \dfrac{h}{mv} = \dfrac{6.626 \times 10^{-34} \times 3600}{0.2 \times 5}$

$= 23.85 \times 10^{-31}$

$= 2.385 \times 10^{-30}$ m

$= 10^{-30}$ m

8. $E = \dfrac{1}{2}mv^2{}_{max} = hf - hf_0 = h\,(f - f_0)$

$E_1 = h\,(f_1 - f_0)$

$E_2 = h\,(f_2 - f_0)$

$f_1 = 5.5 \times 10^8$ MHz $= 5.5 \times 10^{14}$ Hz

$f_2 = 4.5 \times 10^8$ MHz $= 4.5 \times 10^{14}$ MHz

$\dfrac{E_1}{E_2} = \dfrac{f_1 - f_0}{f_2 - f_0}$

$\dfrac{E_1}{E_2} = \dfrac{1}{5}$

$\therefore \dfrac{1}{5} = \dfrac{5.5 \times 10^{14} - f_0}{4.5 \times 10^{14} - f_0}$

$\therefore\ 4.5 \times 10^{14} - f_0 = 27.5 \times 10^{14} - 5f_0$

$\therefore 4f_0 = 23.0 \times 10^{14}$

$\therefore f_0 = \dfrac{23 \times 10^{14}}{4}$

$= 5.75 \times 10^{14}$ Hz

$= 5.75 \times 10^{8}$ Hz

9. $\lambda = \dfrac{\lambda}{p}$ ...(1)

$\therefore\ \lambda + \dfrac{0.25}{100}\lambda = \dfrac{h}{p - p_0}$

$\therefore\ \lambda\dfrac{100.25\lambda}{100} = \dfrac{h}{p - p_0}$ ....(2)

$\dfrac{100.25}{100} = \dfrac{p}{p - p_0}$

$\therefore 100.25p - 100.25\ p_0 = 100p$

$\therefore 0.25p = 100.25\ p_0$

$\therefore p = \dfrac{100.25}{0.25}$

$\therefore p = 401p_0$

14. $T_1 = 10 + 273 = 283K$

$T_2 = 40 + 273 = 313$ K

$\lambda_1 = 0.4\ \mathring{A}$

$\lambda = \dfrac{h}{\sqrt{2mE}}$

$E = \dfrac{3}{2}KT$ હોવાથી $\lambda = \dfrac{h}{\sqrt{3mKT}}$

$\therefore\ \lambda \propto \dfrac{1}{\sqrt{T}}$ $\therefore\ \dfrac{\lambda_2}{\lambda_1} = \sqrt{\dfrac{T_1}{T_2}}$

$\therefore\ \dfrac{\lambda_2}{0.4 \times 10^{-10}} = \sqrt{\dfrac{283}{313}}$

$$\therefore \quad \frac{\lambda_2}{0.4 \times 10^{-10}} = 0.951$$

$$\therefore \quad \lambda_2 = 0.951 \times 0.4 \times 10^{-10}$$

$$\therefore \quad \lambda_2 = 0.38\,\text{Å}$$

$$\therefore \quad = \lambda_2 - \lambda_1$$

$$= 0.38\,\text{Å} - 0.4\,\text{Å} = -0.02\,\text{Å}$$

$$= 2 \times 10^{-2}\,\text{Å} \text{ decreas}$$

15. $\lambda_1 = 4000\,\text{Å} = 4 \times 10^{-7}\,\text{m}$

$$\lambda_2 = 3600\,\text{Å} = 3.6 \times 10^{-7}\,\text{m}$$

$$V_0 e = \frac{hc}{\lambda} - \phi$$

$$\therefore \quad V_{01} e = \frac{hc}{\lambda_1} - \phi \qquad \ldots(1)$$

$$\therefore \quad V_{02} e = \frac{hc}{\lambda_2} - \phi \qquad \ldots(2)$$

$$\therefore \quad V_{02} e - V_{01} e = hc \left[\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right]$$

$$\therefore \quad V_{02} - V_{01} = \frac{hc}{e}\left[\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right]$$

$$= \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{1.6 \times 10^{-19}}$$

$$\left[\frac{10^7}{3.6} - \frac{10^7}{4}\right]$$

$$= \frac{6.62 \times 3}{1.6}\left[\frac{4 - 3.6}{4 \times 3.6}\right]$$

$$\therefore \quad V_{02} - V_{01} = 0.345\text{V}$$

16. $\lambda_1 = 4000\,\text{Å} = 4 \times 10^{-7}\,\text{m}$

$\lambda_2 = 3000\,\text{Å} = 3 \times 10^{-7}\,\text{m}$

$\Delta E = ?$

$$E = \frac{hc}{\lambda} - \phi$$

$$E_1 = \frac{hc}{\lambda_1} - \phi \qquad ...(1)$$

$$E_2 = \frac{hc}{\lambda_2} - \phi \qquad ...(2)$$

$$\therefore\ E_2 - E_1 = hc\left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1}\right)$$

$$= 6.625 \times 10^{-34} \times 3 \times 10^8 \left[\frac{10^7}{3} - \frac{10^7}{4}\right] = 19.956 \times 10^{-19}\left[\frac{1}{12}\right]$$

$$\therefore\ \Delta E = 1.04\,\text{eV}$$

17. $m = 10\,\text{kg},\ \Delta x = 30 \times 10^{-2}\,\text{m}$

$$h = \Delta x = \frac{h}{mv}$$

$$\therefore\ v = \frac{h}{m\Delta x} = \frac{6.625 \times 10^{-34}}{10 \times 30 \times 10^{-2}}$$

$$\therefore\ v = 2.2 \times 10^{-34}\ \frac{m}{s}$$

18. $v = 50\,\text{V}$

$e = 1.6 \times 10^{-19}\,c$

$m_e = 9.1 \times 10^{-31}\,\text{kg}$

$h = 6.62 \times 10^{-34}\ \text{J.s}$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$= \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-34} \times 1.6 \times 10^{-19} \times 50}} = 0.1735 \times 10^{-9} = 1.735 \times 10^{-10}\,\text{m}$$

$$\lambda = 1.735\,\text{Å}$$

19. $\lambda = 0.1\overset{0}{A}$

$$E = \frac{hc}{\lambda} = \frac{6.625 \times 10^{-34} \times 3 \times 10^{8}}{10^{-11}}$$

$$= 19.875 \times 10^{-15}\ J$$

$$1eV = 1.6 \times 10^{-19}\ J$$

$$\therefore\ E = \frac{19.875 \times 10^{-15}}{1.6 \times 10^{-19}}$$

$$= 12.42 \times 10^{4}\ eV$$

20. $E_1 = \frac{hc}{\lambda_1}$

$$E_2 = \frac{nhc}{\lambda_2}$$

$$E_1 = E_2$$

$$\therefore\ \frac{hc}{\lambda_1} = \frac{nhc}{\lambda_2}$$

$$\therefore\ n = \frac{\lambda_2}{\lambda_1} = \frac{8000}{4000} = 2$$

21. $eV_0 = \frac{hc}{\lambda} - \phi$

$$\therefore\ \phi = \frac{hc}{\lambda} - eV_0$$

$$= \left(\frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{3 \times 10^{-7} \times 1.6 \times 10^{-19}} - 1.85\right) eV$$

$$= 4.125 - 1.85\,eV$$

$$\therefore\ \phi = 2.275\,eV$$

$$\phi = \frac{hc}{\lambda_0}$$

$$\therefore\ \lambda_0\ \frac{hc}{\phi}$$

$$= \frac{6.6 \times 10^{-34} \times 3 \times 10^{8}}{2.275 \times 1.6 \times 10^{-19}} = 5.439 \times 10^{-7}\ m = 5439\,\overset{o}{A}$$

22. $p = 1.0\ mW = 10^{-3}\ W$

$\lambda = 632.8\ nm = 632.8 \times 10^{-9}\ m$

$\lambda = 632.8\ nm = 632.8 \times 10^{-9}\ m$

$$P = \frac{nhc}{\lambda}$$

$$\therefore\ n = \frac{p\lambda}{hc}$$

$$= \frac{10^{-3} \times 6.328 \times 10^{-7}}{6.625 \times 10^{-34} \times 3 \times 10^{8}}$$

$$= \frac{6.328 \times 10^{-10}}{19.875 \times 10^{26}}$$

$$= 0.318 \times 10^{16}$$

$$\therefore\ n = 318 \times 10^{15}\ s^{-1}$$

23. $1 = 1.6 \times 10^{-9}\ \frac{W}{m^2}$

$\lambda = 560\,nm = 5.6 \times 10^{-7}\ m$

$r = 2.5 \times 10^{-3}\ m$

$t = 1s,\ n = ?$

$$1 = \frac{E}{At} = \frac{P}{A}$$

$$\therefore\ P = 1A = 1\left(\pi r^2\right) = 1.6 \times 10^{-9} \times 3.14 \times 6.25 \times 10^{-6}$$

$$= 31.4 \times 10^{-15}\ W$$

$$\therefore\ P = \frac{nhc}{\lambda} \qquad \therefore\ n = \frac{p\lambda}{hc} = \frac{31.4 \times 10^{-15} \times 5.6 \times 10^{-7}}{6.62 \times 10^{-34} \times 3 \times 10^{8}}$$

$$\therefore\ n = 8.85 \times 10^{4}$$

25. $P = 4 \times 10^{28}\ W,\ \lambda = 4500\,\mathring{A} = 4.5 \times 10^{-7}\ m$

$$\therefore\ p = nE = nhf = \frac{nhc}{\lambda}$$

$$\therefore\ n = \frac{p\lambda}{hc} = \frac{4 \times 10^{28} \times 4.5 \times 10^{-7}}{6.62 \times 10^{-34} \times 3 \times 10^{8}} = 0.9 \times 10^{47}$$

$$= 9 \times 10^{46}$$

26. $T_1 = 300K$

$T_2 = 1000K$

$m = 4.7 \times 10^{-26}$ kg

$P = 1$ atm

$$\frac{1}{2} mv^2 = \frac{3}{2} KT$$

$$\therefore \ m^2v^2 = p^2 = 3mKT$$

$$\therefore \ p = \sqrt{3mKT}$$

$$\lambda = \frac{h}{p}$$

$$\therefore \ \lambda = \frac{h}{\sqrt{3mKT}}$$

$$\therefore \ \lambda \propto \frac{1}{\sqrt{T}}$$

$$\therefore \ \frac{\lambda_1}{\lambda_2} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{1000}{300}} = \sqrt{\frac{10}{3}}$$

$$\therefore \ \frac{\lambda_1}{\lambda_2} = 1.826$$

27. $\lambda_1 = 3500 \, \text{Å}, \ \lambda_2 = 290 \, \text{nm},$

$h = 6.625 \times 10^{-24}$ J.s

$$\frac{1}{2} mv^2_{max} = \frac{hc}{\lambda} - \phi = eV_0$$

$$\therefore \ V_{01}e = \frac{hc}{\lambda_1} - \phi \qquad \therefore \ V_{02}e = \frac{hc}{\lambda_1} - \phi$$

$$\therefore \ \left(V_{02}e = \frac{hc}{\lambda_2} \right) - \phi \quad \therefore \ (V_{02} - V_{01})e = hc \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

$$\therefore \ V_{02} - V_{01} = \frac{hc}{e} \left[\frac{\lambda_1 - \lambda_2}{\lambda_1 \lambda_2} \right] = 12.42 \left[\frac{0.6}{3.5 \times 2.9} \right]$$

$= 0.7342$

$= 73.42 \times 10^{-2}$ V

28. $\lambda = 6625\,\text{Å}^{\circ}$

$= 6.625 \times 10^{-7}\ \text{m}$

$c = 3 \times 10^{8}\ \frac{m}{s}$

$E = nhf$

$E = \frac{nhc}{\lambda}$

$\therefore\ n = \frac{E\lambda}{hc} = \frac{3 \times 6.625 \times 10^{-7}}{6.625 \times 10^{-34} \times 3 \times 10^{8}}$

$\therefore\ n = 10^{19}$

29. $eV_0 = \frac{hc}{\lambda} - \phi$

$\therefore\ V_0 = \frac{hc}{\lambda c} = \frac{\phi}{e}$

$\therefore\ V_0 = 1031.4\text{V}$

30. $\Delta x = 2r = 2 \times 10^{-15}\ \text{m}$

હવે $\Delta x \cdot \Delta p = \frac{h}{2\pi}$

$\therefore\ \Delta p = \frac{h}{2\pi\ \Delta x} = 0.5274 \times 10^{-19}$

$E = \frac{p^2}{2m}$ $\qquad P = \Delta p$

$= 9.55 \times 10^{9}\ \text{eV} = 9.55 \times 10^{3}\ \text{MeV}$

32. $E_p = \frac{hc}{\lambda}$

$E_p = 19.87 \times 10^{-16}\ .\ \text{J}$

$E_e = \frac{p^2}{2m} = \frac{h^2}{\lambda^2 (2m)}$

$\therefore\ E_e = 2.41 \times 10^{-17}\ \text{J}$

$\therefore\ \frac{E_p}{E_e} = \frac{19.87 \times 10^{-16}}{2.41 \times 10^{-17}}$ $\qquad \therefore\ \frac{E_p}{E_e} = 82.4$

34. $E = K + U$

$$E = \frac{p^2}{2m} - eV_{(x)}$$

$$\therefore \quad p = \left(2mE + eV_{(x)}\right)^{\frac{1}{2}}$$

$$\therefore \quad \lambda = \frac{h}{p} = \frac{h}{\left[2m\left(E + eV_{(x)}\right)\right]^{\frac{1}{2}}}$$

$$\mu = \frac{\lambda_0}{\lambda} = \frac{h}{\sqrt{2mE}} \times \frac{\left[2m\left(E + eV_{(x)}\right)\right]^{\frac{1}{2}}}{h}$$

$$\mu = \left[\frac{E + eV_{(x)}}{E}\right]^{\frac{1}{2}}$$

35. $E = p = nhf \qquad p = \frac{hf}{c}$

$$p = \frac{p\left(\text{उत्सर्ग}\right)}{nc}$$

$$\therefore \quad np = \frac{P}{C}$$

$$\therefore \quad F = \frac{P}{C}$$

$$T\cos\theta = mg \qquad \ldots(1)$$

$$T\sin\theta = \frac{P}{C} \qquad \ldots(2)$$

$$\therefore \quad \frac{T\sin\theta}{T\cos\theta} = \frac{P}{c} \times \frac{1}{mg}$$

$$\therefore \quad \tan\theta = \frac{P}{cmg}$$

$$\therefore \quad \theta = \tan^{-1}\left[\frac{P}{cmg}\right]$$

36. $\frac{1}{2}mv^2_{max} = eV_0 = \frac{hc}{\theta} - \phi$

$= \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{2 \times 10^{-7}}$

$= 4.5 \times 1.6 \times 10^{-19}$

$\therefore \ V_0 = \frac{9.94 \times 10^{-19}}{1.6 \times 10^{-19}} = \frac{4.5 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-19}}$

$= 6.21 - 4.5$

$\therefore \ V_0 = 1.71\,V$

$\frac{1}{2}mv^2_{max} = eV_0 = 1.71 \times 1.6 \times 10^{-19}$

$\therefore \ v^2_{max} = \frac{2 \times 2.74 \times 10^{-19}}{m}$

$\therefore \ v_{max} = \sqrt{\frac{5.48 \times 10^{-19}}{9.11 \times 10^{-31}}}$

$\therefore \ v_{max} = 7.75 \times 10^5 \ \frac{m}{s}$

40. $t = 10\ s, \ N = 11 \times 10^{11}, \ \lambda = 10\,\mathring{A}, \ A = 0.01\,m^2$

$1 = \frac{E}{A.t}$

$E = \frac{Nhc}{\lambda}$

$\therefore \ I = \frac{Nhc}{\lambda At}$

$\therefore \ I = \frac{11 \times 10^{11} \times 6.625 \times 10^{-34} \times 3 \times 10^8}{10 \times 10^{-10} \times 10^{-2} \times 10}$

$= 218.6 \times 10^{-5}$

$= 2.186 \times 10^{-3} \ \frac{W}{m^2}$

41. $\lambda = \dfrac{h}{mv} \quad \therefore \ \lambda_\alpha = \dfrac{h}{m_\infty v_\infty}$

$\lambda_d = \dfrac{h}{m_d v_d}$

$\therefore \ \dfrac{\lambda_\infty}{\lambda_d} = \dfrac{m_d v_d}{m_\infty v_\infty} = \dfrac{2 \times 2v}{4 \times v}$

$\therefore \ \dfrac{\lambda_\infty}{\lambda_d} = 1$

45. (A)

$V = 50 \text{ KV} = 50 \times 10^3 \text{ V}$

$\lambda = \dfrac{h}{\sqrt{2meV}} = \dfrac{6.62 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 50 \times 10^3 \times 1.6 \times 10^{-19}}}$

$= \dfrac{6.62 \times 10^{-34}}{1.207 \times 10^{-22}}$

$= 5.485 \times 10^{-12} \text{ m}$

49. (A)

$\lambda = \dfrac{h}{p} = \dfrac{h}{mv}$

$m = \dfrac{m_0}{\sqrt{1 - \dfrac{v^2}{c^2}}} \qquad \lambda = \dfrac{h\left(\sqrt{1 - \dfrac{v^2}{c^2}}\right)}{m_0 v}$

$v = \dfrac{c}{4}$

$\lambda = \dfrac{\lambda \sqrt{1 - \dfrac{c^2}{16c^2}}}{m_0 \dfrac{c}{4}} = \dfrac{h\sqrt{\dfrac{16c^2 - c^2}{16c^2}}}{m_0 \dfrac{c}{4}}$

$= \dfrac{\sqrt{\dfrac{15}{16}}\, h}{m_0 \dfrac{c}{4}} = \dfrac{4 \times 0.968\, h}{m_0 c} \qquad \therefore \ \lambda = \dfrac{3.87h}{m_0 c}$

50. $\phi = 2.5 \text{ eV} = 2.5 \times 1.6 \times 10^{-19} \text{ J}$

$\lambda = 3000 \overset{o}{A} = 3 \times 10^{-7} \text{ m}$

$h = 6.62 \times 10^{-34} \text{ J.s}$

$c = 3 \times 10^{8} \text{ J. s}$

$\frac{1}{2} mv_{max}^2 = eV_0 = \frac{hc}{\lambda} - \phi$

$\therefore \ V_0 = \frac{hc}{\lambda e} - \frac{\phi}{e}$

52. energy of incident photon is 4.7eV

$E_K = hf - \phi$

$hf = \phi - E_K$

$= 3.5 + 1.2$

$= 4.7 \text{ eV}$

53. $\phi = hf_0 = \frac{hc}{\lambda_0 e} \text{eV}$

$\therefore \ \phi \propto \frac{1}{\lambda_0}$

$\therefore \ \frac{\phi_T}{\phi_N} = \frac{(\lambda_0)_N}{(\lambda_0)_T} = 5460 \times \frac{2.3}{4.5}$

$= 2791 \overset{o}{A}$

$c = f_0 \lambda_0$

$\therefore \ f_0 = \frac{c}{\lambda_0} = \frac{3 \times 10^8}{2791 \times 10^{-10}}$

$= 1.075 \times 10^{15} \text{ Hz}$

$\therefore \ f_0 = 1.075 \times 10^{15} \text{ Hz}$

54. $P = \frac{h}{\lambda} \qquad \therefore \ P \propto \frac{1}{\lambda}$

$\therefore \ \frac{P_1}{P_2} = \frac{\lambda_2}{\lambda_1} = \frac{8000}{4000} = \frac{2}{1}$

$\therefore \ \frac{P_1}{P_2} = 2 : 1$

55. $\frac{1}{2}mv^2_{max} = V_0e = hf - hf_0$

$\therefore\ V_0 = \frac{hf - hf_0}{e} = \frac{(6.2 - 4.2) \times 1.6 \times 10^{-19}}{1.6 \times 10^{-19}}$

$\therefore\ V_0 = 2V$

57. $\frac{1}{2}mv^2_{max} = eV_0$

$\therefore\ eV_0 = 4.0\,eV$

$\therefore\ V_0 = 4.0\ V$

58. $E = hf$

$\therefore\ f = \frac{E}{h} = \frac{66 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}}$

$= 16 \times 10^{15}\ Hz$

61. $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = 2m_0$

$\therefore\ \sqrt{1 - \frac{v^2}{c^2}} = \frac{1}{2}$

$\therefore\ 1 - \frac{v^2}{c^2} = \frac{1}{4}$

$\therefore\ \frac{v^2}{c^2} = \frac{3}{4}$

$\therefore\ \frac{v}{c} = \frac{\sqrt{3}}{2}$

$\therefore\ v = \frac{\sqrt{3}}{2}\,c$

62. 1% ઘટાડો થાય.

$\lambda = \frac{h}{p}$

$\therefore\ \lambda \propto \frac{1}{p}$

$\therefore\ \lambda \propto p^{-1}$

$d\lambda \propto -p^{-2}\, dp$

$\therefore\ d\lambda \propto -\frac{1}{p^2} dp$

$\therefore\ \frac{dp}{\lambda} \times 100 = -\frac{dp}{p^2} \times p \times 100$

$\therefore\ \frac{dp}{\lambda} \times 100 = -1\%$

decreases by 1%

63. $\therefore\ \lambda = \frac{h}{\sqrt{2Vem}}$

$V = \frac{h^2}{2me\lambda^2}$

$= \frac{\left(6.62 \times 10^{-34}\right)^2}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times \left(0.4 \times 10^{-1}\right)^2}$

$= 940.5$

$= 941\,V$

66. $\lambda = \frac{h}{\sqrt{2meV}}$

$V = V_2 - V_1 = 40 - 20 = 20V$

$= \frac{6.62 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 20}}$

$= 0.274 \times 19^{-9}$

$= 2.75\,\mathring{A}$

67. $\Delta x \cdot \Delta n = \frac{h}{2\pi}$

$\therefore\ \Delta x = \frac{h}{2\pi\Delta p} = \frac{6.62 \times 10^{-34}}{6.28 \times 10^{-30}}$

$= 1.054 \times 10^{-4}$ m

$= 10^{-4}$ m

69. $\phi_0 = \dfrac{hc}{\lambda_0}$

$\therefore\ \lambda_0 = \dfrac{hc}{\phi_0}$

$\therefore\ \lambda_0 = \dfrac{6.6 \times 10^{-34} \times 3 \times 10^8}{3 \times 1.6 \times 10^{-19}}$

$= 4.125 \times 10^{-7}\ m$

$= 4125\ \mathring{A}$

70. $\phi = \dfrac{hc}{\lambda_0 e} = \dfrac{6.62 \times 10^{-34} \times 3 \times 10^8}{5 \times 10^{-7} \times 1.6 \times 10^{-19}}$

$= 2.48\ eV$

72. $\phi = hf_0 = \dfrac{hc}{\lambda_0 e}$

$\therefore\ \phi \propto \dfrac{1}{\lambda_0}$

$\therefore\ \dfrac{(\lambda_0)_{\text{ટંગસ્ટન}}}{(\lambda_0)_{\text{સોડિયમ}}} = \dfrac{\phi_{\text{સોડિયમ}}}{\phi_{\text{ટંગસ્ટન}}}$

$\therefore\ (\lambda_0)_{\text{ટંગસ્ટન}} = \dfrac{\phi_{\text{સોડિયમ}}}{\phi_{\text{ટંગસ્ટન}}} \times (\lambda_0)_{\text{સોડિયમ}}$

$= \dfrac{2.3}{4.5} \times 5460\,\mathring{A}$

$= 2791\,\mathring{A}$

74. $\dfrac{\phi_1}{\phi_2} = \dfrac{hf_{01}}{hf_{02}} = \dfrac{f_{01}}{f_{03}}$ $\qquad \therefore\ \dfrac{\phi_1}{\phi_2} = \dfrac{1}{3}$

75. $\phi = hf_0$

$\therefore\ \phi \propto f_0$

$\therefore\ \dfrac{(f_0)_{Na}}{(f_0)_{Fe}} = \dfrac{\phi_{Na}}{\phi_{Fe}} = \dfrac{2.5}{5} = \dfrac{1}{2}$

$= 1 : 2$

76. $\frac{1}{2}mv^2_{max} = eV = 1.6 \times 10^{-19} \times 200 = 3.2 \times 10^{-17}\ J$

78. $\lambda_p = \lambda_\alpha$

$\therefore \frac{h}{m_p v_p} = \frac{h}{m_\alpha v_\alpha}$

$\therefore \frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p}$

$m_\alpha = 4m_p$

$\frac{v_p}{v_\alpha} = \frac{4m_p}{m_p} = 4$

$\therefore \frac{v_p}{v_\alpha} = 4 : 1$

79. $\lambda = \frac{h}{mv}$

$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$

$\lambda = \frac{h \times \sqrt{1 - \frac{v^2}{c^2}}}{m_0 v}$

$v = c\ ,\ \lambda = \frac{h \times \sqrt{1 - \frac{c^2}{c^2}}}{m_0 v}$

$\therefore \lambda = 0$

81. $\left(\frac{1}{2} mv^2_{max}\right)_1 = (hf)_1 - \phi = 1 - 0.5$

$\left(\frac{1}{2} mv^2_{max}\right)_2 = (hf)_2 - \phi = 2.5 - 0.5$

$\therefore \frac{(V^2_{max})_1}{(V^2_{max})_2} = \frac{0.5}{2} = \frac{1}{4}$ $\qquad$ $\therefore \frac{(V^2_{max})_1}{(V^2_{max})_2} = \frac{1}{2} = 1 : 2$

82. $m_p = 1\ q_p = e$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\therefore\ \lambda \propto \frac{1}{\sqrt{mq}}$$

$$\therefore\ \frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\infty\, q_\infty}{m_p\, q_p}}$$

$$= \sqrt{\frac{4 \times 2e}{1 \times e}}$$

$$= \sqrt{8}$$

$$= 2\sqrt{2}$$

$$\therefore\ \frac{\lambda_p}{\lambda_\alpha} = 2\sqrt{2} : 1$$

83. $\Delta x \cdot \Delta p = \frac{h}{2\pi} = \hbar$

$$\Delta x = \lambda$$

$$\Delta p = \frac{\hbar}{\lambda}$$

84. $\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 10^4 \times 1.6 \times 10^{-19}}}$

$$= 0.12 \times 10^{-10}\ m$$

$$\lambda = 0.12\,\text{Å}$$

85. $p = \sqrt{2mK}$

$$\therefore\ \frac{h}{\lambda} = \sqrt{2mK} \qquad \therefore\ K = \frac{h^2}{\lambda^2 \times 2m}$$

$$\therefore\ K = \frac{1}{\lambda^2} \qquad \therefore\ \frac{K_A}{K_B} = \left(\frac{\lambda_B}{\lambda_A}\right)^2 = \left(\frac{\lambda_B}{2\lambda_B}\right)^2$$

$$\therefore\ 2K_A = \frac{K_B}{2} \qquad \therefore\ K_A < \frac{K_B}{2}$$

86. $\dfrac{\text{K.E. of partical}}{\text{K.E. of photon}} = \dfrac{\frac{1}{2}mv^2}{hf} = \dfrac{\frac{1}{2}mv.v}{\frac{hc}{\lambda}} = \dfrac{\frac{1}{2}Pv\lambda}{hc}$

$= \dfrac{\frac{1}{2}\frac{h}{\lambda}\lambda v}{hc} = \dfrac{v}{2c} = \dfrac{2.25\times10^8}{2\times3\times10^8} = \dfrac{3}{8}$

88. $\dfrac{1}{2}mv_m^2 = \dfrac{hc}{\lambda} - \phi$

$= 2 - \phi$ (eV) ...(1)

ಪುನಃ, $\dfrac{1}{2}m4v_m{}^2 = \dfrac{hc}{\frac{3}{4}\lambda} - \phi$

$= \dfrac{4}{3}\times2 - \phi = \dfrac{8}{3} - \phi$

$\therefore\ 4\left(\dfrac{1}{2}mv_m{}^2\right) = \dfrac{8}{3} - \phi$

90. final friquency $\dfrac{1.5}{2}f_0 = 0.75\ f_0 < f_0$

therefore photoelectry current is zero

91. $m = \dfrac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} = \dfrac{m_0}{\sqrt{1-\frac{0.64\ \times\ 9\ \times10^{16}}{9\times10^{16}}}}$

$= \dfrac{m_0}{\sqrt{0.36}} = \dfrac{m_0}{0.6} = \dfrac{10m_0}{6} = \dfrac{5m_0}{3}$

94. $\dfrac{1}{2}mv^2{}_{max} = \dfrac{hc}{\lambda} - \phi$

$= \left(\dfrac{6.62\times10^{-34}\times3.0\times10^8}{3.0\times10^{-7}\times1.6\times10^{-19}} - 1\right)$ eV

$= 3.1375\,\text{eV} = 5.02\times10h - 19$ J

$\therefore\ V_{max} = \sqrt{\dfrac{2\times5.0\times2\times10^{-19}}{9.1\times10^{-31}}}$ $\left(\because m = 9.1\times10^{-31}\ \text{kg}\right)$

$= 1.05\times10^6 = 1\times10^6\ ms^{-1}$ J

95. $\phi = hf_0$

$= \frac{hc}{\lambda_0} J = \frac{hc}{\lambda_0 e} eV$

$\therefore \lambda_0 \propto \frac{1}{0}$

$\therefore \frac{\lambda_0(\text{tungsten})}{\lambda_0(\text{sodium})} = \frac{\varphi(\text{tungsten})}{\varphi(\text{sodium})} = \frac{2.3}{4.5}$

$\therefore \lambda_0 \text{tungsten} = \frac{2.3}{4.5} \times 5460 \overset{o}{A}$

$= 2790.6 = 2791 \overset{o}{A}$

100. $\frac{25}{16} = \frac{l_1}{l_2} = \frac{A_1^2}{A_2^2}$

$\therefore \frac{A_1}{A_2} = \frac{5}{4}$

101. $E = hf = h\frac{C}{\lambda}$

$= \frac{6.62 \times 10^{-34} \times 3.0 \times 10^8}{6.0 \times 10^{-7}}$

$= 3.31 \times 10^{-19} J$

$= \frac{3.31 \times 10^{-19}}{1.6 \times 10^{-19}} eV$

$= 2.06 eV$

103. $hf_0 = \phi$

$\therefore f_0 = \frac{\phi}{h} = \frac{5.3 \times 10^{-19} \times 1.6}{6.62 \times 10^{-34}} Hz$

$= 1.3 \times 10^{15} Hz$

104. $eV_0 = \frac{1}{2} mv^2{}_{max} = 3.8 eV$

$\therefore V_0 = 3.8 V$

$\frac{1}{2} mv^2{}_{max} = eV_0 = hf - \phi = hf - hf_0 = h\left(\frac{c}{\lambda} - \frac{c}{\lambda_0}\right)$

111. $\Delta x = 2r = 2 \times 10^{-15}$ m

$$\Delta x \cdot \Delta p \approx \frac{h}{2\pi}$$

$$\Delta p \approx \frac{h}{2\pi \Delta x}$$

$$= \frac{66 \times 10^{-34}}{2 \times 3.14 \times 2 \times 10h - 15} = 0.5255 \times 10^{-19}$$

$$E = \frac{p^2}{2m} \quad P \approx \Delta p$$

$$= \frac{\left(0.5255 \times 10^{-19}\right)^2}{2 \times 9.1 \times 10^{-31}} J = \frac{\left(0.5255 \times 10^{-19}\right)^2}{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}}$$

$E = 9.48 \times 10^3$ MeV

112. 3.56 μA

$$P = \frac{E}{t}$$

$$P = \frac{n_1 hc}{\lambda} \cdot t$$

$$\therefore \quad n_1 = \frac{p\lambda}{hct} \quad = \frac{2 \times 10^{-9} \times 44 \times 10^{-8}}{6.6 \times 10^{-34} \times 3 \times 10^8 \times 1} = 4.44 \times 10^9$$

$$n = n_1 \text{ of } 0.5\% = 4.4 \times 10^9 \times \frac{0.5}{100}$$

$I = ne = 2.22 \times 10h7 \times 1.6 \times 10h - 19 = 3.552$

$I = 3.56 \times 10^{-6}$ μA

113. The deflection suffered bu chared particle in an electric field is

$$y = \frac{qLED}{mu^2} = \frac{qLED}{\frac{P^2}{m}} \qquad [p = mu]$$

$$\Rightarrow y \propto \frac{qm}{p^2} \Rightarrow y_p : y_d : y_a = \frac{q_p m_p}{p_p^2} : \frac{q_d m_d}{p_d^2}$$

$$\frac{q_a m_a}{p_a^2}$$

Since $= p_a = p_d = p_p$ [Given]

$mp : md : ma = 1 : 2 : 4$ and $q_P : q_d : q_a = 1 : 1 : 2$

$y_p : y_d : y_a = 1 \times 1 : 1 \times 2 : 2 \times 4 = 1 : 2 : 8$

114. As we know in Young's double slit experiment fringe width = separation between two consecutive fringe or dark fringes

$$\beta = \frac{\lambda D}{d}$$

$$\beta = 2y \Rightarrow 2y = \frac{\lambda D}{d}$$

Here, $\lambda = \frac{2yd}{D}$

$$\lambda = \frac{2 \times 1 \times 10^{-3} \times 0.24 \times 10^{-3}}{1.2}$$

$= 4\ \ 10^{-7}\ m = 4000\ \mathring{A}$

Energy of light incident on photo plate

$$E\ (eV) = \frac{12375}{4000} = 3.1eV$$

According to Einstein's photoelectric equation

$$E = W_0 + eV_0 \Rightarrow V_0 = \frac{(E - W_0)}{e}$$

$$= \frac{(3 - 2.22)}{e}\ eV = 0.9V$$

115. $E_{K1} - E_{K2} = hc\left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2}\right) = \frac{hc\,(\lambda_2 - \lambda_1)}{\lambda_1\lambda_2}$

$\therefore\ E_{K1} - E_{K2}$

$$= \frac{(6.62 \times 10^{-34}) \times (3 \times 10^8)(5000 - 2500) \times 10^{-10}}{(5000 \times 10^{-10}) \times (25000) \times 10^{-10}}$$

$= 3.96 \times 10^{-19}$ J.

120. $\phi_0 = hf$

$$\therefore\ f_0 = \frac{\phi_0}{h} = \frac{3.3 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}} \qquad \therefore\ f_0 = 0.8 \times 10^{15} \qquad \therefore\ f_0 = 8.0 \times 10^{14}\ Hz$$

121. $\lambda = \dfrac{12.27}{\sqrt{v}}$

$= \dfrac{12.27}{\sqrt{64}} = \dfrac{12.27}{8} = 1.533$

$\therefore \ \lambda = 1.534 \ \mathring{A}$

122. $K = \dfrac{P^2}{2m}$

$\therefore \ p = \sqrt{2Km}$

$\lambda = \dfrac{h}{p} = \dfrac{h}{\sqrt{2Km}}$

$\therefore \ \lambda \propto \dfrac{1}{\sqrt{m}}$

$\therefore \ \dfrac{\lambda_\alpha}{\lambda_p} = \sqrt{\dfrac{m_p}{m_x}} = \sqrt{\dfrac{1}{4}}$

$\therefore \ \dfrac{\lambda_\alpha}{\lambda_p} = \dfrac{1}{2}$

123. $\lambda = \dfrac{h}{p}$

$\therefore \ h_\alpha = \dfrac{h}{p_\alpha} \ \& \ \lambda_p = \dfrac{h}{p_p}$

$\therefore \ \dfrac{\lambda_\alpha}{\lambda_p} = \dfrac{p_p}{p_\alpha}$ but $P_p = P_\alpha$

125. $K = vq$

$\dfrac{1}{2} mv^2 = Vq, \ m^2v^2 = 2Vqm$

$\therefore \ mv = \sqrt{2Vqm}$

$\lambda = \dfrac{h}{mv} = \dfrac{h}{\sqrt{2Vqm}}$

$\therefore \lambda \propto \frac{1}{\sqrt{qm}}$

$\therefore \frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{q_p\, m_p}{q_\alpha\, m_\alpha}} = \sqrt{\frac{em_p}{2e \times 4m_p}} = \sqrt{\frac{1}{8}}$ $\qquad \therefore \frac{\lambda_\alpha}{\lambda_p} = \frac{1}{2\sqrt{2}}$

131. K.E. $E = \frac{1}{2} mv^2$ or $mv = \sqrt{2mE}$

de-Broglie wavelength, $\lambda = \frac{h}{mv} - \frac{h}{\sqrt{2mE}}$

and momentum, $p = mv = \sqrt{2mE}$

$\therefore \lambda \alpha \frac{h}{\sqrt{mE}}$

$p \alpha \sqrt{mE}$

(A) When $E_a = E_p$ then

$\frac{\lambda_a}{\lambda_p} = \sqrt{\frac{m_p}{m_a}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$ or $\lambda_p > \lambda_a$

$\frac{P_a}{P_p} = \sqrt{\frac{m_a}{m_p}} = \sqrt{\frac{4}{1}} = 2$; So $P_a > P_p$

(B) When $E_a = \frac{1}{4} E_p$,

then $\frac{\lambda_a}{\lambda_p} = \sqrt{\frac{m_p E_p}{m_a\, E_a}} = \sqrt{\frac{1 \times E_p}{4 \times \left(\frac{E_p}{4}\right)}} = 1$

$\frac{P_a}{P_p} = \sqrt{\frac{m_a E_a}{m_p\, E_p}} = \sqrt{\frac{4 \times \frac{E_p}{4}}{1 \times E_p}} = 1$

(C) $E_a = 4E_p$, $\lambda_p > \lambda_a$ & $P_a > P_p$

(D) $v_a = v_p$, $\frac{E_a}{E_p} = \frac{\frac{1}{2} m_a v_a^2}{\frac{1}{2} m_p v_p^2} = \frac{m_a}{m_p} = \frac{4}{1}$; $Ea = 4E_p$

Hence $\lambda_p > \lambda_a$ & $P_a > P_p$

133. $E_{\lambda 1} = 55\text{nm} = \frac{1240}{550}\text{ eV} = 2.25\text{eV}$

$E_{\lambda 2} = 450\,\text{nm} = \frac{1240}{450}\text{eV} = 2.8\text{eV}$

$E_{\lambda 3} = 350\text{nm} = \frac{1240}{350}\text{ eV} = 3.5\text{eV}$

For metal r, λ_3 photoelectron

For metal λ_2 and λ_3 photoelectron

For metal p; all wavelength are able to generate photoelectron.

Hence photoelectric current will be maximum for *p* and least for *r*.

135. $\frac{hv_2}{e} - \frac{w_0}{e} > \frac{hv_1}{e} - \frac{w_0}{e}$ From the graph $v_2 > v_1$

$\Rightarrow \lambda_1 > \lambda_2 \left(\text{as } \lambda\ \alpha\ \frac{1}{v}\right)$

136. $\Rightarrow I \propto \frac{1}{d^2}$ and photo current $i \propto I \Rightarrow i\ \alpha\ \frac{1}{d^2}$

137. $\because\ x\ \alpha\ \frac{1}{d^2}$ The ion whose deflection is less, its velocity will be more, From the curve

$x_1 < x_2 < x_3 < x_4$, therefore $v_1 > v_2 > v_3 > v_4$

142. $I_1 > I_2$ (given) $i_1 > i_2$ $(\because i\ \alpha\ I)$ and stopping potential does not depend upon intensity So its value will be same (V_O).

• • •

Unit - 18
Atom and Nucleus

SUMMARY

1 The distance of the closest approach of an α - particle is given by

$\gamma o = \frac{1}{4\pi E_{o.}} \frac{2Ze^2}{E_K}$ where $E_k = \frac{1}{2} m\upsilon^2$ 9 Z = Atomic number

2. The Impact parameter is given by $b = \frac{1}{4\pi . E_o} \frac{2Ze^2 Cot\left(\theta/2\right)}{m\upsilon^2}$

3 No - of Scattering of α - particle reattered at angle θ is given by

$N = N_o \; ns \; \frac{Ze^2}{m\upsilon^2} \; \frac{1}{\sin^4\left(\theta/2\right)}$

where, No = number of incident α - particles

n = no. of Atoms per unit Volume in the foil

s = thickness of foil

v = speed of α - particle,m = mass of α particle

z = Atomic number of (the element of foil)

e = charge of electron.

4 Angular momentum of electron in Bohr orbit

$mvr = \frac{nh}{2\pi}$, whre n is an integer

5 r_n = radius of nth bohr orbit = $\frac{n^2 h^2 \varepsilon o}{\pi Z e^2 m}$

$r_n \alpha \frac{n^2}{Z}$

6 Energy of electron in n^{th} orbit is

$E_n = -\left(\frac{me^4}{8h^2 \varepsilon_o^{\;2}}\right) \frac{z^2}{n^2} = -13.6 \frac{z^2}{n^2}$ ev $\qquad E_n \alpha \frac{Z^2}{n^2}$

7. different spetral Series Formula for and their gegionls

(i) Lyman series, $\frac{1}{\lambda} = R\left[\frac{1}{1^2} - \frac{1}{n^2}\right]$ where, $n = 2, 3....$

(ultra violet region.)

ii) Balmer series, , $\frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$ where, $n = 3, 4$,.....

(Visible region.)

(iii) Paschen series, , $\frac{1}{\lambda} = R\left[\frac{1}{3^2} - \frac{1}{n^2}\right]$ where, $n = 4, 5$,.....

(Infrared region.)

Braket series, $\frac{1}{\lambda} = R\left[\frac{1}{4^2} - \frac{1}{n^2}\right]$ where, $n = 5, 6$,.....

(Infrared region.)

Pfund series, $\frac{1}{\lambda} = R\left[\frac{1}{5^2} - \frac{1}{n^2}\right]$ where, $n = 6, 7$,.....

(Infrared region.)

8 For, hydrogen atom energy of electron in n^{th} orbit

$E_n = -\frac{13.6}{n^2} eV$

For hydrogen like atom $E_n = -\frac{13.6}{n^2} z^2$ eV

9 ${}^{A}_{Z}X$ or ${}_{Z}X^{A}$

Z = Atomic number = Number of Protons

A = mass number = Number of neucleons

A - Z = N = Number of neutrons

(i) For Isotope nuclei

Z equal and A unequal

e.g. ${}_6c^{12}$ ${}_6c^{13}$ ${}_6c^{14}$

(ii) For, Isotone nuclei

Z unequal, A unqual

A-Z = N = equal

No - of neutrons are equal

e.g. $36^{kr^{86}}$,$37^{Rb^{87}}$

(iii) For, Isobar nuclei

z unequal , A equal

(A-Z) are unequal

e-g ${}_5B^{12}$,${}_6C^{12}$

(iv) For Isomers nucler

Z equal, A equal

But their radio active properies are different

e.g. $_{35}Br^{80}$ pair

10 Radius of nucleus R = Ro (A) $\frac{1}{3}$

Where R_o = Constant

= 1.1 fm to 1.2 fm

A = mass number

11 Mass defect $\Delta m = (zm_p + Nmn) - M$

12 B.E $E_b = (\Delta m)c^2 = \{Zmp + Nmn) - M\}c^2$

13 Unit of energy in nuclear Physics or atomic Physics is eV

$1eV = 1.6 \times 10^{-19} J$

14 unit of mass in nuclear physics or atomic Physics is amu or u

$1amu = 1u = 1.67 \times 10^{-27} Kg$

$1u = 931.48 MeV = 931.48 \times 10^6 eV$

15 B.E Per nucleon is $E_{b_n} = \frac{E_b}{A}$

16 α - Particle = Nucleus of Helium atom

Symbol of α - Particle $_2He^4$

β - Paricle = electron emitted From nucleus i^{eo}

γ - radiation = photon. Symbol γ

β - particle $\rightarrow$ β^- or $_{-1}e^{o}$ o*r* electron

β^+ Particle $\rightarrow$ β^+ or $_{+1}e^{o}$ o*r* positron.

$n \rightarrow p + e^- + \bar{\upsilon}$

$\rho \rightarrow n + e^+ + \upsilon$

Where $\bar{\nu}$ Anti neutrino

ν neutrino

(17) unit of Radio activity

1 Bq = 1 disinteglation/s

1 curie = 3.7 $\times$ 10^{10} disinteglation/s

(18) Activity $I = \frac{dn}{dt} = -\lambda N$

number of Atoms or nucler in a Radio active sample after time is given by

$N = No\, e^{-\lambda t}$ Where λ is radio active decay Constant

Activity $I = Io\, e^{-\lambda t}$

Mass $m = mo\, e^{-\lambda t}$

(19) Half life $T_{\frac{1}{2}} = \frac{0.693}{\lambda}$

Mean life $T = 1/\lambda$ $T_{\frac{1}{2}} = 0.693T$ Or $T = 1.44\, T_{1/2}$

(20) In α - deccay Z decreasc by 2 and mass number A decrease by 4

In β - deccay Z increase by 1 and mass number does not change

In β^+ deccay Z decrease by 1 and mass number does not change

In emission of r - ray or photon, Z and A does not change.

(21) Nuclear Fission.

The spiliting of a heavy nucleus into two or more fragments or moderate and comparable sizes is called nuclear Fission

e.g $_{q2}U^{235} \rightarrow_{56} Ba^{141} +_{36} Kr^{92} + 3_O n^1 + \text{energy}$

(22) Nuclear fusion.

The Process of nuclear fusion consists in the combination of two light nuclei to form steble nucleus of mass less than the total intial mass.

$_1^1H + _1^1H \rightarrow {}_1^2H + e^+ + \upsilon$

$_1H^1 + {}_1^2H \rightarrow {}_2^3He + \gamma$

$_2^3He + {}_2^3He \rightarrow {}_2^4He +_1 H^1 + {}_1^1H$

If any $Q > 0$ Exoergic

$Q < 0$ Endoergic

In any nuclear reactions consveration of momentum, charge and energy is must.

(23) Nucler reaction A (a,b) B or A+a $\rightarrow$ B + b + Q

A = Target Nucleus

a = Projected Particle

B = resulting (product) nucleus

b = emitted Particle

Q = energy equal muss defect

MCQ

For the answer of the following questions choose the correct alternative from among the given ones.

(1) In each of the following question match column -I and column -II select correct Answer.

(A) Bohr atom model	(P) fixed for the atom
(B) Ionisation potential	(q) Nucleus
(C) Rutherford atom modal	(r) stationary orbits
(D) Thomson atom modal	(s) In atom positive and Nagative charge are distrited uniformely.

(A) $a \rightarrow s$, $b \rightarrow r$, $c \rightarrow q$, $d \rightarrow p$

(B) $a \rightarrow r$, $b \rightarrow p$, $c \rightarrow q$, $d \rightarrow s$

(C) $a \rightarrow p$, $b \rightarrow r$, $c \rightarrow s$, $d \rightarrow q$

(D) $b \rightarrow p$, $c \rightarrow q$, $b \rightarrow r$, $d \rightarrow s$

(2) Read the following question and choose correct Answer form given below.

(A) Both assertion and reason are true. Reason is the correct explanation of the Assertion

(B) BOth assertion and reason are true. Reason is not correct explanation of the assertion

(C) Assertion is true but reason is false.

(D) Both Assertion and Reason are false

(i) Assertion :- In a radio-active disintegration, an election is emitted by nucleus.

Reason :- eleccon are always Present in-side the nucleus.

(ii) Assertion :- An election and Positon can annibilate each other creating Photon

Reason:- Electon and Positon form a Particle and anti Particle.

(iii) Assetion:- An isolated radioactive atom may not decay at all what eVer be its half time

Reason:- Readioactive decay is a statistical Phenomena.

(iv) Assertion :- Fragment Produced in the fission of u^{235} are active

Reason:- The fragments have abnormally high Proton to neutron ratio

(3) Each of the following question contain two stalements.choose correct answer form the given below.

(i) statement-I:- Large angle scattering of al-pha Particle led to discovery of atomic nucleus.

state ment-II :- Entire Positive charge of atom is concentrated in the central core.

(A) statement -I and II are true. and statement II is correct explanation of statement-I

(B) statement -I and II are true, but statement-II is not correct explanation of statement I

(C) statement I is true, but state ment II is false.

(D) statement I is false but statement II is true

(ii) statement-I 1 amu=931.48 MeV

statement-II It follows form $E=mc^2$

(iii) statement -I:-half life time of tritium is 12.5 years

statement-II :- The fraction of tritium that remains after 50 years is 6.25%

(iv) statement-I:- Nuclei of different atoms have same size

state ment-II :- $R=Ro(A)^{1/3}$

(4) Match column I and II and chose correct Answer form the given below.

(a) Nuclear fusion	(p) converts some matter into energy
(b) Nuclear fission	(q) generally Possible for muelei low atomic number
(c) β decay	(r) generally Possible for ndcler high atomic number
(d) Exothermic nuclear reaction	(s) Eessentially Proceeds by weak nuclear for(c)

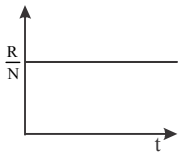
(A) $a \rightarrow p$, $b \rightarrow r$, $c \rightarrow d$, $d \rightarrow q$

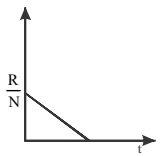
(B) $a \rightarrow d$, $b \rightarrow r$, $c \rightarrow p$, $d \rightarrow a$

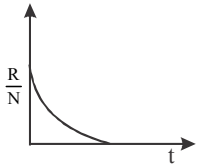
(C) $a \rightarrow q$, $b \rightarrow r$, $c \rightarrow d$, $d \rightarrow p$

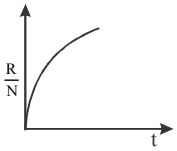
(D) $a \rightarrow r$, $b \rightarrow a$, $c \rightarrow p$, $d \rightarrow a$

(5) A radioative sample has n_o active atom at t=o, at the rate of dissntegration at any time is R and the number of atom is N, then ratio. $\left(\frac{R}{N}\right)$ varies with time (t) as.

(A) 

(B) 

(C) 

(D) 

(6) The transition the state n=4 to n=1 in a hydrogen like atom results in ultraviolet radiation. Infrared radition will be obtained in the transition form

(A) $3 \rightarrow 2$ (B) $5 \rightarrow 4$ (C) $4 \rightarrow 2$ (D) $2 \rightarrow 1$

(7) In Bohr model the hydogen atom, the lowest orbil corresponds to

(A) Intinite energy (B) zero energy

(C) The minimum energy (D) The maximum energy

(8) Energy leVels A , B , C of a certain atom corresponding values of energy i.e $E_A < E_B < E_c$ If $\lambda_1, \lambda_2, \lambda_3$ are wave lengths of radition corresponding to the transition $C \rightarrow B$, $B \rightarrow A$ and $C \rightarrow A$. which of the following is correct

(A) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$ (B) $\lambda_3 = \lambda_1 + \lambda_2$

(C) $\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$ (D) $\lambda_1 + \lambda_2 + \lambda_3 = o$

(9) According to Bohr's theory the radius of electon in an orbit described by Principal quantum number n and atomic number Z, is Proportional to.

(A) $Z^2 n^2$ (B) $\frac{n^2}{Z}$ (C) $\frac{Z^2}{n}$ (D) $\frac{n^2}{z^2}$

(10) The energy of electron in the nth orbit of hydogen atom is expressed as $E_n = -\frac{13.6}{n^2} ev$ The shortest and longest wave length of lyman series will be.

(A) $910 A^o$, $1213 A^o$ (B) $1315 A^\circ$, $1530 A^\circ$

(C) $5463 A^\circ$, $7858 A^\circ$ (D) None of these

(11) Number of spectral lines in hydrogen atom is.

(A) 6 (B) 8 (C) 15 (D) ∞

(12) In the nuclear reaction $X(\eta, \alpha)^7_3 Li$ the atom X will be

(A) $_2He^4$ (B) $_5B^{11}$ (C) $_5B^{10}$ (D) $_5B^9$

(13) The nucleus which has radius one third of the radius of Cs^{189} is

(A) Be^9 (B) F^{19} (C) C^{12} (D) Li^7

(14) A nucleus of $^{210}_{84}Po$ originally at rest emits α-particle with speed v what will be the recoil speed of the daughter nucleus

(A) $\frac{v}{214}$ (B) $\frac{4v}{214}$ (C) $\frac{4v}{206}$ (D) $\frac{v}{206}$

(15) It the radius of a nucleus of mass number 3 is R. then the radius of a nucieus of mass mumber 81 is

(A) 27 R (B) 9 R (C) 3 R (D) $(27)^{\frac{1}{2}}$ R

(16) The nucler of which of following Pairs of nuclei are isotones

(A) $_{34}Se^{74}, _{31}Ca^{71}$ (B) $_{42}Mo^{92}, _{40}Zr^{92}$

(C) $_{38}Sr^{81}, _{38}Sr^{86}$ (D) $_{20}Cd^{40}, _{16}S^{32}$

(17) An electron change its Position from orbit n=4 to the orbit n=2 of an atom the wave length of emitted radition in the form of R (where R is Redburg constanst)

(A) $\frac{16}{7R}$ (B) $\frac{16}{R}$ (C) $\frac{16}{3R}$ (D) $\frac{16}{5R}$

(18) The control rod in a nuclear reactor is made of

(A) uranium (B) Cadmium (C) plutomium (D) grabhite

(19) which of the following isotopes normally fissonable

(A) $_{92}U^{233}$ (B) $_{92}U^{238}$ (C) $_{92}U^{235}$ (D) $_{93}Np^{239}$

(20) which of the following statement is true

(A) $_{78}Pt^{192}$ has 78 neutrons (B) $_{90}Th^{234} \rightarrow _{91}Pa^{234} + _{2}He^{4}$

(C) $_{92}U^{238} \rightarrow _{90}Th^{234} + _{2}He^{4}$ (D) $_{84}Po^{214} \rightarrow _{82}Pb^{210} + \beta^{-}$

(21) A Free neutron decays into a Proton, an electron and

(A) υ (B) $\overline{\upsilon}$ (C) β (D) α

(22) The distance of the closest abbroach of an alpha particle fired at a nuclous with kinctic energy K is ro. The distemce of the closest approach when thw α-particle is fired at the same nucleus with kinetic energy 2k will be.

(A) $\frac{r_o}{2}$ (B) $4r_o$ (C) $\frac{r_o}{4}$ (D) $2r_o$

(23) which of the following series in the spectrum of hydrogen alon lies in the visible legion of the electro magnetic spectrum?

(A) Paschen (B) Lyman (C) Brakett (D) Balmer

(24) If 13.6 eV energy is required to ionige the hydrogen aton the energy required to remove the electron form n=2 state is

(A) Zero (B) 10.2 eV (C) 6.8 eV (D) 3.4 eV

(25) It N_o is the original mass of the substance of halt lift 5 years, the amount of substance left after 15 years is

(A) $\frac{N_o}{16}$ (B) $\frac{N_o}{8}$ (C) $\frac{N_o}{4}$ (D) $\frac{N_o}{2}$

(26) when u-238 nucleus originalluy at lest decay by emitting an α-particle having a speed u the recoil speed of the resi-dual nucleus is.

(A) $\frac{4u}{238}$ (B) $\frac{-4u}{238}$ (C) $\frac{4u}{234}$ (D) $\frac{-4u}{234}$

(27) At a certain instant, a radioactive sample has a decay rate of 5000 dis-interation Per minute. After 5 minuter the decay rate is 1250 dis-interations Per minute. Then the decay constant is (Per-min)

(A) 0.2 ln 4 (B) 0.4 ln 4 (C) 0.4 ln 2 (D) 0.8 ln 2

(28) A nucleus with Z=92 emits the following sequence

$\alpha,\alpha,\beta^-,\beta^-,\alpha,\alpha,\alpha,\alpha,\beta^-,\beta^-,\alpha,\beta^+,\beta^+,\alpha$ The Z of the resulting nucleus is

(A) 76 (B) 78 (C)74 (D) 82

(29) which of the following can not be emitted in radioactive decay of the substance?

(A) Helium-nucleus (B) Electrons (C) Neutrions (D) Proton.

(30) It the radius of $^{27}_{13}Al$ nucleus is 3.6 fm the radius of $^{125}_{52}Te$ nucleus is nearly equal to

(A) 8 fm (B) 6 fm (C) 4 fm (D) 5 fm

(31) which of the following atom has the lowest ionization potentical?

(A) $^{14}_{7}N$ (B) $^{40}_{18}Ar$ (C) $^{133}_{55}cs$ (D) $^{16}_{8}O$

(32) If the binding energy of electron in a hydrogen atom is 13.6 eV, the energy required to remove the elecron form the first state of Li^{2+} is.

(A) 13.6 eV (B) 30.6 eV (C) 122.4 eV (D) 3.4 eV

(33) The ionigation Potential of hydrogen atom is 13.6 eV. An electron in the ground state absords Photon of energy 12.75 eV. How many dirrerent spectral lines can one expect when electron make a down ward transition

(A) 1 (B) 2 (C) 6 (D) 4

(34) A radio-active nucleus $^{A}_{Z}X$ emits 3α-particles and 2 Positrions. the ratio of number of neuleuons to that of Protons in the final nucleus will be

(A) $\frac{A-Z-8}{Z-4}$ (B) $\frac{A-Z-12}{Z-4}$

(C) $\frac{A-Z-4}{Z-8}$ (D) $\frac{A-Z-4}{Z-2}$

(35) An α-particle of energy $\frac{1}{2}mv^2$ bombards by a heavy nuclear target of charge ze. Then the distance of closet approach for the alpha nucleus will be Probprtional to

(A) $\frac{1}{Z^3}$ (B) $\frac{1}{v^4}$ (C) $\frac{1}{m}$ (D) $\frac{1}{v^2}$

(36) when $^{7}_{3}Li$ nucler are bombarded by Proton and the resultant nuclei are $^{8}_{4}Be$, the emitted particle will be

(A) neutron (B) gamma (C) alpha (D) Beta

(37) starting with a samble of Puer cu-66, $\frac{7}{8}$ of it decays into Zn, 15 minules the left of the samble is

(A) 5 min (B) 7.5 min (C) 10 min (D) 15 min

(38) An α-particle of energy 5 MeV is scattered though 180° by a fixed uranium nucleus. The distance of the closest approach nucleus The distance of the closest approach is of the order of

(A) 10^{-8}cm (B) 10^{-12}cm (C) 10^{-10}cm (D) 10^{-15}cm

(39) The binding energy Per nucleon of deutron $\left(^{2}_{1}H\right)$ and Lielium nucleus $\left(^{4}_{2}He\right)$ is 1.1 MeV and 7.0 MeV.resbectively. If two beutron nucler react to form a single helium nucleus, the energy released is

(A)23.6 MeV (B) 26.9 MeV (C) 13.9 MeV (D) 19.2 MeV

(40) The nucleus at rest disintegrate into two nuclear parts which have their velocities in the ratio 2:1 The ratio of their nuclar sizes will be

(A) $3\sqrt{2}:1$ (B) $1:3\sqrt{2}$ (C) $\sqrt{3}:1$ (D) $1:\sqrt{3}$

(41) A radiation of energy E falls normally on a Pertect reflecting surface. The momentum transterred to the surtace is.

(A) $\frac{E}{c}$ (B) $\frac{2E}{c}$ (C) $\frac{E}{c^2}$ (D) Ec

(42) In the following nuclear fusion reaction

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + {}_{o}n^{1}$$

the repalsive potential energy between the two fusing nucler is 7.7×10^{-14}J The Temperature to which the gas must be heated is nearly (Boltzman constmt K=$1.38\times10^{-23}JK^{-1}$)

(A) 10^{3}K (B) 10^{5}K (C) 10^{7}K (D) 10^{9}K

(43) 9 If the binding energy Per nucleon in $^{7}_{3}Li$ and $^{4}_{2}He$ nucler is 5.6 NeV and 7.06 MeV respectively, then in the reaction $P + {}_{3}Li \rightarrow 2\left(^{4}_{2}He\right)$ (P here retrent Proton) enery of Protpn must be

(A) 1.46 MeV (B) 39.2 MeV (C) 17.28 MeV (D) 39.2 MeV

(44) 9 f mo is the mass of an isotope $_{8}{}^{17}0$, mp and mn are the masses of a Proton and neutron respectively, the binding energy of the isotebe is

(A) $(m_o - 8m_p)c^2$ (B) $(m_o - 8m_p - 9m_n)C^2$

(C) $(m_o - 17m_n)c^2$ (D) $m_o c^2$

(45) In gamma ray emission form a nucleus

(A) there is no change in the proton-number and neutron number

(B) Both the number are changes

(C) only Proton number change

(D) only neutron number change

(46) The half life time of a radidactive elements of x is the same as the mean life of another radioactive element Y. Initially they have same number of atoms, then

(A) y will decay faster thean x

(B) x will decay faster then y

(C) x and y will decay at the same rate at all time

(D) x and y will decay at the same rate intially.

(47) which of the following transition in hydrogen atoms emits Photon of highest frequengy?

(A) n=2 to n=6 (B) n=1 to n=2

(C) n=2 to n=1 (D) n=6 to n=2

(48) An electron Passing through a Potential diffencne of 4.9 v colides with a mercury atom and trnasfer it to the first excited state what is trnasfer it to the first excited state. what is the wave length of Photon corresponding to the franition of mercury atom to its normal state.

(A) $2050\,A^{\circ}$ (B) $2935\,A^{\circ}$ (C) $2525\,A^{\circ}$ (D) $2240\,A^{\circ}$

(49) The binding energy Per nucleon for the Parent nucleus is E_1 and that for the daughter nuclei is E_2 then

(A) $E_1 < E_2$ (B) $E_1 > E_2$ (C) $E_1 = E_2$ (D) $E_1 = 2E_2$

(50) The speed of daughter nuclei is

(A) $\frac{(\Delta m)c}{M+\Delta m}$ (B) $C\sqrt{\frac{\Delta m}{m}}$ (C) $C\sqrt{\frac{2(\Delta m)}{M}}$ (D) $C\sqrt{\frac{\Delta m}{M+\Delta m}}$

(51) Light form the discharge tube containing hydrogen atom falls on the surface of a Piece of sodium. The kinetic energy of the fastest photo electrons emitted form sodium is 0.73 eV. The work function for sodium is 1.82 eV. Ionigation Potential of hydrogen is 13.6 v and the mass of hydrogen atom is 1.67×10^{-27} kg.

(i) The energy of Photon causing the Photo electeic emission is

(A) 4.08×10^{-19} J (B) 2.912×10^{-19} J

(C) 1.744×10^{-19} J (D) 1.168×10^{-19} J

(ii) The quantum number of the two leVels in the emission of the Photons are

(A) n=1 , n=3 (B) n=2 , n=4 (C) n=1 , n=4 (D) n=3 , n=4

(iv) In this transition change in the angular momentum of electron is (where h is Plank constanst)

(A) $\frac{h}{2\pi}$ (B) $\frac{h}{\pi}$ (C) $\frac{2h}{\pi}$ (D) $\frac{3h}{2\pi}$

(v) The recoil speed of emitting atom caussing that is at lest before the transition is of the order of

(A) $1\ cms^{-1}$ (B) $10^2\ ms^{-1}$ (C) $10^4\ ms^{-1}$ (D) $1\ ms^{-1}$

(52) A and B are two radioactive substane whose half lives are 1 and 2 years respectively. Initially 10 g of A and 1 g of B is taken. The time after which they will have same quantity remaining is

(A) 3.6 years (B) 7 years (C) 6.6 years (D) 5 years

(53) which of these is a fusion reuction

(A) ${}_1^3H + {}_1^2H = {}_2^4He + {}_0n^1$ (B) ${}_7^{12}C \rightarrow {}_6C^{12} + \beta^+ + g$

(C) ${}_{92}U^{238} \rightarrow {}_{82}^{206}Pb + 8\left({}_2^4He\right) + 6\left({}_{1-}e^{\circ}\right)$ (D) None of these

(54) The activity of a sample of a radio- active material is at time t_1 and A_2 at time t_2 (where $t_2 > t_1$) if T its mean life is then

(A) $A_1 t_1 = A_2 t_2$ (B) $A_2 = A_1 e^{(t_1 - t_2)T}$

(C) $A_1 - A_2 = t_2 - t_1$ (D) $A_2 = A_1 e^{\left(\frac{t1}{t2}\right)T}$

(55) In the following disinteglation series

$$_{92}U^{238} \xrightarrow{\alpha} x \xrightarrow{\beta^-} {}_{Z}y^{A}$$

The value of Z and A respectively will be

(A) 90 , 234 (B) 92 , 236 (C) 88 , 234 (D) 91 , 234

(56) gf $_{92}U^{238}$ undergoes sucesively 8 α decays and 6β decays then resulting nucleus is

(A) Pb^{206} (B) Pb^{208} (C) Pb^{214} (D) None of thes

(57) The energy released by the fission of one unanium atom is 200 MeV. The number of fission Per second required to Produce 3.2 w of Power is

(A) 10^{10} (B) 10^{7} (C) 10^{12} (D) 10^{17}

(58) Radio carbon dating is done by estimating in the specimen

(A) the amount of oridinary carbon still present

(B) the radio of the amounts of $^{14}_{6}C + {}_{6}C^{12}$

(C) the amount of radio carbon still Present

(D) None of these

(59) The enery difference between the first two leVels of hydrogen atom is 10.2 eV. what is the corresponding energy difference for a singly ioniged helium atom?

(A) 10.2 eV (B) 81.6 eV (C) 20.4 eV (D) 40.8 eV

(60) The total energy of the elcton in the first excited state of hydrogen is -3.4 eV. what is the kinatic energy of the electron in this state?

(A) 6.8 eV (B) 3.4 eV (C) -3.4 eV (D) -6.8 eV

(61) The wave length of second line of Balmer series is 486.4 nm. what is the wave length of the first line of lyman saries ?

(A) 364.8 *nm* (B) 729.6 *nm* (C) 121.6 *nm* (D) None of these

(62) The innermost orbit of the hydrogen atom has a radius 0.53 A. what is radius of 2nd orbit is ?

(A) $2.12A^{\circ}$ (B) $1.06A^{\circ}$ (C) $21.2A^{\circ}$ (D) $10.6A^{\circ}$

(63) If a hydrogen atom emits a Photon of wave length, the recoil speed of the atom of mass m is given by

(A) $\frac{h}{m\lambda}$ (B) $\frac{mh}{\lambda}$ (C) $mh\lambda$ (D) $\frac{m\lambda}{h}$

(64) A freshly PrePared radio active source of half time 2 h emits raditation of intenisity which is 64 times the Permissible safe leVel. The minimum time after which is would be possible to work safely with this source is.

(A) 6 h (B) 24 h (C) 12 h (D) 36 h

(65) In Ruthertord experiment, the numer of Particles scttered at 90° angle are 28 Per min. then the number of Particles at the angle 1200 in Per min will be

(A) 25 (B) 12.0 (C) 50 (D) 112

(66) The Rutherford reVolution Per second made by an electron in the first Bohr orbit of hychogen atom is of the order of

(A) 10^{15} (B) 10^{20} (C) 10^{10} (D) 10^{19}

(67) The half time of a radioafive substance is 20 min, difference between Points of time when it is 33% disintegeated and 67% dissintenated is aPProximately

(A) 10 min (B) 20 min (C) 40 min (D) 30 min

(68) The size of the atom is of the order of

(A) $10^{-14}m$ (B) $10^{-10}m$ (C) $10^{-8}m$ (D) $10^{-6}m$

(69) The size of the nucleus is of the order of

(A) $10^{-10}m$ (B) $10^{-14}m$ (C) $10^{-19}m$ (D) $10^{-3}m$

(70) The ratio of atomic volume of nuclear volume is of the order of

(A) 10^{-15} (B) 10^{-10} (C) 10^{15} (D) 10^{-10}

(71) Nucleon is common name for

(A) electron and neutron (B) Proton and neutron

(C) neutron and Positron (D) neufrom and neurtino

(72) The nucler $_7N^{14}$ and $_6C^{13}$ can be desribed as

(A) Isotones (B) Isobars (C) Isotope (D) Isomer

(73) Plutoniumdecays with half life time 24000 yer. if Plutonium is stored after 72000 yes, the fraction of it that remain

(A) $\frac{1}{2}$ (B) $\frac{1}{9}$ (C) $\frac{1}{12}$ (D) $\frac{1}{8}$

(74) The radio of minimum to maximum wave length in Balmer series is

(A) $\frac{1}{4}$ (B) $\frac{5}{36}$ (C) $\frac{3}{4}$ (D) $\frac{5}{9}$

(75) If the binding energy of electron in a hydrogen atom is $13.6\ eV$, the energy required to remove the electron from the second excited state of Li^{++} is

(A) $13.6\ eV$ (B) $3.4\ eV$ (C) $30.6\ eV$ (D) $122.4\ eV$

(76) A radioactive substance decays to $\frac{1}{16}$ th of its initial activity in 40 days. the half life of the radioactive substance expressed in day is

(A) 20 (B) 5 (C) 10 (D) 4

(77) Complete the reaction ${}_0n^1 + {}_{92}U^{235} \rightarrow {}_{56}Ba^{144+} + 3({}_0n^1)$

(A) ${}_{36}Kr^{90}$ (B) ${}_{36}Kr^{89}$ (C) ${}_{36}Kr^{91}$ (D) ${}_{36}Kr^{92}$

(78) ${}_6C^{12}$ absorbs an energetic neutron and emits a β Partical. The resulting nucleus is

(A) ${}_7N^{13}$ (B) ${}_7N^{14}$ (C) ${}_6C^{13}$ (D) ${}_6C^{12}$

(79) two deutrons each of mass m fuse to form helium resulting in release of energy E the mass of helium formed is

(A) $m + \frac{E}{c^2}$ (B) $\frac{E}{mc^2}$ (C) $2m - \frac{E}{c^2}$ (D) $2m + \frac{E}{C^2}$

(80) what Percent of original radio active substans is left after 5 half life time

(A) 3 % (B) 5 % (C) 6% (D) 12 %

(81) In which region of eletromagnetic spectum does the Lyman series of hydrogen atom like

(A) x-ray (B) Intrared (C) visible (D) ultraviolet

(82) In terms of Rydergi constent R. The wave number of first Balmer line is

(A) $\frac{5R}{36}$ (B) $\frac{8R}{9}$ (C) R (D) $\frac{8R}{20}$

(83) The hydrogen atom can give specral lines in the series Lyman, Balmer and Paschen. which of the following statement is correct

(A) Lyman series is in the intrared region

(B) Balmer series is in the ultravioet region

(C) Balmer series is in the visible region

(D) Paschen series is in the visible region.

(84) The ionigation energy of hydrogen atom is 13.6 eV. The ionigation energy of helium atom would be

(A) 27.2 eV (B) 13.6 eV (C) 54.4 eV (D) 6.8 eV

(85) A gamma ray Photon creates an electon- Positon Pair. If the rest mass energy of an electron is 0.5 MeV. and the total kinetrc energy 0.7 MeV, then the energy of the gamma ray Photon must be

(A) 3.9 MeV (B) 1.78 MeV (C) 0.78 MeV (D) 0.16 MeV

(86) The masses of netron and Proton are 1.0087 amu and 1.0073 amu respectively. It the neufron and Protons combins to form binding energy of the helium nucleus will be

(A) 14.2 MeV (B) 28.4 MeV (C) 27.3 MeV (D) 20.8 MeV

(87) Large angle scattering of α - particle could not be explained by

(A) Thomson modal

(B) Ruther Rutherford and Thpmoson modal

(C) Both rutherford and Thomoson modal

(D) Neither Ruthorford modal norThomson modal

(88) The energy of an eleforn in n^{th} orbit of hychogen is $\frac{-13.6}{n^2}$ eV. energy required to exite the electron form the first orbit 4th orbit is

(A) 13.6 (B)3.4 eV (C) 0.85 eV (D) 1.5 eV

(89) The activity of a radioactive sample is measured as no conuts Per minute at t=o and No counts Per minute t=5 min The time (in min) at which activity reduces to half its valus is

(A) $\log e\frac{2}{5}$ (B) $5\log_{10}2$ (C) $5\log_e 2$ (D) $\log_{10}\frac{2}{5}$

(90) A heavy nuecleus at lest breaks into two frigments which fly off with velocities in the ratio 8:1 The ratio of radil of the frigments is

(A) 1:2 (B) 4:1 (C)1:4 (D) 2:1

(91) The binding energy Per nucleon of ${}_8O^{16}$ is 7.97 MeV and that of ${}_8O^{17}$ is 7.75 MeV The energy (in-MeV) required to remove a neutrom from ${}_8O^{17}$ is

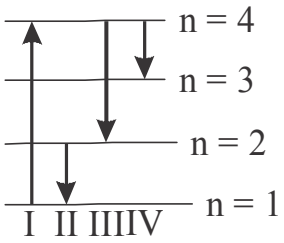
(A) 3.65 (B) 7.86 (C) 3.52 (D) 4.23

(92) The shape of the graph $\ln I \rightarrow t$ is

(A) stright Line (B) Parabolic curve

(C) Hyberbole curve (D) random shabe curve

(93) The fig shows an energy leVes for the electron in a certain aton which transition in a represents the emission of a Photon with most energy?



(A) III (B) Iv (C)I (D) II

(94) The Probability of survival of a radioactive nucleus for one mean life time is

(A) $1-\frac{1}{e}s$ (B) $\frac{1}{e}$ (C) $\frac{2}{e}$ (D) $\frac{3}{e}$

(95) As the electron in Bohr is orbit of hydrogen atom Passes from state n=2 to n=1, the K.E. and Potential energy changes as

(A) Two fold , also two fold

(B) four fold , two fold

(C) four fold , also four fold

(D) two fold , four fold

(96) In the nuclear decay below ${}_{Z}X^{A} \rightarrow {}_{Z+1}Y^{A} \rightarrow {}_{Z-1}B^{A-4} \rightarrow {}_{Z-1}B^{A-4}$

(A) β,α,γ (B) α,β,γ (C) γ,β,α (D) β,γ,α

(97) The wave lenght of the first line of Lyman series for hydrogen atom is equal to that of hydrogen atom is equal to that of second line of Balmar series for a hydrogen like ion. The atomic number Z of hydrogen like ion is

(A) 1 (B) 2 (C) 3 (D) 4

(98) A nucleus ${}_{n}x^{m}$ emist one α-Particle and two $\beta-$ Particle. The resulting nucleus is

(A) ${}_{n-2}Y^{m-4}$ (B) ${}_{n}Y^{m-6}$ (C) ${}_{n}Y^{m-4}$ (D) ${}_{n-4}Y^{m-6}$

(99) Af a certain time, a radio active sample contains 2×10^{20} atoms and disintegration rate is 3×10^{10} atom persec. when 2×10^{15} atoms are Left to decay its disintegation rate will be

(A) $3\times10^{5}\,\text{atom/s}$ (B) $3\times10^{10}\,\text{atom/s}$

(C) $6.6\times10^{1}\,\text{atom/s}$ (D) $2.0\times10^{2}\,\text{atom/s}$

(100) excited hydrogen atom emits a Photon of wave lenght in returninr to the ground state The quantum number n of exilted state is given by

(A) $\sqrt{\frac{\lambda R-1}{\lambda R}}$ (B) $\sqrt{\frac{\lambda R}{\lambda R-1}}$ (C) $\sqrt{\lambda R(R-1)}$ (D) $\lambda R(R-1)$

(101) The radius of Ge nuclide is measured to be twice the radius of ${}^{9}_{4}Be$. The number of nucleons in Ge are

(A) 72 (B) 78 (C)65 (D)80

(102) Match column I and II

column I	column II
(a) sige of nucleus	(P) Z
(b) number of Proton in a nucleus	(q) 10^{-15}m
	(r) (A-Z)
(c) size of Atom	(s) 10^{-10}m
(d) Number of nentrous in a nucles	

(A) a $\rightarrow$ r, b $\rightarrow$ s, c $\rightarrow$ q, d $\rightarrow$ p

(B) a $\rightarrow$ q, b $\rightarrow$ p, c $\rightarrow$ s, d $\rightarrow$ r

(C) a $\rightarrow$ s, b $\rightarrow$ r, c $\rightarrow$ q , d $\rightarrow$ p

(D) b $\rightarrow$ s, c $\rightarrow$ p, c $\rightarrow$ q , d $\rightarrow$ r

KEY NOTE

1	(B) ar, bp, cq, ds	26	A	51	(i) A	73	D	98	C
2	(i) c (ii) b (iii) a (iv) d	27	A		(ii) B	74	D	99	A
3	(i) a (ii) b (iii) c (iv) d	28	B		(iii) B	75	A	100	B
4	(C) aq, br, cs, dp	29	D		(iv) D	76	C	101	A
5	A	31	C	52	C	77	B	102	B
6	B	30	B	53	A	78	A		
7	C	32	B	54	B	79	C		
8	C	33	C	55	D	80	A		
9	B	34	C	56	A	81	D		
10	A	35	C	57	D	82	A		
11	D	36	B	58	C	83	C		
12	D	37	A	59	D	84	C		
13	D	38	B	60	B	85	B		
14	C	39	A	61	C	86	B		
15	C	40	B	62	A	87	A		
16	A	41	B	63	A	88	C		
17	C	42	D	64	B	89	C		
18	B	43	C	65	B	90	A		
19	C	44	B	66	A	91	D		
20	C	45	A	67	B	92	A		
21	B	46	A	68	B	93	D		
22	A	47	C	69	B	94	B		
23	D	48	C	70	C	95	C		
24	D	49	A	71	B	96	A		
25	B	50	C	72	A	97	B		

Hints

(1) * Bohr atom model $\rightarrow$ stationary Orbits

* Ionization Potential $\rightarrow$ Fixed for the atom

* Rutherford atom model $\rightarrow$ Nucleus

* Thomson atom model $\rightarrow$ In atom positive and Negative charge are distaibuted uniformly.

(4) * Nuclear fusion $\rightarrow$ generally Possible for nuclei low atomic number

* Nuclear fission $\rightarrow$ generally Possible for nuclei high atomic number

* β - decay $\rightarrow$ Eessentially Prcoceeds by weak nuclear forces.

* Exothermic nuclear reaction $\rightarrow$ Convets some matter into onergy.

(5) disintegration rate $\frac{dN}{dt} = -\lambda N$

$\therefore R = \lambda N \quad \frac{R}{N} = \lambda =$ Constant

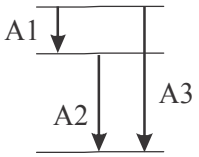
the graph of $\frac{R}{N} \rightarrow t$ will be steaight line.

(6) $5 \rightarrow 4$ transition series of paschen which is fall in infrared region

(7) Ans (C) The minimum energy

$\because$ Orbit energy $E_n = -\frac{13.6}{n^2} = \frac{-13.6}{(1)^2} = -13.6\,ev$

(8) Ans (C) fig From $\frac{hc}{\lambda 1} + \frac{hc}{\lambda 2} = \frac{hc}{\lambda 3}$



$\therefore \frac{1}{\lambda 1} + \frac{1}{\lambda 2} = \frac{1}{\lambda 3}$

$\lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

(9) Bohr radius $r = \frac{n^2 h^2 \varepsilon o}{\pi Z e^2 m}$

$\therefore r \alpha \frac{n^2}{Z}$

(10) For, shortest wave length in Lyman series

$$\frac{1}{\lambda \min} = R\left[\frac{1}{1^2} - \frac{1}{\alpha^2}\right] \Rightarrow \lambda \min = 911\,\mathring{A}$$

For, longest wave rength in Lyman series

$$\frac{1}{\lambda \max} = R\left[\frac{1}{1^2} - \frac{1}{2^2}\right] \Rightarrow \lambda_{max} = 1215\,\mathring{A}$$

(11)

(12) $X(n,\alpha)^{7}_{3}Li$

$^{A}_{Z}X + {}_{o}n^{1} \rightarrow {}^{7}_{3}Li + {}_{2}He^{4}$ By equating Z and A we get

$^{A}_{Z}X \Rightarrow {}^{10}_{5}B$

(13) $R_2 = R_o (A)^{1/3}$

(14) According to conservation of momentum

$$m_1 v_1 + m_2 v_2 = o$$
$$4v_1 + 206\, v_2 = o$$
$$206\ v_2 = -4v_1$$
$$v_2 = \frac{-4v_1}{206}$$

Recoil speed of daughter nucleus is

$$\frac{4v}{206}$$

(15) $R = R_o(3)^{1/3}$ $R^1 = R_0(81)^{1/3}$

$$\frac{R^1}{R} = \left(\frac{81}{3}\right)^{1/3} = (27)^{1/3} = 3$$

$\therefore R^1 = 3R$

(16) Isotones nucler have same neutrons

i.e. A-Z are equal.

From the given pairs $_{34}se^{74}$, $_{31}Ca^{71}$

has same neutruns

(17) $\frac{1}{\lambda} = R\left[\frac{1}{m^2} - \frac{1}{n^2}\right]$

(18)

(19)

(20) ${}_{92}U^{238} \rightarrow {}_{90}Th^{234} + {}_{2}He^{4}$

(21) $n \rightarrow p + e^{-} + \bar{\nu}$

$P \rightarrow n + e^{+} + \nu$

(22) At a distance of closest approach kinetic energy of α- particle = Potentical cnergy of the system

$$\frac{1}{2}mv^2 = K_1 = \frac{Kq_1q_2}{ro}$$

$$\therefore ro = \frac{Kq_1q_2}{K_1}$$

When kinetic energy 219 than

$$ro^1 = \frac{ro}{2}$$

(23)

(24) Inigation potentier (For hydrogen) $E_n = \frac{13.6}{n^2} eV$

(25) $\frac{N}{no} = \left(1/2\right)^{t/T_{\frac{1}{2}}}$

(26) According to conservation of momentum

(27) $N = N_o e^{-\lambda t} \Rightarrow I = I_o e^{-\lambda t}$

$$1250 = 5000\, e^{-\lambda \times 5}$$

$$\frac{1}{4} = e^{-5\lambda}$$

$$4 = e^{-5\lambda}$$

$$\ell n 4 = 5\lambda$$

$$\lambda = \frac{1}{5}\ell n 4 = 0.2\ell n 4$$

(28) Atomic number of final nucleus

= 92 -2 (no.of α - Particle) + 1 (No. of β^- Particle)

- 1(No.of β^+ Particle)

= 92 - 2 $\times$ 8 + 1 $\times$ 4 - 1 $\times$ 2

= 78

(29)

(30) use formula $R = Ro\,(A)^{1/3}$

(31) $^{133}_{55}Cs$ a mucleus having highest Z number, the outer most electron are less in binding with nuclera. That meam lowest binding energy.

(32) Binding energy $= \frac{13.6z^2}{n^2}$

For Li^{2+} Z = 3, n = 2 first exited state

(33) $\Delta E = E_n - E_1$

$E_n = \Delta E + E_1 = 12.75 - 13.6 = -0.85$

$E_n = -\frac{13.6}{n^2}$

$-0.85 = \frac{-13.6}{n^2}$

$n^2 = \frac{13.6}{0.85} = 16 \quad \Rightarrow n = 4$

(34) $^{A}_{Z}X \xrightarrow{3\alpha} {}^{A-12}_{Z-6}X^1 \xrightarrow{2e+} {}^{A-12}_{Z-8}Y$

$\therefore$ Number of newfrons $= (A-12)-(Z-8)$

$= A - 12 - Z + 8$

=A - Z - 4

Number of protom $= Z - 8$

$\frac{\text{No.of neutrons}}{\text{No.of Protons}} = \frac{A-Z-4}{Z-8}$

(35) at distance of closest approach K.E = P.E

$$\frac{1}{2}mv^2 = \frac{1}{4\pi\varepsilon_o}\frac{(ze)(ze)}{ro} \Rightarrow ro = \frac{ze^2}{\pi\varepsilon_o mv^2}$$

$$ro \alpha \, 1/m$$

(36)

(37) $\frac{N}{No} = \left(\frac{1}{2}\right)$ where $n = \frac{t}{T_{\frac{1}{2}}}$

(38)

(39) ${}_1H^2 + {}_1H^2 \rightarrow {}_2He^4$

$\therefore$ B. E of Helium = 4(7)-[2(1.1)+2(1.1)]

= 28-4.4

= 23.6 eV

$\therefore$ energy relewed is 23.6 eV

(40) According to cosveration momentum

$m_1v_1 = m_2v_2$

$$\frac{m_1}{m_2} = \left|\frac{v_2}{v_1}\right| = \frac{1}{2}$$

$$\frac{r_1^3}{r_2^3} = \frac{1}{2}$$

$$\frac{r_1}{r_2} = \left(\frac{1}{3}\right)^{\frac{1}{3}}$$

$\Rightarrow r1 : r2 = 1 : 3\sqrt{2}$

(41) Here, surface is perfacet reflector momentum of incident radiation is E/C momentum of reflected rediation is - E/C change in momentum = $-\frac{E}{c} \quad -\frac{E}{c} =$

$$-\frac{2E}{C}$$

$\therefore$ Momentum transtered to the surface = $+\frac{2E}{C}$

(42) $E = \frac{3}{2}KT$

(43) Energy of proton $({}_1H^1)$ $= 4(7.06)2 - 7(5.6)$

$= 2\times 28.24 - 39.2$

$= 56.48 - 39.2$

$= 17.28$

(44) Ans (B) $B.E = \Delta mc^2 = (mo - 8mp - 9mn)\,C^2$

(45)

(46) $\left(T\,1/2\right)_{\chi} = \frac{0.693}{\lambda x}$ $(T)_Y = \frac{1}{\lambda y}\left(T_{\frac{1}{2}}\right)_x = T_y$

$\frac{0.693}{\lambda x} = 1/\lambda y$ $\qquad \therefore \lambda y = \frac{dx}{0.693}$

$\therefore \left(T_{\frac{1}{2}}\right)_X = (T)y$ $\qquad \lambda y = 1.44\lambda x$

$\frac{0.693}{\lambda x} = \frac{1}{\lambda y}$ $\qquad$ According $N = No\,e^{-\lambda t}$

(47) n = 2 to n = 6 absorbs photon

n = 9 to n = 2 absorbs photon

n = 6 to n = 2 emission of photon

n = 2 to n = 1 emission of photon

$\Delta E = E_2 - E_1 = -\frac{13.6}{36} - \left(-\frac{13.6}{4}\right) = -0.38 + 13.6 = 3.02\,eV.$

$\Delta\sum^1 = E_2 - E_1 = -\frac{13.6}{4} - \left(-\frac{13.6}{1}\right) = -3.4 + 13.6 = 10.2\,ev.$

$\Delta\sum^1 > \Delta E$

$hf' > hf$

(48) energy of electron = ve

$E = 4.9\times 1.6\times 10^{-19}.J$

$\frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$

(49)

(50) $\frac{1}{2}Mv^2 = (\Delta m)C^2$

$V^2 = \frac{2(\Delta m)C^2}{M}$

$V = C\sqrt{\frac{2(\Delta m)}{m}}$

(51) (i) $K_{max} = hf - \varphi$

$hf = K_{mux} + \varphi$

(ii) Correspnding energy leVel for 2.55 eV is n = 2 , n = 4

Ans (B)

(iii) Change Angular momentum = $\frac{4h}{2\pi} - \frac{2h}{2\pi} \Rightarrow \frac{2h}{\pi} - \frac{h}{\pi} \Rightarrow \frac{h}{\pi}$

Ans (B)

(iv) Accoding to conservation of momentum

momentum of photon = momentum of lecoil altom.

$\frac{h}{\lambda} = mu$

$v = \frac{h}{m\lambda} = \frac{hf}{mc} = E/mc$

(52) $\frac{m_1}{mb_1} = \left(\frac{1}{2}\right)^{\frac{t}{T1/2}} = \left(\frac{1}{2}\right)^{\frac{t}{1}} = \frac{1}{2^t}$

$m_1 = \frac{10}{2^+}$ $\quad m_1 = m_2$

$m_2 = \frac{1}{2^{t/2}}$ $\quad \frac{10}{2^t} = \frac{1}{2^{t/2}}$

$10 = (2)^{t/2}$

$\log 10 = t/2 \log 2$

$1.0000 = \frac{t}{2} \times 0.3010$

$t = 6.64$ year $\approx$ 6.6 year

(53) Ans (A) ${}_1^3H + {}_1^2H = {}_2^4He + {}_0n^1$

(54) $A_1 = A_0e^{-t_1/t}$ $\qquad A = A_o^{e-\lambda t} = A = A_0e^{-t/t}$

$$A_0 = \frac{A_1}{e - t_1/T} \Rightarrow$$

$$A_0 = A_1e^{t_1/T}$$

Now, $A_2 = A_0e^{-t_2/T}$

$$= A_1 e^{t_1/T} e^{-t_2/T}$$

$$A_2 = A_1 e(t_1 - t_2)/T$$

(55) ${}_{92}U^{238} \xrightarrow{\alpha} x \xrightarrow{\beta-} {}_ZY^A$

${}_{92}U^{238} \xrightarrow{\alpha} 90^{Th^{234}} \xrightarrow{\beta-} {}_{91}Y^{234}$

(56) $92^{U^{238}} \xrightarrow{8d} {}_{76}X^{206} \xrightarrow{6\beta} {}_{82}Y^{206}$

$\therefore 82^{pb^{206}}$ or $^{pb^{206}}$

(57) Energy released per uranium atom = 200 MeV

$= 200\times10^6 ev = 200\times10^6\times1.6\times10^{-19} = 320\times10^{-13}\,\tau$

For $320x10^{-13}$ $2 \rightarrow 1$ atom

$$32\ J \rightarrow \lambda\ \frac{3.2}{320x10^{-13}} = 10^{11} \text{atoms}$$

(58)

(59) The energy leVel of hydrogen atom corresponding to 10.2 eV is n = 2, n = 1 correponding engery

for the's leVel for helium atom $E = -\frac{13\text{-}6Z^2}{n^2}$

$\Delta E = E_2^1 - E_1^1$

$$\Delta E = -\frac{13.6\times4}{4} - \frac{(-13.6\times4)}{1} = -13.6 + 54.4 = 40.8\,ev$$

(60) For any state $K.E = \frac{1}{8\pi E_o}\frac{Ze^2}{Cn}$ and Cangatomsl

$P.E = -\frac{1}{4\pi\varepsilon o}\ \frac{Ze^2}{rn}$ (Total energy in the stote)

$E = K.E + P.E$

$E = -\frac{1}{8\pi\varepsilon o}\ \frac{Ze2}{ru} = -K.E$

(61) For wave longth of second line of Balmer series

$$\frac{1}{\lambda_{2(B)}} = R\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = R\left[\frac{1}{4} - \frac{1}{16}\right] = \frac{3R}{16}$$

$$\Rightarrow \lambda 2(B) = \frac{16}{3R}$$

For wave length of First line of Luman series

$$\frac{1}{\lambda_{2(L)}} = R\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = R\left[\frac{1}{1} - \frac{1}{4}\right] = \frac{3R}{4}$$

$$\lambda_1(L) = \frac{4}{3R}$$

$$\frac{\lambda_2(B)}{\lambda_1(L)} = \frac{16}{3R} \times \frac{3R}{4} = 4$$

$$\lambda_1(C) = \frac{\lambda_2(B)}{4} = \frac{486.4}{4} = 121.6\,\text{mm.}$$

(62) $r \alpha \frac{n^2}{z}$ For hydrogen atom Z = 1

$r \alpha n^2$

$$\frac{r_1}{r_2} = \left(\frac{1}{4}\right)$$

$$r_2 = 4r_1 = 4 \times 0.53\ \mathring{A} = 2.12\ \mathring{A}$$

(63) According comservation of momentum
momentum of photon = momentum of rocil atom.

$$\frac{h}{\lambda} = m\vartheta$$

$$\therefore \vartheta = \frac{h}{m\lambda}$$

(64)

$$\frac{N}{No} = \left(\frac{1}{2}\right)^{\frac{t}{T\frac{1}{2}}}$$

$$\log 64 = \frac{t}{2}\log 2$$

$$1.8061 = \frac{t}{2} \times 0.3010$$

$$t = \frac{2 \times 1.8061}{0.3010} = 12\text{hrs}$$

(65) Number of scaltering α - particle at angle θ is

$$N = N_o ns\left(\frac{Ze^2}{m\upsilon^2}\right)\frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$$

$$N_1 \alpha \frac{1}{\sin^4\left(\frac{Q}{2}\right)}$$

$$N_2 \alpha \frac{1}{\sin^4\left(\frac{Q}{2}\right)}$$

$$\frac{N_1}{N_2}\frac{\sin^4\left(\frac{Q_1}{2}\right)}{\sin^4\left(\frac{Q_2}{2}\right)} = \frac{\sin^4(45)}{\sin^4(60)} = \frac{\frac{1}{4}}{\frac{9}{16}} = \frac{4}{9}$$

$$N_2 = \frac{4}{9} \times N_1 = \frac{4}{9} \times 28 = 12.44 \approx 12$$

(66) $m\upsilon r = nh/2\pi$

For first Bohr orbit n = 1

$m\vartheta r = h/2\pi$ $\qquad f = \dfrac{h}{4\pi^2 mr^2}$

$mr^2 w = h/2\pi$ $\qquad f = 10^{15}$

$mr^2(2\pi f) = h/2\pi$

(67) $T_1/2 = \dfrac{0.693}{\lambda} \Rightarrow \lambda = \dfrac{0.693}{T_1/2} = \dfrac{0.693}{20} = 0.03465$

$\dfrac{N_1}{N_o} = \left(\dfrac{1}{2}\right)^{t_1/T^{1/2}}$ $\dfrac{N_2}{N_o} = \left(\dfrac{1}{2}\right)^{t_2/T^{1/2}}$

$\dfrac{67}{100} = \left(\dfrac{1}{2}\right)^{\frac{t1}{20}}$ $\dfrac{33}{100} = \left(1/2\right)^{t_2/20}$

$\dfrac{100}{67} = 2^{t_1/20}$

$1.5 = 2^{t_1/20}$

$\log 1.5 = \dfrac{t_1}{20}\log 2$

$0.1761 = \dfrac{t_1}{20} \times 0.3010$

$t_1 = \dfrac{0.1761 \times 20}{0.3010} = 11.7\,\text{min}.$

similary t_2 = 31.7-11.7=20min

$\Delta t = t_2 - t_1 = 31.7 - 11.7 = 20\,\text{min}.$

(68)

(69)

(70)

$$\frac{\text{Volume of atom}}{\text{Volume of nuclear}} = \frac{4\pi\lambda l_1^{\,3}}{\frac{4}{3}\pi\lambda l_2^{\,3}} = \frac{(10^{-10})^3}{(10^{-15})^3} = 10^{15}$$

(71)

(72)

(73)

$$\frac{N}{No}=\left(\frac{1}{2}\right)^{t/T_{1/2}}=\left(\frac{1}{2}\right)\frac{72000}{24000}=\frac{1}{8}$$

$$N=\frac{1}{8}No$$

(74) Use formula $\frac{1}{\lambda}=R\left[\frac{1}{2^2}-\frac{1}{n^2}\right]$

Putiting n = ∞ $\lambda_2=\frac{36}{5R}$

$$\therefore\frac{\lambda 1}{\lambda 2}=\frac{4}{R}\times\frac{5R}{36}=\frac{5}{9}$$

(75) For any atom any state $En=\frac{13.6}{n^2}Z^2$

For Second excited state n = 3 for Li, Z = 3

$$E_2=\frac{13-6\times(3)^2}{(3)^2}=13.6\text{ev}$$

(76)

$$\frac{N}{No}=\left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}}$$

$$\frac{1}{16}=\left(\frac{1}{2}\right)^{40/T_{1/2}}$$

$$16=(2)^{40/T_{1/2}}$$

$$\log 16=\frac{40}{T_{1/2}}\log 2$$

$$1.2041=\frac{40}{T_{1/2}}\times 0.3010$$

$$T_{1/2}=\frac{40\times 0.3010}{1.2041}=9.999\approx 10\,\text{day}$$

(77) A value & Z value

$_{36}kr^{89}$

(78) $_6Cl^2 + _0n^1 \rightarrow _7N^{13} + _{-1}e^o$

(79) $E = (\Delta m)C^2$

$E = (2m - M)C^2$

$\frac{E}{C^2} = 2m - M$

$M = 2m - E/C^2$

(80) at t = o N = No

$t = T_{1/2} \times 1$	$N = No/2$	$\therefore \frac{N}{No} = \frac{1}{16}$
$t = T_{1/2} \times 2$	$N = \frac{No}{4}$	$\frac{N}{No} \times 100 = \frac{1}{32} \times 100$
$t = T_{1/2} \times 3$	$N = \frac{No}{8}$	$= 3.125$
$t = T_{1/2} \times 4$	$N = \frac{N0}{16}$	$\approx 3\%$
$t = T_{1/2} \times 5$	$N = \frac{N0}{32}$	

(81)

(82) Wave number $= \frac{1}{\lambda} = R\left[\frac{1}{2^2} - \frac{1}{3_2}\right] = R\left[\frac{1}{4} - \frac{1}{9}\right] = \frac{5R}{36}$

(83)

(84) Ionigation energy of atom E $= \frac{13.6Z^2}{n^2}$ ev

For helum Z = 2 $= \frac{13.6 \times (2)^2}{(1)^2}$ =54.4 eV

(85) $r \rightarrow e^- + e^+$

The energy of r - ray = Rest mass energy + K. E

= (0.5 + 0.5) + 0.78

= 1.0 + 0.78

= 1.78 eV

(86) Mass degect $\Delta m = (2mp + 2mn) - MHe$

$$\Delta m = (2 \times 1.0087 - 2 \times 1.0073) - 4.0015$$

$$= 4.032 - 4.0015$$

$$\Delta m = 0.030\,amu$$

$$E = \Delta m \times 931.48\,mev$$

$$= 0.0305 \times 931.48 = 28.4Mev$$

(87) Ans (A) Thomson model

(88) Ans (c) 12.75eV

$$\Delta E = E_4 - E_1 = -\frac{13.6}{4} - (-\frac{13.6}{1}) = 0.85 + 13.6 = 12.75\,ev$$

(89) In mean life (T) Rest mass = No/e

$$T = \frac{1}{\lambda} \Rightarrow \lambda = \frac{1}{5}\min^{-1}$$

$$N = No\,e^{-\lambda t}$$

$$\frac{No}{2} = No\,e^{-\frac{1}{5}t} \qquad rn2 = \frac{1}{5}t$$

$$\frac{1}{2} = e^{-}\frac{1}{5}t \qquad t = 5\ln 2$$

$$2^{-1} = e^{-}\frac{1}{5}t \qquad t = 5\log e^{2}$$

$$2^{1} = e\frac{1}{5}t$$

(90) $\frac{v_1}{V_2} = 8:1$ Accotding to conservatron of momentum

$$\frac{v_1}{V_2} = \frac{8}{1} \qquad m_1\vartheta_1 = -m_2\vartheta_2$$

$$\left|\frac{\vartheta_1}{\vartheta_2}\right| = \left|\frac{m_2}{m_1}\right|$$

$$\frac{8}{1} = \frac{\rho V_2}{\rho V_1} = \left(\frac{r_2}{r_1}\right)^3$$

$$\frac{2}{1} = \frac{r_2}{r_1} \therefore \frac{r_1}{r_2} = \frac{1}{2}$$

(91) energy reguired to remove a

neutron $= 17(7.75) - 16(7.97)$

$= 131.75 - 127.52$

$= 4.23\text{ev}$

(92)

(93)

(95)

$\text{kn}\alpha\dfrac{1}{n^2}$ $\qquad$ $\text{un}\alpha\dfrac{1}{n^2}$

$\dfrac{k_1}{k_2} = \dfrac{4}{1}$ $\qquad$ $\dfrac{u_1}{u_2} = \dfrac{4}{1}$

$k1 = 4k_2$ $\qquad$ $u1 = 4u_2$

$\therefore$ K. E. four told $\qquad$ P.E also four told. Ans (c)

(96) ${}_zX^A \xrightarrow{\beta} {}_{z+1}Y^A \xrightarrow{\alpha} {}_{z-1}B^{A-4} \xrightarrow{r} {}_{z-1}B^{A-4}$

Ans (A) β,α,γ

(97) $\dfrac{1}{\lambda} = RZ^2\left[\dfrac{1}{n_2^{\,2}} - \dfrac{1}{n_1^{\,2}}\right] \leftarrow \text{for any atom}$

$$\frac{1}{\lambda_1(L)} = R(1)^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right] = R2^2\left[\frac{1}{4} - \frac{1}{16}\right]\frac{3RZ^2}{4}$$

$$\frac{1}{\lambda_1(L)} = \frac{3R}{4} \Rightarrow \lambda_1(L) = \frac{4}{3R}$$

$$\frac{1}{\lambda_2(B)} = RZ^2\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = R2^2\left[\frac{1}{4} - \frac{1}{16}\right] = \frac{3RZ^2}{16}$$

$$\Rightarrow \lambda_2(B) = \frac{16}{3RZ^2}$$

$$\frac{\lambda_1(L)}{\lambda_2(B)}$$

$$1 = \frac{2^2}{4} \Rightarrow z^2 = 4 \;\; Z = 2$$

(98) ${}_nX^m \xrightarrow{\alpha} {}_{n-2}X^{1\ m-4} \xrightarrow{2\beta-} n\ Y^{m-4}$

(99) $\frac{dN}{df} = dN \Rightarrow I = dN$

$I^1 = \lambda N_1 \qquad I_2 = \lambda N^2$

$\frac{I_1}{N_1} = \lambda \qquad \frac{I_1}{N_2} = \lambda$

$\therefore \frac{I_1}{N_1} = \frac{f_2}{N_2} \Rightarrow I_2 = \frac{I_1}{N_1} \times N_2 = \frac{3\times10^{10}\times2\times10^{15}}{2\times10_{20}} = 3\times10^5$

(100) $\frac{1}{\lambda} R\left[\frac{1}{1^2} - \frac{1}{n^2}\right]$

$\frac{1}{\lambda} = R\left[1 - \frac{1}{n^2}\right]$

$\frac{1}{\lambda} = R - R/n^2$

$\frac{R}{n^2} = R - 1/\lambda = \frac{\lambda R - 1}{\lambda}$

$\frac{R}{n^2} = \frac{\lambda R - 1}{\lambda}$

$n^2 = \frac{\lambda R}{\lambda R - 1}$

$n = \sqrt{\frac{\lambda R}{\lambda R - 1}}$

(101) $R = R_o(a)^{1/3} = 2R = R_0(A)^{1/3}$

$\therefore \frac{1}{2} = \left(\frac{9}{A}\right)^{\frac{1}{3}} \Rightarrow \frac{1}{8} = \frac{9}{A} \Rightarrow \therefore A = 72$

(102)
* Size of nucleus $\rightarrow 10^{-15}$m
* No. of proton in a nucleus $\rightarrow$ Z
* Size of atom $\rightarrow 10^{-10}$m
* No.of neutrin in nucleus $\rightarrow$ (A-Z)

Ans (B) $a \rightarrow q, b \rightarrow P, c \rightarrow s, d \rightarrow r$

Unit - 19
Semiconductor Electronics

SUMMARY

Conductor :-
- Presence of free electrons
- Electrical resistivity is quite less

Insulator :-
- No free electrons
- Very large electrical resistivity

Semi-conductor :-
- Electrical resistance greater than conductor but smallar than insulator
- At ○ K temperature it behaves like perfect insulator (in pure form)

Hole :-
- An empty space, when covalent bond breaks and electron gets escaped.
- It is electron deficiency space called hole
- behaves like positive electric charge.

For intrinsic (pure) semi-conductor :-
- $n_i = n_e = n_h$ where n_i = Intrinsic electrical charge carrier density, n_e = number density of electrons n_h = number density of holes
- Electrical conduction is due to both, electrons and holes

*** Extrinsic semi-conductor :-**

(1) N-type :
- Pentavalent impurity is added
- Majority charge carrier are electrons
- $n_e > n_h$

(2) P-type :
- Trivalent imparity is added
- Majority charge carries are holes
- $n_h > n_e$

*** Valence Band :-** Completely filled (with 4N electrons) lower band is called valence band

*** Forbidden Gap :-** The region above valance band without any available energy levels is called forbidden gap

*** Conduction Band :-** The region above forbidden gap is called conduction band

*** Band Gap (Eg) :-** The difference between minimum energy (Ec) of conduction band and maximum energy (Ev) of valence band is known as band gap energy
- For Insulator Eg > 3eV
- For Conductor Eg = 0
- For semiconductor Eg < 3eV

* **Depletion Region :-**

- A region near the junction which is deplete of respective majority charge carriers.
- Thickness is about $0.5\mu m$

* **Depletion Barrier :-** The varying electrical potential near junction is called depletion barrier ($0.7V$ for Si and $0.3V$ for Ge)

* **Forward Bias :-**

- When P end of PN junction is connected to positive pole of the battery and N end is connected to negative pole of the battery, then such an arrangement is called forward bias.
- Depletion barrier (P.d) and Depletion region (width) is decreased.

* **Reverse Bias :-**

- When P end of PN junction is connected to negative pole of the battery and N end is connected to positive pole of the battery, then such an arrangement is called forward bias.
- Depletion barrier (P.d) and depletion region (width) are increased

* **Brekdown voltage :-** In reverse bias condition of PN junction, for certain voltage, current increases suddenly. This certain voltage is called breakdown voltage.

* **Zener effect :-** Due to smaller width of depletion region even at small reverse bias voltage, electric field becomes strong enough to break covalent bond, giving large number of electron hole pair is called zener effect

* **Avalanche effect :-** Due to large width of depletion region, at only high reverse bias voltage electric field in the depletion region becomes strong enough to break many covalent bonds, giving rise to so many charge carrier is called avalanch effect and diode is called avalanche diode

* **Regulated Power Supply :-** If D.C. output voltage, in a rectifier circuit (or power supply) remains constant with the charge in load current I_L, then such power supply is called reguluted power supply

* **Rectification and Rectifier :-** The process of obtaining D.C. voltage (or current) from A.C. voltage (or current) is called rectification and circuit assembled for this process is called rectifier

* **TRANSISTOR**

- Transistor is a device made of two PN junctions
- Junction between base and emitter is called emitter junction
- Junction between base and collector is called collector junction
- For proper working of transistor, emitter junction should be forward biased and collector junction should be reverse biased
- A.C. parameters for a transistor

(1) Input resistance = $r_i = \left[\frac{\Delta V_{BE}}{\Delta I_B}\right]_{V_{CE}=\text{constant}}$

(2) Output resistance = $r_o = \left[\frac{\Delta V_{CE}}{\Delta I_C}\right]_{I_B = constant}$

(3) A.C. current gain $= A_i = \beta_{ac} = \left[\frac{\Delta Ic}{\Delta I_B}\right]_{V_{CE} = constant}$

(4) Transconductance = $g_m = \frac{\Delta I_C}{\Delta V_{BE}} = \frac{\beta ac}{r_i}$

- D.C. Parameters of a transistor

  (1) $I_E = I_B + I_C$

  (2) Current gain for CB circuit $= \alpha_{dc} = \frac{I_C}{I_E}$

  $\alpha_{dc} < 1$

  (3) Current gain for CE cicuit = $\beta_{dc} = \frac{I_C}{I_B}$

  $\beta_{dc} >> 1$

- Voltage gain for CE Amplifier

$$Av = \frac{\Delta V_{CE}}{\Delta V_{BE}} = -\frac{\beta_{ac}}{r_i} R_L = -g_m R_L$$

- Power gain for CE Amplifier

$$A_p = A_v \times A_i = \beta_{ac}^2 \frac{R_L}{r_i}$$

* Oscillator :- Certain electronic circuits can generated any arbitrary frequency with desired amplitude of voltage and current. Such circuit is known as oscillator.

- Oscillator frequency $= f = \frac{1}{2\pi\sqrt{LC}}$

* Logic gate :-

- The logic circuit, with one or more than one input but only one output is called logic gate.
- Basic logic gates are OR gate, AND gate and NOT gate
- Universal logic gates are NAND gate and NOR gate

* LOGIC GATES

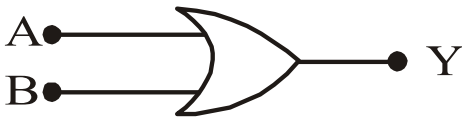
- A digital circuit with one or more input signals but only one output signal is known as logical gate.
- The logic gates are the building blocks of a digital system. Each logic gate follows a certain logical relationship between input and output voltage.
- There are three basic logic gates :
  - OR gate
  - AND gate
  - NOT gate

Truth table

- It is a table that shows all possible input combinations and corresponding output combination for a logic gate.

OR gate

- An OR gate has two or more inputs but only one output.
- It is called OR gate because the output is high if any or all the inputs are high.
- The logic symbol of OR gate is



- The truth table for OR gate is

Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

- The Boolean expression for OR gate is

$$Y = A + B$$

AND gate

- An AND gate has two or more inputs but only one output.
- It is called AND gate because output is high only when all the inputs are high.
- The logic symbol of AND gate is



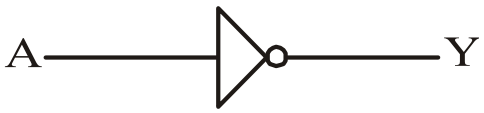
- The truth table for AND gate is

Input		Output
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

- The Boolean expression for AND gate is Y = A.B

NOT gate

- The NOT gate is the simplest of all logic gates. It has only one input and one output.
- NOT gate is also called inverter because it inverts the input.
- The logic symbol of NOT gate is



- The truth table for NOT gate is

Input	Output
A	Y
0	1
1	0

- The Boolean expression for NOT gate is $Y = \overline{A}$

NAND gate

- It is an AND gate followed by a NOT gate.
- The logic symbol for NAND gate is



- The truth table for NAND gate is

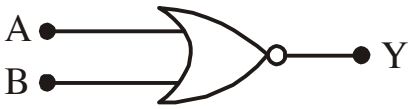
Input		Output
A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

- The Boolean expression for NAND gate is

$$Y = \overline{A \cdot B}$$

NOR gate

- It is an OR gate followed by a NOT gate.
- The logic symbol of NOR gate is



- The truth table for NOR gate is

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

- The Boolean expression for NOR gate is

$$Y = \overline{A + B}$$

Exclusive OR gate or XOR gate

- The logic symbol of XOR gate is



- The truth table for XOR gate is

Input		Output
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

- The Boolean expression for XOR gate is

$$Y = \overline{A} \cdot B + A \cdot \overline{B} = A \oplus B$$

Exlusive NOR gate or XNOR gate

- The logic symbol of XNOR gate is



- The truth table for XNOR gate is

Input		Output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

- The Boolean expression for XNOR gate is

$$Y = A \cdot B + \overline{A} \cdot \overline{B} = \overline{A \oplus B}$$

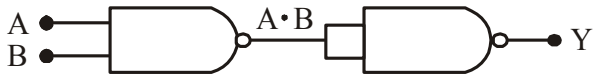
NAND as a universal gate

- NAND gate is called as universal gate because with the repeated use of NAND gate we can construct any basic gate
- NOT gate from NAND gate

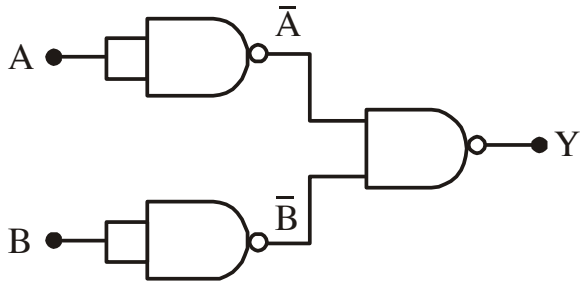


$$Y = \overline{A}$$

- AND gate from NAND gates



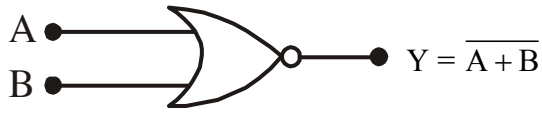
- OR gate from NAND gates



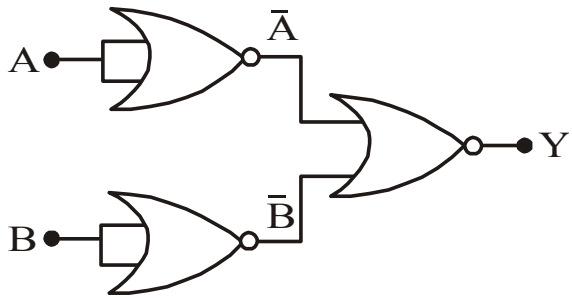
$$Y = \overline{\overline{A} \cdot \overline{B}} = \overline{A} + \overline{B} = A + B$$

NOR gate as a universal gate

- NOR gate is called as universal gate because with the repeated use of NOR gate we can construct any basic gate.
- NOT gate from NOR gate

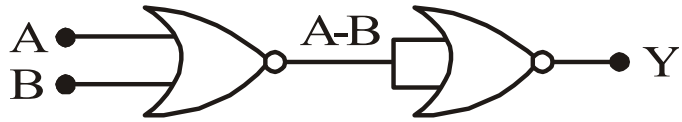


- AND gate from NOR gate



$$Y = \overline{\bar{A} + \bar{B}} = \bar{\bar{A}}.\bar{\bar{B}} = A.B$$

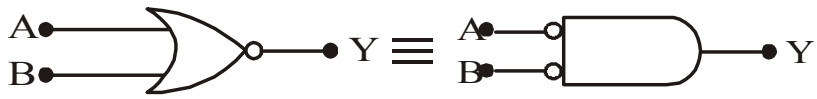
- OR gate from NOR gate



$$Y = \overline{\overline{A + B}} = A + B$$

De Morgan's Theorems

- $\overline{A + B} = \bar{A} \cdot \bar{B}$



NOR gate is equivelent to bubbled AND gate.

- $\overline{A \cdot B} = \bar{A} + \bar{B}$



- NAND is equivalent to bubbled OR gate.

Boolean identities

$A + B = B + A$	$A \cdot B = B \cdot A$
$A + (B + C) = (A + B) + C$	$A \cdot (B \cdot C) = (A \cdot B) \cdot C$
$A \cdot (B + C) = A \cdot B + A \cdot C$	$A + B \cdot C = (A + B) \cdot (A + C)$
$A + 0 = A$	$A \cdot 1 = A$
$A + 1 = 1$	$A \cdot 0 = 0$
$A + A = A$	$A \cdot A = A$
$A + \overline{A} = 1$	$A \cdot \overline{A} = 0$
$\overline{\overline{A}} = A$	$\overline{\overline{A}} = A$
$\overline{A + B} = \overline{A}.\overline{B}$	$\overline{A.B} = \overline{A} + \overline{B}$
$A + A \cdot B = A$	$A \cdot (A + B) = A$
$A + \overline{A} \cdot B = A + B$	$A \cdot (\overline{A} + B) = A \cdot B$

MCQ

For the answer of the following questions choose the correct alternative from among the given ones.

(1) C, Si and Ge have same no. of valence electrons. C is an insulator because energy required to take one electron out from

(A) Si is more (B) C is more (C) Ge is more (D) C is less

(2) Ionization energy of isolated phosphorous atonis 10 eV. Ionization energy of same atom in Si is nearly _____ eV (Relative Permitivity of silicon = 12)

(A) 0.1 (B) 0.2 (C) 0.3 (D) 0.4

(3) By adding _____ impurity in intrinsic semiconductor P type semiconductor is made. charge of these P type semiconductor is _____

(A) trivalent, neutral (B) pentaralent, neutral (C) pentavalent, positive (D) trivalent, negative

(4) Strong overlaping of different atomic orbitals makes

(A) different energy level (B) energy band (C) Conductor (D) Insulators

(5) We can not make p-n junction diode by making P type semi-condutor join with N - type semi-conductor, because

(A) Inter-atomic spacing becomes less than $1A^{o}$

(B) P - type will repel N - type

(C) There will be discontinuity for the flowing charge carriers

(D) semi-conducting properties will be lost

(6) For p-n junction, which statement is incorrect

(A) Donor atoms are depleted of their holes in junction

(B) No net charge exists far from junction

(C) Barrier potential V_B is generated

(D) Energy V_B is to be surmounted before any charge can flow across junction

(7) The intrinsic semi-conductor has :

(A) a finite resistance which does not change with temperature

(B) infinite resistance which decreases with temperature

(C) Finite resistance which decreases with temperature

(D) Finite resistance which does not change with temperature

(8) The behaviour of Ge as semi-conductor is due to width of :

(A) Conduction band being large

(B) Forbidden band being large

(C) Conduction band being small

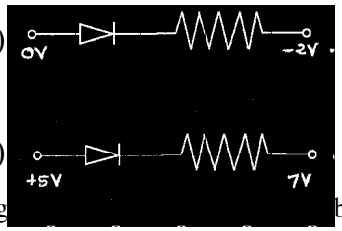
(D) Forbidden band being small and narrow

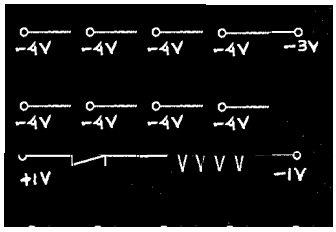
(9) Which of the following is not the advantage of PN junction diode over tube valve ?

(A) Unlimited life (B) No warming-up time after switching

(C) Large efficiency (D) Low consumption of Power

(10) The forward biased diode is

(A) 

(B) 

(C)

(D)

(11) A g ble :

P	Q	R
0	0	1
0	1	1
1	0	1
1	1	0

The gate is :

(A) OR (B) NOR (C) NAND (D) AND

(12) A current gain for a transistor working as CB amplifier is 0.90. If emitter current is 10 mA, then base current is _____.

(A) 1 mA (B) 2mA (C) 0.1 mA (D) 0.2 mA

(13) For a transistor $\frac{I_C}{I_E}=0.96$, then CE current gain is :

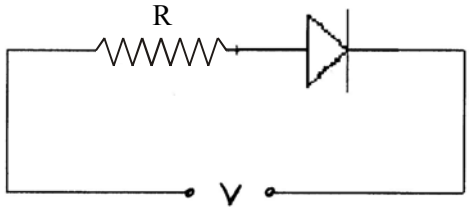
(A) 12 (B) 6 (C) 24 (D) 48

(14) The given truth table is for which logic gate ?

F	F	F	F
F	F	F	F
F	F	F	F
1	1	0	

(A) XOR (B) AND (C) NAND (D) NOR

(15) For the given circuit of ideal P.N junction diode which is correct ?



(A) In F.B, the voltage across R is V

(B) In R.B, the voltage across R is V

(C) In F.B, the voltage across R is -V

(D) In R.B the voltage across R is -V

(16) At ○ K temp, a N - type semi-conductor :

(A) does not have any charge carriers (B) has few holes but no free electrons

(C) few holes and few electrons (D) has equal number of holes and electrons

(17) In Si-crystal, impurity donor atom have valency.

(A) 2 (B) 3 (C) 4 (D) 5

(18) A N-P-N transistor conducts when collector is _____ and emitter is _____ with respect to base.

(A) positive, negative (B) positive, positive

(C) negative, negative (D) negative, positive

(19) A full wave rectifier is operating at 50Hz, 220V the fundamental frequency of ripple will be _____.

(A) 50 Hz (B) 75 Hz (C) 110 Hz (D) 100 Hz

(20) Reverse bias applied on a junction diode :

(A) raises the potential barrier (B) increases majority charge carrier current

(C) lowers the potential barrier (D) increases the temperature of junction

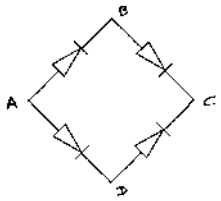
(21) In the figure, the input is across A and C and output is across B and D. The output is

(A) same as input

(B) Halfwave rectified

(C) Fullwave rectified

(D) zero



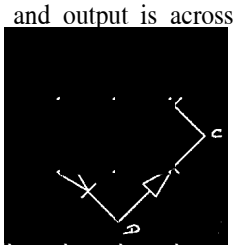
(22) In the figure, the input is across B and D and output is across A and C. The output is

(A) same as input

(B) Halfwave rectified

(C) Fullwave rectified

(D) zero



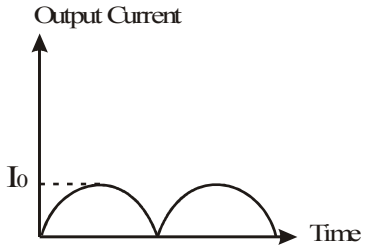
(23) Digital circuits can be made to be respective use of :

(A) AND gate (B) OR gate (C) NOT gate (D) NAND gate

(24) The output current versus time curve of a rectifier is shown in the figure. The avarage value of the output-current is _____ .

(A) 0 (B) $\frac{I_o}{2}$

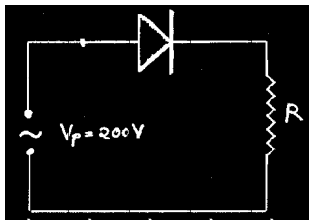
(C) $\frac{2I_O}{\pi}$ (D) I_0



(25) A sinusoidal voltage of peak value 200volts is connected to a diode and resistor R in the circuit shown. If diode is ideal, the r.m.s. voltage across R is _____ volt.

(A) 100 (B) $\frac{200}{\sqrt{2}}$

(C) 200 (D) 280



(26) For a transistor, in a common base configuration the alternating current gain α is given by :

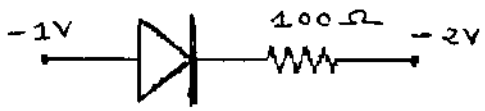
(A) $\left[\frac{\Delta I_C}{\Delta I_B}\right]_{V_C=\text{const.}}$ (B) $\left[\frac{\Delta I_B}{\Delta I_C}\right]_{V_C=\text{const.}}$

(C) $\left[\frac{\Delta I_C}{\Delta I_E}\right]_{V_C=\text{const.}}$ (D) $\left[\frac{\Delta I_E}{\Delta I_C}\right]_{V_C=\text{const.}}$

(27) In a N-P-N transistor circuit, the emitter, collector and base current are respectively I_E, I_C and I_B. The relation between them is _____ .

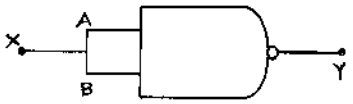
(A) $I_C < I_E < I_B$ (B) $I_B < I_C < I_E$ (C) $I_B > I_C < I_E$ (D) $I_B > I_C > I_E$

(28) Assuming that the junction diode is ideal, the current through the diode is _____ mA



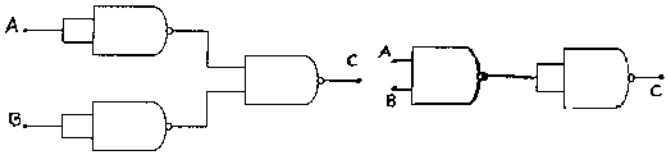
(A) 1 (B) 10 (C) 20 (D) 30

(29) The symbol respresents :



(A) NOT gate (B) OR gate (C) AND gate (D) NOR gate

(30) The combinations of NAND gates shown here under are equivqlent to :



(A) OR gate and NOT gate (B) AND gate and OR gate

(C) AND gate and NOT gate (D) OR gate and AND gate

(31) How many NAND gates are used to form AND gate ?

(A) 1 (B) 2 (C) 3 (D) 4

(32) Ripples are _____ .

(A) A.C. mixed with D.C (B) D.C. mixed with output

(C) D.C. output (D) A.C. output

(33) In an P.N.P transistor circuit, the collector current is 10 mA. If 90% of the electrons emitted reach the collector :

(A) I_E = 9mA (B) I_E = 10mA (C) I_B = 1mA (D) I_B = -1 mA

(34) When a P-type semi-conductor is heated :

(A) number of holes increases while that of electrons decreases

(B) number of electron increases while that of hole decreases

(C) number of electrons and holes remains same

(D) number of electrons and holes increases equally

(35) The depletion layer in PN junction diode is caused by

(A) drift of holes (B) Diffusion of impurity ions

(C) diffusion of charge carriers (D) drift of electrons

(36) The active junction area in a solar cell is _____ as we want _____ power

(A) small, more (B) small, small (C) large, more (D) large, small

(37) The forbidden energy band gap in semi-conductor, conductor and insulator are E_1, E_2 and E_3 respectively. The relation among then is :

(A) $E_1 < E_2 > E_3$ (B) $E_1 > E_2 > E_3$ (C) $E_1 < E_2 < E_3$ (D) $E_1 > E_2 < E_3$

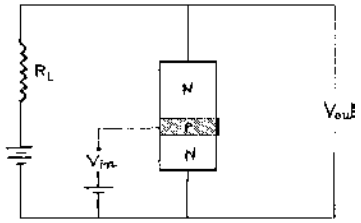
(38) An N-P-N Transistor circuit is shown in figure is

(A) A common base circuit

(B) A common emitter circuit

(C) A common collector circuit

(D) Oscillator circuit



(39) In a common emitter amplifier, output resistance is 5000Ω and input resistance is 1000Ω. If peak value of signal voltage is 1 mV and $\beta=100$, then the peak value of output voltage is _____

(A) 0.1V (B) 0.3V (C) 0.2V (D) 0.5V

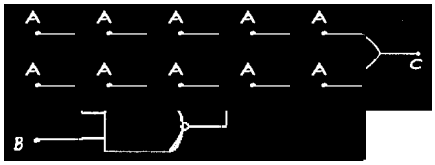
(40) The A.C. current gain of a transistor is 100. If the base current changes by $100\mu A$, What is the charge in collector current ?

(A) 20 mA (B) 30 mA (C) 10 mA (D) $10\mu A$

(41) What is the output of the combination of the gates shown in the fig. below ?

(A) $A+\overline{A\cdot B}$ (B) $A + A\cdot B$

(C) $(A+B)\cdot(\overline{A\cdot B})$ (D) $(A+B)\cdot(\overline{A+B})$



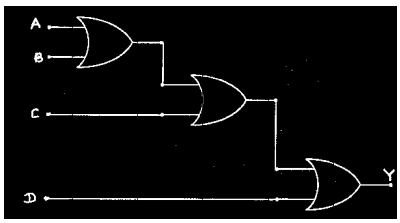
(42) The expression of Y in the following circuit is :

(A) AB + CD

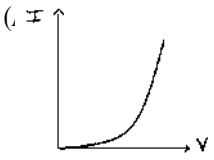
(B) A + BCD

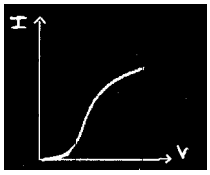
(C) A + B + C + D

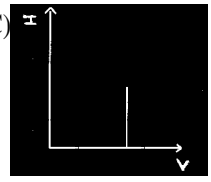
(D) $A\cdot B\cdot C\cdot D$

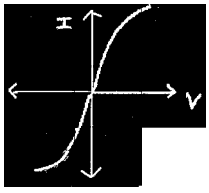


(43) Which of the following figure represents an ideal diode characteristics ?

(A) 

(B) 

(C) 

(D) 

(44) In Ge sample, traces of gallium are added as impurity. The resultant sample would behave like :

(A) a conductor (B) a P-type semiconductor

(C) an N-type semiconductor (D) an insulator

(45) A light emitting diode has a voltage drop of 2V across it when 10mA current is passed. If this LED is to be operated with 6V battery the value of limiting resistor would be _____ .

(A) 400Ω (B) 4000Ω (C) 40kΩ (D) 300Ω

(46) NAND gate is _____ .

(A) A basic gate (B) Not a universal gate

(C) A universal gate (D) Multipurpose gate

(47) The number of holes and electrons in an intrinsic conductors are x and y respectively at room temperature. Which of the following options are true ?

(A) x > y (B) y > x (C) x = y (D) x << y

(48) How will you increase the resistivity of Ge semi-conductor ?

(A) On adding donor impurity (B) On adding acceptor impurity

(C) On making UV light incident on Ge crystal (D) On decreasing the temperature

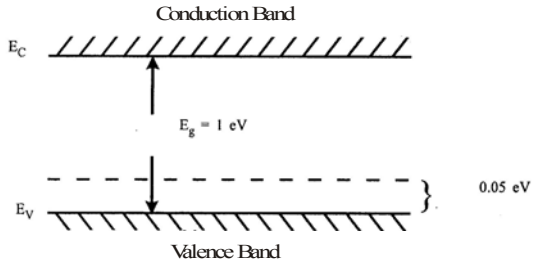
(49) What is type of material, for the energy band diagram shown in the figure ?

(A) N - type semi-conductor

(B) P - type semi-conductor

(C) Insulator

(D) Intrinsic semi-conductor



(50) From the following semi-conductor devices, _____ operates in forward bias only.

(A) Varactar diode (B) Zener diode (C) Light emitting diode (D) photo-diode

(51) _____ device is the odd-one out.

(A) solar-cell (B) Varactor diode (C) Photodiode (D) Zener diode

(52) The value of depletion capacitance _____ on decreasing the reverse bias on varactor diode

(A) decreases (B) increases (C) becomes zero (D) does not change

(53) Which of the following statement is correct for transistor LC oscillator circuit ?

(A) It works with negative feed back

(B) The phase difference between output and input signal is π radian

(C) To start oscillation external signal is required

(D) The frequency of output signal is independent of the components used in feed back circuit

(54) The frequency of output signal of LC oscillator circuit is 100Hz with capacitance value $0.1\mu\text{F}$. If value of capacitance is taken as $0.2\mu\text{F}$, the frequency of output signal _____ .

(A) decreases by $\frac{1}{\sqrt{2}}$ (B) increases by $\frac{1}{\sqrt{2}}$ (C) decreases by $\frac{1}{2}$ (D) increases by $\frac{1}{2}$

(55) The Common Emmiter amplifier has voltage gain equal to 300 and its input signal is $0.5\cos(100\ t)$ volt. The output signal will be equal to _____ .

(A) $150\cos(100t)$ (B) $300\cos(100t)$

(C) $150\cos(100t + \pi)$ (D) $300\cos(100t - \pi)$

(56) Common base current gain of a NPN transistor is 0.99. The input resistance is 1000Ω and load resistance is $10,000\Omega$. The voltage gain in common emitter mode is _____ .

(A) 9900 (B) 99000 (C) 99 (D) 990

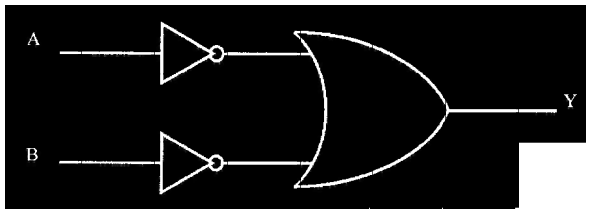
(57) The logic circuit shown in the figure, is the equivalent diagram of which logic gate ?

(A) OR gate

(B) NAND gate

(C) AND gate

(D) NOR gate



(58) In forward bias made, the P.N junction diode resistance will _____ .

(A) infinity (B) zero (C) less (D) more

(59) To obtain OR gate from NOR gate, you will need _____

(A) one NOR gate (B) one NOT gate (C) Two NOT gate (D) one OR gate

(60) For Boolean identities match the pair :

(1) $\overline{\overline{A}}$ (P) $\overline{A}+\overline{B}$

(2) $\overline{A+B}$ (Q) $A\cdot B$

(3) $\overline{A.B}$ (R) $\overline{A}\cdot\overline{B}$

(4) $A\cdot\left(\overline{A}+B\right)$ (S) A

(A) (1) - (S), (2) - (P), (3) - (Q), (4) - (R)

(B) (1) - (S), (2) - (R), (3) - (Q), (4) - (P)

(C) (1) - (S), (2) - (Q), (3) - (P), (4) - (R)

(D) (1) - (S), (2) - (R), (3) - (P), (4) - (Q)

(61) The ratio of concentration of electrons and holes in a semi-conductor is $\frac{7}{5}$ and the ratio of currents is $\frac{7}{4}$, then what is the ratio of their drift velocities ?

(A) $\frac{4}{7}$ (B) $\frac{5}{8}$ (C) $\frac{4}{5}$ (D) $\frac{5}{4}$

(62) In a P-type silicon, which of the following statement is true ?

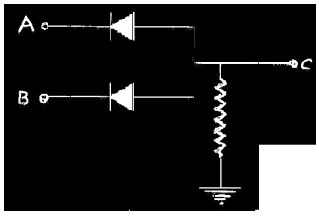
(A) Electrons are majority charge carries and trivalent atoms are the dopants

(B) Electrons are minority charge carries and pentavalent atoms are dopants

(C) Holes are minority charge carriess and pentavalent atoms are dopants

(D) Hole are majority charge carries and trivalent atoms are dopants

(63) In the circuit below A and B represnts two inputs and C represents output. The circuit represents _____ .



(A) NOR gate

(B) NAND gate

(C) AND gate

(D) OR gate

(64) A zener diode used as voltage regulator is connected _____ .

(i) in forward bias (ii) in reverse bias (iii) in parallel with load (iv) in series with load

(A) (i) and (ii) are correct (B) (ii) and (iii) are correct

(C) only (i) is correct (D) only (iv) is correct

Directions : Question numbers (65), (66) and (67) are based on following passage.

PASSAGE : A n-p-n transistor is used in common emitter made in an amplifier circuit. A change of $40\mu A$ in the base current changes the output current by $2mA$ and $0.04V$ in input voltage.

(65) The input resistance is _____ .

(A) $1k\Omega$ (B) 10Ω (C) $10k\Omega$ (D) 100Ω

(66) The current amplification factor is _____ .

(A) 20 (B) 30 (C) 50 (D) 40

(67) If a load of $6k\Omega$ is used, then the voltage gain of the amplifier is _____ .

(A) 100 (B) 200 (C) 300 (D) 400

(68) An amplifier has voltage gain $A_v = 1000$. The voltage gain in dB is _____ .

(A) 20 dB (B) 30 dB (C) 3 dB (D) 60 dB

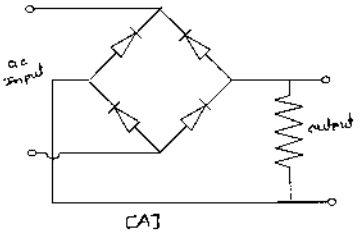
(69) A potential barrier of 0.6V exists across a P-N junction. If the depletion region is $1\mu m$ wide, what is the intensity of electric field in the region ?

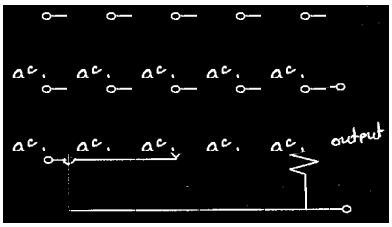
(A) $4\times10^{5}\,Vm^{-1}$ (B) $5\times10^{5}\,Vm^{-1}$ (C) $6\times10^{5}\,Vm^{-1}$ (D) $2\times10^{5}\,Vm^{-1}$

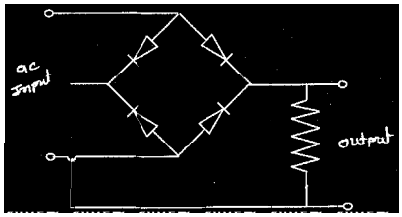
(70) when a PN junction diode is forward biased, then the depletion region is _____ and barrier height is _____ .

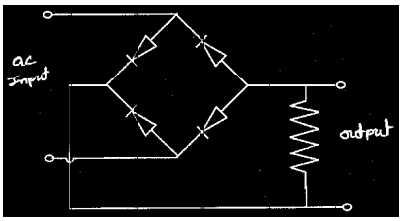
(A) reduced, increases (B) widened, reduced (C) reduced, reduced (D) increased, increased

(71) Which of the following circuit provides full wave rectification ?

(A) 

(B) 

(C) 

(D) 

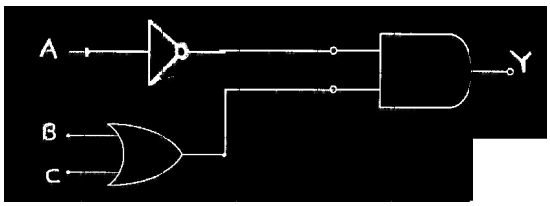
(72) A common- emitter amplifier has a voltage gain of 100, an input impedence of 100Ω and an output impedence of 200Ω .The product of voltage gain and current gain is _____ .

(A) 1000 (B) 3000 (C) 5000 (D) 500

(73) A P-N photodiode is made of a material with a band gap of 2.0ev. The minimum frequency of the radiation that can be absorbed by the material is nearly (Take hc = 1240eVnm)

(A) $5\times10^{14}\,Hz$ (B) $20\times10^{14}\,Hz$ (C) $1\times10^{14}\,Hz$ (D) $10\times10^{14}\,Hz$

(74) The bodean equation for the circuit is



(A) $Y=\overline{A}\cdot B+C$ (B) $Y=\overline{A}\cdot(B+C)$

(C) $Y=\overline{A}\cdot(\overline{B}+\overline{C})$ (D) $Y=\overline{A}\cdot(B+\overline{C})$

(75) A n-p-n transistor circuit has $\alpha=0.985$. If I_c = 9 mA then the value of I_B is _____ .

(A) 0.003mA (B) 0.66mA (C) 0.015mA (D) 0.03mA

(76) For a transistor amplifier, the voltage gain

(A) remains constant for all frequencies

(B) is high at high and low frequencies and constant in the mid-frequency range

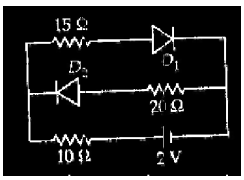
(C) is low at high and low frequencies and constant in the mid-frequency range

(D) None of the above

(77) The current flowing through 10Ω resistor in the circuit shown in the figure is _____ .

(A) 50mA (B) 20mA

(C) 40mA (D) 80mA



(78) The input and outputs from different time intervals are given below for NAND gate

Time interval	Input A	Input B	Output Y
t_1 to t_2	0	1	P
t_2 to t_3	0	0	Q
t_3 to t_4	1	0	R
t_4 to t_5	1	1	S

The value taken by P, Q, R and S are respectively

(A) 1, 0, 1, 1 (B) 0, 1, 0, 0 (C) 0, 1, 0, 1 (D) 1, 1, 1, 0

(79) The manifestation of band structure in solids is due to :

(A) Heisenberg's uncertainty principle

(B) Pauli's exclusion principle

(C) Bohr's correspondence principle

(D) Boltzmann's low

(80) Copper and silicon material is cooled down from 600K to 400K then, resistivity of cooper _____ and silicon _____ .

(A) increases, decreases (B) decreases, increases (C) decreases, decreases (D) increases, increases

(81) Semi-conductor has phospholous as impurity then it will have _____ .

(A) $n_e >> n_h$ (B) $n_e << n_h$ (C) $n_e = n_h$ (D) $n_e = n_h = n_i$

(82) Zener diode is used as _____

(A) Full. wave rectifier (B) amplifier

(C) A.C. voltage regulator (D) D.C. voltage regulator

(83) Break down voltage of a diode is 5V. By which effect this breakdown occurs in diode ?

(A) Only avalanche effect (B) Only zener effect

(C) Avalanche or zener effect (D) None of the above

(84) When NPN transistor is used as an amplifier then _____

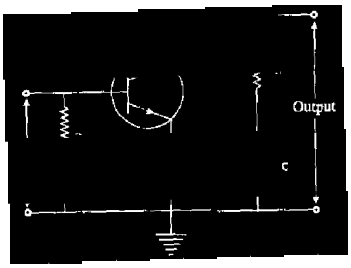
(A) electron moves from base to collector

(B) hole travels from emitter to base

(C) hole goes to emitter from base

(D) electron goes to base from collector

(85) For a given amplifier circuit, to make transistor active as an amplifier, how much value of voltages to be kept for V_{BB} battery and V_{CC} battery ?



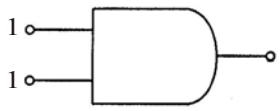
(A) V_{BB} = -1V V_{CC} = +5V

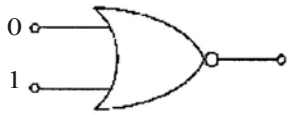
(B) V_{BB} = -1V V_{CC} = -5V

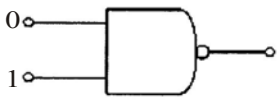
(C) V_{BB} = +1V V_{CC} = +5V

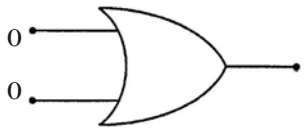
(D) V_{BB} = +1V V_{CC} = -5V

(86) Which of the following logic gate will have output 1 ?

(A) 

(B) 

(C) 

(D) 

(87) For _____ gate, the output is 1 only when both input are 'O'

(A) AND (B) NAND (C) OR (D) NOR

(88) In VLSI circuits more than _____ gates are contained.

(A) 1000 (B) 100 (C) 10 (D) 500

(89) The flow of valence electrons to the left means that holes are flowing.

(A) Left (B) Right (C) Either way (D) None

(90) How many free electrons does a P-type semiconductor contain ?

(A) Many (B) None

(C) Only those produced by thermal energy (D) Same number as holes

(91) Suppose an intrinsic semi-conductor at room temperature has 1 billion free electrons at room temperature. If temperature changes to 75° C, how many holes are there ?

(A) Fewer than 1 billion (B) 1 billion (C) More than 1 billion (D) Impossible to say

(92) Which of the following doesn't fit in the group ?

(A) Conductor (B) Semi conductor (C) Four valence electrons (D) Crystal structure

(93) What kind of device is a diode ?

(A) Bilateral (B) Linear (C) Nonlinear (D) Unipolar

(94) We want a peak load voltage of 40V out of a bridge rectifier, What is the approximate rms value of secondary voltage ?

(A) 0V (B) 14.4V (C) 28.3V (D) 56.6V

(95) The load-current is approximately constant when a zener diode is _____.

(A) Forward biased (B) Reverse biased

(C) Operating in breakdown region (D) Unbiased

(96) When source voltage increases in a zener diode, which of these current remians approximately constant ?

(A) Series current (B) Zener current (C) Load current (D) Total current

(97) The device associated with voltage controlled capacitance is _____.

(A) Light emitting diode (B) Photo diode (C) Varactor diode (D) Zener diode

(98) For normal operation of the transistor, the collector diode has to be _____.

(A) Forward biased (B) Reverse biased

(C) Non conducting (D) Operating in breakdown region

(99) Most of the electrons in the base of N-P-N transistor flow

(A) Out of the base lead (B) Into the collector (C) Into the emitter (D) Into the base supply

Direction for Assertion - Reason type questions

(A) If both Assertion and Reason are true and reason is the correct explanation of assertion.

(B) If both Assertion and Reason are true but Reason is not the correct explanation of assertion.

(C) If Assertion is true but Reason is false

(D) If both assertion and reason are false

(100) A : Intrinsic charge carries are thermally generated

R : Their availability can be easily controlled

(A) (B) (C) (D)

(101) A : Impurity atoms for silicon is selected from third or fourth group

R : These Impurity atoms have same size as that of Si

(A) (B) (C) (D)

(102) A : Photodiode are operated in reverse bias

R : In reverse bias fractional change in minority charge carrier is more

(A) (B) (C) (D)

(103) A : The resistivity of a semi-conductor decreases with temperature

R : At higher temperature more co-valent bond breaks

(A) (B) (C) (D)

(104) A : NAND (or NOR) gates are called digital building blocks

R : The different combination of NAND (or NOR) gates can produce all the basic or complicated gates.

(A) (B) (C) (D)

(105) A : The colour of light emitted by depends on its forward bias.

R : The forward biasing of PN junction diode will increase the width of depletion layer

(A) (B) (C) (D)

(106) A : The ionization energy of isolated phosphorous is very large

R : The ionization energy of phosphorous in lattice is very small

(A) (B) (C) (D)

(107) A : Mostly transistor are used in common emitter configuration

R : Common emitter configuration provide more current gain and small voltage gain

(A) (B) (C) (D)

(108) A : A transistor amplifier circuit in common emitter configuration has low input impednce

R : Base - emitter junction is forward biased

(A) (B) (C) (D)

KEY NOTE

1	B	31	C	61	D	91	C
2	A	32	A	62	D	92	A
3	A	33	C	63	C	93	C
4	B	34	D	64	B	94	C
5	C	35	C	65	A	95	C
6	D	36	C	66	C	96	C
7	C	37	D	67	C	97	C
8	D	38	B	68	D	98	B
9	A	39	D	69	C	99	B
10	A	40	C	70	C	100	C
11	C	41	A	71	D	101	A
12	A	42	C	72	C	102	A
13	C	43	C	73	A	103	B
14	D	44	C	74	B	104	B
15	B	45	A	75	D	105	D
16	A	46	C	76	C	106	B
17	D	47	C	77	D	107	B
18	B	48	D	78	D	108	A
19	D	49	B	79	B		
20	A	50	C	80	B		
21	D	51	A	81	A		
22	C	52	A	82	D		
23	D	53	B	83	C		
24	C	54	A	84	C		
25	B	55	C	85	C		
26	C	56	D	86	C		
27	B	57	B	87	D		
28	B	58	C	88	A		
29	A	59	A	89	B		
30	D	60	D	90	C		

Hints

(2) Out of 5 valence electrons of phosphorous, 4 are shared with Si. Fifth electron can be approximated to revolve around nucleus. Situation is like H-atom

$En \approx -13.6/n^2 \text{ ev}$

For n =1, En = -13.6ev

E inside lattice = $\frac{En}{\varepsilon_1^2} = \frac{-13.6}{12^2} = 0.1\text{ev}$

(12) $\alpha = \frac{I_c}{I_e}$ $\quad \therefore I_c = \alpha I_E = 9\text{mA}$

$I_B = I_E - I_C = 10 - 9 = 1$ mA

(13) $\beta = \frac{\alpha}{1-\alpha}$

(15) Ideal diode has zero resistance in forward bias

(22) Potential At B = Potential at D

(25) $V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{V_p}{\sqrt{2}}$

(33) $I_c = 10\text{mA} = 0.90\ I_E$

$\therefore I_E \approx 11\text{mA}$

$\therefore I_B \approx 1\text{mA}$

(38) Base & Emitter are forward biased & collector is reverse biased with respect to Emitter

$\therefore$ Circuit is Common emitter

(41) $C = A + Y_1$

$Y_1 = \overline{A \cdot B}$

$\therefore C = A + \overline{A \cdot B}$

(51) Only solar-cell generates e.m.f and it does not need bias-voltage

(54) Frequency of LC oscillator is $f = \frac{1}{2\pi\sqrt{LC}}$

(55) Output voltage = voltage gain $\times$ Input voltage.

There will be a phase difference of 180° between input and output signal.

(56) $\beta = \frac{\alpha}{1-\alpha}$ and $A_V = \beta \frac{R_L}{r_i}$

(61) $\nu_d = \frac{I}{nAe}$ $\therefore \nu_d \alpha \frac{I}{n}$

(65) $r_i = \frac{\Delta V_{BE}}{\Delta I_B}$

(66) $\beta = \frac{\Delta I_C}{\Delta I_B}$

(67) $A_V = \beta \frac{R_L}{r_i}$

(68) Voltage gain on dB = $20 \log_{10} A_0$

(69) $E = \frac{v}{d} = \frac{0.6}{10^{-6}}$

(73) $A_p = \frac{Av^2}{\frac{R_L}{r_i}}$

(74) $\lambda = \frac{hc}{E_g}$, $f = \frac{c}{\lambda}$

(76) $\alpha = \frac{I_c}{I_E}$ & $I_E = I_B + I_C$ $\therefore \alpha = \frac{I_C}{I_B + I_C}$

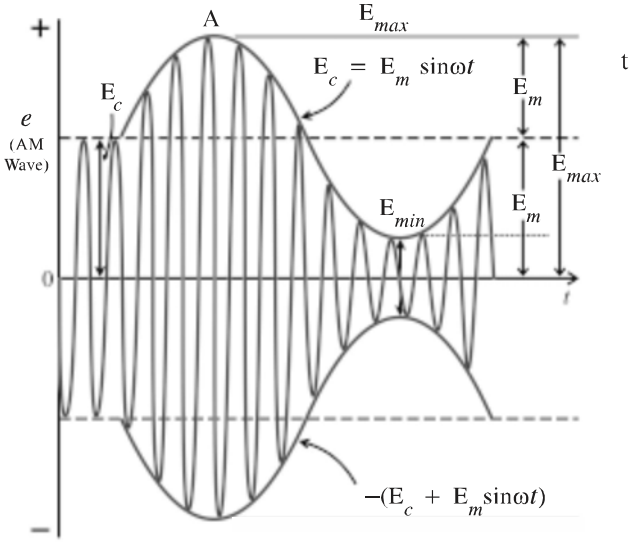
(78) diode D_1 will only conduct $\therefore I = \frac{2}{10+15}$

(95) $V_{rms} = \frac{V_m}{\sqrt{2}}$

Unit - 20
Communication System

SUMMARY

- Main elements of the communication system are :
  (1) Transmitter (2) Transmission channel and (3) Receiver.
- Transducer : The device which transforms one form of energy into another form of energy is called a transducer.
- Transmission Channel : Transmission channel is a link or medium between transmitter and receiver. The signals of information are propagating through this medium and received at receiver.
- Noise : Noise is an unwanted signal. They get admixed with the information signal in the transmission channel and distort the information signal.
- Signal : For a transmission, the information is transformed into electric signal is called a signal. There are two types of signals. (1) Analog signal and (2) Digital signal. Communication Systems 263
- Bandwidth : In communication system the difference of maximum frequency and minimum frequency efficiently transmitted by the system is called the bandwidth of the system. The bandwidth of audio signal is 20 kHz and bandwidth of video signal is 4.2 MHz.
- Modulation : The process of superposing low frequency audio signals on high frequency is called modulation. Low frequency is called the modulating signal and high frequency wave is called the carrier waves. The resultant wave is known as modulated wave. There are three types of the modulation. (1) Amplitude modulation (AM), (2) Frequency modulation (FM) and (3) Phase modulation (PM).
- Amplitude Modulation : A modulation in which the amplitude of the carrier wave is changed according to instantaneous value of the modulating wave is called amplitude modulation. Here, phase and frequency of carrier wave remain unchanged. Modulation Index : The ratio of the amplitude of modulating wave and amplitude of carrier wave is called a modulation index. $m_a = \frac{E_m}{E_c}$ Value of ma is always less than 1.
- Demodulation : The process of separating information signal from carrier wave is called demodulation. This process is performed in the receiver. The circuit which performs the process of demodulation is called a detector circuit.
- Propagation of an Electromagnetic Waves : (1) The wave propagating along the surface of the earth are called ground wave or surface wave. The frequencies up to 2 MHz is propagating through ground waves. (2) The waves returned to the earth after getting reflected by the ionosphere at a height of about 60-400 km is known as sky waves. 2 MHz to 30 MHz frequencies are propagating through the sky waves. (3) The waves from the transmitting antenna, reaching a receiver in a straight line after the reflection from the ground are called space waves.



AM Wave

MCQ

For the answer of the following questions choose the correct alternative from among the given ones.

(1) Type of communication Possible by Modem duplex is .....

(A) Half duplex (B) Simplex (C) Doudle Duplex (D) Full duplex

(2) To tranmit 1Kh_Z frequency signal, wave langth of the signal should be_____m.

(A) 3×10^3 (B) 3×10^5 (C) 3×10^7 (D) 3×10^{10}

(3) What should be the minimum lenght of the antenna copable of emitting audio signal of wave length λ?

(A) $\frac{\lambda}{4}$ (B) $\frac{\lambda}{8}$ (C) λ (D) $\frac{\lambda}{2}$

(4) _____is used as a medium an space communication

(A) Two wire line (B) Space (C) Amplifier (D) Oplical fibre

(5) The band width of Audio signal is

(A) 20 H_Z (B) 20 KH_Z (C) 20 MH_z (D) 20 GH_z

(6) ground waves are polarised at_____direction.

(A) vertcally upward (B) horizontal

(C) tangential direction to the sarface of the earth (D) in all four.

(7) Intenity of electric field of ground waves is in _____ of the distance travelled by it.

(A) Square (B)inverse of root (C) proportion (D) inverse

(8) In radio and television broadcast, the information signal is in the form of

(A) analong signal (B) digital signal

(C) Both analog and digit signals (D) neither ahalig nor digital signal

(9) In light modulation, which charactristie of the carrier light wave is varied?

(A) Amplitude (B) Frequency (C) Phase (D) intensity

(10) Though which mode of communication can radiowaves be sent from one Plate to another?

(A) Ground wave ProPagation (B) sky wave Propagation

(C) SPace wave ProPagation (D) All of the above

(11) Which range of Frequencies is Suitable for sky wave Propagation ?

(A) 1 KH_z to 500 KH_z (B) 1 MH_z to 2 MH_z

(C) 2 MH_z to 20 MH_z (D) Above 30 MH_z

(12) Which of the Following is not a component of communication System ?

(A) Transmitter (B) Transmission channel

(C) Noise (D) Receiver

(13) Device which transform one form of enegy in to another form of energy is called a_____

(A) Transducer (B) Transformer (C) Transponder (D) Transistor

(14) _____is a link between the transmitter and receiver

(A) Microphone (B) Transmission Channel

(C) transducer (D) telephone

(15) _____ in the receiver section are used to reduce the level of noise signal

(A) filter circuit (B) Modulator (C) de modulator (D) Amplifier

(16) The maniman distance a up to which T.V transmission from a T.V. tower of height h can be received is Proportional to

(A) $h^{\frac{1}{2}}$ (B) h (C) $h^{\frac{3}{2}}$ (D) h^2

(17) The energy attenuation in optical fibre is mainly due to

(A) absorption (B) scattering

(C) both a and b (D) neither absorption nor scattering

(18) The electromagnetic waves of frequency 2 MHz to 30 MHz are

(A) in ground wave propagation (B) in sky wave propagation

(C) in micro wave propagation (D) in satellite communication

(19) Coaxial cable offers a band width approximately _____ MHz

(A) 1 (B) 20 (C) 250 (D) 1000

(20) Modulation is used to.....

(A) reduce the band width used

(B) seperate the transmission of different areas

(C) ensure that information may be transmitted to long distance

(D) allow the use of practical antennas

(21) An antenna can transmit _____ radiation with more efficiency.

(A) low frequency (B) high frequency

(C) long wave length (D) None of these

(22) For an amplitade modulated wave, the maximum amplitude is found to be 9V while the minimum amplitude is found to be 3V. The modulation indext is _____

(A) 100% (B) 75% (C) 50% (D) 25%

(23) AM is used for broadcasting because

(A) it is more noise innunerable than other modulation system

(B) it requires less transmitting power compared with other system

(C) it's are avoids receiver complexity....

(D) No other modulation system can provide the neccessary band width faithful tranmission

(24) Frequencies in the UHF range normally propagation by means of

(A) ground waves (B) sky waves (C) surface waves (D) space waves

(25) The area to be covered for T.V telecast is doubled. Then the height to transmitting antenna will have to be

(A) doubled (B) halved (C) quadrupled (D) kept unchanged

(26) Which of the following is related to telecommunication

(A) X-rays (B) r-rays (C) sound wave (D) microwaves

(27) Suppose the amplitade of modulated wave equal to the signal amplitude. Then modulatation is

(A) 10% (B) 25% (C) 50% (D) 100%

(28) Space waves are generally propagated in

(A) UHF (B) VHF (C) HF (D) None of there

(29) If the frequency of 3 KHz has to be transmitted through amplitude modulation. Which of the following frequency should use as a carrier frequency ?

(A) 30 Hz (B) 300 Hz (C) 3000 Hz (D) 3 MHz

(30) The range of frequencies allotted for commercial FM radio broadcast is

(A) 88 to 108 MHz (B)88 to 108 KHz (C) 8 to 88 MHz (D) 88 to 108 GHz

(31) The process of superimposing signal frequency on the carrier wave is known as

(A) transmission (B) reception (C) modulation (D) detection

(32) A T.V tower has a height 150m. What is the population density around the T.V tower if the total population covered is 50 lakh ? (Reduse of earth = 6.4×10^6 m)

(A) 82.6/km^2 (B) 800.6/km^2 (C) 828.6/km^2 (D) 876.6/km^2

(33) In which frequency range, space waves are normally propagated ?

(A) HF (B) VHF (C) UHF (D) SHF

(34) If M_1 & M_2 are the refractive indices of the material of core and cladding of an optical fibre, than the loss of light due to its leakage can be minimized by having

(A) $M_1 > M_2$ (B) $M_1 < M_2$ (C) $M_1 = M_2$ (D) None of these

(35) Through which mode of propagation, the raido waves can be sent from one place to another ?

(A) ground wave propagation (B) sky wave propagation

(C) space wave propagation (D) all of them

(36) A radio can tune to any station in the 7.5 MHz to 12 MHz band. What is the corresponding wave lenght band ?

(A) 400m - 250m (B) 40m - 25m (C) 4m - 2.5m (D) None of these

Assertion Reason type Question

In the following questaion from (37 to 40) statement-1 (Assertion) is followed by statement-2 (Reason) Each question has the following four choice out of which only one choice is correct

(a) Statement-1 is true, statement-2 is true.
Statement-2 is a correct explanation for statement-1.

(b) Statement-1 is true, statement-2 is true, statement-2 is not a correct enplanation for statement-1.

(c) Statement-1 is true, statement-2 is false

(d) Statement-1 is false, statement-2 is true.

(37) Statement-1 :- sky waves are not used in the transmission of T.V signals
Statement-2 :- T.V signals have a frequency range of 400 to 200 MHz
(A) a (B) b (C) c (D) d

(38) Statement-1 :- Micro waves and not radio waves are used in satellite communication.
Statement-2 :-The wavelength of microwaves is much shorter than that of radio waves. Hence micro wave do not disperse or diffract like radio waves.
(A) a (B) b (C) c (D) d

(39) Statement-1 :- sky wave communication is not suitable for frequncies greater than 30 MHz.
Statement-2 :- High frequency signals die out before reaching the ionosphere.
(A) a (B) b (C) c (D) d

(40) Statement-1 :- Modulated wave is received directly on a head phone
Statement-2 :- High frequency waves are beyond audible range.
(A) a (B) b (C) c (D) d

(41) The refractive inden of the ionosphere
(A) increase as we go from the lower to upper layer of the ionosphere
(B) decrease as we go from the lower to upper layer of the ionosphere
(C) remain the same through out the ionosphere
(D) is equal to unity

(42) A T.V tower has a height of 80 m. The maximum distance up to which T.V transmission can be received is equal to _____ . (radius of earth = 6.4×10^{6}m)
(A) 16 km (B) 32 km (C) 80 km (D) 160 km

(43) A receiving station on the ground is receiving a signal of frequency 10 MHz, then the mode of transmission is
(A) ground wave propagation (B) sky wave propagation
(C) both ground wave and sky wave propagation
(D) neither ground wave nor sky wave propagation

(44) On a particular day, the maximum frequency reflected from the ionosphere is 8 MHz. On another day it was found to increase to 9 MHz. The ratio of the minimum electron densities of ionosphere on the two days is

(A) $\sqrt{\frac{8}{9}}$ (B) $\frac{8}{9}$ (C) $\left(\frac{8}{9}\right)^{3/2}$ (D) $\left(\frac{8}{9}\right)^{2}$

(45) Consider an optical communication system operating at $\lambda = 800\,\text{nm}$. Suppose only 1% of the optical source frequency is the available channel bandwidth for optical communication. No. of channels that can be accomodated for transmitting video T.V signals requiring an approximate band width of 4.5 MHz is _____ .

(A) $3.8\times10^{+14}$ (B) 3.8×10^{12} (C) 8.4×10^{5} (D) 1.71×10^{7}

(46) A signal $e_m = 20\sin 2000\,\pi t$ amplitude modulates a carries wave $e_C = 80\sin 200000\,\pi t$. The percantage of modulation is _____

(A) 25% (B) 0.25% (C) 400% (D) 40%

(47) A transmitting antenna at the top of a tower has a height of 50m and the height of the receiving antenna is 32m. The maximum distance between them for satisfactory communication in line of right mode is _____. Given radios of earth R = 6400 km

(A) $25.29\times10^{3}\,\text{km}$ (B) $20.23\times10^{3}\,\text{km}$ (C) 45.5km (D) None of these.

(48) The characteristic impedence of a co-axial cable is between _____ .

(A) 10Ω to 20Ω (B) 20Ω to 30Ω (C) 30Ω to 60Ω (D) 50Ω to 70Ω

(49) The relation between the maximum electron denisity N max and the critical frequency f for the ionosphere can be given as

(A) $f_c = \sqrt{9N_{\max}}$ (B) $f_c = \sqrt{9}\ (N_{\max})$ (C) $f_c = q\sqrt{N_{\max}}$ (D) None of these

(50) In order to cover a circular region of radios 128 km, by a TV transmitter, What must be the height of the transmitting antenna ?

(A) 79 m (B) 1050 m (C) 1280 m (D) 1560 m

(51) The frequency of carries wave emitted from L - C osallator of 10 H inductance and 1 nF capacitor is _____ .

(A) 1592 KHz (B) 1592 Hz (C) 1592 MHz (D) 159.2 Hz

(52) If the band width is $1.6\times10^{12}\,\text{Hz}$, how many communication channels can be obtained with 16 KHz band width radio signal ?

(A) 10^{15} (B) 10^{11} (C) 10^{10} (D) 10^{8}

(53) A transmitter transmit a power of 10 kw when modulation is 50%. What is the power of carrier wave ?

(A) 12 KW (B) 7.0 KW (C) 8.89 KW (D) 5.0 KW

(54) A T.V. tower has a height 200 m. By how much the height of tower be increased to triple its coverage range ?

(A) 1600 m (B) 800 m (C) 1800 m (D) 600 m

(55) In short wave communication wave of which of the following frequencies will be reflected back by the ionospheric layer having electron density $1.23\times10^{12}\,m$

(A) 2 MHz (B) 10 MHz (C) 12 MHz (D) 18 MHz

(56) What is the frequency range of signals that can be transmitted in care of twisted pair of wires ?

(A) 5 MHz to 20 MHz (B) 5 MHz to 10 MHz

(C) 100 Hz to 4 MHz (D) 5 MHz to 20 MHz

(57) What type of medium is used in optical communication ?

(A) guided medium (B) Unguided medium

(C) Both guided and unguided media (D) Neither guided nor unguided media

(58) The maximun electron density of a layer of the ionosphere is $9\times10^{12}m^{-3}$. The critical frequency of this layer is

(A) 9 MHz (B) 6 MHz (C) 3 MHz (D) 3 GHz

(59) Which condition is to be satisfied in an AM detector circuit so that the signal corresponds to an en.... of the carrier wave ?

(A) $T_e << RC$ (B) $T_e >> RC$ (C) $F_c << RC$ (D) $\frac{1}{f_n} << RC$

(60) The transmission of audio signal in radio broadcasting is _____ type.

(A) ASK (B) FSK (C) AM (D) FM

(61) Transmission loss per km in the optical fibre, ured presently is,

(A) 20 dB (B) 2 dB (C) 0.2 dB (D) 0.02 dB

(62) In satellite communication 6 GHz frequency is used for _____ and 4 GHz frequency is used for

(A) uplink, down link (B) down link, up link

(C) modulation, demodulation (D) demodulation, modulation

(63) Assertion : Microwave communication is preferred over optical communication.

Reason : Microwaves provide large number of channels and bandwidth an compared to optical signals.

(A) Both assertion and reason are true and the reason is correct enplanation of assertion

(B) Both assertion and reason are true and the reason is not the correct enplanation of assertion.

(C) Assertion is true but reason is false.

(D) Both assertion and reason are false.

(64) If a radio receiver amphifier all the signal frequencies equally well, it is said to have high

(A) fidality (B) distortion (C) sensibility (D) sensitivity

(65) In frequency modulation

(A) the amplitude of modulated wave varies as frequency of carrier wave.

(B) the frequency of modulated wave varies as amplitude of modulating wave.

(C) The amplitude of modulated wave varies as amplitude of carrier wave.

(D) The frequency of modulated wave varies as frequency of modulating wave.

(66) A silicon optical fibre with a care diameter large enough has a care refractive inder of 1.50 and cladding refractive inder 1.47. The critical angle at the care cladding interface is _____ .

(A) 87.5^{o} (B) 78.5^{o} (C) 38.5^{o} (D) 45°

(67) In satellite communication

(1) The frequency used lies between 5 MHz and 1 MHz

(2) The uplink and downlink frequencies are different.

(3) The orbit of geostationary staellite lies in the equatorial plane at an indication of 0°. In the above statements.

(A) Only 2 and 3 are true (B) all are true

(C) Only 2 is true (D) Only 1 and 3 are true

(68) A signal wave of frequency 12 KHz is modulated with a carrier wave of freaquency 2.51 MHz.The upper and lower side band frequency are respectively.

(A) 2512 Hz and 2508 KHz (B) 2522 KHz and 2488 KHz

(C) 2502 KHz and 2498 KHz (D) 2512 KHz and 2498 KHz

(69) A radio wave has $B_0 = 3.3 \times 10^{-12}\ J/m^3$. What is value of electric energy density ?

(A) $10^{-6}\ V/m$ (B) $10^{-4}\ V/m$ (C) $10^{6}\ V/m$ (D) $10^{4}\ V/m$ Atom and Nucleus

(70) What is the sepeturor sparing in are of co-arial cables ?

(A) 20 km (B) 2 km (C) 200 km (D) 600 km

(71) What is the M.U.F (maximum unuable frequency of an electromagnetic wave aretiacl frequency of $20MH_z$ and incident at an angle of 60° in ionospheric propagation ?

(A) $20\sqrt{2}\ MH_z$ (B) $20\ MH_z$ (C) $\frac{20}{\sqrt{2}}\ MH_z$ (D) $40\ MH_z$

(72) An insulater ared in a transmission line has dierectrie unstant equal to 0.25 What is the velocity factor of the transmission line ?

(A) $\frac{4}{3}$ (B) 2 (C) $\frac{3}{4}$ (D) 4

(73) What is the working principal of optical fibre ?

(A) Refraction (B) Total internal reflection

(C) Dispersion (D) Scattering

(74) For efficient transmission of a 100 MH_Z frequency wave, the minimum lenght of an antenna should be

(A) 3 m (B) $\frac{3}{4}$ m (C) 10 m (D) 100 m

(75) A photo director area light of wavelenght 1400nm Band gap of the Semiconductor used in the photo detector is _____ $(h = 6.63\times10^{-34} JS ; C = 3\times10^{8}\ m/s)$

(A) 0.7eV (B) leV (C) 2 eV (D) 2.5 eV

(76) In modulation procrss radio signal is called

(A) modulating wave (B) Carrier wave

(C) modulated wave (D) transmitting wave

(77) Tank Circuit Contains lnf capcpcitor and 10 microhenry inductor What is the carrier frequency generated by it ?

(A) 1629 MH_z (B) 1592 MH_z (C) 1582 MH_z (D) 156.2 MH_z

(78) The effective power radiated by an antenna varies

(A) inverscly as the lenght of antenna

(B) inverscly as the square of the length of the antenna

(C) directly as the lenght of antenna

(D) directly as the square of the lenght of the antenna

(79) A 1000 KH_z carrier is modulated with 800 KH_z audio signals What are the frequencies of first pair side banch ?

(A) 1000 KH_z and 800 H_z (B) 800 H_z and 999.2 KH_z

(C) 1000.8 KH_z and 999.2 KH_z (D) 1000.8 KH_z and 1000 KH_z

(80) The critical frequency of Flayer is 50 MH_z when angle of incidence is 74° What is the maximum unable frequency ?

(A) 181.5 MH_z (B) 191.5 MH_z (C) 199.5 MH_z (D) 171.5 MH_z

(81) The velocity of e.m. waves in a medium is $2.5\times10^{8}\ ms^{-1}$. what is the dielectric constant of the medium ?

(A) 1.2 (B) 1.44 (C) 0.69 (D) 1.6

(82) How long will it take to transmit the information 30km away in space ?

(A) 0.1 m s (B) 0.1 μ s (C) 0.1 n s (D) 0.1 ps

(83) What is the velocity of electromagnatic wave in a good conductor ?

(A) $3\times10^{8}\ m/s$ (B) more than $3\times10^{8}\ m/s$

(C) High (D) Very low

(84) If `fm' is the modulation frequency in Fm the modulation index is proportional to

(A) fm (B) f_m^2 (C) $\frac{1}{f_m}$ (D) $1/f_m^2$

(85) For 100% modulation, What is the value of minimum amplitude (E_{min}) of the modulated wave ?

(A) Zero (B) 1 (C) 2 (D) 3

(86) When electromagnetic waves enter the ionised layer of ionosphere, the relative permitting of the ionised layer

(A) remain constant (B) appears to decreure

(C) appears to increare (D) chenger randomly

(87) IN which of the following media the wave propagation occurs with out attenation ?

(A) Conductor (B) perfect dielecrei medium

(C) Semi conductor (D) None of the above

(88) The step index fibre has a relative refractive index difference of 0.86% what is the critical angle at the core cladding interface ?

(A) 28.4° (B) 82.4° (C) 24.8° (D) 75.6°

(89) A silicon optical fibre has a core refractive index of 1.50 and a cladding refractive inden 1.47 what is the numerical aperture of the fibre ?

(A) 0.90 (B) 0.60 (C) 0.45 (D) 0.30

(90) Which of the following alternative is not nature Noise ?

(A) Thunder of light (B) Radiation coming from sun.

(C) Radiation coming from stars (D) Radiation produced by flickering of tube light

(91) In optical fibre communication, the APD semi conductor divece is ared

(A) to convert radio single into optical single

(B) to obtain intanxty modulayed signal

(C) to covert optical signal into electrical signal

(D) to covert optical signal into electri signal.

(92) phase difference between space carrent and capacitive carrent is

(A) π rad (B) $\pi/2$ rad (C) $3\pi/2$ rad (D) μ rad

(93) Refractive index of ionosphere is

(A) one (B) more than one (C) less than one (D) Zero

KEYNOTE

1	D	26	D	51	A	76	A
2	B	27	D	52	D	77	B
3	A	28	A	53	C	78	B
4	B	29	D	54	A	79	C
5	B	30	A	55	A	80	A
6	A	31	C	56	C	81	B
7	D	32	C	57	A	82	A
8	A	33	C	58	C	83	D
9	D	34	A	59	A	84	C
10	D	35	D	60	C	85	A
11	C	36	A	61	C	86	B
12	C	37	B	62	A	87	B
13	A	38	A	63	D	88	B
14	B	39	C	64	D	89	D
15	A	40	D	65	B	90	D
16	A	41	B	66	B	91	C
17	C	42	B	67	A	92	A
18	B	43	C	68	D	93	C
19	B	44	D	69	B		
20	A	45	C	70	A		
21	B	46	A	71			
22	C	47	C	72	B		
23	C	48	D	73	B		
24	D	49	C	74	B		
25	A	50	C	75	B		

Unit - 21
Experiment

EXPERIMENT -1

Vernier Callipers

SUMMARY

The least count of venier callipers = M-V .....(1)

Where,

M = The distance between two consecutiue divisions on the main scale.

V = The distance between two consecutive divisions on the vernier scale.

The length of x divisions on vernier scale = the length of y divisions on the main scale.

$\therefore \quad xV = yM$

$$V = \frac{yM}{x}$$

From equation (1) The LC of vernier callipers =

$$= M - \left(\frac{yM}{x}\right) = M - \left(\frac{x\text{-}y}{x}\right)$$

$$= \frac{M}{x}$$

* **Zero Error :**

- When the jaws are made to touch eachother, the zero mark of the main scale and vernier scale may not be with the straight line.
- This gives rise to an error called the zero error.
- Positive zero erro = (No. of the vernier division coinciding with the main scale) X (L.C.)
- Corrected reading = Obeserved reading - Positive zero error.
- Negative Zero error = (No. of vernier division coinciding with the main scale) X (L.C.) 0 (Smallest main scale units)
- Corrected reading = Observed reading + Negavit zero error)

MCQ

1. What is the least count of the vernier callipers ?
   (A) Smallest division on the vernier scale.
   (B) differenceof the smallest division on the mai scale and the smallest division on the vernier scale.
   (C) sum of the smallest division on the main scale and the smallest division on the vernier scale.
   (D) smallest division on the main scale.
2. What is the least count of commonly available vernier ?
   (A) 0.01 cm (B) 0.001 cm
   (C) 0.0001 cm (D) 0.1 cm
3. When the zero mark on the vernier scale lies towards the left side of the zero mark of the main scale, when the jaws are connect, then what will be the zero error ?
   (A) zero error is positive (B) zero error is negative
   (C) zero correction is positive (D) zero error does not exist
4. When the zero mark on the vernier scale lies towards the right side of the zero mark of the main scale, when the jaws are in contact, then what will be the zero error ?
   (A) zero correction in positive (B) zero correction is negative
   (C) zero error in positive (D) zero error does not exist
5. If observed reading is OR, corrected reading is CR, zero error in ZE and zero correction in ZC, then what will be the possibility ?
   (A) CR = OR + ZC and ZE = CR-OR (B) CR = OR + ZE and ZC = CR-OR
   (C) CR = OR - ZC and ZE = OR-CR (D) CR = OR - ZE and ZC = CR-OR
6. When the jaws of a standard vernier are together, the 6th vernier scale division coincides with the 6th main scale division, then what in the zero error ?
   (A) -0.4 mm (B) + 0.6 mm
   (C) -0.6 mm (D) + 0.4 mm
7. When the jaws of a standard vernier are together, the 6th main scale division coincides with the 7th vernier scale division, then what is the zero error ?
   (A) -0.7 mm (B) +0.3 mm
   (C) -0.3 mm (D) +0.7 mm

8. In an anusual vernier, 9 vernier scale divisions coincide with 8 main scale division, then what is the least count of the vernier ?

(A) $\frac{8}{9}$ mm (B) $\frac{1}{9}$ mm

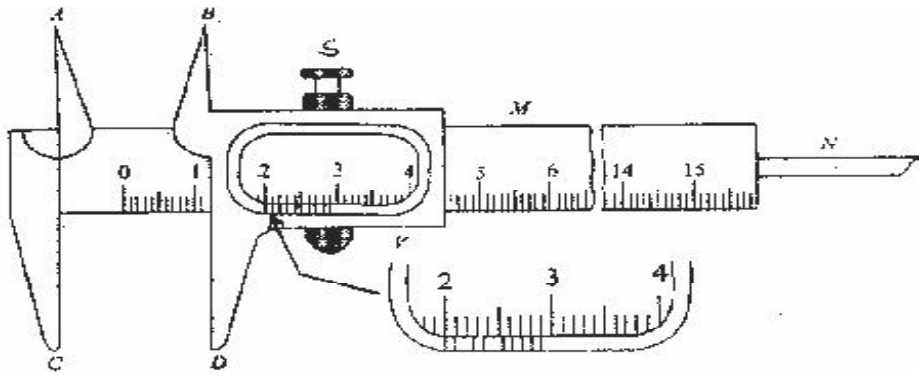
(C) $\frac{1}{17}$ mm (D) $\frac{1}{8}$ mm

9. In an unsual vernier, 10 vernier scale divisions, coinside with 8 main scale divisions, then what is the least count of the vernier ?

(A) 0.1 mm (B) 0.2 mm

(C) 0.8 mm (D) $\frac{1}{8}$ mm

10. Match the two columns :



Column - I	Column - II
(a) Jaws CD	(p) Slide and fix position of vernier scale
(b) Strip N	(q) depth of a calerimeter
(c) Screw S	(r) external diameter of a cylindrical vessel
(d) Jaws AB	(s) Internal diameter of a cylindrical vessel.

(A) a $\rightarrow$ r, b $\rightarrow$ q, c $\rightarrow$ p, d $\rightarrow$ s

(B) a $\rightarrow$ s, b $\rightarrow$ p, c $\rightarrow$ q, d $\rightarrow$ r

(C) a $\rightarrow$ r, b $\rightarrow$ q, c $\rightarrow$ s, d $\rightarrow$ p

(D) a $\rightarrow$ p, b $\rightarrow$ s, c $\rightarrow$ q, d $\rightarrow$ r

11. N divisions on the main scale of a vernier callipers coincides with (N+1) divisions on the vernier scale. If each division on the main scale is of a units, the least of count of instrument is.....

(A) $\frac{a}{N+1}$ (B) $\frac{a}{N-1}$

(C) $\frac{N+1}{a}$ (D) $\frac{N-1}{a}$

12. The edge of a cube is measured using a vernier caliper (9 divisions of the main scale is equal to 10 divisions of vernier scale and 1 main scale division is 1mm). The main scale division reading is 10 and 1 division of vernier scale was found to be coinciding with the main scale. The mass of the cube is 2.736 g. What will be the density in $\frac{g}{cm^3}$ upto correct significant figures ?

(A) $2.66 \times 10^{-3} \frac{g}{cm^3}$ (B) $2.66 \times 10^{3} \frac{g}{cm^3}$

(C) $2.66 \frac{g}{cm^3}$ (D) $2.66 \times 10^{-6} \frac{g}{cm^3}$

KEY NOTE

1 (C)	2 (A)	3 (A)	4 (A)
5 (D)	6 (B)	7 (C)	8 (B)
9 (B)	10 (A)	11(A)	12 (C)

HINT

1. If main scale division = M and
vernier scale division = V
9M = 10 V
(10-1) M = 10V
10M - M = 10V

$10\,(M - V) = M,\ (M - V) = \frac{M}{10} = \text{least count.}$

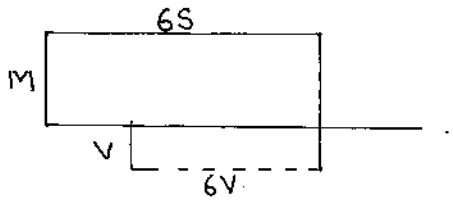
2. Least count $= \frac{M}{n} = \frac{1\,mm}{10} = 0.1\ mm = 0.01\,cm$
3. Here zero error is positive, so zero. correction is negative, as zero correction = - (zero error)
4. Here zero error is negative, so zero correction is positive.
5. Corrected reading = Observed reading - zero error

$\Rightarrow$ CR = OR − ZE

$\Rightarrow$ CR = OR + ZC

$\Rightarrow$ ZC = CR − OR

6. As vernier divisions are smaller then the main scale dividions (9M = 10V), the zero of the vernier must be on the right side of the zero of the main scale. Here zero error is positive.



$\Rightarrow$ Zero error $= 6M - 6V = 6\,(M - V) = 6\,(0.1\text{ mm}) = +\,0.6\,\text{mm}$

7. zero error $= 6M - 7V = (7M - 7V) - M$

$= 7\,(0.1\text{ mm}) - 1\text{mm} = -\,0.3\text{ mm}$

8. $8M = 9V \Rightarrow 9M - M = 9V$

$9\,(M - V) = M$

or $(M - V) = \dfrac{M}{9} = \dfrac{1}{9}\text{ mm}$

9. $8M = 10V \Rightarrow 10M - 2M = 10V,\ 10(M - V) = 2M$

$(M - V) = \dfrac{2M}{10} = 0.2\text{ mm}$

11. (N+1) divisions on the vernier scale = N divisions on main scale

$\therefore$ 1 division on vernier scale $= \dfrac{N}{N+1}$ divisions on main scale

Each division on the main scale in of a units

$\therefore$ 1 division on vernier scale $\left(\dfrac{N}{N+1}\right)$ a units = a' (say)

Least count = 1 main scale division -1 vernier scale division

$$= a - a' = a - \left(\frac{N}{N+1}\right)a = \frac{a}{N+1}$$

12. 1 MSD = 1 mm
9 MSD = 10 VSD
$\therefore$ Least count,

$$LC = 1MSO - 1VSO = 1mm - \frac{9}{10}\text{ mm}$$

$$= \frac{1}{10}\text{ mm}$$

Measure reading of edge $= \text{MSR} + \text{VSR}\ (\text{LC})$

$$= 10 + 1 \ \frac{1}{10} = 10.1 \ \text{mm}$$

Volume of cube $V = (101)^3\ \text{cm}^3 = 1.03\ \text{cm}^3$

[After rounding off upto 3 significant digits, as edge length is measured upto 3 significant digits]

Density of cube $= \dfrac{2736}{1.03} = 2.6563\ \dfrac{\text{g}}{\text{cm}^3} = 2.66\ \dfrac{\text{g}}{\text{cm}^3}$

(After rounding off to 3 significant digits)

EXPERIMENT -2

Micrometer screw gauge :

SUMMARY

- Least count of screw gauge = $\frac{\text{pitch}}{\text{Total number of divisions on circular scale}}$
- Negative zero error :
  If the zero error is negative and the nth division of the circular scale coincides with the line of graduation then
  $$z = (n - 100) \times LC$$
  corrected reading = observed reading + zero correction
- If the zero error is positive and the nth division of the circular scale division concides with the line of graduation then
  $$z = n \times LC$$
  corrected reading = observed reading - zero error.
- If the zero mark of the venier upwards of the line of the graduation then megative zero error arise and zero mark of the verniers downwards then positive zero error aris.

MCQ

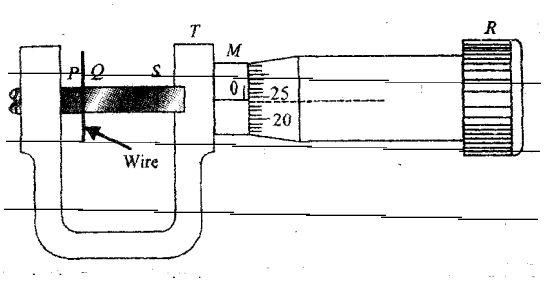
1. When the edge of the circular scale lies to the left of O mark on the main scale, when the stud and spindle touch each other, Then what will be the zero error ?
   (A) zero error is negative (B) zero error is positive
   (C) zero error does not exist (D) zero correction is negative
2. Whe the edge of the circular scale lies to the right of the O mark on the main scale, when the stud and the spindle touch each other,then what will be the zero error ?
   (A) zero correction is negative (B) zero error is negative
   (C) zero error does not exist (D) zero correction is positive
3. When the screw and stud touch each other, the edges of a certain screw gauge is on left of the O mark on the main scale and the 96th division of the circular scale coincides with the circular line of graduation then what is the value of zero error ?
   (A) zero error = + 0.96 mm (B) zero error = – 0.96 mm
   (C) zero error = + 0.04 mm (D) zero error = – 0.04 mm
4. When the screw and stud touch each other, the edge of a certain screw gauge is to the right of the O mark on the main scale and 5th division of the circular scale coincides with the line of graduation, then what is the value of zero error ?
   (A) zero error = + 0.95 mm (B) zero error = – 0.95 mm
   (C) zero error = + 0.05 mm (D) zero error = – 0.05 mm

5. Screw guage A has a pitch of 1 mm and 50 division on its circular scale screw guage B has a pitch of 0.5 mm and 100 divisions on its circular scale. If (LxC) is least count, then which posibility is true ? What is the parilrility ?

(A) $2\left(L \times C\right)_A = \left(L \times C\right)_B$ (B) $\left(L \times C\right)_A = 4\left(L \times C\right)_B$

(C) $\left(L \times C\right)_A = \left(L \times C\right)_B$ (D) $\left(L \times C\right)_A = 2\left(L \times C\right)_B$

6. The screw gauge shown above has a zero error of -0.02 mm and 100 divisions on the circular scale. What is the diameter of wire ?



(A) 0.28 mm (B) 0.22 mm

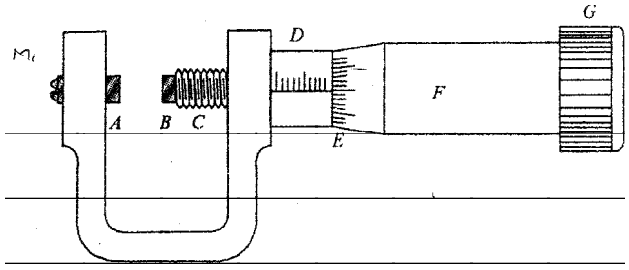
(C) 0.24 mm (D) 0.26 mm

7. The pitch of screw gauge is 1 mm and there are 100 divisions on the circular scale. What measuring the diameter of a wire, the linear scale reads, 1 mm and 47th division on the circular scale coincides with the reference line. The length of the wire is 5.6 cm. What will be the curved surface area (in cm^2) of the wire in appropriate number of significant figures.

(A) 2.6 cm^2 (B) 2.5848 cm^2

(C) 2.585 cm^2 (D) 2.5 cm^2

8. Match column A and B



Column - I	**Column - II**
(a) A	(p) Main scale
(b) B	(q) circular scale
(c) C	(r) stud
(d) D	(s) spindle
(e) E	(t) Ratchet
(f) E	(u) Thimble
(g) G	(v) Screen

(a) $a \rightarrow r, b \rightarrow s, c \rightarrow v, d \rightarrow p, e \rightarrow q, f \rightarrow u, g \rightarrow t$

(b) $a \rightarrow u, b \rightarrow t, c \rightarrow v, d \rightarrow p, e \rightarrow q, f \rightarrow r, g \rightarrow s$

(c) $a \rightarrow v, b \rightarrow p, c \rightarrow r, d \rightarrow s, e \rightarrow t, f \rightarrow q, g \rightarrow u$

(d) $a \rightarrow p, b \rightarrow q, e \rightarrow u, d \rightarrow t, e \rightarrow r, f \rightarrow s, g \rightarrow v$

KEY NOTE

1 (A)	2(A)	3 (D)	4(C)
5 (B)	6 (D)	7(A)	8(A)

HINT

2. Here zero error is positive, so zero correction is negative.
3. Here the zero error is negative.

   $Z = (96 - 100) \times LC = -4 \times 0.01 \text{ mm} = -0.04 \text{ mm}$
4. Here the zero error is positive

   $z = 5 \times L.C = 5 \times 0.01 \text{ mm} = +0.05 \text{ mm}$
5. Least count $= \dfrac{\text{Pithc}}{\text{No. of divisions on circular scale}}$
6. The observed reading is 0.24 mm. The corrected reading = observed reading - zero error.

   $= 0.24 \text{ mm} - (-0.02 \text{ mm}) = 0.26 \text{ mm}$
7. Curved surface area = $2\pi rl$

Experiment - 3

SIMPLE PENDULUM

SUMMARY

- Simple pendulum oscillation consider as a undamped oscillation.

  periodic time $T = 2\pi \sqrt{\frac{l}{g}} = \frac{2\pi}{\omega}$

  Mechanical energy $E = \frac{1}{2} k \left(A_0\right)^2$

- Simple pendulum oscillation consider as a damped oscillatien,

  Periodic time $T = 2\pi \sqrt{\frac{I}{mgl}} = 2\pi \sqrt{\frac{\frac{2}{5}mr^2 + ml^2}{mgl}} \quad (I = M.I)$

  Here $T > 2\pi \sqrt{\frac{l}{g}}$

- Mechanical energy $E = \frac{1}{2} kA_0^2 \left(e^{\frac{-bt}{2m}}\right)^2$

  displacement $x = A_0 e^{\frac{-bt}{2m}} \cos\left(\omega' t + \varphi\right)$ where $\omega' = \sqrt{\frac{k}{m} = \frac{b^2}{4m^2}}$

MCQ

1. Complete the following sentance.
   Time period of oscillatin of a simple pendulum is dependent on.....
   (A) Length of thread (B) initial phase
   (C) amplitude (D) mass of bob
2. Complete the following seutence
   In a damped oscillation of a pendulam .....
   (A) the sum of potential energy and kinetic energy is conserved.
   (B) mechanical energy is not conserved
   (C) the kinetic energy is conserved
   (D) the potential energy is conserved

3. If the time period of undamped oscillation is T and that of damped oscillatin is T^1 ,then what is the relation between T &

(A) $T^1 < T$ (B) nothing can be said, unless the drag force constant is known.

(C) $T^1 = T$ (D) $T^1 > T$

4. The energy dissiplated in a damped oscillation .....

(A) decays exponentially

(B) decay curve will depend on the drag constant

(C) decays linerly

(D) decays following a sine curve with diminishing amplitude.

5. What is the equatin for a damped oscillator, where k and b are constants and x is displacement.

(A) $\frac{md^2x}{dt^2} = kx + \frac{bdx}{dt} = 0$ (B) $\frac{md^2x}{dt^2}$ x kx = $\frac{bdx}{dt}$

(C) $m\frac{d^2x}{dt^2} = kx + \frac{bdx}{dt}$ (D) $\frac{md^2x}{dt^2}$ - kx = b $\frac{dx}{dt}$

6. In a damped oscillatin with damping constant b. The time taken for amplitude of oscillatin to drop to half what is its initial value ?

(A) $\frac{b}{m}$ ln 2 (B) $\frac{b}{2m}$ ln 2

(C) $\frac{m}{b}$ ln 2 (D) $\frac{2m}{b}$ ln 2

7. In a damped oscillatin with damping constant b. The time taken for its mechanical energy to drop to half. What is its value ?

(A) $\frac{b}{m}$ ln 2 (B) $\frac{b}{2m}$ ln 2

(C) $\frac{m}{b}$ ln 2 (D) $\frac{2m}{b}$ ln 2

8. For a pendulum in undamped oscillation, with a bob of mass m and radius r, with a string of length l What is the time period ?

(A) $T = 2\pi \sqrt{\frac{l}{g}}$ (B) T depends on m

(C) $T > 2\pi \sqrt{\frac{l}{g}}$ (D) $T < 2\pi \sqrt{\frac{l}{g}}$

9. In the experiment of simple pendulum, we have taken a thread of 140 cm, and an amplitude of 5 cm to begin with. Here θ to begin with is about.....

(A) 5° (B) 8°

(C) 2° (D) 3°

10. In the experiment we of simple pendulum keep $\theta < 5^o$, so as ensure.....

(A) mass m does not interfere in the time period

(B) g remains constant

(C) the air drag is not too much

(D) $\sin\theta \cong \theta$ where by motion becomes simple harmonic.

KEY NOTE

1 (A)	2 (B)	3 (D)	4 (A)	5 (A)
6 (D)	7 (C)	8 (C)	9 (C)	10 (D)

HINT

1. $T \propto \sqrt{\ell}$
2. knowledge base
3. $T \propto \frac{1}{\omega}$ $\omega = \sqrt{\frac{k}{m}} \Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$

   where as $T' \propto \frac{1}{\omega'}$ where $\omega' = \sqrt{\frac{k}{m} = \frac{b^2}{4m^2}}$

   as $\omega' < \omega \Rightarrow T' > T$
4. $E = E_0\ e^{-\lambda t}$
5. $F = -kx - bv \Rightarrow m\frac{d^2x}{dt^2} + \frac{bdx}{dt} + kx = 0$
6. $A = A_0 e^{\frac{-bt}{m}}$, $t = T_{\frac{1}{2}}$ when, $A = \frac{A_0}{2}$
7. $E = E_0 e^{\frac{-bt}{m}}$ when $E = \frac{E_0}{2}$, $t = T_{\frac{1}{2}}$

8. For an ideal pendulum, $r \to 0$, but for a finite r,

$$T = 2\pi \sqrt{\frac{I}{mgl}} = 2\pi \sqrt{\frac{\frac{2}{5} mr^2 + ml^2}{mgl}}$$

$$T = 2\pi \sqrt{\frac{l}{g}} \text{ where } r \to 0$$

otherwise, $T > 2\pi \sqrt{\frac{l}{g}}$

9. $\theta = \frac{l}{r} = \frac{5cm}{140\ cm} = \frac{1}{28}\ rad = 2.05^o \cong 2^o$

10. IF $\sin\theta \cong \theta$

$F = -mg \sin\theta = -mg\theta$

$\Rightarrow F = -\left(\frac{mg}{l}\right). x$

or $F \propto -x$

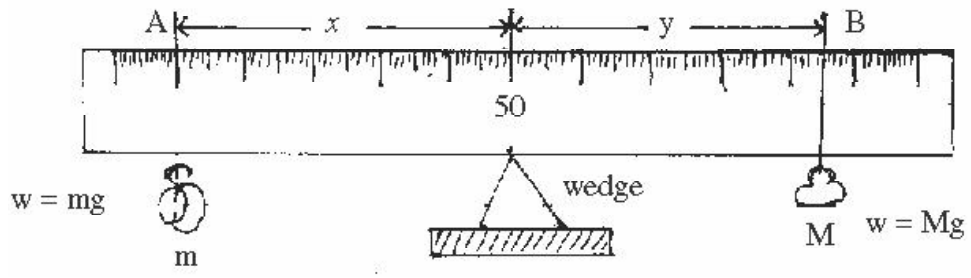
Experiment - 4

METRE SCALE

SUMMARY

Mass of Given Object by the Principle of Moments

A metre rod is supported at its centre of gravity on a wedge as shwon,



As unknown mass m is suspended from the left arm at a distance x and a known standard mass. M is suspended from the right arm at a distance y. The distances are measured from the knife edge. For equilibrium, the moments of forces about the fulcrum or knife edge is equal, so we have,

$$mg \times x = Mg \times y \Rightarrow m = M\left(\frac{y}{x}\right)$$

MCQ

1. When the moment of force is maximum, then what is the angle between force and position vector of the force ?

   (A) 90° (B) 0°

   (C) 30° (D) 45°

2. In the scale and wedge experiment, m situated at x from the wedge and M is situated at y from wedge at equillbrium.Now if m < M, then what is the relation between x and y ?

   (A) x = y

   (B) x < y

   (C) x > y

   (D) nothing can be said about - *x* and y unless values of m and M one know.

3. When the "wedge and scale" experiment is performed at the equator, we get $m = M\left(\frac{y_e}{x_e}\right)$. If the same experiment is performed at the poles, then what is the wright equation ?

(A) $m = M\left(\frac{y_e}{x_e}\right) \cdot \left(\frac{R_e}{R_p}\right)$ (B) $m = M\left(\frac{y_e}{x_x}\right) \cdot \left(\frac{R_p}{R_e}\right)$

(C) $m = M\left(\frac{y_e}{x_x}\right) \cdot \left(\frac{R_p}{R_e}\right)^2$ (D) $m = M\left(\frac{y_e}{x_x}\right)$

Where Re and Rp are the equatorial and polar radius of the earth.

4. A force $2\hat{i} + 3\hat{j}$ acts about an axis at a position vector $\left(\hat{j} + \hat{k}\right)$ from the axis, then what is the torque due to the force about the axis ?

(A) $\sqrt{19}$ Nm (B) $\sqrt{13}$ Nm

(C) $\sqrt{15}$ Nm (D) $\sqrt{17}$ Nm

5. In the experiment of balancing moments, suppose the fulcrum is at the 60 cm mark, and a known mass of 2 kg is used on the longer arm. The greatest mass of m which can be balanced against 2 kg such that the minimum distance of either of the masses from the fulcrum is atleast 10 cm. (Neglect mass of metre scale.) What will be the value of m ?

(A) 12 kg (B) 4 kg

(C) 6 kg (D) 8 kg

6. The wedge is kept below the 60 cm mark on the meter scale. Known masses of 1 kg and 2 kg are hung at the 20 cm and 30 cm mark respectively. Where will a 4 kg mass be hung on the meter scale to balance it ? (Neglect mass of meter scale.)

(A) 85 cm (B) 90 cm

(C) 70 cm (D) 75 cm

7. When a metre scale is balanced above a wedge, 1 kg mass is hung at 10 cm mark and a 2 kg mass is hang at the 85 cm mark. To which mark on the meter scale, the fulcurm be shifted (Neglect mass of meter scale) to balance the scale ?

(A) 70 cm (B) 50 cm

(C) 60 cm (D) 65 cm

KEY NOTE

1(A)	2 (C)	3 (D)	4 (D)
5(A)	6 (A)	7 (C)	

HINT

1. $\tau = Fr\sin\theta$

   $(\sin\theta)_{max} = 1$ for $\theta = 90^o$

2. $m = M\left(\frac{y}{x}\right)$

3. Where we take moments of force at equilibrium, the terms of g cancel on both sides.

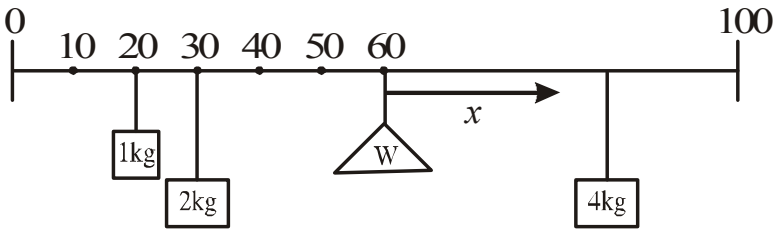
   $mg.x_e = Mg.y_e$

   $m = M.\left(\frac{y_e}{x_e}\right)$

   The mass is indipendent of the grvity at that point ; so mass at the poles will be same as mass at the equator.

4. $\left|\vec{\tau}\right| = \left|\vec{r} \times \vec{F}\right|$

5. $m_{max} = (2kg)\left(\frac{y_{max}}{x_{min}}\right) = 12\,kg.$



6. The anit -clockwise moments due to 1 kg and 2 kg are

   $= (2\text{ kg wt})\ (60\text{-}20)\text{ cm} + (2\text{kg wt})\ (60\text{-}30)\text{ cm}$

   $= (1\text{ kg wt})\,(40)\text{cm} + (2\text{kg wt})\,(30)\text{cm} = 100\text{ kg wt cm.}$

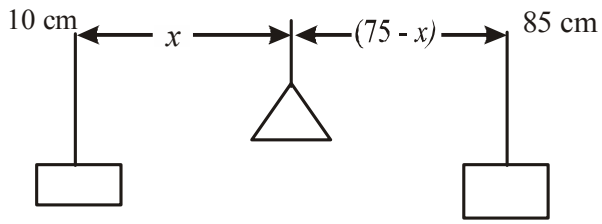
   The clock wise moment due to 4 kg = 4 kg. ωt x x cm

   $\Rightarrow 100 = 4x$ or $x = 25$ cm

   So the 4 kg mass must be hung at (60 cm + x) = (60 cm + 25 cm)

   = 85 cm mark to balance the scale

7.



Balancing moment $1x = 2\ (75 - x) \Rightarrow 3x = 150$ or $x = 50$ cm

Fulcrum is at 10 cm + 50 cm = 60 cm mark.

Experiment - 5

To determine Young's modulus of elasticity of the material of a metalic wire.

Where a wire of length L is stratched by a force F. Its cross section is A and it extends by ***l.***

SUMMARY

Young's Modulus $Y = \frac{F.L}{A.l} = \frac{mgL}{\pi r^2 l}$

Where r is radius wire

MCQ

1. Arrange rubber, steel and glass in the order of decreasing elasticity.
   (A) glass, steel, rubber (B) rubber, glass, steel
   (C) glass, rubber, steel (D) steel, glass, rubber
2. The following wires A, B, C, D, are made of the same material with diff. length and diameter. Which of these will have the largest extension, when the same tension is applied ?
   (A) length = 200 cm, diameter = 2 mm (B) length = 50 cm, diameter = 0.5 mm
   (C) length = 300 cm,diameter = 3 mm (D) length = 100 cm, diameter = 1 mm
3. The compressibility of a substance equals.....
   (A) $\frac{\Delta V}{PV}$ (B) $\frac{P \Delta V}{V}$
   (C) $\frac{V}{P\Delta V}$ (D) $\frac{PV}{\Delta V}$
4. Two rods of different materials having coefficients of thermal expansions $\propto_1$, $\propto_2$ and Young's moduli y_1, y_2 respectively are fixed between two rigid walls. The rods are heated such that they undergo the same increase in temperature. There is no bending of the rod. if $\alpha_1 : \alpha_2 = 2 : 3$ the thermal stresses developed in the rod are equal provided $y_1 : y_2$ equals.
   (A) 3:2 (B) 2:3
   (C) 4:9 (D) 1:1
5. A uniform rod of length L and density ρ is being pulled along A smooth floor with a horizontal acceleration α what is the magnitude of the stress at the trausverse cross-section through the mid-point of the rod ?
   (A) $\frac{1}{3} L\rho\,\alpha$ (B) $L\rho\,\alpha$
   (C) $\frac{2}{3} L\rho\,\alpha$ (D) $\frac{L\rho\,\alpha}{2}$

6. A solid cylindrical steel column is 4 m long and 9 cm in diameter . $\left(Y_{steel} = 1.9 \times 10^{11}\ Nm^{-2}\right)$. The decrease in length of the column, while carrying a load of 80000 kg is.....

(A) 3.2 mm (B) 1.8 mm

(C) 4.4 mm (D) 2.6 mm

7. A solid sphere of radius R made of a material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass M is placed on the pisten to compress the liquid, the fractional change in the radius of the sphere $\frac{\delta R}{R}$ is.....

(A) $\frac{Mg}{3KA}$ (B) $\frac{Mg}{KA}$

(C) $\frac{Mg}{4KA}$ (D) $\frac{Mg}{2KA}$

8. In a Searle's experiment, the diameter of the wire as measured by a screw gauge of least count 0.1 m, is 110.0 cm. When a weight of 50N is suspended from the wire, the extension is measured to 0.125 cm by a micrometer of least count 0.001 cmthe maximum error in the measurement of Young's modulus of the material of the wire is.....N/m^2 ?

(A) 1.09×10^{12} (B) 1.09×10^{10}

(C) 3.09×10^{10} (D) 3.09×10^{12}

KEY NOTE

1 (C)	2 (B)	3 (A)	4 (A)
5 (D)	6 (D)	7 (A)	8 (B)

HINT

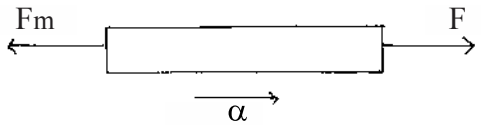
2. $y = \frac{F \cdot L}{Al} \Rightarrow l = \frac{F \cdot L}{\left(\frac{\pi d^2}{4}\right) y}$ so $l \propto \left(\frac{L}{d^2}\right)$

$l_A > l_B > l_C > l_D$

3. $\frac{1}{k} = \frac{\Delta V}{PV}$

4. $\left(\frac{F}{A}\right) = y \left(\frac{\ell}{L}\right) = y \propto \Delta T$

5. $F = M\alpha = (LA\rho)\alpha$



$$F - Fm = \left(\frac{LA\rho}{2}\right)\alpha \quad \Rightarrow F_m = \left(\frac{LA\rho}{2}\right)\alpha$$

$$\text{stress at mid-point} = \frac{Fm}{A} = \left(\frac{L\rho\alpha}{2}\right)$$

6. $y = \frac{FL}{Al}$

7. $$K = \frac{p}{\left(\frac{dV}{V}\right)} = \frac{Mg}{A\left(\frac{dV}{V}\right)} \Rightarrow \frac{dV}{V} = \frac{Mg}{KA}$$

Now, $V = \frac{4}{3}\pi R^3$

$$\ln V = \ln\left(\frac{4}{3}\pi\right) + 3 \ln R$$

Differentiatin $\frac{dV}{V} = \frac{3dR}{R} = \frac{Mg}{KA} \Rightarrow \frac{dR}{R} = \frac{mg}{3KA}$

8. Young's modulus of elasticity is given by

$$Y = \frac{Stress}{Strain} = \frac{F/A}{l/A} = \frac{FL}{LA} = \frac{FL}{l\left(\frac{\pi d^2}{4}\right)}$$

Substituting the values, we get

$$Y = 2.24 \times 10^{11} \frac{N}{m^2}$$

Now $\frac{\Delta y}{y} = \frac{\Delta L}{L} + \frac{\Delta l}{l} + 2\frac{\Delta d}{d}$ substituting the values,

$$\frac{\Delta y}{y} = 0.0489$$

Experiment - 6

Determine the surface tension by capillary rise method & ifect of detergents.

SUMMARY

- Weight of meniscus = $\left(\frac{\pi \ r^3}{3}\right)\rho g$ (if r is small)

  Where, ρ is the density of liquid.
  r is the radius of capillary

  Surface tension $T = \frac{rh\rho g}{2 \ \cos\theta}$

  $T \propto r\,h$ (ρ, θ & g are constant) ρ = density

  h = height of liquid

  θ = angle of contact

  Pressure different in liquid drop,

  $$P_i - P_o = \frac{2T}{R}$$ where R ; radius of drop.

  Pressure different in ditergen bubble,

  $$P_i - P_o = \frac{4T}{R}$$

MCQ

1. The amount of energy evolved when eight droplets of mercury (surface tension 0.55 Nm^{1}) of radius 1 mm each combine into one drop is .....

   (A) 18 μJ (B) 24 μJ

   (C) 28 μJ (D) 16 μJ

2. To what height can mercury be filled in a ressel without any leakage if there is a pin hole of diameter 0.1 mm at the bottom of the vessel. (Density of mercury = 13.6×10^3 kg m^{-3} , surface tension of mercury = $550 \times 10h^{-3}$ ρNm^{-1} , Angle of contact with the vessel for mercury = 0°)

   (A) 16.5 cm (B) 18.5 cm

   (C) 20 cm (D) 12.5 cm

3. A spherical soap bubble of radius 2 cm attached to the outside of a spherical bubble of radius 4 cm. Then what is the radius of the common surface ?

   (A) 3.5 cm (B) 4 cm

   (C) 4.5 cm (D) 3 cm

4. By how much depth will the surface of a liquid be depressed in a glass tube of radius 0.2 mm if the angle of contact of the liquid is 135^0 and the surface tension is 0.547 Nm^{-1} ? (Density of the liquid is kg m^{-3})

(A) 3 cm (B) 4 cm

(C) 5 cm (D) 2 cm

5. Complete the sentence

If the height of a capillary is smaller than the height to which water should rise, then.....

(A) Water stops gets depressed in the capillary below the water surface.

(B) Water rises up to the height of the capillary and meniscus becomes less concave

(C) Water does not rise in such a capillary at all.

(D) Water states flowing out like a fountain.

KEY NOTE

1 (C) 2 (A) 3 (B) 4(A) 5 (B)

HINT

1. Using the formula

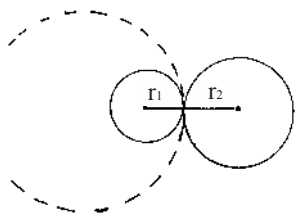
$W = T\Delta A$

2. Mercury will start leaking when $\rho gh = \frac{2T}{r}$

$$h_{min} = \frac{2T}{\rho gr} = \frac{4T}{\rho\ g\ d}$$

3. $$P_2 - P_1 = 4T\left(\frac{1}{r_2} - \frac{1}{r_1}\right) = \frac{4T}{r}$$

$$\frac{1}{r} = \frac{1}{r_2} - \frac{1}{r_1} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4} \Rightarrow r = 4 \text{ cm}$$



4. $$h = \frac{2T\cos\theta}{\rho.\ g.\ r}$$

5. $$T = \frac{h\rho\ g\ r}{2\ \cos\ \theta}$$

If h is not sufficient, then θ changes h takes on the value of the length of the tube.

Experiment - 7

Coefficient of viscosity of a given viscos liquid by measuring terminal velocity of a given spherical body.

SUMMARY

Stoke's force $F = 6\pi\eta r v$

Where,
η = coefficient of viscocity
r = radius of sphere
v = velocity of spheres

$$\eta = \frac{2}{9} r^2 g \left(\frac{\rho - \sigma}{v_t} \right)$$

Where ,
η = viscosity of liquid
r = radius of sphere
ρ = density of sphere
σ = density of liquid
v_t = terminal velocity

Correction of terminal velocity

$$\upsilon_t = \upsilon_0 \left[1 + \frac{2.4r}{R} \right]$$ Where, R = Radius of vessel

v_0 = velocity of sphere in viscous fluid in vessels.

MCQ

1. The velocity of a small ball of mass m and density d_1 when dropped in a container filled with glycerine becomes constants after sometime. What is the viscous force acting on the ball ? (density of glyeerine is d_2)

(A) $mg\left(1 - \frac{d_1}{d_2}\right)$ (B) $mg\left(1 - \frac{d_2}{d_1}\right)$

(C) $mg\left(\frac{d_1}{d_1 - d_2}\right)$ (D) $mg\left(\frac{d_2}{d_1 - d_2}\right)$

2. A boat of area 10 m^2 floating on the surface of a river is made to move horizontaly with a speed of 2 ms^{-1} by applying a tangential force. If the river is 1m deep and the water in contact with the bed is stationary, what is the tangential force needed to keep the boat moving ? (viscosity of water 0.01 poise)

(A) 0.04 N (B) 0.05 N

(C) 0.02 N (D) 0.03 N

3. A steeel ball of diameter 3 mm falls through glycerine and covers a distance of 25 cm in 10S. The specific gravity of steel and glycerine are 7.8 and 1.26 respectively. The viscosity of glycerine is about..... pa^{-s}

(A) 1.3 (B) 1.5

(C) 0.8 (D) 1.0

4. A steel ball of diameter 3.2 mm falls under gravity through an oil of density 920 kg m^3 and viscosity 1.64 Ns m^{-2}. The density of steel. may be taken as 7820 kg m^3. What is the terminal velocity of the ball ?

(A) 3.45 cms^{-1} (B) 4.25 cms^{-1}

(C) 1.85 cm^{-1} (D) 2.35 cms^{-1}

5. 64 equal drops of water are falling through air with a steady velocity V_O, if the drops coalesce, what is their new velocity ?

(A) $4\ v_0$ (B) $2^{-\frac{1}{3}}\ v_0$

(C) $2\ v_0$ (D) $2^{\frac{1}{3}}\ v_0$

6. The C.G.S. unit of coefficient of viscosity is poise andthe SI unit is Pa-S. What is the relation between the two ?

(A) 1 Pa-s = 10^3 poise (B) 1 Pa-s = 10^4 poise

(C) 1 Pa-s = 10 poise (D) 1 Pa-s = 100 poise

7. The terminal velocity of fall of lead shots in the cylindrical vessel is influenced by the nearness of the walls of the vessel. Hence the velocity in formula is divided by a factor $\left(1 + \frac{2.4r}{R}\right)$. Where r is the radius of the lead shots and R the radius of the vessel. This is called .....

(A) Ladnberg correction (B) Velocity correction

(C) End-correction (D) Rydberg correction

8. A bubble of air 2 mm in diameter rises in a liquid of viscocity 0.075 SI units and density 1350 kg m^{-3}. The terminal velocity of the bubble is about..... m/s

(A) 4×10^{-2} (B) 5×10^{-2}

(C) 2×10^{-2} (D) 3×10^{-2}

KEY NOTE

1 (B)	2 (C)	3 (A)	4 (D)
5 (A)	6 (C)	7 (A)	8 (A)

HINT

1. Viscous force = Weight - Buoyant force

$$= Vd_1g - Vd_2g = Vd_1g\left[1 - \frac{d_2}{d_1}\right] = mg\left(1 - \frac{d_2}{d_1}\right)$$

2. Velocity gradient = $\frac{dv}{dx}$

$$F = -\eta A \frac{dv}{dx}$$

Tangential force is equal and opposite to the viscous force

3. $\eta = \frac{2r^2}{9}\frac{(\rho - \sigma)\, g}{v_1}$

4. $v_t = \frac{2}{9}\frac{r^2\ (\rho - \sigma)\ g}{\eta}$

5. $v = kr^2$

When drops coalesce, their volume is constant

$R^3 = 64r^3$ or $R = 4r$

$$\frac{v}{v_0} = \frac{R^2}{r^2}$$

8. $v_t = \frac{2}{9}\left(\frac{r^2\ (\rho - \sigma)\ g}{\eta}\right)$

Experiment - 8

Newton's Law of cooling :

SUMMARY

- The rate of loss of heat by the body,

$$= -\frac{dH}{dt} \propto \theta - \theta_0$$

$$[\theta > \theta_0]$$

$$\therefore -\frac{dH}{dt} = k\ (\theta - \theta_0)$$

Where,

θ = temperature of body

θ_0 = temperature of surrounding

$H = msd\theta$ where,

m = mass of body

s = specific heat

$d\theta$ = difference of temperature in body

- $\frac{d\theta}{(\theta - \theta_0)} = -\frac{k}{ms}\,dt$

$\therefore \ln(\theta - \theta_0) = -\left(\frac{-k}{ms}t + c\right)$ where, c = constant of integration

Emissive power,

$E = \in \sigma AT^4$ where,

$\in$ = emissivity of body

σ = stefan's constant

- Nature of graph $(\theta - \theta_0) \rightarrow t$ is exponentially decreasing with time
- Nature of graph $\frac{d\theta}{dt} \rightarrow \theta$ is straight line with positive slope.
- Nature of graph $\ell n\,(\theta - \theta_0) \rightarrow t$ is straight line with negative slope.

MCQ

1. A liquid takes 5 minute to cool from 80^0 C to 50^0 C. The temperature of the surrounding is 20^0C. What is the time it will take to cool from 60^0C to 30^0C ?

   (A) 9 min (B) 5 min

   (C) 12 min (D) 6 min

2. Two spheres of the same materialhave radii 1 m and 4 m and temperatures 2000 k and 4000 k respectively. If the energy radiated by the spheres are E_1 and E_2 respectively then find ratio of $\frac{E_1}{E_2}$

   (A) 1:256 (B) 256:1

   (C) 16:1 (D) 1:16

3. A body cools in 5 minute from 60^0 C to 40^0 C.The temperature of the surroundings is 10^0 C. What is its temperature after the next 5 minute ?

   (A) 32^0 C (B) 24^0 C

   (C) 28^0 C (D) 30^0 C

4. What is the units of emissive power in stefan's law ?

   (A) Ws^{-1} (B) W

   (C) Ws^{-1} m^{-2} (D) Wm^{-2}

5. A sphere, a cube and a thin circular plate are all made of the same material, have the same mass and are initially heated to a temperature of 200^0C.Arrange then in the ascending order of rate of cooling.

   (A) sphere, disc, cube (B) sphere, cube, disc

   (C) cube, sphere, disc (D) disc, cube, sphere

KEY NOTE

1 (A) 2 (D) 3 (C) 4 (D) 5(B)

HINT

1. Using the equation

$$\frac{\theta_1 - \theta_2}{t} = K\left(\frac{\theta_1 + \theta_2}{2} - \theta_0\right)$$ $\theta_0 =$ where, = tempreture of surrounding

2. $$\frac{E_1}{E_2} = \left(\frac{R_1}{R_2}\right)^2 \left(\frac{T_1}{T_4}\right)^4$$

3. According to newton's law approximately,

$$\frac{\theta_1 - \theta_2}{t} = K\left(\frac{\theta_1 + \theta_2}{2} - \theta_0\right)$$

$160 - 4\theta = 20 + \theta \Rightarrow \theta = 28^0$ C

4. Emissive power is defined as the radiant energy emitted per sec per unit area of the surface.

Hence $[E] = \left[\frac{P}{A}\right] \Rightarrow$ Unit of $E = Wm^{-2}$

5. $\frac{d\theta}{dt} \propto A$ now, for a given mass,sphere has the least surface area and circular plate will have the maximum surface area. Hence the sphere will cool the slowest and the disc the fastest.

Experiment - 9

Speed of sound in air at room temperature using a resonance tube.

SUMMARY

If resonance tube is one side open & one side close.

For first harmonic, $\frac{\lambda}{4} = l_1 + \in$ where, $\in$ is the end correction

For second harmonic & first overton,

$\frac{3\lambda}{4} = l_2 + \in$

There of $\lambda = (2l_2 - l_1)$

- Velocity of sound

$v = f\lambda = 2f\ (2l_2 - l_1)$ Where, f = frequency of tunning fork

End correction $\in = \frac{l_2 - 3l_1}{2}$

- $\nu \propto \sqrt{T}$ where, T = temperature in Kelvin

MCQ

1. A tube, closed at one end and containg air, produces, when excited,the fundamental note of frquency 512 Hz. If the tube is opened at both ends what is the fundamental freuency that can be excited (in Hz) ?

   (A) 256 (B) 1024

   (C) 128 (D) 512

2. An open pipe is suddenly closed at one endwith the result that the frequency of third harmoni of the closed pipe is found to be higher by 100 Hz than the fundamental frequency of the open pipe. What is the fundamental frequency of the open pipe ?

   (A) 240 (B) 200

   (C) 300 (D) 480

3. Two vibrating strings of the same material but of lengths Land 2L have radii 2r and r respectively. They are stretched under the same tension. Both the string vibrate in their fundamental modes. The one of length L with frequency v_1 and the other with frequency v_2. What is the ratio υ_1/υ_2 ?

   (A) 8 (B) 2 (C) 1 (D) 4

4. A closed organ pipe of length L and an open organ pipe contain gases of densities n_1 and n_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. What is the length of the open organ pipe ?

(A) $\frac{4L}{3}\sqrt{\frac{\rho_1}{\rho_2}}$ (B) $\frac{L}{3}$

(C) $\frac{4L}{3}\sqrt{\frac{\rho_2}{\rho_1}}$ (D) $\frac{4L}{3}$

5. An open pipe is in resonance in 2nd harmonic with frequency f_1. Now one end of the tube is closed and frequency is increased to f_2 such that the resonance again occurs in nth harmonic. Choose the correct option.

(A) $n = 5,\ f_2 = \frac{5}{4} f_1$ (B) $n = 3,\ f_2 = \frac{3}{4} f_1$

(C) $n = 5,\ f_2 = \frac{3}{4} f_1$ (D) $n = 3,\ f_2 = \frac{5}{4} f_1$

6. A tuning fork of 512 Hz is used to produce resonance in a resonance tube experiment. The level of water at first resonance is 30.7 cm and at second resonance is 63.2 cm. What is the error in calculating velocity of sound ? Asume the speed of sound 330 m/s.

(A) $58\ \frac{cm}{s}$ (B) $204.1\ \frac{cm}{s}$

(C) $280\ \frac{cm}{s}$ (D) $110\ \frac{cm}{s}$

7. A hollow pipe of length 0.8 m is closed at one end. At its open end a 0.5 m long uniform string is vibrating in its second harmonic and it rosonates with the fundamentals frequency of the pipe if the tension in the wire is 50 N and the speed of sound is 320 ms^{-1}, what is the mass of the string.

(A) 20 g (B) 5 g

(C) 40g (D) 10 g

8. An air column in a pipe, which is closed at one end will be in resonance with a ribrating tuning fork of frequency 264 Hz, What is the length of the column if it is in cm ? (speed of sound in air = 330 m/s)

(A) 62.50 (B) 15.62

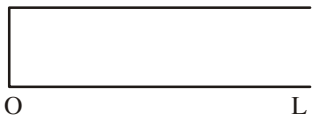
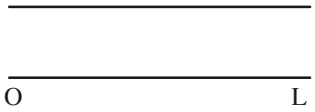
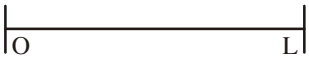
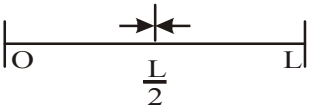
(C) 125 (D) 93.75

9. In a resonance tube experiment, the first resonance is obtained for 10 cm of air column and the second for 32 cm. The end correction for this apparatus is equal to .....

(A) 1.9 cm (B) 0.5 cm

(C) 2 cm (D) 1.0 cm

10. Column I shows four systems, each of the same length L, for producing standing waves. The lowest possible natural frequency of a system is called its fundamentals frequency. Whose wave length is denotes as λ_f . Match each system with statements given in column II describing the nature and wave length of the standing waves.

Column - I	**Column - II**
(a) Pipe closed at one end 	(p) Longitudinal waves
(b) Pipe open at both ends 	(q) Transverse waves
(c) Stretched wire clamped at both ends 	(r) $\lambda_f = L$
(d) Stretched wire clamped at both ends and at mid-point 	(s) $\lambda_f = 2L$ (t) $\lambda_f = 4L$

(A) a $\rightarrow$ p, b $\rightarrow$ p,s, c $\rightarrow$ q, d $\rightarrow$ q,r

(B) a $\rightarrow$ p, b $\rightarrow$ q,r c $\rightarrow$ p,s, d $\rightarrow$ q

(C) a $\rightarrow$ q, b $\rightarrow$ r,s, c $\rightarrow$ p, d $\rightarrow$ r

(D) a $\rightarrow$ r, b $\rightarrow$ q,r, c $\rightarrow$ p,s, d $\rightarrow$ q

KEY NOTE

1 (B)	2 (B)	3 (C)	4 (C)	5 (A)
6 (C)	7 (D)	8 (D)	9 (D)	10 (A)

HINT

1. $f_0 = \frac{v}{4\ell} \Rightarrow f_0 = \frac{v}{2\ell} = 2f_C$

2. Fundamental frequency of open pipe is $f_1 = \frac{v}{2\ell}$ and Frequency of third harmonic of closed pipe will be $f_2 = 3\left(\frac{v}{4\ell}\right)$

   Given that $f_2 - f_1 = 100$

3. Fundamental frequency is given by

   $v = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$

   (with both the end fixed)

   $\therefore$ Fundamental frequency

   $v \propto \frac{1}{l\sqrt{\mu}}$ (for same tension in both strings)

   where,

   μ = mass per unit length of wire

   $= \rho . A$

   $= \rho\left(\pi r^2\right)$ or $\sqrt{\mu} \propto r$

   $\therefore \quad v \propto \frac{1}{rl}$

   $\therefore \quad \frac{v_1}{v_2} = \left(\frac{r_2}{r_1}\right)\left(\frac{l_2}{l_1}\right) = \left(\frac{r}{2r}\right)\left(\frac{2L}{L}\right) = 1$

4. $f_C = f_0$ (both first overtone)

   $\Rightarrow 3\left(\frac{v_c}{4L}\right) = 2\left(\frac{v_0}{2l_0}\right)$

   $\therefore \quad \ell_0 = \frac{4}{3}\left(\frac{v_0}{v_C}\right)L = \frac{4}{3}\sqrt{\frac{\rho_1}{\rho_2}}\,L$ as $v \propto \frac{1}{\sqrt{\rho}}$

5. $f_1 = \frac{v}{l}$ (2nd harmonic of open pipe)

$f_2 = n\left(\frac{v}{4l}\right)$ (n^{th} harmonic of closed pipe)

Here, n is odd and $f_2 > f_1$ it is possible when n = 5 because with n = 5

$$f_2 = \frac{5}{4}\left(\frac{v}{l}\right) = \frac{5}{4} f_1$$

6. $\frac{\lambda}{2} = (63.2 - 30.7)\text{cm}$ or $\lambda = 0.65$ m

$\therefore$ speed of sound observed, $v_0 = f\lambda$

7. --- knowledge based ---

8. For closed organ pipe

$f = n\left(\frac{v}{4l}\right)$ Where n = 1, 3,5,.....

$l = \frac{nv}{4f}$

9. End correction = $\frac{l_2 - 3l_1}{2}$

Experiment - 10

Specific heat capacity of a given (i) solid and (ii) liquid by method of mixtures.

SUMMARY

$\Delta\theta = ms\,\Delta\theta$ Where,

$\Delta\theta$ = rise or fall in the temperature thebody,

$\Delta\theta$ = gain or lost amount of heat

m = mass of body

s = specific heat of body material

Heat lost = Heat gain

$$m_S . S_S . (\theta_S - \theta_e) = (m_W . S_S + m_C . S_C)(\theta_e - \theta_W)$$

Where,

S_s = Specific heat of solid

S_w = specific heat of water

S_c = specific heat of calorimeter

θs = temperature of solid when heated

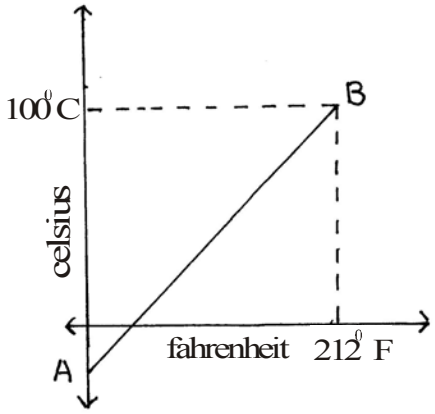
θ_w = initial temperature of water

θ_e = equillibrium temperature of mixture finally

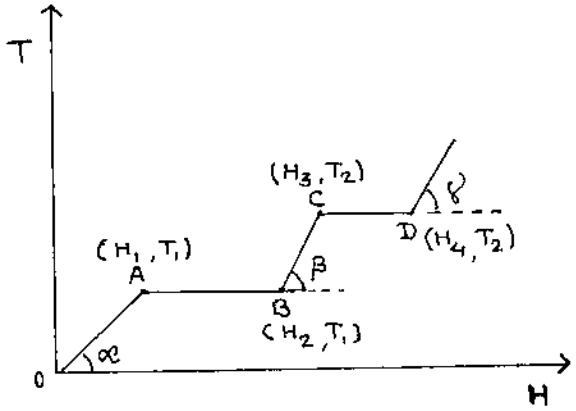
m_s, m_w and m_c is masses of solid, water and calorimeter.

MCQ

1. Amount of heat required to raise the temperature of a body through 1k is called its
   (A) water equivalent (B) Thermal capacity
   (C) entropy (D) specific heat
2. The graph AB shown in figure is a plot of temperature of body in degree celsius and degree Fahrenheit, Then.....
   (A) Slope line AB is $\frac{9}{5}$
   (B) Slope of line AB is $\frac{5}{9}$
   (C) Slope of line AB is $\frac{1}{9}$
   (D) Slope of line AB is $\frac{3}{9}$



3. The graph shows the variation of temperature (T) of one kilogram of a material with the heat (H) supplied to it, At O, the substance is in the solid state. From the graph, we can conclude that...



(A) T_2 is the melting point of the solid.

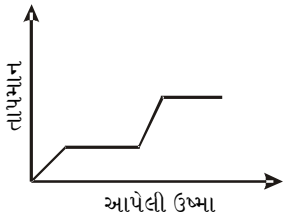
(B) BC represents the change of state from solid to liquid.

(C) (H_2-H_1) represents the latent heat of fusion of the substance.

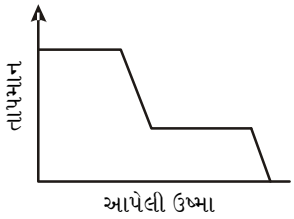
(D) (H_3-H_1) represents the latent heat of vaporization of the liquid.

4. A block of ice at -10^0C is slowly heated and converted to steam at 100^0C. Which of the following curves represents the phenomenon qualitatively.

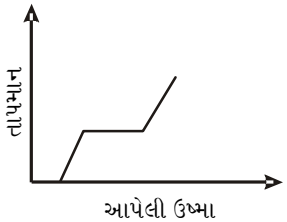
(A)



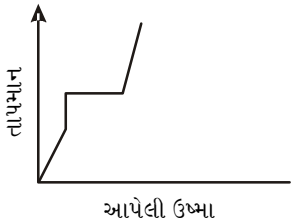
(B)



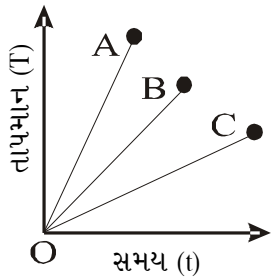
(C)



(D)



5. Which the substances A, B or C has the highest specific heat in The temperature Vs time graph is shown.



(A) A

(B) B

(C) C

(D) All have equal specific heat.

6. A centigrade and a Fahrenhait thermometer one dipped in boiling water. The water temperature is lowered until the Fahrenheit thermometer registers 140^0. What is the fall in temperatures as registered by the centigrade thermometer ?

(A) 30^0 (B) 40^0 (C) 60^0 (D) 80^0

7. 100 g ice at 0^0C placed in 100 g water at 100^0C. The final temperature of the mixture will be..... (Latent heat of ice is 80 Cal/g, and specific heat of water is 1 Cal/g C^0)

(A) 10^0C (B) 20^0C

(C) 30^0C (D) 50^0C

8. On a hot day at Ahmedabad a trucker loaded 37,000 L of diesel fuel. He delivered the disel at Shrinagar (Kashmir) Where the temperature was lower then that of Ahmedabad by 23 k. How many liters did he deliver ?For diesel $\gamma = 3 \propto = 9.50 \times 10^{-4}\ C^{o^{-1}}$

(Neglect the thermal expansion / Contration of steel tank of the trunk)

(A) 808 L (B) 36,190 L

(C) 37808 L (D) 37,000 L

9. For which value of the temperature will the values of Fahrenhit scale and Kelvin scale be equal ?

(A) 459.67 (B) 574.32

(C) -32 (D) 100

10. The temperautre of equal masses of three different liquids x, y, z are 12^0C, 19^0C and 28^0C respectively. The temperature when X and Y aremixed is 16^0C and when Y and Z are mixed is 23^0 C what is the temperature when X and Z are mixed ?

(A) 21.6^0 C (B) 18.5^0 C

(C) 23.25^0 C (D) 20.3^0 C

11. Two liquids of equal volume are throughly mixed. If their specific heat are C_1, C_2, temperatures θ_1, θ_2 and densities d_1, d_2 respectively. What is the final temperature of the mixture ?

(A) $\dfrac{d_1c_1\theta_1 + d_2c_2\theta_2}{d_1\theta_1 + d_2\theta_2}$ (B) $\dfrac{c_1\theta_1 + c_2\theta_2}{d_1c_1 + d_2c_2}$

(C) $\dfrac{d_1\theta_1 + d_2\theta_2}{c_1\theta_1 + c_2\theta_2}$ (D) $\dfrac{d_1c_1\theta_1 + d_2c_2\theta_2}{d_1c_1 + d_2c_2}$

KEY NOTE

1 (B)	2 (B)	3 (C)	4 (A)
5 (C)	6 (B)	7 (A)	8 (B)
9 (B)	10 (D)	11(D)	

HINT

1. $\theta = m.C.\Delta\theta$, if $\Delta\theta = 1k$

2. $\frac{C}{5} = \frac{F-32}{9} \Rightarrow C = \frac{5}{9}F - \frac{160}{9}$

   Equating above equation with standard equation of line.

   $y = mx + c$ we get slope of the line AB is $m = \frac{5}{9}$

3. Since in the rigion AB temperature is constant therefore at this temperature phase of the material changes from solid to liquid and (H_2 - H_1) heat will be absorb by the material. This heat is known as the heat of melting of the solid.

   Similarity in the region CD temperature is constant therefore at this temperature phase of the material changes from liquid to gas and (H_4-H_3) heat will be absorbed by the material. This heat as known as the heat of vaporisation of the liquid.

4. Initially on heating temperature rises from -10^{0}C to 0^{0}C. Then ice melts and temperature doesnot rise. After the whole ice has melted, temperature begins to rise untile it reaches 100^{0}C. Then it becomes constante, as at the boiling point will not rise.

5. If we draw a line parallel to the time axis then it cuts the given graph at three different points. Corresponding points on the times axis shows that

   $t_C > t_B > t_A \Rightarrow C_C > C_B > C_A$

6. $\frac{\Delta T_C}{100} = \frac{\Delta T_F}{180}$

7. Let temperature of mixture = T

   $\therefore$ Heat absorbed by ice = Heat lost by water

   $\therefore$ Heat required to melt ice + heat required to acquire Temperature T of water of ice

   = Heat lost by water

   $\therefore$ $mL + mc(T-0) = mc(100-T)$

8. V = 37000 L ΔT = 27K. As temperature decreases in volume.

   $\Delta V = 3\alpha V\Delta T = 9.5 \times 10^{-4} \times 37000 \times 23$

   = 808 L decreases

   $\therefore$ Diesel supplied to shrinagar

   = 37000 - 808

   =36192L =36190L

10. $T_F = \frac{9}{15}\left[T_K - 273\right] + 32$

11. When X and Y are mixed

heat lost by Y = Heat gained by X

$ms_y\ (19\text{-}16) = ms_X\ (16\text{-}12)$

Similarly when Y and Z are mixed

$ms_y\ (23\text{-}19) = ms_z\ (28\text{-}23)$

Mixing x and z yields,

$ms_X\ (T - 12^o) = ms_Z\ (28 - T)$

Mixing X and Z yields.

Solved all equations

12. $mC_1\ (\theta_1 - \theta_e) = mC_2\ (\theta_e - \theta_2)$

$Vd_1C_1\ (\theta_1 - \theta_e) = Vd_2C_2\ (\theta_e - \theta_2)$

$\Rightarrow\ d_1c_1\theta_1 - (d_1c_1)\theta_e = (d_2c_2)\,\theta_e - d_2c_2\theta_2$

Experiment - 11

To find resistivity of the material of a given wire using metre-bridge.

SUMMARY

- According to wheastone bridge principle, at the null point.

$$\frac{R}{X} = \frac{\lambda l}{\lambda(100 - l)} \Rightarrow X = \frac{R}{l}(100 - l)$$

$$\text{Resistivity} = \frac{XA}{l}$$

Where,

A = cross section area

X = resistance

l = length of resistor

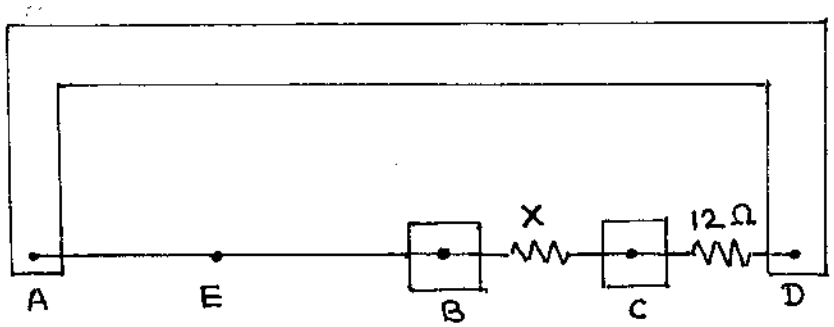
$$\rho = \frac{x.\ \pi D^2}{4l}$$

Where,

D = diameter of resister

MCQ

2. A thin uniform wire AB of length 1m, an unknown resistance X and a resistance of 12Ω are connected by thick conducting strips as shown in figure. A battery and a galvanometer (with a sliding jockey connected to it) are also available. Connections are to be measure the unknown resistance X using the principle of wheatstone bridge. The appropriate connections are.



(E is the balance point for wheastone bridge)

(A) battery across EB and galvanometer across BC

(B) battery across EC and galvanometer across BD

(C) battery across BD and galvanometer across EC

(D) battery across BC and galvanometer across CD

1. In meter bridge expriment, A thin uniform wire AB of lenth 1 m and unknown resistance x and a resistance of 12Ω are connected.

In the above question, after appropriate conneditions are made, it is found that no deflection takes places in the galvanometer wher the sliding jockey touches the wire at a distance of 60 cm from A. What is the value of the resistance X ?

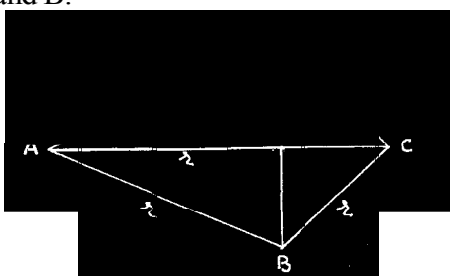
(A) 18Ω (B) Ω

(C) 16Ω (D) 4Ω

3. A wire is in the form of a tetrahedron shown in figure. The resistance of each wire is r. What is the resistance of the frame between the corners A and B.

(A) $\frac{2r}{3}$ (B) 2r

(C) r (D) $\frac{r}{2}$



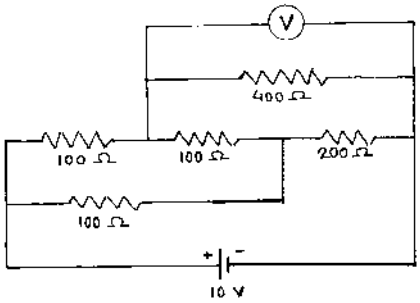
4. For the electrical circuit shown in the figure, the potential difference across the resistor of 400Ω as will be measured by the voltmeter V of resistance 400 is.....

(A) $\frac{10}{3}$V

(B) 4V

(C) $\frac{20}{3}$V

(D) 5V



5. In a simple metre-bridge circuit, the both gaps are bridge by coils P and Q having the smaller resistance. A balance is obtained when the jockey key makes contact at a point of the bridge wire 40 cm from the P end. On shunting the coil Q with a resistance of 50Ω the balance point is moved through 10 cm. What are the resistance of P and Q ?

(A) $\frac{100}{3}\Omega, \frac{100}{2}\Omega$ respectively (B) $\frac{50}{3}\Omega, \frac{50}{2}\Omega$ respectively

(C) $\frac{25}{3}\Omega, \frac{25}{2}\Omega$ respectively (D) $\frac{75}{3}\Omega, \frac{75}{2}\Omega$ respectively

6. What is the resistance of an open key ?

(A) $\propto$ (B) Can't be determined

(C) 0 (D) depends on the other resistances in the circuit

7. What is the unit of temperature coefficient of resistance ?

(A) $\Omega^{-1}\,{}^{o}C$ (B) $\Omega^{1}\,{}^{o}C^{-1}$

(C) ${}^{o}C^{-1}$ (D) $\Omega^{0}\,{}^{o}C^{-1}$

KEY NOTE

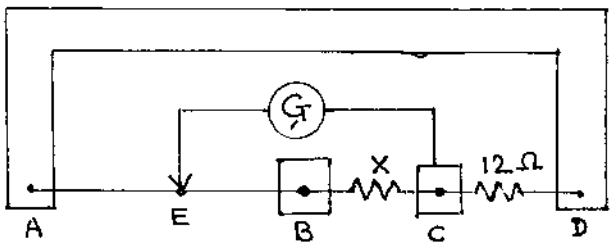
1 (C)	2 (B)	3 (D)	4 (C)
5 (B)	6 (A)	7 (C)	

HINT

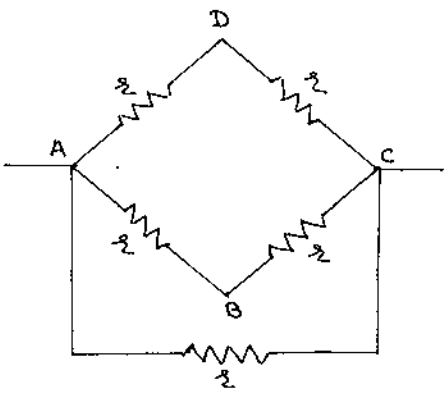
1. No deflection in the galvanometer means wheastone bridg is balanced,

$$\text{So} \quad \frac{x}{12} = \frac{R_{BJ}}{R_{AJ}} = \frac{40}{60}$$

2. The appropriate connections for wheastone bridge are as shown below.

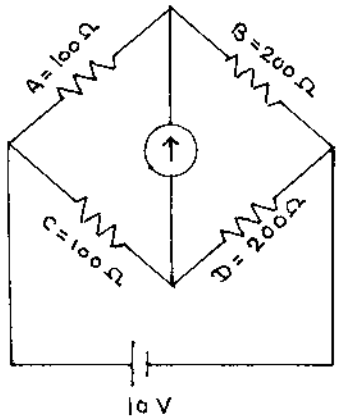


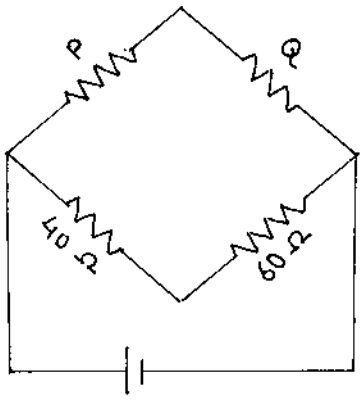
3. On redrawing the circuit, it becomes



4. The given circuit actually forms a balanced wheastone bridge (including the voltmeter) as shown below.

Voltmeter measures voltage across B as $\frac{200}{300}$ (10V)



5. 

Case 1 : $\frac{P}{40} = \frac{Q}{60}$ ....(1)

On shunting Q with 50 Ω the resistance of the arm containing Q comesdown, so length also must shorten to keep the ratio same.

$$\Rightarrow \frac{P}{(40+10)} = \frac{\left(\frac{50Q}{50+Q}\right)}{(60-10)} \Rightarrow P = \frac{50}{50+Q} \quad \text{....(2)}$$

Solving (1) and (2) we get $P = \frac{50}{3}\ \Omega$ and $\frac{50}{2}\ \Omega$

6. $R = \frac{V}{I}$ i.e. $I = 0 \Rightarrow R =$ infinite

7. Kwnoledge base

Experiment - 12

To find resistance of a given wire using Ohm's law.

SUMMARY

$R=\frac{V}{I}$

If the current is stedy for different voltage across to resistor R, the graph nature of $V \rightarrow I$ is straight line with positive slope.(i.e. temperature of resistor is constant)

Slope of graph $V \rightarrow I$ gives values of resistor.

MCQ

1. Which one of the following is not Ohm's law ?

   (J = current density, E = Electric field, ρ = resistivity and σ conductivity)

   (A) $J=\sigma E$ (B) $J=\rho E$

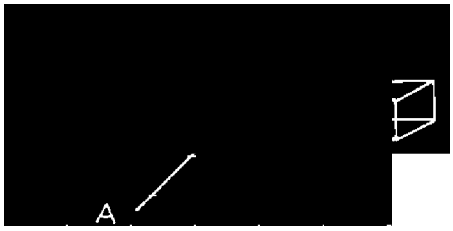
   (C) $I=V/R$ (D) $E=\rho J$

2. The external resistance of a circuit is η times higher than the internal resistance of the source. The ratio ofthe potential difference across the terminals of the source to its emf is.....

   (A) $\frac{\eta-1}{\eta}$ (B) $\frac{1}{\eta}$

   (C) $\frac{\eta}{1+\eta}$ (D) η

3. The dimensions of a conductor of specific resistance ρ are shown below :



   What is its resistance across AB ?

   (A) $\frac{\rho b}{ac}$ (B) $\frac{\rho a}{bc}$

   (C) $\frac{\rho ab}{c}$ (D) $\frac{\rho c}{ab}$

4. A wire has resistance R_O, What will its resistance if it is stretched to double, What is its length ?

(A) 4 Ro (B) 3 Ro

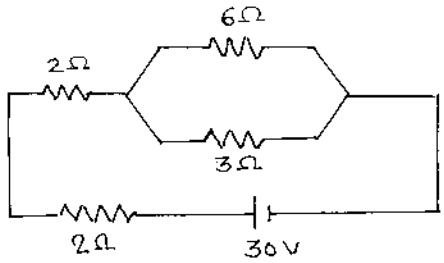
(C) Ro (D) 2 Ro

5. The temperature coefficient of resistance of a wire is $1.25 \times 10^{-3}/^0c$ At 300 K, its resistance is What is the temperature at which its resistance becomes 2Ω ?

(A) 827^0 C (B) 1127^0 C

(C) 800^0 C (D) 1100^0 C

6. The current taken from the 30 V supply and the current through the 6Ω resistor are respectively.



(A) 6A,2A (B) 6A, $\frac{5}{6}$A

(C) 5A, $\frac{5}{3}$A (D) 5A, $\frac{5}{6}$A

7. If a copper wire is stretched to make it $\eta = 0.1\%$ longer, then what is the percentage change in its resistance ?

(A) 0.2 % (B) 0.3 %

(C) 0.4 % (D) 0.1 %

8. The reciprocal of resistivity is called .....

(A) mho-m (B) conductivity

(C) retentivity (D) conductance

KEY NOTE

1 (B)	2 (C)	3 (A)	4 (A)
5 (A)	6 (C)	7 (A)	8 (B)

HINT

1. $I = \frac{V}{R}$ is obviously Ohm's law.

   $\Rightarrow J = \sigma E$ is still another form for Ohm's law.

2. $R = \eta.r$, $V = IR$, $\varepsilon = I(R + r)$

3. $R = \frac{\rho l}{A}$

   The cross-sectional area along AB is 'ac' $\Rightarrow R = \frac{pb}{ac}$

4. $R \propto (n)\,\ell$

5. $R = (1 + \alpha \triangle T)$ and $T_f = T_0 + \Delta T$

6. A current in the main circuit will divide in the inverse ratio of resistances, while splitting in the $6\,\Omega$ and $3\,\Omega$ parallel combination.

   So, $I_{6\Omega} = 5A \times \left(\frac{3\Omega}{3\Omega + 6\Omega}\right)$

7. $\frac{dR}{R} = \frac{2d\ell}{\ell}$ (As $\eta = 0.1\%$ is small, dl = 0.1 %)

8. The reciprocal of resistivity (ρ) (specific resistance) is conducting (σ) = kwodedge base

Experiment - 13

Potentio meter :

SUMMARY

(i) Comparison of emf of two primary cells.
(ii) internal resistance of a cell.

Potential different between the two points,

$V \propto \ell$,

$\Rightarrow V = k\ell$ Where,

l = length of wire at null point.

k = potential gradient

A lower value of k makes potentiometer more sensitive and accurate.

Comparison of emf E_1 and E_2 of primary cell.

$E_1 \propto l_1 \Rightarrow E_1 = kl_1$

$E_2 \propto l_2 \Rightarrow E_2 = kl_2$

$\therefore \frac{E_1}{E_2} = \frac{l_1}{l_2}$

Internal resistance of a cell

$$r = R\left(\frac{l_1 - l_2}{l_2}\right)$$

Where,

l_1 = length of the poteutiometer wire to the point where a balance point is obtained in an open circuit. i.e. K_2 is open.

l_2 = length of the potentiometer wire to the point where a balance point is obtained and current through shauted resistance R and K_2 closed.

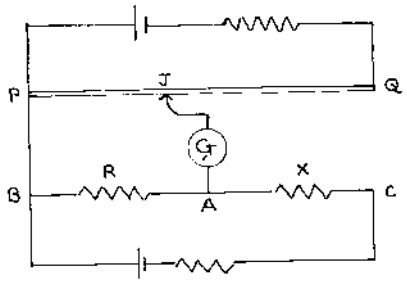
MCQ

1. A 10 m wire potentiometer is connected to an accumulator of steady voltage. A 7.8 m length of it balances the emf of a cell on 'open-circuit'. When cell delivers current through a conductor of resistance 10 Ω it is balanced against 7.0 m of the same poleutomelic. What is the internal resistance of the cell ?

   (A) 1.24Ω (B) 1.36Ω (C) 1.14Ω (D) 1Ω

2. A potentiometer is preferred over a voltmeter to measure the emf of a cell because.....
   (A) The material of the potentiometer wire has a low temperature coefficient of resistance.
   (B) emf measured by potentiometer is more accurate because the cell is open-circuit.
   (C) potetial gradients can be varied in a poteutiometer using a rheostat.
   (D) Potentiometer is more sensitive than voltmeter.

3. Circuit for the measurement of resistance by potentiometer is shown. The galvonometer is first connected at point-A and zero-deflection is observed at length PJ=10 cm. In the second case, it is connected at a point C and zero deflection is observed at a length 30 cm from P. Then what is the unknown resistance X ?



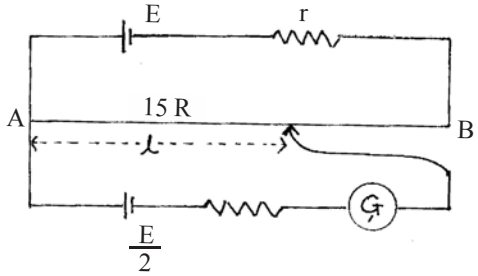
(A) $\frac{R}{2}$

(B) $\frac{R}{3}$

(C) 2R

(D) 3R

4. Consider the potentiometer circuit shown. The potentiometer wire is 600 cm long and having restance 15r At what distance from the point A should the jockey touch the wire to get zero deflection in the galvonometer ?



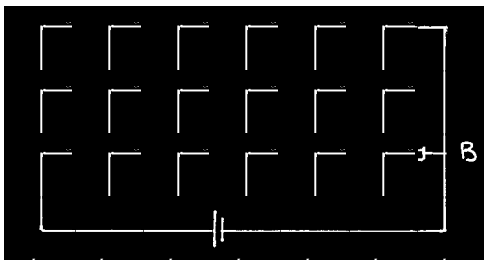
(A) 450 cm

(B) 320 cm

(C) 420 cm

(D) 360 cm

5. In the experiment of potentiometer wire AB is 100 cm long shown in figure When AC = 40 cm, no deflection occurs in the galvanometer. What is the value of R ?



(A) 15 Ω

(B) 18 Ω

(C) 12 Ω

(D) 14 Ω

6. If the deflection of the galvanometer is m the same direction at both the ends of the potentio meter wire, then point out the statement which is difinetly not the possible reason for the.

(A) current flows is the same direction in both cases through the galvanometer.

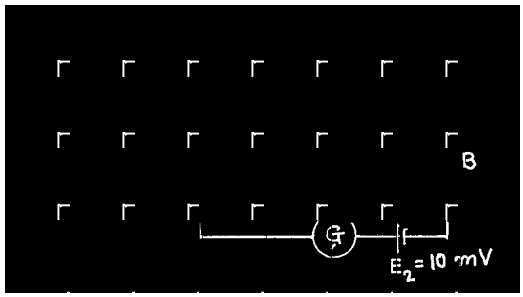
(B) Connecting wire in the galvonometer circuit is broken.

(C) The possitive terminals of all the cells are not connected at one point.

(D) The potential difference between the ends of the wire is less than the emf of the cell which is to be measured.

7. In experimant of the potentiometer wire AB of length 100 cm has a resistance of 10Ω . It is connected in series with a resistance R and a cell of emf 2 volts and of negligible internal resistance.

A source of emf 10 mV is balanced against a length of 40 cm of the poteatio meter wire. What is the value of the external resistance ?



(A) 900Ω

(B) 820Ω

(C) 790Ω

(D) 670Ω

8. A potentiometer wire, which is 4 m long is connected to the terminals of a battery of steady voltage. A leclanche cell gives a null point at 1m if the length of the potentiometer wire be increased by 1 m, What is the new new position of the null point ?

(A) 1.25 m (B) 1.4 m

(C) 1.75 m (D) 1.2 m

9. A battery of 2V and internal resistance 1 is used to send a current through a potentiometer wire of length 200 cm and resistance 4 Ω What is the potential gradieut of the wire ?

(A) 8×10^{-3} V cm^{-1} (B) 4×10^{-3} V cm^{-1}

(C) 6×10^{-3} V cm^{-1} (D) 2×10^{-3} V cm^{-1}

KEY NOTE

1 (C)	2 (B)	3 (C)	4 (B)	5 (A)
6 (B)	7 (C)	8 (A)	9 (A)	

HINT

1. Using the formula desired and used in the experiment

$$r = R\left(\frac{l_1 - l_2}{l_2}\right)$$

2. Potentiometer measures the emf at null deflection of the galvanometer.

$\varepsilon - Ir = V$

When $I = 0, \Rightarrow V = \varepsilon$

3. Let potential gradient along PQ is k, then

$V_{BA} = kl_1 = 10k, \; V_{BC} = kl_2 = 30k$

$$\frac{V_{BA}}{V_{BC}} = \frac{R}{R + X} = \frac{10}{30}$$

4. Current in $AB = \frac{\varepsilon}{16r}$

$$V_{AB} = I_{AB}(15r) = \frac{\varepsilon}{16r} \times 15r = \frac{15}{16}.\varepsilon$$

Now if the balancing length is l, then $\left(\frac{15\,\varepsilon}{16}\right)\ \left(\frac{l}{600}\right) = \frac{\varepsilon}{2}$

or $l = \frac{600 \times 16}{15 \times 2} = 320\ cm$

5. Use wheastone bridge principle
6. If connecting wire in the galvanometer circuit is broken, the galvanometer will show null deflection, as no current passes through it.
7. As potentiometer is balanced, $V_{CB} = E_2$

$$= \frac{R_{CB}.E_1}{R + R_{AB}}$$

8. Emf of cell = Potential gradient x null point on potentiometer wire,

Hence, $\frac{V_0}{4} \times 1 = \frac{V_0}{5} \times l$

9. $i = \frac{2v}{1\Omega+4\Omega} = \frac{2}{5}\ A = 0.4A$

Potential difference across wire $= (0.4A)(4\Omega) = 1.6V$

$$\text{Potential gradient} = \frac{\text{Poteutial diff. across wire}}{\text{lenght of potentiometer wire}}$$

$$= \frac{1.6v}{200\,cm} = 8 \times 10^{-3}\ \frac{V}{cm}$$

Experiment - 14

Resistance and figure of merit of a galvanometer by half-deflection method.

SUMMARY

- Galvanometer constant, $K = \left(\frac{c}{NAB}\right)\theta$

  Where, θ = deflection of galvanometer coil.
  c = t wist constant of spring
  N = number of turn of coil
  A = area of galvanometer coil
  B = radial magnetic field

- Figure of merit,

$$K = \frac{I}{\theta} = \frac{C}{NAB} = K \text{ (Galvanometer constant)}$$

- Current sensitivity,

$$S_I = \frac{\theta}{I} = \frac{NAB}{C}$$

- Voltage sensitivity,

$$S_v = \frac{\theta}{V} = \frac{NAB}{CR}$$ Where,

R = coil resistance

- Figure of merit

$$K = \frac{E}{(R+G)\theta}$$

Where, G = resistance of galvanometer coil
R = series resistance
θ = diflection in galvanometer

- Galvanometer resistance $G = \frac{RS}{R-S}$ Where, S = resistance of shant
- Shunt

$$S = \frac{I_g \,.\, G}{(I - I_g)} = \frac{G}{n-1}$$ Where,

$$n = \frac{I}{I_g}$$

Value of high series resistance,

$$R_S = \frac{V}{I_g} - G = [(n-1)\,G]$$ Where, $n = \frac{V}{V_g}$

MCQ

1. A moving coil galvanometer has 150 equal divisions, Hs current sensitivity is 10 divisions per milliampere and voltage sensitivity is 2 divisions per millivolt. In order that each divisions reads 1 V. What will be the resistance in ohms needed to be connected in series with the coil ?

   (A) 10^3 (B) 99995

   (C) 9995 (D) 10^5

2. A galvonometer of resistance 200 Ω gives full scale deflection for a current of 10^{-3} A. To convert it into an anmeter capable of measuring upto 1A. What resistance should be connected in parallel with it ?

   (A) $2 \times 10^{-1}\ \Omega$ (B) $2\ \Omega$

   (C) $2 \times 10^{-3}\ \Omega$ (D) $2 \times 10^{-6}\ \Omega$

3. A gahanometer with resistance 100 Ω is converted into an ammeter with a resistance of 0.1 . The galvanometer shows full scale deflection with current of 100 μA. Then what will be the minimum current in the circuit for fall scale deflection of galvanometer ?

   (A) 0.1001 mA (B) 100.1 mA

   (C) 1000.1 mA (D) 1.001 mA

4. The range of a galvanometer of resistance G ohm is V volt. The resistance required to be connected in series with it in order to convert it into voltmeter of range nV volt will be.....

   (A) $\frac{G}{n}$ (B) nG

   (C) (n-1)G (D) $\frac{G}{(n-1)}$

5. The seave of a galvanometer of resistance 100 Ω contains 25 divisions. It gives a deflection of 1 division on passing a current of 4×10^{-4} A. The resistance in ohm to be added to it. so that it may become a voltmeter of range 2.5 V is .....

   (A) 250 (B) 300

   (C) 150 (D) 100

6. A galvanometer of resistance 200 Ω gives full scale deflection with 15 milli-aupere current. In order to convert it into a 15V range voltmeter. What is the value of resistance connected in series ?

   (A) 1000 Ω (B) 800 Ω

   (C) 2500 Ω (D) 1500 Ω

7. A galvanometer, having a resistance of $50\,\Omega$, gives a fall scale deflection for a current of 0.05A. The length in metre of a resistance wire of are of cross-secion 2.97×10^{-2} cm^2 that can be used to convertt the galvanometer into an ammeter which can read a maximum of 5A current is (specific resistance of wire $= 5 \times 10^{-7}\ \Omega$ m)

(A) 1.5 (B) 6

(C) 8 (D) 3

8. An ammeter of range 5A is to be converted into an ammeter of range 10V. If the resistance of ammeter be 0.1 ,then what resistance should be connected in series with it ?

(A) 4.9 Ω (B) 2.1 Ω

(C) 1.1 Ω (D) 1.9 Ω

9. What is the relation between figure of merit (k) and current sensitivity (S_I) ?

(A) $S_I = k^{-1}$ (B) $S_I = \frac{k}{2}$

(C) $S_I = kV$ (D) $S_I = (k)I$

10. One microammeter has a resistance of 100 Ω and a full scale range of 50 μ A. It can be used as a voltmeter or as a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combinations (s).

(A) 5 mA range with 1 Ω resistance in parallel

(B) 10 mA range with 1 Ω resistance in parallel

(C) 10 V range with 200 k Ω resistance in series

(D) 50 V range with 10 k Ω resistance in series

KEY NOTE

1 (C)	2 (A)	3 (B)	4 (C)	5 (C)
6 (B)	7 (D)	8 (D)	9 (A)	10 (A)

HINT

1. $I_g = \dfrac{V}{(R+G)}$

2. $S = \left(\dfrac{Ig}{I - Ig}\right) G$

4. $nV = Ig\,(G + R)$

5. For full scale deflection, $Ig = nk$

$$R = \frac{V}{Ig} - G$$

6. $R = \left(\dfrac{V}{Ig}\right) - G$

7. $(I - Ig)\,S = Ig\,G$

$$R = \rho\left(\frac{\ell}{A}\right)$$

8. $R = \left(\dfrac{V}{I} - G\right)$

9. $S_I = \dfrac{I}{\theta}$ Where, as $k = \dfrac{\theta}{I} \Rightarrow S_I = \dfrac{1}{k}$

10. $I_g = \dfrac{V}{(R + G)}$

Experiment - 15

Focal length of (i) convex mirror (ii) concane mirror (iii) convex lens

SUMMARY

- For mirror,

$$f=\frac{R}{2} \text{ where ,}$$

R = Radius of curvahour
f = focal length of mirror

- Gaussian equation :

$$\frac{1}{f}=\frac{1}{u}+\frac{1}{v} \text{ or } f=\frac{uv}{u+v}$$

Where,
u = distance of object
v = distance of image

- For concave mirror f is always negative.
  Convex mirror f is positive.

Information obtained regarding the position and types images of the object placed at different positions canbe tabulated as below :

No.	Position of the object	Position of	Type of the	Size of the image
1.	At some distance from centre of curvature C (Beyond C)	Between the principle focus (F) and curvature (C)	Real and inverted	Smaller than the object
2.	On centre of curvature (C)	On centre of curvature (C)	Real and inverted	Same as that of object
3.	Between the centre of curvature (C) & the principle focus (F)	Away centre of curvature (C)	Real and inverted	Magnified
4.	Between the principle focus (F) and the pole (P)	Behind the mirror	Virtual and erect	Magnified

For mirror,

Nature of the graph of $v \rightarrow u$ is hiperbolla and nature of graph of $\frac{1}{v} \rightarrow \frac{1}{u}$ is straight line.

For

Lens :

- Gausses equation

$$\frac{1}{f} = \frac{-1}{u} + \frac{1}{v} \text{ or } f = \frac{uv}{u-v}$$

Position, Nature and Size is shown in table when object is put in different position infront of a convex on principal axis.

Sr. No.	Position of object	Position	Image Nature	size
1.	At infinity	At focus F	Real - inverted	Point size
2.	Beyond 2F	Between F and 2F on the opposite side	Real - inverted	Small as that of object
3.	At 2F	At 2F on opposite side of a object	Real-inverted	Small as that of object
4.	Between F and 2F	Beyond 2F on	Real-Inverted	Langer than object CEn large
5.	At F	At infinity	Imaginary inverted	Extremely enlarged as that of object
6.	Between F and optical centre (O)	Behind lens, same side of a object	Virtual - Erect	Enlarged as that of object

MCQ

1. A concave lense of focal length f forms an image which is n times the size of the object. What is the distance of the object from the lents ?

(A) $(1+n)\,f$ (B) $(1-n)\,f$

(C) $\left(\frac{1-n}{n}\right)f$ (D) $\left(\frac{1+n}{n}\right)f$

2. A convex lens of focal length f is placed somewhere in between an object and a screen. The distance between the object and the screen is x. If the numerical value of the magnification product by the lens is m, What is the focal length of the lens ?

(A) $\frac{mx}{(m-1)^2}$ (B) $\frac{mx}{(m+1)^2}$

(C) $\frac{(m-1)^2}{m}\,x$ (D) $\frac{(m+1)^2}{m}x$

3. A convex lens of focal length f produces a real image x times the size of an object, Then what is the distance of the object fromthe lens ?

(A) $(x+1)\,f$ (B) $(x-1)\,f$

(C) $\left\{\frac{x+1}{x}\right\}f$ (D) $\left\{\frac{x-1}{x}\right\}f$

4. A thin lens has focal length f, and its aperture has diameter d. It forms an image of intensity I. Now, the central part of the aperture upto diameter $\frac{d}{2}$ is blocked by an opeque paper. The focal length and image intensity will change to.....

(A) f and $\frac{3I}{4}$ (B) $\frac{3f}{4}$ and $\frac{I}{2}$

(C) f and $\frac{I}{4}$ (D) $\frac{f}{2}$ and $\frac{I}{2}$

5. The distance between object and the screen is D. Real images of an object are formed on the screen two positions of a lens seperated by a distance d. What will be the ratio between the sizes of two images ?

(A) $\frac{D^2}{d^2}$ (B) $\frac{D}{d}$

(C) $\sqrt{\frac{D}{d}}$ (D) $\frac{\left(D-d^2\right)}{\left(D+d\right)^2}$

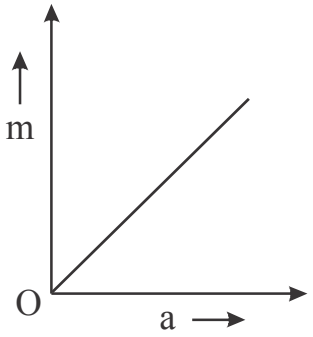
6. A spherical mirror forms an erect image three times the linear size of the object. If the distance between the object and the image is 80 cm, What is the focal length of the mirror ?

(A) 30 cm (B) 40 cm

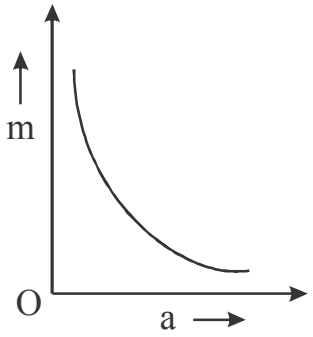
(C) -15 cm (D) 15 cm

7. Which of the following graphs is the magnifications of a real image against the distance from the focus of a concave mirror ?

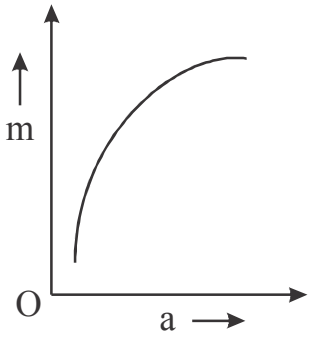
(A)



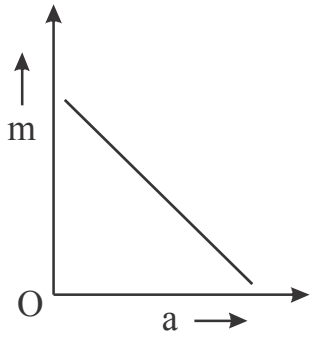
(B)



(C)



(D)



8. A short linear object of length L lies on the axis of a spherical mirror of focal length of f at a distance u from the mirror. Its image has an axial length L` equal to .....

(A) $L\left[\frac{f}{(u-f)}\right]^2$ (B) $L\left[\frac{(u-f)}{f}\right]^2$

(C) $L\left[\frac{(u+f)}{f}\right]^{\frac{1}{2}}$ (D) $L\left[\frac{f}{u-f}\right]^{\frac{1}{2}}$

9. A concave mirror of focal length f produces an images n times the size of the object. If the image is real then What is the distance of the object from the mirror ?

(A) $(n+1)f$ (B) $\left[\frac{n-1}{n}\right]f$

(C) $(n-1)f$ (D) $\left[\frac{n+1}{n}\right]f$

10. An object is placed at a distance of $\frac{f}{2}$ the from a convex lens the image will be ...

(A) at f, real and inverted

(B) at , $\frac{3f}{2}$ real and inverted

(C) at one of the foci, virtual and double its size

(D) at 2f, virtual and erect.

KEY NOTE

1 (C)	2 (B)	3 (C)	4 (A)	5 (D)
6 (A)	7 (B)	8 (A)	9 (D)	10 (C)

HINT

1. For concave lens

$\therefore\ u = \left(\frac{1-n}{n}\right)f$

2. $x = u + \upsilon$

$m = \frac{f}{(f+u)} = \frac{(f-\upsilon)}{f}$

$$u = \frac{-(m+1)}{m} f$$

$\upsilon = (m+1) f$ solved the equation

3. $v = f\left(1 - \frac{v}{u}\right) = f(1-x)$

$$\therefore u = f\frac{(1+x)}{x}$$

4. Due to blocking of central part, focal lenght does not change. However, theintensity decreases. The amount of light crossing the lens decreases by a factor of

$$\frac{\pi\left(\frac{d}{2}\right)^2}{\pi d^2} = \frac{1}{4}, \text{ Hence } I' = I - \frac{I}{4} = \frac{3I}{4}$$

5. $m_1 = \frac{I_1}{0} = \frac{\upsilon}{u} = \frac{D+d}{D-d}$

$$m_2 = \frac{I_2}{0} = \frac{\upsilon}{u} = \frac{D+d}{D-d} \quad \therefore \quad \frac{I_1}{I_2} = \frac{(D+d)^2}{(D-d)^2} \quad \text{or} \quad \frac{I_2}{I_1} = \frac{(D-d)^2}{(D+d)^2}$$

6. Knoledge base

7. $\frac{f}{u-f} = \frac{f}{a} = m$

So, m is inversely proportional to a. Hence the graph will be a rectangular hyperbola.

8. for a spherical mirrer

$$v = \frac{uf}{u-f} \quad \text{..... (1)}$$

The value of u for the two ends of the object of length are

$$u_1 = \left(u - \frac{L}{2}\right) \text{ and } u_2 = \left(u + \frac{L}{2}\right) \quad \text{..... (2)}$$

solved the eq^n. (1) and (2)

9. $u = \left(\frac{n+1}{n}\right) f$

10. $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ Here, $u = \frac{-f}{2}$

Experiment - 16

Using the parallax method, plot of angle of deviation Vs angle of incidence for a triangular prism. Reflective index of the material of the prism,

SUMMARY

$$n = \frac{\sin\left(\frac{A + \delta m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$n = \frac{\sin i}{\sin r}$ Where,

i = angle of incident

r = angle of refraction

$i + e = A + \delta$ Where,

A = Prism of angle

δ = deviation

If prism of angle is small,

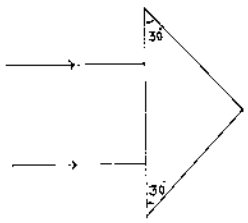
$\delta = A\,(n-1)$

If critical angle is C, reflective index

$n = \frac{1}{\sin C}$ $\quad A = r_1 + r_2$

MCQ

1. Two parallel light rays are incident at one surface of a prism of refractive index 1.5 as shown in figure. What is the angle between the rays as they emerge ?



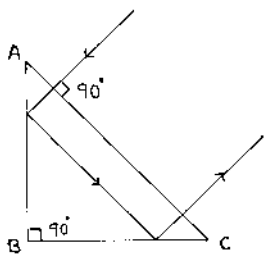
(A) 49^0

(B) 45^0

(C) 30^0

(D) 37^0

2. A ray falls on a prism ABC (AB = BC) and travels as shown in the figure. The minimum refractive index of the prism material should be .



(A) $\sqrt{2}$ (B) $\sqrt{3}$

(C) 1.5 (D) $\frac{4}{3}$

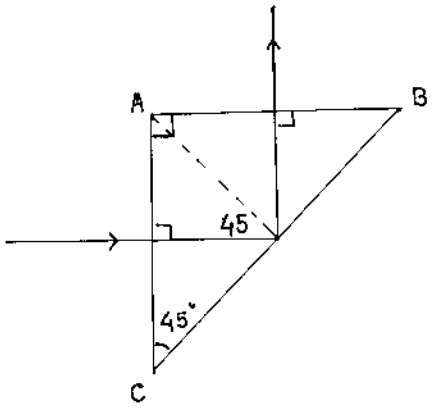
3. A light ray is incident perpendicular to one face of 90^0 prism and is totally internally reflected at the glass-air interface. If the angle of reflection is 45^0. We conclude that the refractive index.....

(A) $n > \sqrt{2}$

(B) $n > \frac{1}{\sqrt{2}}$

(C) $n < \sqrt{2}$

(D) $n < \frac{1}{\sqrt{2}}$



4. Angle of prism is A and its one surface is silvered. Light rays falling at an angle at incidence 2A on first surface return back through the samepath after suffering reflection at second silvered surface. What is the refractive index of material ?

(A) tan A (B) 2 sin A

(C) 2 cos A (D) $\cos \frac{A}{2}$

5. The minimum deviation produced by a glass prism of angle 60^0 is 30^0. If the velocity of light in vaccum is $3x10^8$ m/s. Then what is the velocity of light in glass in m/s ?

(A) 2.72×10^8 (B) 3.1×10^8

(C) 2.9×10^8 (D) 2.121×10^8

6. An equilateral prism deviates a ray through 45^0 for two angles of incidence differing by 20^0. What is the n of the prism?

(A) 1.467 (B) 1.567

(C) 1.65 (D) 1.5

7. The minimum angle of deviation of a prism of refractive index 1.732 is equal to its refracting angle. What is the angle of prism ?

(A) 45^0 (B) 30^0

(C) 60^0 (D) 40^0

8. There is a prism with refractive index equal to $\sqrt{2}$ and the refracting angle equal to 30^0. One of the refracting surfaces of the prism is polished. A beam of monochromatic light will retrace its path if its angle of incidence over the refracting surface of the prism is.....

(A) 45^0 (B) 0^0

(C) 60^0 (D) 30^0

9. A ray of light is incident normally on one of the faces of a prism of apex 30^0 and $n = \sqrt{2}$ What is the angle of deviation of the ray ?

(A) 45^0 (B) 30^0

(C) 15^0 (D) 60^0

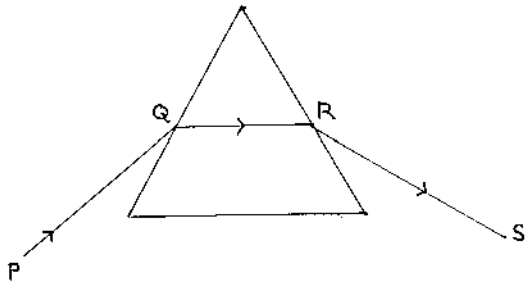
10. A ray of light is incident on an equilateral galss prism placed on a horizontal table. For minimum deviation which of the following is true ?

(A) RS is horizontal

(B) either PQ or RS is horizontal

(C) QR is horizontal

(D) PQ is horizontal

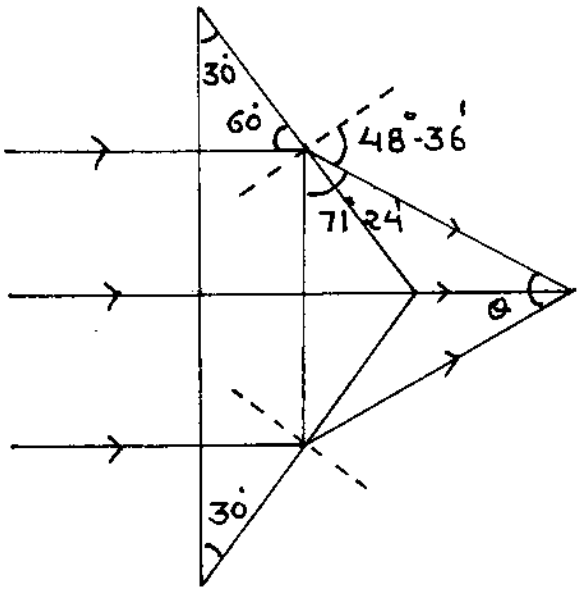


KEY NOTE

1 (D)	2 (A)	3 (A)	4 (C)	5 (D)
6 (B)	7 (C)	8 (A)	9 (C)	10 (C)

HINT

1. From the figure



$$\frac{\sin 30^o}{\sin r} = \frac{1}{n}$$

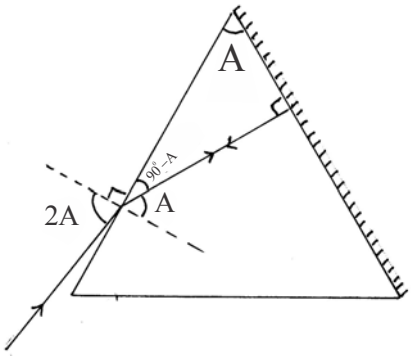
2. $n = \frac{1}{\sin C}$

3. For total internal reflection at glass air interface critical angle C must be less than 45^0.

Now $n = \frac{1}{\sin C}$

4. From figure, r = A

$\therefore n = \frac{\sin i}{\sin r}$



5. $n = \frac{\sin\left(\frac{A + \delta m}{2}\right)}{\sin\frac{A}{2}}$ Velocity of light in glass $= \frac{C_{air}}{n}$

6. $\delta = i_1 + i_2 - A$

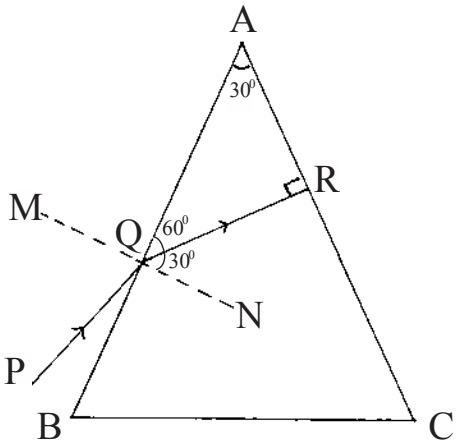
Now $n = \frac{\sin i_1}{\sin r_1} = \frac{\sin i_2}{\sin\left(60^o - r_1\right)}$

7. $n = \sin\left[\frac{A+\delta m}{2}\right]$, take $\delta m = r_1 = \frac{A}{2}$

8. $n = \frac{\sin i}{\sin r}$

$\therefore \sin i = n \times \sin r$

From the figure angle of refraction $r = 30^0$



9. Here $i = 0$ and $r_1 = 0$ Now, $A = r_1 + r_2$ $n = \frac{\sin e}{\sin r_2}$

10. When the prism is in the position of minimum deviation, the ray inside the prism is parallel to the base of the prism. So, QR is parallel to base.

Experiment - 17

Refractive index of a glass slab using a travelling microscope.

SUMMARY

Reflective index

$$n = \frac{\text{real thickness of slab}}{\text{apperent thickness of slab}}$$

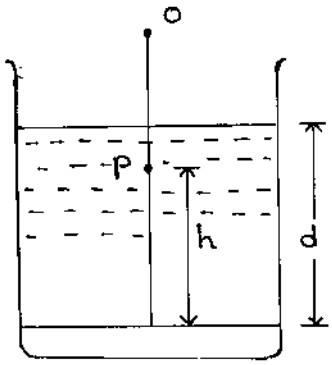
$$\frac{\text{Apparent depth}}{\text{real depth}} = \frac{n\ (\text{observer})}{n\ (\text{object})}$$

Absolute reflective index

$$n = \frac{\text{veloaly of light in vacuum}}{\text{veloa lg of light in medium}} = \frac{c}{v}$$

MCQ

1. A plane mirror is placed at the bottom of a tank containing a liquid of refractive index n. P is a small object at a height h above the mirror. An observer O, vertically above P, outside the liquid sees P and its image in the mirror. The apparent distance between these two will be.



(A) $\frac{2h}{n}$

(B) $2h\,(n-1)$

(C) $h\left[1+\left(\frac{1}{n}\right)\right]$

(D) $\frac{2n}{h}$

2. A veseel of depth t is half filled with oil of refractive index n_1 and the other half is filled with water (refractive index n_2). The apparent depth of the vessel when viewed from above is .....

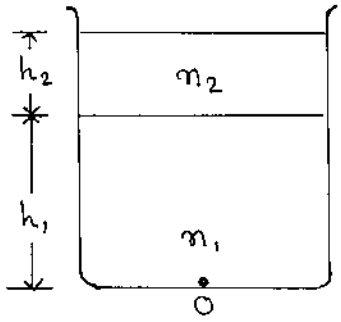
(A) $\frac{2t\,(n_1 - n_2)}{n_1 n_2}$

(B) $\frac{2t\,(n_1 + n_2)}{n_1 n_2}$

(C) $\frac{t\,(n_1 - n_2)}{2n_1 n_2}$

(D) $\frac{t\,(n_1 + n_2)}{2n_1 n_2}$

3. A tank contains two different liquids which do not mix with each other as shown in figure. The lower and upper liquids are at depths h_1 and h_2 respectively. An object O is located at the bottom. When seen vertically from above, locate the position of image of O.



(A) $\frac{h_1}{n_2} - \frac{h_2}{n_1}$ (B) $\frac{h_1}{n_2} + \frac{h_2}{n_1}$

(C) $\frac{h_1}{n_1} + \frac{h_2}{n_2}$ (D) $\frac{h_1}{n_1} - \frac{h_2}{n_2}$

4. A bird in air looks at a fish vertically below it and inside water, h_1 is the height of the bird above the surface of water and h_2, the depth of the fish below the surface of water. If refractive index of water with respect to air be n, then what is the distance of the fish observed by the bird ?

(A) $n_1h_1 + nh_2$ (B) $nh_1 + h_2$

(C) $h_1 + \frac{h_2}{n}$ (D) $h_1 + h_2$

5. Monochromatic light of wave length traveling in a medium of reflactive index n_1 euters a denser medium of refractive index n_2. What is the wave length in the second medium ?

(A) $\lambda_1\left(\frac{n_2}{n_1}\right)$ (B) $\frac{\lambda_1(n_2 - n_1)}{n_1}$

(C) $\frac{\lambda_1(n_2 - n_1)}{n_2}$ (D) $\lambda_1\left(\frac{n_1}{n_2}\right)$

6. Light travels through a glass plate of thickness t and having refractive index n. If C be the velocity of light in vacuum. What is the time taken by the light of travel this thickness of glass ?

(A) $\frac{t_C}{n}$ (B) tnC

(C) $\frac{nt}{C}$ (D) $\frac{t}{nC}$

7. A beam of light is converging towards a point I on a screen. A plane parallel plate of glass whose thickness is in the direction of beam = t, refractive index = n is introduced in the path of the beam. The convergence point is shifted by .....

(A) $t\left(1 + \frac{1}{n}\right)$ nearer (B) $t\left(1 - \frac{1}{n}\right)$ nearer

(C) $t\left(1 + \frac{1}{n}\right)$ away (D) $t\left(1 - \frac{1}{n}\right)$ away

8. The velocity of light in glass whose refractive index with respect to air is 1.5 is $2x10^8$ m/s. In a certain liquid, the velocity of light is found to be $2.5x10^8$ m/s. What is the refractive index of the liquid with respect to air?

(A) 1.44 (B) 0.80

(C) 1.20 (D) 0.64

KEY NOTE

1 (A)	2 (D)	3 (C)	4 (C)
5 (D)	6 (C)	7 (B)	8 (C)

HINT

1. The image of p will be formed at a distance h below the mirror. Apparent depth of p is $x_1 = \left(\frac{d-h}{n}\right)$

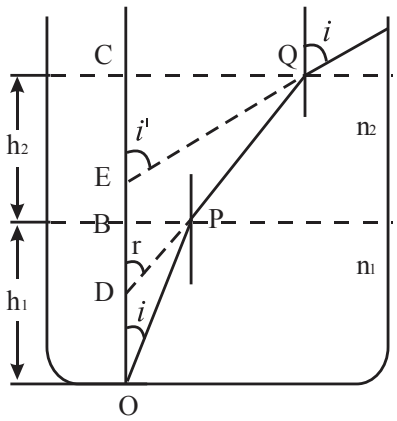
   Apparent depth of the image of p $x_2 = \left(\frac{d+h}{n}\right)$

   Apparent distance between p and its image $= x_2 - x_1$

   $= \frac{2h}{n}$

2. The appartent depth is given by $I' = \frac{t_1}{n_1} + \frac{t_2}{n_2}$

3. According to snell's Law at the point P.



   $n_1 \sin i = n_2 \sin r$

   $\therefore \frac{n_1}{n_2} = \frac{\sin r}{\sin i} = \frac{\tan r}{\tan i} = \frac{h_1}{BD}$

   $\therefore BD = hi.\frac{n_2}{n_1}$

   Similarly at the point Q, $CE = \frac{BD + BC}{n_2} = \frac{h_1}{n_1} + \frac{h_2}{n_2}$

4. Apperaht position of fish as seen by bird is $= h_1 + \frac{h_2}{n}$

5. $\lambda_1 = \frac{c}{n_1 v}$ and $\lambda_2 = \frac{c}{n_2 v}$

6. $n = \frac{\text{velocity of light in vacaum}}{\text{velocity of light in glass plate}}$

$\text{Time taken} = \frac{\text{dis tan ce}}{\text{velocity}} = \frac{t}{\left(\frac{c}{n}\right)} = \frac{nt}{c}$

7. When a glass plate of thickness t is introduced, then the optical path increases by $\frac{t}{n}$. So, the convergance point shifts nears by $t - \frac{t}{n} = t\left(1 - \frac{1}{n}\right)$

8. $n_g = \frac{C}{v_g}$ and $n_l = \frac{C}{v_l}$

$n_l = ng \frac{v_g}{v_l}$

Experiment - 18

To draw the I-V characteristic curve of a p-n junction in forward - bias and reverse - bias.

SUMMARY

Junction resistance for forward - bias.

$$r_f = \frac{\Delta V_F}{\Delta I_F}$$

Junction resistance for reservse - bias

$$r_R = \frac{\Delta V_F}{\Delta I_F}$$

MCQ

1. When P-N junction diode is forward based, then.....
   (A) Both the depletion region and barrier height are reduced.
   (B) Both depletion region and barrier height are increased.
   (C) The deplection region is winded and barrier height is reduced.
   (D) The depletion region is reduced and barrier height is increased.
2. A P-N junction is said to be forward based when
   (A) a magnetic field is applied in the region of junction.
   (B) a potential difference is applied across P and N regions making P region negative and N region positive.
   (C) not potential difference is applied across P and N regions.
   (D) a potential difference is applied across P and N regions making P region positive and N region negative.
3. In a P-N junction, there is no appreciable current if.....
   (A) a potential difference is applied across the junction
   (B) it is impossible
   (C) P-section is a made positive and N-section negative
   (D) a potential difference is applied across junction making P section netagive and N-section positive.
4. What is the resistance of P-N junction diode in forward biasing ?
   (A) zero (B) high
   (C) infinity (D) a few ohms
5. When P-N junction diode is in forward biased condition, the flow of current is mainly due to.....
   (A) both by drift and diffusion of eletrons (B) the drift of electrons
   (C) the diffusion of electrons (D) none of the a above

6. The reverse biasing in juction diode.....
   (A) increase the potential barier
   (B) increases the number of minority change carriers
   (C) increases the number of majority change carriers
   (D) decreases the potential diode
7. When a P-N junction diode is reverse biased .....
   (A) height of the potential barriers decreases
   (B) no change in the current takes place
   (C) electrons andholes move away from the junction deplection region.
   (D) electrons and holes one attracted towards each other and move towards the deplection region.
8. The electrical resistance resistance of depletion layer is large because.....
   (A) it contains electrons as change carriers
   (B) it has holes as change carriers.
   (C) it has no change carriers.
   (D) It has large number of change carriers
9. The number of minority carriers crossing the junction of diode depends primarily on the.....
   (A) magnitude of potential barrier
   (B) magnitude of the forward bias barrier.
   (C) rate of thermal generation of electron hole pair.
   (D) concentration of doping impurities.
10. In a semiconductor diode, bazzier potential offers opposition to only.....
   (A) free electrons in N region
   (B) holes in the P region
   (C) minority carriers in both regions
   (D) majority carriers in both regions.

KEY NOTE

1 (A)	2 (D)	3 (D)	4 (D)	5 (C)
6 (A)	7 (C)	8 (C)	9 (C)	10 (D)

Experiment - 19

To draw the characteristics of a zener diode and to determine its reverse breakdown voltage.

SUMMARY

Zener diode

Circuit parameters

V_I = Input (reverse blas) voltage

V_0 = Output voltage = $R_L I_L$

R_I = Input resistance

R_L = Load resistance

I_I = Input current (reverse current)

I_Z = zener diode current

I_L = Load current

Relations between parameters :

$I_L = I_I - I_Z$

$V_O = V_I - R_I I_I$

$V_O = R_L I_L$

At break-down, increase of V_I increased I_I by large amount,

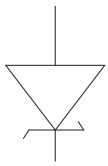
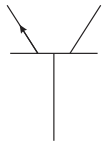
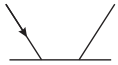
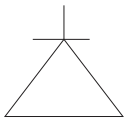
so that $V_O = V_I - R_I I_I$ becomes constout.

This constant value of V_O which is the reverse breakdown voltage, is called zener voltage V_O

$V_O = V_I - R_I I_I$

MCQ

1. In the zener diode, at VZ, the breakdown voltage.....
   (A) a large change in voltage produces an insignificant change in the current.
   (B) a large change in current produces an insignificant change in the voltage.
   (C) a small change in current causes a small change in voltage.
   (D) a small change in current can cause a large change in the voltage.
2. The zener voltage of a zener diode is kept at a desired value by.....
   (A) adjusting the input voltage
   (B) adjusting the input current
   (C) by connecting an appropriate resistance in series
   (D) changing the level of doping
3. The emission of electrons from the host atoms due to the high electric field is known as...
   (A) avalanche field emmission (B) breakdown field emission
   (C) zener field emission (D) internal field emission

4. To get a constant dc voltage from the dc unregulated output of a rectifier. We use.....
   (A) Oscillator (B) ampilifier
   (C) zener diode (D) pacifier
5. Zener diodes are represented as
   (A)  (B) 
   (C)  (D) 
6. Zener diodes are used as
   (A) amplifiers (B) voltage regulators
   (C) oscillators (D) half-wave rectifiers
7. For the same density of impurity atoms, Zener voltage is.....
   (A) same for both Ge and Si
   (B) higher for Ge than for Si
   (C) higher for silicon than for germanium
   (D) none of the above
8. When p-n junction is reverse biased.....
   (A) no current flows
   (B) majority carriers move towards the junction
   (C) minority carriers move towards the junction
   (D) both majority and minority carriers move away from the junction
9. What constitutes the reverse current ?
   (A) holes in both p and n type.
   (B) free electrons in both p and n type
   (C) free electrons in p-type and holes in n-type
   (D) holes in p-type and free electrons in n-type.
10. Avalanche breakdown in a semiconductor diode happend when.....
   (A) forward bias exceeds a certain value
   (B) forward current exceeds a certain value
   (C) reverse bias exceed a certain value
   (D) the potential barrier is reduced to zero

KEY NOTE

1 (B)	2 (D)	3 (D)	4 (C)	5 (A)
6 (B)	7 (C)	8 (C)	9 (C)	10 (C)

Experiment - 20

To study the characteristics of a common emitter n-p-n (or p-n-p) transistor and to find out the values of current and voltage gains.

SUMMARY

For common emitter transistor.

Input resistance, $R_I = \frac{\Delta V_b}{\Delta I_b}$

Output resistance, $R_0 = \frac{\Delta V_C}{\Delta I_C}$

Resistance gain, $= \frac{R_0}{R_I}$

Current gain, $\beta = \frac{\Delta I_c}{\Delta I_b}$

Voltage gain = current gain $\times$ Resistance gain

$$Av = \beta . \frac{R_0}{R_I}$$

MCQ

1. In a common emitter amplifier, using output resistance of 5000Ω and input resistance of 2000Ω, if the peak value of input signal voltage is 10 mV and $\beta = 50$ then what is the peak value of output voltage ?

   (A) 2.5×10^{-4} Volt (B) 5×10^{-6} Volt

   (C) 1.25 Volt (D) 125 Volt

2. A transistor is connected in common emitter configuration. The collector supply is 8V and the voltage drop across a resistor of 800 in the collector circuit is 0.5 V. If the current gain factor α is 0.96, then base current will be.....

   (A) 27 μA (B) 26 μA

   (C) 25 μA (D) 24 μA

3. In a transistor, a change of 8.0 mA in the emitter current produces a change of 7.8 mA in the collector current. What change in the base current is necessary to produce the same change in the collector current ?

   (A) 300 μA (B) 400 μA

   (C) 200 μA (D) 100 μA

4. In a transistor amplifier, $\beta = 62$, $R_L = 5000\ \Omega$ and internal resistance of the transistor is $500\ \Omega$. What will be the ratio of power amplification to voltage amplification ?

(A) 61 (B) 62

(C) 60 (D) 100

5. The current gain of a transistor in common base mode is 0.99. To change the emitter current by 5 mA. What will be the necessary change in collector current ?

(A) 5.1 mA (B) 4.95 mA

(C) 2.45 mA (D) 0.196 mA

6. In a common base amplifire circuit, calculate the change in base current if that in the collector current is 2mA and $\alpha = 0.98$

(A) 980 mA (B) 2 mA

(C) 0.04 mA (D) 1.96 mA

7. The current gain of a transistor is 100. If the base current changes by 200 ηA , What is the change in collector current ?

(A) 20 mA (B) 200 mA

(C) 2 mA (D) 0.2 mA

8. Which of the following the correct relationship between two current gains $\propto$ and β in a transistor ?

(A) $\beta = \dfrac{1 + \alpha}{\alpha}$ (B) $\beta = \dfrac{\alpha}{1 + \alpha}$

(C) $\beta = \dfrac{1 + \beta}{\beta}$ (D) $\beta = \dfrac{\beta}{1 + \beta}$

9. What is the relation of the transistor current ?

(A) $I_B = I_C + I_E$ (B) $I_C = I_E + I_B$

(C) $I_E = I_C + I_B$ (D) $I_C = I_E + I_B$

KEY NOTE

1 (C)	2 (B)	3 (C)	4 (B)	5 (B)
6 (C)	7 (A)	8 (D)	9 (C)	

HINT

1. The voltage gain of common emitter amplifier

$$A = \beta \left(\frac{R_L}{R_i}\right), \quad A = \frac{\text{Output Voltage}}{\text{Input Voltage}}$$

2. $\beta = \frac{\alpha}{1-\alpha}, \therefore I_B = \frac{I_C}{\beta}$

3. $\alpha = \frac{\Delta I_C}{\Delta I_B}, \qquad \beta = \frac{\alpha}{1-\alpha}, \qquad \beta = \frac{\Delta I_C}{\Delta I_B}$

4. Voltage amplification $Av = \frac{BR_L}{R_C} \Rightarrow A_P = \frac{B^2 R_L}{R_C}$

5. $\frac{\Delta I_C}{\Delta I_E} = \alpha$

6. $\beta = \frac{\alpha}{1-\alpha}$

7. $\Delta I_C = \beta \times I_b$

8. $\beta = \frac{\alpha}{1-\alpha}$

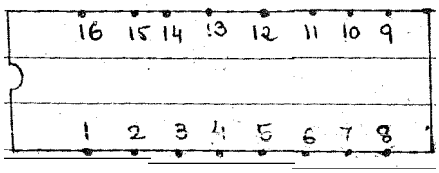
9. Knowledge Base

Experiment - 21

To identify a diode, an LED, a transistor, an IC, a resistor and a capacitor from a mixed collection of such items.

SUMMARY

- For identification, appearance and working of each item will have to be considered.
- A diode is a two terminaldevice. It conducts when forward biased and does not conduct when reverse biased. It does not emit light while conducting.
- A LED (light emitting diode) is also a two terminal device. It also conducts when forward biased and does not conduct when reverse biased. It emits light while conducting.
- A transistor is a three terminal device. The terminals represent emitter (E), base (B) and collector (C).
- An IC (intergrated circuit) is a multi terminal device in form of a chip.



- A resistor is a two terminal device. It conducts when either forward biased or reverse biased. (Infact there is no forward or reverse bias for a resistor). It conducts even when operated with A.C. voltage.
- A capacitor is also a two terminal device. It does not conduct when either forward biased or reverse biased (Hence it does not conduct with D.C. voltage). However, it conducts with A.C. voltage.
- Match column A and B type Questions

  In the column A some name of electronic component given and in the column B number of its terminal match each other.

Column A		Column B	
(1)	IC (Intigrated circuit)	(a)	Three
(2)	transistor	(b)	One
(3)	LED	(c)	Three or more than three
(4)		(d)	Two

(A) $1 \rightarrow c$, $2 \rightarrow a$, $3 \rightarrow d$

(B) $1 \rightarrow d$, $2 \rightarrow c$, $3 \rightarrow b$

(C) $1 \rightarrow c$, $2 \rightarrow b$, $3 \rightarrow a$

(D) $1 \rightarrow b$, $2 \rightarrow a$, $3 \rightarrow c$

In the experiment identify adiode, an LED,a transistor, an IC, a resistor and a capacitor from a mixed collection of such items.When the switch on the baterry eliminator the movement of the multimeter polnter given in column A. and electronic componants name given in column B.

	Column A		Column B
(1)	If pointer moves when voltage is applied in one way and does not move when reversed and there is no light emission	(a)	a LED
(2)	If pointer moves when voltage is applied in one way and does not move when reverse and there is light emission	(b)	a capacitor
(3)	If pointer moves when voltage is applied in one way and also when reversed	(c)	a diode
(4)	If pointer does not move when voltage is applied in one way and also when reversed	(d)	a resistor
		(e)	a transistor

(A) $1 \rightarrow a$, $2 \rightarrow d$, $3 \rightarrow e$, $4 \rightarrow c$

(B) $1 \rightarrow b$, $2 \rightarrow c$, $3 \rightarrow d$, $4 \rightarrow a$

(C) $1 \rightarrow c$, $2 \rightarrow a$, $3 \rightarrow d$, $4 \rightarrow b$

(D) $1 \rightarrow d$, $2 \rightarrow b$, $3 \rightarrow a$, $4 \rightarrow d$

KEY NOTE

1 (A) 2 (C)

N O T E S

N O T E S

N O T E S

N O T E S

N O T E S

N O T E S

N O T E S

N O T E S

N O T E S

N O T E S

Gujarat Secondary and Higher Secondary Education Board

Sector 10-B, Near Old Sachivalaya, Gandhinagar-382 010.